2nd September 2020 - Shift 1

Physics

Q. 1 An amplitude modulated wave is represented by the expression
$v_m = 5\,(1 + 0.6\cos 6280t)\sin\left(211 \times 10^4\, t\right)\ volt$
The minimum and maximum amplitudes of the amplitude modulated waves are, respectively:

Option 1:
$\frac{3}{2}\,V, 5\,V$

Option 2:
$2\,V, 8\,V$

Option 3:
$5\,V, 8\,V$

Option 4:
$3\,V, 5\,V$

Correct Answer:
$2\,V, 8\,V$

Solution:
$\mathrm{V_m} = 5(1 + 0.6\cos 6280\mathrm{t})\sin\left(211 \times 10^4\mathrm{t}\right)$
$\mathrm{V_m} = [5 + 3\cos 6820\mathrm{t}]\sin\left(211 \times 10^4\mathrm{t}\right)$
$\mathrm{V_{max.}} = 5 + 3 = 8$
$\mathrm{V_{min.}} = 5 - 3 = 2$

Q. 2 Magnetic materials used for making permanent magnets (P) and magnets in a transformer (T) have different properties of the following, which property best matches for the type of magnet required?

Option 1:
T: Large retentivity, small coercivity

Option 2:
P: small retentivity, Large coercivity

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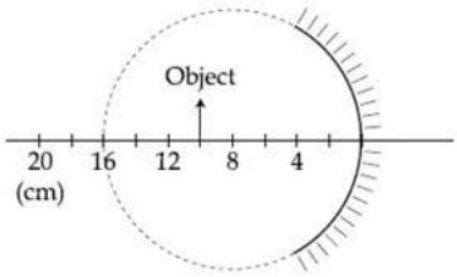
Option 3:
T: Large retentivity, Large coercivity

Option 4:
P: Large retentivity, Large coercivity

Correct Answer:
P: Large retentivity, Large coercivity

Solution:
Materials suitable for permanent magnets should have high retentivity, high coercivity, and high permeability.

Q. 3



A spherical mirror is obtained as shown in the figure from a hollow glass sphere. If an object is positioned in front of the mirror, what will be the nature and magnification of the image of the object? (Figure drawn as schematic and not to scale)

Option 1:
Inverted, real and magnified

Option 2:
Erect, virtual and magnified

Option 3:
Erect, virtual and Unmagnified

Option 4:
Inverted, real and Unmagnified

Correct Answer:
Inverted, real and Unmagnified

Solution:
Given (from the figure)

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$f = \frac{-8}{2} = -4 \text{ cm}$

$u = -10 \text{ cm}$

Now using

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} + \left(\frac{1}{-10}\right) = \frac{1}{-4}$

$\frac{1}{v} = \frac{1}{10} - \frac{1}{4}$

$\frac{1}{v} = \frac{4-10}{40}$

$v = \frac{40}{-6}$

$v = \frac{-20}{3}$

$m = \frac{-v}{u}$

$m = \frac{-\left(-\frac{20}{3}\right)}{-10} \Rightarrow m = \frac{-2}{3}$

So

the image will be real, inverted, and unmagnified.

Q. 4 The least count of the main scale of a vernier callipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scle. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale.When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is (in cm) : (VSD is vernier scale division)

Correct Answer:
3.07

Solution:

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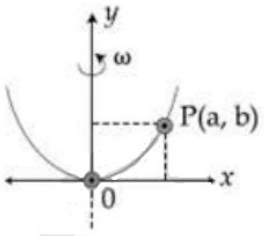
$$\text{Least count } = 1 \text{ mm or } 0.01 \text{ cm}$$
$$\text{Zero error } = 0 + 0.01 \times 7 = 0.07 \text{ cm}$$
$$\text{Reading } = 3.1 + (0.01 \times 4) - 0.07$$
$$= 3.1 + 0.04 - 0.07$$
$$= 3.1 - 0.03$$
$$= 3.07 \text{ cm}$$

Q. 5 A bead of mass m stays at point P(a,b) on a wire bent in the shape of a parabola $y = 4Cx^2$ and rotating with angular speed w (see figure). The value of w is neglect friction):



Option 1:
$2\sqrt{2gC}$

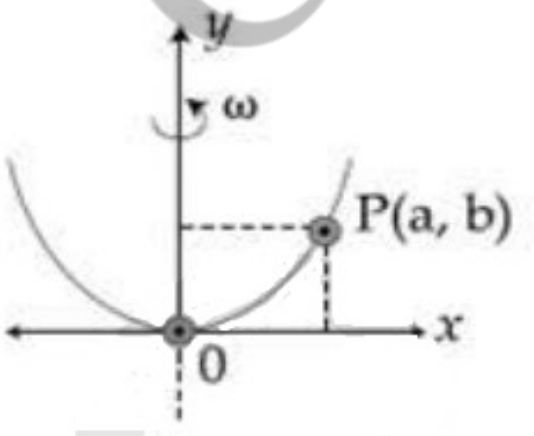
Option 2:
$2\sqrt{gC}$

Option 3:
$\sqrt{\frac{2gC}{ab}}$

Option 4:
$\sqrt{\frac{2g}{C}}$

Correct Answer:
$2\sqrt{2gC}$

Solution:



First Draw FBD for point P

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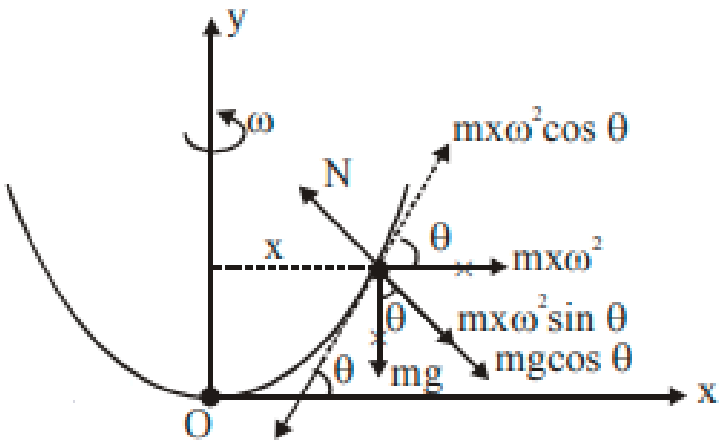

$For\ y = 4Cx^2$

$\Rightarrow \frac{dy}{dx} = 8Cx = tan\theta$

Along tangential direction of P

$mx\omega^2 \cos\theta = mg\sin\theta$

$x\omega^2 = g\tan\theta$

$x\omega^2 = g \cdot \frac{dy}{dx}$

$x\omega^2 = g \cdot (8Cx)$

$\omega^2 = 8gC$

$\omega = 2\sqrt{2gC}$

Q. 6

A particles of mass m with an initial velocity $u\hat{i}$ collides perfectly elastically with a mass 3m at rest. It moves with a velocity $v\hat{j}$ after collision, then, v is given by:

Option 1:

$v = \sqrt{\frac{2}{3}}u$

Option 2:

$v = \frac{u}{\sqrt{3}}$

Option 3:

$v = \frac{u}{\sqrt{2}}$

Option 4:

$v = \frac{1}{\sqrt{6}}u$

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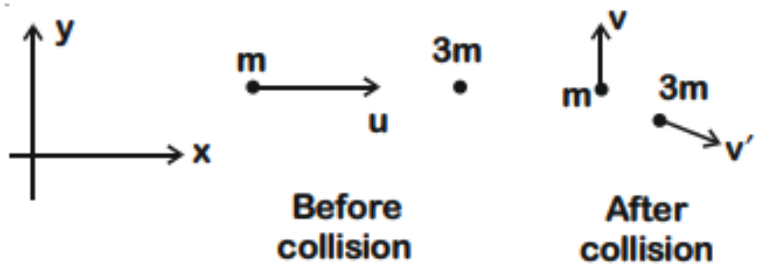
Correct Answer:

$v = \frac{u}{\sqrt{2}}$

Solution:



From momentum conservation

$\overrightarrow{P}_i = \overrightarrow{P}_f$

$m(ui) + 3\ m(0) = mvj + 3\ mv\vec{'}$

$mui - mvj = 3\ mv\vec{'}$

$\vec{v'} = \frac{ui - vj}{3}$

or $|v'| = \frac{\sqrt{u^2 + v^2}}{3}$

or $v'^2 = \frac{u^2 + v^2}{9} \ldots..(1)$

As collision is perfectly elastic hence

$k_i = k_j$

$\frac{1}{2}mu^2 + \frac{1}{2}3m * 0^2 = \frac{1}{2}mv^2 + \frac{1}{2} * 3m * v'^2$

$\Rightarrow u^2 = v^2 + 3v'^2$

Using equation (1)

$u^2 = v^2 + 3\frac{(u^2+v^2)}{9}$

$\Rightarrow 3u^2 = 3v^2 + u^2 + v^2$

$\Rightarrow 2u^2 = 4v^2$

$v = \frac{u}{\sqrt{2}}$

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Q. 7

Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity $\rho_C, \rho_T, \rho_M \; and \; \rho_A$ respectively. Then:

Option 1:
$\rho_C > \rho_A > \rho_T$

Option 2:
$\rho_M > \rho_A > \rho_C$

Option 3:
$\rho_A > \rho_T > \rho_C$

Option 4:
$\rho_A > \rho_M > \rho_C$

Correct Answer:
$\rho_M > \rho_A > \rho_C$

Solution:
$\rho_M > \rho_T > \rho_A > \rho_C$

So answer is

$\rho_M > \rho_A > \rho_C$

Q. 8

Two identical strings X and Z made of the same material have tension T_X and T_Z in them. If their fundamental frequency are 450 Hz and 300 Hz, respectively, then the ratio T_X/T_Z is :

Option 1:
2.25

Option 2:
0.44

Option 3:
1.25

Option 4:
1.5

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Correct Answer:
2.25

Solution:
the velocity of the wave in the string is given by-

$$\nu = \sqrt{\frac{T}{\mu}}$$

So

$$\mathrm{f} = \frac{1}{2\ell}\sqrt{\frac{\mathrm{T}}{\mu}}$$

For identical string l and μ will be same

$$So \quad \mathrm{f} \propto \sqrt{\mathrm{T}}$$

$$\frac{450}{300} = \sqrt{\frac{\mathrm{T_x}}{\mathrm{T_y}}}$$

$$\Rightarrow \frac{\mathrm{T_x}}{\mathrm{T_y}} = \frac{9}{4} = 2.25$$

Q. 9 A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Assuming the gases to be ideal and the oxygen bond to be rigide, the total internal energy(in units of RT) of the mixture is :

Option 1:
15

Option 2:
13

Option 3:
20

Option 4:
11

Correct Answer:
15

Solution:

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$$u_t = \frac{f_1 n_1 RT}{2} + \frac{f_2 n_2 RT}{2}$$

$$Using\ f_1 = 5\ and\ f_2 = 3$$

$$u_t = \frac{5}{2} \times 3RT + \frac{3 \times 5RT}{2} = 15RT$$

Q. 10 Train A and B are running on a parallel tracks in the opposite direction with speeds of 36 Km/hour and 72 Km/hour respectively. A person is walking in train A in the direction opposite to its motion with a speed of 1.8 km/hour. Speed (in ms^{-1}) of this person as observed from train B will close to :(take the distance between the tracks as negligible)

Correct Answer:
29.5

Solution:
The velocity of a man with respect to ground

$$\vec{V}_{m/g} = \vec{V}_{m/A} + \vec{V}_A = -1.8 + 36$$

The velocity of man w.r.t. B

$$\vec{V}_{m/B} = \vec{V}_m - \vec{V}_B = -1.8 + 36 - (-72) = 106.2\ \text{km/hr} = 29.5\ \text{m/s}$$

Q. 11 A beam of protons with speed $4 \times 10^5\ ms^{-1}$ enters a uniform magnetic field of 0.3 T at an angle of 60^0 to the magnetic field. The pitch of the resulting helical path of protons is close to: (in cm)
$(Mass\ of\ the\ : proton = 1.67 \times 10^{-27} kg, charge\ of proton = 1.69 \times 10^{-19} C$

Correct Answer:
4

Solution:

$$\text{Pitch} = \frac{2\pi m}{qB} v \cos\theta$$

$$\text{Pitch} = \frac{2(3.14)\left(1.67 \times 10^{-27}\right) \times 4 \times 10^5 \times \cos 60}{(1.69 \times 10^{-19})(0.3)}$$

$$\text{Pitch} = 0.04\ \text{m} = 4\ \text{cm}$$

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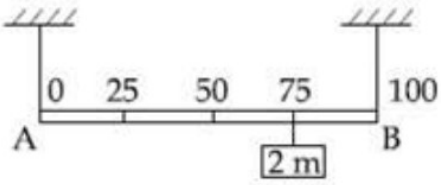
Q. 12



Shown in the figure is rigid and uniform one meter long rob AB held in horizontal position by two strings tried to its ends and attached to the ceiling. The rod is of mass 'm' and has another weight of mass 2 m hung at a distance of 75 cm from A. The tension in the string at A is :

Option 1:
0.5 mg

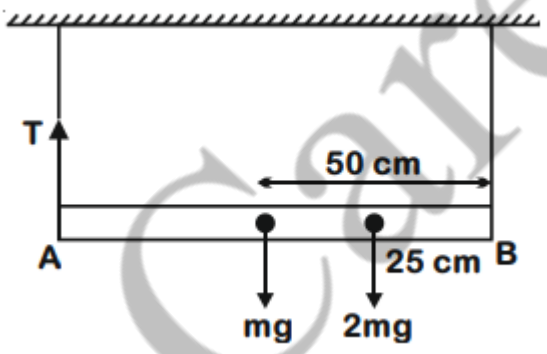
Option 2:
2 mg

Option 3:
0.75 mg

Option 4:
1 mg

Correct Answer:
1 mg

Solution:



As shown in the figure

The tension in the string at A is T

So

$$\text{As Net torque about } B = 0$$
$$\Rightarrow T \times 100 = mg \times 50 + 2mg \times 25$$
$$\Rightarrow T = mg$$

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Q. 13 A plane electromagnetic wave has a frequency of $2.0 \times 10^{10} Hz$ and its energy density is $1.02 \times 10^{-8}\ J/m^3$ in vacuum. The amplitude of the magnetic field (in nT) of the wave is close to $\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \frac{Nm^2}{C^2} and\ speed\ of\ light = 3 \times 10^8 ms^{-1}\right)$

Correct Answer:
160

Solution:

$$\text{Energy density } = \frac{\mathrm{dU}}{\mathrm{dV}} = \frac{\mathrm{B}_0^2}{2\mu_0}$$

$$1.02 \times 10^{-8} = \frac{\mathrm{B}_0^2}{2 \times 4\pi \times 10^{-7}}$$

$$\mathrm{B}_0^2 = (1.02 \times 10^{-8}) \times (8\pi \times 10^{-7})$$
$$\Rightarrow \mathrm{B}_0 = 16 \times 10^{-8}\ \mathrm{T} = 160\mathrm{nT}$$

Q. 14 A cylindrical vessel containing a liquid is rotated about its axis so that the liquid rises at its sides as shown in the figure. The radius of the vessel is 5 cm and the angular speed of rotation is $\omega\ rad\ s^{-1}$.The difference in the height, h (in cm) of liquid at the center of vessel and at the side will be:

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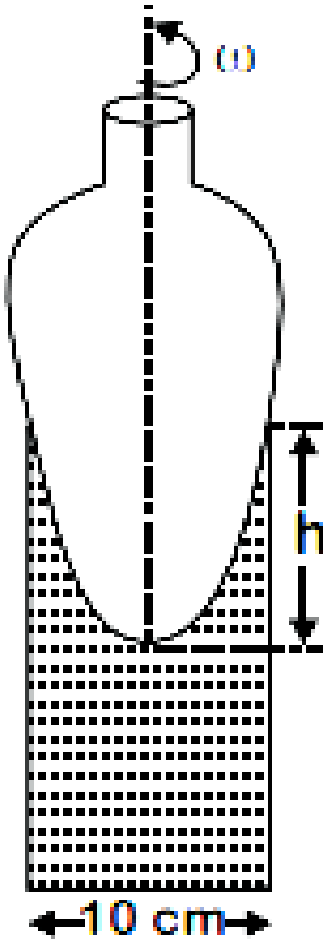

Option 1:

$\frac{2\omega^2}{25g}$

Option 2:

$\frac{5\omega^2}{2g}$

Option 3:

$\frac{25\omega^2}{2g}$

Option 4:

$\frac{2\omega^2}{5g}$

Correct Answer:

$\frac{25\omega^2}{2g}$

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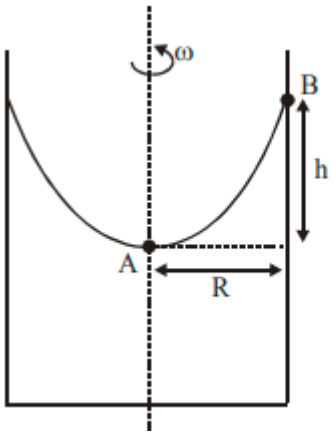
Solution:



Applying Bernoulli's equation from A to B

$$P_0 + \rho \cdot \frac{V^2}{2} - \rho g h = P_0$$

$$P_0 + \rho \cdot \frac{\omega^2 R^2}{2} - \rho g h = P_0$$

$$\Rightarrow \frac{\rho R^2 \omega^2}{2} = \rho g h$$

$$\Rightarrow \mathrm{h} = \frac{\mathrm{R}^2 \omega^2}{2\ \mathrm{g}} = (5)^2 \frac{\omega^2}{2\ \mathrm{g}} = \frac{25}{2} \frac{\omega^2}{\mathrm{g}}$$

Q. 15 In a reactor, 2 kg of $92U^{235}$ fuel is fully used up in 30 days. The energy released per fission is 200 MeV. Given that the Avogadro number, $N = 6.023 \times 10^{26}$ per kilo mole and $1\ eV = 1.6 \times 10^{-19}\ J$. The power output (in MW) of the reactor is close to:

Correct Answer:
60

Solution:

Number of uranium atoms in 2 kg=$\frac{2 \times 6.023 \times 10^{26}}{235}$

energy from one atom is $200 \times 10^6\ e.v.$

hence total energy from 2 kg uranium$= \frac{2 \times 6.023 \times 10^{26}}{235} \times 200 \times 10^6 \times 1.6 \times 10^{-19}$ J

And 2kg uranium is used in 30 days hence this energy is received in 30 days hence energy received per second or power is

$$\text{Power } = \frac{2 \times 6.023 \times 10^{26} \times 200 \times 10^6 \times 1.6 \times 10^{-19}}{235 \times 30 \times 24 \times 3600}$$

$$Power = 63.2 \times 10^6\ Watt \ \ or \ \ 63.2\ Mega\ \ Watt$$

So 60 MW is the answer as it is closest to 63.2 MW

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Q. 16 If speed V, area A and Force F are chosen as fundamental units, then the dimension of Young's modulus will be:

Option 1:

FA^2V^{-1}

Option 2:

FA^2V^{-3}

Option 3:

FA^2V^{-2}

Option 4:

$FA^{-1}V^0$

Correct Answer:

$FA^{-1}V^0$

Solution:

Let

$$Y = F^x A^y V^z$$

So

$$\mathrm{M^1\ L^{-1}\ T^{-2}} = \left[\mathrm{MLT^{-2}}\right]^{\mathrm{x}}\left[\mathrm{L^2}\right]^{\mathrm{y}}\left[\mathrm{LT^{-1}}\right]^{\mathrm{z}}$$
$$\mathrm{M^1\ L^{-1}\ T^{-2}} = [\mathrm{M}]^{\mathrm{x}}[\mathrm{L}]^{\mathrm{x+2y+z}}[\ \mathrm{T}]^{\mathrm{-2x-z}}$$

comparing the power of M and L and T

$$\mathrm{x} = 1 \ldots (1)$$
$$x + 2y + \mathrm{z} = -1 \ldots..(2)$$
$$-2x - \mathrm{z} = -2 \ldots (3)$$

after solving

$$x = 1$$
$$y = -1$$
$$\mathrm{z} = 0$$

$$Y = FA^{-1}V^0$$

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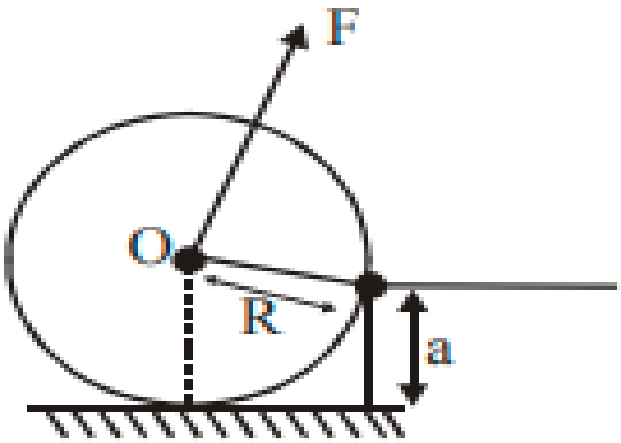
Q. 17 A uniform cylinder of mass M and radius R is to be pulled over a step of height a (a < R) by applying a force F at its center' O' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is :



Option 1:

$Mg\sqrt{1-\left(\frac{R-a}{R}\right)^2}$

Option 2:

$Mg\sqrt{\left(\frac{R}{R-a}\right)^2-1}$

Option 3:

$Mg\frac{a}{R}$

Option 4:

$Mg\sqrt{1-\frac{a^2}{R^2}}$

Correct Answer:

$Mg\sqrt{1-\left(\frac{R-a}{R}\right)^2}$

Solution:

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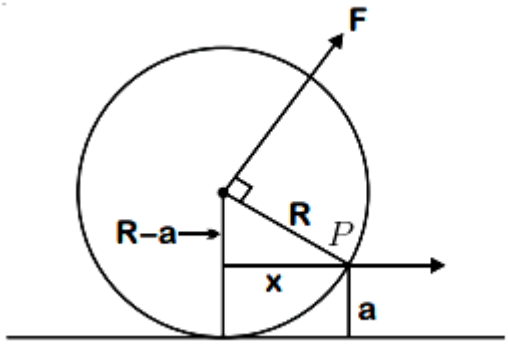

For minimum force to topple

Let the net torque about point P tends to zero

i.e

$(\tau)_P = 0$
$F_{min}.R. - \mathrm{mgx} = 0$

And $x = \sqrt{R^2 - (R-a)^2}$

So

$\mathrm{F_{min}} = \mathrm{mg}\frac{\mathrm{x}}{\mathrm{R}}$

$\mathrm{F_{min}} = \mathrm{mg}\sqrt{1 - \left(\frac{\mathrm{R-a}}{\mathrm{R}}\right)^2}$

Q. 18 Interference fringes are observed on a screen by illuminating two thin slits 1 mm apart with a light source $(\lambda = 632.8\ nm)$. The distance between the screen and the slits is 100 cm. If a bright fringe is observed on a screen at a distance of 1.27 mm from the central bright fringe, then the path difference between the waves, which are reaching this point from the slits is close to:

Option 1:
$1.27\ \mu m$

Option 2:
$2.87\ nm$

Option 3:
$2\ nm$

Option 4:
$2.05\ \mu m$

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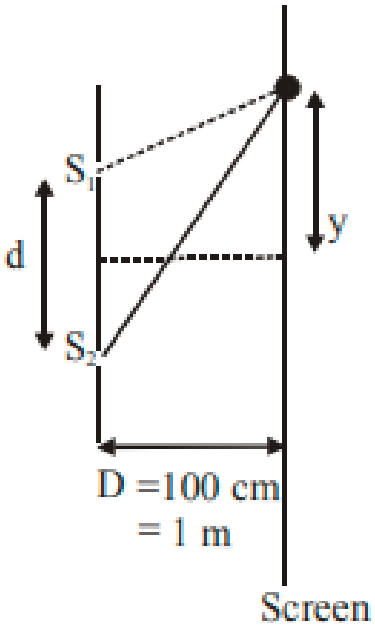
Correct Answer:

$1.27\ \mu m$

Solution:



$$y = \frac{nD\lambda}{d}$$

or

$$n = \frac{yd}{D\lambda} = \frac{1.27 \times 10^{-3} \times 10^{-3}}{1 \times 632.8 \times 10^{-9}} = 2$$

So Path difference $\Delta x = n\lambda$

$= 2 \times 632.8nm$

$= 1265.6nm$

$= 1.27\mu m$

Q. 19 The mass density of a spherical galaxy varies as $\frac{K}{r}$ over a large distance 'r' from its center. In that region, a small star is in a circular orbit of radius R. Then the period of revolution, T depends on R as:

Option 1:

$T^2 \propto R$

Option 2:

$T^2 \propto R^3$

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Option 3:

$T^2 \propto \frac{1}{R^3}$

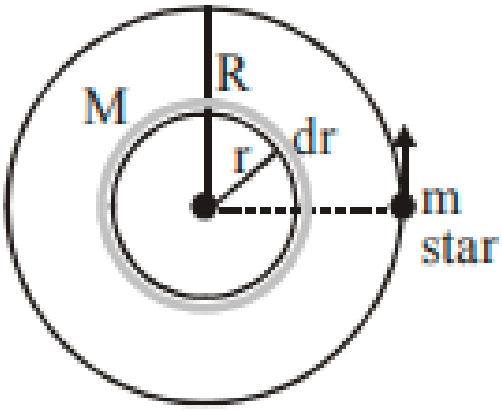
Option 4:

$T \propto R$

Correct Answer:

$T^2 \propto R$

Solution:



$\mathrm{dm} = \rho \mathrm{dv}$

$\mathrm{dm} = \left(\frac{\mathrm{k}}{\mathrm{r}}\right)\left(4\pi \mathrm{r}^2 \mathrm{dr}\right)$

$\mathrm{dm} = 4\pi \mathrm{krdr}$

$\mathrm{M} = \int_0^{\mathrm{R}} \mathrm{dm} = \int_0^{\mathrm{R}} 4\pi \mathrm{krdr}$

$\mathrm{M} = 4\pi \mathrm{k}[\frac{\mathrm{r}^2}{2}]_0^{\mathrm{R}}$

$\mathrm{M} = 2\pi \mathrm{k}\left(\mathrm{R}^2 - 0\right)$

$\mathrm{M} = 2\pi \mathrm{kR}^2$

for circular motion, gravitational force will provide the required centripetal force

i.e

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$\frac{GMm}{R^2} = \frac{mv^2}{R}$

$\frac{G\left(2\pi kR^2\right)m}{R^2} = \frac{mv^2}{R}$

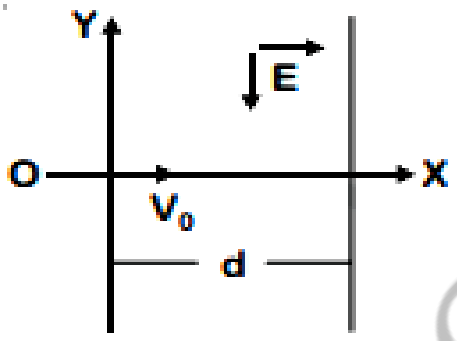
$\Rightarrow v = \sqrt{2\pi GkR}$

And Time period $= T = \frac{2\pi R}{v}$

$T = \frac{2\pi R}{(\sqrt{2\pi GkR})} \propto \sqrt{R}$

So $T^2 \propto R$

Q. 20 A charged particle (mass m and charge q) moves along X axis with velocity V_o. When It passes through the origin it enters a region having uniform electric field $\vec{E} = -E\hat{j}$ which extends upto x=d. Equation of path of electron in the region x > d is :



Option 1:

$y = \frac{qEd}{mV_0^2}(x-d)$

Option 2:

$y = \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$

Option 3:

$y = \frac{qEd}{mV_0^2}x$

Option 4:

$y = \frac{qEd^2}{mV_0^2}x$

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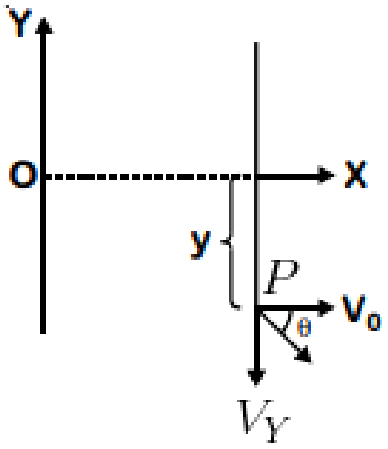
Correct Answer:

$$y = \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$$

Solution:



$For \quad x \leq d$

Along x-axis

$x = V_0 t$
$v_x = V_0$
$a_x = 0$

So for x=d

$$t = \frac{d}{V_0}$$

Now for y-axis

$$F = qE = ma$$

$$\Rightarrow a = \frac{qE}{m}$$

So

$$y = \frac{1}{2}a_y t^2$$

$$v_y = a_y t$$

$$a_y = \frac{qE}{m}$$

So at point P

Along x

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$v_{xp} = V_0$

$x_p = d$

Along y

$$v_{yp} = a_y t = \frac{qE}{m} * \frac{d}{V_0}$$

$$y_p = \frac{1}{2} a_y t^2 = \frac{1}{2} * \frac{qE}{m} * (\frac{d}{V_0})^2$$

So now for $x \geq d$

$$y = -y_p - (v_{yp} t) ....(1)$$

$$x = x_p + (v_{xp} t)$$

$$\Rightarrow t = \frac{x - x_p}{v_{xp}} .....(2)$$

Put equation (2) in equation (1)

We get

$$y = -y_p - (v_{yp} * (\frac{x - x_p}{v_{xp}}.))$$

$$y = -(\frac{1}{2} * \frac{qE}{m} * (\frac{d}{V_0})^2) - ((\frac{qE}{m} * \frac{d}{V_0}) * (\frac{x - d}{V_0}))$$

$$y = (\frac{qE}{m} * \frac{d}{(V_0)^2}) * (-\frac{d}{2} - (x - d))$$

$$y = (\frac{qE}{m} * \frac{d}{(V_0)^2}) * (\frac{d}{2} - x)$$

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Q. 21 A circular coil of radius 10 cm is placed in a uniform magnetic field of $3.0 \times 10^{-5} T$ with its plane perpendicular to the field initially. It is rotated at constant angular speed about an axis along the diameter of coil and perpendicular to magnetic field so that it undergoes half of rotation in 0.2s. The maximum value of EMF induced $(in\ \mu V)$in the coil will be close to the integer_____

Correct Answer:
15

Solution:

Given

$r = 0.1\ \mathrm{m}$

$and \quad \frac{\mathrm{T}}{2} = 0.2\mathrm{sec} \Rightarrow \quad \mathrm{T} = 0.4\ \mathrm{sec}$

$\mathrm{B} = 3 \times 10^{-5}\mathrm{T}$

At any time

$\text{flux } = \phi = BA\cos\omega t$

$$|\mathrm{emf}| = \left|\frac{\mathrm{d}\phi}{\mathrm{dt}}\right| = |\mathrm{BA}\omega\sin\omega\mathrm{t}|$$

$$(\mathrm{emf})_{\max} = \mathrm{BA}\omega = \mathrm{BA}\frac{2\pi}{\mathrm{T}}$$

$$= \frac{3 \times 10^{-5} \times \pi \times (0.1)^2 \times 2\pi}{0.4}$$

$$= \frac{6\pi^2}{4} \times 10^{-6} \qquad (\ Use\ \pi^2 \simeq 10\ )$$

$$= 15 \times 10^{-6}$$

$$= 15\mu\mathrm{V}$$

Q. 22 An engine takes in 5 moles of air at 20°C and 1 atm, and compresses it adiabaticaly to 1/10th of the orginal volume. Assuming air to be a diatomic ideal gas made up of rigid molecules, the change in its internal energy during this process comes out to be X kj. The value of X to the nearest integer is _____

Option 1:
11

Option 2:
46

Option 3:
35

Option 4:
28

Correct Answer:
46

Solution:

$\text{Given Gas is Diatomic}$

$So\ \mathrm{f} = 5$

$\gamma = 7/5$

$\mathrm{T_i} = \mathrm{T} = 273 + 20 = 293\ \mathrm{K}$

$\mathrm{V_i} = \mathrm{V}$

$\mathrm{V_f} = \mathrm{V}/10$

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For adiabatic process

$$\mathrm{TV}^{\gamma-1} = \text{ constant}$$
$$\mathrm{T}_1\ \mathrm{V}_1^{\gamma-1} = \mathrm{T}_2\ \mathrm{V}_2^{\gamma-1}$$
$$\mathrm{T}.\mathrm{V}^{7/5-1} = \mathrm{T}_2 \left(\frac{\mathrm{V}}{10}\right)^{7/5-1}$$

$$\Rightarrow \mathrm{T}_2 = \mathrm{T} * 10^{2/5}$$

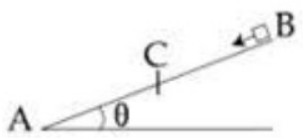
So

$$\Rightarrow T_2 = 293(10)^{2/5} = 735.98 = 736\ \mathrm{K}$$
$$\therefore \Delta T = 443\ \mathrm{K}$$

Now

$$\Delta U = \frac{\mathrm{nfR}\ (\mathrm{T}_2 - \mathrm{T}_1)}{2} = \frac{5 \times 5 \times \frac{25}{3} \times 443}{2} = 46.038 * 10^3\ J \approx 46\ \mathrm{kJ}$$

Q. 23



A small block starts slipping down from a point B on an inclined plane AB, which is making an angle θ with the horizontal. section BC is smooth and the remaining section CA is rough with a coefficient of friction μ. It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If BC = 2AC, the coefficient of friction is given by $\mu = k\ \tan\theta$. The value of k is _____

Option 1:
1

Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
3

Solution:

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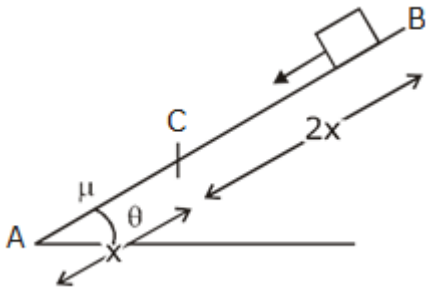


from work energy theorem

$W_g + W_f = \Delta KE$

$(mg\sin\theta * 3x) - (\mu mg\cos\theta * x) = 0 - 0$

$3\tan\theta = \mu$

$K = 3$

Q. 24 A $5\ \mu F$ capacitor is charged fully by a 220 V supply. It is then disconnected from the supply and is connected in series to another uncharged $2.5\ \mu F$ capacitor. If the energy change during the charge redistribution is $\frac{X}{100}J$ then value of X to the nearest integer is _____

Option 1:
1

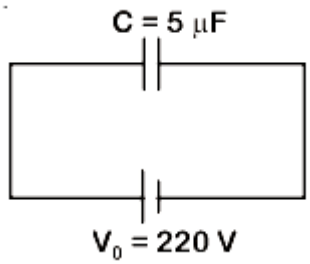
Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
4

Solution:
Initially



$Q_0 = CV_0$

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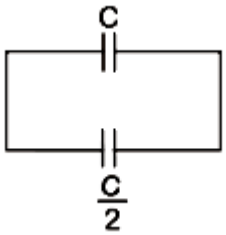
Now battery is disconnected and a capacitor of $\frac{C}{2}$ is connected to C then



$$\text{Total charge } = \mathbf{Q}_0$$

$$C_{eq.} = \frac{3C}{2}$$

$$\therefore \Delta E = \frac{Q_0^2}{2C} - \frac{2Q_0^2}{3C\times 2} \Rightarrow \Delta E = \frac{Q_0^2}{2C}\left(\frac{1}{3}\right)$$

$$\Delta E = \frac{1}{6} \times 220 \times 220 \times 5 \times 10^{-6} \approx 4 \times 10^{-2} \text{ J}$$

$$\therefore \quad X = 4$$

Q. 25 When radiation of wavelength λ is used to illuminate a metallic surface, the stopping potential is V. When the same surface is illuminated with radiation of wavelength 3λ, the stopping potential is V/4 . If the threshold wavelength for the metallic surface is $n\lambda$ then value of n will be_____

Correct Answer:
9

Solution:

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + eV \quad \ldots\ldots (i)$$

$$\frac{hc}{3\lambda} = \frac{hc}{\lambda_0} + \frac{e \cdot V}{4} \quad \ldots\ldots (ii)$$

$(multiply\ equation(ii)\ by\ 4)$

$we\ get\ \frac{4hc}{3\lambda} = \frac{4hc}{\lambda_0} + eV \quad \ldots (iii)$

$Now\ From\ (i)\ \&\ (iii)\ we\ get$

$$\frac{hc}{\lambda} - \frac{hc}{\lambda_0} = \frac{4hc}{3\lambda} - \frac{4hc}{\lambda_0}$$

$$-\frac{hc}{3\lambda} = -\frac{3hc}{\lambda_0}$$

$$\Rightarrow 9\lambda = \lambda_0$$

$$n = 9$$

Chemistry

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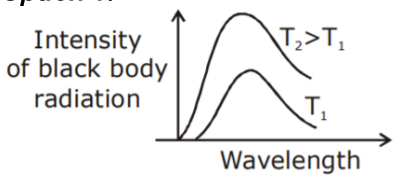
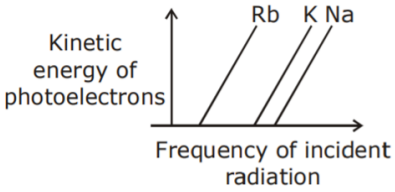
Q. 1 The figure that is not a direct manifestation of the quantumnature of atom is :

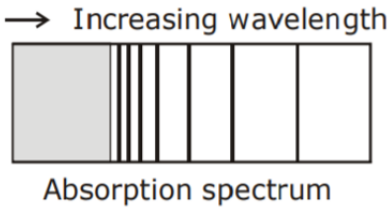
Option 1:



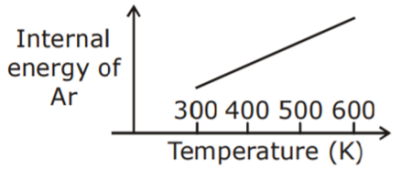
Option 2:



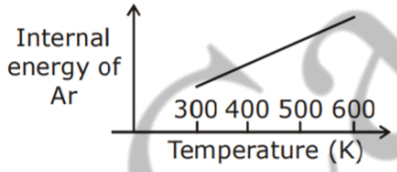
Option 3:



Option 4:



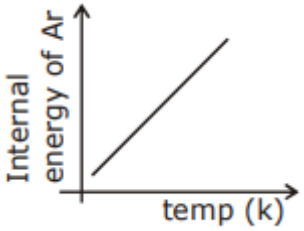
Correct Answer:



Solution:

Photoelectric effect (option 2), atomic spectrum (option 3), and Black body radiations (option 1) may be explained by quantum theory. As on increasing temperature, all the values of internal energy becomes possible, it is not directly explained from quantum theory.

The internal energy of 'Ar' or any gas, has nothing to do with Quantum nature of atom hence



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Therefore, Option 4 is correct.

Q. 2 Consider that d^6 metal ion (M^{2+}) forms a complex with aqua ligands, and the spin only magnetic moment of the complex is 4.90 BM. The geometry and the crystal field stabilization energy of the complex is :

Option 1:
octahedral and $-2.4\Delta_0 + 2P$

Option 2:
tetrahedral and $-0.6\Delta_t$

Option 3:
octahedral and $-1.6\Delta_0$

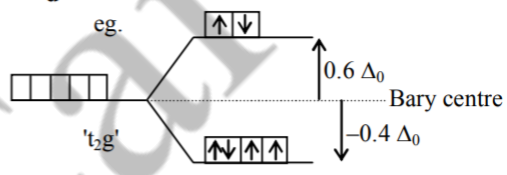
Option 4:
tetrahedral and $-1.6\Delta_t + 1P$

Correct Answer:
tetrahedral and $-0.6\Delta_t$

Solution:
If spin only magnetic moment of the complex is 4.90 BM, it means number of unpaired electrons should be 4.

(A) In octahedral complex: $[M(H_2O)_6]^{2+}$

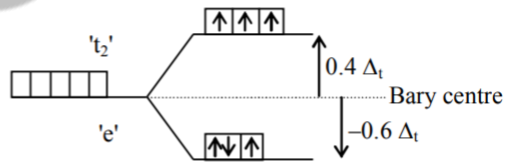
d^6



C.F.S.E. = $(-0.4\ \Delta_0) \times 4 + (+0.6\ \Delta_0) \times 2 + 0 \times P$
$= -0.4\ \Delta_0$

(B) In tetrahedral complex: $[M(H_2O)_4]^{2+}$

d^6



C.F.S.E. = $(-0.6\ \Delta_t) \times 3 + (+0.4\ \Delta_t) \times 3 + 0 \times P$
$= -0.6\ \Delta_t$

Option matching: $\text{tetrahedral and } -0.6\Delta_t$

Therefore, Option 2 is correct.

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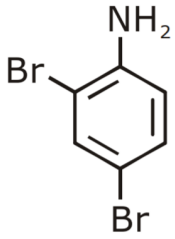
Q. 3 In Carius method of estimation of halogen, 0.172 g of an organic compound showed presence of 0.8 g of bromine. Which of these is the correct structure of the compound ?

Option 1:

$H_3C - Br$

Option 2:



Option 3:

Option 4:

$H_3C - CH_2 - Br$

Correct Answer:

Solution:
Given,

mass of organic compound = 0.172 gm

mass of Bromine = 0.08 gm

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carius method

$$\text{mass \% of 'Br'} = \frac{0.08}{0.172} \times 100 = \frac{8000}{172} = 46.51\%$$

$$\text{option (1) mass \%} = \frac{80}{95} \times 100$$

$$\text{(2) mass \%} = \frac{2 \times 80 \times 100}{252}$$

$$\text{(3) mass \%} = \frac{1 \times 80 \times 100}{80 + 72 + 6 + 14} = \frac{8000}{172}\%$$

$$\text{(4) mass \%} = \frac{1 \times 80 \times 100}{109}\%$$

Option (3) matches with the given mass percentage value

Therefore, Option 3 is correct.

Q. 4 For octahedral Mn(II) and tetrahedral Ni(II) complexes, consider the following statements :

(I) both the complexes can be high spin.

(II) Ni (II) complex can very rarely below spin.

(III) with strong field ligands, Mn (II) complexes can be low spin.

(IV) aqueous solution of Mn (II) ions is yellow in colour.

The correct statements are :

Option 1:
(I) and (II) only

Option 2:
(I),(III) and (IV) only

Option 3:
(I), (II) and (III) only

Option 4:
(II), (III) and (IV) only

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Correct Answer:
(I), (II) and (III) only

Solution:
(I) Under weak field ligand, octahedral Mn(II) (Mn^{2+} $[Ar]3d^5$) and tetrahedral Ni(II) both complexes are high spin complex.

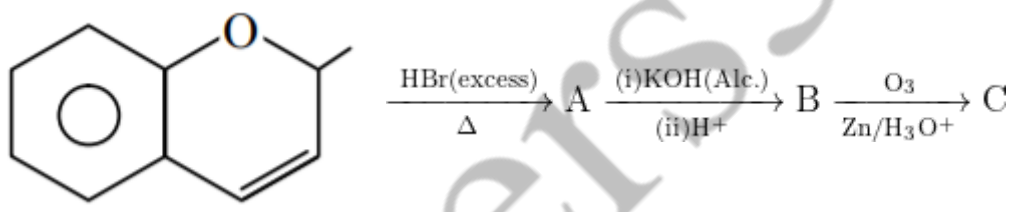
(II) Tetrahedral Ni(II) complex can very rarely be low spin because square planar (under strong ligand) complexes of Ni(II) are low spin complexes.

(III) With strong field ligands Mn (II) complexes can be low spin because they have less number of unpaired electrons (unpaired electron = 1) While with weak field ligands Mn(II) complexes can be high spin because they have more number of unpaired electron (unpaired electron = 5)

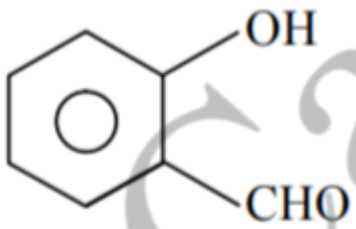
(IV) Aqueous solution of Mn(II) ions is pink in colour.

Therefore, Option 2 is correct.

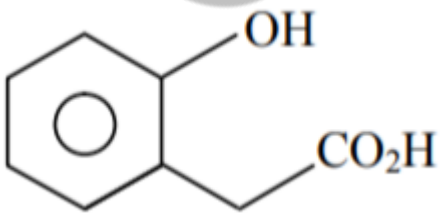
Q. 5 The major aromatic product C in the following reaction sequence will be :



Option 1:



Option 2:



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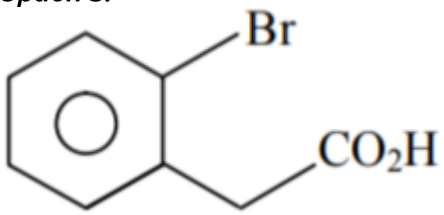
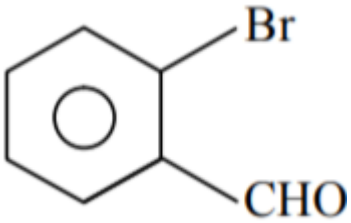
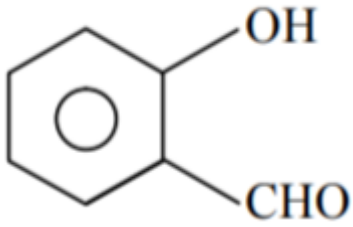
Option 3:



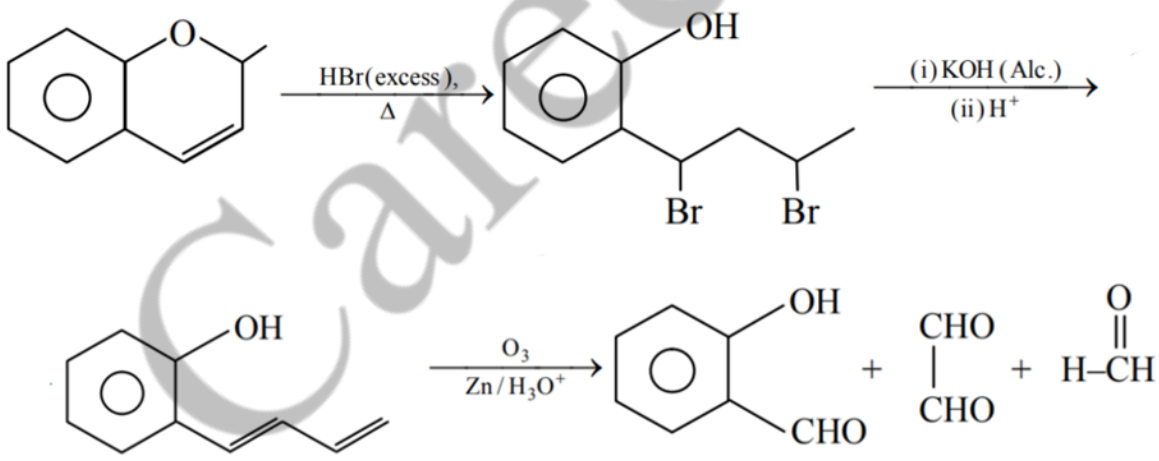
Option 4:



Correct Answer:



Solution:
Reaction will be-



Therefore, the Option 1 is correct.

Q. 6 Which of the following compounds will show retention in configuration on nucleophilic substitution by OH^- ion?

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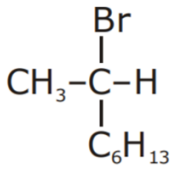
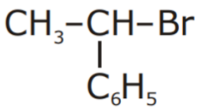
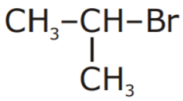
Option 1:



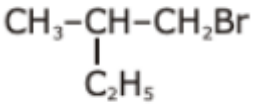
Option 2:



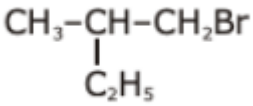
Option 3:



Option 4:



Correct Answer:



Solution:

The reactions will be -

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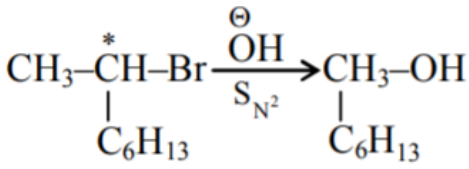
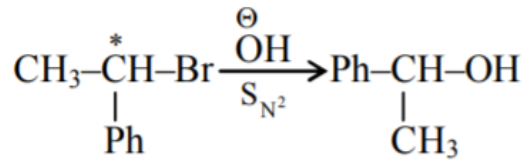
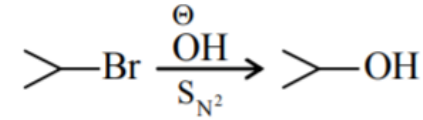
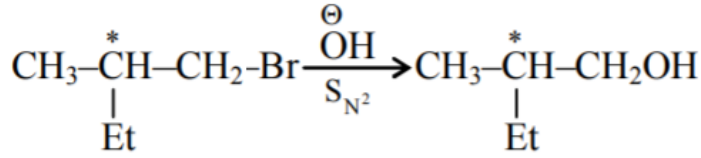
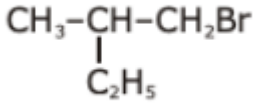
(1) 

(2) 

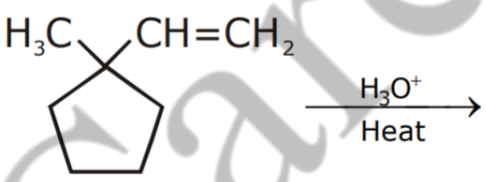
(3) 

(4) 

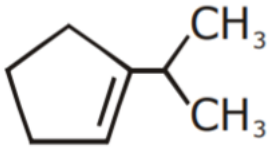
In  attack of OH^- is not on chiral carbon, it is on the carbon adjacent to chiral carbon, so the configuration of chiral carbon remains same.

Therefore, Option 4 is correct.

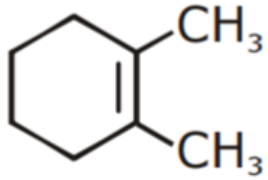
Q. 7 The major product in the following reaction is :



Option 1:



Option 2:



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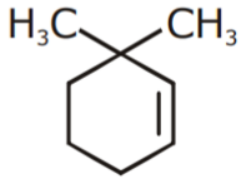
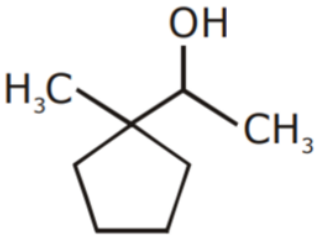
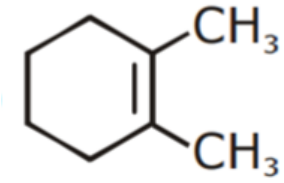
Option 3:



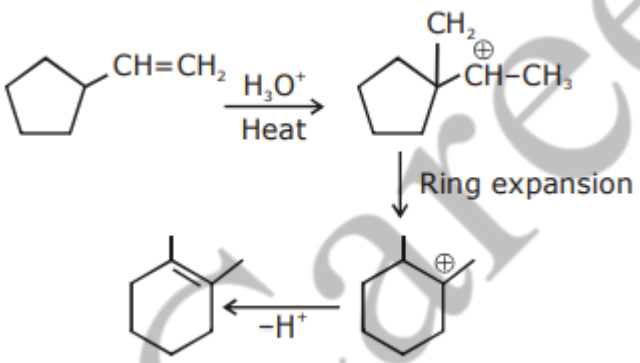
Option 4:



Correct Answer:



Solution:
The reaction will be -



Therefore, Option 2 is correct.

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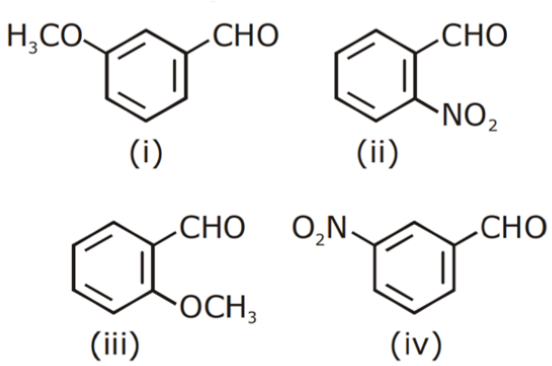
Q. 8 The increasing order of the following compounds towards HCN addition is :



Option 1:
(i) < (iii) < (iv) < (ii)

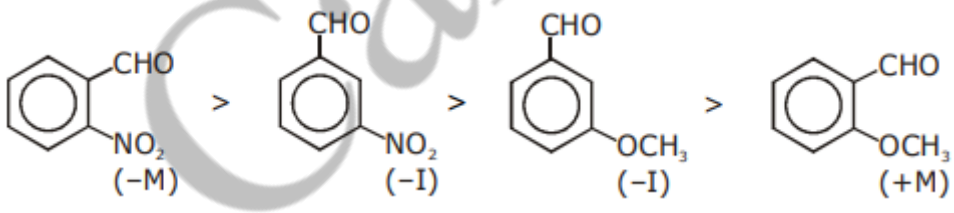
Option 2:
(iii) < (iv) < (i) < (ii)

Option 3:
(iii) < (i) < (iv) < (ii)

Option 4:
(iii) < (iv) < (ii) < (i)

Correct Answer:
(iii) < (i) < (iv) < (ii)

Solution:
In HCN, CN^- is acts as nucleophile, attack first that -CHO group which has maximum positive charge. The magnitude of the (+ve) charge increases by -M and -I group. So reactivity order will be



Therefore, Option 3 is correct.

Q. 9 In general, the property (magnitudes only) that shows an opposite trend in comparison to other properties across a period is :

Option 1:
Ionization enthalpy

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Option 2:
Electronegativity

Option 3:
Electron gain enthalpy

Option 4:
Atomic radius

Correct Answer:
Atomic radius

Solution:
In general across a period atomic radius decreases while ionisation enthalpy, electron gain enthalpy and electronegativity increases because effective nuclear charge (Z_{eff}) increases

Therefore, Option 4 is correct.

Q. 10 Consider the following reactions :

$$\text{(i) Glucose} + \text{ROH} \xrightarrow{\text{dry HCl}} \text{Acetal} \xrightarrow[(CH_3CO)_2O]{\text{x eq. of}} \text{acetyl derivative}$$

$$\text{(ii) Glucose} \xrightarrow{Ni/H_2} \text{A} \xrightarrow[(CH_3CO)_2O]{\text{y eq.of}} \text{acetyl derivative}$$

$$\text{(iii) Glucose} \xrightarrow[(CH_3CO)_2O]{\text{z eq.of}} \text{acetyl derivative}$$

'x' , 'y' and 'z' in these reactions are respectively.

Option 1:
5,4 and 5

Option 2:
4,6 and 5

Option 3:
4,5 and 5

Option 4:
5,6 and 5

Correct Answer:
4,6 and 5

Solution:

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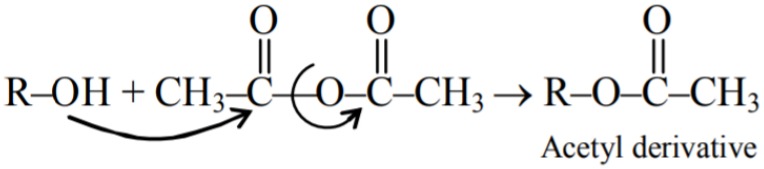
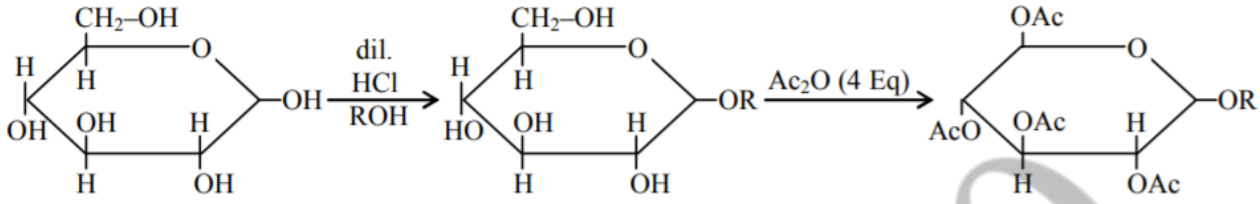
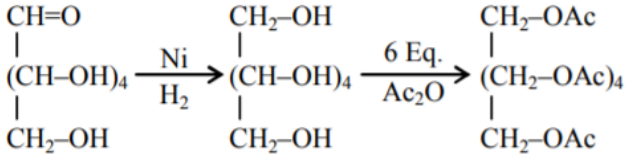
Due to the presence of –OH group in Glucose the reaction is-



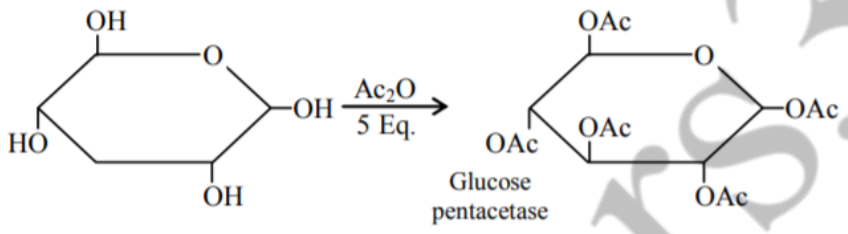
(i)



(ii)



(iii)



$(CH_3CO)_2O$ reacts with –OH group to form acetyl derivative, so as the no. of –OH group no. of eq. of $(CH_3CO)_2O$ will be used

So, x = 4 y = 6 z = 5

Q. 11 The metal mainly used in devising photoelectric cells is :

Option 1:
Na

Option 2:
Li

Option 3:
Rb

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Option 4:
Cs

Correct Answer:
Cs

Solution:
Cs used in photoelectric cell as it has least ionization energy.

Q. 12 For the following Assertion and Reason, the correct option is

Assertion (A) : When Cu (II) and sulphide ions are mixed, they react together extremely quickly to give a solid.

Reason (R): The equilibrium constant of $\mathrm{Cu^{2+}(aq) + S^{2-}(aq) \rightleftharpoons CuS(s)}$ is high because the solubility product is low.

Option 1:
(A) is false and (R) is true.

Option 2:
Both (A) and (R) are false.

Option 3:
Both (A) and (R) are true but (R) is not the explanation for (A).

Option 4:
Both (A) and (R) are true and (R) is the explanation for (A).

Correct Answer:
Both (A) and (R) are true but (R) is not the explanation for (A).

Solution:
(A) is (R) true &
(R) is not the explanation of (A)

The rate of chemical reaction has nothing to do with the value of the equilibrium constant.

Therefore, Option 3 is correct.

Q. 13 On heating compound (A) gives a gas (B) which is a constituent of air. This gas when treated with H_2 in the presence of a catalyst gives another gas (C) which is basic in nature. (A) should not be :

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Option 1:
NaN_3

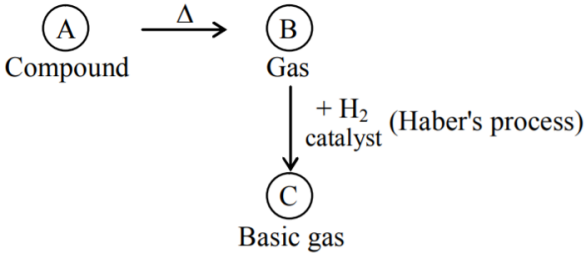
Option 2:
$Pb(NO_3)_2$

Option 3:
$(NH_4)_2Cr_2O_7$

Option 4:
NH_4NO_2

Correct Answer:
$Pb(NO_3)_2$

Solution:



Basic gas (C) must be ammonia (NH_3). It means (B) gas should be N_2 which is formed by the heating of compound (A).

(1) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2\uparrow + Cr_2O_3 + 4H_2O\uparrow$

(2) $Pb(NO_3)_2 \xrightarrow{\Delta} PbO + 2NO_2\uparrow + \frac{1}{2}O_2\uparrow$

(3) $2NaN_3 \xrightarrow{\Delta} 2Na + 3N_2\uparrow$

(4) $NH_4NO_2 \xrightarrow{\Delta} N_2\uparrow + 2H_2O\uparrow$

$Pb(NO_3)_2$ is not giving Nitrogen gas.

Q. 14 An open beaker of water in equilibrium with water vapour is in a sealed container. When few grams of glucose are added to the beaker of water, the rate at which water molecules:

Option 1:
leaves the vapour increases

Option 2:
leaves the solution increases

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Option 3:
leaves the solution decreases

Option 4:
leaves the vapour decreases

Correct Answer:
leaves the vapour increases

Solution:
With the addition of solute in solvent, surface area for vapourisation decreases causes a lowering in vapour pressure.

To maintain the equilibrium vapour increases while vapour pressure decreases.

Q. 15 The statement that is not true about ozone is :

Option 1:
in the stratosphere, CFCs release chlorine free radicals (Cl) which reacts with O_3 to give chlorine dioxide radicals.

Option 2:
in the atmosphere, it is depleted by CFCs.

Option 3:
in the stratosphere, it forms a protective shield against UV radiation.

Option 4:
it is a toxic gas and its reaction with No gives NO_2

Correct Answer:
in the stratosphere, it forms a protective shield against UV radiation.

Solution:
In the stratosphere, CFCs release chlorine-free radicals ($Cl^{\cdot}$).

$$CF_2Cl_2(\ g) \xrightarrow{UV} Cl(g) + CF_2Cl(g)$$

Which reacts with O_3 to give chlorine oxide ClO radical not chlorine dioxide ClO_2 radical.

$$Cl(g) + O_3(\ g) \rightarrow ClO(g) + O_2(\ g)$$

Q. 16 Which of the following is used for the preparation of colloids ?

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Option 1:
Ostwald Process

Option 2:
Van Arkel Method

Option 3:
Bredig's Arc Method

Option 4:
Mond Process

Correct Answer:
Bredig's Arc Method

Solution:
Bredig's Arc method is used to form metal colloids.

Q. 17 If AB_4 molecules is a polar molecule, a possible geometry of AB_4 is :

Option 1:
Square pyramidal

Option 2:
Tetrahedral

Option 3:
Rectangular planar

Option 4:
Square planar

Correct Answer:
Square pyramidal

Solution:
(1) If AB_4 molecule is a square pyramidal then it has one lone pair and their structure should be

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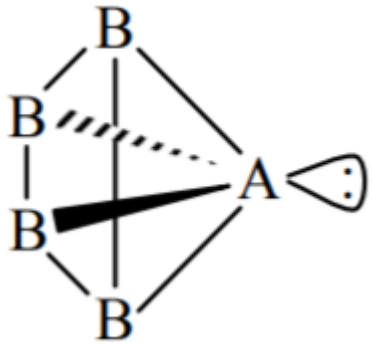
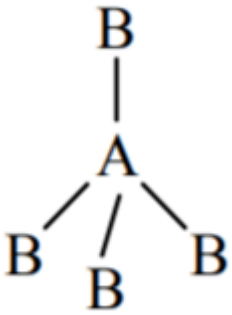


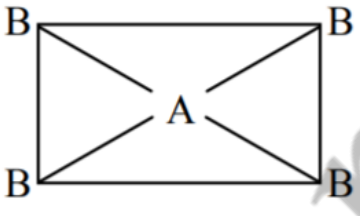
and it should be polar because dipole moment of lone pair of 'A' never be cancelled by others.

(2) If AB_4 molecule is a tetrahedral then it has no lone pair and their structure should be



and it should be non polar due to perfect symmetry.

(3) If AB_4 molecule is a rectangular planar then



it should be non-polar because the vector sum of dipole moment is zero.

(4) If AB_4 molecule is a square planar then

it should be non-polar because the vector sum of the dipole moment is zero.

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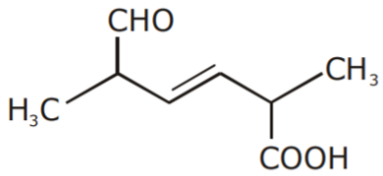
Q. 18 The IUPAC name for the following compound is :



Option 1:
2,5-dimethyl-5-carboxy-hex-3-enal

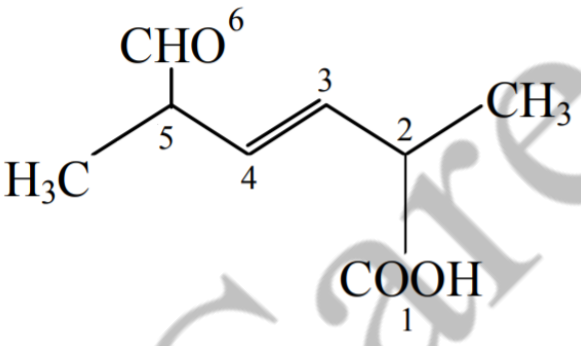
Option 2:
2,5-dimethyl-6-carboxy-hex-3-enal

Option 3:
2,5-dimethyl-6-oxo-hex-3-enoic acid

Option 4:
6-formyl-2-methyl-hex-3-enoic acid

Correct Answer:
2,5-dimethyl-6-oxo-hex-3-enoic acid

Solution:



IUPAC Name: 2, 5-Dimethyl-6-oxo-hex-3-enoic acid

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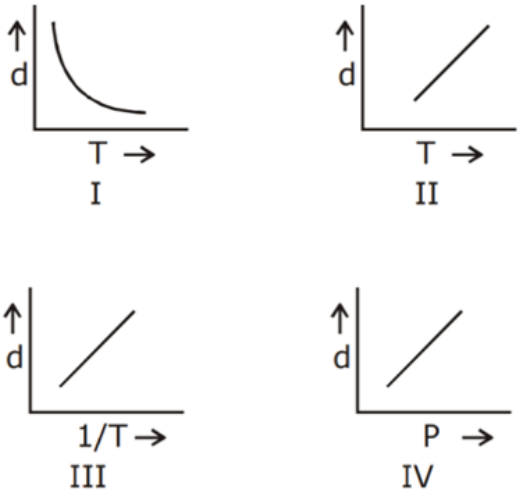
Q. 19 Which one of the following graphs is not correct for ideal gas ?



d = Density, P = Pressure, T = Temperature

Option 1:
I

Option 2:
II

Option 3:
IV

Option 4:
III

Correct Answer:
II

Solution:
For ideal Gas

$$d = \frac{P \times M}{RT}$$

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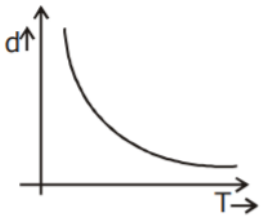
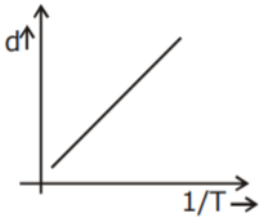
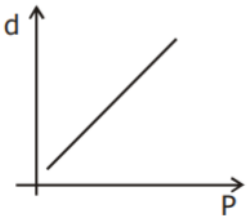
d v/s T → Hyperbolic



d v/s $\frac{1}{T}$ → St. line



d v/s p → St line



'II' Graph is incorrect.

Q. 20 While titrating dilute HCl solution with aqueous NaOH, which of the following will not be required ?

Option 1:
Burette and porcelain tile

Option 2:
Pipette and distilled water

Option 3:
Clamp and phenolphthalein

Option 4:
Bunsen burner and measuring cylinder

Correct Answer:
Bunsen burner and measuring cylinder

Solution:
Bunsen Burner & measuring cylinder are not Required.

As titration is already on exothermic process

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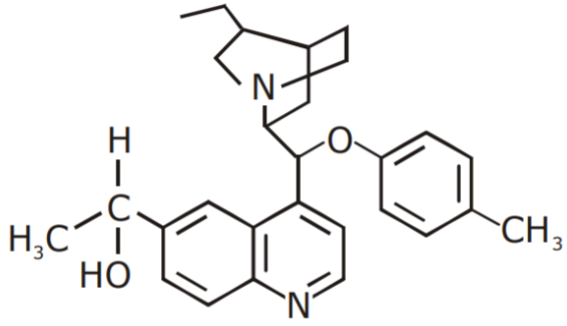
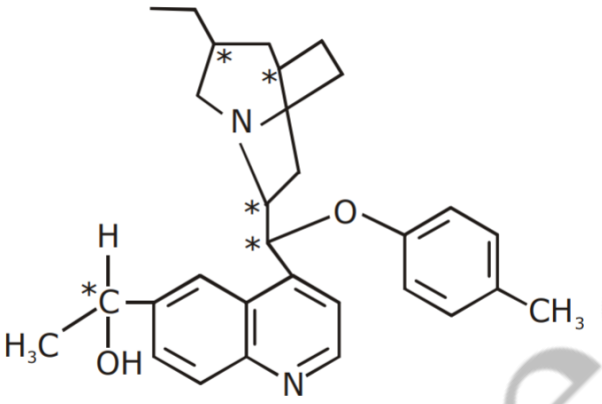
Q. 21 The number of chiral carbons present in the molecule given below is _____.



Correct Answer:
5

Solution:
Total chiral carbon = 5



Q. 22 The oxidation states of iron atoms in compounds (A), (B) and (C), respectively, are x, y and z. The sum of x, y and z is_____.

$Na_4[Fe(CN)_5(NOS)]$ (A) $Na_4[FeO_4]$ (B) $[Fe_2(CO)_9]$ (C)

Correct Answer:
6

Solution:

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(A) $Na_4[\overset{x}{Fe}(CN)_5(NOS)]$

$(+1)4 + x + (-1)5 + (-1)\,1 = 0$

$\boxed{x = +2}$

(B) $Na_4[\overset{y}{Fe}O_4]$

$(+1)4 + y + (-2)4 = 0$

$\boxed{y = +4}$

(C) $[\overset{z}{Fe}_2(CO)_9]$

$2z + 0 \times 9 = 0$

$\boxed{z = 0}$

so $(x + y + z) = +2 + 4 + 0$

$= 6$

Q. 23 The Gibbs energy change (in J) for the given reaction at $[Cu^{2+}] = [Sn^{2+}] = 1\ M$ and 298 K is :

$$Cu(s) + Sn^{2+}(aq.) \rightarrow Cu^{2+}(aq.) + Sn(s);$$

$$\left(E^0_{Sn^{2+}|Sn} = -0.16\ V, E^0_{Cu^{2+}|Cu} = 0.34V, \text{Take F=96500 C mol}^{-1}\right)$$

Correct Answer:
96500

Solution:

$$Cu(s) + Sn^{+2}(aq) \rightleftharpoons Cu^{+2}(aq) + Sn(s)$$

$$E^0 = -0.16 - 0.34$$
$$= -0.50$$

$$\Delta G^0 = -nFE^0$$
$$= -2 \times 96500 \times (-0.5)$$
$$= +96500$$

$$\Delta G = \Delta G^0 + RT\ell nQ$$
$$= 96500 + \frac{25}{3} \times 298 \times 2.303 \log(1)$$

$$\Delta G = 96500 \text{ Joules}$$

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Q. 24 The mass of gas adsorbed, x, per unit mass of adsorbate, III, was measured at various pressures, p. A graph between $log\frac{x}{m}$ and log p gives a straight line with slope equal to 2 and the intercept equal to 0.4771. The value of $\frac{x}{m}$ at a pressure of 4 atm is : (given log 3 = 0.4771)

Correct Answer:
48

Solution:

$$\frac{x}{m} = kp^x \qquad \ldots\ldots(i)$$

$$\Rightarrow \underbrace{\log \frac{x}{m}}_{y} = \underbrace{\log k}_{c} + \underbrace{x}_{\text{slope}} \underbrace{\log p}_{x}$$

Given $c = \log k = 0.4771$ or $k = 3$

Slope $x = 2$

put in eq. (1) $\frac{x}{m} = 3 \times (4)^2 \Rightarrow 48$

Q. 25 The internal energy change (in J) when 90 g of water undergoes complete evaporation at 100° C is ____

(Given : ΔH_{vap} for water at 373 K = 41 kJ / mol, R = 8.314 JK^{-1} mol^{-1})

Correct Answer:
189494

Solution:

$H_2O(\ell) \rightleftharpoons H_2O(g)$ 90 gm of H_2O

Given moles of H_2O is 5

$\Delta H = \Delta U + \Delta n_g RT$

$5 \times 41000\ J = \Delta U + 1 \times 8.314 \times 373 \times 5$

$\Delta U = 189494.39 \approx 189494$ Joule

Maths

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Q. 1 The domain of the function $f(x)=\sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$ is $(-\infty,-a]\cup[a,\infty)$.

then a is equal to:

Option 1:

$\frac{\sqrt{17}}{2}$

Option 2:

$\frac{\sqrt{17}-1}{2}$

Option 3:

$\frac{1+\sqrt{17}}{2}$

Option 4:

$\frac{\sqrt{17}}{2}+1$

Correct Answer:

$\frac{1+\sqrt{17}}{2}$

Solution:

$f(x)=\sin\left(\frac{|x|+5}{x^2+1}\right)$

For domain :

$-1\leq\frac{|x|+5}{x^2+1}\leq 1$

$Since\ |x|+5\ \&\ x^2+1$ is always positive

$So,\ \frac{|x|+5}{x^2+1}\geq 0\forall x\in\mathbb{R}$

So for domain:

$\frac{|x|+5}{x^2+1}\leq 1$

$\Rightarrow |x|+5\leq x^2+1$

$\Rightarrow 0\leq x^2-|x|-4$

$\Rightarrow 0\leq\left(|x|-\frac{1+\sqrt{17}}{2}\right)\left(|x|-\frac{1-\sqrt{17}}{2}\right)$

$\Rightarrow |x|\geq\frac{1+\sqrt{17}}{2}$ or $|x|\leq\frac{1-\sqrt{17}}{2}$ (Rejected)

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$$\Rightarrow x \in \left(-\infty, -\frac{1+\sqrt{17}}{2}\right] \cup \left[\frac{1+\sqrt{17}}{2}, \infty\right)$$

$$\text{So, } a = \frac{1+\sqrt{17}}{2}$$

Q. 2 *If* $R = \{(x, y) : x, y \in \mathbb{Z}, x^2 + 3y^2 \leq 8\}$ is a relation on the set of integers Z, then the domain of R^{-1} is:

Option 1:
$\{-2, -1, 1, 2\}$

Option 2:
$\{0, 1\}$

Option 3:
$\{-2, -1, 0, 1, 2\}$

Option 4:
$\{-1, 0, 1\}$

Correct Answer:
$\{-1, 0, 1\}$

Solution:
$\mathrm{R} = \{(\mathrm{x}, \mathrm{y}) : \mathrm{x}, \mathrm{y} \in \mathrm{z}, \mathrm{x}^2 + 3\mathrm{y}^2 \leq 8\}$

For domain of R^{-1}

Collection of all integral of y's

For $x = 0, 3y^2 \leq 8$

$\Rightarrow \mathrm{y} \in (-1, 0, 1)$

Q. 3 Box I contains 30 cards numbered 1 to 30 and Box II contains 20 cards numbered 31 to 50. A box is selected at random and a card is drawn from it. The number on the card is found to be a non-prime number. The probability that the card was drawn from Box I is:

Option 1:
$\frac{2}{3}$

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Option 2:

$\frac{8}{17}$

Option 3:

$\frac{4}{17}$

Option 4:

$\frac{2}{5}$

Correct Answer:

$\frac{8}{17}$

Solution:

$\text{LetB}_1\text{be the event where Box-I is selected and B}_2\text{ where box-II selected}$

$\mathrm{P}\left(\mathrm{B}_1\right)=\mathrm{P}\left(\mathrm{B}_2\right)=\frac{1}{2}$

Let E be the event where the selected card is non-prime.

For B1 : Prime numbers: {2, 3, 5, 7, 11, 13, 17, 19, 23, 29}

For B2 : Prime numbers: {31, 37, 41, 43, 47}

$$\mathrm{P(E)}=\mathrm{P}\left(\mathrm{B}_1\right)\times\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{B}_1}\right)+\mathrm{P}\left(\mathrm{B}_2\right)\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{B}_2}\right)$$

$$=\frac{1}{2}\times\frac{20}{30}+\frac{1}{2}\times\frac{15}{20}$$

$\text{Required probability:}$

$$P\left(\frac{B_1}{E}\right)=\frac{\frac{1}{2}\times\frac{20}{30}}{\frac{1}{2}\times\frac{20}{30}+\frac{1}{2}\times\frac{15}{20}}=\frac{\frac{2}{3}}{\frac{2}{3}+\frac{3}{4}}=\frac{8}{17}$$

Q. 4 Area (in sq. units) of the region outside $\frac{|x|}{2}+\frac{|y|}{3}=1$ and inside the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1\ is$:

Option 1:

$6\left(\pi-2\right)$

Option 2:

$3\left(\pi-2\right)$

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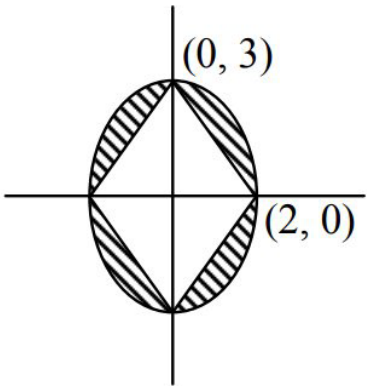
Option 3:
$3\,(4-\pi)$

Option 4:
$6\,(4-\pi)$

Correct Answer:
$6\,(\pi-2)$

Solution:
$\frac{|x|}{2}+\frac{|y|}{3}=1$
$\frac{x^2}{4}+\frac{y^2}{9}=1$



$\text{Area of Ellipse} = \pi ab = 6\pi$
$\text{Required area,} = \pi \times 2 \times 3 - (\text{Area of quadrilateral})$
$= 6\pi - \frac{1}{2}6 \times 4$
$= 6\pi - 12 = 6(\pi - 2)$

Q. 5 $Let\ y = y(x)$ be the solution of the differential equation,

$$\frac{2+\sin x}{y+1}.\frac{dy}{dx} = -\cos x, y > 0, y(0) = 1. If\ y(x) = a\ and\ \frac{dy}{dx} at\ x = \pi\ is\ b,$$

then the ordered pair (a,b) is equal to:

Option 1:
$\left(2, \frac{3}{2}\right)$

Option 2:
$(1, -1)$

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Option 3:
$(1, 1)$

Option 4:
$(2, 1)$

Correct Answer:
$(1, 1)$

Solution:

$$\frac{2+\sin x}{y+1}\frac{dy}{dx} = -\cos x, y > 0$$

By integrating both sides:

$$\Rightarrow \frac{dy}{y+1} = \frac{-\cos x}{2+\sin x}dx$$

$$\Rightarrow \int \frac{dy}{y+1} = \int \frac{-\cos x}{2+\sin x}dx$$

$$\ell n|y+1| = -\ln|2+\sin x| + \ln K$$

$$\Rightarrow y+1 = \frac{K}{2+\sin x}(y+1>0)$$

$$\Rightarrow y(x) = \frac{K}{2+\sin x} - 1$$

Given $y(0) = 1 \Rightarrow K = 4$

So, $y(x) = \frac{4}{2+\sin x} - 1$

$$a = y(\pi) = 1$$

$$b = \frac{dy}{dx}\Big]_{x=\pi} = \frac{-\cos x}{2+\sin x}(y(x)+1)\Big]_{x=\pi} = 1$$

So, $(a, b) = (1, 1)$

Q. 6 $Let\ \alpha > 0, \beta > 0\ be\ such\ that\ \alpha^3 + \beta^2 = 4.$ If the maximum value of the term independent of x in the binomial expansion of $\left(\alpha x^{\frac{1}{9}} + \beta x^{-\frac{1}{6}}\right)^{10}\ is\ 10k,$ then k is equal to :

Option 1:
336

Option 2:
352

Option 3:
84

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Option 4:
176

Correct Answer:
336

Solution:

Let t_{r+1} denotes

$\mathrm{r}+1^{\text{th}}$ term of $\left(\alpha \mathrm{x}^{1/9}+\beta \mathrm{x}^{-1/6}\right)^{10}$

$$t_{r+1}={}^{10}C_r\alpha^{10-r}(x)^{\frac{10-r}{9}}\cdot\beta^r x^{-\frac{r}{6}}$$
$$={}^{10}C_r\alpha^{10-r}\beta^r(x)^{\frac{10-r}{9}-\frac{r}{6}}$$

If t_{r+1} is independent of x

$$\frac{10-r}{9}-\frac{r}{6}=0\Rightarrow r=4$$

maximum value of t_5 *is* 10 K $(given)$

$\Rightarrow {}^{10}\mathrm{C}_4\alpha^6\beta^4$ is maximum

By AM $\geq$ GM (for positive numbers)

$$\frac{\frac{\alpha^3}{2}+\frac{\alpha^3}{2}+\frac{\beta^2}{2}+\frac{\beta^2}{2}}{4}\geq\left(\frac{\alpha^6\beta^4}{16}\right)^{\frac{1}{4}}$$

$\Rightarrow \alpha^6\beta^4\leq 16$

So, $10\ \mathrm{K}={}^{10}\mathrm{C}_4 16$

$\Rightarrow \mathrm{K}=336$

Q. 7 Let A be a 2 × 2 real matrix with entries from {0, 1} and $|\mathrm{A}|\neq 0$ Consider the following two statements :

(P) If $\mathrm{A}\neq \mathrm{I}_2$, then |A| = –1

(Q) If |A| = 1, then tr(A) = 2,

where I_2 denotes 2 × 2 identity matrix and tr(A) denotes the sum of the diagonal entries of A. Then:

Option 1:
(P) is false and (Q) is true

Option 2:
Both (P) and (Q) are false

Option 3:
(P) is true and (Q) is false

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Option 4:
Both (P) and (Q) are true

Correct Answer:
(P) is false and (Q) is true

Solution:
$|A| \neq 0$

$\text{For (P)} : A \neq I_2$

$\text{So, } A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$|A| \text{ can be } -1 \text{ or } 1$

So (P) is false

For (Q): |A| = 1

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$\Rightarrow \text{tr}(A) = 2$

$\Rightarrow Q \text{ is true}$

Q. 8 The sum of the first three terms of a G.P. is S and their product is 27. Then all such S lie in:

Option 1:
$(-\infty, -9] \cup [3, \infty)$

Option 2:
$[-3, \infty)$

Option 3:
$(-\infty, -3] \cup [9, \infty)$

Option 4:
$(-\infty, 9]$

Correct Answer:
$(-\infty, -3] \cup [9, \infty)$

Solution:

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Let three terms of G.P. are $\frac{a}{r}, a, ar$

product $= 27$

$\Rightarrow a^3 = 27 \Rightarrow a = 3$

$S = \frac{3}{r} + 3r + 3$

For r > 0

$$\frac{\frac{3}{r} + 3r}{2} \geq \sqrt{3^2} \qquad \text{(By AM-GM)}$$

$$\Rightarrow \frac{3}{r} + 3r \geq 6 \qquad \ldots(1)$$

For $r < 0 \quad \frac{3}{r} + 3r \leq -6 \qquad \ldots(2)$

From (1) & (2)

$S \in (-\infty, -3] \cup [9, \infty)$

Q. 9 *If* $|x| < 1, |y| < 1$ and $x \neq y$, then the sum of infinite terms of the following series $(x + y) + \left(x^2 + xy + y^2\right) + \left(x^3 + x^2y + xy^2 + y^3\right) + ...is :$

Option 1:

$\frac{x + y - xy}{(1 + x)(1 + y)}$

Option 2:

$\frac{x + y + xy}{(1 + x)(1 + y)}$

Option 3:

$\frac{x + y - xy}{(1 - x)(1 - y)}$

Option 4:

$\frac{x + y + xy}{(1 - x)(1 - y)}$

Correct Answer:

$\frac{x + y - xy}{(1 - x)(1 - y)}$

Solution:

Sum of an infinite GP

If a is the first term and r is the common ratio of a G.P. Then,

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$$S_\infty = \frac{a}{1-r}$$

S_∞ is the sum to infinite terms of the G.P.

Now,

$|x| < 1, |y| < 1, x \neq y$
$(x+y) + \left(x^2+xy+y^2\right) + \left(x^3+x^2y+xy^2+y^3\right) + \ldots\ldots$

By multiplying and dividing x – y :

$$\frac{(x^2-y^2)+(x^3-y^3)+(x^4-y^4)+\ldots.}{x-y}$$
$$= \frac{(x^2+x^3+x^4+\ldots\ldots)-(y^2+y^3+y^4+\ldots.)}{x-y}$$
$$= \frac{\frac{x^2}{1-x}-\frac{y^2}{1-y}}{x-y}$$
$$= \frac{(x^2-y^2)-xy(x-y)}{(1-x)(1-y)(x-y)}$$
$$= \frac{x+y-xy}{(1-x)(1-y)}$$

Q. 10 Let $\alpha \; and \; \beta$ be the roots of the equation,
$5x^2+6x-2=0. If \; S_n = \alpha^n + \beta^n, n = 1,2,3,...,then$:

Option 1:
$6S_6 + 5S_5 = 2S_4$

Option 2:
$6S_6 + 5S_5 = 2S_4 = 0$

Option 3:
$5S_6 + 6S_5 = 2S_4$

Option 4:
$5S_6 + 6S_5 = 2S_4 = 0$

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Correct Answer:
$5S_6 + 6S_5 = 2S_4$

Solution:
α and β are roots of $5x^2 + 6x - 2 = 0$

$\Rightarrow 5\alpha^2 + 6\alpha - 2 = 0$

$\Rightarrow 5\alpha^{n+2} + 6\alpha^{n+1} - 2\alpha^n = 0 \qquad \ldots(1)$

(By multiplying α^n)

Similarly

$5\beta^{n+2} + 6\beta^{n+1} - 2\beta^n = 0 \qquad \ldots(2)$

Add (1) and (2)

$5\ S_{n+2} + 6\ S_{n+1} - 2\ S_n = 0$

For $n = 4$

$5\ S_6 + 6\ S_5 = 2\ S_4$

Q. 11 Let S be the set of all $\lambda \in R$ for which the system of the linear equation.

$2x - y + 2z = 2$

$x - 2y + \lambda z = 4$

$x + \lambda y + z = 4$

has no solution. Then the set S is

Option 1:
contains more than two elements.

Option 2:
is an empty set.

Option 3:
is a singleton.

Option 4:
contains exactly two elements.

Correct Answer:
contains exactly two elements.

Solution:

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$2x - y + 2z = 2$

$x - 2y + \lambda z = -4$

$x + \lambda y + z = 4$

For no solution:

$$D = \begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & \lambda \\ 1 & \lambda & 1 \end{vmatrix} = 0$$

$\Rightarrow 2(-2 - \lambda^2) + 1(1 - \lambda) + 2(\lambda + 2) = 0$

$\Rightarrow -2\lambda^2 + \lambda + 1 = 0$

$= \lambda = 1, -\frac{1}{2}$

$$D_x = \begin{vmatrix} 2 & -1 & 2 \\ -4 & 2 & \lambda \\ 4 & \lambda & 1 \end{vmatrix} = 2 \begin{vmatrix} 1 & -1 & 2 \\ -2 & -2 & \lambda \\ \lambda & \lambda & 1 \end{vmatrix}$$

$= 2(1 + \lambda)$

Which is not equal to zeros for

$$\lambda = 1, -\frac{1}{2}$$

Q. 12 A line parallel to the straight line $2x - y = 0$ is tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ at the point (x_1, y_1). Then $x_1^2 + 5y_1^2$ is equal to:

Option 1:
6

Option 2:
8

Option 3:
10

Option 4:
5

Correct Answer:
6

Solution:

Slope of tangent is 2, Tangent of the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ at the point (x_1, y_1) is

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$$\frac{xx_1}{4} - \frac{yy_1}{2} = 1 \quad (T = 0)$$

$$\text{Slope: } \frac{1}{2}\frac{x_1}{y_1} = 2 \Rightarrow x_1 = 4y_1$$

(x_1, y_1) lies on the parabola

$$\Rightarrow \frac{x_1^2}{4} - \frac{y_1^2}{2} = 1$$

From above equations

$$\frac{(4y_1)^2}{4} - \frac{y_1^2}{2} = 1 \Rightarrow 4y_1^2 - \frac{y_1^2}{2} = 1$$

$$\Rightarrow 7y_1^2 = 2 \Rightarrow y_1^2 = \frac{2}{7}$$

$$\text{Now } x_1^2 + 5y_1^2 = (4y_1)^2 + 5y_1^2$$

$$= (21)y_1^2 = 21 \times \frac{2}{7} = 6$$

Q. 13 If the tangent to the curve $y = x + \sin\ y$ at a point (a,b) is parallel to the line joining $\left(0, \frac{3}{2}\right)\ and\ \left(\frac{1}{2}, 2\right), then:$

Option 1:

$b = a$

Option 2:

$|b - a| = 1$

Option 3:

$|b + a| = 1$

Option 4:

$b = \frac{\pi}{2} + a$

Correct Answer:

$|b - a| = 1$

Solution:

Slope of tangent to the curve y = x + sin y

$$\text{at } (a, b) \text{ is } \frac{2 - \frac{3}{2}}{\frac{1}{2} - 0} = 1$$

$$\Rightarrow \left.\frac{dy}{dx}\right]_{x=a} = 1$$

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$$\frac{dy}{dx} = 1 + \cos y \cdot \frac{dy}{dx} (\text{ from equation of curve })$$

$$\left.\frac{dy}{dx}\right]_{x=a} = 1 + \cos b \cdot \left.\frac{dy}{dx}\right]_{x=a}$$

$$\Rightarrow \cos b = 0$$

$$\Rightarrow \sin b = \pm 1$$

$$\text{Now, from curve } y = x + \sin y$$

$$b = a + \sin b$$

$$\Rightarrow |b - a| = |\sin b| = 1$$

Q. 14 Let $X = \{x \in \mathbb{N} : 1 \leq x \leq 17\}$ and $Y = \{ax + b : x \in X \text{ and } a, b \in \mathbb{R}, a > 0\}$. If mean and variance of elements of Y and 17 and 216 respectively then $a + b$ is equal to:

Option 1:
7

Option 2:
-7

Option 3:
-27

Option 4:
9

Correct Answer:
-7

Solution:

$$\sigma^2 = \text{ variance}$$

$$\mu = \text{ mean}$$

$$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \mu)^2}{n}$$

$$\mu = 17$$

$$\Rightarrow \frac{\sum_{x=1}^{17} (ax + b)}{17} = 17$$

$$\Rightarrow 9a + b = 17 \quad \ldots\ldots (1)$$

$$\sigma = 216$$

$$\Rightarrow \frac{\sum_{x=1}^{17} (ax + b - 17)^2}{17} = 216$$

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$$\Rightarrow \frac{\sum_{x=1}^{17} a^2(x-9)^2}{17} = 216$$
$$\Rightarrow a^2 81 - 18 \times 9a^2 + a^2 3 \times (35) = 216$$
$$\Rightarrow a^2 = \frac{216}{24} = 9 \Rightarrow a = 3(a > 0)$$
$$\Rightarrow \text{ From } (1), b = -10$$
$$\text{So, } a + b = -7$$

Q. 15 The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$

Option 1:
$\frac{1}{2}\left(1-i\sqrt{3}\right)$

Option 2:
$\frac{1}{2}\left(\sqrt{3}-i\right)$

Option 3:
$-\frac{1}{2}\left(\sqrt{3}-i\right)$

Option 4:
$-\frac{1}{2}\left(1-i\sqrt{3}\right)$

Correct Answer:
$-\frac{1}{2}\left(\sqrt{3}-i\right)$

Solution:

The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$

$$= \left(\frac{1+\sin\left(\frac{\pi}{2}-\frac{5\pi}{18}\right)+i\cos\left(\frac{\pi}{2}-\frac{5\pi}{18}\right)}{1+\sin\left(\frac{\pi}{2}-\frac{5\pi}{18}\right)-i\cos\left(\frac{\pi}{2}-\frac{5\pi}{18}\right)}\right)^3$$
$$= \left(\frac{1+\cos\frac{5\pi}{18}+i\sin\frac{5\pi}{18}}{1+\cos\frac{5\pi}{18}-i\sin\frac{5\pi}{18}}\right)^3$$
$$= \left(\frac{2\cos^2\frac{5\pi}{36}+2i\sin\frac{5\pi}{36}\cos\frac{5\pi}{36}}{2\cos^2\frac{5\pi}{36}-2i\sin\frac{5\pi}{36}\cdot\cos\frac{5\pi}{36}}\right)^3$$

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$$= \left(\frac{\cos\frac{5\pi}{36} + i\sin\frac{5\pi}{36}}{\cos\frac{5\pi}{36} - i\sin\frac{5\pi}{36}}\right)^3$$
$$= \left(\frac{e^{i5\pi/36}}{e^{-i5\pi/36}}\right) = \left(e^{i5\pi/18}\right)^3$$
$$= \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}$$
$$= -\frac{\sqrt{3}}{2} + \frac{i}{2}$$

Q. 16 The contrapositive of the statement "If I reach the station in time, then I will catch the train' is:

Option 1:
If I do not reach the station in right time,
then I will catch the train.

Option 2:
If I do not reach the station in right time,
then I will not catch the train.

Option 3:
If I will catch the train, then I reach the station in time.

Option 4:
If I will not catch the train, then I do not reach the station in time.

Correct Answer:
If I will not catch the train, then I do not reach the station in time.

Solution:
Let p and q denotes statement

p : I reach the station in time.

q : I will catch the train.

Contrapositive of p $\rightarrow$ q

is $\sim q \rightarrow \sim p$

$\sim q \rightarrow \sim p$: I will not catch the train, then I do not reach the station in time.

Q. 17 If p(x) be a polynomial of degree three that has a local maximum value 8 at x=1 and a local minimum value 4 at x=2; then p(0) is equal to:

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Option 1:
6

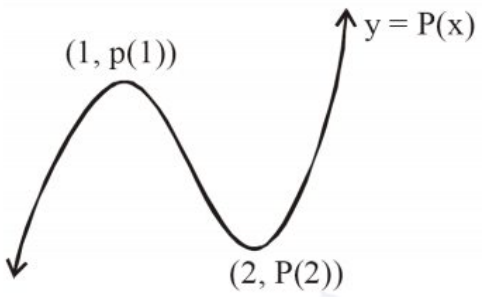
Option 2:
-12

Option 3:
-24

Option 4:
12

Correct Answer:
-12

Solution:



Since p(x) has relative extreme at x = 1 & 2

so p'(x) = 0 at x = 1 & 2

⇒ p'(x) = A(x – 1) (x – 2)

$$\Rightarrow p(x) = \int A\left(x^2 - 3x + 2\right) dx$$

$$p(x) = A\left(\frac{x^3}{3} - 3x^2 + 2x\right) + C \qquad \ldots(1)$$

P(1) = 8

from (1)

$$8 = A\left(\frac{1}{3} - \frac{3}{2} + 2\right) + C$$

$$\Rightarrow 8 = \frac{5\ \mathrm{A}}{6} + \mathrm{C} \Rightarrow 48 = 5\ \mathrm{A} + 5\mathrm{C}$$

$$\mathrm{P}(2) = 4$$

$$\Rightarrow 4 = \mathrm{A}\left(\frac{8}{3} - 6 + 4\right) + \mathrm{C}$$

$$\Rightarrow 4 = \frac{2\ \mathrm{A}}{3} + \mathrm{C} \Rightarrow 12 = 2\ \mathrm{A} + 3\mathrm{C}$$

From above

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C = –12 So P(0) = C = -12

Q. 18 If a function f(x) defined by

$$f(x)=\begin{cases} ae^2+be^{-x}, & -1\le x<1 \\ cx^2, & 1\le x\le 3 \\ ax^2+2cx, & 3<x\le 4 \end{cases}$$

be continuous for some a, b, c ϵ R and $f'(0)+f'(2)=e$, then the value of a is:

Option 1:

$$\frac{1}{e^2-3e+13}$$

Option 2:

$$\frac{e}{e^2-3e-13}$$

Option 3:

$$\frac{e}{e^2+3e+13}$$

Option 4:

$$\frac{e}{e^2-3e+13}$$

Correct Answer:

$$\frac{e}{e^2-3e+13}$$

Solution:

$$f(x)=\begin{cases} ae^x+be^{-x}, & -1\le x<1 \\ cx^2, & 1\le x\le 3 \\ ax^2+2cx, & 3<x\le 4 \end{cases}$$

For continuity at x = 1

$$\lim_{x\to 1^-} f(x)=\lim_{x\to 1^+} f(x)$$
$$\Rightarrow ae+be^{-1}=c$$
$$\Rightarrow b=ce-ae^2$$

For continuity at x = 3

$$\lim_{x\to 3^-} f(x)=\lim_{x\to 3^+} f(x)$$
$$\Rightarrow 9c=9a+6c$$
$$\Rightarrow c=3a$$

$$f(0)+f'(2)=e\Rightarrow (ae^x-be^x)_{x=0}+(2cx)_{x=2}=e$$
$$\Rightarrow a-b+4c=e$$

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$$a - 3ae + ae^2 + 12a = e$$
$$\Rightarrow a\left(e^2 + 13 - 3e\right) = e$$
$$\Rightarrow a = \frac{e}{e^2 - 3e + 13}$$

Q. 19 The plane passing through thr points (1,2,1), (2,1,2) and parallel to the line, $2x = 3y, z = 1$ also passes through the point:

Option 1:
$(0, 6, -2)$

Option 2:
$(-2, 0, 1)$

Option 3:
$(0, -6, 2)$

Option 4:
$(2, 0, -1)$

Correct Answer:
$(-2, 0, 1)$

Solution:
Two points on the line (L)

$\frac{x}{3} = \frac{y}{2}, z = 1$ are $(0, 0, 1)$ & $(3, 2, 1)$

So dr's of the line is (3, 2, 0)

Line passing through (1, 2, 1), parallel to L and coplanar with given plane is

$\vec{r} = \hat{i} + 2\hat{j} + \hat{k} + t(3\hat{i} + 2j), t \in R(-2, 0, 1)$ satisfies the line (for t = –1)

(–2, 0, 1) lies on a given plane.

Q. 20 Let P(h,k) be a point on the curve $y = x^2 + 7x + 2$, nearest to the line, $y = 3x - 3$,Then the equation of the normal to the curve at P is:

Option 1:
$x + 3y + 26 = 0$

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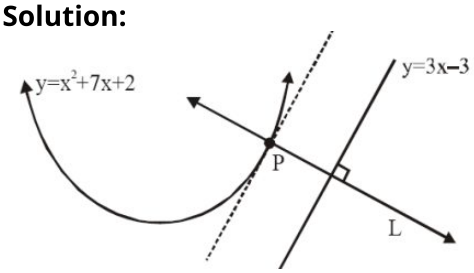
Option 2:
$x + 3y - 26 = 0$

Option 3:
$x - 3y - 11 = 0$

Option 4:
$x - 3y + 22 = 0$

Correct Answer:
$x + 3y + 26 = 0$

Solution:



Let L be the common normal to the parabola

$y = x^2 + 7x + 2$ and line $y = 3x - 3$

$\Rightarrow$ slope of tangent of $y = x^2 + 7x + 2$ at $P = 3$

$\Rightarrow \left.\frac{dy}{dx}\right]_{\text{for P}} = 3$

$\Rightarrow 2x + 7 = 3 \Rightarrow x = -2 \Rightarrow y = -8$

So $P(-2, -8)$

Normal at P : x + 3y + C = 0

⇒C = 26 (P satifies the line)

Normal : x + 3y + 26 = 0

Q. 21 If $\lim_{x \to 1} \frac{x + x^2 + x^3 + ... + x^n - n}{x - 1} = 820, (n \epsilon N)$

then the value of n is equal to _____

Correct Answer:
40

Solution:

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$$\lim_{x\to 1}\frac{x+x^2+\ldots+x^2-n}{x-1}=820$$
$$\Rightarrow \lim_{x\to 1}\left(\frac{x-1}{x-1}+\frac{x^2-2}{x-1}+\ldots\frac{x^n-1}{x-1}\right)=820$$
$$\Rightarrow 1+2+\ldots\ldots+n=820$$
$$\Rightarrow n(n+1)=2\times 820$$
$$\Rightarrow n(n+1)=40\times 41$$
$$\text{Since } n\in\mathbb{N},\ \text{so } n=40$$

Q. 22 $The\ integral \int_0^2 ||\,x-1\,|-x|\,dx\ is\ equal\ to$_____.

Correct Answer:
1.5

Solution:
$$\int_0^2 |x-1|-x\mid dx$$
$$\text{Let } f(x)=||x-1|-x|$$
$$=\begin{cases}1, & x\geq 1\\ |1-2x|, & x\leq 1\end{cases}$$
$$\int_0^{1/2}(1-2x)dx+\int_{1/2}^1(2x-1)+\int_0^2 1dx$$
$$=\left[x-x^2\right]_0^{1/2}+\left[x^2-x\right]_{1/2}^1+[x]_1^2$$
$$=3/2=1.5$$

Q. 23 The number of integral values of k for which the line, $3x+4y=k$ intersects the circle, $x^2+y^2-2x-4y+4=0$

at two distinct point is_____.

Correct Answer:
9

Solution:
$$\text{Circle } x^2+y^2-2x-4y+4=0$$
$$\Rightarrow (x-1)^2+(y-2)^2=1$$

Center = (1,2) ; Radius = 1

Line 3x + 4y – k = 0 intersects the circle at two distinct points.

distance of centre from the line < radius

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$\Rightarrow \left|\frac{3\times 1+4\times 2-\mathrm{k}}{\sqrt{3^2+4^2}}\right|<1$

$\Rightarrow |11-k|<5$

$\Rightarrow 6<\mathrm{k}<16$

$\Rightarrow \mathrm{k}\in\{7,8,9,\ldots\ldots 15\}$ since $\mathrm{k}\in \mathrm{I}$

Number of K is 9

Q. 24 If the letter of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then the position of the word 'MOTHER' is _____

Correct Answer:
309

Solution:
$\mathrm{MOTHER}:\mathrm{E,H,M,O,R,T}$

So position of word MOTHER in dictionary

Starting with E: 5!

Starting with H: 5!

Starting with ME: 4!

Starting with MH: 4!

Starting with MOE: 3!

Starting with MOH: 3!

Starting with MOR: 3!

Starting with MOTE: 2!

Next word is MOTHER: 1

Total: 309

Q. 25 Let $\vec{a},\vec{b}\,and\,\vec{c}$ be three unit vectore such that $\left|\vec{a}-\vec{b}\right|^2+|\vec{a}-\vec{c}|^2=8.Then\ \left|\vec{a}+2\vec{b}\right|^2+\left|\vec{a}+2\vec{c}\right|^2$ is equal to__

Correct Answer:
2

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Solution:

$|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$

$|\vec{a} - \vec{b}|^2 + |\vec{a} - \vec{c}|^2 = 8$

$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{a}|^2 + |\vec{c}|^2 - 2\tilde{a} \cdot \vec{c} = 8$

$\Rightarrow 4 - 2(\tilde{a} \cdot \vec{b} + \tilde{a}.\tilde{c}) = 8 \Rightarrow \tilde{a} \cdot \tilde{b} + \tilde{a} \cdot \tilde{c} = -2$

$|\vec{a} + 2\vec{b}|^2 + |\vec{a} + 2\vec{c}|^2$

$= |\vec{a}|^2 + 4|\vec{b}|^2 + 4\vec{a} \cdot \vec{b} + |\vec{a}|^2 + 4|\vec{c}|^2 + 4\tilde{a}.\tilde{c}$

$= 10 + 4(\tilde{a} \cdot \vec{b} + \tilde{a} \cdot \tilde{c})$

$= 10 - 8$

$= 2$

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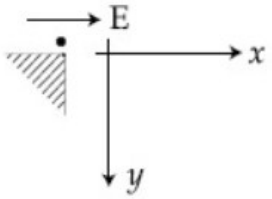
2nd September Shift-2

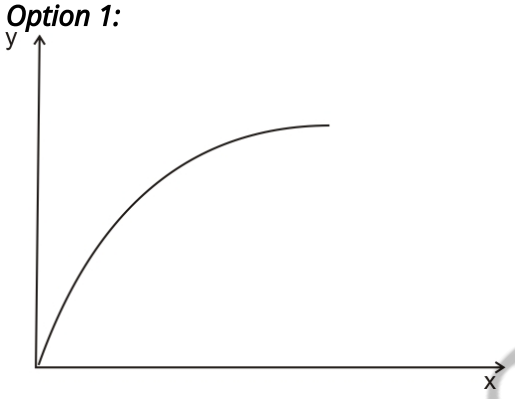
Physics

Q. 1 A small point mass carrying some positive charge on, it is redirected from the edge of the table. There is a uniform electric field in this region in the horizontal direction. Which of the following options then correctly describe the trajectory of the mass ?

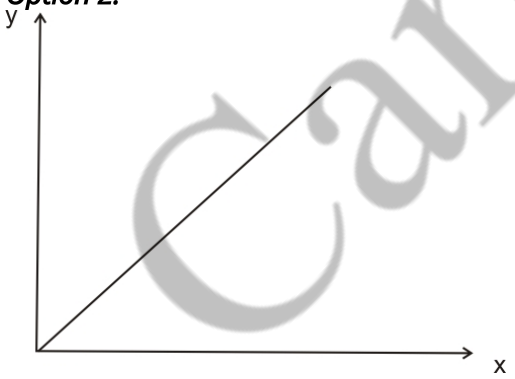
(where we have drawn schematically and are not to scale)



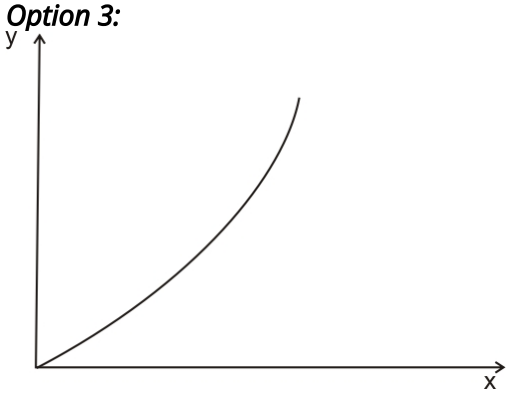
Option 1:



Option 2:



Option 3:



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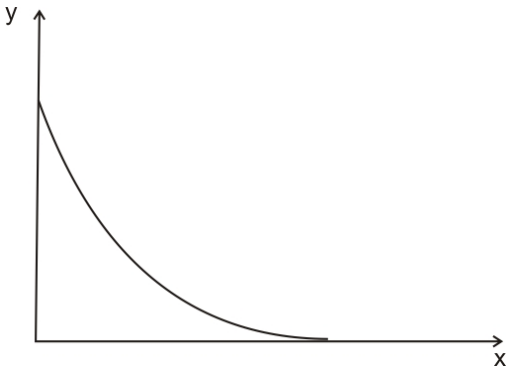
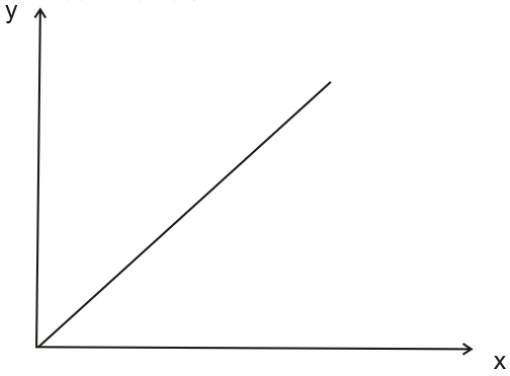
Option 4:



Correct Answer:



Solution:

$$y = \frac{1}{2}gt^2 = k_1t^2$$

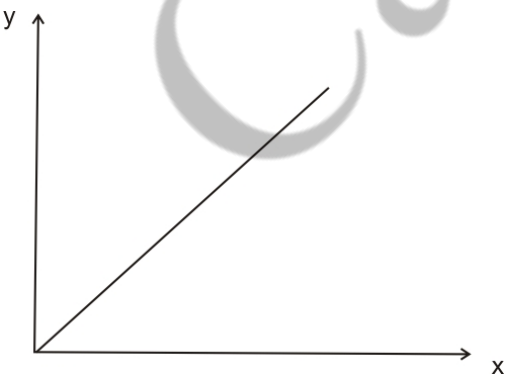
$$x = \frac{1}{2}\frac{qE}{m}t^2 = k_2t^2$$

$$y = \frac{k_1}{k_2}x = kx$$

$$where\ k_1\ and\ k_2\ and\ k\ are\ some\ constant$$

So we get a linear equation in y and x.

So most appropriate graph will be



Q. 2 In a hydrogen atom, the electron makes the transition from $(n+1)^{th}$level to the n^{th} level. If $n >>> 1$ the frequency of radiation emitted is proportional to

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Option 1:
$\frac{1}{n}$

Option 2:
$\frac{1}{n^4}$

Option 3:
$\frac{1}{n^3}$

Option 4:
$\frac{1}{n^2}$

Correct Answer:
$\frac{1}{n^3}$

Solution:
An energy gap, $\Delta E = hv$

Here, h is Planck's constant
therefore,
Frequency= ν

$$\nu = \frac{\Delta E}{h} = k\left[\frac{1}{(n)^2} - \frac{1}{(n+1)^2}\right] \Rightarrow \nu = \frac{k(2n+1)}{n^2(n+1)^2}$$

$since \;\; n >>> 1$
$So \; (n+1)^2 \approx n^2$

$$\Rightarrow \nu \propto \frac{1}{n^3}$$

Q. 3 The uniform circular discs are rotating in the same direction around their common axis passing through their centers. The moment of inertia and angular velocity of the first disc are $0.1 \; kgm^2$ and $10 \; rad.s^{-1}$ respectively, While those for the second one are $0.2 \; kg.m^2$ and $5 \; rad.s^{-1}$ respectively. At some instant, they get stuck together and start rotating as a single system about their common axis with some angular speed. The kinetic energy of the combined system is

Option 1:
$\frac{10}{3} \; J$

Option 2:
$\frac{20}{3} \; J$

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Option 3:

$\frac{2}{3}\ J$

Option 4:

$\frac{5}{3}\ J$

Correct Answer:

$\frac{20}{3}\ J$

Solution:

By angular momentum conservation

$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$

$0.1 \times 10 + 0.2 \times 5 = (0.3) \times \omega$

$\omega = \frac{20}{3}$

$K.E. = \frac{1}{2}I\omega^2 = \frac{1}{2}(0.1 + 0.2)\left[\frac{20}{3}\right]^2 = \frac{20}{3}J$

Q. 4 An inductive coil has a resistance of 100 Ω. When an a.c. signal of frequency 1000 Hz is fed to the coil, the applied voltage leads the current by 45^0 What is the inductance of the coil?

Option 1:

6.7 x 10^{-7} H

Option 2:

1.1 x 10^{-1} H

Option 3:

1.1 x 10^{-5} H

Option 4:

1.6 x 10^{-3} H

Correct Answer:

1.6 x 10^{-3} H

Solution:

The applied voltage leads the current by 45°

$i.e., \frac{X_L}{R} = \tan 45^\circ$

$X_L = R$

$\Rightarrow 2 \times 3.14 \times 1000L = 100$

$\Rightarrow 6280L = 100$

$\Rightarrow L = \frac{25}{1570} = 0.01592 \cong 16mH$

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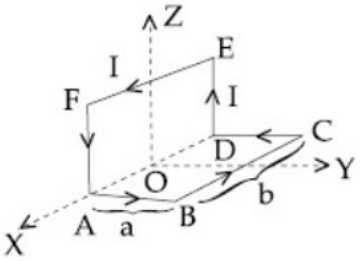
Q. 5 A single piece of wire carrying a current I is bent in the shape ABCDEF as shown where the rectangle ABCDA and ADEFA are perpendicular to each other.If the sides of the rectangles are of the length a and b then the magnitude and direction of magnetic moment of the loop ABCDEFA



Option 1:

$\sqrt{a}abI,\ along\ (\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}})$

Option 2:

$abI,\ along\ (\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}})$

Option 3:

$abI,\ along\ (\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}})$

Option 4:

$\sqrt{2}abI,\ along\ (\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}})$

Correct Answer:

$\sqrt{2}abI,\ along\ (\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}})$

Solution:

$M_1 = IA\hat{k} = Iab\hat{k}$ and $M_2 = IA\hat{j} = Iab\hat{j}$

$|M_{net}| = \sqrt{(M_1)^2+(M_2)^2} = \sqrt{(I^2a^2b^2)+(I^2a^2b^2)} = \sqrt{2}Iab$

And the direction of net magnetic moment is along $\vec{u}$ where $\vec{u} = \hat{j}+\hat{k}$

So $\vec{M_{net}} = \sqrt{2}Iab\hat{u} = \sqrt{2}Iab * \frac{\vec{u}}{|\vec{u}|} = \sqrt{2}Iab * \frac{(\hat{k}+\hat{j})}{\sqrt{2}}$

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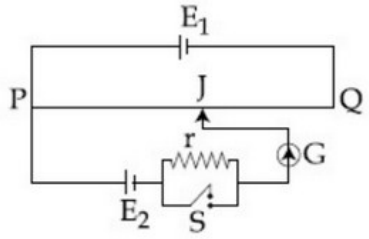
Q. 6 Potentiometer wire PQ of length 1m is connected to a standard cell E1. Another cell of E2 of emf 1.02v is connected with a resistance 'r' and with a switch 'S' as shown in the circuit diagram. With switch S is open, the null position is obtained at a distance of 49 cm from Q. Calculate the potential gradient in the potentiometer wire is:-



Option 1:
0.02 V/cm

Option 2:
0.04 V/cm

Option 3:
0.03 V/cm

Option 4:
0.01 V/cm

Correct Answer:
0.02 V/cm

Solution:

Length of potentiometer wire $l = 1\text{m}$
Emf of standard cell $= E_1$
Emf of cell $E_2 = 1.02v/s$
Given Null point from Q is 49 cm
Null point from $P = 100 - 49 = 51\text{cm} = 0.51\text{m}$
Potential gradient of the wire $V = \dfrac{1.02}{0.51}V = 2V/m = 0.02V/cm$

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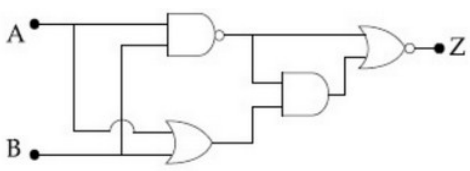
Q. 7 In the following digital circuit, what will be the output at Z when the input (A,B) are (1,0), (0,0), (1,1), (0,1)?



Option 1:
(1,0,1,1)

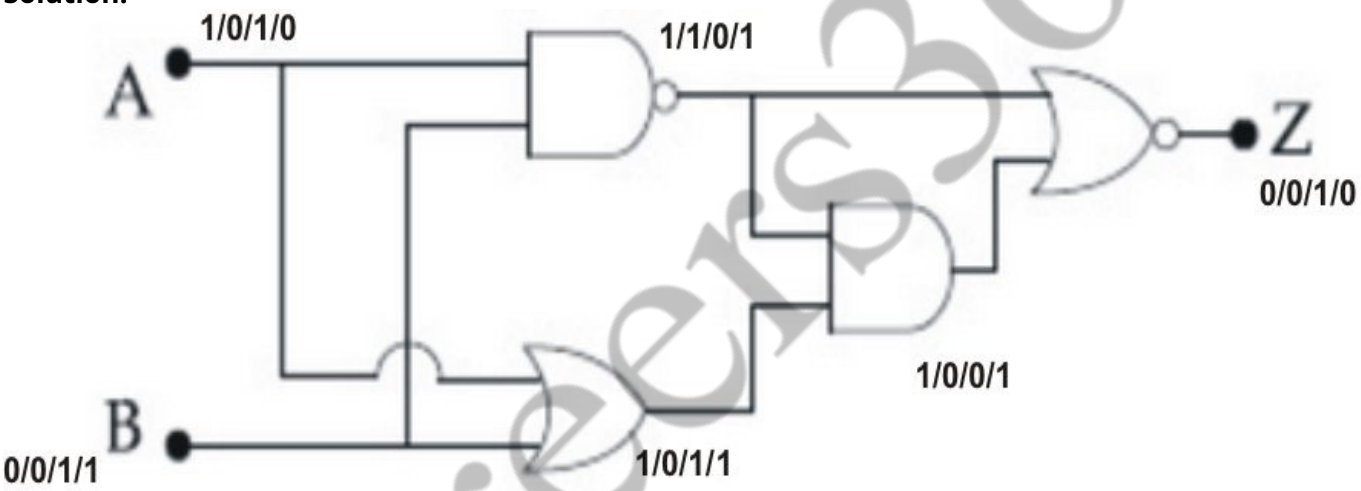
Option 2:
(0,1,0,0)

Option 3:
(0,0,1,0)

Option 4:
(1,1,0,1)

Correct Answer:
(0,0,1,0)

Solution:



A	B	Z
1	0	0
0	0	0
1	1	1
0	1	0

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Q. 8 If momentum P, area A, and time T are taken to be fundamental quantities, then the dimensional formula for energy is:-

Option 1:
$PA^{1/2}T^{-1}$

Option 2:
P^2AT^{-2}

Option 3:

$PA^{-1}T^{-2}$

Option 4:

$P^{\frac{1}{2}}AT^{-1}$

Correct Answer:

$PA^{1/2}T^{-1}$

Solution:

Let energy,

$U \propto P^a A^b T^c$

$U = kP^a A^b T^c \ldots\ldots (i)$

$where$ k is dimensionless constant of proportionality.

$From(i)$

$\left[ML^2T^{-2}\right] = \left(MLT^{-1}\right)^a (L^2)^b T^c = M^a L^{a+2b} T^{-a+c}$

Applying the principle of homogeneity of dimensions,

$we\ get$

$a = 1,$

$a + 2b = 2 \Rightarrow b = \dfrac{2-a}{2} = \dfrac{2-1}{2} = \dfrac{1}{2}$

$-\mathrm{a} + \mathrm{c} = -2 \Rightarrow \mathrm{c} = -2 + \mathrm{a} = -2 + 1 = -1$

$\therefore from\ (i), U = k\left[P^1 A^{1/2} T^{-1}\right]$

Q. 9 The particle is moving 5 times as fast as an electron, the ratio of the $\lambda$of the particle to that of the electron is 1.878 x 10^{-4}, then the mass of the particle is close to

Option 1:

9.7 x 10^{-28} kg

Option 2:

4.8 x 10^{-27} kg

Option 3:

1.2 x 10^{-28} kg

Option 4:

9.1 x 10^{-31} kg

Correct Answer:

9.7 x 10^{-28} kg

Solution:

$\dfrac{\lambda_e}{\lambda_p} = \dfrac{\frac{h}{m_e v_e}}{\frac{h}{m_p v_p}}$

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It is given that

$$\frac{\lambda_p}{\lambda_e} = 1.878 \times 10^{-4}$$

$m_e = 9.109 \times 10^{-31}$ kg

Also given that

$V_p = 5\ V_e$

After putting the values -

$m_p = 9.7 \times 10^{-28}$ kg

Q. 10 The height h at which the weight of a body will be the same as that at the same depth h from the surface of the earth is:

Option 1:

$$\frac{R}{2}$$

Option 2:

$$\frac{\sqrt{5}R - R}{2}$$

Option 3:

$$\frac{\sqrt{3}R - R}{2}$$

Option 4:

$$\frac{\sqrt{5}R - 2R}{2}$$

Correct Answer:

$$\frac{\sqrt{5}R - R}{2}$$

Solution:
Since,

weight at hieght h = weight at depth d

So, $mg_h = mg_d$

So, $g_h = g_d$

So,

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$$\frac{g_o R^2}{(R+h)^2} = \frac{g_o(R-d)}{R}$$

Where,

g_o = acceleration due to gravity at the surface of earth

h= height

d = Depth

R = Radius of the earth

On solving,

$$\frac{R^2}{(R+h)^2} = \frac{(R-d)}{R}$$

$$\Rightarrow R^3 = (R^2 + h^2 + 2Rh)(R-d)$$

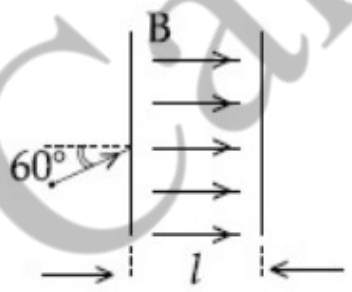
On Solving this we get -

$$h^2 + hR - R^2 = 0$$

So

$$\Rightarrow h = \frac{\sqrt{5}R - R}{2}$$

Q. 11 A magnetic field of 0.3T is present in a region and a proton enters the region with a velocity 4 x 10^5 m/s making an angle of 60^0 with the field. If the proton completes 10 revolutions by the time it crosses the region as shown in the figure, then l is close to (mass of proton=1.67 x 10^{-27} kg, a charge of proton=1.6 x 10^{-19} C, Answer in meter)



Correct Answer:
0.44

Solution:
Given

m = 1.67 x 10^{-27} kg

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v = 4 x 10^5 m/s

q = 1.6 x 10^{-19} C

B = 0.3 T

Using

$T = \frac{2\pi m}{qB}$ and

$$pitch = v\cos 60^\circ \times T = v\cos 60^\circ \times \frac{2\pi m}{qB}$$

we get

$$\begin{aligned} l &= 10 \times \text{ pitch} \\ &= 10 \times v\cos 60^\circ \times \frac{2\pi m}{qB} \\ &= 10 \times v\cos 60^\circ \times \frac{2\pi m}{qB} \\ &= 10 \times v \times \frac{1}{2} \times \frac{2\pi m}{qB} \end{aligned}$$

$$l = \frac{10v\pi m}{qB} = \frac{10 \times 4 \times 10^5 \times 3.14 \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.3} \cong 0.44m$$

Q. 12 A 10μF capacitor is charged to a potential difference of 50V and is connected to another uncharged capacitor in parallel. Now the common potential difference becomes 20 volts. The capacitance of the second capacitor is

Option 1:
10μF

Option 2:
30μF

Option 3:
20μF

Option 4:
15μF

Correct Answer:
15μF

Solution:

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First case:
$Q = 10 \times 50 = 500\mu C$
Second case:
$500 = (C + 10) \times 20$
$\Rightarrow C = 15\mu F$

Q. 13 An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statement are true?

(A) The mean free path of the molecules decreases.
(B) the mean collision time between the molecules decreases.
(C) the mean free path remains unchanged.
(D) the mean collision time remain unchanged.

Option 1:
(A) and (B)

Option 2:
(A) and (D)

Option 3:
(C) and (D)

Option 4:
(B) and (C)

Correct Answer:
(B) and (C)

Solution:
As Mean Free Path is given as

$$\lambda = \frac{1}{\sqrt{2}\pi N d^2}$$

where

d = Diameter of the molecule,
N = number of molecules per unit volume.

So Mean Free Path is independent of temperature so it will remain unchanged.

And relaxation time decreases as temperature increases.

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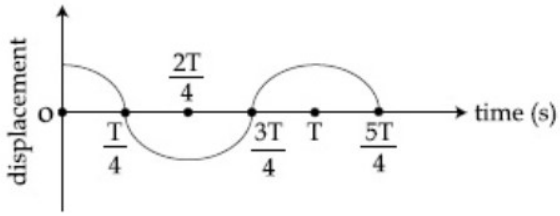
Q. 14 The displacement time graph of a particle executing S.H.M. is given in figure :(sketch is schematic and not to scale)



Which of the following statement is/are true for this motion?

(A) The force is zero at $t = \frac{3T}{4}$

(B) The acceleration is maximum at t=T

(C) The speed is maximum at $t = \frac{T}{4}$

(D) The P.E. is equal to K.E. of the oscillation at $t = \frac{T}{2}$

Option 1:
(A) and (D)

Option 2:
(A), (B) and (C)

Option 3:
(B), (C) and (D)

Option 4:
(A), (B) and (D)

Correct Answer:
(A), (B) and (C)

Solution:
From graph equation of SHM

$$X = ACos\omega t$$

A)

At $t = \frac{3T}{4}$, the particle is at the mean position.

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So The force is zero at $t = \frac{3T}{4}$

B)

at T particle is at the extreme position so acceleration is maximum

C)

The speed is maximum at $t = \frac{T}{4}$

because at $t = \frac{T}{4}$ particle is at the mean position.

D)

$\mathrm{KE} = \mathrm{PE}$
$\frac{1}{2}\mathrm{k}\left(\mathrm{A}^2 - \mathrm{x}^2\right) = \frac{1}{2}\mathrm{kx}^2$
$\mathrm{A}^2 = 2\mathrm{x}^2$
$\mathrm{x} = \frac{+\mathrm{A}}{\sqrt{2}}$
$Using\ \frac{\mathrm{A}}{\sqrt{2}} = \mathrm{A}cos\omega t$
$\mathrm{t} = \mathrm{T}/8$

So A, B and C are correct

Q. 15 A capillary tube made of glass of radius 0.15 mm is dipped vertically in a beaker filled with methylene iodided (surface tension = 0.05 Nm^{-1}, density = 667 kg m^{-3}) Which rises to height h in the tube. It is observed that the two tangents drawn from liquid-glass interfaces (from opp. sides of the capillary) make and angle of 60^o with one another. Then h is close to (g=10 ms^{-2}).

Option 1:
0.172 m

Option 2:
0.049 m

Option 3:
0.087 m

Option 4:
0.137 m

Correct Answer:
0.087 m

Solution:

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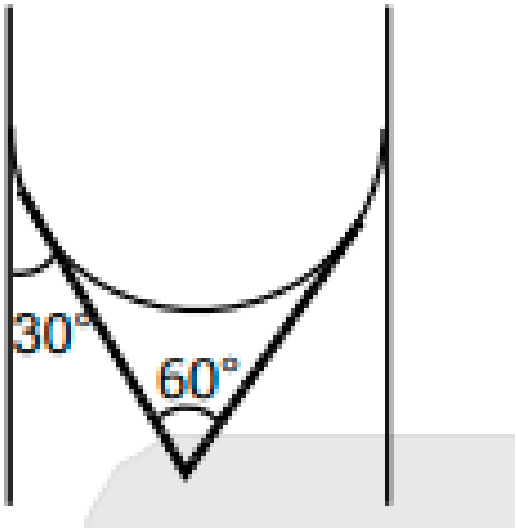

So Angle of contact=30^0

So using

$$h=\frac{2T\cos\theta}{r\rho g}=\frac{2\times0.05\times\left(\frac{\sqrt{3}}{2}\right)}{0.15\times10^{-3}\times667\times10}=0.087\ \text{m}$$

Q. 16 In young's double slit experiment, 16 fringes are observed in a certain segment of the screen when light of wavelength 700 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen would be:

Option 1:
30

Option 2:
28

Option 3:
18

Option 4:
24

Correct Answer:
28

Solution:

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$$y = \frac{D\lambda}{d}$$
$$\text{or } n_1\frac{D\lambda_1}{d} = n_2\frac{D\lambda_2}{d}$$
$$n_1\lambda_1 = n_2\lambda_2$$
$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1}$$
$$n_2 = n_1 \cdot \frac{\lambda_1}{\lambda_2} \Rightarrow 16 \times \frac{700}{400} = 28$$

Q. 17 A heat engine is involved with exchange of heat of 1915 J, -40 J, +125 J and -QJ, during one cycle achieving an efficiency of 50.0%. The value of Q is:

Option 1:
640 J

Option 2:
980 J

Option 3:
400 J

Option 4:
40J

Correct Answer:
980 J

Solution:
$$\eta = \frac{W}{\Sigma Q_+} = \frac{Q_1+Q_2+Q_3+Q_4}{Q_1+Q_3}$$

$$0.5 = \frac{1915-40+125-Q}{1915+125}$$

$$1020 = 1915 - 40 + 125 - Q$$

$$Q = 2000 - 1020 = 980$$

Q. 18 When the temperature of a metal wire is increased from 0°C to 10°C, its length increases by 0.02%. The percentage change in its mass density will be closest to:

Option 1:
3

Option 2:
0.06

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Option 3:
0.36

Option 4:
300

Correct Answer:
0.06

Solution:
$\Delta \ell = \ell \alpha \Delta t$

$\alpha = \frac{\Delta \ell}{\ell \Delta T} = \frac{0.02}{100 \times 10}; \quad \alpha = 2 \times 10^{-5}$

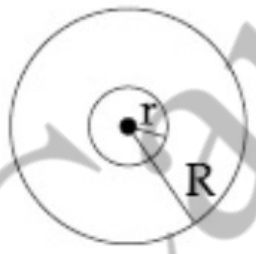
Now

$$\rho = \frac{\mathrm{M}}{\mathrm{V}} \Rightarrow \left|\frac{\Delta \rho}{\rho} \times 100\right| = \left|\frac{\Delta V}{V} \times 100\right|$$

$As\ \frac{\Delta V}{V} = \gamma \times \Delta T;$
$\Rightarrow\ Using\ \gamma = 3\alpha = 6 \times 10^{-5}$
$\Rightarrow\ \ \frac{\Delta V}{V} \times 100 = (6 \times 10^{-5} \times 10 \times 100) = 6 \times 10^{-2}$

So density decrease by 0.06%

Q. 19 A change Q is distributed over two concentric conducting thin spherical shells radii r and R (R>r). If the surface charge densities on the two shells are equal, the electric potential at the common centre is ;



Option 1:
$\frac{1}{4\pi\epsilon_0}\frac{(2R+r)}{(R^2+r^2)}Q$

Option 2:
$\frac{1}{4\pi\epsilon_0}\frac{(R+r)}{2\,(R^2+r^2)}Q$

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Option 3:

$$\frac{1}{4\pi\epsilon_0}\frac{(R+r)}{(R^2+r^2)}Q$$

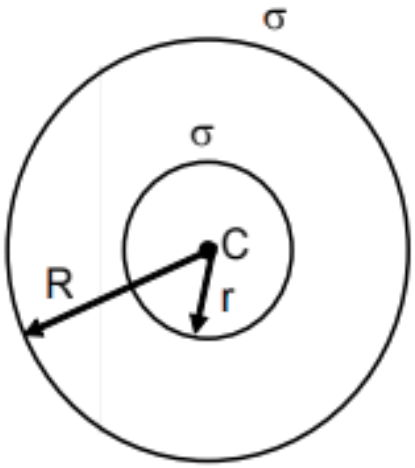
Option 4:

$$\frac{1}{4\pi\epsilon_0}\frac{(R+2r)\,Q}{2\,(R^2+r^2)}$$

Correct Answer:

$$\frac{1}{4\pi\epsilon_0}\frac{(R+r)}{(R^2+r^2)}Q$$

Solution:



$$\mathrm{Q_1} = \sigma 4\pi \mathrm{r}^2$$
$$\mathrm{Q_2} = \sigma 4\pi \mathrm{R}^2$$

Where Q_1= charge on inner shell and Q_2= charge on the outer shell

and $Q = Q_1 + Q_2$

So

$$Q = \sigma 4\pi\,(r^2 + R^2)$$
$$\sigma = \frac{Q}{4\pi(r^2+R^2)}$$

$$\mathrm{V_C} = \frac{\mathrm{KQ_1}}{\mathrm{r}} + \frac{\mathrm{KQ_2}}{\mathrm{R}} = \frac{\mathrm{K}\sigma 4\pi \mathrm{r}^2}{\mathrm{r}} + \frac{\mathrm{K}\sigma 4\pi \mathrm{R}^2}{\mathrm{R}} = \mathrm{K}\sigma 4\pi(\mathrm{r}+\mathrm{R})$$

$$V_C = \frac{\mathrm{KQ}4\pi(\mathrm{r}+\mathrm{R})}{4\pi\,(\mathrm{r}^2+\mathrm{R}^2)} = \frac{\mathrm{KQ}(\mathrm{r}+\mathrm{R})}{(\mathrm{r}^2+\mathrm{R}^2)} = \frac{\mathrm{Q}(\mathrm{r}+\mathrm{R})}{4\pi\epsilon_0(\mathrm{r}^2+\mathrm{R}^2)}$$

Q. 20 In a plane electromagnetic wave, the directions of electric field and magnetic field are represented by $\hat{k}\ and\ 2\hat{i} - 2\hat{j}$ respectively. What is the unit vector along direction of propagation of the wave.

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Option 1:

$\frac{1}{\sqrt{2}}\left(\hat{j}+\hat{k}\right)$

Option 2:

$\frac{1}{\sqrt{5}}\left(2\hat{i}+\hat{j}\right)$

Option 3:

$\frac{1}{\sqrt{5}}\left(\hat{i}+2\hat{j}\right)$

Option 4:

$\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$

Correct Answer:

$\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$

Solution:

$As \quad \vec{E} \times \vec{B} \parallel \vec{C}$

$\hat{E} \times \hat{B} = \frac{1}{\sqrt{2}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$

$\Rightarrow \hat{C} = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Q. 21 The wire of density 9 X 10^{-3} kg cm^{-3} is stretched between two clamps 1 m apart. The resulting strain in the wire is 4.9 X 10^{-4} . The lowest frequency of the transverse vibration in the wire is (Young's modulus of wire Y=9 X 10^{10} Nm^{-2}) (to the nearest integer),.....

Option 1:

3

Option 2:

35

Option 3:

49

Option 4:

135

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Correct Answer:
35

Solution:
Fundamental frequency in the string is given as

$$f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}} = \frac{1}{2\ell}\sqrt{\frac{T}{\rho A}} = \frac{1}{2\ell}\frac{\sqrt{Y\Delta\ell}}{\rho\ell}$$

$$f = \frac{1}{2\ell}\sqrt{\frac{Y\Delta\ell}{\rho\ell}}$$

$$(Using \quad \tfrac{\Delta\ell}{\ell} = 4.9 \times 10^{-4})$$

$$f = \frac{1}{2}\sqrt{\frac{9\times10^{10}\times4.9\times10^{-4}}{9\times10^{-3}\times10^{-6}}} = 35$$

Q. 22 A light ray enters a solid glass sphere of refractive index $\mu = \sqrt{3}$ at an angle of incidence 60^o . The ray is both reflected and refracted at the farther surface of the sphere. The angle (in degress) between the reflected and refracted rays at this surface is .....

Option 1:
30

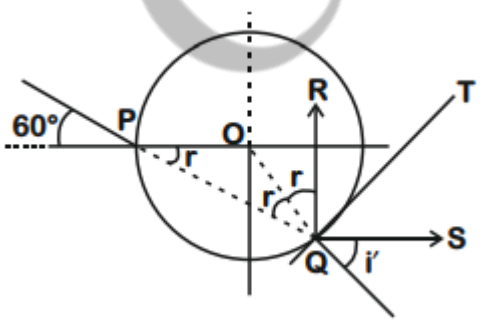
Option 2:
60

Option 3:
45

Option 4:
90

Correct Answer:
90

Solution:



In the above figure

QR is a reflected ray and QS is refracted ray. QT is tangent and OQ is normal.

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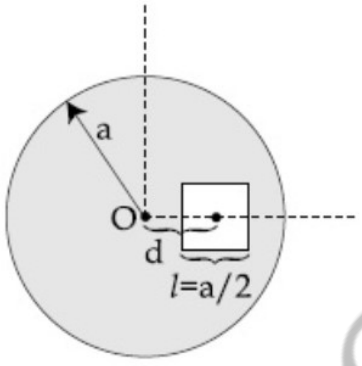
$$\text{At P, } \mu \cdot \sin 60^\circ = \sqrt{3}\sin r$$
$$\Rightarrow \quad \frac{\sqrt{3}}{2} = \sqrt{3}\sin r$$
$$\Rightarrow \sin r = \frac{1}{2} \quad \therefore \quad r = 30^\circ$$

Similarly

$$At \;\; Q$$
$$\sqrt{3}\sin r = \sin i'$$
$$\therefore i' = 60^\circ$$
$$\therefore \; Angle\; between \;\; QR \;\; and \;\; QS$$
$$\Rightarrow (\angle \mathrm{RQS}) = 90^\circ$$

Q. 23 A square shaped hole of side $l = \frac{a}{2}$ is carved out at a distance $d = \frac{a}{2}$ from the centre 'O' of a uniform circular disk of radius a. If the distance of the centre of mass of the remaining portion from O is $-\frac{a}{X}$, value of X (to the nearest integer)is .....



Option 1:
2

Option 2:
5

Option 3:
23

Option 4:
56

Correct Answer:
23

Solution:

6. **Position of centre of mass when some mass is removed**

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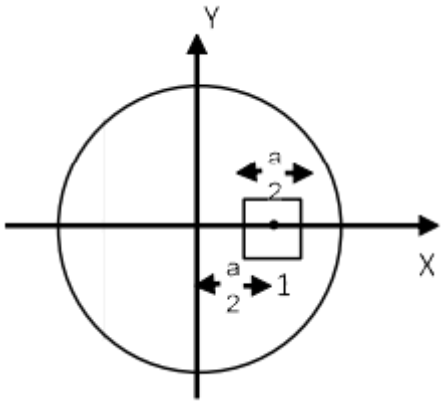
$$\vec{r}_{cm} = \frac{m_1\vec{r_1} - m_2\vec{r_2}}{m_1 - m_2}$$

Where $m_1$is value of whole mass and $\vec{r_1}$is position of centre of mass for whole mass.Similarly m_2 & $\vec{r_2}$ are values for mass which has been removed.

From question



Let the density is σ

So

$$\mathrm{m_1} = \pi \mathrm{a}^2\sigma$$

And

$$m_2 = \frac{a^2}{4} \cdot \sigma$$

And $r_1 = 0 \;\; and \;\; r_2 = \frac{a}{2}$

So

$$X_{cm} = \frac{(\sigma\pi a^2 \times 0) - (\sigma\frac{a^2}{4} \times \frac{a}{2})}{\sigma\pi a^2 - \frac{\sigma a^2}{4}}$$

$$X_{cm} = \frac{-a^3/8}{\left(\pi - \frac{1}{4}\right)a^2} = \frac{-a}{2(4\pi - 1)} = \frac{-a}{8\pi - 2} = -\frac{a}{23}$$

So X=23

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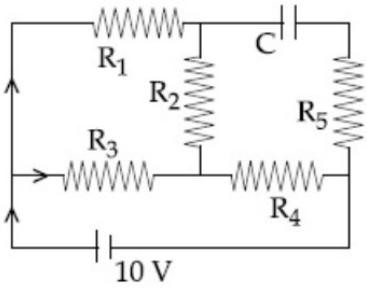
Q. 24 An ideal cell of emf 10V is connected in circuit shown in figure. Each resistance is 2Ω. The potential difference (in V) across the capacitor when it is fully charged is .....



Option 1:
2

Option 2:
4

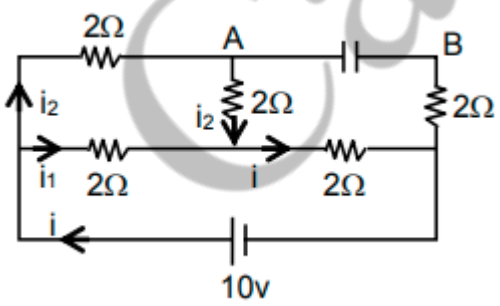
Option 3:
6

Option 4:
8

Correct Answer:
8

Solution:
when the capacitor is fully charged it behaves as an open circuit (voltage drop with no current)

So the new circuit diagram is given below.



So

$$i = \frac{10}{\frac{4}{3}+2} = \frac{10\times 3}{10} = 3\text{Amp}$$
$$\Rightarrow i_1 = 2\ \text{A}\ \&\ i_2 = 1\ \text{A}$$
$$\Rightarrow\ V_{AB} = 1\times 2 + 3\times 2 = 8\ \text{V}$$

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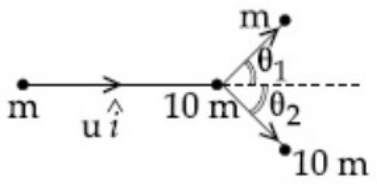
Q. 25 A particle of mass m is moving along the x-axis with an initial velocity $u\hat{i}$. It collides elastically with a particle of mass 10m at rest and then moves with half of its initial kinetic energy (see figure). If $\sin\theta_1 = \sqrt{n}\sin\theta_2$ then value of n is .....



Option 1:
1

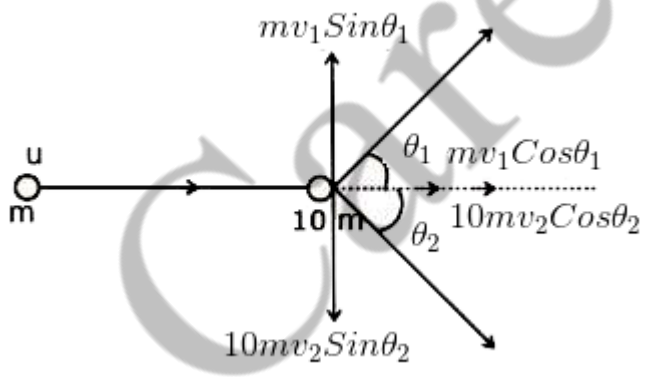
Option 2:
10

Option 3:
20

Option 4:
30

Correct Answer:
10

Solution:



$Given$

$\frac{1}{2}mv_1^2 = \frac{1}{2}\left(\frac{1}{2}\mathrm{mu}^2\right)$

$v_1^2 = \frac{u^2}{2}$

$v_1 = \frac{u}{\sqrt{2}} \quad ....(i)$

Also

Apply energy conservation

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$\frac{1}{2}\mathrm{mu}^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}(10\ \mathrm{m}) \times v_2^2$
$\frac{1}{2} \times 10\ \mathrm{m} \times v_2^2 = \frac{1}{2} \times \frac{1}{2}\mathrm{mu}^2$
$v_2^2 = \frac{u^2}{20}\ \text{or}\ , v_2 = \frac{u}{\sqrt{20}}$

Now

From momentum conservation in the perpendicular direction of the initial motion

$mv_1 \sin\theta_1 = 10mv_2 \sin\theta_2$

$\frac{u}{\sqrt{2}} \sin\theta_1 = 10 \times \frac{u}{\sqrt{20}} \sin\theta_2$

$\sin\theta_1 = \frac{10 \sin\theta_2}{\sqrt{10}}$

$\sin\theta_1 = \sqrt{10} \sin\theta_2$
$\therefore n = 10$

Chemistry

Q. 1 The one that is not expected to show isomerism is:

Option 1:
$[Ni\,(NH_3)_2\,Cl_2]$

Option 2:
$[Ni\,(NH_3)_4\,(H_2O)_2]^{+2}$

Option 3:
$Pt\,(NH_3)_2\,Cl_2$

Option 4:
$[Ni\,(en)_3\,]^{2+}$

Correct Answer:
$[Ni\,(NH_3)_2\,Cl_2]$

Solution:
$Ni(NH_3)_2Cl_2$ is expected to have sp^3 hybridization and tetrahedral geometry. Hence, it will not show isomerism.
$[Pt(NH_3)_2Cl_2]$ and $[Ni(NH_3)_4(H_2O)_2]^{2+}$ can show cis-trans isomerism.
$[Ni(en)_3]^{2+}$ will show optical isomerism.

Therefore, **Option(1) is correct.**

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Q. 2 Match the type of interaction in column A with the distance dependence of their interaction energy in column B.

A	B
i) ion-ion	a) $\frac{1}{r}$
ii) dipole-dipole	b) $\frac{1}{r^2}$
iii) London- dispersion	c) $\frac{1}{r^3}$
	d) $\frac{1}{r^4}$

Option 1:
i-a, ii-c, iii-d

Option 2:
i-a, ii-b, iii-d

Option 3:
i-b, ii-d, iii-c

Option 4:
i-a, ii-b, iii-c

Correct Answer:
i-a, ii-c, iii-d

Solution:
(I) ion-ion interaction energy $\propto$ 1/r

(II) dipole-dipole interaction energy $\propto 1/r^3$ for stationary molecules
dipole-dipole interaction energy $\propto 1/r^6$ for rotating molecules

(III) London-dispersion energy $\propto 1/r^6$

Hence, the correct combination is:
(I)-A, (II)-C, (III)-d

Therefore, **Option(a) is correct.**

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Q. 3 The correct observation in the following reactions is :

$$\text{Sucrose} \xrightarrow[\text{cleavage (hydrolysis)}]{\text{Glycosidic bond}} \text{A} + \text{B} \xrightarrow[\text{reagent}]{\text{seliwanoff}}$$

Option 1:
Gives no colour

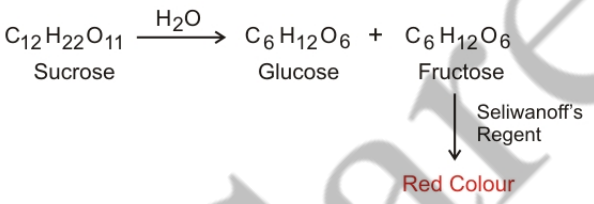
Option 2:
Formation of Blue colour

Option 3:
Formation of Red colour

Option 4:
Formation of Violet colour

Correct Answer:
Formation of Red colour

Solution:
The reaction occurs as follows:



- Seliwanoff's test is useful to distinguish aldose and ketose sugars.
- Fructose gives red colour with Seliwanoff's reagent.

Therefore, **Option(3) is correct.**

Q. 4 Cast iron is used for the manufacture of:

Option 1:
pig iron, scrap iron and steel

Option 2:
wrought iron, pig iron and steel

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Option 3:
wrought iron and steel

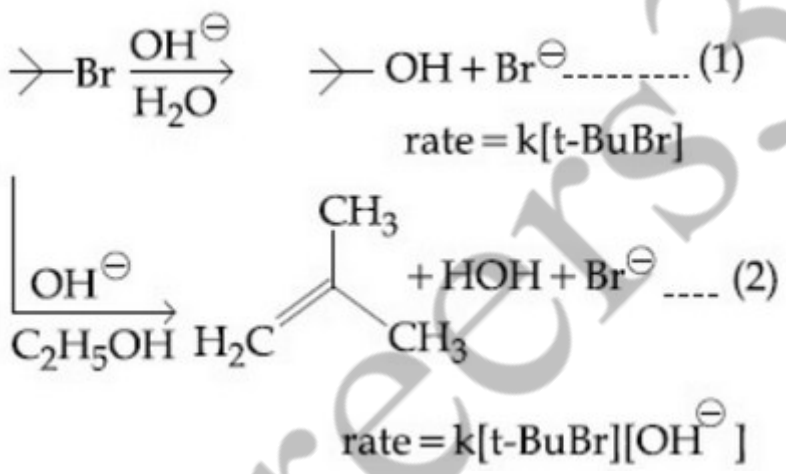
Option 4:
wrought iron and pig iron

Correct Answer:
wrought iron and steel

Solution:
Cast iron is prepared from pig iron and scrap iron. It is used to prepare wrought iron and steel.

Therefore, **Option(3) is correct.**

Q. 5 Consider the reaction sequence given below:



Which of the following statements is true:

Option 1:
changing the concentration of base will have no effect on reaction (1)

Option 2:
changing the base from OH^- to OR^- will have no effect on reaction (2)

Option 3:
changing the concentration of base will have no effect on reaction (2)

Option 4:
Doubling the concentration of base will double the rate of both the reactions

Correct Answer:
changing the concentration of base will have no effect on reaction (1)

Solution:

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According to the rate law given in the question,

Reaction (1) occurs by the SN_1 mechanism while Reaction (2) occurs by the E_2 mechanism.

Changing the concentration of base will not affect the rate of Reaction (1) as the concentration of base does not appear in the rate law expression.

All other statements are incorrect.

Therefore, **Option(1) is correct.**

Q. 6 Amongst the following statements regarding absorption, those that are valid are:

Option 1:
ΔH becomes less negative as absorption proceeds.

Option 2:
on a given absorbent, ammonia is absorbed more than nitrogen gas

Option 3:
on adsorption, the residual force acting along the surface of the adsorbent increases

Option 4:
with increase in temperature, the equilibrium concentration of adsorbate increases

Correct Answer:
on a given absorbent, ammonia is absorbed more than nitrogen gas

Solution:
(a) ΔH becomes less negative as adsorption proceeds.

(b) As NH_3 is more easily liquifiable, it is adsorbed to a greater extent than N_2.

(c) On adsorption, the residual forces acting along the surface of the adsorbent **decreases**.

(d) With the increase in temperature, the equilibrium concentration of adsorbate **decreases.**

Thus, statements (a) and (b) are correct.

Therefore, **Option(2) is correct.**

Q. 7 The number of subshells associated with $n = 4$ and $m = -2$ quantum numbers is:

Option 1:
2

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Option 2:
4

Option 3:
8

Option 4:
16

Correct Answer:
2

Solution:
We have
n = 4 and m = -2

As m can take values from -l to +l and l can take values from 0 to (n-1).

Thus, possible combinations of (n, l, m) are (4, 3, -2) and (4, 2, -2)

Thus, 2 subshells are possible.

Therefore, **Option(1) is correct.**

Q. 8 The shape/structure of $[XeF_5]^-$ and XeO_3F_2 respectively are:

Option 1:
riangular bipyramidal and pentagonal planar

Option 2:
Octahedral and square pyramidal

Option 3:
Pentagonal planer and trigonal bipyramidal

Option 4:
trigonal bipyramidal and trigonal bipyramidal

Correct Answer:
Pentagonal planer and trigonal bipyramidal

Solution:
The structures are given below:

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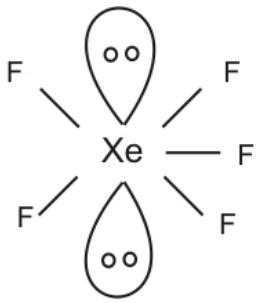

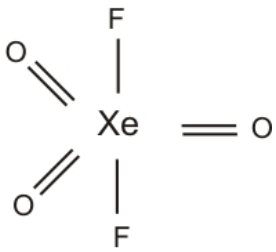

$[XeF_5]^-$ $[XeO_3F_2]$

Geometry: Pentagonal bipyramidal **Geometry:**Trigonal bipyramidal

Shape: Pentagonal planar **shape:** Trigonal bipyramidal

Q. 9 Three elements X, Y, and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is:

Option 1:
Z < Y < X

Option 2:
Y < X < Z

Option 3:
X < Z < Y

Option 4:
X < Y < Z

Correct Answer:
X < Y < Z

Solution:
As we move from left to right in a period, the acidic nature of oxide increases.
Thus, the correct order of atomic number is:
Z > Y > X

Therefore, **Option(4) is correct.**

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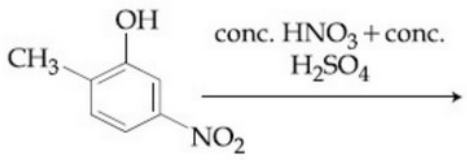
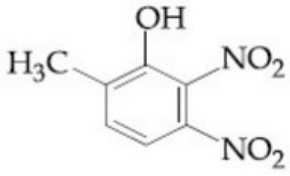
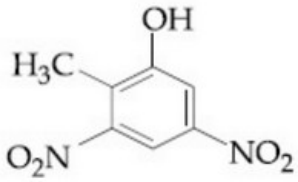
Q. 10 The major product of the following reaction is:



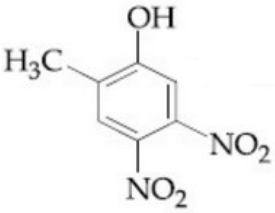
Option 1:



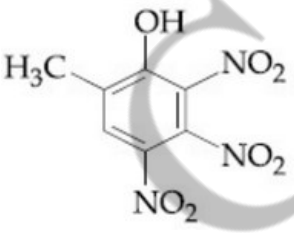
Option 2:



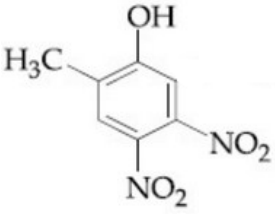
Option 3:



Option 4:



Correct Answer:



Solution:
The reaction occurs as follows:

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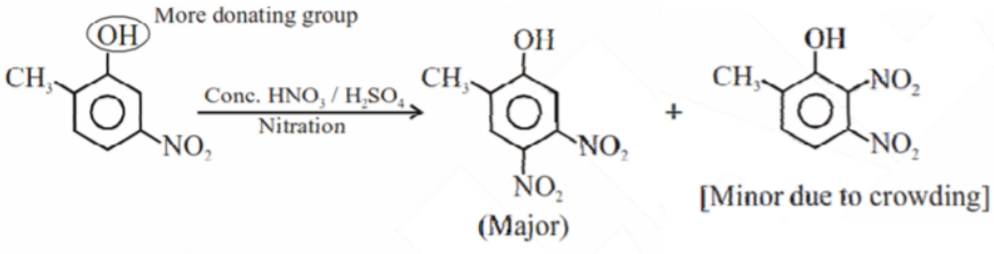


- -OH and CH_3 is electron donating group where -OH is stronger.
- $-NO_2$ is an electron withdrawing group.
- Due to srtrong donating tendency of -OH, attack of electrophile ($-NO_2^+$) will be governed by -OH and $-NO_2^+$ will attack at para position w.r.t -OH.

Therefore, **Option(3) is correct.**

Q. 11 If you spill a chemical toilet cleaning liquid on your hand, your first aid would be:

Option 1:
aqueous NH_3

Option 2:
gaseous $NaHCO_3$

Option 3:
aqueous NaOH

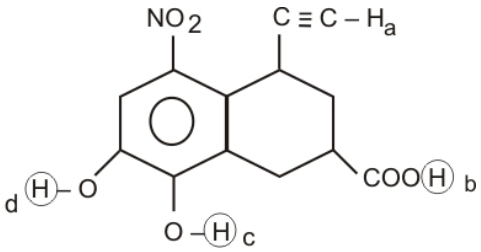
Option 4:
Vinegar

Correct Answer:
gaseous $NaHCO_3$

Solution:
Chemical toilet cleaning liquid contains acid and hence the first aid would be a weak base like $NaHCO_3$.

Therefore, **Option(2) is correct.**

Q. 12 Arrange the following labeled hydrogens in decreasing order of acidity:



Option 1:
c > b > a > d

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Option 2:
b > a > c> d

Option 3:
b > c > d > a

Option 4:
c > b > d > a

Correct Answer:
b > c > d > a

Solution:
Stable the anion, stronger the acid.

Thus, K_a: b > c > d > a

Anion of C is stabilized by -M effect of $-NO_2$ and benzene but anion of d is only stabilized by -M effect of benzene.

Therefore, **Option(3) is correct.**

Q. 13 Two elements A and B have similar chemical properties. They dont form solid hydrogen carbonate but react with nitrogen to form nitrides. A and B, respectively, are:

Option 1:
Cs and Ba

Option 2:
Na and Rb

Option 3:
Li and Mg

Option 4:
Na and Ca

Correct Answer:
Li and Mg

Solution:
Li and Mg show diagonal relationship. they do not form solid hydrogen carbonate and react with nitrogen to form nitride.

Therefore, **Option(3) is correct.**

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Q. 14 The size of a raw mango shrinks to a much smaller size when kept in a conectrated salt solution which one of the following process can explain this?

Option 1:
Reverse osmosis

Option 2:
Diffusion

Option 3:
Osmosis

Option 4:
Dialysis

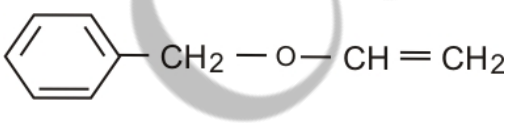
Correct Answer:
Osmosis

Solution:
Size of mango shrinks when it is kept in concentrated salt solution. This is due to osmosis in which the water molecules move out from the mango into the salt solution causing the mango to shrink.

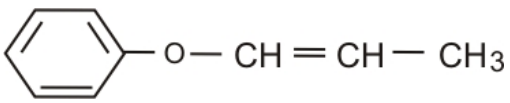
Therefore, **Option(3) is correct.**

Q. 15 An organic compound 'A' ($C_9H_{10}O$) when treated with conc HI undergoes cleavage to yeaild precipate with $AgNO_3$ where at 'C' tantomerizes to 'D'. 'D' gives positive iodoform test. 'A' could be?

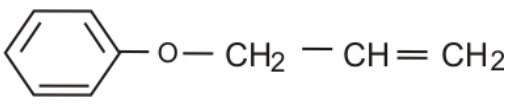
Option 1:



Option 2:



Option 3:



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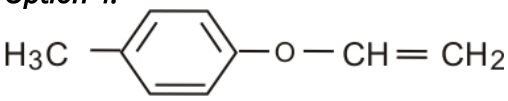
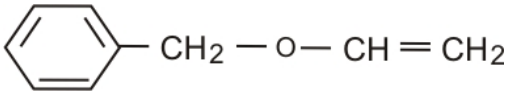
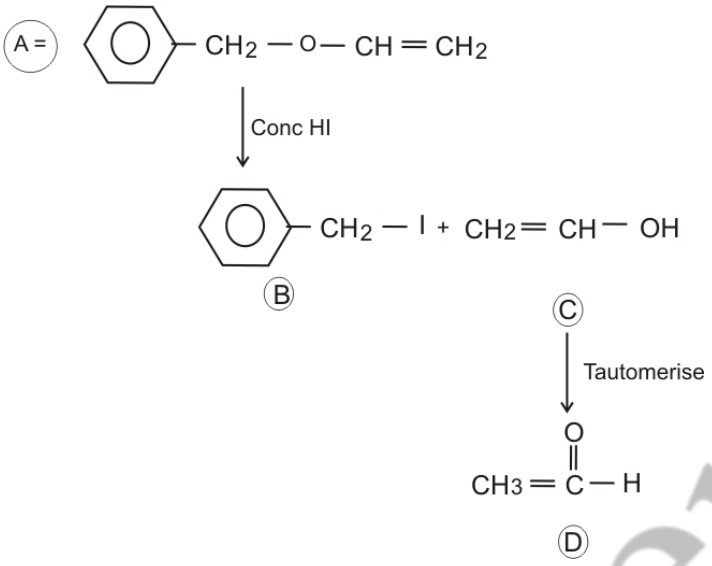
Option 4:



Correct Answer:



Solution:
The reactions occur as follows:



Therefore, **Option(1) is correct.**

Q. 16 Two compunds A and B with some molecular formula (C_3H_6O) undergo Grignand's reactions with methylmagnesium bromide to give products C and D show the following chemical tests.

Test	**C**	**D**
Ceric ammonium nitrate test	Positive	Positive
Lucas test	Turbidity obtained after five minutes	Turbidity obtained immedietely
Iodoform test	Positive	Negative

C and D respectively are:

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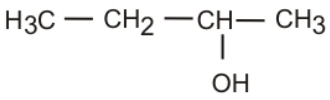
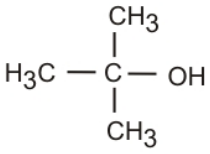
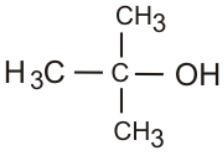
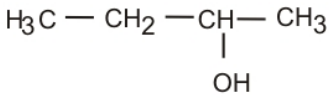
Option 1:

C = 

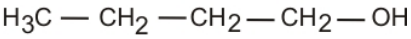
D = 

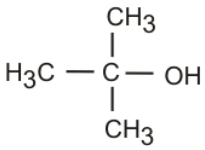
Option 2:

C = 

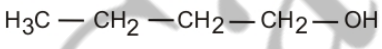
D = 

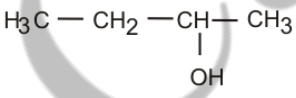
Option 3:

C = 

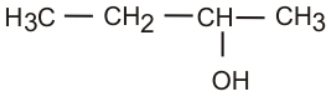
D = 

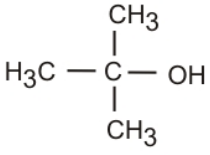
Option 4:

C = 

D = 

Correct Answer:

C = 

D = 

Solution:

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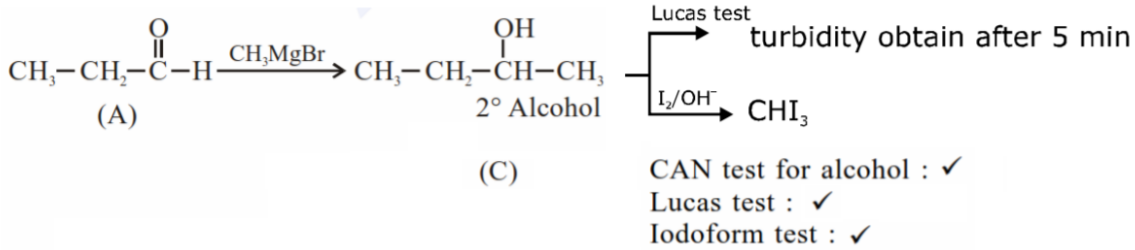
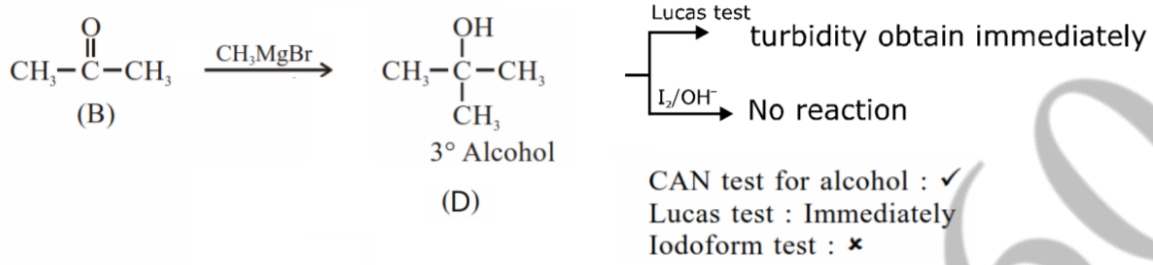
The reactions occur as follows:



(A) → (C) (2° Alcohol)

CAN test for alcohol : ✓
Lucas test : ✓
Iodoform test : ✓



(B) → (D) (3° Alcohol)

CAN test for alcohol : ✓
Lucas test : Immediately
Iodoform test : ×

Therefore, Option 1 is correct.

Q. 17 The molecular geometry of SF_6 is octahedral . What is the geometry SF_4 of including lone pairs of electrons (if any) ?

Option 1:
Trigonal Bipyramidal

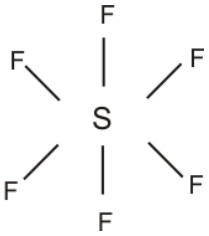
Option 2:
Square planar

Option 3:
Tetrahedral

Option 4:
Pyramidal

Correct Answer:
Trigonal Bipyramidal

Solution:
The geometry of SF_6 is given as octahedral as given below:



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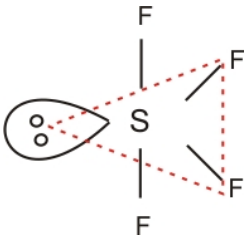
Further, The geometry of SF_4 is given as octahedral as given below:



Thus, the geometry of SF_4 including lone pair of electrons is **Trigonal bipyramidal.**

Therefore, **Option(1) is correct.**

Q. 18 The results given in the below were obtained during kinetic studies of the following reaction : $2A + B \rightarrow C + D$

Experiment	$\frac{(A)}{molL^{-1}}$	$\frac{(B)}{molL^{-1}}$	$\frac{(initial\,rate)}{molL^{-1}min^{-1}}$
i	0.1	0.1	6.00×10^{-3}
II	0.1	0.2	2.40×10^{-2}
III	0.2	0.1	1.20×10^{-2}
IV	X	0.2	7.20×10^{-2}
v	0.3	Y	2.880×10^{-1}

X and Y in the given table are respectively :

Option 1:
$0.4, 0.4$

Option 2:
$0.4, 0.3$

Option 3:
$0.3, 0.3$

Option 4:
$0.3, 0.4$

Correct Answer:
$0.3, 0.4$

Solution:
The rate law is given as:

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Rate law = $k[A][B]^2$

Case I:

$$\text{Thus, } \frac{7.2 \text{ x } 10^{-2}}{6 \text{ x } 10^{-3}} = \frac{k[X][2B]^2}{k[A][B]^2}$$

$$\text{Thus, } 12 = \frac{[X] \text{ x } [4B]}{[A] \text{ x } [B]^2}$$

$$\Rightarrow \frac{X}{A} = 3$$

$$\Rightarrow \frac{X}{A} = 3$$

Case II:

$$\text{Thus, } \frac{2.88 \text{ x } 10^{-1}}{6 \text{ x } 10^{-3}} = \frac{k[3A][Y]^2}{k[A][B]^2}$$

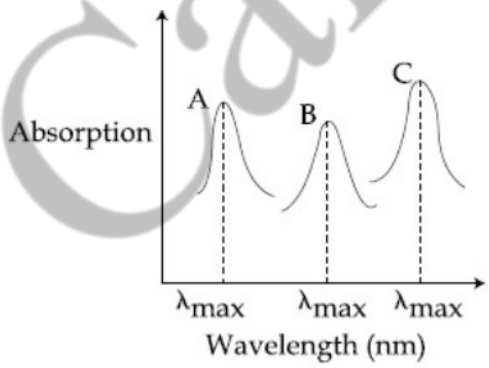
$$\Rightarrow 48 = \frac{3Y^2}{B^2}$$

$$\text{Thus, } Y = 4B$$
$$Y = 0.4$$

Thus, X and Y in the given table are: 0.3 and 0.4 respectively.

Therefore, **Option(4) is correct.**

Q. 19 Simplified adsorption spectrum of there complexed $(i), (ii) and (iii)\ of, M^{nt} ion$ are provided below ; their λ_{max} values are marked as $A, BandC$ respectively. The correct match between the complexed and their λ_{max} value is :



$[M(NCS)_6]^{(-6+n)}$

$[MF_6]^{(-6+n)}$

$[M(NH_3)_6]^{n+}$

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Option 1:
A-ii , B-i , C-iii

Option 2:
A-ii , B-iii , C-i

Option 3:
A-i , B-ii , C-iii

Option 4:
A-iii , B-i , C-ii

Correct Answer:
A-iii , B-i , C-ii

Solution:
(i) $[M(NCS)]^{(-6+n)}$: NCS is a medium filed ligand. Thus, splitting will be required medium amount of energy. Thus λ_{max} will be medium.
(ii) $[MF_6]^{(-6+n)}$: F is a weak field ligand. Thus, splitting will be low and low energy will be required. Thus, λ_{max} will be highest.

(iii) $[M(NH_3)]^{n+}$: NH_3 is strong filed ligand. Thus, splitting will be high and high energy will be required. Thus, λ_{max} will be lower.

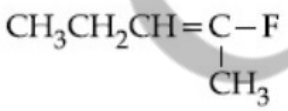
Thus, the correct sequence is:

A-(iii), B-(i), C-(ii)

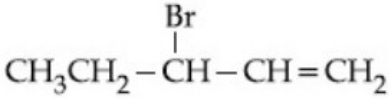
Therefore, **Option(4) is correct.**

Q. 20 The major product obtained from E_2 elemination of $3\text{-bromo-2-Fluoropentane}$ is:

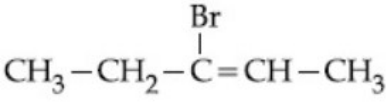
Option 1:



Option 2:



Option 3:



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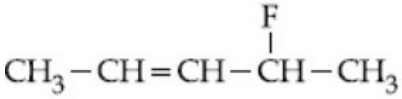
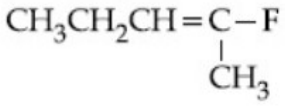
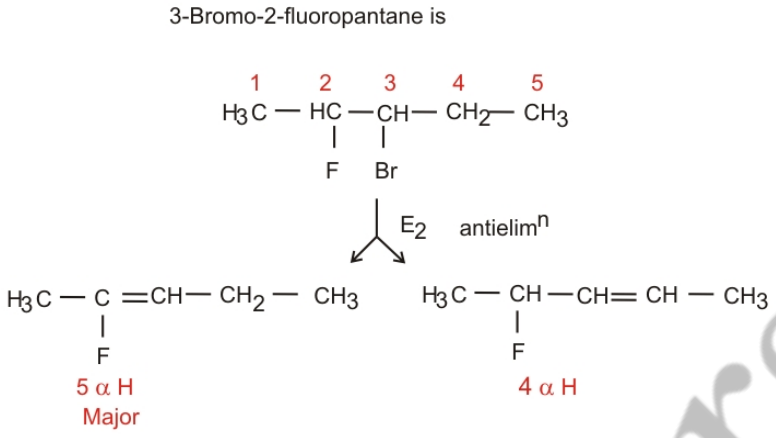
Option 4:



Correct Answer:



Solution:
The reaction occurs as follows:



Therefore, the Option 1 is correct.

Q. 21 The oxidation states of transition metal atoms in $K_2Cr_2O_7, KMnO_4$ and K_2FeO_4, respectively are X, Y and Z , the sum of X, Y and Z is _____

Correct Answer:
19

Solution:
In $K_2Cr_2O_7$, Cr is in +6, thus, x = 6

In $KMnO_4$, Mn is in +7, thus, y = 7

In K_2FeO_4, Fe is in +6, thus, z = 6

Thus, we have:

x + y + z = 6 + 7 + 6 = 19

Therefore, the correct answer is 19.

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Q. 22 The work function of sodium metal is $4.41 \times 10^{-19} j$, if photons of wavelength 300 nm are incident on the metal , the Kinetic Energy of the ejected electrons will be $(h = 6.63 \times 10^{-54} js; c = 3 \times 10^{8} ms)$ _____ $\times 10^{-21} J$

Correct Answer:
222

Solution:
We have:

W_o = 4.41 x 10^{-19}J

$hv = hv_o + K.E$

Thus, we have:

$$\frac{hc}{\lambda} = 4.41 \text{ x } 10^{-19} + \text{K.E}$$

$$\text{K.E} = \frac{hc}{\lambda} - 4.41 \text{ x } 10^{-19}$$

$$\text{K.E} = \frac{6.63 \text{ x } 10^{-34} \text{ x } 3 \text{ x } 10^{8}}{300 \text{ x } 10^{-9}} - 4.41 \text{ x } 10^{-19}$$

$$\text{K.E} = 6.63 \text{ x } 10^{-34} \text{ x } 10^{15} - 4.41 \text{ x } 10^{-19}$$

$$\text{K.E} = 222 \text{ x } 10^{-21} \text{J}$$

Thus, the correct answer is 222.

Q. 23 For the disproportionation reaction

$2Cu^{+}(aq) \rightleftharpoons Cu(s) + Cu^{2+}(aq) at, 298K,$ lnK (where K is the equillibrium Constant) is _____ $\times 10^{-1}$

Given $\left(E^{0}_{\frac{Cu^{2+}}{Cu^{+}}} = 0.16V, E^{0}_{\frac{Cu^{+}}{Cu}} = 0.52V, \frac{RT}{F} = 0.025 \right)$

Correct Answer:
144

Solution:
We have given:

$$\text{Cu}^{2+} \overset{0.16\text{V}}{\rightarrow} \text{Cu}^{+} \overset{0.52}{\rightarrow} \text{Cu}$$

Thus, $E^{o}_{\text{disproportionation}}$ = 0.52 - 0.16 = 0.36V

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Now, we know:

ΔG= -nFE^o = -nRTlnK

$$\frac{\mathrm{RT}}{\mathrm{F}}\ln\mathrm{K} = 1\,\mathrm{x}\,0.36$$

$$\ln\mathrm{K} = \frac{0.36}{0.025}$$

Thus, lnK = 14.4

Thus, lnK = 144 x 10^{-1}

Q. 24 The heat of Combustion of ethanol into Carbon dioxide and water is $-327Kcal$ at constant pressure . The heat evolved (in cal) at constant volume and $27^{\circ}C$ (if all gases behave ideally) is _____

$R = 2calmol^{-1}K^{-1}$

Correct Answer:
-326400

Solution:
The reaction occurs as follows:

$$\mathrm{C_2H_5OH} + 3\mathrm{O_2} \rightarrow 2\mathrm{CO_2} + 3\mathrm{H_2O}$$

ΔH = -327kcal

Δn = 1

Now,

ΔH = ΔU + ΔnRT

ΔU = ΔH - ΔnRT

$$\Delta\mathrm{U} = -327 + \frac{1\mathrm{x}2\mathrm{x}300}{1000}$$

ΔU = -326.4kcal

Therefore, correct answer is -326.4kcal.

Ans = -326400.00

Q. 25 The ratio of the masses percentages of $'C'and,'H'$ and $'C'and,'O'$ of a saturated acyclic organic compound $'X'$ are 4:1 and 3:4 respectively . Then , the moles of oxygen gas required for complete combustion of two moles of organic compound $'X'$ is_____

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Correct Answer:
5

Solution:
We have

Mass ratio of C:H = 4:1

Mass ratio of C:O = 3:4

Thus, mass ratio of C:H:O = 12:3:16

Mass ratio of C:H:O = 1:3:1

Thus, empirical formula = CH_3O

Now, since $(CH_3O)_n$ is saturated acyclic. Hence only possible value of n is 2.

Thus, the compound is $C_2H_6O_2$.

Now, the reaction occurs as follows:

$$C_2H_6O_2 + \frac{5}{2}O_2 \rightarrow 2CO_2 + 3H_2O$$

Thus, 5 moles of O_2 are required for combustion of 2 moles of the organic compound.

Maths

Q. 1 Let $A = \{X = (x, y, z)^2 : Px = 0, and, x^2 + y^2 + z^2 = 1\}$ where $P = \begin{pmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{pmatrix}$ then Set A contains :

Option 1:
exactly two elements

Option 2:
is an empty set ,

Option 3:
is a singletan set ,

Option 4:
contains more than 2 elements .

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Correct Answer:
exactly two elements

Solution:

$$\text{Given } P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix},$$

$$|P| = 1(3\times(-1)-(-4)\times 9)-2((-1)\times(-2)-(-4)\times 1)+1(9\times(-2)-3\times 1)$$

$$|P| = 0$$

Here $|P| = 0$

also given $PX = 0$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\left.\begin{array}{l} x+2y+z=0 \\ \Rightarrow -2x+3y-4z=0 \\ x+9y-z=0 \end{array}\right\} D=0, \text{ so system have infinite many solutions,}$$

By solving these equation

$$l \text{ we get } x = \frac{-11\lambda}{2}; y = \lambda; z = \frac{7\lambda}{2}$$

$$\text{Also given, } x^2+y^2+z^2 = 1$$

$$\Rightarrow \left(\frac{-11\lambda}{2}\right)^2 + \lambda^2 + \left(\frac{7\lambda}{2}\right)^2 = 1$$

$$\Rightarrow \lambda = \pm\frac{1}{\sqrt{\frac{121}{4}+1+\frac{49}{4}}}$$

So, there are 2 values of λ

Therefore, there are 2 solution set of (x, y, z).

Q. 2 Let $f : R \to R$ be a function which statisfies $f(x+y) = f(x)+f(y)$, $x, y \epsilon R, if\, f(1) = 2, and g(n) = \sum_{k=1}^{n-1} f(k), n \epsilon N$ then the value of n for which $g(n) = 20, is:$

Option 1:
4

Option 2:
5

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Option 3:
9

Option 4:
20

Correct Answer:
4

Solution:

$f(x+y)=f(x)+f(y)$

$f(1)=2$

$g(x)=\sum_{k=1}^{n-1} f(k) \quad n \in N$

$g(n)=20$

$f(1)=f(1)+f(0)$

$f(2)=2f(2)=4$

$f(3)=8f(2)+f(2)=6$

$f(4)=8$

$f(5)=f(3)+f(2)=6+4=10$

$f(x)=2x$

$\Rightarrow 2\times\frac{n(n+1)}{2}=20$

$\Rightarrow 4\times 5=20\Rightarrow n=4$

Q. 3

$a, b, c \epsilon R$ non -zero and satisfies $a^3+b^3+c^3=2$ if the matrix $A=\begin{Bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{Bmatrix}$ satisfies $A^TA=I$ then the value of abc can be ;

Option 1:
$\frac{-1}{3}$

Option 2:
$\frac{1}{3}$

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Option 3:

$\frac{2}{3}$

Option 4:

3

Correct Answer:

$\frac{1}{3}$

Solution:

$$A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$$

$$A^T = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$$

$$A \cdot A^T = I$$

$$\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} \times \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a^2+b^2+c^2 & ab+bc+ca & ab+bc+ca \\ ab+bc+ca & a^2+b^2+c^2 & ab+bc+ca \\ ab+bc+ca & ab+bc+ca & a^2+b^2+c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow a^2+b^2+c^2 = 1$ and $ab+bc+ca = 0$

$(a+b+c)^2 = 1 \Rightarrow (a+b+c) = \pm 1$

$a^3+b^3+c^3 = (a+b+c)\left[a^2+b^2+c^2-ab-bc-ac\right]+3abc$

$2 = 1 \times [\pm 1] + 3abc$

$abc = 1 \quad \text{or} \quad abc = \frac{1}{3}$

Q. 4 Equation of normal to the curve . $y = (1+x)^{2y} + \cos^2(\sin^{-1} x)$ at x=0 is

Option 1:

$y + 4x = 2$

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Option 2:
$x + 4y = 8$

Option 3:
$y = 4x + 2$

Option 4:
$2y + x = 4$

Correct Answer:
$2y + x = 4$

Solution:
$y = (1 + x)^2 + \cos^2(\sin^{-1} x)$
$\frac{dy}{dx} = 2(1 + x) + 2\cos(\sin^{-1} x) \cdot (-\sin(\sin^{-1} x))\frac{1}{\sqrt{1 - x^2}}$

$\frac{dy}{dx}|_{x=0} = 2$
slope of normal
$\frac{dx}{dy} = -\frac{1}{2}$
$y - y_1 = -\frac{1}{2}(x - x_1)$
at x=0, $y = 2$
$y - 2 = -\frac{1}{2}(x - 0)$
$2y - 4 = -x$

Q. 5 Tautology ?

Option 1:
$(\sim q) \vee (p \wedge q) \rightarrow q$

Option 2:
$(\sim p) \wedge (p \vee q) \rightarrow q$

Option 3:
$(q \rightarrow p) \vee \sim (p \rightarrow q)$

Option 4:
$(p \rightarrow q) \wedge (q \rightarrow p)$

Correct Answer:
$(\sim p) \wedge (p \vee q) \rightarrow q$

Solution:

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Truth table of $(\sim p) \wedge (p \vee q) \rightarrow q$

$\mathbf{p}$	$\mathbf{q}$	$((\neg \mathbf{p} \wedge (\mathbf{p} \vee \mathbf{q})) \rightarrow \mathbf{q})$
F	F	$\mathbf{T}$
F	T	T
T	F	T
T	T	T

Q. 6 Imaginary part of $\left(3+2\sqrt{-54}\right)^{\frac{1}{2}}-\left(3-2\sqrt{-54}\right)^{\frac{1}{2}}$ can be:

Option 1:
6

Option 2:
$\sqrt{6}$

Option 3:
$-\sqrt{6}$

Option 4:
$-2\sqrt{6}$

Correct Answer:
$-2\sqrt{6}$

Solution:
$(3+2\sqrt{-54})^{1/2}-(3-2\sqrt{-54})^{1/2}$

$(3+2\cdot 3\sqrt{6}i)^{1/2}-(3-2\cdot 3\sqrt{6}i)^{1/2}$

$[3^2+(\sqrt{6}i)^2+2\cdot 3\sqrt{6}i]^{1/2}-[3^2+(\sqrt{6}i)^2-2\cdot 3\sqrt{6}i]^{1/2}$

$\left|3+\sqrt{6}i\right|-\left|3-\sqrt{6}\right|$

$b_1=\pm\sqrt{6}$ and $b_2=\pm\sqrt{6}$

So the value of $b_1+b_2=2\sqrt{6}$ or $-2\sqrt{6}$ or 0

Q. 7 For some $\theta\epsilon\left(0,\frac{\pi}{2}\right)$ if the eccentricity of the hyperbola $x^2-y^2\sec^2\theta=10$ is $\sqrt{5}$ times the eccentricity of the ellipse , $x^2\sec^2\theta+y^2=5,$ then the length of the latus rectum of the ellipse is

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Option 1:
$\sqrt{30}$

Option 2:
$\frac{4\sqrt{5}}{3}$

Option 3:
$\frac{2\sqrt{5}}{7}$

Option 4:
$\frac{2\sqrt{5}}{6}$

Correct Answer:
$\frac{4\sqrt{5}}{3}$

Solution:
equation of hyperbola

$$x^2 - \frac{y^2}{\cos^2\theta} = 10$$

equation of ellipse

$$\frac{x^2}{\cos^2\theta} + y^2 = 5$$

$$e_h = \sqrt{1 + \frac{\cos^2\theta}{1}}, \quad e_e = \sqrt{1 - \frac{\cos^2\theta}{1}}$$

$$1 + \cos^2\theta = 5\sin^2\theta$$

$$2 = 6\sin^2\theta$$

$$\sin^2\theta = 1/3$$

$$\text{LR of ellipse} = \frac{2 \times \frac{2}{3} \times \cdot 5}{\sqrt{5}} = \frac{4}{3}\sqrt{5}$$

Q. 8 Let E^c denote the complement of an event E, E_1, E_2, E_3 be any pair wise independent events with $P(E) > 0$ and $P(E_1 \cap E_2 \cap E_3) = 0$ then $P\left(\frac{E_2 \cap E_3}{E_1}\right)$ equal to:

Option 1:
$P(E_3^c) - P(E_2^c),$

Option 2:
$P(E_3^c) - P(E_2),$

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Option 3:
$P(E_2^c) - P(E_3),$

Option 4:
$P(E_3) - P(E_2^c),$

Correct Answer:
$P(E_2^c) - P(E_3),$

Solution:

$$P(\bar{E}_2 \cap \bar{E}_3 / E_1) = \frac{P(\bar{E}_2 \cap \bar{E}_3 \cap E_1)}{P(E_1)} = \frac{P(\bar{E}_2)P(\bar{E}_2)P(E_1)}{P(E_1)} = P(\bar{E}_2)P(\bar{E}_3)$$

$$= P(\bar{E}_2)\{1 - P(E_3)\} = P(\bar{E}_2) - P(\bar{E}_2)P(E_3)$$

$$= P(\bar{E}_2) - \{1 - P(E_2)\}P(E_3) = P(\bar{E}_2) - P(E_3) + P(E_2)P(E_3)$$

now given that E, E_1, E_2, E_3 are independent and $P(E_1 \cap E_2 \cap E_3) = 0$

$\Rightarrow P(E_1)P(E_2)P(E_3) = 0$ but $P(E_1) \neq 0$

$\text{hence,} \quad P(\bar{E}_2 \cap \bar{E}_3 / E_1) = P(\bar{E}_2) - P(E_3)$

Q. 9 Area of an equilateral triangle inscribed in the parabola $y^2 = 8x$ with one of its vertices on the vertex of this parabola is:

Option 1:
$128\sqrt{3}$

Option 2:
$192\sqrt{3}$

Option 3:
$64\sqrt{3}$

Option 4:
$256\sqrt{3}$

Correct Answer:
$192\sqrt{3}$

Solution:

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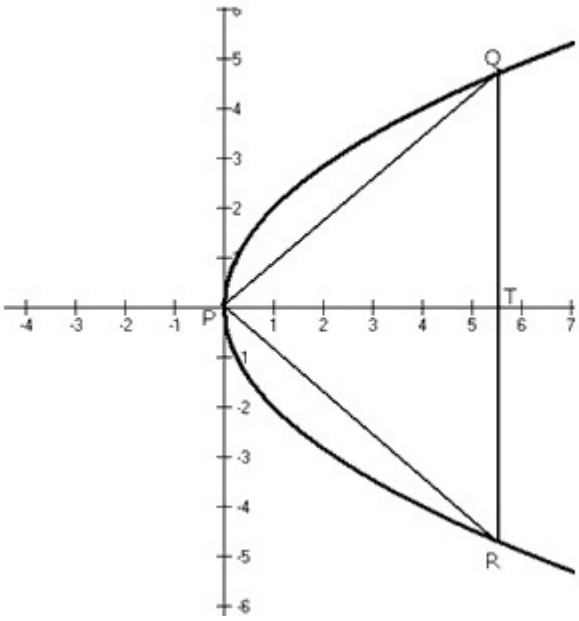

Let the two vertices of the triangle be Q and R. Points Q and R will have the same x - coordinate =k(say)

Now in the right triangle PRT, right-angled at T.

$$\tan 30^\circ = \frac{\mathrm{RT}}{\mathrm{k}} \Rightarrow \frac{1}{\sqrt{3}} = \frac{\mathrm{RT}}{\mathrm{k}} \Rightarrow \mathrm{RT} = \frac{\mathrm{k}}{\sqrt{3}} \Rightarrow \mathrm{R}\left(\mathrm{k}, \frac{\mathrm{k}}{\sqrt{3}}\right)$$

Now R lies on the parabola: $y^2 = 4ax$

$$\Rightarrow \left(\frac{k}{\sqrt{3}}\right)^2 = 4a(k)$$

$$\Rightarrow \frac{k}{3} = 4a$$

$$\Rightarrow k = 12a$$

$$\text{Length of side of the triangle } = 2(\mathrm{RT}) = 2 \cdot \frac{\mathrm{k}}{\sqrt{3}} = 2 \cdot \frac{(12\mathrm{a})}{\sqrt{3}} = 8\sqrt{3}\mathrm{a} = 16\sqrt{3}$$

$$\text{Area of Equilateral triangle is } = \frac{\sqrt{3}}{4}a^2 = 192\sqrt{3}$$

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Q. 10 Sum of first 11 terms of an A.P $a_1, a_2, \ldots$ is 0 $(a_1 \neq 0)$ then the sum of the A.P $a_1, a_3, a_5, \ldots, a_{23}$ is ka_1 where k is equal to:

Option 1:

$\frac{121}{10}$

Option 2:
$\frac{-121}{10}$

Option 3:
$\frac{72}{5}$

Option 4:
$\frac{-72}{5}$

Correct Answer:
$\frac{-72}{5}$

Solution:
$S_n = \frac{n}{2}(2a_1 + (n-1)d)$

$S_{11} = \frac{11}{2}(2a_1 + 10d) = 0$
$a_1 = -5d$

Now required series is AP with 12 terms, a_1 as first term and 2d as common difference,

$$\begin{aligned} S_{12} &= \frac{12}{2}(2a_1 + 22d) \\ &= (a_1 + 11d) \times 12 \\ &= 6d \times 12 \\ &= 72d = -\frac{72}{5}a \end{aligned}$$

Q. 11 Let $f(x)$ is a quadratic polynomial such that $f(-1) + f(-2) = 0$ If the roots of $f(x) = 0$ s 3, then the other root lies in:

Option 1:
(-3,-1)

Option 2:
(0,2)

Option 3:
(1,3)

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Option 4:
(-1,0)

Correct Answer:
(-1,0)

Solution:
$f(x) = a(x-3)(x-\alpha)$

$f(2) = a(\alpha - 2)$

$f(-1) = 4a(1+\alpha)$

$f(-1) + f(2) = 0 \Rightarrow a(\alpha - 2 + 4 + 4a) = 0$

$a \neq 0 \Rightarrow 5\alpha = -2$

$\alpha = \frac{2}{5} = -0.4$

$\alpha \in (-1, 0)$

Q. 12 S is sum of first 9 terms $(x+ka) + (x^2 + (k+2)a) + (x^3 + (k+4)a) + (x^4 + (k+6)a) + \ldots$ where $a \neq 0$ and $x \neq 1$.If $S = \frac{x^{10} - x + 45a(x-1)}{x-1}$ then k is?

Option 1:
3

Option 2:
2

Option 3:
-3

Option 4:
-5

Correct Answer:
-3

Solution:
$S = (x+ka) + \left(x^2 + (k+2)a\right) + \left(x^3 + (k+4)a\right) + \left(x^4 + (k+6)a\right) + \ldots$

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$= \left(x + x^2 + x^3 + - - - x^9\right) + a(k + (k+2) + (k+4) + \cdots (k+16))$

$First\ series\ is\ GP\ and\ second\ series\ becomes\ an\ AP\ after\ separating\ k\ terms\ .So,$

$= \dfrac{x\left(x^9 - 1\right)}{(x-1)} + a[9k + 72]$

Now, comparing this with $S = \dfrac{x^{10} - x + 45a(x-1)}{x-1}$

$\therefore 9k + 72 = 45$

$k + 8 = 5$

$k = -3$

Q. 13 If a curve $y = f(x)$, passing through the points (0,2) is the solution of the differential equation $2x^2\mathrm{dy} = (2xy + y^2)\mathrm{dx}$ then $f\left(\dfrac{1}{2}\right)$ is equal to:

Option 1:

$\dfrac{1}{1 + \log_e 2}$

Option 2:

$1 + \log_e 2$

Option 3:

$\dfrac{-1}{1 + \log_e 2}$

Option 4:

$\dfrac{1}{-1 - \log_e 2}$

Correct Answer:

$\dfrac{1}{1 + \log_e 2}$

Solution:

$2x^2 dy = \left(2xy + y^2\right) dx$

$\frac{dy}{dx} = \frac{2xy + y^2}{2x^2}$

$\frac{dy}{dx} = \frac{2\left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2}{2}$

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$$\frac{y}{x} = t$$

$y = tx \Rightarrow \frac{dy}{dx} = t + x\frac{dt}{dx}$

$t + x\frac{dt}{dx} = t + \frac{t^2}{2}$

$\Rightarrow \int \frac{2}{t^2} dt = \int \frac{dx}{x}$

$\Rightarrow -\frac{2}{t} = \ln x + c$

$\Rightarrow -\frac{2x}{y} = \ln x + c$

$-\frac{2\times 1}{2} = 0 + c \Rightarrow c = -1$

$-\frac{2x}{y} = \ln x - 1$

$\text{at } x = 1/2$

$$\frac{-2 \times \frac{1}{2}}{y} = \ln\left(\frac{1}{2}\right) - 1$$

$$-\frac{1}{y} = \ln\left(\frac{1}{2}\right) - 1 \Rightarrow y = \frac{1}{1 + \ln 2}$$

Q. 14 If the equation $\cos^4\theta + \sin^4\theta + \lambda = 0$ does have real solutions for θ then λ lies in the interval:

Option 1:

$\left(\frac{-5}{4}, -1\right)$

Option 2:

$\left(\frac{-1}{2}, \frac{-1}{4}\right]$

Option 3:

$\left[\frac{-3}{2}, \frac{-5}{4}\right]$

Option 4:

$\left[-1, \frac{-1}{2}\right]$

Correct Answer:

$\left[-1, \frac{-1}{2}\right]$

Solution:

$\cos^4\theta + \sin^4\theta + \lambda = 0$

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$$f(\theta) = \cos^4\theta + \sin^4\theta$$

$$= \left(\cos^2\theta + \sin^2\theta\right)^2 - 2\sin^2\theta\cos^2\theta$$

$$= 1 - \frac{\sin^2 2\theta}{2}$$

$$\sin^2 2\theta \in [0, 1]$$

$$\frac{\sin^2 2\theta}{2} \in \left[\frac{1}{2}, 1\right]$$

$$\therefore \lambda \in \left[-1, -\frac{1}{2}\right]$$

Q. 15 Consider a region $R = \left\{(x, y) \in \mathbb{R}^2 : x^2 \leq 4 \leq 2x\right\}$. if a line $y = \alpha$ divided the region R into two equal parts, then which of the following is true?

Option 1:

$3\alpha^2 - 8\alpha + 8 = 0$

Option 2:

$\alpha^2 - 6\alpha + 16 = 0$

Option 3:

$\alpha^3 - 6\alpha^{\frac{3}{2}} - 16 = 0$

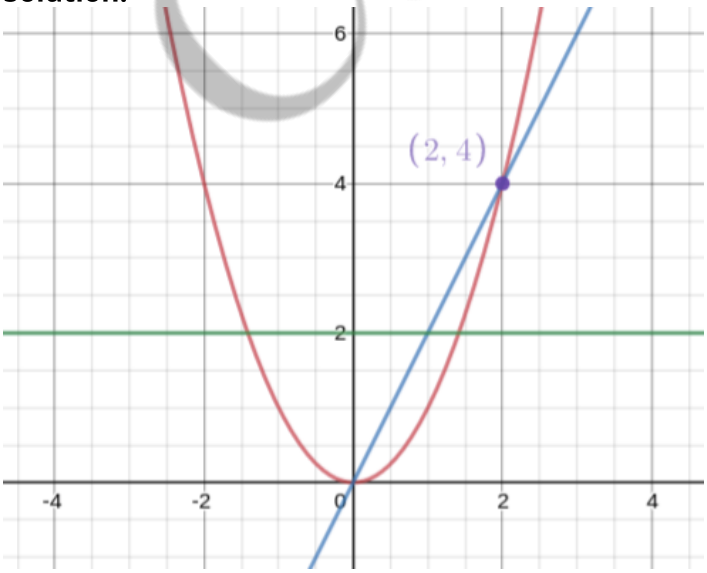
Option 4:

$3\alpha^2 - 8\alpha^{\frac{3}{2}} + 8 = 0$

Correct Answer:

$3\alpha^2 - 8\alpha^{\frac{3}{2}} + 8 = 0$

Solution:



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$x^2 \leq y \leq 2x$

$\int_0^{\alpha} \left(\sqrt{y} - \frac{y}{2}\right) dy = \frac{1}{2}\int_0^2 (2x - x^2)\, dx$

$\frac{2}{3}y^{3/2} - \frac{y^2}{4}\Big|_0^{\alpha} = \frac{1}{2}\left[x^2 - \frac{x^3}{3}\right]_0^2$

$\Rightarrow \frac{2}{3}\alpha^{3/2} - \frac{\alpha^2}{4} = \frac{1}{2}\left[4 - \frac{8}{3}\right]$

$\Rightarrow \frac{8\alpha^{3/2} - 3\alpha^2}{12} = \frac{1}{2} \times \frac{4}{3}$

$\Rightarrow 8\alpha^{3/2} - 3\alpha^2 = 8$

Q. 16 Let n > 2 be an integer. Suppose that there are n Metro station in a city located along a circular path. Each pair of stations is connected by a straight track only. Further, each pair of nearest station is connected by blue line, whereas all remaining pair of stations are connected by red line. If the number of red lines is 99 times the number of blue lines, then the value of n is :

Option 1:
101

Option 2:
201

Option 3:
200

Option 4:
199

Correct Answer:
201

Solution:

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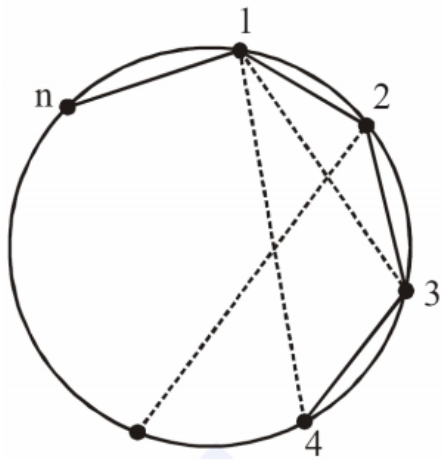

Number of blue lines = Number of sides = n

Number of red lines = number of diagonals = ${}^{n}C_{2} - n$

$${}^{n}C_{2} - n = 99n \Rightarrow \frac{n(n-1)}{2} - n = 99n$$

$$\frac{\mathrm{n}-1}{2} - 1 = 99 \Rightarrow \mathrm{n} = 201$$

Q. 17 $Let\, f : (-1, \infty) \to R\ be\ defined\ by\ f(0) = 1 and\ f(x) = \frac{1}{x}\log_e(1+x), x \neq 0$

Then the function f :

Option 1:
Decreases in (-1,0) and increases in (0,∞)

Option 2:
increases in (-1,∞).

Option 3:
increases in (-1,0) and decreases in (0,∞)

Option 4:
decreases in (-1,∞).

Correct Answer:
decreases in (-1,∞).

Solution:

$$f'(x) = \frac{\frac{x}{1+x} - \ln(1+x)}{x^2} = \frac{x - (1+x)\ln(1+x)}{x^2(1+x)}$$

Here ${}_{x}2$ is always positive

Hence

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$f'(x)$ depends on

$$\frac{x-(1+x)\ln(1+x)}{(1+x)}$$

$$\frac{1+x-(1+x)\ln(1+x)-1}{(1+x)}$$

$$-\left(\frac{1}{1+x}+\ln(1+x)-1\right)$$

$$-\left(\frac{1}{1+x}+\ln(1+x)-\ln e\right)$$

$$-\left[\frac{1}{1+x}+\ln\left(\frac{1+x}{e}\right)\right]$$

Which is negative foe all values of $x \in (-1,\infty)$

Q. 18 $\lim\limits_{x\to 0}\int tan\left(\frac{\pi}{4}+x\right)^{v_x}$ is equal to:

Option 1:
2

Option 2:
1

Option 3:
e^2

Option 4:
e

Correct Answer:
e^2

Solution:

$$\begin{aligned}\lim_{x\to 0}\tan\left(\frac{\pi}{4}+x\right)^{1/x} &= \lim_{x\to 0}\left(\frac{\tan(\pi/4)+\tan x}{1-\tan(\pi/4)\tan x}\right)^{1/x}\\ &= \lim_{x\to 0}\left(\frac{1+\tan x}{1-\tan x}\right)^{1/x}\\ &= \lim_{x\to 0}\left(1+\frac{2\tan x}{1-\tan x}\right)^{1/x}\\ &= e^{\lim_{xn\to 0}\left(\frac{2\tan x}{1-\tan x}\right)\left(\frac{1}{x}\right)}\\ &= e^2\end{aligned}$$

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Q. 19 $y = \sum_{k=1}^{6} k \cos^{-1}\left\{\frac{3}{5}\cos kx - \frac{4}{5}\sin kx\right\}$ then $\frac{dy}{dx}$ at $x = 0$ is_____

Correct Answer:
91

Solution:

$$y = \sum_{k=1}^{6} k \cos^{-1}\left\{\frac{3}{5}\cos kx - \frac{4}{5}\sin kx\right\}$$

$$= \sum_{k=1}^{6} k \cos^{-1}(\cos(kx + \alpha))$$

$$\text{where } \cos\alpha = \frac{3}{5}$$

$$= \sum_{k=1}^{6} k(kx + \alpha)$$

$$= \frac{6 \times 7 \times 13 \times x}{6} + \frac{6 \times 7}{2}\alpha$$

$$= 91x + 21\alpha$$

$$\frac{dy}{dx} = 91$$

Q. 20 Let [t] denote the greater integer less than or equal to t,than the value of $\int_{2}^{1} |2x - [3x]|\, dx$ is _____

Option 1:
1

Option 2:
1.5

Option 3:
2

Option 4:
0.5

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Correct Answer:
1

Solution:

$$\int_1^2 |2x-[3x]|dx$$

$$\Rightarrow \int_1^{4/3} |2x-3|dx+\int_{4/3}^{5/3} |2x-4|dx+\int_{5/3}^{6} |2x-5|dx$$

$$=\frac{2}{9}+\frac{3}{9}+\frac{4}{9}=1$$

Q. 21 If the variance of the terms in on increasing AP. $b_1, b_2, b_3.....b_4$ is 90 then common difference of this A.P is ___

Option 1:
-3

Option 2:
6

Option 3:
3

Option 4:
Both (a) and (c)

Correct Answer:
3

Solution:

$$\sum \frac{(xi-\mu)^2}{n}=90$$

$$\mu=\frac{\frac{11}{2}[2a+10d]}{11}=a+5d$$

$$\sum \frac{(xi-\mu)^2}{n}=\frac{\sum xi^2}{n}-\mu^2$$

$$=\frac{\sum(a+(r-1)d)^2}{11}-(a+5d)^2$$

$$=a^2+\frac{10\times 11\times 21}{6\times 11}d^2+\frac{2ad\times 10\times 11}{2\times 11}-\left(a^2+25d^2+10ad\right)$$

$$10d^2=90$$

$$d=\pm 3$$

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But it is given that AP is increasing

Hence d = 3

Q. 22 Let the position vector S of point A and B be $\hat{i}+\hat{j}+\hat{k}$ and $\hat{2i}-\hat{j}-\hat{3k}$ respectively. A point 'p' divides the line segment AB intervally in the ratio $\lambda:1(\lambda,0)$ if O is the origin and $\overrightarrow{OB}.\overrightarrow{OP}-3\left|\overrightarrow{OA}\times\overrightarrow{OP}\right|^2=6$ then λ is equal to _____

Option 1:
8

Option 2:
0.8

Option 3:
6

Option 4:
0.6

Correct Answer:
0.8

Solution:

$$\overrightarrow{OA}\times\overrightarrow{OP}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 1 & 1\\ \frac{2\lambda+1}{\lambda+1} & 1 & \frac{3\lambda+1}{\lambda+1}\end{vmatrix}=\frac{2\lambda+1}{\lambda+1}\hat{i}+\frac{-\lambda}{\lambda+1}\hat{j}+\frac{-\lambda}{\lambda+1}\hat{k}=\frac{6\lambda^2+1}{(\lambda+1)^2}$$

$$\Rightarrow \frac{14\lambda+6}{\lambda+1}-3\times\frac{(6\lambda^2+1)}{(\lambda+1)^2}=6$$

$$\Rightarrow 10\lambda^2-8\lambda=0$$

$$\Rightarrow \lambda=0,\frac{8}{10}=0.8\Rightarrow\lambda=0.8$$

Q. 23 For a positive integer n, $(1+x)^n$ is expanded in increasing power of x, if three consecutive coefficient in this expansion are in the ratio $2:5:12$, then n is equal to = _____

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Option 1:
124

Option 2:
120

Option 3:
118

Option 4:
120

Correct Answer:
118

Solution:
Let three consecutive terms be ${}^nC_{r-1},\ {}^nC_r,\ {}^nC_{r+1}$

So,

$$\frac{{}^nC_{r-1}}{{}^nC_r} = \frac{2}{5}$$

$$\frac{r}{n-r+1} = \frac{2}{5} - - - (i)$$

And

$$\frac{{}^nC_r}{{}^nC_{r+1}} = \frac{5}{12}$$
$$\frac{r+1}{n-r} = \frac{5}{12} \ldots\ldots\ldots\ldots (ii)$$

From euqation (i) and (ii)

n = 118

Q. 24 A plane passing through the point $(3, 1, 1)$ contains two lines whose direction ratios are $1, -2, 2$ and $2,\ 3,\ -1$ respectively. If this plane also passes through the point $(\alpha,\ -3,\ 5)$, the α is equal to :

Option 1:
10

Option 2:
-10

Option 3:
5

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Option 4:

-5

Correct Answer:

5

Solution:

Hence normal is $\perp^{r}$ to both the lines so normal vector to the plane is

$\overrightarrow{n} = (\hat{i} - 2\hat{j} + 2\hat{k}) \times (2\hat{i} + 3\hat{j} - \hat{k})$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 2 & 3 & -1 \end{vmatrix} = \hat{i}(2-6) - \hat{j}(-1-4) + \hat{k}(3+4)$$

$\vec{n} = -4\hat{i} + 5\hat{j} + 7\hat{k}$

Now equation of plane passing through $(3, 1, 1)$ is

$\Rightarrow -4(x-3) + 5(y-1) + 7(z-1) = 0$

$\Rightarrow -4x + 12 + 5y - 5 + 7z - 7 = 0$

$\Rightarrow -4x + 5y + 7z = 0$

Plane is also passing through $(\alpha, -3, 5)$ so this point satisfies the equation of plane so put in equation (1)

$-4\alpha + 5 \times (-3) + 7 \times (5) = 0$

$\Rightarrow -4\alpha - 15 + 35 = 0$

$\Rightarrow \alpha = 5$

Q. 25 The set of all possible values of θ in the interval $(0, \pi)$ for which the points $(1, 2)$ and $(\sin\theta, \cos\theta)$ lie on the same side of the line $x + y = 1$ is :

Option 1:

$\left(0, \frac{\pi}{4}\right)$

Option 2:

$\left(0, \frac{3\pi}{4}\right)$

Option 3:

$\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

Option 4:

$\left(0, \frac{\pi}{2}\right)$

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Correct Answer:

$\left(0,\ \frac{\pi}{2}\right)$

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3rd September Shift 1

Physics

Q. 1 A charged particle carrying charge $1\mu C$ is moving with velocity $\left(2\hat{i}+3\hat{j}+4\hat{k}\right) ms^{-1}$. If an external MF of $\left(5\hat{i}+3\hat{j}-6\hat{k}\right) \times 10^{-3}T$ exists in the region where the particle is moving then the force on the particle is $\underset{F}{\rightarrow} \times 10^{-9}N$ The vector $\underset{F}{\rightarrow}$ is

Option 1:
$0.3\hat{i}+0.32\hat{j}-0.09\hat{k}$

Option 2:
$-30, 32, -9$

Option 3:
$-300, +320, -90$

Option 4:
$-3, +3.2, -0.9$

Correct Answer:
$-30, 32, -9$

Solution:
$given,\ q=1\mu C=1\times 10^{-6}C$
$\vec{v}=2\hat{i}+3\hat{j}+4\hat{k}$
$\vec{B}=(5\hat{i}+3\hat{j}-6\hat{k})\times 10^{-3}T$
We know that,
$\vec{F}=q(\vec{v}\times\vec{B})$
$\Rightarrow \vec{F}\times 10^{-9}=10^{-6}[(-18-12)\hat{i}-(-12-20)\hat{j}+(6-15)\hat{k}]\times 10^{-3}$
$\Rightarrow \vec{F}=-30\hat{i}+32\hat{j}-9\hat{k}$

So, option 2 is correct.

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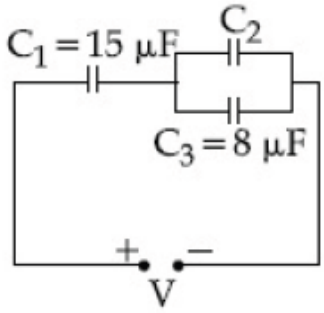
Q. 2 In the circuit shown in the figure the total charge is $750\mu F$ and the voltage across capacitor C_2 is 20 V. Then the charge on the capacitor C_2 is.



Option 1:
$450\ \mu C$

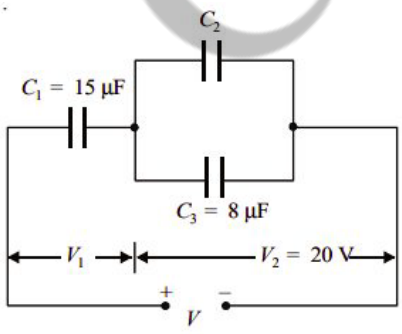
Option 2:
$590\ \mu C$

Option 3:
$160\ \mu C$

Option 4:
$650\ \mu C$

Correct Answer:
$590\ \mu C$

Solution:



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Voltage across $C_1 = V_1 = \dfrac{q}{C_1} = \dfrac{(750 \times 10^{-6})}{15 \times 10^{-6}} = 50\text{V}$
Applied voltage, $V = V_1 + V_2 = 50 + 20 = 70\text{V}$
Charge on $C_3 = C_3V_2 = \left(8 \times 10^{-6}\right) 20 = 160 \times 10^{-6} = 160\mu\text{C}$
$\therefore\ Charge\ on\ C_2 = 750 - 160 = 590\mu\text{C}$

So option 2 is correct.

Q. 3 In a Young's double slit exp. light of 500nm ia used to produce an interference pattern. When the distance b/w the slits is 0.05mm, the angular width(in degree) of the fringes formed on the distance screen close to :

Option 1:
0.17

Option 2:
0.57

Option 3:
1.7

Option 4:
0.07

Correct Answer:
0.57

Solution:
$given\ \lambda = 500nm$
$d = 0.05mm$
$\theta(in\ radian) = \dfrac{\lambda}{d} = \dfrac{500 \times 10^{-9}}{0.05 \times 10^{-3}} = 0.01$
$Now,\ \theta(indegree) = 0.01 \times \dfrac{180}{\pi} = 0.57^o$

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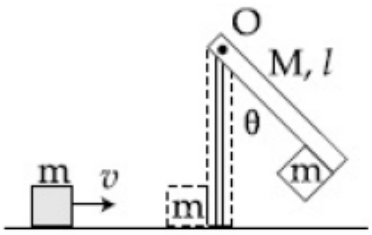
Q. 4 A block of mass 1kg slides with velocity $v = 6m/s$ on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as a result of the collision making angle θ before momentarily coming to rest. If the rod has mass M=2kg and length 1m. The value of θ is approx (take $g = 10m/s^2$).



Option 1:
63^o

Option 2:
55^o

Option 3:
69^o

Option 4:
49^o

Correct Answer:
63^o

Solution:
Angular momentum conservation

$$mvr = \left[\frac{Ml^2}{3} + ml^2\right]\omega$$

$$\Rightarrow 1 \times 6 \times 1 = \left[\frac{2}{3} + 1\right]\omega$$

$$\Rightarrow \omega = \frac{18}{5} rad/s$$

Applying energy conservation

$$\frac{1}{2}I\left[\frac{18}{5}\right]^2 + Mg\frac{l}{2} = \frac{1}{2}I(0) + Mg\left[\frac{l}{2} + \frac{l}{2}(1 - cos\theta)\right] + mgl(1 - cos\theta)$$

$$\Rightarrow I = \frac{Ml^2}{3} + ml^2 = \frac{5}{3}$$

$$\Rightarrow (1 - cos\theta)20 = \frac{1}{2} \times \frac{5}{3} \times \left[\frac{18}{5}\right]^2$$

$$\Rightarrow \theta \approx 63^o$$

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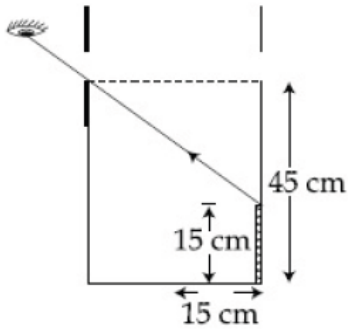
Q. 5 An observer can see through a small hole on the side of the jar (radius $15\ cm$) at a point of height of $45\ cm$. When the jar is filled with a liquid up to a height of $30\ cm$ the same observer can see the edge at the bottom of the jar. If the R.I. of the liquid is $\frac{N}{100}$, where N is an integer the value of N is



Option 1:
5

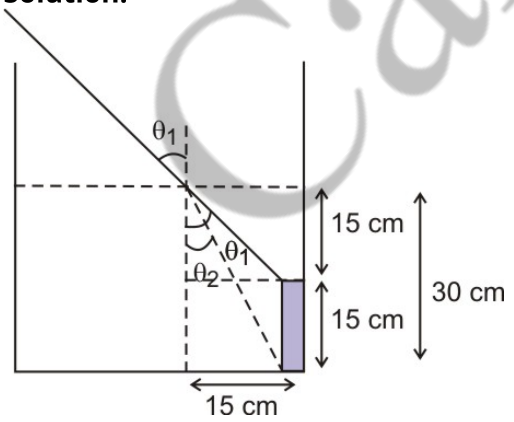
Option 2:
46

Option 3:
158

Option 4:
96

Correct Answer:
158

Solution:



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$tan\theta_1 = \frac{15}{15} = 1$

$\Rightarrow \theta_1 = 45^o$

$tan\theta_2 = \frac{15}{30} \Rightarrow tan\theta_2 = \frac{1}{2} \Rightarrow \theta_2 = \tan^{-1}\frac{1}{2}$

$Sin\theta_2 = \frac{1}{\sqrt{5}}$

$\frac{Sin\theta_1}{Sin\theta_2} = N \Rightarrow \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{5}}} = N$

$\Rightarrow N = \frac{\sqrt{5}}{\sqrt{2}} = 1.58$

$N = \frac{158}{100}$

$So, N = 158$

Q. 6 A cricket ball of mass $0.15\ kg$ is thrown vertically up, by a bowling machine, so that it rises to a maximum height of 20 m after leaving the machine. If the part pushing the ball applies a constant force F on the ball and moves horizontally a distance of $0.2\ m$ while launching the ball the value of F (is N). is $(g = 10\ m/s^2)$

Option 1:
10

Option 2:
100

Option 3:
150

Option 4:
400

Correct Answer:
150

Solution:
Work done by bowling machine = max. P.E. of the ball

$F \times s = mgh$

$F = \frac{mgh}{s} = \frac{0.15 \times 10 \times 20}{0.2} = 150N$

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Q. 7 A person of $80\ kg$ mass is standing on a circular platform of mass $200\ kg$ rotating about its axis at 5 rpm (revolutions per minute). The person now starts moving towards the center of the platform. What will be the rotational speed in rpm of the platform when the person reaches its center.

Option 1:
3

Option 2:
6

Option 3:
9

Option 4:
12

Correct Answer:
9

Solution:
Applying conservation of angular momentum -

$$I_1N_1 = I_2N_2$$

Now

$$I_1 = M_1R^2 + \frac{M_2R^2}{2} = 80 \times R^2 + \frac{200 \times R^2}{2}$$

$$N_1 = 5\ rpm$$

When the person reaches center -

$$I_2 = \frac{M_2R^2}{2} = \frac{200 \times R^2}{2}$$

So,

$$I_1N_1 = I_2N_2$$

$$\frac{360 \times R^2}{2} \times 5 = \frac{200 \times R^2}{2} \times N_2$$

$$N_2 = 9\ \ rpm$$

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Q. 8 When a long glass capillary tube of radius $0.015\ cm$ is dipped in a liquid, the liquid rises to a height of $15\ cm$ within it. If the contact angle b/w the liquid and glass to close to 0°, the surface tension of the liquid, in milli newton m^{-1} is
$(\ For\ liquid\ \ \rho = 900\ kg/m^3,\ and\ use\ g = 10m/s^2)$

Answer in closest integer.

Option 1:
3

Option 2:
101

Option 3:
57

Option 4:
38

Correct Answer:
101

Solution:
As,

$$\mathrm{h} = \frac{2\mathrm{s}\cos\theta}{\mathrm{r}\rho\mathrm{g}}$$

$$0.15 = \frac{2\mathrm{s}\cos 0^{o}}{0.015 \times 10^{-2} \times 900 \times 10}$$

Surface tension=$s = 0.10125\ N/m = 101.25\ \ milli - Newton/meter$

So the closet integer is 101

Q. 9 A bakelite breaker has volume capacity of $500\ cc$ at $30^{o}c$ when it is partially filled with V_m volume (at $30^{o}c$) of mercury, it is found that the unfilled volume of the beaker remains constant as temp is varied. If $y_{(beaker)} = 6 \times 10^{-6}\ (^{\circ}C)^{-1}$ and $y_{(mercury)} = 1.5 \times 10^{-4}\ (^{\circ}C)^{-1}$, Where y is the cooefficient of volume expansion then $V_m(in\ cc)$ is close to -

Option 1:
10

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Option 2:
20

Option 3:
30

Option 4:
40

Correct Answer:
20

Solution:

$$V_b = V_m + V_R$$
$$\Rightarrow \Delta V_b = \Delta V_m + \Delta V_R$$
$$\Rightarrow \gamma_b \times \Delta T_b \times V_b = \gamma_m \times \Delta Tm \times V_m \qquad [\because \Delta V_R = 0]$$
$$\Rightarrow \left[\frac{\gamma_b}{\gamma_m}\right] V_b = V_m$$
$$\Rightarrow V_m = 20cc$$

Q. 10 A satellite is moving in a low nearly circular orbit around the earth. Its radius is roughly to that of earth's radius R_e. By firing rockets attached to it its speed is instanteneously increased in the direction of its motion so that it becomes $\sqrt{3/2}$ times larger. Due to this the farthest distance frim the centre of the earth that the satellite reaches is R. value of R is:

Option 1:
$4\ R_e$

Option 2:
$2.5\ R_e$

Option 3:
$3\ R_e$

Option 4:
$2\ R_e$

Correct Answer:
$3\ R_e$

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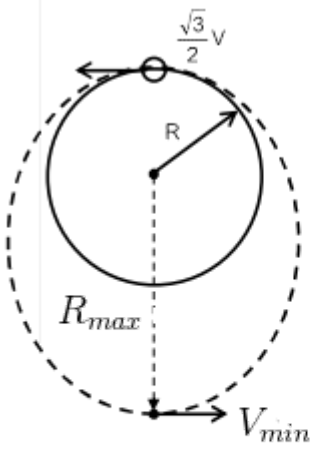
Solution:



Initially at A

$$\frac{GM_em}{R_e^2} = \frac{mV_0^2}{R_e}$$

$$V_0 = \sqrt{\frac{GM_e}{R_e}}$$

Finally at A,

$$V_1 = \sqrt{\frac{3}{2}}V_0 = \sqrt{\frac{3GM_e}{2R_e}}$$

From energy conversation

$$-\frac{\mathrm{GM_em}}{\mathrm{R_e}} + \frac{1}{2}\,\mathrm{m}\left(\sqrt{\frac{3}{2}}\,\mathrm{V}\right)^2 = -\frac{\mathrm{GM_em}}{\mathrm{R_{max}}} + \frac{1}{2}\mathrm{mV_{min}^2}$$

From angular momentum conversation

$$\sqrt{\frac{3}{2}}\mathrm{VR_e} = \mathrm{V_{min}R_{max}}$$

Eliminating $V_{\min}$ from equation (i) and (ii) we get

$$R_{\max} = 3R_e$$

Q. 11 Using a screw ganged pitch $0.1\ cm$ and 50 divisions of its circular scale, the thickness of an object is measured. It should correctly be recorded as :

Correct Answer:
2.124

Solution:
Pitch of the circular scale $\mathrm{P} = 0.1\mathrm{cm}$

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Number of divisions on a circular scale $N = 50$

So least count of screw gauge $LC = \frac{P}{N} = \frac{0.1}{50}\text{cm} = 0.002\text{cm}$

So any measurement will be integral Multiple of LC
From the option, we can say that The answer will be 2.124 cm

Q. 12 A uniform thin rope of length $12\ m$ and mass $6\ kg$ has vertically support and a block of mass $2\ kg$ is attached to its free end. A transverse short wave train of wavelength 6cm is produced at the lower end of the rope. What is the wavelength of the wave train (in cm) when it reaches top of the rope

Option 1:
3

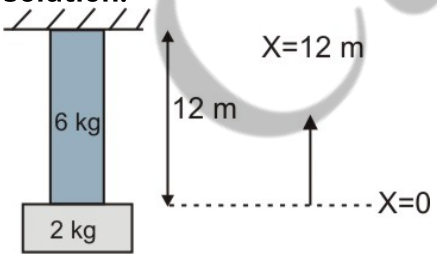
Option 2:
6

Option 3:
12

Option 4:
9

Correct Answer:
12

Solution:



$\mu = \frac{6}{12} = 0.5$

using $V = \frac{T}{\mu}$

At a distance x from the bottom

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$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{(2g + \mu x g)}{\mu}} = \sqrt{\frac{20 + 5x}{0.5}}$$

at x=0, $V_1 = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{20}{0.5}} = \sqrt{40}$

and $\lambda_1 = 6\ cm$

$$f = \frac{V_1}{\lambda_1} = \frac{\sqrt{40}}{6}$$

Now at x=12 cm

$$V_0 = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{(2g + \mu x g)}{\mu}} = \sqrt{\frac{20 + 5x}{0.5}} = \sqrt{\frac{20 + (5 * 12)}{0.5}} = \sqrt{\frac{80}{0.5}}$$

$$\Rightarrow V_0 = 2\sqrt{40}$$

So, $\lambda_0 = \frac{V_0}{f} = \frac{2\sqrt{40}}{\frac{\sqrt{40}}{6}} = 12cm$

Q. 13 Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere, respectively. the ratio of volumes will be:

Option 1:
4:1

Option 2:
0.8:1

Option 3:
8:1

Option 4:
2:1

Correct Answer:
8:1

Solution:

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$\text{Outside pressure} = 1atm$
$\text{Pressure inside first bubble} = 1.01atm$
$\text{Pressure inside second bubble} = 1.02atm$
$\text{Excess pressure in 1st bubble} \Delta P_1 = 1.01 - 1 = 0.01\text{atm}$
$\text{Excess pressure in 2nd bubble} \Delta P_2 = 1.02 - 1 = 0.02\text{atm}$

$$\Delta P \propto \frac{1}{r} \Rightarrow r \propto \frac{1}{\Delta P}$$

$$\Rightarrow \frac{r_1}{r_2} = \frac{\Delta P_2}{\Delta P_1} = \frac{0.02}{0.01} = \frac{2}{1}$$

$$since\ V = \frac{4}{3}\pi r^3$$

$$\therefore \quad \frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{2}{1}\right)^3 = \frac{8}{1}$$

So, option 3 is correct

Q. 14 When the wavelength of radiation falling on metal is changed from $500\ nm$ to $200\ nm$ the maximum kinetic energy of the photoelectrons becomes there times larger. The wave function (in eV) of the metal is close to

---

Correct Answer:
0.61

Solution:

$$\text{K.E.} = \text{hv} - \Phi$$

$$\frac{\text{hc}}{\lambda_1} = \phi + (KE)_1 \cdot (\text{i})$$

$$\frac{\text{hc}}{\lambda_2} = \phi + (KE)_2 (\text{ii})$$

from (i) - (ii)

$$\text{hc}\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) = (KE_2 - KE_1) = 3KE_1 - KE_1 = 2KE_1$$

$$KE_1 = \frac{\text{hc}}{2}\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)$$

Now, $\phi = \dfrac{hc}{\lambda_1} - KE_1$

So,
$$\phi = \frac{hc}{\lambda_1} - (\frac{\text{hc}}{2}\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)) = \frac{3hc}{2\lambda_1} - \frac{hc}{2\lambda_2}$$

$$\Rightarrow \phi = \frac{3 * 1240}{2 * 500} - \frac{1240}{2 * 200} = 3.7 - 3.1 = 0.6\ eV$$

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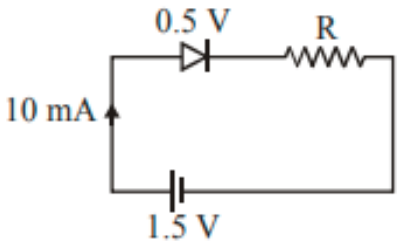
Q. 15 When a diode is forward biased it has a voltage drop of $0.5\ V$. The safe unit of current through the diodes is $10\ mA$. If a battery of emf is $1.5\ V$ is used in the circuit, the value of minimum resistance (in ohms) to be connected in series with the diode so that the current doesn't exceed the safe limit is

Correct Answer:
100

Solution:



$$I_{max} = 10\ mA = 10^{-2}\ A$$

$$V_R = 1.5 - 0.5 = I_{max} R$$

$$R = \frac{1}{I_{max}} = 100\Omega$$

Q. 16 The magnitude of Magnetic field (in SI units) at the cube of a hexagonal shape coil of side 10cm, 50 turns and carrying current I (ampere) in units of $\frac{\mu_0 I}{\pi}$ is

Option 1:
$250\sqrt{3}$

Option 2:
$50\sqrt{3}$

Option 3:
$5\sqrt{3}$

Option 4:
$500\sqrt{3}$

Correct Answer:
$500\sqrt{3}$

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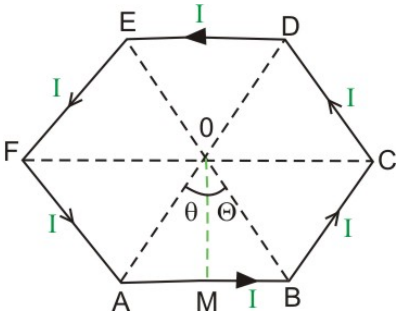
Solution:



$$B_{AB} = \frac{\mu_0}{4\pi} \times \frac{I}{r}(\sin\theta + \sin\theta)$$

$$B_{AB} = \frac{\mu_0}{4\pi} \times \frac{2I}{r}\sin\theta \ldots (1)$$

Here,

$$\theta = 30^o$$

and OM=r

$$\text{Also from right angled } \Delta AMO, we\ have$$
$$\tan\theta = \frac{AM}{OM},$$
$$or, \tan 30^\circ = \frac{(a/2)}{r}$$

$$\Rightarrow r = \frac{a}{2\tan 30^\circ} = \frac{\sqrt{3}a}{2}$$

$$r = \frac{\sqrt{3}a}{2} \ldots (2)$$

From (1) and (2)

$$B_{AB} = \frac{\mu_0}{4\pi} \times \frac{2I\sin 30^\circ}{\sqrt{3}a/2} = \frac{\mu_0}{4\pi} \times \frac{2I}{\sqrt{3}a}$$

Hence magnetic field due to the whole loop is -

$$B_{Loop} = 6 \times B_{AB}$$

$$B_{Loop} = \frac{\mu_0 I\sqrt{3}}{\pi a}$$

Here,

$$n = 50, a = 0.1\ m$$

So $B_{net} = n * \frac{\mu_0 I\sqrt{3}}{\pi a} = 50 * \frac{\mu_0 I\sqrt{3}}{\pi * 0.1} = 500\sqrt{3} * \frac{\mu_0 I}{\pi}$

So, the answer will be -

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$500\sqrt{3}$

Q. 17 A balloon filled with helium $(32^{\circ}Cand1.7atm)$ burst. immediately afterwards the expansion of helium can be consider as

Option 1:
Irreversible Isothermal

Option 2:
Irreversible Adiabatic

Option 3:
Reversible Adiabatic

Option 4:
Reversible Isothermal

Correct Answer:
Irreversible Adiabatic

Solution:
Since process is very fast so, there will be no time to exchange heat between the system and the surroundings. SO this process is adiabatic.

Now when the gas comes out after bursting, a large amount of entropy will generate because the volume expansion is done in the ideal gas and since it is a free expansion, so temperature remain same. So it is irreversible.

So the Process is irreversible adiabatic.

Q. 18 A 750 $H_2, 20V(rms)$ source is connected to a ressitance of 100Ω an inductance of 0.18034 and a capacitance of $10\mu F$ all in series. The time (in seconds) in which the resistance (heat capacity $2J/^{\circ}C$) will get heated by $10^{\circ}C$ (assume no loss of to the surrounding) is close to

Correct Answer:
348

Solution:

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$\omega = 2\pi \times 750 = 1500\pi$
$\mathrm{R} = 100\Omega, \mathrm{L} = 0.1803\mathrm{H}, \mathrm{C} = 10\mu\mathrm{F}$
$\mathrm{z} = \sqrt{100^2 + \left(\omega\mathrm{L} - \frac{1}{\omega\mathrm{c}}\right)^2} = 834\Omega$

$\mathrm{I_{rms}} = \frac{V_{rms}}{Z} = \frac{20}{834} = 0.024\mathrm{A}$

$\mathrm{P} = I_{rms} \times V_{rms} \times cos\theta = 0.024 \times 20 \times \frac{R}{Z} = 0.058$
$Now,\ P \times t = S\Delta\theta$
$Here\ S = thermal\ capacity$

$\therefore t = \frac{S\Delta\theta}{P} = \frac{2\times10}{0.058} = 348s$

Q. 19 The magnetic field of a plane electromagnetic wave is

$\overrightarrow{B} = 3 \times 10^{-8} sin\left(200\pi\left(y + ct\right)\right)\hat{i}\ T$ where $c = 3 \times 10^8 m/s$ is the speed of the light. The corresponding Electric field is

Option 1:
$\overrightarrow{E} = 9sin\left(200\pi\left(y + ct\right)\right)\hat{k}\ V/m$

Option 2:
$\overrightarrow{E} = -10^{-6} sin\left(200\pi\left(y + ct\right)\right)\hat{K}\ V/m$

Option 3:
$\overrightarrow{E} = -9sin\left(200\pi\left(y + ct\right)\right)\hat{K}\ V/m$

Option 4:
$\overrightarrow{E} = 3 \times 10^{-8} sin\left(200\pi\left(y + ct\right)\right)\hat{K}\ V/m$

Correct Answer:
$\overrightarrow{E} = -9sin\left(200\pi\left(y + ct\right)\right)\hat{K}\ V/m$

Solution:
We know $|\mathrm{E}| = |\mathrm{B}| \times |\mathrm{C}|$
where E *and* B are the amplitude of electric and magnetic field intensities in an electromagnetic wave.
Given $|\mathrm{B}| = 3 \times 10^{-8}$
$\therefore |\mathrm{E}| = |\mathrm{B}| \times |\mathrm{C}| = 3 \times 10^{-8} \times 3 \times 10^8 = 9\ V/m$

As,

$\overrightarrow{\mathbf{B}} = 3 \times 10^{-8} \sin(200\pi(y + ct))\hat{\mathrm{i}}\ \mathrm{T}$

Now,

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and direction of wave propagation is given as

$(\overrightarrow{E} \times \overrightarrow{B}) \| \overrightarrow{C}$

$\hat{B} = \hat{i} \quad \& \quad \hat{C} = -\hat{j}$

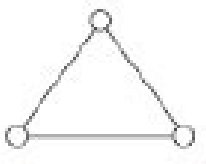
$-\hat{j} = \hat{E} \times \hat{i},$

$\therefore \hat{E} = -\hat{K}$

So,

$\therefore \tilde{E} = -9 \sin(200\pi(y + ct))\hat{k}$

Q. 20 Consider a gas of triatomic molecules. The molecules are assumed to be triangular and made of massless rigid rods whose vertices are occupied by atoms. The internal energy of a mole of the gas at temperature T is



Option 1:

$\frac{5}{2}RT$

Option 2:

$\frac{3}{2}RT$

Option 3:

$\frac{9}{2}RT$

Option 4:

$3\ RT$

Correct Answer:

$3\ RT$

Solution:

Degree of freedom of the non-linear triatomic molecules is - 6

As we know that the internal energy of the triatomic gas is -

$\frac{f}{2}.R.T$

Now f = 6

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So, Internal energy -

$$\frac{6}{2}.R.T = 3RT$$

Q. 21 Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}Rand\frac{1}{3}R$ have $12\mu C$ and $-3\mu C$ charges respectively and are at a large distance from each other. They are now connected by a conducting wire. A long time after this is done the charges on S_1 andS_2 are respectively

Option 1:
$+4.5\mu C\ and\ +4.5\mu C$

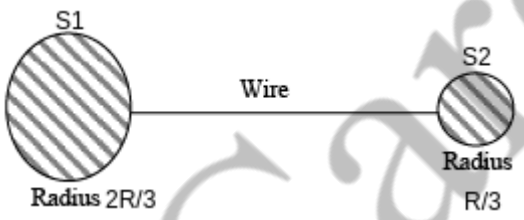
Option 2:
$+4.5\mu C\ and\ -4.5\ \mu C$

Option 3:
$+3\mu C\ and\ +6\ \mu C$

Option 4:
$+6\mu C\ and\ +3\ \mu C$

Correct Answer:
$+6\mu C\ and\ +3\ \mu C$

Solution:



After connecting with wires, there will be charge flow such that the potential of both spheres are equal.

Let the flow of charge be q from (1) to (2)

So, final charge on $(1) = (12 - q)\mu C$

and, final charge on $(2) = (-3 + q)\mu C$

So,

$$V_1 = V_2$$

$$\frac{k(12-q)}{(2R/3)} = \frac{k(q-3)}{(R/3)}$$

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$$12 - q = 2q - 6$$
$$3q = 18$$
$$\Rightarrow q = 6\mu C$$
$$So,$$
$$Q_1 = 12 - 6 = 6\mu c$$
$$Q_2 = -3 + 6 = 3\mu C$$

Q. 22 Moment of inertia of a cylinder of mass M and length L and radius R about an axis passing through its centre and perpendicular to the axis of the cylinder is $I = M\left(\frac{R^2}{4} + \frac{L^2}{12}\right)$ If such a cylinder is to be make for given mass of a material, the ratio $\frac{L}{R}$ for it to have minimum possible I is

Option 1:
$\frac{2}{3}$

Option 2:
$\frac{3}{2}$

Option 3:
$\sqrt{\frac{2}{3}}$

Option 4:
$\sqrt{\frac{3}{2}}$

Correct Answer:
$\sqrt{\frac{3}{2}}$

Solution:
$$I = m\left[\frac{R^2}{4} + \frac{L^2}{12}\right]$$

Given,

$$M = \rho(\pi R^2 L)$$

So,

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$$I = M\left[\frac{R^2}{4} + \frac{M^2}{12(\rho\ \pi R^2)^2)}\right]$$

For minimum I–

$$\frac{dI}{dR} = 0$$

$$\Rightarrow M[\frac{2R}{4} + \frac{-4M^2}{12\rho^2\pi^2R^5}] = 0$$

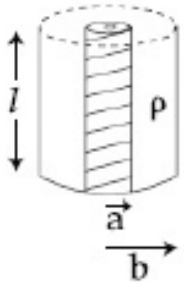
$$R = (2/3)^{1/6} \times (\frac{M}{\rho\pi})^{1/3} \ldots (1)$$

$$L = \frac{M}{\rho\pi(\frac{2}{3})^{1/3}(\frac{M}{\rho\pi})^{2/3}} \ldots (2)$$

So, (2) divided by (1) will give

$$\frac{L}{R} = \sqrt{\frac{3}{2}}$$

Q. 23 Model a torch battery of length L to be a made up of a thin cylindrical bar of radius 'a' and con-centric thin cylindrical shell of radius 'b' filled in b/w with an electrolyte of resistivity ρ. If the battery is connected to a resistance R, the maximum joule heating in R will take place for



Option 1:

$$R = \frac{\rho}{2\pi l}\left(\frac{b}{a}\right)$$

Option 2:

$$R = \frac{\rho}{2\pi l}\ ln\ \left(\frac{b}{a}\right)$$

Option 3:

$$R = \frac{\rho}{\pi l}\ ln\ \left(\frac{b}{a}\right)$$

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Option 4:

$R = \frac{2\rho}{\pi l}\, ln\, \left(\frac{b}{a}\right)$

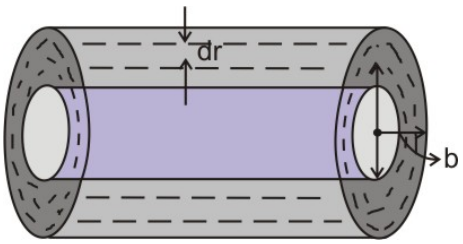
Correct Answer:

$R = \frac{\rho}{2\pi l}\, ln\, \left(\frac{b}{a}\right)$

Solution:

For maximum power lost or maximum joule heating

Internal Resistance= External Resistance



$$dR = \frac{\rho\, dr}{2\pi r l}$$

$$R = \text{Internal resistance} = \int dR = \rho \int_a^b \frac{dr}{2\pi r l} = \frac{\rho}{2\pi l} \ln\left(\frac{b}{a}\right)$$

Q. 24 In the radioactive material fraction of active material remaining after time t is $\frac{9}{16}$. The fraction that was remaining after t/2 is

Option 1:

$\frac{4}{5}$

Option 2:

$\frac{3}{5}$

Option 3:

$\frac{3}{4}$

Option 4:

$\frac{7}{8}$

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Correct Answer:

$\frac{3}{4}$

Solution:

$\frac{9}{16} = \left(\frac{1}{2}\right)^{\frac{t}{T}}$

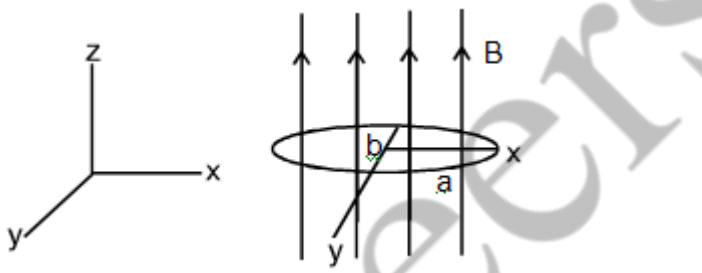
$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{t}{2T}}$

$\left(\frac{N}{N_0}\right)^2 = \left(\frac{1}{2}\right)^{t/T}$

$\text{or } \frac{N^2}{N_0^2} = \frac{9}{16} \text{ or } \frac{N}{N_0} = \frac{3}{4}$

So, option 3 is correct.

Q. 25 An elliptical loop having resistance R of semi-major axis a, and semi-minor axis b is placed in a Magnetic field as shown in the figure. If the loop is rotated about the x-axis with angular frequency ω, the average power loss in the loop due to Joule heating is



Option 1:

$\frac{\pi^2 a^2 b^2 B^2 \omega^2}{2R}$

Option 2:

$Zero$

Option 3:

$\frac{\pi a b B \omega}{R}$

Option 4:

$\frac{\pi^2 a^2 b^2 B^2 \omega^2}{R}$

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Correct Answer:

$$\frac{\pi^2 a^2 b^2 B^2 \omega^2}{2R}$$

Solution:

$\phi = B.A = BAcos\theta = \pi abBcos\theta$

$$\varepsilon = -\frac{d\Phi}{dt} = -\frac{d(\pi abBcos\theta)}{dt} = \pi aBsin\theta\frac{d\theta}{dt} = \pi aB\omega bsin\theta$$

$$i_{max} = \frac{\varepsilon_{max}}{R} = \frac{\pi abB\omega}{R}$$

$$\because P_{avg} = \frac{i_{max}^2 R}{2}$$

$$\therefore \text{avg power loss in the loop due to joule heating is} = \frac{\pi^2 a^2 b^2 B^2 \omega^2}{2R}$$

Chemistry

Q. 1 An acidic buffer is obtained by mixing:

Option 1:
100 mL of 0.1 M CH_3COOH and 100 mL of 0.1 M NaOH

Option 2:
100 mL of 0.1 M HCl and 200 mL of 0.1 M CH_3COONa

Option 3:
100 mL of 0.1 M CH_3COOH and 200 mL of 0.1 M NaOH

Option 4:
100 mL of 0.1 M HCl and 200 mL of 0.1 M NaCl

Correct Answer:
100 mL of 0.1 M HCl and 200 mL of 0.1 M CH_3COONa

Solution:
An acidic buffer contains a weak acid and its salt.

Here,

Given the weak acid is CH_3COOH and its salt will be CH_3COONa

So,

100 mL of 0.1 M HCl = 10 meq HCl

200 mL of 0.1 M CH_3COONa = 20 meq of CH_3COONa

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After mixing 10 meq of HCl react with 10 meq of CH_3COONa then 10meq of CH_3COOH form.

But from other options, the same amount of weak acid and its salt cannot be obtained.

Finally, 10meq of CH_3COOH and 10 meq of CH_3COONa will be present.

So upon mixing 100ml of 0.1M HCl and 200ml of 0.1M CH_3COONa, a buffer solution will be obtained.

Therefore, Option(2) is correct.

Q. 2 The atomic number of the element unnilennium

Option 1:
109

Option 2:
102

Option 3:
108

Option 4:
119

Correct Answer:
109

Solution:
The atomic number of unnilennium is 109.

Word root: un - 1
nil - 0
ennium - 9

Therefore, Option(1) is correct.

Q. 3 Let C_{NaCl} and C_{BaSO_4} be a conductance (in S) measured for saturated aqueous solutions of $NaCl$ and $BaSO_4$ espectively at a temperature T. Which of the following is false?

Option 1:
ionic mobilities of ions from both salts increases with T

Option 2:
$C_{BaSO_4}(T_2) > C_{BaSO_4}(T_1)$ for $T_2 > T_1$

Option 3:
$C_{NaCl}(T_2) > C_{NaCl}(T_1)$ for $T_2 > T_1$

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Option 4:
$C_{Nacl} >> C_{BaSO_4}$ at a given T

Correct Answer:
$C_{NaCl}(T_2) > C_{NaCl}(T_1)$ for $T_2 > T_1$

Solution:
We know with the increase in temperature, the mobility increases as well as the conductance increases.

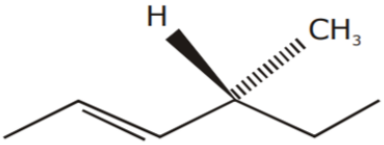
NaCl is completely soluble salt while $BaSO_4$ is sparingly soluble.

So, with the increase in temperature, the conductance of NaCl will be the same due to complete soluble. It already reached its max mobility.

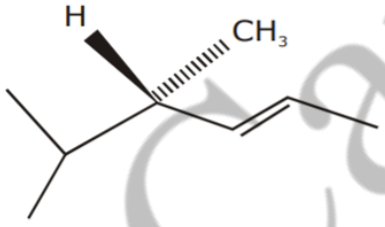
Therefore, Option(3) is correct.

Q. 4 Which of the following compounds produces an optically inactive compound on hydrogenation -

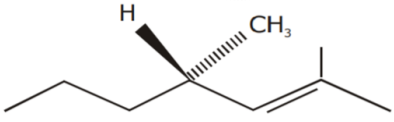
Option 1:



Option 2:



Option 3:



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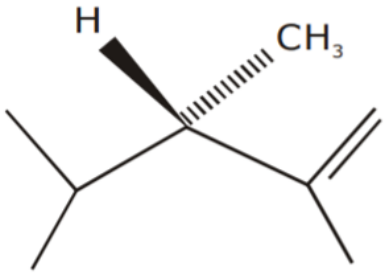
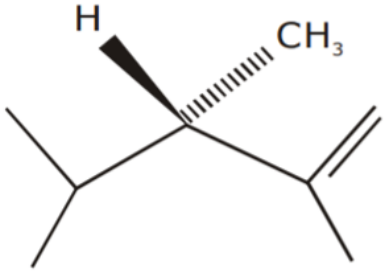
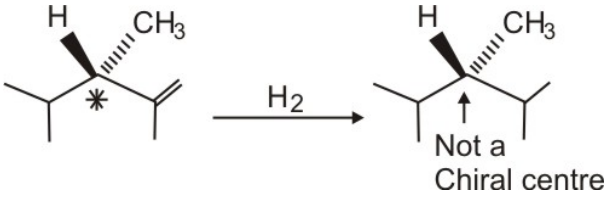
Option 4:



Correct Answer:



Solution:
The reaction occurs as follows:



Thus, after hydrogenation, there will be no chiral centre.

Therefore, Option(4) is correct.

Q. 5 Aqua regia is used for dissolving noble metal the gas evolved in this process it:

Option 1:
N_2

Option 2:
N_2O_3

Option 3:
NO

Option 4:
N_2O_5

Correct Answer:
NO

Solution:

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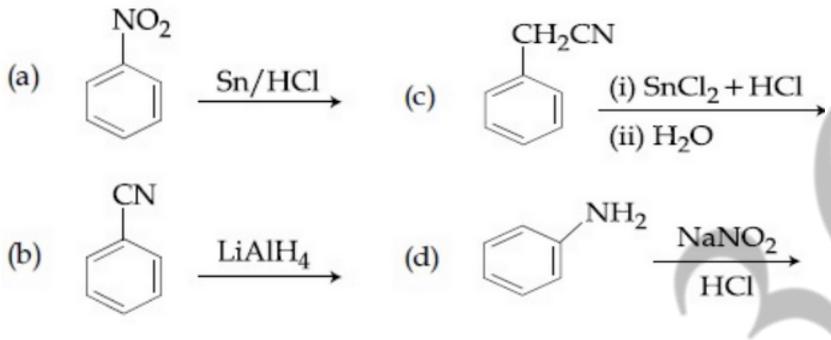
Aqua regia is used to dissolve noble metals like Au, Pt, etc. The gas evolved in the process is NO. The reactions occur as follows:

$$Au + 4H^+ + NO_3^- + 4Cl^- \rightarrow AuCl_4^- + NO + 2H_2O$$

$$3Pt + 16H^+ + 4NO_3^- + 18Cl^- \rightarrow 3PtCl_6^{2-} + 4NO + 8H_2O$$

Therefore, Option(3) is correct.

Q. 6 The Kjeldahl method of Nitrogen estimation fails for which of the following reaction products:



Option 1:
c & d

Option 2:
a & d

Option 3:
a c d

Option 4:
b c

Correct Answer:
c & d

Solution:
The reactions occur as follows:

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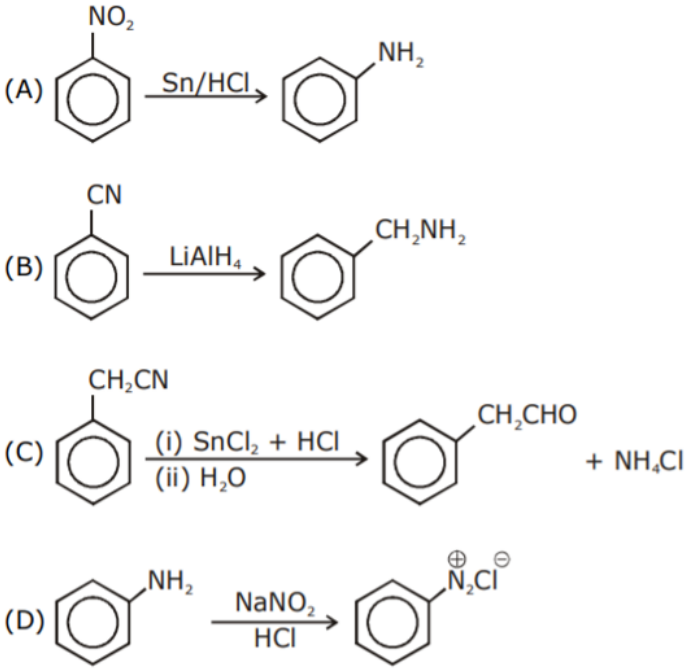

Diazo compound and inorganic nitrogen can't be estimated by the Kjeldahl method.

Options(c) and (d) will not respond to the Kjeldahl method.

Therefore, Option(1) is correct.

Q. 7 Thermal power plants can lead to:

Option 1:
Acid rain

Option 2:
Blue baby syndrome

Option 3:
Ozone layer depletion

Option 4:
Eutrophication

Correct Answer:
Acid rain

Solution:

While burning of fossil fuels such as coal and oil in thermal power plants, some gases such as SO_2 and NO_2 form after oxidation. Then the reaction between the water and these gases leads to acid rain.

Therefore, Option(1) is correct.

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Q. 8 The mechanism of S_N1 reaction is given as

$$R-X \longrightarrow \underset{\text{Ion pair}}{R^{\oplus}X^{\ominus}} \longrightarrow \underset{\text{Separated ion pair}}{R^{\oplus} \,||\, X^{\ominus}} \xrightarrow{Y^{\ominus}} R-Y+X^{\ominus}$$

(Solvent)

A student writes general characteristics based on the given mechanism as :

(a) The reaction is favoured by weak nucleophiles.

(b) $R^{\oplus}$ would be easily formed if the substituents are bulky

(c) The reaction is accompanied by racemization.

(d) The reaction is favoured by non-polar solvents.

Which observations are correct-

Option 1:
a & b

Option 2:
a & c

Option 3:
a,b,& c

Option 4:
b & d.

Correct Answer:
a,b,& c

Solution:
SN_1 reactions are favoured by weak nucleophiles. In SN_1 reactions, C^+ formation takes place which is established by alkyl groups and the reaction is accompanied by racemisation. This reaction is favoured by polar protic solvents. Hence, observations (a), (b) and (c) are correct.

Therefore, Option(3) is correct.

Q. 9 Of the species NO, NO^+, NO^{2+} and NO^- the one with the minimum bond strength is

Option 1:
NO^+

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Option 2:
NO

Option 3:
NO^{2+}

Option 4:
NO^{-}

Correct Answer:
NO^{-}

Solution:
Bond order of NO^{2+} = 2.5
Bond order of NO^{+} = 3
Bond order of NO = 2.5
Bond order of NO^{-} = 2

$$\text{Bond order} \propto \text{bond strength}$$

Thus, NO^{-} has the minimum bond strength.

Therefore, Option(4) is correct.

Q. 10 It is true that

Option 1:
A second order reaction is always a multistep reaction

Option 2:
A zero order reaction is a multistep reaction

Option 3:
A first order reaction is always a single step reaction

Option 4:
A zero order reaction is a single step reaction

Correct Answer:
A zero order reaction is a multistep reaction

Solution:
A zero-order reaction is always a complex or a multistep reaction as the molecularity of any elementary reaction can never be zero.

Therefore, Option(2) is correct.

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Q. 11 If the boiling point of H_2O is 373 K the boiling point of H_2S will be :

Option 1:
less than 300K

Option 2:
equal to 373K

Option 3:
more than 373K

Option 4:
Greater than 300 K but less than 373 K

Correct Answer:
less than 300K

Solution:
The boiling point of H_2S < Boiling point of H_2O

The boiling point of H_2S is 213K.

So, less than 300 K, is correct.

Therefore, Option(1) is correct.

Q. 12 The antifertility drug "Novestrol" can react with

Option 1:
$ZnCl_2/HCl; FeCl_3\ Alocholic\ HCN$

Option 2:
$Br_2/water; ZnCl_2/HCl; FeCl_3$

Option 3:
$Alcoholic\ HCN\ ; NaOCl; ZnCl_2/HCl$

Option 4:
$Br_2/water; ZnCl_2/HCl; NaOCl$

Correct Answer:
$Br_2/water; ZnCl_2/HCl; FeCl_3$

Solution:

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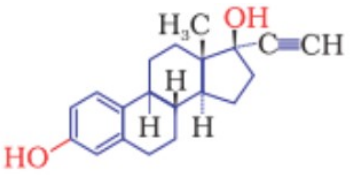


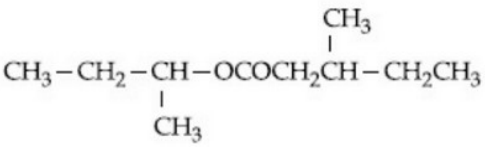
Ethynylestradiol (novestrol)

Alkyne group will react with Br_2/water. Tertiary alcohol will react with $ZnCl_2$/HCl. Phenolic group will react with $FeCl_3$.

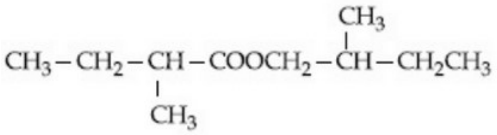
Therefore, Option(2) is correct.

Q. 13 An organic compound (A) is molecular formula $C_{10}H_{20}O_2$ was hydrolysed with dilute sulphuric acid to give a carboxylic acid (B) and a alcohol (C). Oxidation of (C) with CrO_3–H_2SO_4 produced (B). What structure are not possible fo (A)?

Option 1:



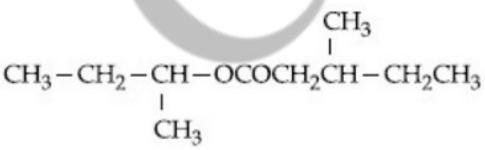
Option 2:



Option 3:

Option 4:

Correct Answer:



Solution:
Option 3 does not have 10 carbons so, it cannot be A.

Now,

The reaction occurs as follows:

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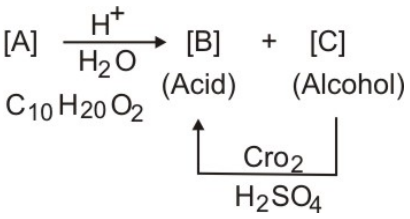
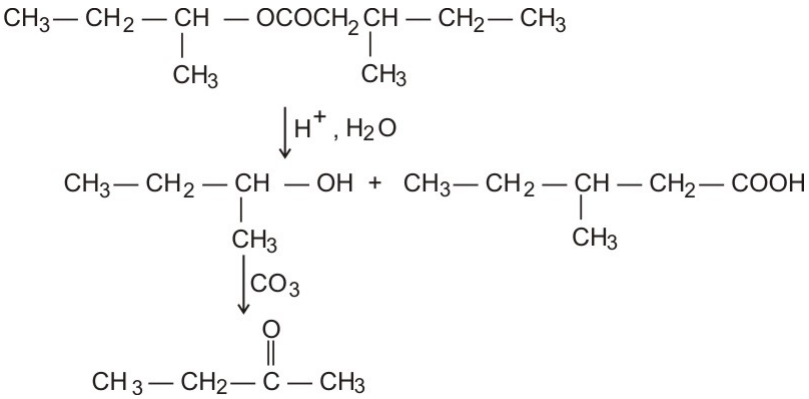


So, the ester (I) cannot be (A) because of the Oxidation of (C) with CrO_3-H_2SO_4 **not making** (B).



Therefore, Option(1) is correct.

Q. 14 Henry's constant (in K bar) for tour gases α, β, γ and δ in water at 298 K is given below:

	α	β	γ	δ
K_H	50	2	2×10^{-5}	0.5

(density of water = 10^3 kg/m^3 at 298 K) This table implies that

Option 1:
α has the highest solubility in water at a given pressure

Option 2:
Solubility of γ at 308 K is lower than at 298 K

Option 3:
The pressure of a 55.5 molal solution of γ s 1 bar

Option 4:
The pressure of a 55.5 molal solution of Δ is 250 bar

Correct Answer:
The pressure of a 55.5 molal solution of Δ is 250 bar

Solution:
(1) $K_H \downarrow$ solubility $\uparrow$ and lowest K_H is for γ .

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So, γ has the greatest solubility as it has the least value of Henry's constant.

(2) On increasing temperature solubility of gases decreases, solubility of γ is greater at 298K than 308K although that is not implied by the table.

(3) Now,

We know this formula.

$$P_{\gamma} = K_H X_Y$$

$$P_{\gamma} = 2 \times 10^{-5} \times \frac{55.5}{55.5 + \frac{1000}{18}} = 10^{-5} \text{ kbar}$$

$$P_{\gamma} = 10^{-2} \text{ bar} \neq 1 \text{ bar}$$

$$(4)\ P_{\delta} = K_H X_{\delta}$$

$$P_{\delta} = 0.5 \times \frac{55.5}{55.5 + \frac{1000}{18}} = 0.25 \text{ kbar } = 250 \text{ bar}$$

The Correct Statement: The pressure of a 55.5 molal solution of δ is 250 bar

Therefore, Option(4) is correct.

Q. 15 Tyndall effect is observed when:

Option 1:
The diameter of dispersed pan is much larger than the wavelength of light used.

Option 2:
The diameter of dispersed particles is much smaller than the wavelength of light used

Option 3:
The refractive index of dispersed phase is greater than that of the dispersion medium

Option 4:
The diameter of dispersed particles is similar to the wavelength of light used

Correct Answer:
The diameter of dispersed particles is similar to the wavelength of light used

Solution:
For the Tyndall effect to be observed, the refractive indices of the dispersed phase and dispersion medium should be greatly differing in magnitude and the diameter of the dispersed particles is similar to the wavelength of light used.

Therefore, Option(4) is correct.

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Q. 16 The complex that can show optical activity is:

Option 1:
$\text{trans} - \left[\text{Cr}\left(\text{Cl}_2(\text{ox})_2\right]^{3-}\right.$

Option 2:
$\text{trans-}\left[\text{Fe}\left(\text{NH}_3\right)_2\left(\text{CN}\right)_4\right]^-$

Option 3:
$cis - \left[\text{CrCl}_2(\text{ox})_2\right]^{3-}\left(\text{ox} = \text{ oxalate }\right)$

Option 4:
$cis - \left[\text{Fe}\left(\text{NH}_3\right)_2\left(\text{CN}\right)_4\right]^-$

Correct Answer:
$cis - \left[\text{CrCl}_2(\text{ox})_2\right]^{3-}\left(\text{ox} = \text{ oxalate }\right)$

Solution:
The complex that can show optical activity is:

$cis - \left[\text{CrCl}_2(\text{ox})_2\right]^{3-}\left(\text{ox} = \text{ oxalate }\right)$

The structure is given below:

This compound has 2 forms, i.e, d, and l. Hence it is optically active.

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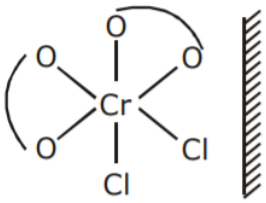
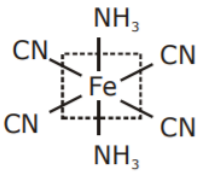


Nonsuper imposable
M.I.

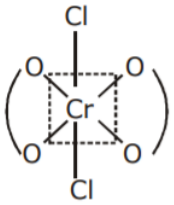
optically active

$cis-\left[CrCl_2(ox)_2\right]^{3-}$ (ox = oxalate)



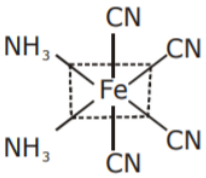
POS optically inactive

$trans-\left[Fe(NH_3)_2(CN)_4\right]^-$



→ POS optically inactive

$trans-\left[Cr(Cl_2)(ox)_2\right]^{3-}$



→ POS opticaly inactive

$cis-\left[Fe(NH_3)_2(CN)_4\right]^-$

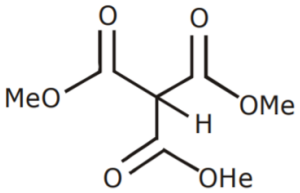
Therefore, Option(3) is correct.

Q. 17 Which one of the following compounds possesses the most acidic hydrogen ?

Option 1:

$H_3C-C\equiv C-H$

Option 2:



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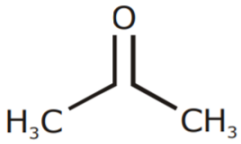
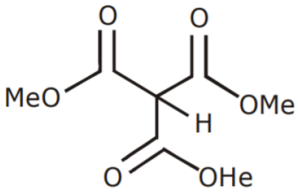
Option 3:

Option 4:

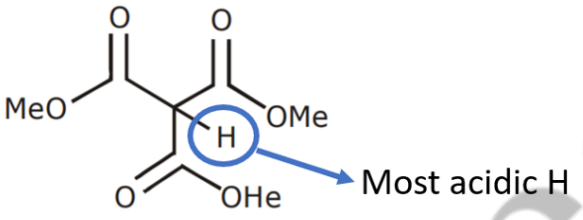


Correct Answer:



Solution:

Among these, Option 2 has the most acidic H, Due to the presence of the 3 (COOR) group. More resonating structure more stability of negative charge.



Therefore, Option(2) is correct.

Q. 18 Glycerol is separated in soap industries by

Option 1:

Fractional Distillation

Option 2:

Differential extraction

Option 3:

Distillation under reduced press.

Option 4:

Stead Distillation

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Correct Answer:
Distillation under reduced press.

Solution:
Glycerol is separated in the soap industry by distillation under reduced pressure because in simple distillation very high temperature is required which might decompose the component.

Therefore, Option(3) is correct.

Q. 19 The electronic spectrum of $[\mathrm{Ti}\,(\mathrm{H_2O})_6]^{3+}$ shows a single bread peak with a max at 20,300 cm^{-1} The crystal field stabilization energy(CFSE) of the complex ion in kJ/mol, is:

(1 kJ/mol =83.7 cm^{-1})

Option 1:
145.5

Option 2:
242.5

Option 3:
83.7

Option 4:
97

Correct Answer:
97

Solution:
We have: $[Ti(H2O)6]^{3+}$

Gives a peak with maxima at 20,300cm^{-1}

The crystal field stabilization energy is given as:

$$\Delta_o = hc\bar{\nu}$$

$$\Delta_o = 6.625 \times 10^{-34} \times 3 \times 10^8 \times 20300 = 4 \times 10^{-19}\mathrm{J/molecule}$$

$$\Delta_0 = 6.023 \times 10^{23} \times 4 \times 10^{-19}\ \mathrm{J/\ molecule}$$

$$\Delta_o = 243005\ \mathrm{J/mol} = 243\ \mathrm{kJ/mol}$$

$$\mathrm{CFSE} = 0.4\Delta_o = 0.4 \times 243 = 97\ \mathrm{kJ/mol}$$

Therefore, Option(4) is correct.

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Q. 20 In a molecule of pyrophosphoric acid, the number of P-OH, P = O, and P – O – P bonds/moiety (ies) respectively are :

Option 1:
2, 4 and 1

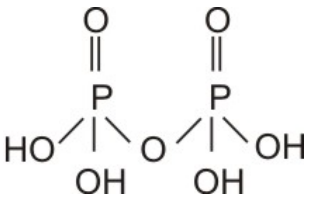
Option 2:
3, 3 and 3

Option 3:
4, 2 and 0

Option 4:
4, 2 and 1

Correct Answer:
4, 2 and 1

Solution:
Pyrophosphoric acid = $H_2P_2O_7$



P-OH bonds are 4

P=O bonds are 2

P-O-P bond is 1

Therefore, Option(4) is correct.

Q. 21 Element with molar mass $2.7 \times 10^{-2}\ \text{kg mol}^{-1}$ from cubic unit cell with edge length 405 pm If its density is $2.7 \times 10^{3}\ \text{kg m}^{-3}$, the radius of the element is approx $\times 10^{-12}\ \text{m}$ (to the nearest integer).

Correct Answer:
143

Solution:
Molar mass = $2.7 \times 10^{-2}\ \text{kg mol}^{-1}$

a(edge length) = 405 pm = 405 x 10^{-12}m

Density = 2.7 x 10^{3} kgm^{-3}

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Unit cell = cubic

Z = ?

Now, we know:

$$\mathrm{d} = \frac{\mathrm{Zxm}}{\mathrm{a^3xN_A}} = 2.7 \times 10^3 \mathrm{kgm}^{-3} = \frac{Z \times 2.7 \times 10^{-2}}{(\mathrm{a})^3 \times 6.022 \times 10^{23}}$$

On solving the above equation, we get:
Z = 4 Thus, the cell is FCC.

$$\text{Hence. a } = 2\sqrt{2}\,\mathrm{r}$$

$$\Rightarrow \mathrm{a} = \frac{405}{2\sqrt{2}} = 143\mathrm{pm}$$

Therefore, the correct answer is 143.

Q. 22 The volume strength of 8.9 M H_2O_2 solution calculated at 273 K and 1 atm is _____. (R = 0.0821 L atm K^{-1} mol^{-1}) rounded off to the nearest integer)

Correct Answer:
100

Solution:
For H_2O_2, we know:

$$M = \frac{N}{2}$$

Thus, N = 2M

Thus, $\mathrm{N} = 2 \times 8.9 = 17.8$

Now we know that:

Volume strength of $H_2O_2 = 5.6 \times \text{normality} = 5.6 \times 17.8$

$= 99.68 \text{ or } 100$

Thus, the volume strength of 8.9 M H_2O_2 solution will be around 100.

Ans =100

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Q. 23 The photoelectric current from Na (work function, w_0 = 2.3 eV) is stopped by the output voltage of the cell

Pt(s) | H_2(g, 1 bar) | HCl (aq., pH = 1) | AgCl(s) | Ag(s).

The pH of aq HCl required to stop the photoelectric current from K(w_0 = 2.25 eV) , all other candidate remaining the same is _____$\times 10^{-2}$ (to the nearest interger)

$$\text{Given } 2.303\frac{RT}{F} = 0.06 \text{ V}; E^0_{AgCl|Ag|Cl^-} = 0.22 \text{ V}$$

Correct Answer:
142

Solution:
$$Pt(s) \mid H_2(g) \mid HCl(aq) \mid AgCl(s) \mid Ag(s)$$
$$A: H_2 \longrightarrow 2H^+ + 2e^-$$

$$C: AgCl + e^- \longrightarrow Ag + Cl^-$$
Adding both these reactions, we get:

$$H_2 + 2AgCl \longrightarrow 2H^+ + 2Ag + 2Cl^-$$

$$Q = \frac{[H^+]^2[Cl^-]^2}{P_{H_2}}$$

$$\therefore E = E^o - \frac{0.06}{2}\log(0.1)^2(0.1)^2$$

$$\Rightarrow 0.22 - 0.06\log(10^{-2})$$

$$\Rightarrow 0.34V$$

Thus, KE of photoelectron $= 0.34eV$

Thus, KE of photon$= (2.3 + 0.34)eV = 2.64eV$

Now, for K:

KE of photoelectron $= (2.64 - 2.25)eV = 0.39V$

Thus, E of second cell $= 0.39V$

Thus, $0.39 = 0.22 - 0.06\log[H^+][Cl^-]$

$\Rightarrow 0.39 = 0.22 - 0.06 \times 2 \times \log[H^+]$
$\Rightarrow 0.39 = 0.22 + 0.12pH$
$\Rightarrow pH = 0.014 = 1.4 \times 10^{-2}$

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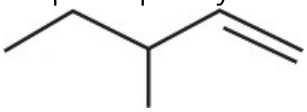
Thus, the pH value is 141.6×10^2
Thus, the correct answer is 142.

Q. 24 The total number of monohalogenated organic products in the following (including stereoisomers) reaction is ___

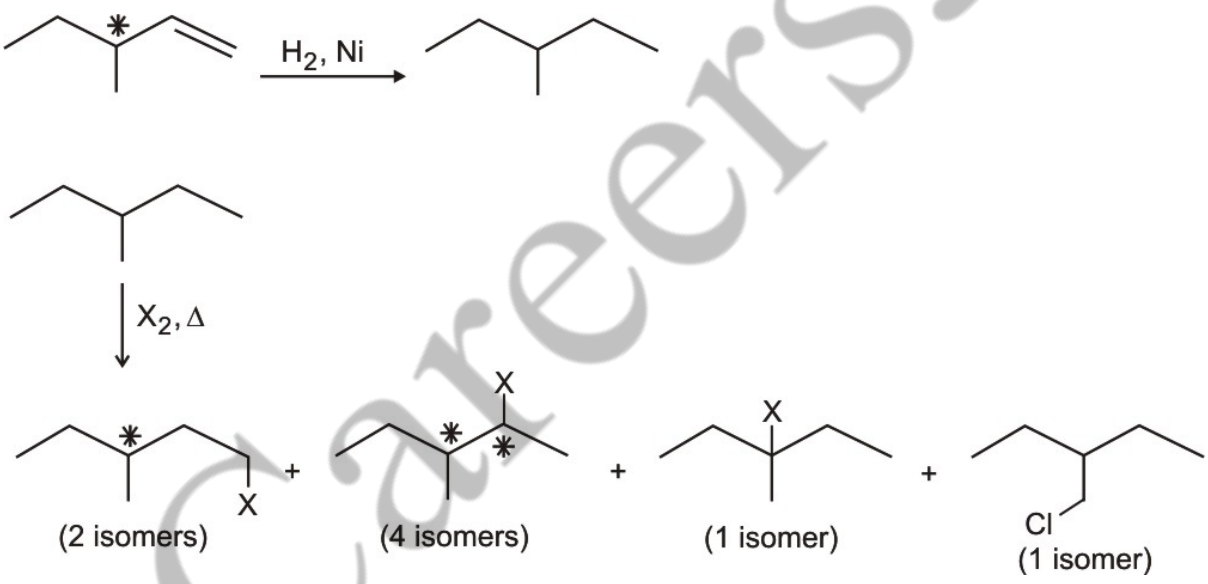
$$\underset{\text{(Simplest optically active alkene)}}{\text{A}} \xrightarrow[\text{(ii) } X_2/\Delta]{\text{(i) } H_2/Ni/\Delta}$$

Correct Answer:
8

Solution:
Simplest optically active alkene is:



It undergoes the reactions as given below:



Hence, total number of isomers produced is 8.

Q. 25 The mole fraction of glucose $(C_6H_{12}O_6)$ in an aqueous binary solution is 0.1. The mass percentage of water in it to the nearest integer is-

Correct Answer:
47

Solution:

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Mole fraction of glucose, in $C_6H_{12}O_6 = 0.1$

Now, lets assume we have 10 moles of solution.

Thus, moles of $C_6H_{12}O_6 = 1$

Ans, moles of $H_2O = 9$

Now, the mass of water = number of moles x molar mass

Thus, the mass of water $= 9 \times 18 = 162\ \text{g}$

Mass of $C_6H_{12}O_6 = 180\ gm$

Now, mass percent of water is given as:

$$\text{Mass \% of water} = \frac{\text{Mass of water}}{\text{Mass of water + mass of glucose}} \text{ x } 100$$

$$\text{Mass \% of water} = \frac{162}{180 + 162} \text{ x } 100$$

$$\text{Mass \% of water} = 47.3\%$$

Thus, the correct answer is 47.

Maths

Q. 1 If $y^2 + \log_e(\cos^2 x) = y, x\epsilon \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then :

Option 1:
$y''(0) = 0$

Option 2:
$|y'(0)| + |y''(0)| = 1$

Option 3:
$|y''(0)| = 2$

Option 4:
$|y'(0)| + |y''(0)| = 3$

Correct Answer:
$|y''(0)| = 2$

Solution:
$y^2 + log_e(\cos^2 x) = y$

$y^2 + 2\log_e(\cos x) = y$

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$$2yy' - 2\frac{\sin x}{\cos x} = y'$$

$$2yy' - 2\tan x = y'$$

$$\therefore y'(0) = 0$$

$$2(y')^2 + 2yy'' - 2\sec^2 x = y''$$

$$\therefore |y''(0)| = 2$$

Q. 2 $2\pi - \left(\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{16}{65}\right)$ is equal to:

Option 1:
$\frac{\pi}{2}$

Option 2:
$\frac{5\pi}{4}$

Option 3:
$\frac{3\pi}{2}$

Option 4:
$\frac{7\pi}{4}$

Correct Answer:
$\frac{3\pi}{2}$

Solution:

$$2\pi - \left(\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{16}{65}\right)$$

$$= 2\pi - \left(\sin^{-1}\left(\frac{4}{5}\times\frac{12}{13}\times\frac{5}{13} \mp \frac{3}{5}\right) + \sin^{-1}\frac{16}{65}\right)$$

$$= 2\pi - \left(\sin^{-1}\left(\frac{63}{65}\right) + \sin^{-1}\left(\frac{16}{65}\right)\right)$$

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$$= 2\pi - \left(\cos^{-1}\left(\frac{16}{65}\right) + \sin^{-1}\left(\frac{16}{65}\right)\right)$$

$$= 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$$

Q. 3 If the first term of an A.P. is 3 and the sum of its first 25 terms is equal to the sum of its next 15 terms, then the common difference of this A.P. is:

Option 1:
$\frac{1}{6}$

Option 2:
$\frac{1}{5}$

Option 3:
$\frac{1}{4}$

Option 4:
$\frac{1}{7}$

Correct Answer:
$\frac{1}{6}$

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Solution:
Sum of n terms of an AP

$S_n = \frac{n}{2}[2a + (n-1)d]$
OR
$S_n = \frac{n}{2}[a + l]$
$a \rightarrow$ first term
$d \rightarrow$ common difference
$n \rightarrow$ number of terms

Given a=3

$S_{1-25} = S_{26-40}$

$\frac{25}{2}[a + a + 24d] = \frac{15}{2}[a + 25d + a + 39d]$

(Using second form of the formula)

$10a + 120d = 6a + 192d$

$4a = 72d$

$d = \frac{12}{72} = \frac{1}{6}$

Q. 4 A hyperbola having the transverse axis of length $\sqrt{2}$ has the same foci as that of the ellipse $3x^2 + 4y^2 = 12$, then this hyperbola does not pass through which of the following points?

Option 1:
$\left(\frac{1}{\sqrt{2}}, 0\right)$

Option 2:
$\left(-\sqrt{\frac{3}{2}}, 1\right)$

Option 3:
$\left(1, -\frac{1}{\sqrt{2}}\right)$

Option 4:
$\left(\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}\right)$

Correct Answer:
$\left(\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}\right)$

Solution:

Given equation of ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$

$a^2 = 4, b^2 = 3 \therefore e = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}$

Thus foci of ellipse $(\pm 1, 0)$

$\Rightarrow \pm 1 = \sqrt{a^2 + b^2} \Rightarrow b^2 = \frac{1}{2}$

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or $b^2 = 1 - a^2 = 1 - \frac{1}{2} = \frac{1}{2}$

Now the equation of hyperbola

$$\frac{x^2}{\frac{1}{2}} - \frac{y^2}{\frac{1}{2}} = 1 \Rightarrow 2x^2 - 2y^2 = 1$$

Point $\left(\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}\right)$

Q. 5 The proposition $p \rightarrow \sim (p \wedge -q)$ is equivalent to:

Option 1:
p

Option 2:
$(\sim p) \vee q$

Option 3:
$(\sim p) \wedge q$

Option 4:
$(\sim p) \vee (\sim q)$

Correct Answer:
p

Solution:
$p \rightarrow \sim (p \wedge -q)$

P	q	$(\sim p) \wedge q$	$p \rightarrow \sim (p \wedge -q)$
T	T	F	T
F	T	F	T
F	F	F	T

Q. 6 Let P be a point on the parabola, $y^2 = 12x$ and N be the foot of the perpendicular drawn from P on the axis of the parabola. A line is now drawn through mid-point M of PN, parallel to its axis which meets the parabola at Q. If the y-intercept of the NQ is $\frac{4}{3}$, then:

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Option 1:
PN=4

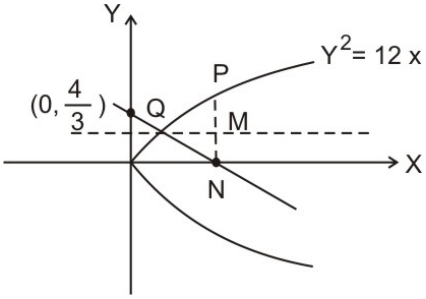
Option 2:
$MQ = \frac{1}{3}$

Option 3:
$MQ = \frac{1}{4}$

Option 4:
PN=3

Correct Answer:
$MQ = \frac{1}{4}$

Solution:



$P = (3t^2, 6t), N = (3t^2, 0)$

$M = (3t^2, 3t), Q = \left(\frac{3}{4}t^2, 3t\right)$

$\therefore t = \frac{1}{3}$\

$\therefore \text{MQ} = \frac{9}{4}\text{t}^2 = \frac{1}{4}$ and $\text{PN} = 6\text{t} = 2$

Q. 7 Consider the two sets:

$A = m \epsilon R$: both the roots of $x^2 - (m+1)x + m + 4 = 0$ are real} and

$B = [-3, 5]$

Which of the following is not true?

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Option 1:
$A - B = (-\infty, -3) \cup (5, \infty)$

Option 2:
$A \cap B = -3$

Option 3:
$B - A = 3, 5\}$

Option 4:
$A \cup B = R$

Correct Answer:
$A - B = (-\infty, -3) \cup (5, \infty)$

Solution:
$x^2 - (m+1)x + m + 4 = 0$

$b^2 - 4ac \geqslant 0$

$\Rightarrow (m+1)^2 - 4(m+4) \geqslant 0$

$\Rightarrow m^2 - 2m - 15 \geqslant 0$

$\Rightarrow (m-5)(m-3) \geqslant 0$

$\Rightarrow m\epsilon(-\infty, 3) \cup (5, \infty)$

Q. 8 If α and β are the roots of the equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2px + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to:

Option 1:
$\frac{9}{4}(9 + q^2)$

Option 2:
$\frac{9}{4}(9 - q^2)$

Option 3:
$\frac{9}{4}(9 + p^2)$

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Option 4:

$\frac{9}{4}(9-p^2)$

Correct Answer:

$\frac{9}{4}(9-p^2)$

Solution:

$x^2+px+2=0 \quad \alpha,\beta$

$2x^2+2px+1=0 \quad \frac{1}{\alpha},\frac{1}{\beta}$

$\alpha+\beta=-p, \alpha\beta=2$

$\frac{1}{\alpha}+\frac{1}{\beta}=-q, \frac{1}{\alpha\beta}=\frac{1}{2}$

$\left(\alpha-\frac{1}{\alpha}\right)\left(\beta-\frac{1}{\beta}\right)\left(\alpha+\frac{1}{\beta}\right)\left(\beta+\frac{1}{\alpha}\right)$

$=\left(\alpha\beta-\frac{\alpha}{\beta}-\frac{\beta}{\alpha}+\frac{1}{\alpha\beta}\right)\left(\alpha\beta+2+\frac{1}{\alpha\beta}\right)$

$=\left(\frac{5}{2}-\frac{\alpha^2+\beta^2}{\alpha\beta}\right)\left(\frac{9}{2}\right)$

$=\left(\frac{5}{2}-\frac{p^2-4^{-1}}{2}\right)\left(\frac{9}{2}\right)$

$=\frac{9}{4}(9-p^2)$

Q. 9

If $\Delta=\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix}=Ax^2+Bx^2+Cx+D$, then B+C is equal to:

Option 1:

-1

Option 2:

1

Option 3:

-3

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Option 4:
9

Correct Answer:
-3

Solution:

$$\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix}$$

$R_3 \to R_3 - R_1 - R_2$

$$= \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 0 & -1 & 3x-8 \end{vmatrix}$$

$R_2 \to R_2 - R_1$

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 0 & -1 & 3x-8 \end{vmatrix}$$

$$(x-1)\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 1 & 1 & 1 \\ 0 & -1 & 3x-8 \end{vmatrix}$$

$$(x-1)\left[(x-2)-(3x-4)+(3x-8)(x-2-2x+3)\right]$$

$$(x-1)\left[(-2x+2)+(3x-8)(-x+1)\right]$$

$$(x-1)\left[-3x^2+9x-6\right]$$

$$B = 3+9 = 12$$

$$c = -6-9 = -15$$

$$B+C = -3$$

Q. 10 The area (in sq. units) of the region $\left\{(x,y): 0 \le y \le x^2+1, 0 \le y \le x+1, \frac{1}{2} \le x \le 2\right\}$ is:

Option 1:
$\frac{23}{16}$

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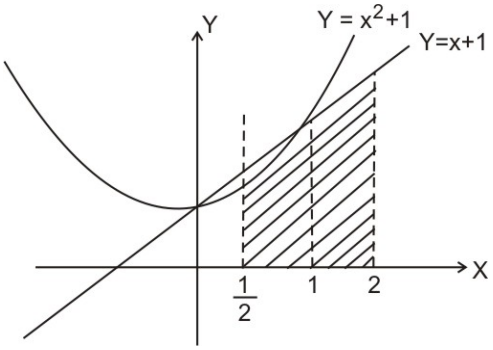
Option 2:
$\frac{79}{24}$

Option 3:
$\frac{79}{16}$

Option 4:
$\frac{23}{6}$

Correct Answer:
$\frac{79}{24}$

Solution:



Required area

$$\int_{\frac{1}{2}}^{1}(x^2+1)dx+\int_{1}^{2}(x+1)dx$$

$$\frac{19}{24}+\frac{5}{2}=\frac{79}{24}$$

Q. 11 The foot of the perpendicular drawn from the point (4,2,3) to the line joining the points (1,-2,3) and (1,1,0) lies on the plane:

Option 1:
2x +y-z =1

Option 2:
x-2y +z=1

Option 3:
x-2y+z =1

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Option 4:
x+2y -z =1

Correct Answer:
2x +y-z =1

Solution:
x=1, $y = -3\lambda - 2, z = 3 + 3\lambda$

L = (0,3,-3)

$$P = (1, -3\lambda - 3, 3 + 3\lambda)$$

A = (4,2,3)

$$AP = (3, 4 - 3\lambda, -3\lambda)$$

$$\vec{AP}.\vec{L} = 0$$

$$3.0 + (4 + 3\lambda).(-3) + (3\lambda).(-3) = 0$$

$$0 - 12 + 9\lambda + 9\lambda = 0$$

$$\lambda = \frac{2}{3}$$

$$P = (1, 0, 1)$$

Q. 12 The solution curve of the differential equation $(1 + e^{-x})(1 + y^2)\frac{dy}{dx} = y^2$. which passes through the point (0,1), is:

Option 1:
$$y^2 + 1 = y\left\{\log_e\left(\frac{1 - e^{-x}}{2}\right) + 2\right\}$$

Option 2:
$$y^2 + 1 = y\left\{\log_e\left(\frac{1 - e^{x}}{2}\right) + 2\right\}$$

Option 3:
$$y^2 = 1 + y\left\{\log_e\left(\frac{1 - e^{x}}{2}\right) + 2\right\}$$

Option 4:
$$y^2 = 1 + y\left\{\log_e\left(\frac{1 - e^{-x}}{2}\right) + 2\right\}$$

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Correct Answer:

$$y^2 = 1 + y\left\{\log_e\left(\frac{1-e^{-x}}{2}\right) + 2\right\}$$

Solution:

$$\int \frac{(1+y^2)}{y^2}dy = \int \frac{dx}{(1+e^{-x})}$$

$$\int \left(\frac{1}{y^2}+1\right)dy = \int \frac{e^x}{(1+e^x)}dx$$

$$y - \frac{1}{y} = \ln(1+e^x) + c$$

Passing through (0,1)

$$0 = \ln 2 + c$$

$$\Rightarrow c = \ln\frac{1}{2}$$

$$\Rightarrow y - \frac{1}{y} = \ln\left(\frac{1+e^x}{2}\right)$$

$$\Rightarrow y^2 = 1 + y\ln\left(\frac{1+e^x}{2}\right)$$

Q. 13 A dice is thrown thrown two times and the sum of the scores apperaing on the dice is observed to be multiple of 4. Then the conditional probability that the score 4 has appeared alteast once is:

Option 1:

$\frac{1}{4}$

Option 2:

$\frac{1}{3}$

Option 3:

$\frac{1}{8}$

Option 4:

$\frac{1}{9}$

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Correct Answer:
$\frac{1}{9}$

Solution:
A = event of sum being multiple of 4

B = event that 4 has appeared atleast once

$$P(A) = \frac{9}{36} = \frac{1}{4}$$

$$P(B) = \frac{11}{36}$$

$$P(A \cap B) = \frac{1}{36}$$

$$P\frac{B}{A} = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{36}}{\frac{9}{36}} = \frac{1}{9}$$

Q. 14 $\int_{-\pi}^{\pi} |\pi - |x||\, dx$ is equal to:

Option 1:
$\sqrt{2}\pi^2$

Option 2:
$2\pi^2$

Option 3:
π^2

Option 4:
$\frac{\pi^2}{2}$

Correct Answer:
π^2

Solution:
$$\int_{-\pi}^{\pi} |x - |x||\, dx$$

$$= 2\int_{0}^{\pi} |x - |x||\, dx$$

$$= 2\int_{0}^{\pi} |\pi - |x||\, dx$$

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$= 2\int_0^{\pi}(\pi - x)dx$

$2\left(\pi x - \frac{x^2}{2}\right)_0^{\pi} = \pi^2$

Q. 15 For the frequency distribution:

Variate (x) : $x_1 \quad x_2 \quad x_3 \quad .....x_{15}$

Frequency (f): $f_1 \quad f_2 \quad f_3.....f_{15}$

where $0 < x_1 < x_2 < x_3 < ..... < x_{15} = 10$ and $\sum_{i=1}^{15} f_i > 0$, the standard deviation cannont be:

Option 1:
4

Option 2:
1

Option 3:
6

Option 4:
2

Correct Answer:
6

Solution:
$\because \sigma^2 \leq \frac{1}{4}(\mathrm{M} - \mathrm{m})^2$

Where M and m are upper and lower bounds of values of any random variable.

$\because \sigma^2 \leq \frac{1}{4}(10 - 0)^2$
$\Rightarrow 0 < \sigma < 5$
$\therefore \sigma \neq 6$

Q. 16 The lines $\vec{r} = (\hat{i} - \hat{j}) + l(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + m(\hat{i} + \hat{j} - \hat{k})$

Option 1:
do not intersect for any values of l and m

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Option 2:
intersect for all values of l and m

Option 3:
intersect for the values when l=2 and $m = \frac{1}{2}$

Option 4:
intersect for the values when l=2 and m=2

Correct Answer:
do not intersect for any values of l and m

Solution:
$\vec{r} = (\hat{i} - \hat{j}) + l(2\hat{i} + \hat{k})$

$= (2l + 1)\hat{i} - \hat{j} + l\,\hat{k}$

$\vec{r} = (2\hat{i} - \hat{j}) + m(\hat{i} + \hat{j} - \hat{k})$

$= (m + 2)\hat{i} + (m - 1)\hat{j} - m\hat{k}$

For intersection

$$1 + 2\ell = 2 + \mathrm{m} \quad \ldots(i)$$
$$-1 = m - 1 \quad \ldots(ii)$$
$$\ell = -\mathrm{m} \quad \ldots(iii)$$
$$\text{from (ii) } m = 0$$

$$\text{from (iii) } \ell = 0$$

These values of m and l do not satisfy equation (1). Hence the two lines do not intersect for any values of l and m

---

Q. 17 Let [t] denote the greatest interger $\leq t$. If for some $\lambda \in R - [0, 1]$, $\lim_{x \to 0} \left| \frac{1 - x + |x|}{\lambda - x + [x]} \right| = L$, then L is equal to:

Option 1:
12

Option 2:
2

Option 3:
$\frac{1}{2}$

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Option 4:
0

Correct Answer:
12

Solution:

$$\lim_{x\to 0}\left|\frac{1-x+|x|}{\lambda-x+[x]}\right|=L$$

$$LHL=\lim_{x\to 0}\left|\frac{1-2x}{\lambda-x-1}\right|=\frac{1}{|\lambda-1|}$$

$$RHL=\lim_{x\to 0}\left|\frac{1}{\lambda-x-0}\right|=\frac{1}{|\lambda|}$$

$$\therefore \frac{1}{|\lambda-1|}=\frac{1}{|\lambda|}\Rightarrow \lambda=\frac{1}{2}$$

$$\therefore L=2$$

Q. 18 The value of $\left(2.{}^1P_0-3.{}^2P_1+4.{}^3P_2-\ldots.\text{ upto } 51^{th}\text{ term}\right)+(1!-2!+3!-\ldots.\text{ upto}51^{th}\text{ term})$ is equal to:

Option 1:
$1-51(51)!$

Option 2:
$1+(51)!$

Option 3:
$1+(52)!$

Option 4:
1

Correct Answer:
$1+(52)!$

Solution:

$$\left(2.{}^1P_0-3.{}^2P_1+4.{}^3P_2-\ldots.\text{ upto } 51^{th}\text{ term}\right)+(1!-2!+3!-\ldots.\text{ upto}51^{th}\text{ term})$$

$$\therefore \mathrm{S}=(2\times 1!-3\times 2!+4\times 3!\ldots\ldots.52.51!)+(1!-2!+3!\ldots\ldots\ldots\ldots.(51)!)$$

$$=[2!-3!+4!-5!+\ldots\ldots 52!]+[1!-2!+3!\ldots\ldots\ldots+51!]$$

$$=1!+52!$$

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Q. 19 The function of , $f(x)=(3x-7x)x^{\frac{2}{3}}, x\epsilon R$, is increasing for all x lying in:

Option 1:
$(-\infty,0)\cup\left(\frac{14}{15},\infty\right)$

Option 2:
$(-\infty,0)\cup\left(\frac{3}{7},\infty\right)$

Option 3:
$\left(-\infty,\frac{14}{15}\right)$

Option 4:
$\left(-\infty,\frac{14}{15}\right)\cup(0,\infty)$

Correct Answer:
$(-\infty,0)\cup\left(\frac{14}{15},\infty\right)$

Solution:
$f(x)=(3x-7)x^{\frac{2}{3}}$

$f(x)=3x^{\frac{5}{3}}-7x^{\frac{2}{3}}$

$f'(x)=5x^{\frac{2}{3}}-\frac{14}{3}x^{\frac{1}{3}}$

$5x^{\frac{2}{3}}-\frac{14}{3}x^{\frac{1}{3}}>0$

$\frac{15x-14}{3x^{\frac{1}{3}}}>0$

$\therefore \mathrm{f'(x)}>0\forall \mathrm{x}\in(-\infty,0)\cup\left(\frac{14}{15},\infty\right)$

Q. 20 If the number of integral terms in the expansion of $(3^{\frac{1}{2}}+5^{\frac{1}{8}})^n$ is excatly 33, then the least value of n is:

Option 1:
264

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Option 2:
128

Option 3:
256

Option 4:
248

Correct Answer:
256

Solution:
$(3^{\frac{1}{2}}+5^{\frac{1}{8}})^n$

$T_{r+1}={}^nC_r 3^{\frac{n-r}{2}}.5^{\frac{r}{8}}$

Clearly r should be a multiple of 8.

There are exactly 33 integral terms possible values of r can be 0, 8, 16, ..., (32 × 8)

Least value of n = 256.

n=256

Q. 21 The value of $(\frac{1}{3})^{\log_9\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}\ldots\ldots \text{ to } \infty\right)}$ is equal to:

Correct Answer:
4

Solution:
$(\frac{1}{3})^{\log_9\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}\ldots\ldots to\infty\right)}$

$\frac{1}{3}+\frac{1}{3^2}\ldots\ldots is\ a\ GP$

$S_\infty=\frac{a}{1-r}, a=\frac{1}{3}, r=\frac{1}{3}$

$S_\infty=\frac{\frac{1}{3}}{1-\frac{1}{3}}=\frac{\frac{1}{3}}{\frac{2}{3}}=\frac{1}{2}$

Now, $(\frac{1}{3})^{\log_9(\frac{1}{2})}$

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$= (\frac{1}{2})^{\log_9(\frac{1}{3})}$

$= (\frac{1}{2})^{-2}$

$= 4$

Q. 22 Let $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}, x \epsilon R$ and $A^4 = [a_{ij}]$. If $a_{11} = 109$, then a_{22} is equal to____

Option 1:
10

Option 2:
9

Option 3:
90

Option 4:
None of the above

Correct Answer:
10

Solution:

$A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix} \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$

$A^3 = \begin{bmatrix} x^2+2x & x^2+1 \\ x^2+1 & x \end{bmatrix}$

$A^4 = \begin{bmatrix} x^2+2x & x^2+1 \\ x^2+1 & x \end{bmatrix} \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$

$A^4 = \begin{bmatrix} x^4+2x^2+x^2+1 & x^3+2x \\ x^3+2x & x^2+1 \end{bmatrix}$

$x^4 = 3x^2 + 1 = 109$

$x = 3$

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$x^2 + 1 = 10$

Q. 23 The diameter of the circle, whose centre lies on the line x+y=2 in the first qaudrant and which touches both the lines x=3 and y=2, is_____

Option 1:
1

Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
3

Solution:
The center of the circle = (k,2-k) circle is pass via point (0,2) and (3,0)

$$\frac{k-3}{\sqrt{1^2+0^2}} = \left|\frac{2-k-2}{\sqrt{1^2+0^2}}\right|$$

$$|k-3| = |k|$$

$$|k-3| = \pm k$$

$$k = \frac{3}{2}$$

centre =$\left(\frac{3}{2}, \frac{1}{2}\right)$

diameter (x) = $2 \times |k-3| = 2 \times \left|\frac{3}{2} - 3\right| = 2 \times \frac{3}{2} = 3unit$

Q. 24 If $\left(\frac{1+i}{1-i}\right)^{\frac{m}{2}} = \left(\frac{1+i}{1-i}\right)^{\frac{n}{3}} = 1, (m, n \epsilon N)$ then the greatest common divisior of the least values of m and n is_____

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Option 1:
4

Option 2:
2

Option 3:
3

Option 4:
5

Correct Answer:
4

Solution:

$$\left(\frac{1+i}{1-i}\right)^{\frac{m}{2}}=\left(\frac{1+i}{1-i}\right)^{\frac{n}{3}}=1$$

$$\left(\frac{1+i}{1-i}\times\frac{-1-i}{-1-i}\right)^{\frac{m}{2}}$$

$$=(i)^{\frac{m}{2}}$$

$$\frac{m}{2}=4\Rightarrow m=8$$

$$\left(\frac{1+i}{1-i}\times\frac{-1-i}{-1-i}\right)^{\frac{n}{3}}$$

$$=(-i)^{\frac{n}{3}}$$

$$\frac{n}{3}=4\Rightarrow n=12$$

$$\therefore GCD=4$$

Q. 25 If $\lim_{x\to 0}\left\{\frac{1}{8}\left(1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}\cos\frac{x^2}{4}\right)\right\}=2^{-k}$,then the value of k is_____

Option 1:
2

Option 2:
4

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Option 3:
6

Option 4:
8

Correct Answer:
8

Solution:

$$\lim_{x\to 0}\left\{\frac{1}{8}\left(1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}\cos\frac{x^2}{4}\right)\right\}$$

$$=\lim_{x\to 0}\left\{\frac{1}{8}\left[\left(1-\cos^2\frac{x^2}{4}\right)\left(1-\cos\frac{x^2}{2}\right)\right]^2\right\}$$

$$\cos 2\theta = 1-2\sin^2\theta \Rightarrow 2\sin^2\theta = 1-\cos 2\theta$$

$$1-\cos\frac{x^4}{4}=2\sin\frac{x^2}{8}; 1-\cos\frac{x^2}{2}=2\sin\frac{x^2}{4}$$

$$=\lim_{x\to 0}\left\{\frac{1}{x^8}\left(2\sin^2\frac{x^2}{8}\right)\left(2\sin\frac{x^2}{4}\right)\right\}$$

$$=\lim_{x\to 0}\left\{\frac{4}{x^8}\left(\sin\frac{x^2}{8}\right)^2\left(\sin\frac{x^2}{4}\right)^2\right\}$$

$$=\lim_{x\to 0}\left\{\frac{4}{16\times 64}\left(\frac{\sin\frac{x^2}{4}}{\frac{x^2}{4}}\right)^2\left(\frac{\sin\frac{x^2}{4}}{\frac{x^2}{4}}\right)^2\right\}$$

$$\frac{1}{4\times 64}=2^{-k}$$

$$2^{-2}\times 2^{-6}=2^{-k}K=8$$

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3rd September Shift 2

Physics

Q. 1 Concentric metallic hollow spheres of radii R and $4R$ hold charges Q_1 and Q_2 respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference $V(R) - V(4R)$ is :

Option 1:
$\frac{3Q_1}{16\pi\varepsilon_0 R}$

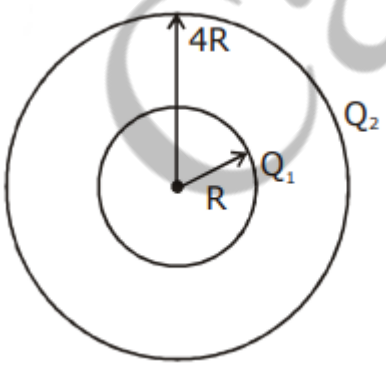
Option 2:
$\frac{3Q_2}{4\pi\varepsilon_0 R}$

Option 3:
$\frac{Q_2}{4\pi\varepsilon_0 R}$

Option 4:
$\frac{3Q_1}{4\pi\varepsilon_0 R}$

Correct Answer:
$\frac{3Q_1}{16\pi\varepsilon_0 R}$

Solution:



$\sigma = \frac{Q_1}{4\pi R^2} = \frac{Q_2}{4\pi 16R^2}$

$16Q_1 = Q_2$

$V_R - V_{4R} = \frac{KQ_1}{R} + \frac{KQ_2}{4R} - \frac{KQ_1}{4R} - \frac{KQ_2}{4R} = \frac{3KQ_1}{4R} = \frac{3Q_1}{16\pi\varepsilon_0 R}$

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Q. 2 The electric field of a plane electromagnetic wave propagating along the x-direction in a vacuum is $\vec{E} = E_0\,\hat{j}\cos(\omega t - kx)$. The magnetic field $\vec{B}$, at the moment, is :

Option 1:

$\vec{B} = \frac{E_0}{\sqrt{\mu_0\epsilon_0}}\cos(kx)\hat{k}$

Option 2:

$\vec{B} = E_0\sqrt{\mu_0\epsilon_0}\cos(kx)\hat{j}$

Option 3:

$\vec{B} = E_0\sqrt{\mu_0\epsilon_0}\cos(kx)\hat{k}$

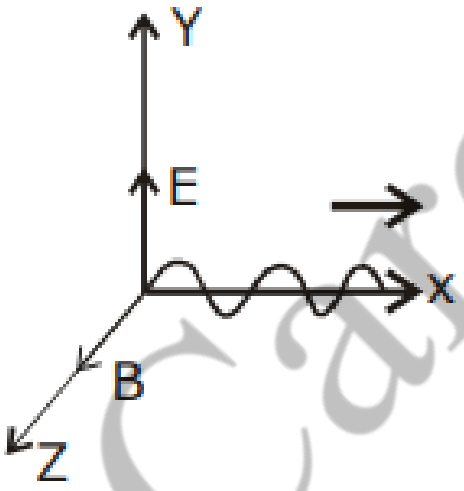
Option 4:

$\vec{B} = \frac{E_0}{\sqrt{\mu_0\epsilon_0}}\cos(kx)\hat{j}$

Correct Answer:

$\vec{B} = E_0\sqrt{\mu_0\epsilon_0}\cos(kx)\hat{k}$

Solution:



$E = E_0\cos(\omega t - kx)\hat{j}$

$E_0 = B_0 C$

$B_0 = \frac{E}{C} = \frac{E_0}{\frac{1}{\sqrt{\mu_0\epsilon_0}}}$

$B_0 = E_0\sqrt{\mu_0\epsilon_0}$

$\mathrm{B} = \mathrm{E}_0\sqrt{\mu_0\epsilon_0}\cos(\omega\mathrm{t} - \mathrm{kx})\hat{\mathrm{k}}$

$\text{at } \mathrm{t} = 0$

$\overrightarrow{\mathrm{B}} = \mathrm{E}_0\sqrt{\mu_0\,\epsilon_0}\cos(\mathrm{kx})\hat{\mathrm{k}}$

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Q. 3 A block of mass m attached to a massless spring is performing oscillatory motion of amplitude $'A'$ on a frictionless horizontal plane. If half of the mass of the block breaks off when it is passing through its equilibrium point, the amplitude of oscillation for the remaining system become fA. The value of f is :

Option 1:
$\frac{1}{\sqrt{2}}$

Option 2:
1

Option 3:
$\frac{1}{2}$

Option 4:
$\sqrt{2}$

Correct Answer:
$\frac{1}{\sqrt{2}}$

Solution:
Let the amplitude of oscillation for the remaining system is A_1

then

$$\text{At equilibrium position}$$

$$\mathrm{V_0 = \omega_0\ A = \sqrt{\frac{K}{m}} A} \ldots\ldots (1)$$

$$\mathrm{V_0 = \omega A_1 = \sqrt{\frac{K}{\frac{m}{2}}}\ A_1} \ldots\ldots (2)$$

$$\text{from equation (1) and (2)}$$

$$\text{we get} \quad \mathrm{A_1 = \frac{A}{\sqrt{2}}}$$

Q. 4 A particle is moving unidirectionally on a horizontal plane under the action of a constant power supplying energy source. The displacement (s) - time (t) graph that describes the motion of the particle is (graphs are drawn schematically and are not to scale) :

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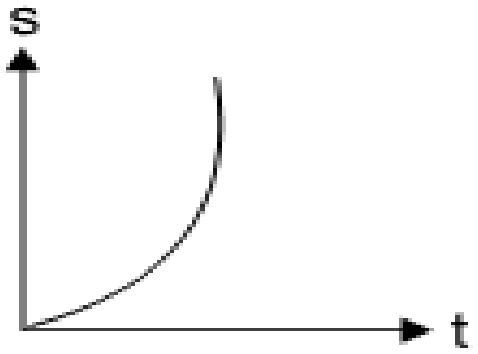
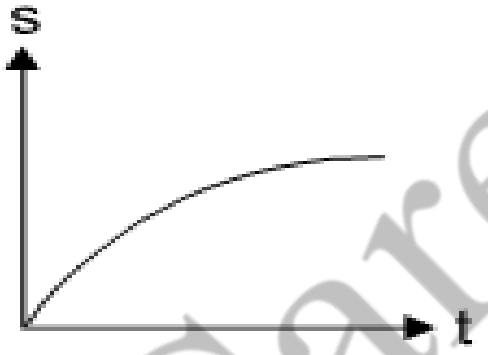
Option 1:

Option 2:



Option 3:



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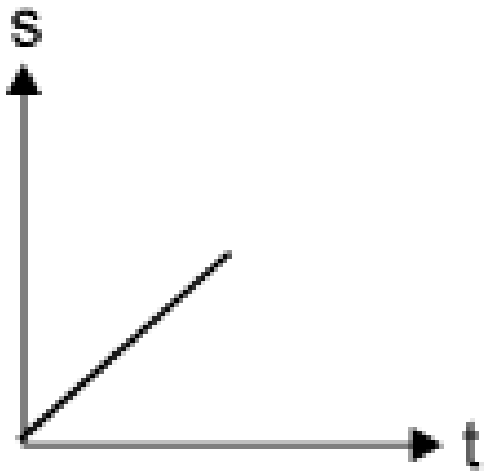
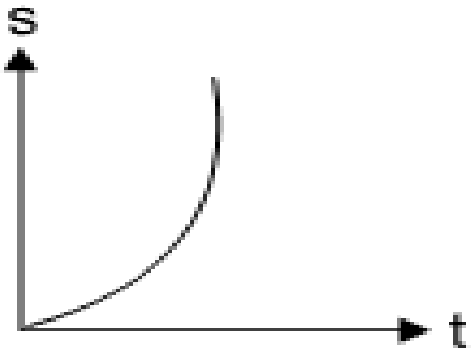
Option 4:



Correct Answer:



Solution:

$\frac{dK}{dt} = P = costant$

$\Rightarrow K = Pt = \frac{1}{2}mV^2$

$\therefore V = \sqrt{\frac{2Pt}{m}} = \frac{ds}{dt}$

$\Rightarrow S = \sqrt{\frac{2P}{m}} * \frac{2}{3} * t^{\frac{3}{2}}$

i.e $s \ \alpha \ t^{\frac{3}{2}}$

So the correct graph is

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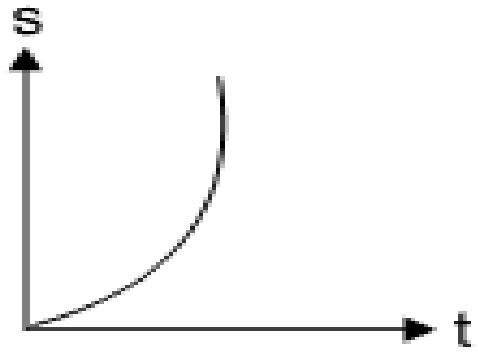

Q. 5 A uniform magnetic field B exists in a direction perpendicular to the plane of a square loop made of a metal wire. The wire has a diameter of $4\ mm$ and a total length of $30\ cm$. The magnetic field changes with time at a steady rate $dB/dt = 0.032\ Ts^{-1}$. The induced current (in A) in the loop is close to (Resistivity of the metal wire is $1.23 \times 10^{-8}\ \Omega m$).

Correct Answer:
0.61

Solution:

$$\text{Area of loop } = l^2/16$$

$$\varepsilon = -\frac{\mathrm{d}\phi}{\mathrm{dt}} = \left(\frac{\mathrm{dB}}{\mathrm{dt}}\right)\frac{\ell^2}{16}$$

$$\mathrm{I} = \frac{\left(\frac{\mathrm{dB}}{\mathrm{dt}}\right)\frac{\ell^2}{16}\times\pi\mathrm{r}^2}{\rho\ell}$$

$$I = \frac{30}{4} \times \frac{30}{4} \times \frac{10^{-4}\times 0.032\times 4\times 10^{-6}\times\pi}{1.23\times 10^{-8}\times 30\times 10^{-2}}$$

$$\mathrm{I} = 612.68 \times 10^{-3} = 0.61\ \mathrm{A}$$

Q. 6 The mass density of a planet of radius R varies with the distance r from its centre as $\rho(r) = \rho_0\left(1 - \frac{r^2}{R^2}\right)$. Then the gravitational field is maximum at :

Option 1:
$r = \sqrt{\frac{3}{4}}R$

Option 2:
$r = R$

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Option 3:

$r = \frac{1}{\sqrt{3}}R$

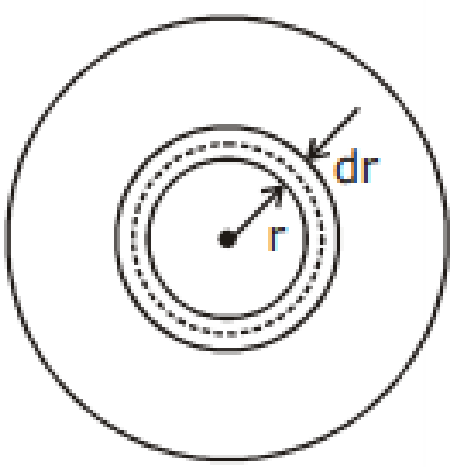
Option 4:

$r = \sqrt{\frac{5}{9}}R$

Correct Answer:

$r = \sqrt{\frac{5}{9}}R$

Solution:



$$\mathrm{dm} = \rho \mathrm{dv}$$

$$\rho(r) = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$$

$$\therefore \quad m = \int_0^r \rho(r) \times 4\pi r^2 dr$$

$$= 4\pi\rho_0 \left(\frac{r^3}{3} - \frac{r^5}{5R^2}\right)$$

$$\mathrm{E_g} = \frac{\mathrm{GM}}{\mathrm{r}^2}$$

$$\mathrm{E_g} = \mathrm{G}4\pi\rho_0 \left(\frac{\mathrm{r}}{3} - \frac{\mathrm{r}^3}{5\mathrm{R}^2}\right)$$

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$$\frac{dE_g}{dr} = \frac{1}{3} - \frac{3r^2}{5R^2} = 0$$

$$\frac{1}{3} = \frac{3}{5}\frac{r^2}{R^2}$$

$$r = \sqrt{\frac{5}{9}}R$$

Q. 7 A metallic sphere cools from $50^\circ C$ to $40^\circ C$ in $300\ s$. If atmospheric temperature around is $20^\circ C$, then the sphere's temperature after the next 5 minutes will be close to :

Option 1:
$31^\circ C$

Option 2:
$33^\circ C$

Option 3:
$28^\circ C$

Option 4:
$35^\circ C$

Correct Answer:
$33^\circ C$

Solution:

$As\ \ \frac{\Delta T}{\Delta t} = k\left[\frac{T_f+T_i}{2} - T_0\right]$

$From\ question$

$\frac{50-40}{300} = k\left[\frac{90}{2} - 20\right] ....(1)$

$\frac{40-T}{300} = k\left[\frac{40+T}{2} - 20\right] ....(2)$

$Taking\ ratio\ of\ equation\ (1)\ and (2)\ we\ get$

$\frac{10}{40-T} = \left[\frac{50}{40+T-40}\right]$

$\mathrm{T} = 200 - 5\ \mathrm{T}$

$6\ \mathrm{T} = 200$

$\Rightarrow\ \mathrm{T} = 33^0\mathrm{C}$

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Q. 8 The radius R of a nucleus of mass number A can be estimated by the formula $R = (1.3 \times 10^{-15})A^{\frac{1}{3}}$. It follows that the mass density of a nucleus is of the order of : $(M_{prot.} \cong M_{neut.} \simeq\ 1.67 \times 10^{-27} kg)$

Option 1:
$10^{3}\ kg\ m^{-3}$

Option 2:
$10^{10}\ kg\ m^{-3}$

Option 3:
$10^{24}\ kg\ m^{-3}$

Option 4:
$10^{17}\ kg\ m^{-3}$

Correct Answer:
$10^{17}\ kg\ m^{-3}$

Solution:

$$\rho_{\text{nucleus}} = \frac{\text{mass}}{\text{volume}}$$
$$= \frac{\mathrm{A}(1.67 \times 10^{-27}\mathrm{kg})}{(4/3)\pi \mathrm{r}_0^3\ \mathrm{A}}$$
$$= \frac{3(1.67 \times 10^{-27} kg)}{4\pi \mathrm{r}_0^3}$$
$$= 2.3 \times 10^{17}\ \mathrm{kg/m^3} \approx 10^{17}\ \mathrm{kg/m^3}$$

Q. 9 If a semiconductor photodiode can detect a photon with a maximum wavelength of $400\ nm$, then its band gap energy (in eV) is :

Planck's constant $h = 6.63 \times 10^{-34} J.s.$

Speed of light $c = 3 \times 10^{8}\ m/s$

Correct Answer:
3.1

Solution:

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For photodiode to detect

$$E = \frac{hc}{\lambda} > (\text{ band gap energy })$$

$$\Rightarrow (\text{ band gap energy })_{max} = \mathrm{hc}/\lambda_{\max}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{400 \times 10^{-9}} = 5 \times 10^{-19}\ \mathrm{J} = 3.1\mathrm{eV}$$

Q. 10 Two light waves having the same wavelength λ in vacuum are in phase initially. Then the first wave travels a path L_1 through a medium of refractive index n_1 while the second wave travels a path of length L_2 through a medium of refractive index n_2. After this the phase difference between the two waves is :

Option 1:

$$\frac{2\pi}{\lambda}\left(\frac{L_2}{n_1} - \frac{L_1}{n_2}\right)$$

Option 2:

$$\frac{2\pi}{\lambda}\left(\frac{L_1}{n_1} - \frac{L_2}{n_2}\right)$$

Option 3:

$$\frac{2\pi}{\lambda}(n_1L_1 - n_2L_2)$$

Option 4:

$$\frac{2\pi}{\lambda}(n_2L_1 - n_1L_2)$$

Correct Answer:

$$\frac{2\pi}{\lambda}(n_1L_1 - n_2L_2)$$

Solution:

As we know

Phase difference = $\frac{2\pi}{\lambda}$*(path difference)

S0

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$\lambda_{n1} = \frac{\lambda}{n_1}$

$\lambda_{n2} = \frac{\lambda}{n_2}$

$(\Delta\phi)_1 = \frac{2\pi}{\lambda_{n1}} * L_1$

$(\Delta\phi)_2 = \frac{2\pi}{\lambda_{n2}} * L_2$

$\Delta\phi = \Delta\phi_1 - \Delta\phi_2 = \frac{2\pi}{\lambda}(n_1L_1 - n_2L_2)$

Q. 11 Two resistors $400\ \Omega$ and $800\ \Omega$ are connected in series across a $6\ V$ battery. The potential difference measured by a voltmeter of $10\ k\Omega$ across $400\ \Omega$ resistor is close to :

Option 1:
$2\ V$

Option 2:
$1.8\ V$

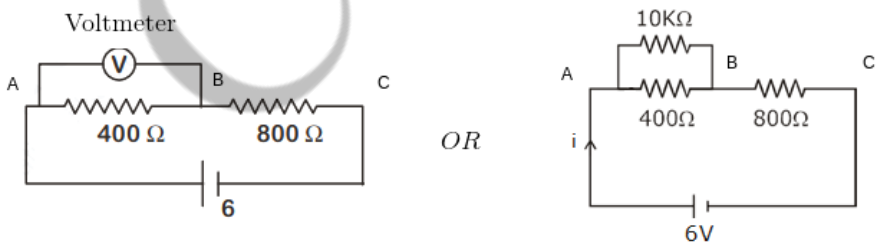
Option 3:
$2.05\ V$

Option 4:
$1.95\ V$

Correct Answer:
$1.95\ V$

Solution:
From the question

The circuit diagram is drawn as shown below



$i = \frac{6}{800 + (\frac{40 \times 10000}{400 + 10000})}$

$i = \frac{6}{1184.61} = 0.00506$

$V_{AB} = 6 - (800 \times 0.00506) = 6 - 4.05 = 1.95$

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Q. 12 To raise the temperature of a certain mass of gas by $50^{\circ}C$ at a constant pressure, 160 calories of heat is required. When the same mass of gas is cooled by $100^{\circ}C$ at constant volume,240 calories of heat is released. How many degrees of freedom does each molecule of this gas have (assume gas to be ideal) ?

Option 1:
5

Option 2:
6

Option 3:
3

Option 4:
7

Correct Answer:
6

Solution:
$As \quad \mathrm{Q} = \mathrm{nC_p}\Delta\mathrm{T}$
$160 = \mathrm{nC_p}(50)....(1)$
$240 = \mathrm{nC_v}100....(2)$

$\frac{C_p}{C_v} = \gamma....(3)$

So from equation (1) and (2) and (3)

we get

$$\frac{C_p}{2C_v} = \frac{160}{240} = \frac{\gamma}{2}$$
$$\therefore \gamma = \frac{4}{3} \text{ and } \mathrm{f} = \frac{2}{\gamma - 1} = 6$$

Q. 13 Which of the following will NOT be observed when a multimeter (operating in resistance measuring mode) probes connected across a component, are just reversed ?

Option 1:
Multimeter shows an equal deflection in both case i,e. before and after reversing the probes if the chosen component is resistor.

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Option 2:
Multimeter shows NO deflection in both case i,e. before and after reversing the probes if the chosen component is capacitor.

Option 3:
Multimeter shows a deflection, accompanied by a splash of light out of connected component in one direction and NO deflection on reversing the probes if the chosen component is LED.

Option 4:
Multimeter shows NO deflection in both case i,e. before and after reversing the probes if the chosen component is metal wire.

Correct Answer:
Multimeter shows NO deflection in both case i,e. before and after reversing the probes if the chosen component is capacitor.

Solution:
1.No matter, how we connect the resistance across multimeter It shows the same deflection.

2.Multimeter shows deflection when it connects with a capacitor.

So option 2 is the answer.

3.If we assume that LED has negligible resistance then the multimeter shows no deflection for the forward bias but when it connects in the reverse direction, it break down occurs so splash of light out.

4.The resistance of the metal wire may be taken zero, so no deflection in the multimeter.

Q. 14 A calorimeter of water equivalent $20\ g$ contains $180\ g$ of water at $25^{\circ}C$. 'm' grams of steam at $100^{\circ}C$ is mixed in it till the temperature of the mixture is $31^{\circ}C$. The value of 'm' is close to (Latent heat of water $= 540\ cal\ g^{-1}$, specific heat of water $= 1\ cal\ g^{-1}\ ^{\circ}C^{-1}$)

Option 1:
2

Option 2:
4

Option 3:
3.2

Option 4:
2.6

Correct Answer:
2

Solution:
As we know

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$$Q = mc\Delta T$$

and as

Heat lost by steam = heat gained by the water

$$[180 \times 1 \times (31-25)] + [20 \times (31-25)] = [m \times 540] + [m \times 1 \times (100-31)]$$
$$\Rightarrow (180 \times 6) + (20 \times 6) = 540m + 100m - 31m$$
$$\Rightarrow 1080 + 120 = 640m - 31m$$
$$\Rightarrow 1200 = 609m$$
$$\Rightarrow m = \frac{1200}{609} = 1.97 \approx 2$$

Q. 15 A block of mass $1.9\ kg$ is at rest at the edge of a table, of height $1\ m$. A bullet of mass $0.1\ kg$ collides with the block and sticks to it. If the velocity of the bullet is $20\ m/s$ in the horizontal direction just before the collision then the kinetic energy just before the combined system strikes the floor, is [Take $g = 10\ m/s^2$. Assume there is no rotational motion and loss of energy after the collision is negligable.]

Option 1:
$20\ J$

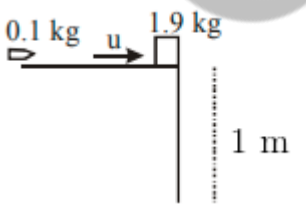
Option 2:
$21\ J$

Option 3:
$19\ J$

Option 4:
$23\ J$

Correct Answer:
$21\ J$

Solution:



Let the velocity of the combined system after a collision is V

$$\text{Applying law of conservation of linear momentum}$$
$$\Rightarrow 0.1 \times 20 = (1.9 + 0.1)\text{V}$$
$$\Rightarrow 2 = 2\ \text{V}$$
$$\Rightarrow V = 1\ \text{m/sec}$$

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So

At A

$$\mathrm{KE} = (\tfrac{1}{2}) * \mathrm{mv}^2 = (\tfrac{1}{2}) * 2 * (1)^2 = 1 \text{ J}$$
$$PE = mgh = 2 \times 10 \times 1 = 20 \ J$$
$$\mathrm{TE} = \mathrm{KE} + \mathrm{mgh} = 1 + 20 = 21 \text{ J}$$

As Total energy is conserved

So

$$(\mathrm{TE})_{initial} = (\mathrm{TE})_{final}$$
$$\therefore (\mathrm{TE})_{final} = 21 \ J$$

Q. 16 Amount of solar energy received on the earth's surface per unit area per unit time is defined a solar constant. Dimension of solar constant is :

Option 1:
ML^2T^{-2}

Option 2:
ML^0T^{-3}

Option 3:
$M^2L^0T^{-1}$

Option 4:
MLT^{-2}

Correct Answer:
ML^0T^{-3}

Solution:
Let the solar constant= S

So

$$\mathrm{S} = \frac{E}{A * T} = \frac{\mathrm{P}}{\mathrm{A}} = \frac{\mathrm{ML}^2\ \mathrm{T}^{-3}}{\mathrm{L}^2} = \mathrm{ML}^0\mathrm{T}^{-3}$$

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Q. 17 A perfectly diamagnetic sphere has a small spherical cavity at its centre, which is filled with a paramagnetic substance.

The whole system is placed in a uniform magnetic field $\vec{B}$. Then the field inside the paramagnetic substance is :

Option 1:
$\vec{B}$

Option 2:
zero

Option 3:
much large than $\left|\vec{B}\right|$ and parallel to $\vec{B}$

Option 4:
much large than $\left|\vec{B}\right|$ but opposite to $\vec{B}$

Correct Answer:
zero

Solution:
A perfect diamagnetic substance will completely expel the magnetic field. Therefore, there will be no magnetic field inside the cavity of the sphere. Hence the paramagnetic substance kept inside the cavity will experience no force.

Q. 18 Hydrogen ion and singly ionized helium atom are accelerated, from rest, through the same potential difference. The ratio of final speeds of hydrogen and helium ions is close to :

Option 1:
$1:2$

Option 2:
$10:7$

Option 3:
$2:1$

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Option 4:

$5 : 7$

Correct Answer:

$2 : 1$

Solution:

$K.E = qV$

$K.E = \frac{1}{2}mv^2 = qV$

$\Rightarrow \quad v = \sqrt{\frac{2qV}{m}}$

As Voltage (V) is same for both

$v \propto \sqrt{\frac{q}{m}}$

$\Rightarrow \frac{v_H}{v_{\text{He}}} = \frac{\sqrt{\frac{e}{m}}}{\sqrt{\frac{e}{4m}}} = \frac{2}{1}$

Q. 19 Two sources of light emit X-rays of wavelength $1\ nm$ and visible light of wavelength $500\ nm$, respectively. Both the sources emit light of the same power $200\ W$. The ratio of the number density of photons of X-rays to the number density of photons of the visible light of the given wavelengths is :

Option 1:

$\frac{1}{500}$

Option 2:

250

Option 3:

$\frac{1}{250}$

Option 4:

500

Correct Answer:

$\frac{1}{500}$

Solution:

The energy of each photon is given by

$E = h\nu = \frac{hc}{\lambda} \ \ (in\ Joule)$

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And

n (the number of photons emitted per second) is given as

$$n = \frac{Power\ of\ source\ (W\ or\ \frac{J}{sec})}{Energy\ of\ each\ photon(J)} = \frac{P}{E}\ (sec^{-1})$$

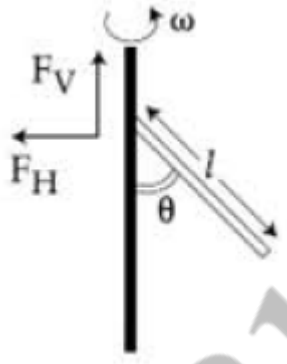
So $P = nE$

$$Here, P = 100W, \lambda_1 = 1nm, \lambda_2 = 500\text{nm}$$

Let $n_1, n_2 =$ no. of photons X -rays and visible light emitted from the two sources

$$\therefore P = n_1\frac{hc}{\lambda_1} = n_2\frac{hc}{\lambda_2}$$
$$\Rightarrow \frac{n_1}{\lambda_1} = \frac{n_2}{\lambda_2}$$
$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2} = \frac{1}{500}$$

Q. 20



A uniform rod of length 'l' is pivoted at one of its ends on a vertical shaft of negligible radius. When the shaft rotates at angular speed ω the rod makes an angle θ with it (see figure).To find θ equate the rate of change of angular momentum (direction going into the paper) $\frac{ml^2}{12}\omega^2 \sin\theta\cos\theta$ about the centre of mass (CM) to the torque provided by the horizontal and vertical forces F_H and F_V about the CM. The value of θ is then such that :

Option 1:

$$\cos\theta = \frac{2g}{3l\omega^2}$$

Option 2:

$$\cos\theta = \frac{g}{2l\omega^2}$$

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Option 3:

$$\cos\theta = \frac{g}{l\omega^2}$$

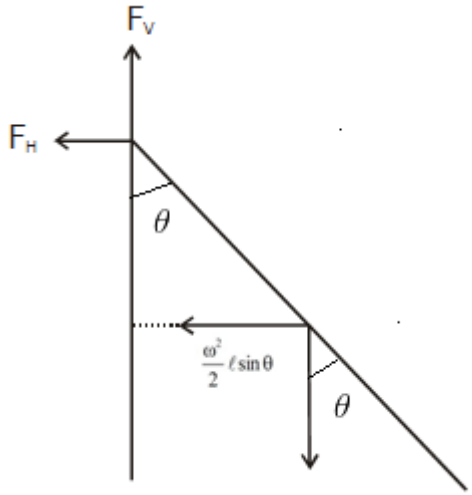
Option 4:

$$\cos\theta = \frac{3g}{2l\omega^2}$$

Correct Answer:

$$\cos\theta = \frac{3g}{2l\omega^2}$$

Solution:



$$F_V = mg$$
$$F_H = m\omega^2\frac{\ell}{2}\sin\theta$$

$$\therefore \tau_{net} \text{ about COM} = F_v \cdot \frac{\ell}{2}\sin\theta - F_H\left(\frac{\ell}{2}\cos\theta\right) = \frac{m\ell^2}{12}\omega^2\sin\theta\cos\theta$$

$$\Rightarrow mg\left(\frac{\ell}{2}\sin\theta\right) - m\omega^2\left(\frac{\ell}{2}\sin\theta\right)\left(\frac{\ell}{2}\cos\theta\right) = m\omega^2\left(\frac{\ell^2}{12}\right)(\sin\theta)(\cos\theta)$$

$$\Rightarrow \frac{g\ell}{2} - \frac{\omega^2\ell^2}{4}\cos\theta = \frac{\ell^2}{12}\omega^2\cos\theta$$

$$\Rightarrow \frac{g\ell}{2} = \omega^2\ell^2\cos\theta\left(\frac{1}{12} + \frac{1}{4}\right)$$

$$\Rightarrow \frac{g\ell}{2} = \frac{\omega^2\ell^2\cos\theta}{3}$$

$$\Rightarrow \cos\theta = \frac{3g}{2\omega^2\ell}$$

Q. 21 If minimum possible work is done by a refrigerator in converting 100 grams of water at $0°C$ to ice, how much heat (in calories) is released to the surroundings at temperature $27°C$ (Latent heat of ice $= 80\ Cal/gram$) to the nearest integer ?

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Option 1:
8791

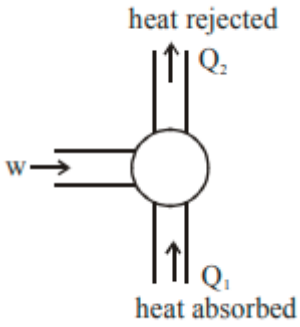
Option 2:
309

Option 3:
4567

Option 4:
21

Correct Answer:
8791

Solution:



$$w + Q_1 = Q_2$$
$$w = Q_2 - Q_1$$

$$\text{C.O.P.} = \frac{Q_1}{w} = \frac{Q_1}{Q_2 - Q_1} = \frac{273}{300-273} = \frac{Q_1}{W}$$

$$w = \frac{27}{273} \times 80 \times 100 \times 4.2$$

$$Q_2 = w + Q_1$$

$$Q_2 = \left(\frac{27}{273} \times 80 \times 100 \times 4.2\right) + (80 \times 100 \times 4.2)$$

$$Q_2 = \frac{300}{273} \times 80 \times 100 = 8791.2\text{cal} \approx 8791\text{cal}$$

Q. 22 When an object is kept at a distance of $30\ cm$ from a concave mirror, the image is formed at a distance of $10\ cm$ from the mirror. If the object is moved with a speed of $9\ cms^{-1}$, the speed $(in\ cms^{-1})$ with which image moves at that instant is _____.

Option 1:
1

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Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
1

Solution:
For concave mirror

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$And\ |\mathrm{m}|=\left|-\frac{\mathrm{v}}{\mathrm{u}}\right|=\frac{10}{30}=\frac{1}{3}$$

$$Nowusing$$

$$v_{\text{image}}=v_{\text{object}}\ \left(m^2\right)$$

$$\Rightarrow v_{\text{image}}=9\times\frac{1}{9}=1\ \text{cm/s}$$

Q. 23 A block starts moving up an inclined plane of inclination 30° with an initial velocity of v_0. It comes back to its initial position with velocity $\frac{v_0}{2}$. The value of the coefficient of kinetic friction between the block and the inclined plane is close to $\frac{I}{1000}$, The nearest integer to I is _____.

Option 1:
346

Option 2:
12345

Option 3:
7689

Option 4:
12341

Correct Answer:
346

Solution:
Let the length of the inclined plane as S (as shown in the figure)

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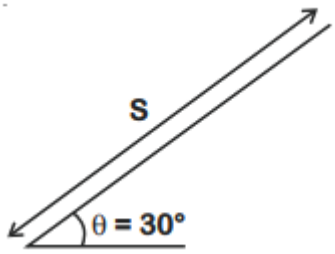
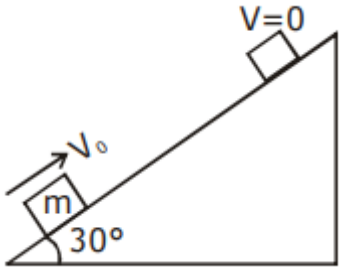


For upward motion



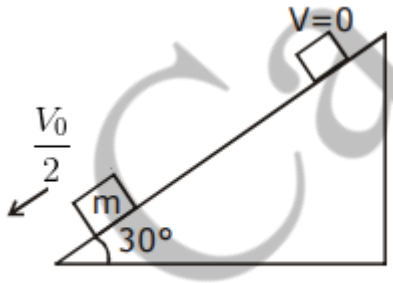
from work energy theorem

$$W_f = k_f - k_i$$

So

$$\frac{1}{2}mv_0^2 = \left(mg\frac{S}{2} + \mu mg\frac{\sqrt{3}}{2}S\right) ....(1)$$

For downward motion



from work energy theorem

$$W_f = k_f - k_i$$

$$\frac{1}{2}\frac{mv_0^2}{4} = \left(mg\frac{S}{2} - \mu mg\frac{\sqrt{3}}{2}\cdot S\right) .....(2)$$

From equation (1) and (2)

we get

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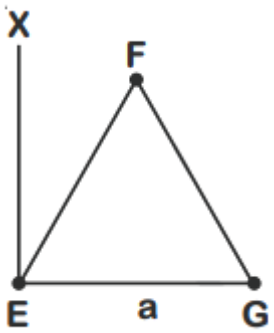
$4 = \frac{(1+\mu\sqrt{3})}{(1-\mu\sqrt{3})}$

$\Rightarrow 4 - 4\mu\sqrt{3} = 1 + \mu\sqrt{3}$
$\Rightarrow 3 = 5\mu\sqrt{3}$

$\therefore \mu = \frac{\sqrt{3}}{5} \approx \frac{346.41}{1000}$
$I = 346$

Q. 24 A massless equilateral triangle EFG of side $'a'$ (As shown in the figure) has three particles of mass m situated at its vertices. The moment of inertia of the system about the line EX perpendicular to EG in the plane of EFG is $\frac{N}{20}ma^2$ where N is an integer. The value of N is _____.



Option 1:
5

Option 2:
10

Option 3:
20

Option 4:
25

Correct Answer:
25

Solution:

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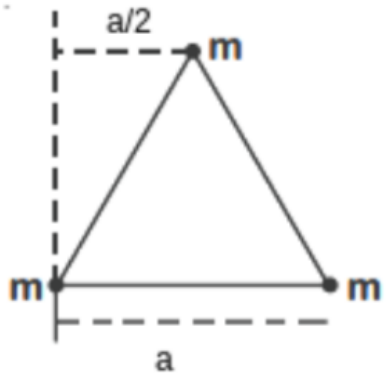

The moment of inertia of the system about the line passing through EX is

$$I = I_E + I_F + I_G$$

As we know for point mass m at separation r from the axis of rotation is $I = mr^2$

Point masses separation from the axis of rotation at position E, F & G are 0, a/2, a respectively

$$I = m(0)^2 + m(a/2)^2 + m(a)^2 = \frac{5ma^2}{4} = \frac{25ma^2}{20}$$

comparing with, $\frac{N}{20}ma^2$

N=25

Q. 25 A galvanometer coil has 500 turns and each turn has an average area $3 \times 10^{-4} m^2$. If a torque of $1.5\ Nm$ is required to keep this coil parallel to a magnetic field when a current of $0.5\ A$ is flowing through it, the strength of the field (in T) is _____.

Correct Answer:
20

Solution:
As coil is parallel to magnetic field, the Area vector or Magnetic Moment is perpendicular to magnetic field i.e., $\theta = 90^0$

Torque applied due to magnetic field when angle between magnetic moment and magnetic field θ is

$$\tau = MB\sin\theta = NIAB\sin\theta$$

$N = 500 turns$
$A = 3 \times 10^{-4} m^2$
$\tau_{external} = 1.5 Nm$
$I = 0.5A$

$\tau = 1.5 = 500 \times 0.5 \times 3 \times 10^{-4} B \sin\ 90^0$
$B = 20T$

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Chemistry

Q. 1 Among the statements (I–IV), the correct ones are:
(I) Be has smaller atomic radius compared to Mg.
(II)Be has higher ionization enthalpy than Al.
(III) Charge/radius ratio of Be is greater than that of Al.
(IV) Both Be and Al form mainly covalent compounds.

Option 1:
(I), (II) and (IV)

Option 2:
(I), (III) and (IV)

Option 3:
(II), (III) and (IV)

Option 4:
(I), (II) and (III)

Correct Answer:
(I), (II) and (IV)

Solution:
(I) Be has a smaller atomic radius compared to Mg, in a group atomic radius increases.
(II) Be has higher ionization enthalpy than Al due to fully filled s-block orbital of Be.
(III) Charge/radius ratio of Be is greater than that of Al, due to Be being small.
(IV) Only Al form mainly covalent compounds, Be forms ionic compounds.

(I), (II), and (IV) are correct.

Therefore, Option 1 is correct.

Q. 2 An ionic micelle is formed on the addition of :

Option 1:
Liquid diethyl ether to aqueous $NaCl$ solution.

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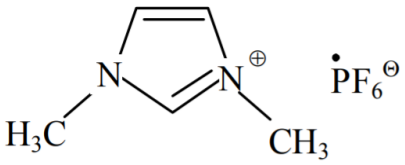
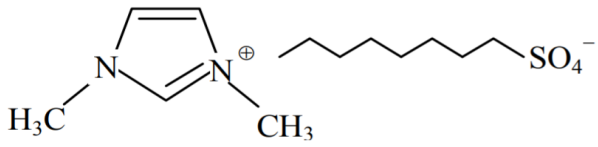
Option 2:
excess water to liquid

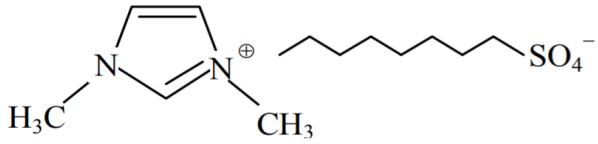


Option 3:
excess water to liquid



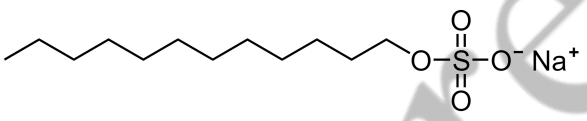
Option 4:
Sodium stearate to pure toluene.

Correct Answer:
excess water to liquid

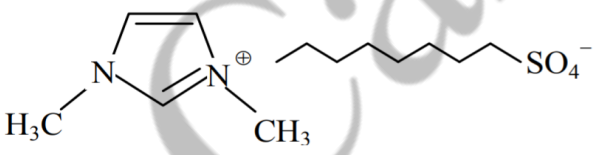


Solution:
ionic micelles formed by the addition of water to soap.

Soap -



Soap is present in



Q. 3 The increasing order of the reactivity of the following compounds in nucleophilic addition reaction is :

Propanal, Benzaldehyde, Propanone, Butanone

Option 1:
Benzaldehyde < Butanone < Propanone < Propanal

Option 2:
Butanone < Propanone < Benzaldehyde < Propanal

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Option 3:
Propanal < Propanone < Butanone < Benzaldehyde

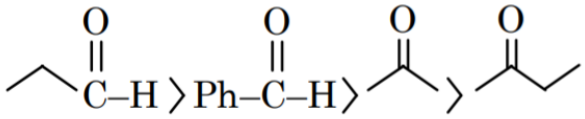
Option 4:
Benaldehyde < Propanal < Propanone < Butanone

Correct Answer:
Butanone < Propanone < Benzaldehyde < Propanal

Solution:
Rate of Nucleophilic addition-

1. Aldehyde > Ketone

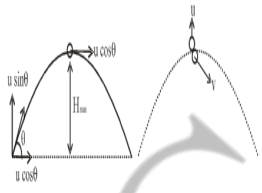
2. Aliphatic aldehyde > Aromatic aldehyde



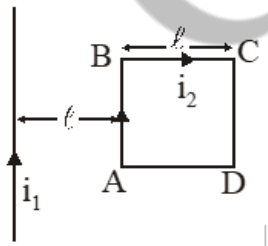
Order : Butanone < Propanone < Benzaldehyde < Propanal

Q. 4 100 mL of 0.1 M HCl is taken in a beaker and to it 100 mL of 0.1 M NaOH is added in steps of 2 mL and the pH is continuously measured. Which of the following graphs correctly depicts the change in pH?

Option 1:



Option 2:



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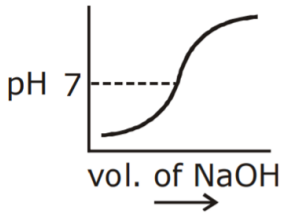
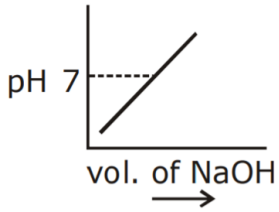
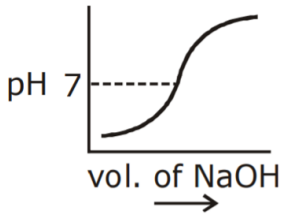
Option 3:



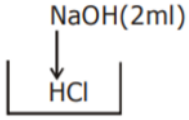
Option 4:



Correct Answer:

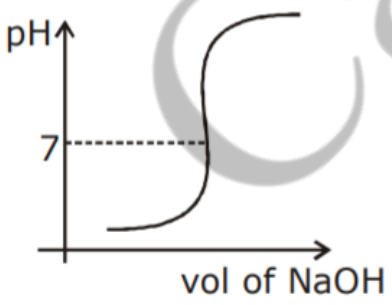


Solution:



initially pH will be acidic < 7
at eq pH pH = 7
& finally pH will be basic > 7

Steep rise in pH around the equivalence point for titration of a strong acid with a strong base.



Q. 5 A mixture of one mole each of H_2, He, and O_2 each are enclosed in a cylinder of volume V at temperature T. If the partial pressure of H_2 is 2 atm,the total pressure of the gases in the cylinder is :

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Option 1:
6 atm

Option 2:
38 atm

Option 3:
14 atm

Option 4:
22 atm

Correct Answer:
6 atm

Solution:
According to Dalton's law of partial pressure

$$p_i = x_i \times P_T$$

Where

$$p_i = \text{ partial pressure of the } i^{th} \text{ component}$$

$$x_i = \text{ mole fraction of the } i^{th} \text{ component}$$

$$p_T = \text{ total pressure of mixture}$$

$$\Rightarrow 2 \text{ atm} = \left(\frac{n_{H_2}}{n_{H_2} + n_{He} + n_{O_2}} \right) \times p_T$$

$$\Rightarrow p_T = 2 \text{ atm} \times \frac{3}{1} = 6 \text{ atm}$$

Q. 6 Consider the hypothetical situation where the azimuthal quantum number, l, takes values $0, 1, 2, ...n + 1$, where n is the principal quantum number. Then, the element with atomic number :

Option 1:
9 is the first alkali metal

Option 2:
13 has a half-filled valence subshell

Option 3:
8 is the first noble gas

Option 4:
6 has a $2p$-valence subshell

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Correct Answer:
13 has a half-filled valence subshell

Solution:
GIven l= 0 to (n + 1)

Now for n = 1,
l = 0, 1, 2

Orbitals :

1s , l =0

1p , l = 1

1d , l = 2

And for

n = 2

l = 0, 1, 2, 3

$\text{Orbitals} : 2s, 2p, 2d, 2f$

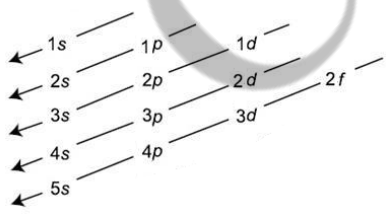
And for

n = 3

l = 0, 1, 2, 3,4

$\text{Orbitals} : 3s, 3p, 3d, 3f, 3g$

Now, in order to wire electronic configuration, we need to apply (n + l) rule.



Energy order : 1s < 1p < 2s < 1 d < 2p < 3s < 2d ...

1) 9 : $1s^2\ 1p^6\ 2s^1$ is the first alkali metal because after losing one electron, it will achieve the first noble gas configuration

2) 13 : $1s^2\ 1p^6\ 2s^2\ 1d^3$ is **not** half filled. **(Showing incorrect statement)**

3) 8 : $1s^2\ 1p^6$ is the first noble gas.

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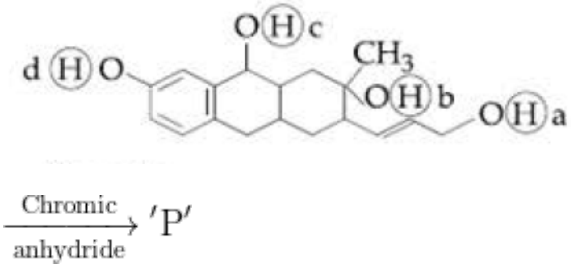
4) 6 : $1s^2\ 1p^4$ has 1p valence subshell.

Therefore, Option 2 is correct.

Q. 7



$\xrightarrow[\text{anhydride}]{\text{Chromic}}$ 'P'

The product $'P'$ gives positive ceric ammonium nitrate test. This is because of the presence of which of these $-OH$ group(s) ?

Option 1:
(b) only

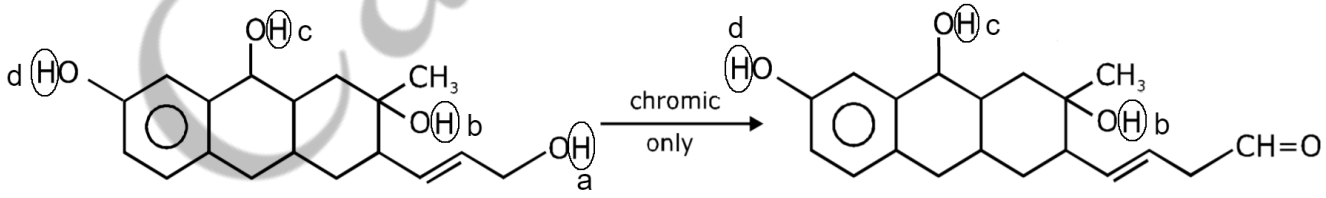
Option 2:
(c) and (d)

Option 3:
(d) only

Option 4:
(b) and (d)

Correct Answer:
(b) only

Solution:



After the reaction, in the product, only b (-OH) will be remaining to react with the ceric nitrate ion to form the complex.

Therefore, Option 1 is correct.

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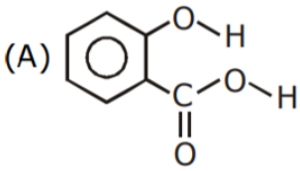
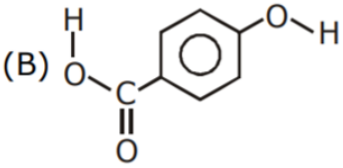
Q. 8 Consider the following molecules and statements related to them :

(A) 

(B) 

(a) (B) is more likely to be crystalline than (A)

(b) (B) has higher boiling point than (A)

(C) (B) dissolves more readily than (A) in water

Identify the correct option from below :

Option 1:
(a), (b) and (c) are true

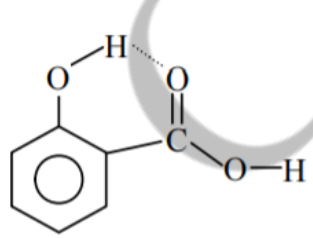
Option 2:
(a) and (c) are true

Option 3:
only (a) is true

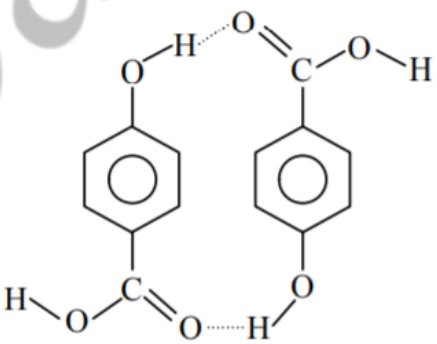
Option 4:
(b) and (c) are true

Correct Answer:
(a), (b) and (c) are true

Solution:



O-salicylic acid
intra molecular
H-bonding



p-salicylic acid
inter molecular
H-bonding

(a) B will be more crystalline due to more intermolecular interactions hence more efficient packing.
(b) B will have a higher boiling point due to higher intermolecular interactions.
(c) B will be more soluble in water than A as B will have more extent of H-bonding in water. ({Solubility data : O-salicylic acid = 2g/L ; P-salicylic acid = 5g/L)

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So all three statements are correct.

Therefore, Option 1 is correct.

Q. 9 The incorrect statement is :

Option 1:
Magnate and permanganate ions are tetrahedral

Option 2:
In manganate and permanganate ions, the π-bonding takes place by overlap of p-orbitals of oxygen and d-orbitals of manganese

Option 3:
Manganate and permanganate ions are paramagnetic

Option 4:
Manganate ion is green in colour and permanganate ion is purple in colour

Correct Answer:
Manganate and permanganate ions are paramagnetic

Solution:
(1) Both have d^3 s hybridisation hence both have tetrahedral geometry.

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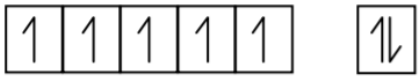
Maganate $\Rightarrow$ MnO_4^{2-},
Permanganate $\Rightarrow$ MnO_4^-

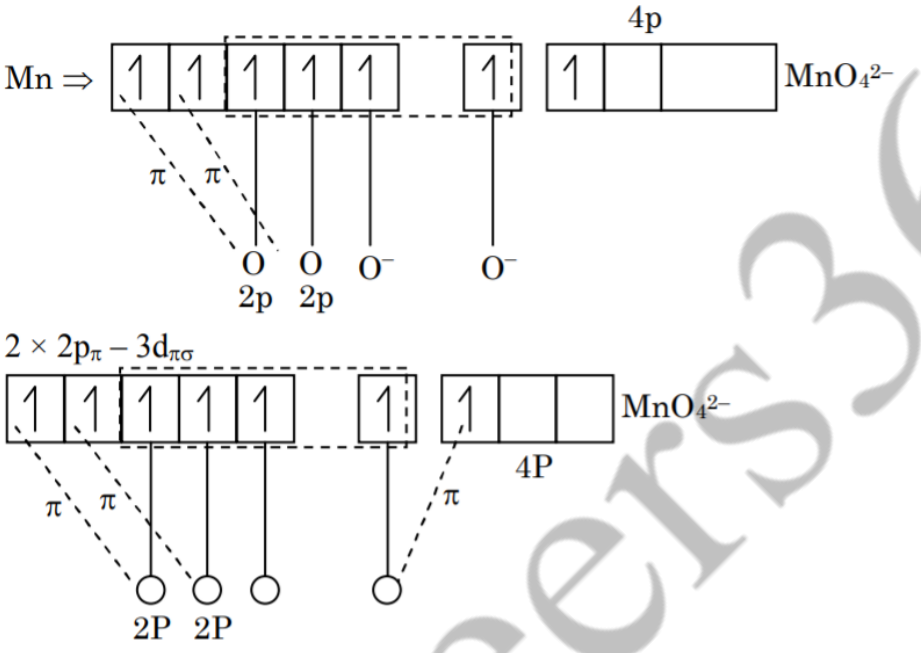
hybridisation of Mn $\Rightarrow$ d^3s | hybridisation of Mn $\Rightarrow$ d^3s

Mn $\Rightarrow$ 

$3d^5$ $4s^2$

After excitation

(2)



So, $2 \times 2P_\pi - 3d_\pi$
$1 \times 2P_\pi - 4P_\pi$

(3) Manganate contains 1 unpaired electron hence it is paramagnetic. whereas permanganate contains no unpaired electrons hence it is **diamagnetic**. **(incorrect statement)**

(4) $MnO_4^{2-} \Rightarrow$ Green
$MnO_4^- \Rightarrow$ purple/violet

Therefore, Option 3 is correct.

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Q. 10 The incorrect statement(s) among (a) - (d) regarding acid rain is (are) :

(a) It can corrode water pipes.

(b) It can damage structures made up of stone.

(c) It cannot cause respiratory ailments in animals

(d) It is not harmful for trees.

Option 1:
(a), (c) and (d)

Option 2:
(c) only

Option 3:
(a), (b) and (d)

Option 4:
(c) and (d)

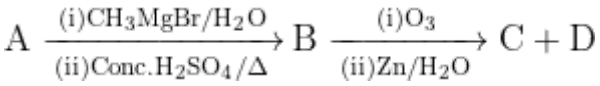
Correct Answer:
(c) and (d)

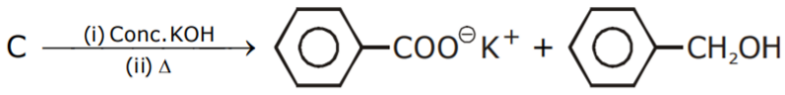
Solution:
(1) Acid rain corrodes water pipes resulting in the leaching of heavy of heavy metals such as iron, lead, and copper into the drinking water.
(2) Acid rain damages buildings and other structures made of stone or metal.
(3) It causes respiratory ailments in human beings and animals.
(4) It is harmful to agriculture, trees, and plants as it washes down the nutrients needed for its growth.

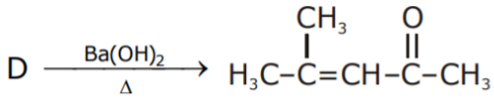
The incorrect statement (c) and (d).

Therefore, Option 4 is correct.

Q. 11 The compound A in the following reactions is :







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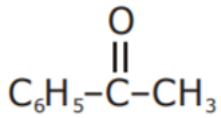
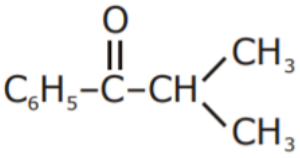
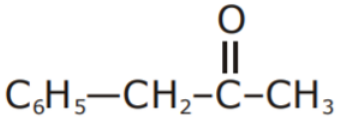
Option 1:



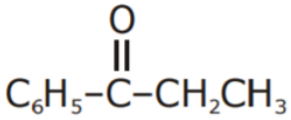
Option 2:



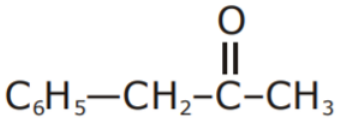
Option 3:



Option 4:

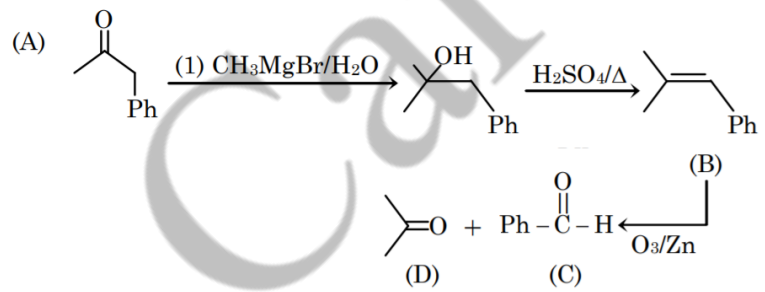


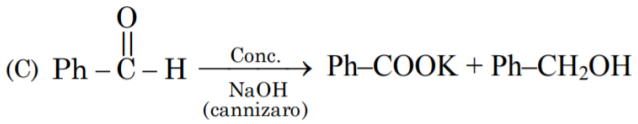
Correct Answer:

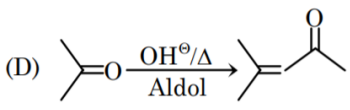


Solution:

The reaction will be -







Therefore, Option 3 is correct.

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Q. 12 The strengths of 5.6 volume hydrogen peroxide (of density 1 g/mL) in terms of mass percentage and molarity (M), respectively, are:

(Take molar mass of hydrogen peroxide as 34 g/mol)

Option 1:
1.7 and 0.5

Option 2:
0.85 and 0.25

Option 3:
1.7 and 0.25

Option 4:
0.85 and 0.5

Correct Answer:
1.7 and 0.5

Solution:
Strength of 5.6 volume means that 5.6 is Volume strength.

Volume strength = 11.2 X molarity

$$\Rightarrow \text{molarity} = \frac{5.6}{11.2} = 0.5$$

Assuming 1 litre solution;
mass of solution = 1000 ml X 1 g/ml = 1000 g
mass of solute = moles X molar mass
= 0.5 mol X 34 g/mol
= 17 gm

$$\Rightarrow \text{mass}\,\% = \frac{17}{1000} \times 100 = 1.7\%$$

Therefore, Option 1 is correct.

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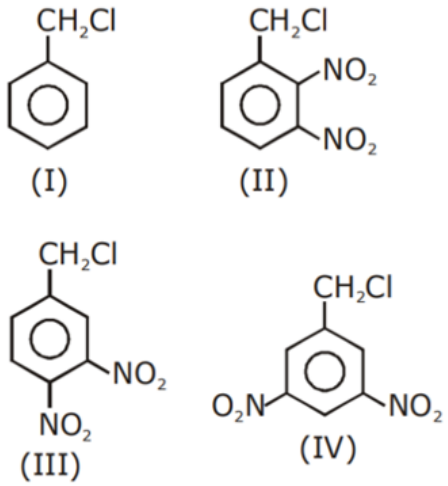
Q. 13 The decreasing order of reactivity of the following compounds towards nucleophilic substitution (S_N2) is :



Option 1:
(II) > (III) > (I) > (IV)

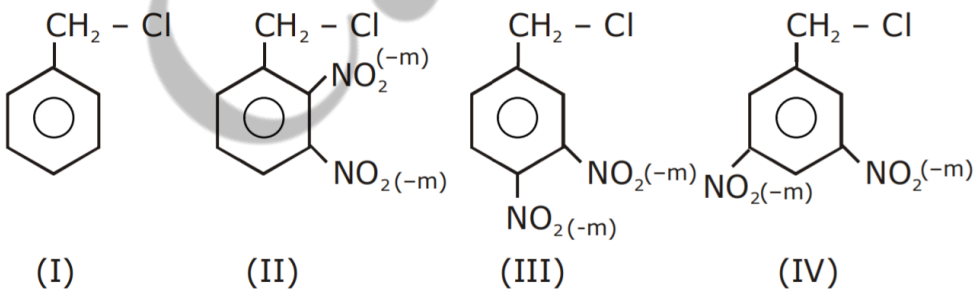
Option 2:
(II) > (III) > (IV) > (I)

Option 3:
(III) > (II) > (IV) > (I)

Option 4:
(IV) > (II) > (III) > (I)

Correct Answer:
(II) > (III) > (IV) > (I)

Solution:



So,

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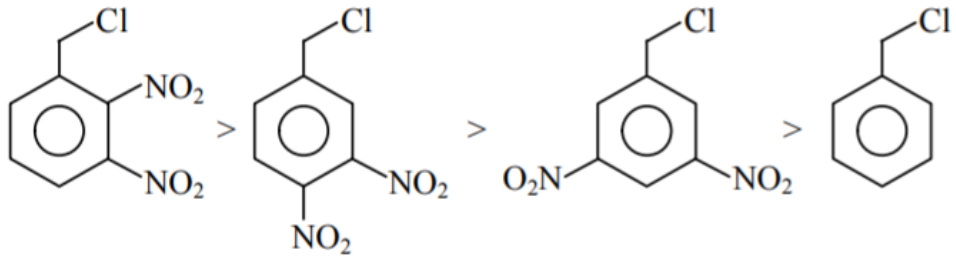

–I & – R | less –I & –R rate of SN^2 reaction | Only 2 – 1

Order will be (II) > (III) > (IV) > (I)

Therefore, Option 2 is correct.

Q. 14 For the reaction $2A + 3B + \frac{3}{2}C \rightarrow 3P$, Which statement is correct ?

Option 1:

$$\frac{dn_A}{dt} = \frac{3}{2}\frac{dn_B}{dt} = \frac{3}{4}\frac{dn_C}{dt}$$

Option 2:

$$\frac{dn_A}{dt} = \frac{dn_B}{dt} = \frac{dn_C}{dt}$$

Option 3:

$$\frac{dn_A}{dt} = \frac{2}{3}\frac{dn_B}{dt} = \frac{4}{3}\frac{dn_C}{dt}$$

Option 4:

$$\frac{dn_A}{dt} = \frac{2}{3}\frac{dn_B}{dt} = \frac{3}{4}\frac{dn_C}{dt}$$

Correct Answer:

$$\frac{dn_A}{dt} = \frac{2}{3}\frac{dn_B}{dt} = \frac{4}{3}\frac{dn_C}{dt}$$

Solution:

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For $aA + bB \rightarrow cC$;

$$\frac{-1}{a}\frac{d[A]}{dt} = \frac{-1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt}$$

$$2A + 3B + \frac{3}{2}C \longrightarrow 3P$$

$$\therefore \quad \frac{-1}{2}\frac{d[A]}{dt} = \frac{-1}{3}\frac{d[B]}{dt} = \frac{-2}{3}\frac{d[C]}{dt} = \frac{1}{3}\frac{d[p]}{dt}$$

$$ROR = \frac{1}{2}\left[\frac{-d[n_A]}{dt}\right] = \frac{1}{3}\left[\frac{-d[n_B]}{dt}\right] = \frac{2}{3}\left[\frac{-d[n_c]}{dt}\right] = \frac{1}{3}\left[\frac{+d[n_p]}{dt}\right]$$

$$\left[\frac{-dn_A}{dt}\right] = \frac{2}{3}\left[\frac{-d[n_B]}{dt}\right] = \frac{4}{3}\left[\frac{-d[n_p]}{dt}\right]$$

Therefore, Option 3 is correct.

Q. 15 The five successive ionization enthalpies of an element are 800, 2427, 3658, 25024, and 32824 kJ/mol. The number of valence electrons in the element is :

Option 1:
5

Option 2:
4

Option 3:
3

Option 4:
2

Correct Answer:
3

Solution:
The fourth & Fifth I.E. are very high (periodic properties) indicating the presence of three valence shell electrons.

So valence electrons are 3.

Therefore, Option 3 is correct.

Q. 16 Complex A has a composition of $H_{12}O_6Cl_3Cr$. If the complex on treatment with $conc.H_2SO_4$ loses $13.5\ ^o/_o$ of its original mass, the correct molecular formula of A is :

[Given : atomic mass of $Cr = 52\ amu$ and $Cl = 35\ amu$]

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Option 1:
$[Cr(H_2O)_6]\,Cl_3$

Option 2:
$[Cr(H_2O)_3Cl_3]\cdot 3H_2O$

Option 3:
$[Cr(H_2O)_5Cl]\,Cl_2\cdot H_2O$

Option 4:
$[Cr(H_2O)_4Cl_2]\,Cl\cdot 2H_2O$

Correct Answer:
$[Cr(H_2O)_4Cl_2]\,Cl\cdot 2H_2O$

Solution:
The complex on treatment with conc.H_2SO_4 loses its mass, which means that water will react with conc.H_2SO_4 and get separate from the complex.

Let x molecule of water are lost then

$$\%\text{ mass of water} = \frac{\text{x}\times 18}{(18+6\times +35\times 3+52)}\times 100$$

$$13.5 = \left[\frac{\text{x}\times 18}{6\times 18+3\times 35+52}\right]\times 100$$

$$\text{x} = \frac{265\times 13.5}{18\times 100}\simeq 2$$

We have to maintain 6 donating ligands inside the complex.

so, complex is $[\text{Cr}\,(\text{H}_2\text{O})_4\,\text{Cl}_2]\,.2\text{H}_2\text{O}$

Therefore, Option 4 is correct.

Q. 17 The d-electron configuration of $[Ru(en)_3]\,Cl_2$ and $[Fe(H_2O)_6]\,Cl_2$, respectively are :

Option 1:
$t_{2g}^6\ \ e_g^0\ and\ t_{2g}^6\ \ e_g^0$

Option 2:
$t_{2g}^4\ \ e_g^2\ and\ t_{2g}^6\ \ e_g^0$

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Option 3:

$t_{2g}^6 \ \ e_g^0 \ and \ t_{2g}^4 \ \ e_g^2$

Option 4:

$t_{2g}^4 \ \ e_g^2 \ and \ t_{2g}^4 \ \ e_g^2$

Correct Answer:

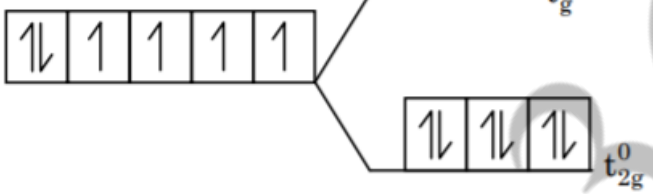
$t_{2g}^6 \ \ e_g^0 \ and \ t_{2g}^4 \ \ e_g^2$

Solution:

$[Ru(en)_3]\ Cl_2$ $\quad$ $R_u \Rightarrow$ 4d series

en $\Rightarrow$ chelating ligand

CN = 6, octahedral splitting $\quad$ hence laye splitting of d-subshell

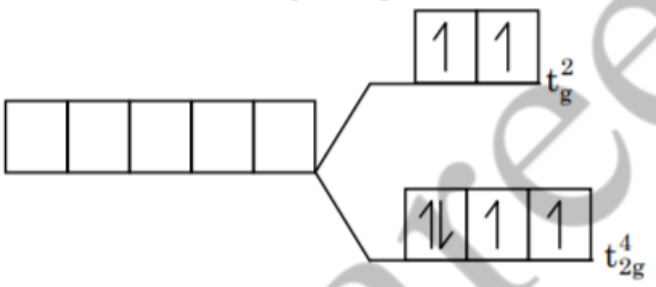
$R_u^{+2} \Rightarrow [Kr]4d^6 5s^0$ 

$[Fe(H_2O)_6]Cl_2 \Rightarrow H_2O \Rightarrow$ Weak filled ligand

$Fe^{+2} \Rightarrow [Ar]\ 3d^6 4s^0$

less plitting

CN = 6 octahedral splitting



Therefore, Option 3 is correct.

Q. 18 Three isomers A, B and C (mol. formula $C_8H_{11}N$) give the following results :

$$\text{A and C} \xrightarrow{\text{Azotization}} \text{P} + \text{Q} \xrightarrow[\text{(ii) Oxidation } (KMnO_4 + H^+)]{\text{(i) Hydrolysis}} \begin{matrix}\text{R(product of A)} \\ + \\ \text{S(product of C)}\end{matrix}$$

R has lower boiling point than S

$\text{B} \xrightarrow{C_6H_5SO_2Cl}$ alkali-insoluble product

A, B and C, respectively are :

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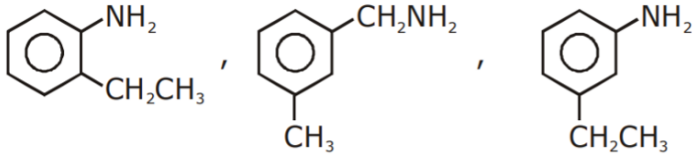
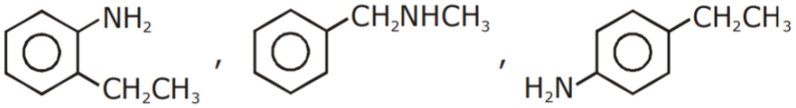
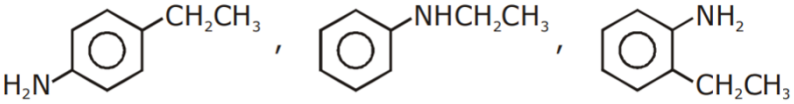
Option 1:



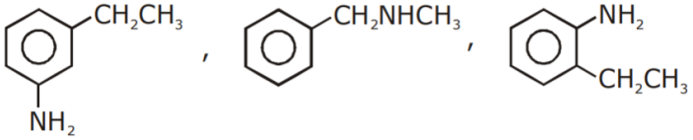
Option 2:



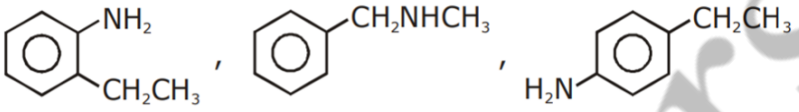
Option 3:



Option 4:

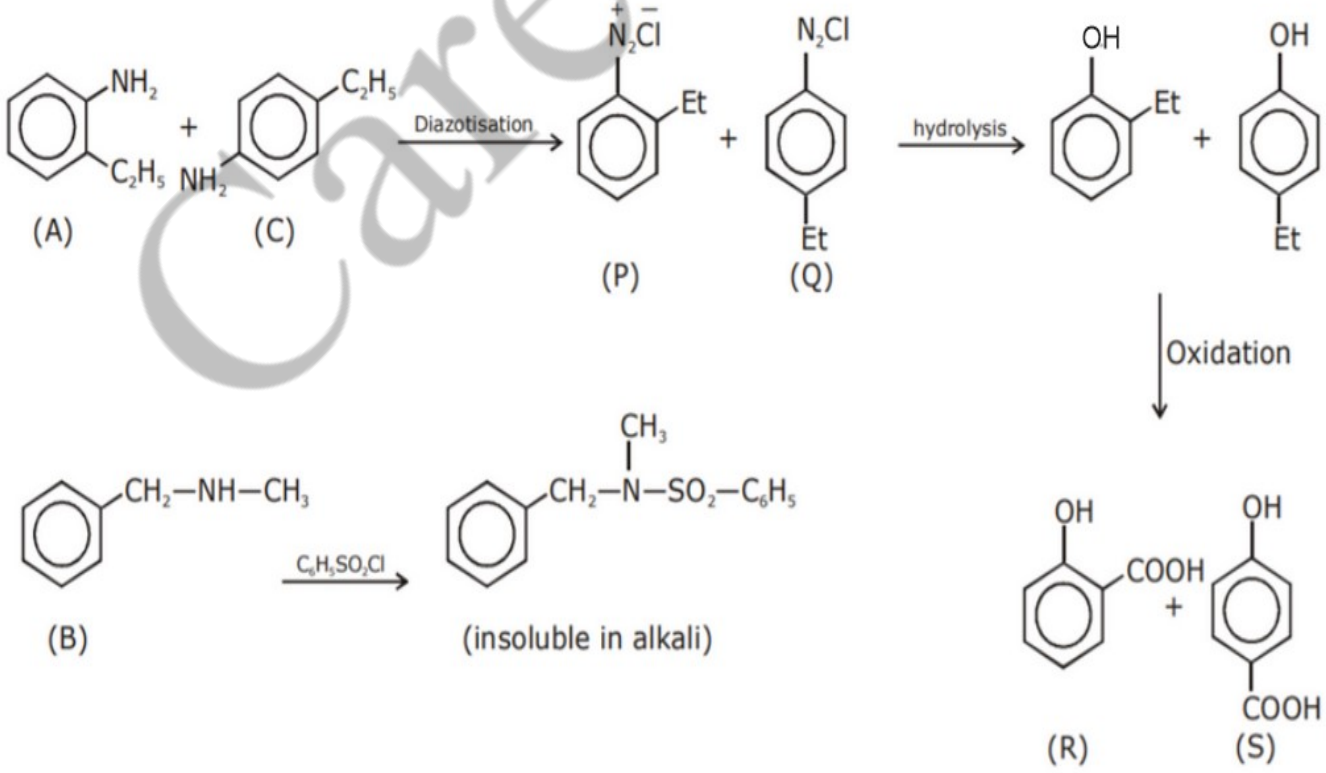


Correct Answer:



Solution:

The reaction will be -



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R (Intramolecular H-bonding) has a lower boiling point than S (Intermolecular H-bonding) because of different types of H-bonding.

Therefore, Option 2 is correct.

Q. 19 Match the following drugs with their therapeutic actions :

(i) Ranitidine	(a) Antidepressant
(ii) Nardil(Phenelzine)	(b) Antibiotic
(iii) Chloramphenicol	(c) Antihistamine
(iv) Dimethane(Brompheniramine)	(d) Antacid
	(e) Analgesic

Option 1:
(i)-(a); (ii)-(c); (iii)-(b); (iv)-(e)

Option 2:
(i)-(d); (ii)-(a); (iii)-(b); (iv)-(c)

Option 3:
(i)-(e); (ii)-(a); (iii)-(c); (iv)-(d)

Option 4:
(i)-(d); (ii)-(c); (iii)-(a); (iv)-(e)

Correct Answer:
(i)-(d); (ii)-(a); (iii)-(b); (iv)-(c)

Solution:
Drugs with their therapeutic actions :

Ranitidine $\longrightarrow$ Antacid
Nardil (Phenelzine) $\longrightarrow$ Antidepressant
Chloramphenicol $\longrightarrow$ Antibiotic
Dimethane (Brompheniramine) $\longrightarrow$ Antihistamine

Therefore, Option 2 is correct.

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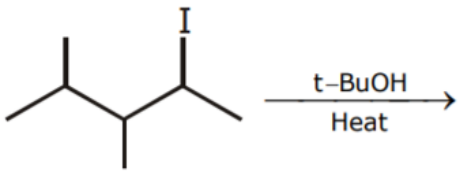
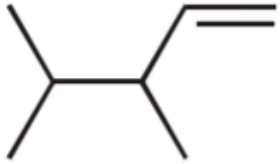
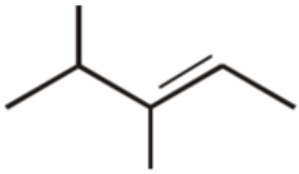
Q. 20 The major product in the following reaction is :



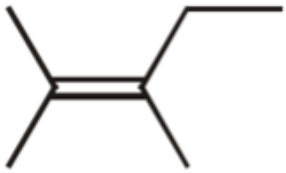
Option 1:



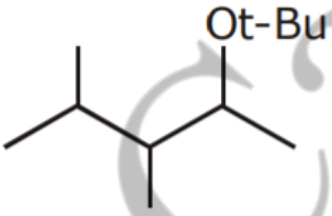
Option 2:



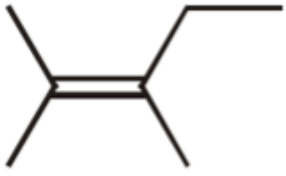
Option 3:



Option 4:



Correct Answer:



Solution:

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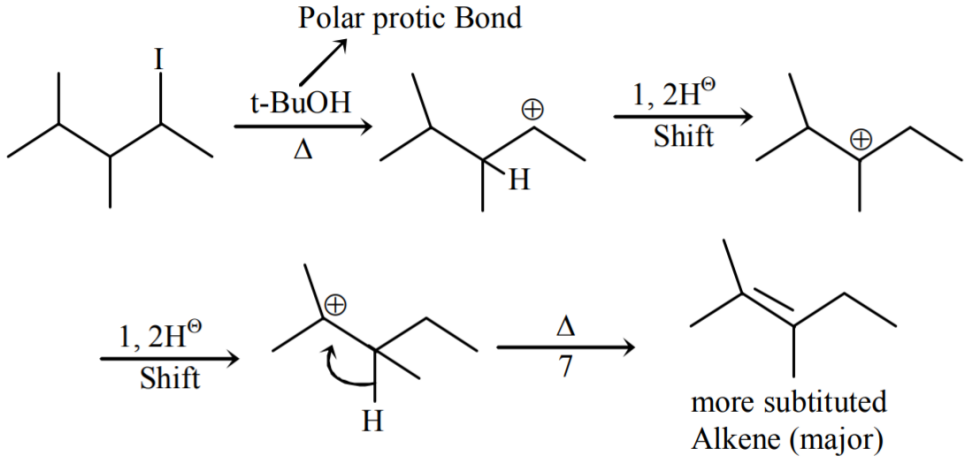
The reaction will be -



Therefore, Option 3 is correct.

Q. 21 If $250\ cm^3$ of an aqueous solution containing $0.73\ g$ of a protein A is isotonic with one litre of another aqueous solution containing $1.65\ g$ of a protein B, AT $298\ K$, the ratio of the molecular masses of A and B is _____$\times 10^{-2}$ (to the nearest integer).

Correct Answer:
177

Solution:
Let molar mass of protein A = x g/mol
Let molar mass of protein B = y g/mol

$$\pi_{\mathrm{A}} = \text{ osmotic pressure of protein A} = \frac{\left(\frac{0.73}{\mathrm{x}}\right)}{0.25}\mathrm{RT}$$

$$\pi_{\mathrm{B}} = \text{ osmotic pressure of protein B} = \frac{\left(\frac{1.65}{\mathrm{y}}\right)}{1}\mathrm{RT}$$

Now, $\pi_{\mathrm{A}} = \pi_{\mathrm{B}}$

$$\left(\frac{0.73}{\mathrm{x} \times 0.25}\right)\mathrm{RT} = \left(\frac{1.65}{\mathrm{y}}\right)\mathrm{RT}$$

$$\left(\frac{\mathrm{x}}{\mathrm{y}}\right) = \frac{0.73}{0.25 \times 1.65} = 1.769 \cong 1.77$$

$$\frac{\mathrm{x}}{\mathrm{y}} = 177 \times 10^{-2}$$

Ans = 177

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Q. 22 An acidic solution of dichromate is electrolyzed for 8 minutes using $2A$ current . As per the following equation $Cr_2O_7^{2-} + 14\ H^+ + 6\ e^- \rightarrow 2\ Cr^{3+} + 7\ H_2O$ The amount of Cr^{3+} obtained was $0.104\ g$. The efficiency of the process(in $^o/_o$) its (Take : $F = 96000\ C$, At. mass of chromium = 52) _____.

Correct Answer:
60

Solution:

$$\left[\text{ moles of } Cr^{3+}\right] \times 3 = \frac{8 \times 60 \times 2}{96000}$$

$$\text{moles of } Cr^{3+} = \frac{8 \times 4}{9600} = \frac{1}{300} \text{ mol}$$

$$\text{mass of } Cr^{3+} = \frac{52}{300} \text{ g}$$

$$\%\text{ efficiency } = \frac{\text{Actual obtained Amt}}{\text{Theo. obtained Amt}} \times 100$$

$$\%\text{ efficiency } = \frac{\frac{0.104}{52}}{300} \times 100$$

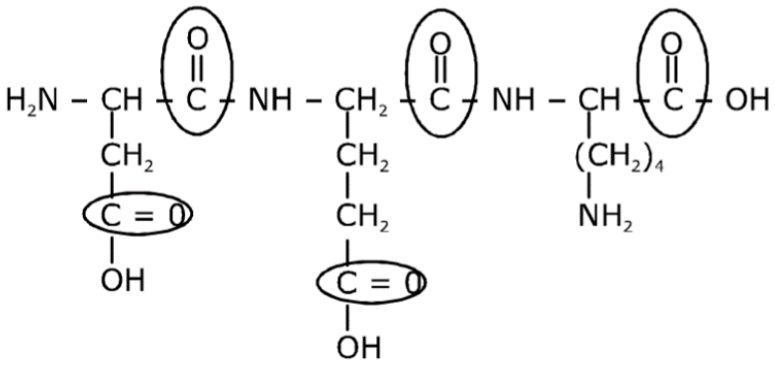
$$\%\text{ efficiency } = 30 \times \frac{104}{52} = 60\%$$

Ans = 60

Q. 23 The number of $>C=O$ groups present in a tripeptide $Asp - Glu - Lys$ is _____.

Correct Answer:
5

Solution:
The number of $>C=O$ groups present in a tripeptide $Asp - Glu - Lys$ is 5.



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Ans = 5

Q. 24 6.023×10^{22} molecules are present in $10\ g$ of a substance $'x'$. The molarity of a solution containing $5\ g$ of substance $'x'$ in $2\ L$ solution is _____$\times 10^{-3}$.

Correct Answer:
25

Solution:

$$\text{moles} = \frac{\text{number of molecules}}{6 \times 10^{23}} = \frac{\text{given mass}}{\text{molar mass}}$$

$$\text{molar mass} = \frac{\text{given mass} \times 6 \times 10^{23}}{\text{number of molecules}}$$

$$\text{molar mass} = \frac{10 \times 6.023 \times 10^{23}}{6.023 \times 10^{22}} = 100\ \text{g/mol}$$

$$\text{molarity} = \frac{\text{moles of solute}}{\text{volume of sol}^{\text{n}}(\ell)} = \frac{(5/100)}{2} = 0.025$$

$$\text{M} = 25 \times 10^{-3}\ \text{mol/lit}$$

Ans = 25

Q. 25 The volume (in mL) of 0.1 N NaOH required to neutralise 10 mL of 0.1 N phosphinic acid is .....

Correct Answer:
10

Solution:

$$\text{H}_3\text{PO}_2 + \text{NaOH} \rightarrow \text{NaH}_2\text{PO}_2 + \text{H}_2\text{O}$$

$$\frac{\text{n}_{\text{H}_3\text{PO}_2}\ \text{reacted}}{1} = \frac{\text{n}_{\text{NaOH}}\ \text{reacted}}{1}$$

$$\frac{0.1 \times 10}{1} = 0.1 \times \text{V}_{\text{NaOH}}$$

$$\text{V}_{\text{NaOH}} = 10\text{ml}$$

Ans = 10

Maths

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Q. 1 Let A be a 3×3 matrix such that

$$adj\ A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$B = adj(adj\ A)$.

If $|A| = \lambda$ and $\left|(B^{-1})^T\right| = \mu$, then the ordered pair,

$(|\lambda|, \mu)$ is equal to :

Option 1:
$\left(3, \frac{1}{81}\right)$

Option 2:
$\left(9, \frac{1}{9}\right)$

Option 3:
$(3, 81)$

Option 4:
$\left(9, \frac{1}{81}\right)$

Correct Answer:
$\left(3, \frac{1}{81}\right)$

Solution:

$$\mathrm{C} = \text{ adj. } \mathrm{A} = \begin{vmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{vmatrix}$$

$$|C| = |\operatorname{adj} A| = 2(0+4) + 1 \cdot (1-2) + 1 \cdot (2-0) = 9$$

$$|\operatorname{adj} \mathrm{A}| = |\mathrm{A}|^{n-1} = |\mathrm{A}|^2 = 9 = 9$$
$$\lambda = |\mathrm{A}| = \pm 3$$
$$|\lambda| = 3$$

B = adj (adj A) = adj C

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$$l|\mathrm{B}| = |\operatorname{adj}\mathrm{C}| = |\mathrm{C}|^2 = 81$$
$$\left|\left(\mathrm{B}^{-1}\right)^{\mathrm{T}}\right| = |\mathrm{B}|^{-1} = \frac{1}{81}$$
$$(|\lambda|, \mu) = \left(3, \frac{1}{81}\right)$$

Q. 2 The set of all real values of λ for which the quadratic equations,

$\left(\lambda^2+1\right)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval $(0, 1)$ is :

Option 1:
$(0, 2)$

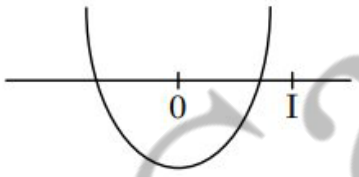
Option 2:
$(2, 4)$

Option 3:
$(1, 3]$

Option 4:
$(-3, -1)$

Correct Answer:
$(1, 3]$

Solution:
If exactly one root in (0, 1) then



$\Rightarrow \mathrm{f}(0) \cdot \mathrm{f}(1) < 0$
$\Rightarrow 2\left(\lambda^2 - 4\lambda + 3\right) < 0$
$\Rightarrow 1 < \lambda < 3$

Now for $\lambda = 1$

$2x^2 - 4x + 2 = 0$
$(x-1)^2 = 0$
$x = 1, 1$

So both roots doesn't lie between (0, 1)

$\therefore \lambda \neq 1$

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Again for $\lambda = 3$

$10x^2 - 12x + 2 = 0$

$\Rightarrow x = 1, \frac{1}{5}$

So if one root is 1 then second root lie between (0, 1) so $\lambda = 3$

$\therefore \lambda \in (1, 3]$

Q. 3 If $\int \sin^{-1}\left(\sqrt{\frac{x}{1+x}}\right) dx = A(x)\tan^{-1}(\sqrt{x}) + B(x) + C$, where C is a constant of integration, then the ordered pair $(A(x),\ B(x))$ can be :

Option 1:
$(x+1,\ -\sqrt{x})$

Option 2:
$(x+1,\ \sqrt{x})$

Option 3:
$(x-1,\ -\sqrt{x})$

Option 4:
$(x-1,\ \sqrt{x})$

Correct Answer:
$(x+1,\ -\sqrt{x})$

Solution:

$\int \sin^{-1}\left(\sqrt{\frac{x}{1+x}}\right) dx = A(x)\tan^{-1}(\sqrt{x}) + B(x) + C,$

Put $x = \tan^2\theta \Rightarrow dx = 2\tan\theta\sec^2\theta d\theta$

$\int \theta \cdot (2\tan\theta \cdot \sec^2\theta)\, d\theta$

Now do the integration by parts

$\theta \rightarrow$ 1st function

$2\tan\theta\sec^2\theta \rightarrow$ 2nd function

$= \theta \cdot \tan^2\theta - \int \tan^2\theta d\theta$

$= \theta \cdot \tan^2\theta - \int (\sec^2\theta - 1)\, d\theta$

$= \theta\,(1 + \tan^2\theta) - \tan\theta + C$

$= \tan^{-1}(\sqrt{x})(1 + x) - \sqrt{x} + C$

$A(x) = 1 + x, \quad B(x) = -\sqrt{x}$

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Q. 4 If the term independent of x in the expansion of $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is k, then $18\ k$ is equal to :

Option 1:
5

Option 2:
9

Option 3:
7

Option 4:
11

Correct Answer:
7

Solution:
General term in this expansion is

$$T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(-\frac{1}{3x}\right)^r$$

$$T_{r+1} = {}^9C_r \left(\frac{3}{2}\right)^{9-r} \left(-\frac{1}{3}\right)^r x^{18-3r}$$

For independent of x

18 - 3r = 0

r = 6

$$\therefore \quad T_7 = {}^9C_6 \left(\frac{3}{2}\right)^3 \left(-\frac{1}{3}\right)^6 = \frac{21}{54} = k$$

$$\therefore \quad 18k = \frac{21}{54} \times 18 = 7$$

Q. 5 Let $x_i (1 \le i \le 10)$ be ten observations of a random variable X. If $\sum_{i=1}^{10} (x_i - p) = 3$ and $\sum_{i=1}^{10} (x_i - p)^2 = 9$ where $0 \ne p \in \mathbb{R}$, then the standard deviation of these observations is :

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Option 1:
$\sqrt{\frac{3}{5}}$

Option 2:
$\frac{4}{5}$

Option 3:
$\frac{9}{10}$

Option 4:
$\frac{7}{10}$

Correct Answer:
$\frac{9}{10}$

Solution:

$$\text{Variance } = \frac{\Sigma\left(x_i - p\right)^2}{n} - \left(\frac{\Sigma\left(x_i - p\right)}{n}\right)^2$$

$$= \frac{9}{10} - \left(\frac{3}{10}\right)^2 = \frac{81}{100}$$

$$\text{S.D.} = \frac{9}{10}$$

Q. 6 If $x^3dy + xy\ dx = x^2dy + 2y\ dx;\ y(2) = e$ and $x > 1$, then $y(4)$ is equal to :

Option 1:
$\frac{3}{2} + \sqrt{e}$

Option 2:
$\frac{3}{2}\sqrt{e}$

Option 3:
$\frac{1}{2} + \sqrt{e}$

Option 4:
$\frac{\sqrt{e}}{2}$

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Correct Answer:

$\frac{3}{2}\sqrt{e}$

Solution:

$x^3dy + xydx = x^2dy + 2ydx$

$\Rightarrow dy\left(x^3 - x^2\right) = dx(2y - xy)$

$\Rightarrow -\int \frac{1}{y}dx = \int \frac{x-2}{x^2(x-1)}dx$

$\Rightarrow -\ell ny = \int \left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{(x-1)}\right)dx$

Where A = 1, B = + 2, C = –1

$\Rightarrow -\ell ny = \ell nx - \frac{2}{x} - \ln(x-1) + \lambda$

$\Rightarrow y(2) = e$

$\Rightarrow -1 = \ln 2 - 1 - 0 + \lambda$

$\therefore \lambda = -\ln 2$

$\Rightarrow \ell \mathrm{ny} = -\ell \mathrm{nx} + \frac{2}{\mathrm{x}} + \ln(\mathrm{x}-1) + \ln 2$

Now put $x = 4$ in equation

$\Rightarrow \ell \mathrm{ny} = -\ln 4 + \frac{1}{2} + \ln 3 + \ln 2$

$\Rightarrow \quad \ell \mathrm{ny} = \ln\left(\frac{3}{2}\right) + \frac{1}{2}\ell \mathrm{ne}$

$\Rightarrow \mathrm{y} = \frac{3}{2}\sqrt{\mathrm{e}}$

Q. 7 If z_1, z_2 are complex numbers such that $Re(z_1) = |z_1 - 1|, \quad Re(z_2) = |z_2 - 1|$ and $arg(z_1 - z_2) = \frac{\pi}{6}$, then $Im(z_1 + z_2)$ is equal to :

Option 1:

$\frac{2}{\sqrt{3}}$

Option 2:

$2\sqrt{3}$

Option 3:

$\frac{\sqrt{3}}{2}$

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Option 4:

$\frac{1}{\sqrt{3}}$

Correct Answer:

$2\sqrt{3}$

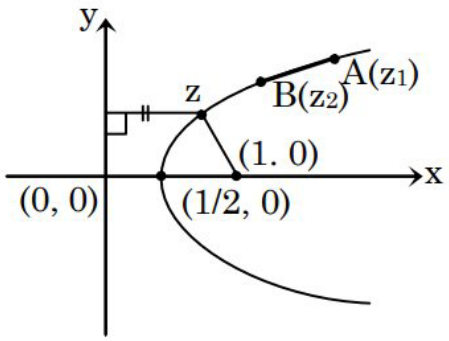
Solution:

Re(z) = |z – 1|

$$\Rightarrow x = \sqrt{(x-1)^2 + (y-0)^2} \quad (x > 0)$$
$$\Rightarrow y^2 = 2x - 1 = 4 \times \frac{1}{2}\left(x - \frac{1}{2}\right)$$

a parabola with focus (1, 0) & directrix as imaginary axis.

$$\therefore \quad \text{Vertex} = \left(\frac{1}{2}, 0\right)$$



$\text{A}(z_1)$ & $\text{B}(z_2)$ are two points on it such that slope of $\text{AB} = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$

$\left(\arg(z_1 - z_2) = \frac{\pi}{6}\right)$

for $y^2 = 4ax$

Let $A\left(at_1^2, 2at_1\right)$ & $B\left(at_2^2, 2at_2\right)$

$$m_{AB} = \frac{2}{t_1 + t_2} = \frac{4a}{y_1 + y_2} = \frac{1}{\sqrt{3}}$$

$\left(\text{Here } a = \frac{1}{2}\right)$

$$\Rightarrow y_1 + y_2 = 4a\sqrt{3} = 2\sqrt{3}$$

Q. 8 $\lim\limits_{x \to a} \frac{(a+2x)^{1/3} - (3x)^{1/3}}{(3a+x)^{1/3} - (4x)^{1/3}} (a \neq 0)$ is equal to :

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Option 1:

$\left(\frac{2}{3}\right)^{\frac{4}{3}}$

Option 2:

$\left(\frac{2}{3}\right)\left(\frac{2}{9}\right)^{\frac{1}{3}}$

Option 3:

$\left(\frac{2}{9}\right)^{\frac{4}{3}}$

Option 4:

$\left(\frac{2}{9}\right)\left(\frac{2}{3}\right)^{\frac{1}{3}}$

Correct Answer:

$\left(\frac{2}{3}\right)\left(\frac{2}{9}\right)^{\frac{1}{3}}$

Solution:

$$L=\lim_{h\to 0}\frac{(a+2(a+h))^{1/3}-(3(a+h))^{1/3}}{(3a+a+h)^{1/3}-(4(a+h))^{1/3}}$$

$$=\lim_{h\to 0}\frac{(3a)^{1/3}\left(1+\frac{2h}{3a}\right)^{1/3}-(3a)^{1/3}\left(1+\frac{h}{a}\right)^{1/3}}{(4a)^{1/3}\left(1+\frac{h}{4a}\right)^{1/3}-(4a)^{1/3}\left(1+\frac{h}{a}\right)^{1/3}}$$

$$=\lim_{h\to 0}\left(\frac{3^{1/3}}{4^{1/3}}\right)\left[\frac{\left(1+\frac{2h}{9a}\right)-\left(1+\frac{h}{3a}\right)}{\left(1+\frac{h}{12a}\right)-\left(1+\frac{h}{3a}\right)}\right]$$

$$=\left(\frac{3}{4}\right)^{1/3}\frac{\left(\frac{2}{9}-\frac{1}{3}\right)}{\left(\frac{1}{12}-\frac{1}{3}\right)}=\left(\frac{3}{4}\right)^{1/3}\left(\frac{8-12}{3-12}\right)$$

$$=\left(\frac{3}{4}\right)^{1/3}\left(\frac{-4}{-9}\right)=\frac{4^{1-\frac{1}{3}}}{3^{2-\frac{1}{3}}}=\frac{4^{2/3}}{3^{5/3}}$$

$$L=\frac{(8\times 2)^{1/3}}{(27\times 9)^{1/3}}=\frac{2}{3}\left(\frac{2}{9}\right)^{1/3}$$

Q. 9 Let p, q, r be three statements such that the truth value of $(p\wedge q)\to(\sim q\vee r)$ is F. Then the truth values of p, q, r are respectively :

Option 1:

T, F, T

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Option 2:
T, T, T

Option 3:
F, T, F

Option 4:
T, T, F

Correct Answer:
T, T, F

Solution:
$(p \wedge q) \rightarrow (\sim q \vee r) = \text{ false}$
when $(p \wedge q) = T$
and $\quad (\sim q \vee r) = F$

So $(p \wedge q) = T$ is possible when $p = q = \text{ true}$
$\therefore \sim q = \text{ False } (q = \text{ true })$

So $(\sim q \vee r) = \text{ False is possible if } r \text{ is false}$
$\therefore \quad \mathrm{p = T, q = T, r = F}$

Q. 10 If the sum of the series $20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} + ...upto\ n^{th}$ term is 488 and the n^{th} term is negative, then :

Option 1:
$n = 60$

Option 2:
n^{th} term is -4

Option 3:
$n = 41$

Option 4:
n^{th} term is $-4\frac{2}{5}$

Correct Answer:
n^{th} term is -4

Solution:
Sum of n terms of an AP

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$S_n = \frac{n}{2}[2a + (n-1)d]$
OR
$S_n = \frac{n}{2}[a + l]$
$a \rightarrow$ first term
$d \rightarrow$ common difference
$n \rightarrow$ number of terms

In this Question,

$$S = \frac{100}{5} + \frac{98}{5} + \frac{96}{5} + \frac{94}{5} + \ldots . \text{nterms}$$

$$S_n = \frac{n}{2}\left(2 \times \frac{100}{5} + (n-1)\left(\frac{-2}{5}\right)\right) = 488$$

$n(100 - n + 1) = 488 \times 5$
$n^2 - 101n + 488 \times 5 = 0$
$n = 61, 40$

n = 40 is rejected as 40th term is positive for this series. So, n = 61

$T_n = a + (n-1)d = \frac{100}{5} - \frac{2}{5} \times 60$
$= 20 - 24 = -4$

Q. 11

Let e_1 and e_2 be the eccentricities of the ellipse, $\frac{x^2}{25} + \frac{y^2}{b^2} = 1 (b < 5)$ and the hyperbola, $\frac{x^2}{16} - \frac{y^2}{b^2} = 1$ respectively satisfying $e_1 e_2 = 1$. If α and β are the distances between the foci of the ellipse and the foci of the hyperbola respectively, then the ordered pair (α, β) is equal to :

Option 1:
$(8, 12)$

Option 2:
$\left(\frac{20}{3}, 12\right)$

Option 3:
$\left(\frac{24}{5}, 10\right)$

Option 4:
$(8, 10)$

Correct Answer:
$(8, 10)$

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Solution:

For ellipse $\frac{x^2}{25}+\frac{y^2}{b^2}=1 \quad (b<5)$

Let e_1 is eccentricity of ellipse

$\therefore \quad b^2=25\left(1-e_1^2\right) \ldots\ldots(1)$

For Hyperbola

$\frac{x^2}{16}-\frac{y^2}{b^2}=1$

Let e_2 is eccentricity of hyperbola.

$\therefore \quad b^2=16\left(e_2^2-1\right) \quad \ldots\ldots(2)$

From (1) and (2)

$25\left(1-e_1^2\right)=16\left(e_2^2-1\right)$

Now $e_1 \cdot e_2=1 \quad$ (given)

$\therefore \quad 25\left(1-e_1^2\right)=16\left(\frac{1-e_1^2}{e_1^2}\right)$

or

$e_1=\frac{4}{5} \quad \therefore e_2=\frac{5}{4}$

distance beween foci = 2ae

$\therefore$ distance for ellipse $=2 \times 5 \times \frac{4}{5}=8=\alpha$

distance for hyperbola $=2 \times 4 \times \frac{5}{4}=10=\beta$

$\therefore (\alpha, \beta) \equiv (8,10)$

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Q. 12 If the value of the integral $\int_0^{\frac{1}{2}} \frac{x^2}{(1-x^2)^{\frac{3}{2}}} dx$ is $\frac{k}{6}$, then k is equal to :

Option 1:

$2\sqrt{3}-\pi$

Option 2:

$2\sqrt{3}+\pi$

Option 3:

$3\sqrt{2}+\pi$

Option 4:

$3\sqrt{2}-\pi$

Correct Answer:

$2\sqrt{3}-\pi$

Solution:

$$\int_0^{1/2}\frac{((x^2-1)+1)}{(1-x^2)^{3/2}}dx$$

$$\int_0^{1/2}\frac{dx}{(1-x^2)^{3/2}}-\int_0^{1/2}\frac{dx}{\sqrt{1-x^2}}$$

$$\int_0^{1/2}\frac{x^{-3}}{(x^{-2}-1)^{3/2}}dx-\left(\sin^{-1}x\right)_0^{1/2}$$

$$\text{Let } x^{-2}-1=t^2\Rightarrow x^{-3}dx=-tdt$$

$$\int_\infty^{\sqrt{3}}\frac{-tdt}{t^3}-\frac{\pi}{6}=\int_{\sqrt{3}}^{\infty}\frac{dt}{t^2}-\frac{\pi}{6}=\frac{1}{\sqrt{3}}-\frac{\pi}{6}=\frac{k}{6}$$

$$k=2\sqrt{3}-\pi$$

Q. 13 Suppose $f(x)$ is a polynomial of degree four, having critical points at $-1,0,1$. If $T=\{x\,\epsilon\,R\,|\,\backslash right f(x)=f(0)\}$, then the sum of squares of all the elements of T is :

Option 1:

4

Option 2:

6

Option 3:

2

Option 4:

8

Correct Answer:

4

Solution:

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$f'(x) = x(x+1)(x-1) = x^3 - x$

$\int df(x) = \int x^3 - x dx$

$f(x) = \frac{x^4}{4} - \frac{x^2}{2} + C$

$f(x) = f(0)$

$\frac{x^4}{4} - \frac{x^2}{2} = 0$

$x^2\left(x^2 - 2\right) = 0$

$x = 0, 0, \sqrt{2}, -\sqrt{2}$

$x_1^2 + x_2^2 + x_3^2 = 0 + 2 + 2 = 4$

Q. 14 If the surface area of a cube is increasing at a rate of $3.6\ cm^2/sec$, retaining its shape ; then the rate of change of its volume (in cm^2/sec) , when the length of a side of the cube is $10\ cm$, is :

Option 1:

18

Option 2:

10

Option 3:

20

Option 4:

9

Correct Answer:

9

Solution:

$\frac{d}{dt}\left(6a^2\right) = 3.6 \Rightarrow 12a\frac{da}{dt} = 3.6$

$\text{a}\ \frac{da}{dt} = 0.3$

$\frac{dv}{dt} = \frac{d}{dt}\left(a^3\right) = 3a\left(a\frac{da}{dt}\right)$

$= 3 \times 10 \times 0.3 = 9$

Q. 15 Let $a, b, c \in R$ be such that $a^2 + b^2 + c^2 = 1$. If $a\cos\theta = b\cos\left(\theta + \frac{2\pi}{3}\right) = c\cos\left(\theta + \frac{4\pi}{3}\right)$,

Where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ is :

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Option 1:
$\frac{\pi}{2}$

Option 2:
$\frac{2\pi}{3}$

Option 3:
$\frac{\pi}{9}$

Option 4:
0

Correct Answer:
$\frac{\pi}{2}$

Solution:

$$\cos\phi = \frac{\overline{p}\cdot\overline{q}}{|\overline{p}||\overline{q}|} = \frac{ab+bc+ca}{a^2+b^2+c^2} = \frac{\Sigma ab}{1}$$
$$= abc\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$$
$$= \frac{abc}{\lambda}\left(\cos\theta + \cos\left(\theta+\frac{2\pi}{3}\right)+\cos\left(\theta+\frac{4\pi}{3}\right)\right)$$
$$= \frac{abc}{\lambda}\left(\cos +2\cos(\theta+\pi)\cos\frac{\pi}{3}\right)$$
$$= \frac{abc}{\lambda}(\cos\theta-\cos\theta) = 0$$
$$\phi = \frac{\pi}{2}$$

Q. 16 The probability that a randomly chosen $5-digit$ number is made from exactly two digits is :

Option 1:
$\frac{135}{10^4}$

Option 2:
$\frac{121}{10^4}$

Option 3:
$\frac{150}{10^4}$

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Option 4:

$\frac{134}{10^4}$

Correct Answer:

$\frac{135}{10^4}$

Solution:

First Case : Choose two non-zero digits ${}^9C_2$

Now, number of 5 -digit numbers containing both digits $= 2^5 - 2$

Second Case : Choose one non-zero and one zero as digit ${}^9C_1$.

Number of 5 -digit numbers containing one non zero and one zero both $= (2^4 - 1)$

Required prob.

$$= \frac{({}^9C_2 \times (2^5 - 2) + {}^9C_1 \times (2^4 - 1))}{9 \times 10^4}$$

$$= \frac{36 \times (32 - 2) + 9 \times (16 - 1)}{9 \times 10^4}$$

$$= \frac{4 \times 30 + 15}{10^4} = \frac{135}{10^4}$$

Q. 17 If a ΔABC has vertices $A(-1, 7),\ B(-7, 1)$ and $C(5, -5)$, then its orthocentre has coordinates :

Option 1:

$\left(-\frac{3}{5}, \frac{3}{5}\right)$

Option 2:

$(-3, 3)$

Option 3:

$\left(\frac{3}{5}, -\frac{3}{5}\right)$

Option 4:

$(3, -3)$

Correct Answer:

$(-3, 3)$

Solution:

Let orthocentre is $H(x_0, y_0)$

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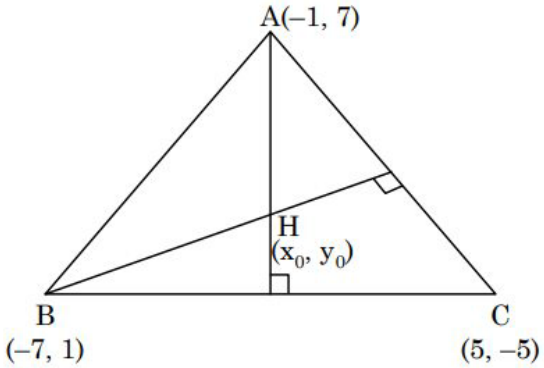

$$m_{AH} \cdot m_{BC} = -1$$

$$\Rightarrow \left(\frac{y_0 - 7}{x_0 + 1}\right)\left(\frac{1+5}{-7-5}\right) = -1$$

$$\Rightarrow 2x_0 - y_0 + 9 = 0 \qquad \ldots(1)$$

$$\text{and } m_{BH} \cdot m_{AC} = -1$$

$$\Rightarrow \left(\frac{y_0 - 1}{x_0 + 7}\right)\left(\frac{7-(-5)}{-1-5}\right) = -1$$

$$\Rightarrow x_0 - 2y_0 + 9 = 0 \qquad \ldots(2)$$

Solving Rq (1) and (2)

we get

$$(x_0, y_0) \equiv (-3, 3)$$

Q. 18 Let R_1 and R_2 be two relations defined as follows :

$R_1 = \left[(a, b) \in R^2; a^2 + b^2 \in Q\right]$ and

$R_2 = \left[(a, b) \in R^2 : a^2 + b^2 \notin Q\right]$ where Q is the set of all rational numbers. Then :

Option 1:
Neither R_1 nor R_2 is transitive.

Option 2:
R_2 is transitive but R_1 is not transitive.

Option 3:
R_1 is transitive but R_2 is not transitive.

Option 4:
R_1 and R_2 are both transitive.

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Correct Answer:
Neither R_1 nor R_2 is transitive.

Solution:
Let $a^2+b^2 \in Q \& b^2+c^2 \in Q$
eg. $a=2+\sqrt{3} \& b=2-\sqrt{3}$
$a^2+b^2=14 \in Q$
Let $c=(1+2\sqrt{3})$
$b^2+c^2=20 \in Q$

But $a^2+c^2=(2+\sqrt{3})^2+(1+2\sqrt{3})^2 \notin Q$
for R_2 Let $a^2=1, b^2=\sqrt{3} \& c^2=2$
$a^2+b^2 \notin Q \& b^2+c^2 \notin Q$
But $a^2+c^2 \in Q$

Q. 19 The plane which bisects the line joining the points $(4,-2,3)$ and $(2,4,-1)$ at right angles also passes through the point :

Option 1:
$(4,0,1)$

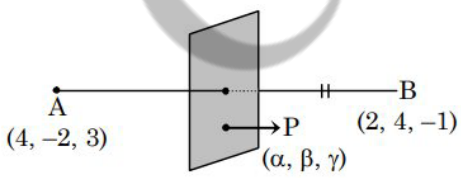
Option 2:
$(0,-1,1)$

Option 3:
$(4,0,-1)$

Option 4:
$(0,1,-1)$

Correct Answer:
$(4,0,-1)$

Solution:



$$PA = PB$$
$$\Rightarrow PA^2 = PB^2$$
$$\Rightarrow (\alpha-4)^2+(\beta+2)^2+(\gamma+3)^2 = (\alpha-2)^2+(\beta-4)^2+(\gamma+1)^2$$
$$\Rightarrow -4\alpha+12\beta-8\gamma=-8$$
$$\Rightarrow 2x-6y+4z=4$$

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(4,0,-1) lies in this plane

Q. 20 Let the latus rectum of the parabola $y^2 = 4x$ be the common chord to the circles C_1 and C_2 each of them having radius $2\sqrt{5}$. Then, the distance between the centres of the circles C_1 and C_2 is :

Option 1:
$8\sqrt{5}$

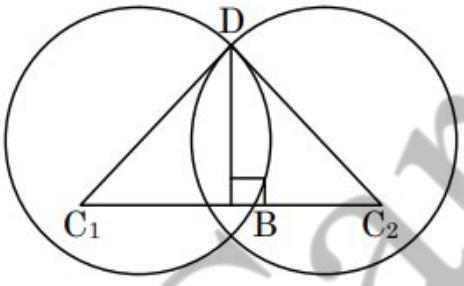
Option 2:
8

Option 3:
12

Option 4:
$4\sqrt{5}$

Correct Answer:
8

Solution:
Length of latus rectum = 4



$$DB = 2$$
$$C_1B = \sqrt{(C_1D)^2 - (DB)^2} = 4$$
$$C_1C_2 = 8$$

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Q. 21 The total number of 3-digit numbers, whose sum of digits is 10, is _____.

Correct Answer:
54

Solution:
Let three-digit number is xyz

$x + y + z = 10; x \geq 1, y \geq 0, z \geq 0 \quad \ldots(1)$

Let $\mathrm{T} = \mathrm{x} - 1 \Rightarrow \mathrm{x} = \mathrm{T} + 1$ where $\mathrm{T} \geq 0$

Put in (1)

$\mathrm{T} + \mathrm{y} + \mathrm{z} = 9;\ 0 \leq \mathrm{T} \leq 8, 0 \leq \mathrm{y}, \mathrm{z} \leq 9$

No. of non negative integral solution
$= {}^{9+3-1}\mathrm{C}_{3-1} - 1 \quad (\text{when } \mathrm{T} = 9) = 55 - 1 = 54$

Q. 22 Let a plane P contains two lines $\vec{r} = \hat{i} + \lambda(\hat{i} + \hat{j}), \lambda \in R$ and $\vec{r} = -\hat{j} + \mu(\hat{j} - \hat{k}), \mu \in R$. If $Q(\alpha, \beta, \gamma)$ is the foot of the perpendicular drawn from the point $M(1, 0, 1)$ to P, then $3(\alpha + \beta + \gamma)$ equals _____.

Correct Answer:
5

Solution:
Sol. Dr's normal to plane

$$= \begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} = -\hat{i} + \hat{j} + \hat{k}$$

Equation of plane

$-1(x-1) + 1(y-0) + 1(z-0) = 0$
$x - y - z - 1 = 0 \quad \ldots(1)$

$$\text{Now } \frac{\alpha - 1}{1} = \frac{\beta - 0}{-1} = \frac{\gamma - 1}{-1} = \frac{(1 - 0 - 1 - 1)}{3}$$
$$\frac{\alpha - 1}{1} = \frac{\beta}{-1} = \frac{\gamma - 1}{-1} = \frac{1}{3}$$
$$\alpha = \frac{4}{3}, \beta = -\frac{1}{3}, \gamma = \frac{2}{3}$$
$$3(\alpha + \beta + \gamma) = 3\left(\frac{4}{3} - \frac{1}{3} + \frac{2}{3}\right) = 5$$

Q. 23 If the tangent to the curve, $y = e^x$ at a point (c, e^c) and the normal to the parabola, $y^2 = 4x$ at the point $(1, 2)$ intersect at the same point on the x-axis, then the value of c is _____.

Option 1:
2

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Option 2:
4

Option 3:
-2

Option 4:
-4

Correct Answer:
4

Solution:

$y = e^x \Rightarrow \frac{dy}{dx} = e^x$

$m = \left(\frac{dy}{dx}\right)_{(c,e^c)} = e^c$

$\text{Tangent at } (c, e^c)$

$y - e^c = e^c(x - c)$

it intersects x-axis

$\text{Put} \quad y = 0 \Rightarrow x = c - 1$

$\text{Now } y^2 = 4x \Rightarrow \frac{dy}{dx} = \frac{2}{y} \Rightarrow \left(\frac{dy}{dx}\right)_{(1,2)} = 1$

$\Rightarrow \text{ Slope of normal } = -1$

Equation of normal y – 2 = – 1(x – 1)

x + y = 3 it intersect x-axis

$\text{Put } y = 0 \Rightarrow x = 3$

Points are same

$\Rightarrow x = c - 1 = 3$

$\Rightarrow c = 4$

Q. 24 If m arithmatic means $(A.Ms)$ and three geometric means $(G.Ms)$ are inserted between 3 and 243 such that 4^{th} A.M. is equal to 2^{nd} G.M., then m is equal to _____.

Correct Answer:
39

Solution:
Insertion of n-Arithmetic Mean Between a and b

Let $A_1, A_2, A_3...., A_n$ be n arithmetic mean between two numbers a and b. Then,
$a, A_1, A_2, A_3...., A_n, b$ is an A.P. Let d be the common difference of this A.P. Clearly, this A.P. contains n + 2 terms.

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$\therefore b = (n+2)^{th}$ term

$\Rightarrow b = a + ((n+2)-1)d = a + (n+1)d$

$\Rightarrow d = \dfrac{b-a}{n+1}$

now,

$A_1 = a + d = \left(a + \dfrac{b-a}{n+1}\right)$

$A_2 = a + 2d = \left(a + \dfrac{2(b-a)}{n+1}\right)$

...

...

$A_n = a + nd = \left(a + n\left(\dfrac{b-a}{n+1}\right)\right)$

Insertion of n-Geometric Mean Between a and b

Let $G_1, G_2, G_3...., G_n$ be n geometric mean between two numbers a and b. Then, $a, G_1, G_2, G_3...., G_n, b$ is an G.P. Clearly, this G.P. contains n + 2 terms.

now, $b = (n+2)^{th}$ term $= ar^{n+2-1}$

$\therefore r = \left(\dfrac{b}{a}\right)^{\frac{1}{n+1}}$

[where, r = common ratio]

$\therefore G_1 = ar,\ G_2 = ar^2,\ G_3 = ar^3, ..., Ga_n = ar^n$

$\Rightarrow G_1 = a\left(\dfrac{b}{a}\right)^{\frac{1}{n+1}},\ G_2 = a\left(\dfrac{b}{a}\right)^{\frac{2}{n+1}},\ G_3 = a\left(\dfrac{b}{a}\right)^{\frac{3}{n+1}}$ .....

$G_n = a\left(\dfrac{b}{a}\right)^{\frac{n}{n+1}}$

-

$3,\ A_1,\ A_2 \ldots\ldots A_m, 243$

$d = \dfrac{243-3}{m+1} = \dfrac{240}{m+1}$

Now $3, G_1, G_2, G_3, 243$

$r = \left(\dfrac{243}{3}\right)^{\frac{1}{3+1}} = 3$

$\therefore\ A_4 = G_2$

$\Rightarrow\ a + 4d = ar^2$

$3 + 4\left(\dfrac{240}{m+1}\right) = 3(3)^2$

$m = 39$

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Q. 25 Let S be the set of all integer solutions, (x, y, z), of the system of equations

$$x - 2y + 5z = 0$$

$$-2x + 4y + z = 0$$

$$-7x + 14y + 9z = 0$$

such that $15 \leq x^2 + y^2 + z^2 \leq 150$. Then, the number of elements in the set S is equal to _____.

Option 1:
6

Option 2:
4

Option 3:
10

Option 4:
8

Correct Answer:
8

Solution:
The system of equations are

$$x - 2y + 5z = 0$$

$$-2x + 4y + z = 0$$

$$-7x + 14y + 9z = 0$$

$$\Delta = \begin{vmatrix} 1 & -2 & 5 \\ -2 & 4 & 1 \\ -7 & 14 & 9 \end{vmatrix} = 0$$

Let $x = k$

$\Rightarrow$ Put in (1)&(2)

$$k - 2y + 5z = 0$$

$$-2k + 4y + z = 0$$

$$z = 0, y = \frac{k}{2}$$

$\therefore x, y, z$ are integer

$\Rightarrow$ k is even integer

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Now$x = k, y = \frac{k}{2}, z = 0$

put in condition

$$15 \leq k^2 + \left(\frac{k}{2}\right)^2 + 0 \leq 150$$

$$12 \leq k^2 \leq 120$$

$\Rightarrow k = \pm 4, \pm 6, \pm 8, \pm 10$

$\Rightarrow$ Number of element in $S = 8$

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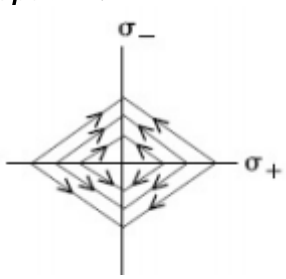
4th September Shift 1

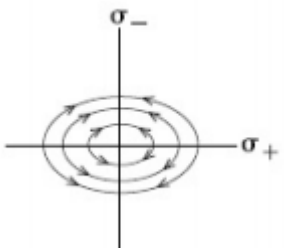
Physics

Q. 1 Two charged thin infinite plane sheets of uniform surface charge density $\sigma_+ \; and \; \sigma_-$, where $|\sigma_+| > |\sigma_-|$, intersects at a right angle. Which of the following best represents the electric field lines for this system:

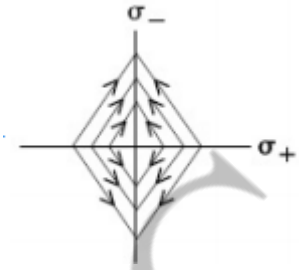
Option 1:



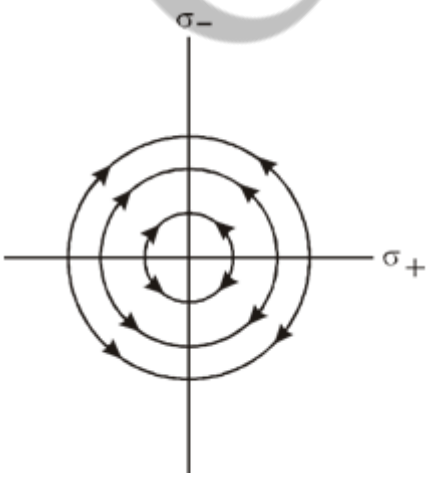
Option 2:



Option 3:



Option 4:



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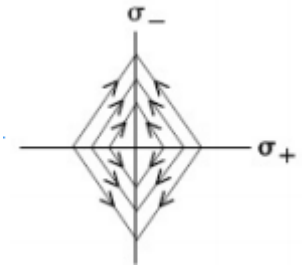
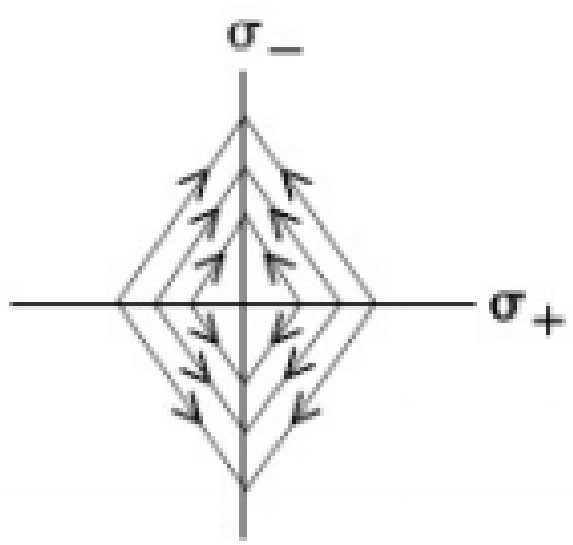
Correct Answer:



Solution:

Thin infinite uniformly charged planes produce a uniform electric field. And as positive charge density is bigger in magnitude so its field along Y direction will be bigger than the field of negative charge in the X direction.



Q. 2 A small bar magnet placed with its axis at 30° with the external field of 0.06T experiences a torque of 0.018 Nm. The minimum work required to rotate it from its stable to unstable equilibrium position is:

Option 1:

$6.4 \times 10^{-2} J$

Option 2:

$9.2 \times 10^{-3} J$

Option 3:

$7.2 \times 10^{-2} J$

Option 4:

$11.2 \times 10^{-3} J$

Correct Answer:

$7.2 \times 10^{-2} J$

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Solution:

Here, $\theta = 30^\circ, \tau = 0.018\ \mathrm{N-m}, \mathrm{B} = 0.06\ \mathrm{T}$

And Torque on a bar magnet

$$\tau = MB\sin\theta$$
$$\Rightarrow 0.018 = \mathrm{M} \times 0.06 \times \sin 30^\circ$$
$$\Rightarrow 0.018 = \mathrm{M} \times 0.06 \times \frac{1}{2}$$
$$\Rightarrow \mathrm{M} = 0.6\ \mathrm{A-m^2}$$

Now using $U = -MB\cos\theta$
Position of stable equilibrium $(\theta = 0^0)$
$\mathrm{U_i} = -\mathrm{MB}$

Position of unstable equilibrium $(\theta = 180^0)$
$\mathrm{U_f} = \mathrm{MB}$

$$\text{Using work done } : \Delta U$$
$$\Rightarrow W = 2MB$$
$$\Rightarrow W = 2 \times 0.6 \times 0.06$$
$$\Rightarrow W = 7.2 \times 10^{-2}\ \mathrm{J}$$

Q. 3 The specific heat of water = $4200 Jkg^{-1}K^{-1}$ and the latent heat of ice = $3.4 \times 10^5 Jkg^{-1}$. 100 grams of ice at $0^\circ C$ is mixedwith 200g of water at $25^\circ C$. The amount of ice that will melt as the temperature of water reaches $0^\circ C$ is close to (in grams):

Option 1:
61.7

Option 2:
63.8

Option 3:
69.3

Option 4:
64.6

Correct Answer:
61.7

Solution:
Here the water will provide heat for ice to melt therefore

$$i.e\ \ \mathrm{m_w s_w \Delta\theta = m_{ice}\, L_{ice}}$$
$$\Rightarrow \mathrm{m_{ice}} = \frac{0.2 \times 4200 \times 25}{3.4 \times 10^5} = 0.0617\ \mathrm{kg} = 61.7\mathrm{gm}$$

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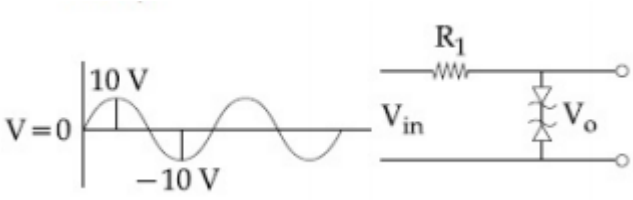
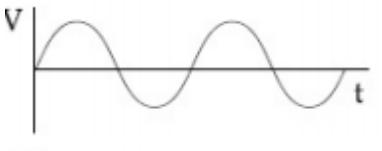
The remaining ice will remain un-melted.

Q. 4 Take the breakdown voltage of the zener diode used in the given circuit as 6V. For the input voltage shown in the figure below, the time variation of the output voltage is :

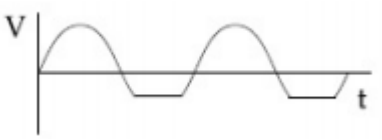
(graphs drawn schematic and not to scale)



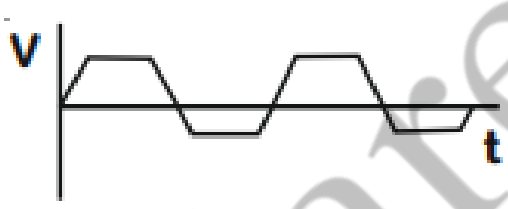
Option 1:



Option 2:

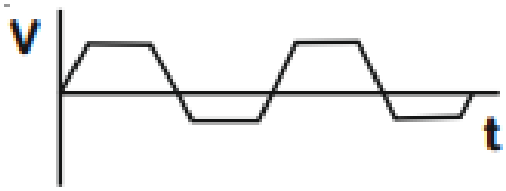


Option 3:



Option 4:

Correct Answer:



Solution:

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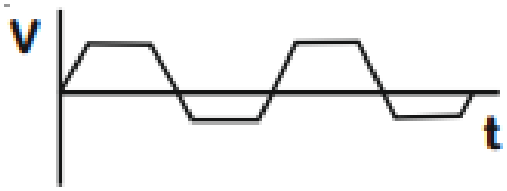
As there are two Zener diodes in reverse polarity, so if one is in forward bias the other will be in reverse bias and above 6 V the reverse bias will too be in conduction mode. Therefore when the voltage is more than 6 V the output will be constant. And when it is less than 6 V it will follow the input voltage.

so the correct graph is given as



Q. 5 Match the C_p/C_v ratio ideal gases with different types of molecules:

Molecular type	C_p/C_v
(A) monoatomic	I- 7/5
(B) diatomic rigid molecules	II- 9/7
(C) diatomic non- rigid molecules	III- 4/3
(D) triatomic rigid molecules	IV - 5/3

Option 1:
A-IV , B-II, C-I, D-III

Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-II, B-III, C-I, D-IV

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:

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$$\gamma = \frac{C_p}{C_v} = 1 + \frac{2}{f}$$

where 'f is degree of freedom

$(A) Mono\ \ atomic$

$\mathrm{f} = 3, \gamma = 1 + \frac{2}{3} = \frac{5}{3}$

$(B) Diatomic\ \ rigid\ \ molecules,$

$\mathrm{f} = 5, \gamma = 1 + \frac{2}{5} = \frac{7}{5}$

$(C) Diatomic\ \ non - rigid\ \ molecules$

$\mathrm{f} = 7, \gamma = 1 + \frac{2}{7} = \frac{9}{7}$

$(D) Triatomic\ \ rigid\ \ molecules$

$\mathrm{f} = 6, \gamma = 1 + \frac{2}{6} = \frac{4}{3}$

Q. 6 A beam of plane polarised light of large cross-sectional area of uniform intenisty of 3.3 Wm^{-2} falls normally on a polariser (cross -sectional area $3 \times 10^{-4} m^2$) which rotates about its axis with an angular speed of 31.4 rad/s. The energy of light passing through the polariser per revolution, is close to:

Option 1:

$1.1 \times 10^{-5} J$

Option 2:

$1.0 \times 10^{-4} J$

Option 3:

$1.5 \times 10^{-4} J$

Option 4:

$5.0 \times 10^{-4} J$

Correct Answer:

$5.0 \times 10^{-4} J$

Solution:

Given

$\text{Intensity, } I = 3.3 \mathrm{Wm}^{-2}$

$\text{Area } = A = 3 \times 10^{-4}\ \mathrm{m}^2$

$\text{Angular speed } = \omega = 31.4 \mathrm{rad/s}$

$\text{now as we know } < \cos^2\theta >= \frac{1}{2}, \text{ in one time period}$

$\therefore \text{ Average energy } = \mathrm{I}_0\ \mathrm{A} \times \frac{1}{2} = \frac{(3.3)(3 \times 10^{-4})}{2} \simeq 5 \times 10^{-4}\ \mathrm{J}$

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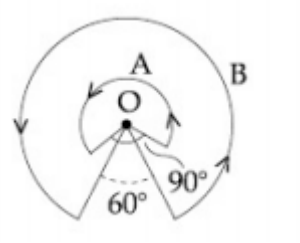
Q. 7 A wire A, bent in the shape of an arc of a circle, carrying a current of 2A and having a radius of 2cm and another wire B, also bent in the shape of an arc of a circle, carrying a current 3A and having a radius of 4cm, are placed as shown in the figure. The ratio of the magnetic fields due to the wires A and B at the common center O is:



Option 1:
4:6

Option 2:
6:4

Option 3:
2:5

Option 4:
6:5

Correct Answer:
6:5

Solution:

Given

$i_A = 2A, r_A = 2\ cm, \theta_A = 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$

$i_B = 3A, r_B = 4\ cm, \theta_B = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

$B = \frac{\mu_0 I \theta}{4\pi R}$

$\frac{B_A}{B_B} = \frac{I_A}{I_B} \times \frac{\theta_A R_B}{\theta_B R_A} = \frac{6}{5}$

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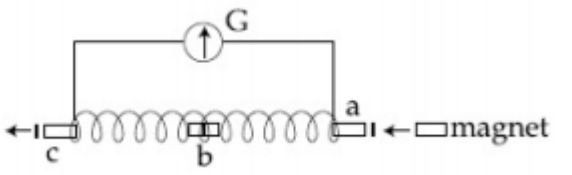
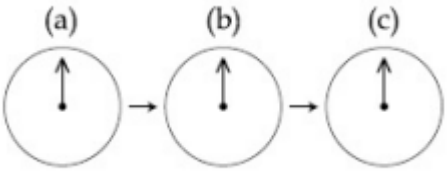
Q. 8 A small bar magnet is moved through a coil at a constant speed from end to other end. Which of the following series of observations will be seen on the galvanometer G attached to the coil?



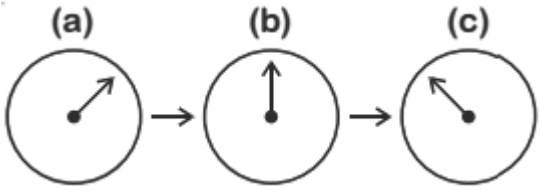
Three positions shown describe:

(a) the magnet's entry (b) magnet is completely inside (c) magnet's exit

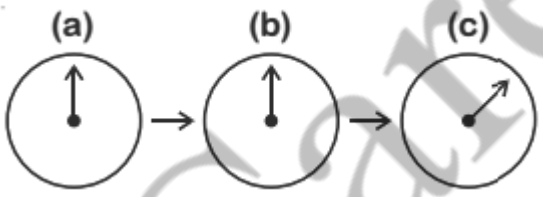
Option 1:



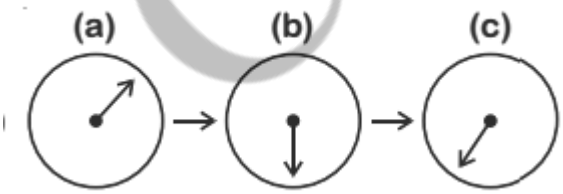
Option 2:



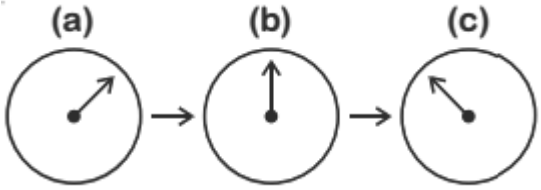
Option 3:



Option 4:



Correct Answer:



Solution:

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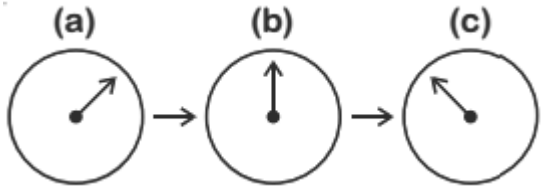
When a bar magnet is entering with constant speed. flux will change and e.m.f. is induced, so the galvanometer will deflect in a positive direction.

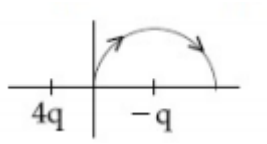
When the magnet is completely inside, the flux will not change, so the reading of the galvanometer will be zero.

When the bar magnet is making on exit, again flux will change and on e.m.f. is induced in opposite direction to not of (a), so galvanometer will deflect in a negative direction.

So the answer is



Q. 9 A two point charge 4q and -q are fixed on the x-axis at $x = -\frac{d}{2}$ and $x = \frac{d}{2}$, respectively. If a third point charge 'q' is taken from the origin to x= d along the semicircle as shown in the figure, the energy of the charge will:



Option 1:

increased by $\frac{3q^2}{4\pi\epsilon_0 d}$

Option 2:

increase by $\frac{2q^2}{3\pi\epsilon_0 d}$

Option 3:

decreases by $\frac{q^2}{4\pi\epsilon_0 d}$

Option 4:

decrease by $\frac{4q^2}{3\pi\epsilon_0 d}$

Correct Answer:

decrease by $\frac{4q^2}{3\pi\epsilon_0 d}$

Solution:

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Initial and final potential energy are,

$$U_i = \frac{1}{4\pi\varepsilon_0}\left[\frac{4q^2}{\left(\frac{d}{2}\right)} - \frac{q^2}{\left(\frac{d}{2}\right)}\right]$$

$$U_f = \frac{1}{4\pi\varepsilon_0}\left[\frac{4q^2}{\left(\frac{3\ d}{2}\right)} - \frac{q^2}{\left(\frac{d}{2}\right)}\right]$$

$$U_f - U_i = \Delta U = \left(\frac{1}{4\pi\varepsilon_0}\cdot\frac{4q^2}{(3\ d/2)}\right) - \left(\frac{1}{4\pi\varepsilon_0}\cdot\frac{4q^2}{(d/2)}\right)$$

$$= \frac{4q^2}{4\pi\varepsilon_0}\left(\frac{2}{3d} - \frac{2}{d}\right)$$

$$= (-)\frac{4q^2}{3\pi\varepsilon_0\cdot d}$$

the energy of the charge will decrease by $\frac{4q^2}{3\pi\epsilon_0 d}$

Q. 10 A battery of 3.0 V is connected to a resistor dissipating 0.5W of power. If the terminal voltage of the battery is 2.5 V, the power dissipated within the internal resistance is:

Option 1:
0.50W

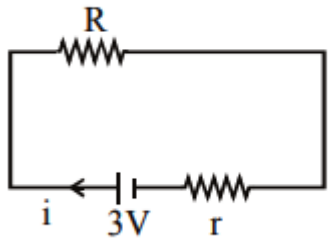
Option 2:
0.072 W

Option 3:
0.10 W

Option 4:
0.125 W

Correct Answer:
0.10 W

Solution:



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$P_R = 0.5W$
$\Rightarrow P_R = \mathrm{i^2R} = 0.5\ \mathrm{W}$
Also,

$\mathrm{V} = \mathrm{E} - \mathrm{ir}$
$\Rightarrow 2.5 = 3 - \mathrm{ir}$
$\Rightarrow ir = 0.5$
Power dissipated across 'r' $= P_r = i^2 r$
Now

iR =2.5 .....(1)
ir =0.5 .....(2)
From equation (1) and (2)

we get

$\frac{R}{r} = 5$

Now,

$\frac{P_R}{P_r} = \frac{i^2 R}{i^2 r} \Rightarrow \frac{P_R}{P_r} = \frac{R}{r} \Rightarrow \frac{P_R}{P_r} = 5$

$\Rightarrow P_r = \frac{P_R}{5}$

$\Rightarrow P_r = \frac{0.50}{5}$
$\Rightarrow P_r = 0.10\ \mathrm{W}$

Q. 11 For transverse wave travelling along a straight line, the distance between two peaks (crests) is 5m, while the distance between one crests and one trough is 1.5m. The possible wavelengths (in m) of the waves are:

Option 1:
1,3.5,.....

Option 2:
$\frac{1}{1}, \frac{1}{3}, \frac{1}{5}, \ldots\ldots\ldots\ldots$

Option 3:
1,2,3,.....

Option 4:
$\frac{1}{2}, \frac{1}{4}, \frac{1}{6}\ldots\ldots$

Correct Answer:
$\frac{1}{1}, \frac{1}{3}, \frac{1}{5}, \ldots\ldots\ldots\ldots$

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Solution:
Given trough to crest distance=1.5 m

So $(2n_1+1)\frac{\lambda}{2}=1.5$

and crest to crest distance= 5 m

So

$n_2\lambda=5$

S0

from these two pieces of information, we get

$$\frac{1.5}{5}=\frac{(2n_1+1)}{2n_2}$$
$$\Rightarrow 3n_2=10n_1+5$$

And since $n_1 \quad and \ n_2$ are integers

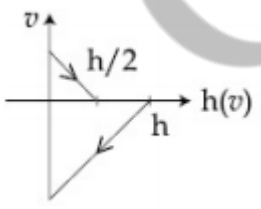
So

$$n_1=1, n_2=5 \rightarrow \lambda=1$$
$$n_1=4, n_2=15 \rightarrow \lambda=1/3$$
$$n_1=7, n_2=25 \rightarrow \lambda=1/5$$

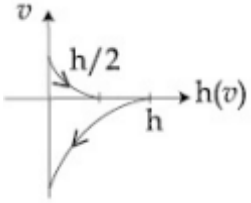
Q. 12 A tennis ball is released from a height h and after freely falling on a wooden floor, it rebounds and reaches height $\frac{h}{2}$. The velocity versus height of the ball during its motion may be represented graphically by:

(graphs are drawn schematically and not to the scale)

Option 1:



Option 2:



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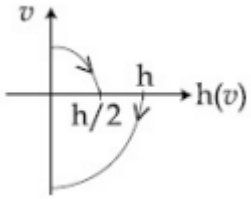
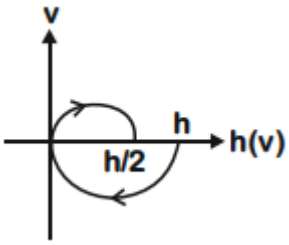
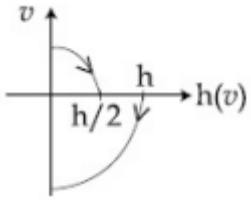
Option 3:



Option 4:



Correct Answer:



Solution:
Velocity at the ground (means zero height) is non-zero therefore the option 4 is incorrect.

The velocity versus height curve will be parabolic as for motion under gravity.

i.e The velocity versus height curve is non-linear therefore option 1 is also incorrect.

$$\mathrm{v}^2 = 2\mathrm{gh}$$

$$\mathrm{v}\frac{\mathrm{dv}}{\mathrm{dh}} = 2\ \mathrm{g} = \mathrm{const}$$

$$\Rightarrow \frac{\mathrm{dv}}{\mathrm{dh}} = \frac{\mathrm{constant}}{\mathrm{v}}$$

Here we can see the slope is very high when velocity is low therefore at the maximum height the slope should be very large which is in option 3 and as velocity increases slope must decrease therefore option 3 is correct.

Q. 13 Particle A of mass $m_A = \frac{m}{2}$ moving along the axis with velocity v_0 collides eleastically with another particle B at rest having mass $m_B = \frac{m}{3}$. If both particles move along the x-axis after the collision, the change $\Delta\lambda$ in de-Brogile wavelength of particle A, in terms of its de-Brogile wavelength (λ_0) before collision is:

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Option 1:

$\Delta\lambda = \frac{3}{2}\lambda_0$

Option 2:

$\Delta\lambda = \frac{5}{2}\lambda_0$

Option 3:

$\Delta\lambda = 2\lambda_0$

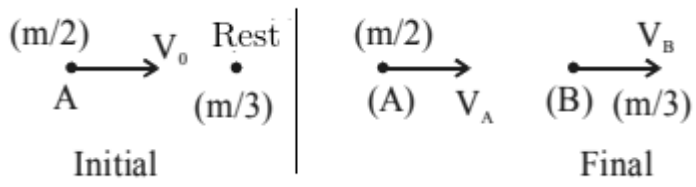
Option 4:

$\Delta\lambda = 4\lambda_0$

Correct Answer:

$\Delta\lambda = 4\lambda_0$

Solution:



Applying momentum conservation

$$\frac{m}{2} \times V_0 + \frac{m}{3} \times (0) = \frac{m}{2} V_A + \frac{m}{3} V_B$$

$$\Rightarrow \frac{V_0}{2} = \frac{V_A}{2} + \frac{V_B}{3} \quad \ldots(1)$$

Since, collision is elastic (e=1)

$$e = 1 = \frac{V_B - V_A}{V_0}$$

$$\Rightarrow V_0 = V_B - V_A \ldots(2)$$

From equation (1) and (2)

we get

$$V_A = \frac{V_0}{5}$$

Now, De-Broglie wavelength of A before collision:

$$\lambda_0 = \frac{h}{m_A V_0} = \frac{h}{\left(\frac{m}{2}\right) V_0}$$

$$\Rightarrow \lambda_0 = \frac{2h}{mV_0}$$

Final De-Broglie wavelength

$$\lambda_f = \frac{h}{m_A V_0} = \frac{h}{\frac{m}{2} \times \frac{V_0}{5}}$$

$$\Rightarrow \lambda_f = \frac{10\ h}{mV_0}$$

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$\text{Now } \Delta\lambda = \lambda_f - \lambda_0$
$\Delta\lambda = \frac{10\ h}{mV_0} - \frac{2\ h}{mV_0}$
$\Rightarrow \Delta\lambda = \frac{8\ h}{mv_0}$
$\Rightarrow \Delta\lambda = 4 \times \frac{2\ h}{mv_0}$
$\Rightarrow \Delta\lambda = 4\lambda_0$

Q. 14 Starting from the origin at time t=0, with initial velocity $5\hat{j} ms^{-1}$, a particle moves in the x-y plane with constant accelertaion of $\left(10\hat{i} + 4\hat{j}\right) ms^{-2}$. At time t, its coordinates are $(20m, y_0\ m)$. The values of t and y_0 are , respectively:

Option 1:
2 s and 18 m

Option 2:
4 s and 52 m

Option 3:
5 s and 24 m

Option 4:
5 s and 25 m

Correct Answer:
2 s and 18 m

Solution:
Given

$\vec{a} = \left(10\hat{i} + 4\hat{j}\right) ms^{-2}$

$\vec{u} = 5\hat{j} ms^{-1}$

$\text{And final coordinates } (20, y_0) \text{ in time } t$

So

For x-axis

$S_x = u_x t + \frac{1}{2} a_x t^2$
$20 - 0 = 0 + (\frac{1}{2} \times 10 \times t^2)$
$t = 2\text{sec}$

For y-axis

$S_y = u_y \times t + \frac{1}{2} a_y t^2$
$y_0 = (5 \times 2) + (\frac{1}{2} \times 4 \times 2^2) = 18\ m$

So answer is 2 s and 18 m

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Q. 15 On the x-axis and at a distance x from the origin, the gravitational field due to mass distribution is given by $\frac{Ax}{(x^2+a^2)^{\frac{3}{2}}}$ in the x-direction. The magnitude of gravitational potential on the x-axis at a distance x, taking its value to be zero at infinity, is:

Option 1:
$\frac{A}{(x^2+a^2)^{\frac{1}{2}}}$

Option 2:
$\frac{A}{(x^2+a^2)^{\frac{3}{2}}}$

Option 3:
$A(x^2+a^2)^{\frac{1}{2}}$

Option 4:
$A(x^2+a^2)^{\frac{3}{2}}$

Correct Answer:
$\frac{A}{(x^2+a^2)^{\frac{1}{2}}}$

Solution:
Given
$E_G = \frac{Ax}{(x^2+a^2)^{3/2}},\ V_\infty = 0$
$Using$
$\int_{V_\infty}^{V_x} dV = -\int_\infty^x \vec{E}_G \cdot \vec{d}_x$

$V_x - V_\infty = -\int_\infty^x \frac{Ax}{(x^2+a^2)^{3/2}} dx$

put $x^2 + a^2 = z$
$2xdx = dz$

So

$$V_x - 0 = -\int_\infty^x \frac{Adz}{2(z)^{3/2}} = \left[\frac{A}{z^{1/2}}\right]_\infty^x = \left[\frac{A}{(x^2+a^2)^{1/2}}\right]_\infty^x$$

$$V_x = \frac{A}{(x^2+a^2)^{1/2}} - 0 = \frac{A}{(x^2+a^2)^{1/2}}$$

Q. 16 Dimensional formula for thermal conductivity is (here K denotes the temperature)

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Option 1:
$MLT^{-2}K$

Option 2:
$MLT^{-2}K^{-2}$

Option 3:
$MLT^{-3}K$

Option 4:
$MLT^{-3}K^{-1}$

Correct Answer:
$MLT^{-3}K^{-1}$

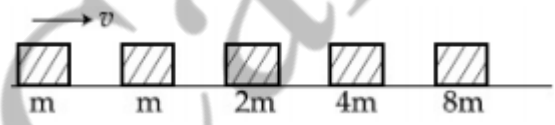
Solution:

$$\text{as} \quad \frac{\mathrm{dQ}}{\mathrm{dt}} = \mathrm{kA}\frac{\mathrm{dT}}{\mathrm{dx}}$$

$$\Rightarrow \mathrm{k} = \frac{\left(\frac{\mathrm{dQ}}{\mathrm{dt}}\right)}{\mathrm{A}\left(\frac{\mathrm{dT}}{\mathrm{dx}}\right)}$$

$$\Rightarrow [\mathrm{k}] = \frac{\left[\mathrm{ML}^2\ \mathrm{T}^{-3}\right]}{[\mathrm{L}^2]\left[\mathrm{KL}^{-1}\right]} = \left[\mathrm{MLT}^{-3}\ \mathrm{K}^{-1}\right]$$

Q. 17 Blocks of masses m, 2m, 4m, and 8m are arranged in a line on a frictionless floor. Another block of a mass m, moving with speed v along the same line (see figure) collides with mass m in a perfectly inelastic manner. All the subsequent collisions are also perfectly inelastic. By the time the last block of a mass 8m starts moving the total energy loss is p% of the original energy. Value of 'p; is close to :



Option 1:
77

Option 2:
94

Option 3:
37

Option 4:
87

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Correct Answer:
94

Solution:
All collisions are perfectly inelastic, so after the final collision, all blocks are moving together
So let the final velocity be $_{\mathrm{v}'}$, so on applying momentum conservation:

$$\mathrm{mv} = 16\mathrm{mv}'$$
$$\Rightarrow \mathrm{v}' = \mathrm{v}/16$$

Now

initial energy=$\mathrm{E_i} = \frac{1}{2}\mathrm{mv}^2$

Final energy =$\mathrm{E_f} = \frac{1}{2} \times 16\ \mathrm{m} \times \left(\frac{\mathrm{v}}{16}\right)^2 = \frac{1}{2} \times\ \mathrm{m} \times \left(\frac{\mathrm{v}^2}{16}\right)$

$$\text{Energy loss : } \mathrm{E_i} - \mathrm{E_f} = \tfrac{1}{2}\mathrm{mv}^2 - \tfrac{1}{2}\ \mathrm{m}\tfrac{\mathrm{v}^2}{16} = \tfrac{1}{2}\mathrm{mv}^2\left[1 - \tfrac{1}{16}\right] = \tfrac{1}{2}\mathrm{mv}^2\left[\tfrac{15}{16}\right]$$

$$\%\mathrm{p} = \frac{\text{Energy loss}}{\text{Original energy}} \times 100 = \frac{\frac{1}{2}mv^2\left[\frac{15}{16}\right]}{\frac{1}{2}mv^2} \times 100 = 93.75\%$$

$$\Rightarrow \text{ Value of } P \text{ is close to } 94$$

Q. 18 choose the correct option relating wavelengths of the different parts of electromagnetic wave spectrum:

Option 1:
$\lambda_{visible} < \lambda_{microwaves} < \lambda_{radiowaves} < \lambda_{x-rays}$

Option 2:
$\lambda_{radiowaves} > \lambda_{microwaves} > \lambda_{visible} > \lambda_{x-rays}$

Option 3:
$\lambda_{x-rays} < \lambda_{microwaves} < \lambda_{radiowaves} < \lambda_{visible}$

Option 4:
$\lambda_{visible} > \lambda_{x-rays} > \lambda_{radiowaves} > \lambda_{microwaves}$

Correct Answer:
$\lambda_{radiowaves} > \lambda_{microwaves} > \lambda_{visible} > \lambda_{x-rays}$

Solution:
$\lambda_{radiowaves} > \lambda_{microwaves} > \lambda_{visible} > \lambda_{x-rays}$

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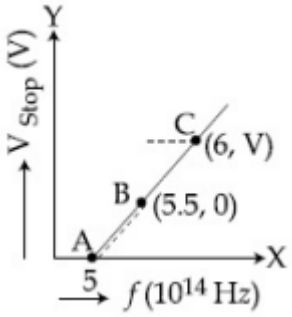
Q. 19 Given figure shows few data points in a photo electric efferct experiment for a certain metal. The minimum energy (in eV) for ejection of electron from its surface is: (Plank's constant h = $6.62 \times 10^{-34} J.s$)



Correct Answer:
2.27

Solution:

$$hv = \phi + \mathrm{eV_s}$$

$$\text{at B,} \quad \mathrm{V_s} = 0, \quad v = 5.5$$

$$\Rightarrow \quad \mathrm{h} \times 5.5 \times 10^{14} = \phi$$

$$\phi = \frac{6.62\times10^{-34}\times5.5\times10^{14}}{1.6\times10^{-19}}\mathrm{eV} = 2.27\mathrm{eV}$$

Q. 20 An air bubble of a radius 1cm in water has an upward acceleration 9.8 cm s^{-2}. The density of water is 1gm cm^{-3} and water offers negligible drag force on the bubble. The mass of the bubble is (g=980 cm/s^{2})

Option 1:
4.51 gm

Option 2:
3.15 gm

Option 3:
4.15 gm

Option 4:
1.52 gm

Correct Answer:
4.15 gm

Solution:

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According to Newtons 2nd law

B-mg=ma

$$m = \frac{B}{g + a}$$

Given

$$\text{Volume } = \mathrm{V} = \tfrac{4\pi}{3}\mathrm{r}^3 = \tfrac{4\pi}{3} \times (1)^3 = 4.19 \text{ cm}^3$$
$$\mathrm{a} = 9.8 \text{ cm/s}^2$$

And

$$B = \mathrm{V}\rho_w \mathrm{g}$$

So

$$\mathrm{m} = \frac{(\mathrm{V}\rho_w \mathrm{g})}{\mathrm{g}+\mathrm{a}} = \frac{\mathrm{V}\rho_\mathrm{w}}{1+\frac{\mathrm{a}}{\mathrm{g}}} = \frac{(4.19)\times 1}{1+\frac{9.8}{980}} = \frac{4.19}{1.01} = 4.15\mathrm{gm}$$

Q. 21 A closed vessel contains 0.1mole of a monoatomic ideal gas at 200K. If 0.05 mole of the same gas at 400 K is added to it, the final equilibrium tempertaure (in K) of the gas in the vessel will be close to _____

Option 1:
10

Option 2:
267

Option 3:
35

Option 4:
100

Correct Answer:
267

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Solution:
The work done on gas is zero. (As the vessel is closed)

and Assuming the heat supplied to the gas is zero. (as no information on the heat supplied to the gas is given in the question)

So The total internal energy of gases remains the same.

$$u_1 + u_2 = u_1' + u_2'$$
$$\Rightarrow \ (0.1)\ C_v(200) + (0.05)C_v(400) = (0.15)C_vT$$
$$\Rightarrow T = \frac{800}{3}k = 266.67k = 267\ k$$

Q. 22 In the line of spectra of the hydrogen atom, the difference between the largest and the shortest wavelengths of the Lyman series is 304 $\dot{A}$. The corresponding difference for the Paschen is series in $\dot{A}$ is_____ (Give answer in closest integer)

Option 1:
10553

Option 2:
10567

Option 3:
9876

Option 4:
122234

Correct Answer:
10553

Solution:
$$\lambda = \frac{c}{\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}$$
for Lyman series

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$$\lambda_1 = \frac{c}{\frac{1}{1^2} - \frac{1}{\infty^2}} = c(n = \infty \text{ to } n = 1)$$

$$\lambda_2 = \frac{c}{\frac{1}{1^2} - \frac{1}{2^2}} = \frac{4c}{3}(n = 2 \text{ to } n = 1)$$

$$\Delta\lambda = \lambda_2 - \lambda_1 = \frac{c}{3} = 304\ A^0$$
$$\Rightarrow c = 912\ A^0$$

for Paschen series

$$\lambda_1 = \frac{c}{\frac{1}{3^2} - \frac{1}{\infty^2}} = 9c \quad (n = \infty \text{ to } n = 3)$$

$$\lambda_2 = \frac{c}{\frac{1}{3^2} - \frac{1}{4^2}} = \frac{144c}{7} \quad (n = 4 \text{ to } n = 3)$$

$$\Delta\lambda = \lambda_2 - \lambda_1 = \frac{144c}{7} - 9c = \frac{81c}{7}$$
$$\Delta\lambda = \frac{81 \times 912}{7} = 10553.14\ A^0 \approx 10553\ A^0$$

Q. 23 In a compound microscope, the magnified virtual image is formed at a distance of 25cm from the eye-piece. The focal length of its objective lens is 1cm. If the magnification is 100 and the tube length of the microscope is 20cm, then the focal length of eyepiece lens (in cm) is:_____(Give answer up to 2 decimal points)

Option 1:
4.48

Option 2:
100

Option 3:
90

Option 4:
80

Correct Answer:
4.48

Solution:

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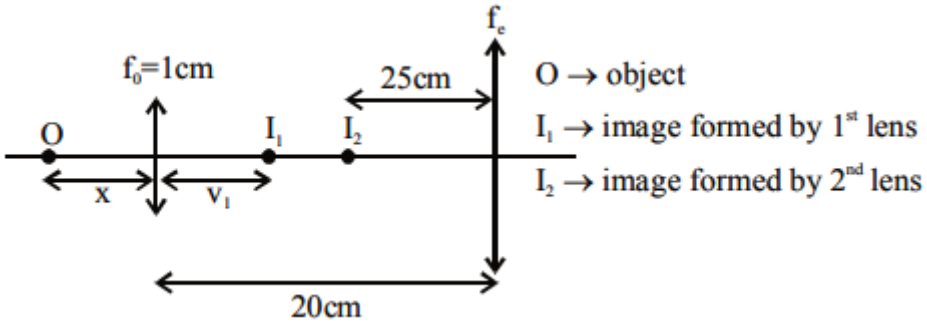

As Lens formula is given as

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

For first lens

$$\Rightarrow \frac{1}{v_1}-\frac{1}{-x}=\frac{1}{1} \Rightarrow v_1=\frac{x}{x-1}$$

$$\text{also magnification } |m_1|=\left|\frac{v_1}{u_1}\right|=\frac{1}{x-1}$$

$$\text{for } 2^{nd} \text{ lens this is acting as object}$$

$$\text{so } u_2=-(20-v_1)=-\left(20-\frac{x}{x-1}\right)$$

$$\text{and } v_2=-25 \text{ cm}$$

$$\text{angular magnification}=|m_2|=\left|\frac{D}{u_2}\right|=\frac{25}{|u_2|}$$

$$\text{Total magnification } m=m_1m_2=100$$

$$\Rightarrow \left(\tfrac{1}{x-1}\right)\left(\tfrac{25}{20-\frac{x}{x-1}}\right)=100$$

$$\tfrac{25}{20(x-1)-x}=100 \Rightarrow 1=80(x-1)-4x$$

$$\Rightarrow 76x=81 \Rightarrow x=\tfrac{81}{76}$$

$$\Rightarrow u_2=-\left(20-\frac{81/76}{81/76-1}\right)=\frac{-19}{5}$$

Now apply the lense formula for the eyepiece

$$\frac{1}{-25}-\frac{1}{-19/5}=\frac{1}{f_e}$$

$$\Rightarrow f_e=\frac{25\times 19}{106}\approx 4.48 \text{ cm}$$

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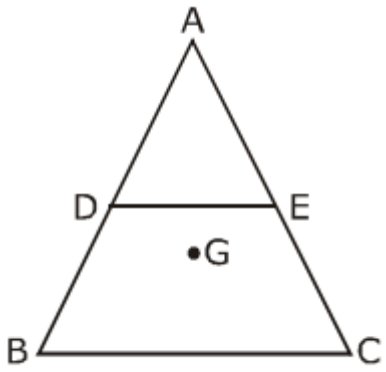
Q. 24 ABC is a plane lamina of the shape of an equilateral triangle. D, E are midpoints of AB, AC and G is the centroid of the lamina. Moment of inertia of lamina about an axis passing through G and perpendicular to the plane ABC I_0. If the part ADE is removed, the moment of inertia of the remaining part about the same axis is $\frac{NI_0}{16}$ where N is an integer. Value of N is _____



Option 1:
11

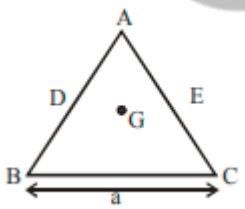
Option 2:
9

Option 3:
8

Option 4:
7

Correct Answer:
11

Solution:



Let the side of the triangle is a and mass is m

$$\text{MOI of plate ABC about centroid}$$

$$I_0 = \frac{m}{3}\left(\left(\frac{a}{2\sqrt{3}}\right)^2 \times 3\right) = \frac{ma^2}{12}$$

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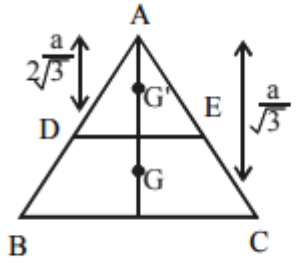
Triangle ADE is also an equilateral triangle of side $\frac{a}{2}$.

Let the moment of inertia of triangular plate ADE about its centroid (G') is I_1 and mass is m_1

$$\mathrm{m_1} = \frac{\mathrm{m}}{\frac{\sqrt{3}\mathrm{a}^2}{4}} \times \frac{\sqrt{3}}{4}\left(\frac{\mathrm{a}}{2}\right)^2 = \frac{\mathrm{m}}{4}$$

$$\mathrm{I_1} = \frac{\mathrm{m_1}}{12}\left(\frac{\mathrm{a}}{2}\right)^2 = \frac{\mathrm{m}}{4\times 12}\frac{\mathrm{a}^2}{4} = \frac{\mathrm{ma}^2}{192}$$



$$\text{from figure distance } \mathrm{GG'} = \frac{\mathrm{a}}{\sqrt{3}} - \frac{\mathrm{a}}{2\sqrt{3}} = \frac{\mathrm{a}}{2\sqrt{3}}$$

so MOI of part ADE about centroid G is

$$\mathrm{I_2} = \mathrm{I_1} + \mathrm{m_1}\left(\frac{\mathrm{a}}{2\sqrt{3}}\right)^2 = \frac{\mathrm{ma}^2}{192} + \frac{\mathrm{m}}{4}\cdot\frac{\mathrm{a}^2}{12} = \frac{5\mathrm{ma}^2}{192}$$

now MOI of remaining part

$$I_{re} = \frac{ma^2}{12} - \frac{5ma^2}{192} = \frac{11ma^2}{12\times 16} = \frac{11I_0}{16}$$

$$\Rightarrow N = 11$$

Q. 25 A circular disc of mass M and radius R is rotating about its axis with angular speed ω_1. Another stationary disc having radius $\frac{R}{2}$ and same mass M is dropped co-axially on the rotating disc. Gradually both the discs attain constant angular speed ω_2. The energy lost in the process is p% of the initial energy. Value of p is_____

Option 1:
10

Option 2:
20

Option 3:
30

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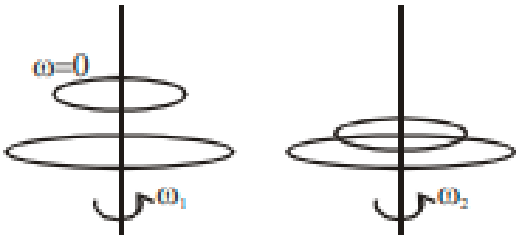
Option 4:
40

Correct Answer:
20

Solution:



Let the moment of inertia of bigger disc is $I = \frac{MR^2}{2}$

$\Rightarrow$ MOI of small disc $= I_2 = \frac{M\left(\frac{R}{2}\right)^2}{2} = \frac{I}{4}$

by angular momentum conservation

$\mathrm{I}\omega_1 + \frac{\mathrm{I}}{4}(0) = \mathrm{I}\omega_2 + \frac{\mathrm{I}}{4}\omega_2 \Rightarrow \omega_2 = \frac{4\omega_1}{5}$

initial kinetic energy $=\mathrm{K}_1 = \frac{1}{2}\mathrm{I}\omega_1^2$

final kinetic energy

$\mathrm{K}_2 = \frac{1}{2}\left(I + \frac{I}{4}\right)\left(\frac{4\omega_1}{5}\right)^2 = \frac{1}{2}I\omega_1^2\left(\frac{4}{5}\right)$

$\mathrm{P}\% = \frac{\mathrm{K}_1 - \mathrm{K}_2}{\mathrm{K}_1} \times 100\% = \frac{1 - 4/5}{1} \times 100 = 20\%$

Chemistry

Q. 1 The elements with atomic number 101 and 104 belongs to, respectively:

Option 1:
Group 11 and Group 4

Option 2:
Actinoids and Group 6

Option 3:
Actinoids and Group 4

Option 4:
Group 4 and Actinoids

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Correct Answer:
Actinoids and Group 4

Solution:
$Z = 101 \rightarrow [R_n]^{86}\, 7s^2 5f^{13}$ —— It is Actinoids

$Z = 104 \rightarrow [R_n]^{86}\, 7\, s^2 5f^{14} 6d^2$ —— It is 4th group element

So, Actinoids and Group 4, are correct respectively.

Therefore, the correct option is (3).

Q. 2 Which of the following will react with $CHCl_3 + \text{ alc. } KOH$?

Option 1:
Adenine and Proline

Option 2:
Thymine and Proline

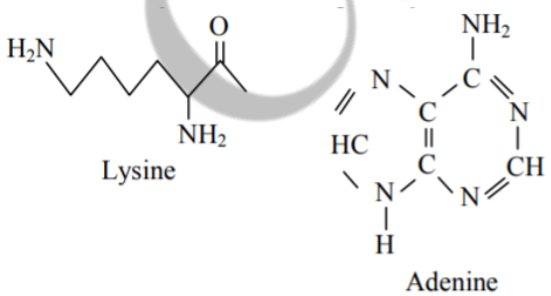
Option 3:
Adenine and Lysine

Option 4:
Adenine and Thymine

Correct Answer:
Adenine and Lysine

Solution:
$CHCl_3 + \text{ alc. } KOH$ reacts with those compounds which have $-NH_2$ group.

Only Lysine and Adenine have $-NH_2$ group.



Therefore, the correct option is (3).

Q. 3 The number of isomers possible for $[Pt(en)(NO_2)_2]$ is:

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Option 1:
2

Option 2:
4

Option 3:
1

Option 4:
3

Correct Answer:
2

Solution:
$[Pt(en)(NO_2)_2]$ does not show G.I. as well as optical isomerism -

1. Geometrical isomerism does not show due to the steric hindrance of en ligand.

2. Optical isomerism does not show due to the presence of a plane of symmetry.

But, this complex will have three linkage isomers as follows -

1. $[Pt(en)\ (NO_2)_2]$

2. $[Pt(en)\ (NO_2)\ (ONO)]$

3. $[Pt(en)(ONO)_2]$

So, other than $[Pt(en)\ (NO_2)_2]$ this, there are 2 isomer is possible.

Ans = 2

Therefore, the correct option is (1).

Q. 4 The region in the electromagnetic spectrum where the Balmer series lines appear is:

Option 1:
Visible

Option 2:
Microwave

Option 3:
Infrared

Option 4:
Ultraviolet

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Correct Answer:
Visible

Solution:
Balmer series give visible lines for H-atom.

As lines of Balamer series belongs to both UV as well visible region of EM spectrum. However most appropriate should be visible region.

Therefore, the correct option is (1).

Q. 5 For one mole of an ideal gas, which of these statements must be true?

(a) U and H each depends only on temperature.

(b) Compressibility factor z is not equal to 1

(c) $C_{p,m} - C_{v,m} = R$

(d) $dU = C_V dT$ for any process

Option 1:
(a) and (c)

Option 2:
(b), (C) and (d)

Option 3:
(c) and (d)

Option 4:
(a), (c) and (d)

Correct Answer:
(a), (c) and (d)

Solution:
For ideal gas
U = f(T), H = f(T)
Z = 1 (Compressibility factor z is equal to 1)
$C_p - C_v = R$
$dU = C_v dT$

So, a, c, and d are correct

Therefore, the correct option is (4).

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Q. 6 What are the functional groups present in the structure of maltose?

Option 1:
One ketal and one hemiketal

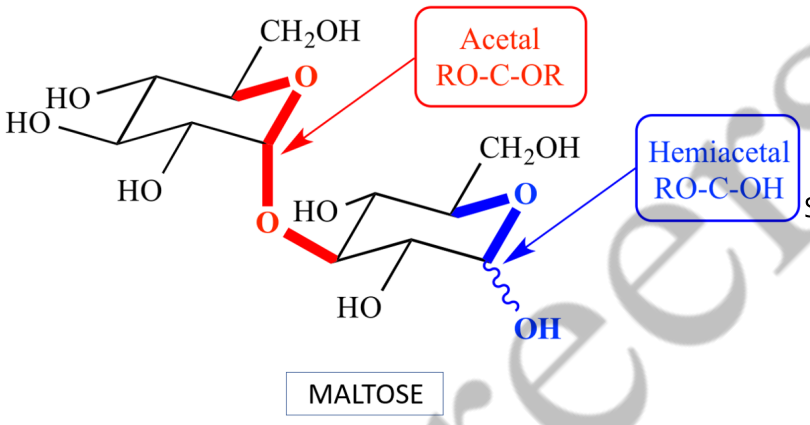
Option 2:
Two acetals

Option 3:
One acetal and one hemiacetal

Option 4:
One acetal and one ketal

Correct Answer:
One acetal and one hemiacetal

Solution:

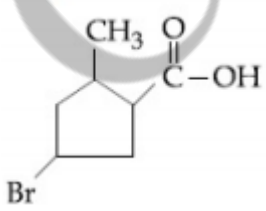


MALTOSE

So, One acetal and one hemiacetal functional groups are present in the structure of maltose.

Therefore, the correct option is (3).

Q. 7 The IUPAC name of the following comound is:



Option 1:
5-Bromo-3-methylcyclopentanoic acid

Option 2:
4-Bromo-2-methylcyclopentane carboxylic acid

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Option 3:
3-Bromo-5-methylcyclopentanoic acid

Option 4:
3-Bromo-5-methylcyclopentane carboxylic acid

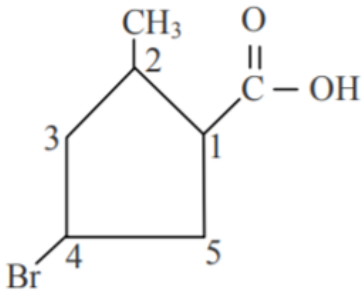
Correct Answer:
4-Bromo-2-methylcyclopentane carboxylic acid

Solution:
-COOH group has more seniority than others.

At 2 branched carbon preferred.

At 4 there is Br

So, counting will be



So, IUPAC Name

4-Bromo-2-methylcyclopentane carboxylic acid

Therefore, the correct option is (2).

Q. 8 One combination of Li, Na and K in excess of air, the major oxides formed, respectively are:

Option 1:
Li_2O_2, Na_2O_2 and K_2O_2

Option 2:
Li_2O, Na_2O_2 and KO_2

Option 3:
Li_2O, Na_2O and K_2O_2

Option 4:
Li_2O, Na_2O_2 and K_2O

Correct Answer:
Li_2O, Na_2O_2 and KO_2

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Solution:
The formed oxides will be -

$$\mathrm{Li} + \mathrm{O_2}\ (\text{excess}) \rightarrow \mathrm{Li_2O}\ (\text{Major oxides})\mathrm{Na} + \mathrm{O_2}\ (\text{excess}) \rightarrow \mathrm{Na_2O_2}\ (\text{Major oxides})$$
$$\mathrm{K} + \mathrm{O_2}\ (\text{excess}) \rightarrow \mathrm{KO_2}\ (\text{Major oxides})$$

Therefore, the correct option is (2).

Q. 9 On heating, lead (II) nitrate gives a brown gas (A). The gas (A) on cooling changes to colourless soild/liquid (B). (B) on heating with NO changes to a blue soild (C). The oxidation number of nitrogen in solid (C) is:

Option 1:
+5

Option 2:
+2

Option 3:
+3

Option 4:
+4

Correct Answer:
+3

Solution:
The reactions will be -

$$\mathrm{Pb(NO_3)_2} \xrightarrow{\Delta} \mathrm{PbO} + \underset{\substack{\text{Brown gas}\\(A)}}{2\mathrm{NO_2}} + \frac{1}{2}\mathrm{O_2(g)}$$

$$\mathrm{NO_2} \xrightarrow{\text{Cooling}} \underset{(B)}{\mathrm{N_2O_4}}$$

$$\mathrm{N_2O_4} + \mathrm{NO} \xrightarrow{\Delta} \underset{\substack{\text{Blue Solid}\\(C)}}{\mathrm{N_2O_3}}$$

The oxidation number of nitrogen in N_2O_3 , given in NCERT is +3.

Therefore, the correct option is (3).

Q. 10 For equilibrium $\mathrm{A} \rightleftharpoons \mathrm{B}$, the variation of the rate of the forward (a) and reverse (b) reaction with the time is given by:

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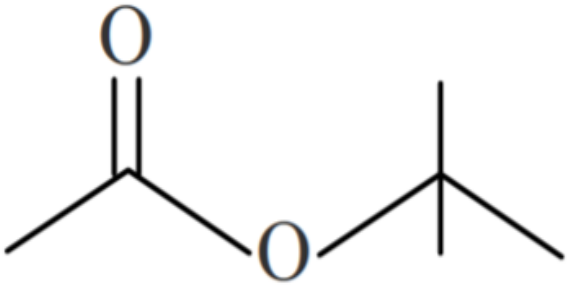
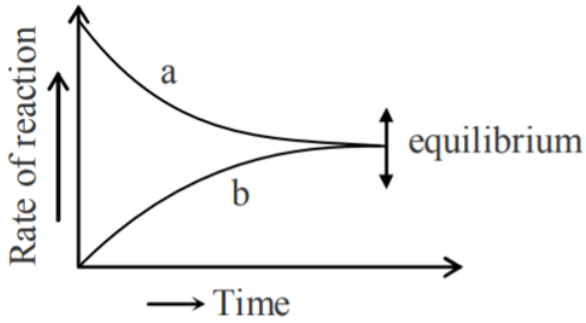
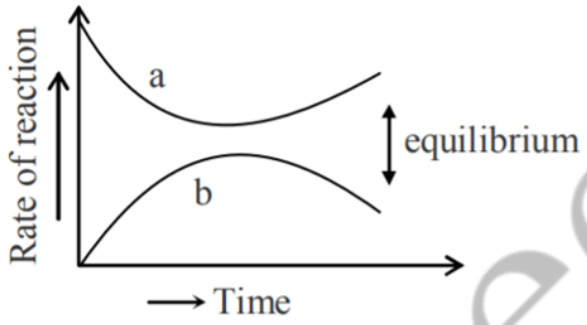
Option 1:



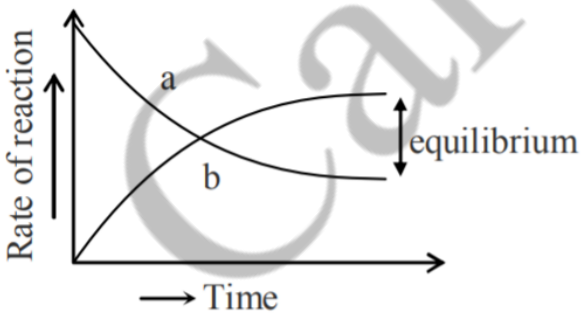
Option 2:



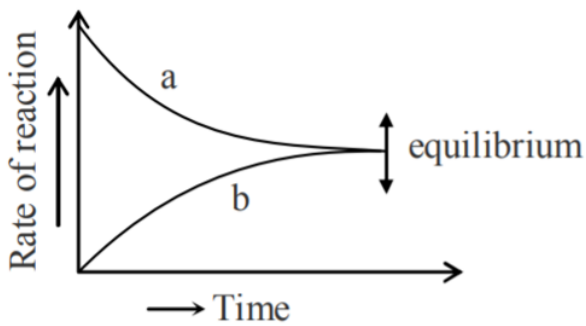
Option 3:



Option 4:



Correct Answer:



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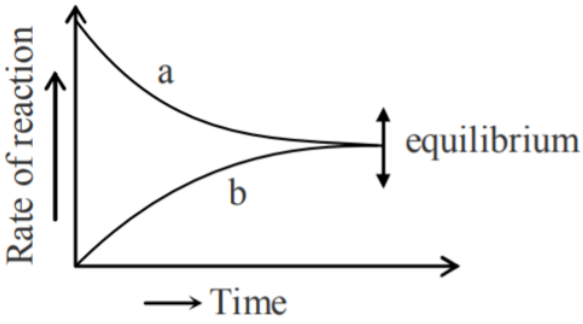
Solution:
At equilibrium

Rate of forward = Rate of backward

So, a = b Hence

The correct graph will be -



Therefore, the correct option is (2).

Q. 11 Among statements (a)-(d) the correct ones are:

(a) Lime stone is decomposed to CaO during extraction of iron from its oxides.

(b) In the extraction of sliver, sliver is extracted as an anionic complex.

(c) Nickel is purified by Mond's process.

(d) Zr and Ti are purified by Van Arkel method.

Option 1:
a,b,c and d

Option 2:
a,c and d only

Option 3:
b,c, and d only

Option 4:
c and d only

Correct Answer:
a,b,c and d

Solution:
$(a)\ CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ { In Blast furnace } lime stone

(b) Ag form cyanide complex $[Ag(CN)_2]^-$ during cyaride process
$Ag/Ag_2S + 2CN^- \rightarrow [Ag(CN)_2]^-$

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(c) Ni is purified by Mond's process.

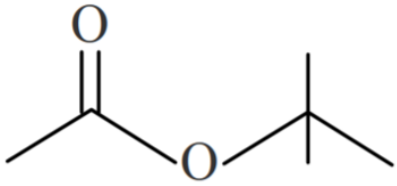
(d) Zr and Ti are purified by Van Arkel method.

All (a), (b), (c), (d) are correct statements.

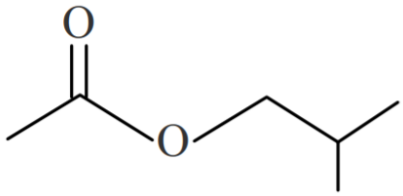
Therefore, the correct option is (1).

Q. 12 An organic compound (A) (molecular formula $C_6H_{12}O_2$) was hydrolysed with dil. H_2SO_4 to give carboxylic acid (B) and an alcohol (C). 'C' gives white turbidity immediately when treated with anhydrous $ZnCl_2$ and conc. HCl. The organic compound (A) is:

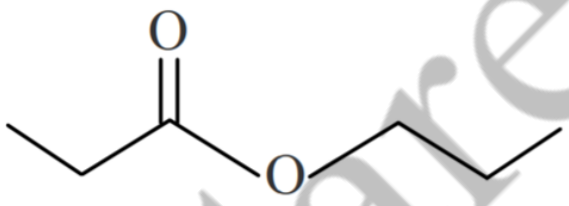
Option 1:



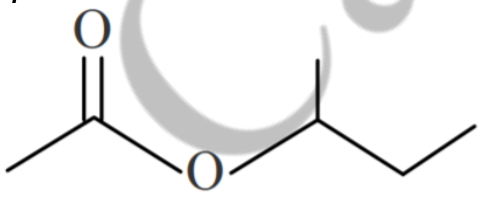
Option 2:



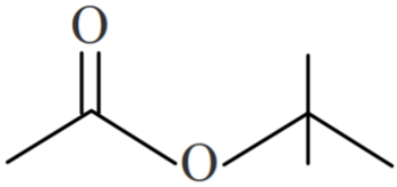
Option 3:



Option 4:



Correct Answer:



Solution:
Tertiary Alcohol gives white turbidity immediately when treated with anhydrous $ZnCl_2$ and conc. HCl.

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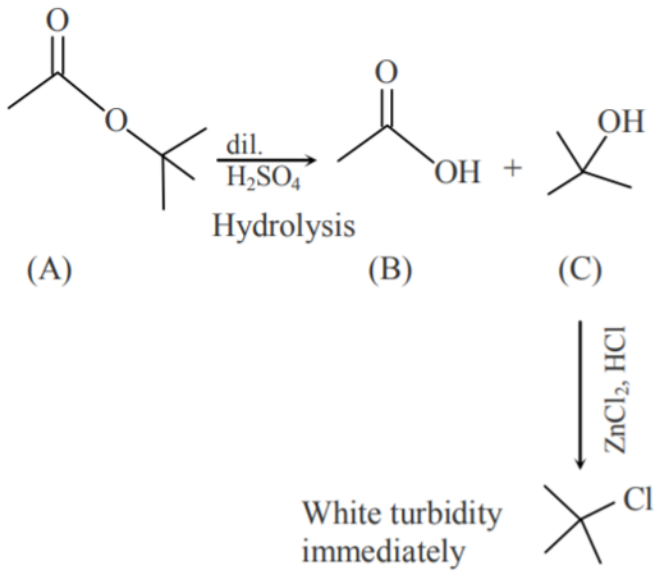
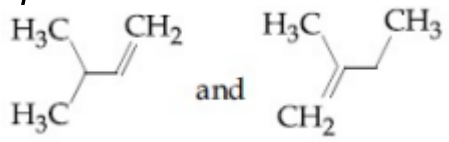
Only option 1 will give Tertiary Alcohol.



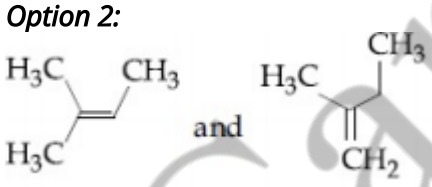
Therefore, the correct option is (1).

Q. 13 When neopentyl alcohol is heated with an acid, it slowly converted into a 85:15 mixture of alkenes A and B, respectively. What are these alkenes?

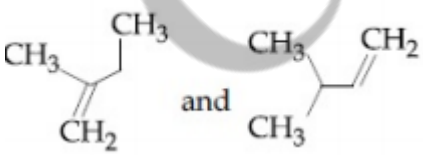
Option 1:



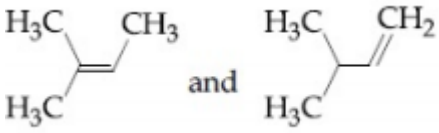
Option 2:



Option 3:



Option 4:



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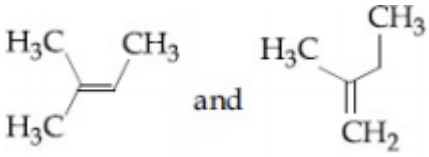
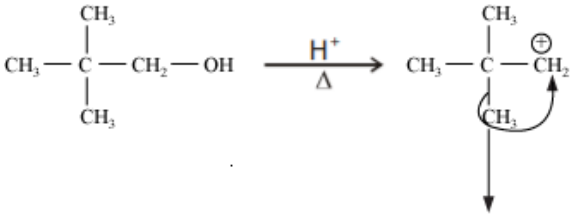
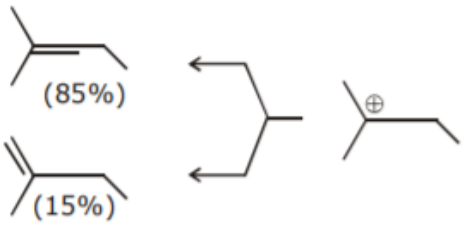
Correct Answer:



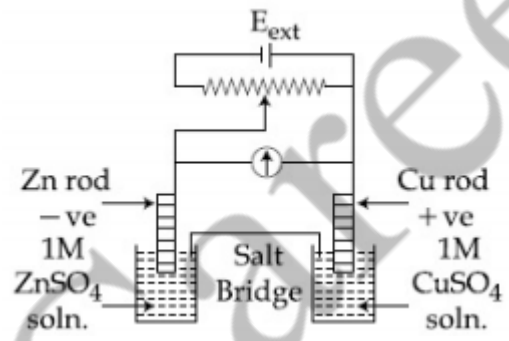
Solution:
The reaction of neopentyl alcohol heated with an acid is,





Therefore, the correct option is (2).

Q. 14



$E^0_{Cu^{2+}|Cu} = +0.34\ V$

$E^o_{Zn^{2+}|Zn} = -0.76\ V$

Identify the incorrect statement from the options below for the above cell:

Option 1:
If E_{ext} > 1.1 V, e^- flows from Cu to Zn

Option 2:
If E_{ext} > 1.1 V, Zn dissolves at Zn electrode and Cu deposits at Cu electrode

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Option 3:
If E_{ext} < 1.1 V Zn dissolves at anode and Cu deposits at cathode

Option 4:
If E_{ext} = 1.1 V, no flow of e^- or current occurs

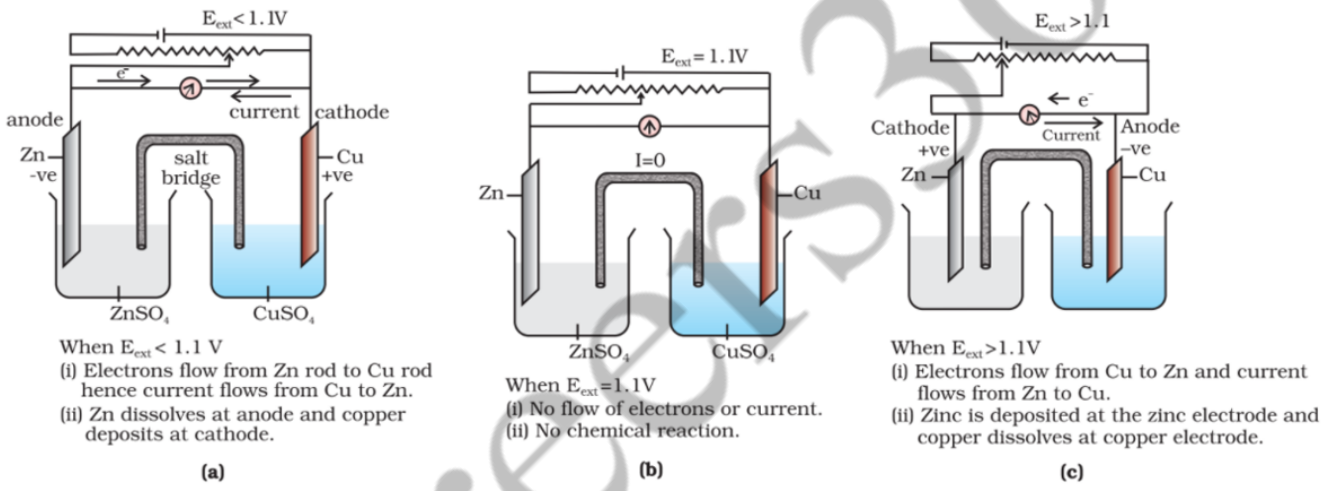
Correct Answer:
If E_{ext} > 1.1 V, Zn dissolves at Zn electrode and Cu deposits at Cu electrode

Solution:
We know,

$$E^o_{cell} = E^o_{right} - E^o_{left}$$

$$E^o_{cell} = 0.34 - (-0.76)$$

$$E^o_{cell} = 1.10 \text{ volt}$$

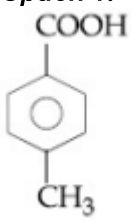


Incorrect statement - If E_{ext} > 1.1 V, Zn dissolves at Zn electrode and Cu deposits at Cu electrode

Therefore, the correct option is (2).

Q. 15 [P] on the treatment with $Br_2/FeBr_2$ in CCl_4 produced a single isomer $C_8H_7O_2Br$ while heating [P] with sodalime gave toluene. The compound [P] is:

Option 1:



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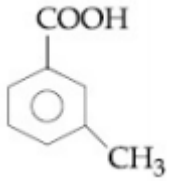
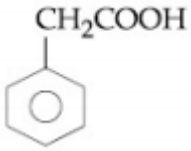
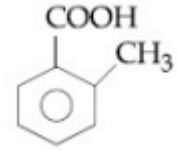
Option 2:



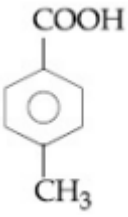
Option 3:



Option 4:

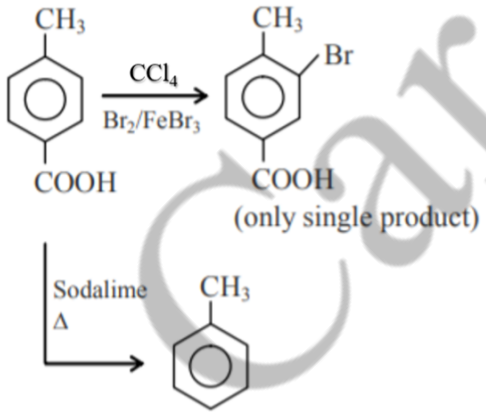


Correct Answer:



Solution:

P will be p-Methylbenzoic acid. The reaction will be -



Therefore, the correct option is (1).

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Q. 16 Match the following:

(i) Foam	(a) smoke
(ii) Gel	(b) cell fluid
(iii) Aerosol	(c) jellies
(iv) Emulsion	(d) rubber
	(e) froth
	(f) milk

Option 1:
i-b, ii-c, iii-e,iv-d

Option 2:
i-d,-ii-b,iii-a,iv-e

Option 3:
i-e,ii-c,iii-a,iv-f

Option 4:
i-d, ii-b ,iii-e, iv-f

Correct Answer:
i-e,ii-c,iii-a,iv-f

Solution:
We know,

Foam → Froth
Gel → Jellies
Aerosol → Smoke
Sol → Cell fluids
Solid sol → rubber

The correct match will be -

i-e,ii-c,iii-a,iv-f

Therefore, the correct option is (3).

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Q. 17 The pair in which both the species have the same magnetic moment (spin only) is:

Option 1:
$[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$

Option 2:
$[Co(OH)_4]^{2-}$ and $[Fe(NH_3)_6]^{2+}$

Option 3:
$[Mn(H_2O)_6]^{2+}$ and $[Cr(H_2O)]^{2+}$

Option 4:
$[Cr(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$

Correct Answer:
$[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$

Solution:
For solving need to know the concept of WFL (weak field ligand) and SFL (strong field ligand).

Electronic configuration and unpaired electrons of given complexes-

Complex	e^- configuration	no. of unpaired e^-
$[Mn(H_2O)_6]^{2+}$	↑ ↑ eg	5
WFL	↑ ↑ ↑ t_2g	
$[Cr(H_2O)_6]^{2+}$	↑ eg	4
WFL	↑ ↑ ↑	
$[COCl_4]^{2-}$	↑ ↑ ↑ t_2	3
Tetrahedral	↑↓ ↑↓ e	
$[Fe(H_2O)_6]^{2+}$	↑ ↑ eg	4
WFL	↑↓ ↑ ↑ t_2g	
$[Co(OH)_4]^{2-}$	↑ ↑ ↑ t_2	3
WFL	↑↓ ↑↓ e	
Tetrahedral	↑ ↑	4
$[Fe(NH_3)_6]^{2+}$	↑↓ ↑ ↑	

Thus complex $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$ have the same no. of unpaired e^- and hence same magnetic moment (spin only).

Therefore, the correct option is (1).

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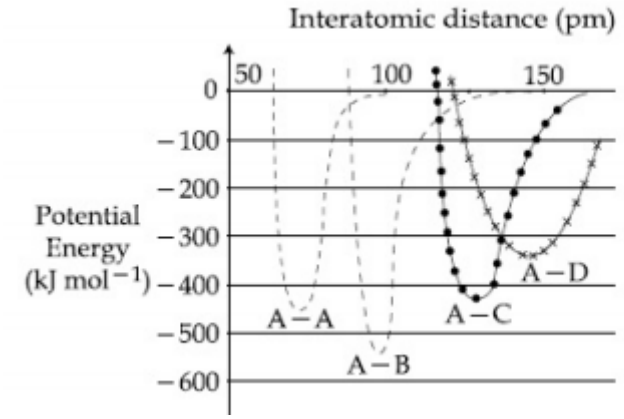
Q. 18 The intermolecular potenial energy for the molecules A,B,C and D given below suggest that:



Option 1:
A-D has the shortest bond length.

Option 2:
A-A has the largest bond enthalpy.

Option 3:
D is more electronegative than other atoms.

Option 4:
A-B has the stiffest bond.

Correct Answer:
A-B has the stiffest bond.

Solution:
From the given graph, potential energy of A-B molecule is minimum. Thus A-B bond is most stable and have strongest bond amongst these.

B → Most electronegative

D → Least electronegative

A-A → Shortest bond length

A-B → Largest bond enthalpy

Therefore, the correct option is (4).

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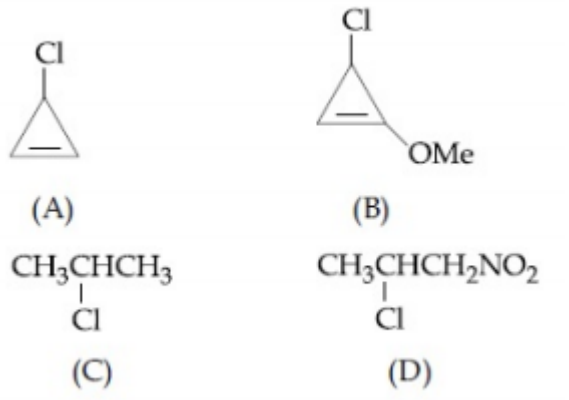
Q. 19 The decreasing order of reactivity of the following organic molecules towards $AgNO_3$ solution is:



Option 1:
(C) > (D) > (A) > (B)

Option 2:
(A) > (B) > (D) > (C)

Option 3:
(A) > (B) > (C) > (D)

Option 4:
(B) > (A) > (C) > (D)

Correct Answer:
(B) > (A) > (C) > (D)

Solution:
The reactions of organic molecules towards $AgNO_3$ solution are

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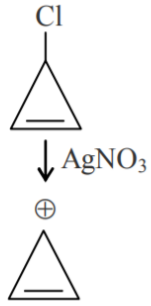
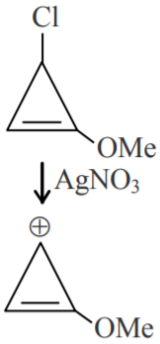
(A) 

Aromatic

(B) 

Aromatic + OMe increases stability

(C) $CH_3 - \underset{\underset{Cl}{|}}{CH} - CH_3 \xrightarrow{AgNO_3} CH_3 - \overset{\oplus}{C}H - CH_3$
(2° Cation)

(D) $CH_3 - \underset{\underset{Cl}{|}}{CH} - CH_2 - NO_2 \xrightarrow{AgNO_3} CH_3 - \overset{\oplus}{C}H - CH_2 - NO_2$
2° Cation and (–I of NO_2)

Order or stability (B) > (A) > (C) > (D)

Therefore, the correct option is (4).

Q. 20 The ionic radii of O^{2-}, F^-, Na^+ and Mg^{2+} are in order:

Option 1:
$F^- > O^{2-} > Na^+ > Mg^{2+}$

Option 2:
$O^{2-} > F^- > Na^+ > Mg^{2+}$

Option 3:
$Mg^{2+} > Na^+ > F^- > O^{2-}$

Option 4:
$O^{2-} > F^- > Mg^{2+} > Na^+$

Correct Answer:
$O^{2-} > F^- > Na^+ > Mg^{2+}$

Solution:
We know these about ions-

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	O^{-2}	F^-	Na^+	Mg^{2+}
z	8	9	11	12
e^-	10	10	10	10
$\frac{z}{e}$	0.8	0.9	1.1	1.2

We know as (z/e) ratio increases size decreases.

Thus correct ionic radii order is

$$O^{-2} > F^- > Na^+ > Mg^{2+}$$

Therefore, the correct option is (2).

Q. 21 A 20.0 mL solution containing 0.2 g impure H_2O_2 reacts completely with 0.316 g of $KMnO_4$ in acidic solution. The purity of H_2O_2 (in%) is _____ (mol.wt.of H_2O_2 =34, mol. wt. of $KMnO_4$ = 158)

Correct Answer:
85

Solution:
The reaction will be-

$$H_2O_2 + KmnO_4 \rightarrow Mn^{+2} + O_2$$

In above-balanced reaction

Electron transfer of H_2O_2 (n) = 2 (-2 to 0 of Oxygen)

Electron transfer of $KmnO_4$ (n) = 5 (+7 to +2 of Mn)

We know this formula:

$$\text{Number of Eq. of } H_2O_2 = \text{Number of Eq. of } KMnO_4$$

(Mole X n-factor) of H_2O_2 = (Mole X n-factor) of $KmnO_4$

$$x \times 2 = \frac{0.316}{158} \times 5$$

$$x = 5 \times 10^{-3} \text{ mol}$$

Now

Mass = Mole X Molar mass

$$\text{mass}_{H_2O_2} = 5 \times 10^{-3} \times 34 = 0.17\text{gm}$$

We know,

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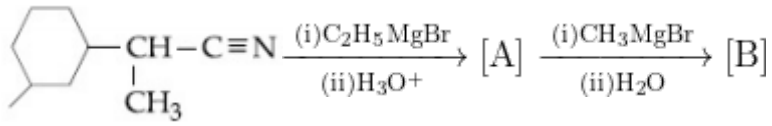
$$\% \text{ purity } = \frac{\text{mass of pure product}}{\text{mass of impure product obtained}} \times 100\%$$

$$\% \text{ purity of } H_2O_2 = \frac{0.17}{0.2} \times 100 = 85\ \%$$

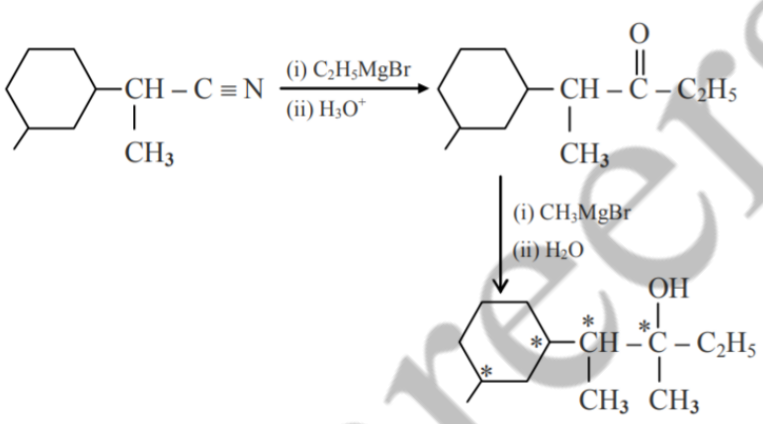
Ans = 85

Q. 22 The number of Chiral centres present in [B] is _____



Correct Answer:
4

Solution:
The reaction will be -



From the above reaction in product B, 4 chiral centers are present.

Ans = 4

Q. 23 The mass of ammonia in grams produced when 2.8 kg of dinitrogen quantitatively reacts with 1 kg of dihydrogen is_____

Correct Answer:
3400

Solution:
Given,

2.8 kg of dinitrogen quantitatively reacts with 1 kg of dihydrogen.

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N_2	+	$3H_2$	→	$2NH_3$
$\frac{2.8}{28}$ K mol		$\frac{1}{2}$ Kmol		
= 0.1 K mol		0.5 K mol		–
0		0.2 K mol		0.2 K mol

From above, the mole of ammonia = 0.2 kMol

So, mass of ammonia = mole X molar mass

mass of ammonia = 0.2 (kMol) X 17 (g/Mol)

mass of ammonia = 3.4 kg = 3400 g

Ans = 3400

Q. 24 In 75% of a first order reaction was completed in 90 minutes, 60% of the same reaction would be completed in approximately (in mimutes) _____

(Take log 2= 0.30; log 2.5 = 0.40)

Correct Answer:
60

Solution:
We know this formula for first-order reaction.

$$\log \frac{[A]_0}{[A]} = \frac{kt}{2.303}$$

75% of reaction is completed in 90 min means 25% of $[A]_o$ is left

So, $[A] = 25\%$ of $[A]_o = 0.25\ [A]_o$

So,

$$\log \frac{[A]_o}{0.25 \times [A]_o} = \frac{k \times 90}{2.303}$$

$$\log \frac{[A]_o}{0.25 \times [A]_o} = \frac{k \times 90}{2.303}$$

$$\log \frac{10}{2.5} = \frac{k \times 90}{2.303}$$

$$\log 10 - \log\ 2.5 = \frac{k \times 90}{2.303}$$

$$1 - 0.4 = \frac{k \times 90}{2.303}$$

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$$k = \frac{0.6 \times 2.303}{90}$$

Now, 60% of the same reaction means k will be the same, and 40% of $[A]_o$ is left.

So,

$$\log \frac{[A]_o}{0.4 \times [A]_o} = \frac{k \times t}{2.303}$$

$$\log \frac{10}{0.4} = \frac{k \times t}{2.303}$$

$$\log 10 - 2\log 2 = \frac{k \times t}{2.303}$$

$$1 - 0.6 = \frac{k \times t}{2.303}$$

$$0.4 = \frac{\frac{0.6 \times 2.303}{90} \times t}{2.303}$$

$$t = \frac{0.4 \times 90}{0.6} = 60 \text{ min}$$

Ans = 60

Q. 25 A 300 K, the vapour pressure of a solution containing 1 mole of n-hexane and 3 moles of n-heptane is 550 mm of Hg. At the same temperature, if one more mole of n-heptane is added to this solution, the vapour pressure of the solution increases by 10mm of Hg. What is the vapour pressure in mm Hg of n-heptane in its pure state _____?

Correct Answer:
600

Solution:
At 300 K, the vapour pressure of a solution containing 1 mole of n-hexane and 3 moles of n-heptane is 550 mm of Hg.

We know,

$$P_t = P_A^o \times X_A + P_B^o \times X_B$$

1) 1 mole of n-hexane and 3 moles of n-heptane is 550 mm of Hg.

So,

$$550 = P_A^o \times \frac{1}{4} + P_B^o \times \frac{3}{4}$$

$$2200 = P_A^o + 3P^o{}_B \quad \text{.....(i)}$$

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2) At the same temperature, if one more mole of n-heptane is added to this solution, the vapour pressure of the solution increases by 10 mm of Hg.

So,

$$550 + 10 = \mathrm{P_A^o} \times \frac{1}{4+1} + \mathrm{P_B^o} \times \frac{3+1}{4+1}$$

$$2800 = \mathrm{P_A^o} + 4\mathrm{P^o}_{\mathrm{B}} \quad \text{.....(ii)}$$

From (i) and (ii),

$$\mathrm{P_B^o} = 600, \mathrm{P_A^o} = 400$$

The vapour pressure in mm Hg of n-heptane in its pure state is 600 mm Hg.

Ans =600

Maths

Q. 1 If $1 + (1 - 2^2.1) + (1 - 4^2.3) + (1 - 6^2.5) + ... + (1 - 20^2.19) = \alpha - 220\beta$, then an ordered pair (α, β) is equal to:

Option 1:
(10,97)

Option 2:
(11, 103)

Option 3:
(10,103)

Option 4:
(11, 97)

Correct Answer:
(11, 103)

Solution:
As we have learnt,

$$\sum_{r=1}^{n} n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{r=1}^{n} n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Now,

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$$1+\left(1-2^2\cdot 1\right)+\left(1-4^2\cdot 3\right)+\ldots..+\left(1-20^2\cdot 19\right)$$
$$=11-\left(2^2\cdot 1+4^2\cdot 3+\ldots..+20^2\cdot 19\right)$$
$$=11-2^2\cdot\sum_{r=1}^{10}r^2(2r-1)=11-4\left(\sum_{r=1}^{10}2r^3-\sum_{r=1}^{10}r^2\right)$$
$$=11-4\left(\frac{110^2}{2}-35\times 11\right)$$
$$=11-220(103)$$
$$\Rightarrow \alpha=11,\beta=103$$

Q. 2 Let $y=y(x)$ be the solution of the differential equation.

$xy'-y=x^2(x\cos x+\sin x), x>0$. If $y(\pi)=\pi$, then $y''\left(\frac{\pi}{2}\right)+y\left(\frac{\pi}{2}\right)$ is equal to:

Option 1:
$2+\frac{\pi}{2}$

Option 2:
$1+\frac{\pi}{2}+\frac{\pi^2}{4}$

Option 3:
$2+\frac{\pi}{2}+\frac{\pi^2}{4}$

Option 4:
$1+\frac{\pi}{2}$

Correct Answer:
$2+\frac{\pi}{2}$

Solution:
$$x\frac{dy}{dx}-y=x^2(x\cos x+\sin x), x>0$$
$$\frac{dy}{dx}-\frac{y}{x}=x(x\cos x+\sin x)\Rightarrow \frac{dy}{dx}+Py=Q$$
$$\text{so, I.F.}=e^{\int -\frac{1}{x}dx}=\frac{1}{|x|}=\frac{1}{x}(x>0)$$
$$\text{Thus, }\frac{y}{x}=\int\frac{1}{x}(x(x\cos x+\sin x))dx$$
$$\Rightarrow \frac{y}{x}=x\sin x+C$$

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$\because y(\pi) = \pi \Rightarrow C = 1$

$\text{so, } y = x^2 \sin x + x \Rightarrow (y)_{\pi/2} = \frac{\pi^2}{4} + \frac{\pi}{2}$

$\text{Also, } \frac{dy}{dx} = x^2 \cos x + 2x \sin x + 1$

$\Rightarrow \frac{d^2y}{dx^2} = -x^2 \sin x + 4x \cos x + 2 \sin x$

$\Rightarrow \left.\frac{d^2y}{dx^2}\right|_{\frac{\pi}{2}} = -\frac{\pi^2}{4} + 2$

$\text{Thus, } y_{(\frac{\pi}{2})} + \frac{d^2y}{dx^2}\,|_{\frac{\pi}{2}} = \frac{\pi}{2} + 2$

Q. 3 The value of $\sum_{r=0}^{20} {}^{50-r}C_6$ is equal to:

Option 1:
$^{51}C_7 - {}^{30}C_7$

Option 2:
$^{50}C_7 - {}^{30}C_7$

Option 3:
$^{50}C_6 - {}^{30}C_6$

Option 4:
$^{51}C_7 + {}^{30}C_7$

Correct Answer:
$^{51}C_7 - {}^{30}C_7$

Solution:
As we have learnt

$^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r$

Now,

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$$\sum_{r=0}^{20} {}^{50-r}C_6 = {}^{50}C_6 + {}^{49}C_6 + {}^{48}C_6 + \ldots + {}^{30}C_6$$

$$= {}^{50}C_6 + {}^{49}C_6 + \ldots.. + {}^{31}C_6 + \left({}^{30}C_6 + {}^{30}C_7\right) - {}^{30}C_7$$

$$= {}^{50}C_6 + {}^{49}C_6 + \ldots.. + \left({}^{31}C_6 + {}^{31}C_7\right) - {}^{30}C_7$$

$$= {}^{50}C_6 + {}^{50}C_7 - {}^{30}C_7$$

$$= {}^{51}C_7 - {}^{30}C_7$$

Q. 4 Two vertical poles AB=15m and CD=10m are standing apart on a horizontal ground with the points A and C on the ground. If P is the point of intersection of BC and AD, then the height of P (in m) above the line AC is:

Option 1:
20/3

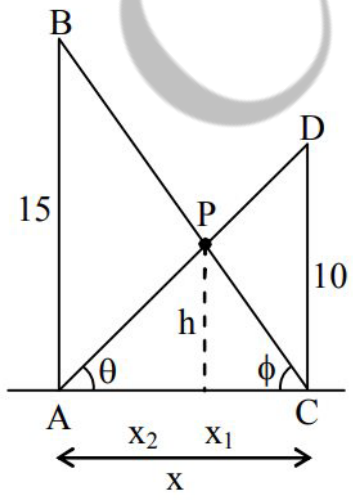
Option 2:
5

Option 3:
10/3

Option 4:
6

Correct Answer:
6

Solution:



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$$\tan\theta = \frac{10}{x} = \frac{h}{x_2} \Rightarrow x_2 = \frac{hx}{10}$$

$$\tan\phi = \frac{15}{x} = \frac{h}{x_1} \Rightarrow x_1 = \frac{hx}{15}$$

$$\text{Now, } x_1 + x_2 = x = \frac{hx}{15} + \frac{hx}{10}$$

$$\Rightarrow 1 = \frac{h}{10} + \frac{h}{15} \Rightarrow h = 6$$

Q. 5 A triangle ABC lying in the first quadrant has two vertices as A (1,2) and B(3,1). If $\angle BAC = 90°$, and ar $(\Delta ABC) = 5\sqrt{5}$ sq. units, then the abscissa of vertex C is:

Option 1:
$1 + \sqrt{5}$

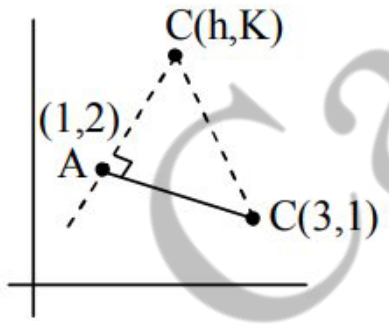
Option 2:
$1 + 2\sqrt{5}$

Option 3:
$2 + \sqrt{5}$

Option 4:
$2\sqrt{5} - 1$

Correct Answer:
$1 + 2\sqrt{5}$

Solution:



$$\left(\frac{K-2}{h-1}\right)\left(\frac{1-2}{3-1}\right) = -1 \Rightarrow K = 2\ h$$

$$\sqrt{5}|\ h-1| = 10$$

$$\because [\Delta ABC] = 5\sqrt{5}$$

$$\Rightarrow \frac{1}{2}(\sqrt{5})\sqrt{(h-1)^2 + (K-2)^2} = 5\sqrt{5}$$

$$\Rightarrow h = 2\sqrt{5} + 1(\ h > 0)$$

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Q. 6 The mean and variance of 8 observations are 10 and 13.5, respectively. If 6 of these observations are 5,7,10,12,14,15, then the absolute difference of the remaining two observations is:

Option 1:
9

Option 2:
5

Option 3:
3

Option 4:
7

Correct Answer:
7

Solution:

$\overline{x} = 10$

$\Rightarrow \bar{x} = \dfrac{63 + a + b}{8} = 10$

$\Rightarrow a + b = 17 \qquad \ldots(1)$

Since, variance is independent of origin. So, we subtract 10 from each observation. So,

$\sigma^2 = 13.5 = \dfrac{79 + (a-10)^2 + (b-10)^2}{8} - (10-10)^2$

$\Rightarrow a^2 + b^2 - 20(a+b) = -171$

$\Rightarrow a^2 + b^2 = 169 \qquad \ldots(2)$

From (1) & (2);

$a = 12 \; and \; b = 5$

Q. 7 Let [t] denote the greatest interger $\leq t$. Then the equation in x, $[x]^2 + 2[x+2] - 7 = 0$ has:

Option 1:
exactly two solutions

Option 2:
exactly four intergral solution

Option 3:
no intergal solution

Option 4:
infinitely many solutions

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Correct Answer:
infinitely many solutions

Solution:
$[x]^2 + 2[x+2] - 7 = 0$
$\Rightarrow [x]^2 + 2[x] + 4 - 7 = 0$
$\Rightarrow [x] = 1, -3$
$\Rightarrow x \in [1, 2) \cup [-3, -2)$

Q. 8 The integral $\int \left(\frac{x}{x\sin + \cos x}\right)^2 dx$ is equal to (where C is constant of integration):

Option 1:
$\tan x - \frac{x \sec x}{x \sin x + \cos x} + C$

Option 2:
$\sec x \frac{x \tan x}{x \sin x + \cos x} + C$

Option 3:
$\sec x - \frac{x \tan x}{x \sin x + \cos x} + C$

Option 4:
$\tan x + \frac{x \sec x}{x \sin x + \cos x} + C$

Correct Answer:
$\tan x - \frac{x \sec x}{x \sin x + \cos x} + C$

Solution:
$$\int \left(\frac{x}{x\sin x + \cos x}\right)^2 dx = \int \left(\frac{x}{\cos x}\right) \cdot \frac{x \cos x dx}{(x \sin x + \cos x)^2}$$
$$= \frac{x}{\cos x}\left(-\frac{1}{x \sin x + \cos x}\right) + \int \left(\frac{\cos x + x \sin x}{\cos^2 x}\right)\left(\frac{1}{x \sin x + \cos x}\right) dx$$
$$= -\frac{x \sec x}{x \sin x + \cos x} + \int \sec^2 x dx$$
$$= -\frac{x \sec x}{x \sin x + \cos x} + \tan x + C$$

Q. 9 If $A = \begin{bmatrix} \cos\theta & i \sin\theta \\ i \sin\theta & \cos\theta \end{bmatrix}, \left(\theta = \frac{\pi}{24}\right)$ and $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $i = \sqrt{-1}$, then which of the following is not true?

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Option 1:
$0 \leq a^2 + b^2 \leq 1$

Option 2:
$a^2 - d^2 = 0$

Option 3:
$a^2 - c^2 = 1$

Option 4:
$a^2 - b^2 = \frac{1}{2}$

Correct Answer:
$a^2 - b^2 = \frac{1}{2}$

Solution:
$A^2 = \begin{pmatrix} \cos 2\theta & i\sin 2\theta \\ i\sin 2\theta & \cos 2\theta \end{pmatrix}$

Similarly, $A^5 = \begin{pmatrix} \cos 5\theta & i\sin 5\theta \\ i\sin 5\theta & \cos 5\theta \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$(1) a^2 + b^2 = \cos^2 5\theta - \sin^2 5\theta = \cos 10\theta = \cos 75^\circ$

$(2) a^2 - d^2 = \cos^2 5\theta - \cos^2 5\theta = 0$

$(3) a^2 - b^2 = \cos^2 5\theta + \sin^2 5\theta = 1$

$(4) a^2 - c^2 = \cos^2 5\theta + \sin^2 5\theta = 1$

Q. 10 Let x_0 be the point of local maxima of $f(x) = \vec{a}.\left(\vec{b} \times \vec{c}\right)$, where

$\vec{a} = x\hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = -2\hat{i} + x\hat{j} - \hat{k}$ and $\vec{c} = 7\hat{i} - 2\hat{i} + x\hat{k}$. Then the value of $\vec{a}.\vec{b} + \vec{b}.\vec{c} + \vec{c}.\vec{a}$ at $x = x_0$ is:

Option 1:
-4

Option 2:
-30

Option 3:
14

Option 4:
-22

Correct Answer:
-22

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Solution:

$$f(x) = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} x & -2 & 3 \\ -2 & x & -1 \\ 7 & -2 & x \end{vmatrix} = x^3 - 27x + 26$$

$f'(x) = 3x^2 - 27 = 0 \Rightarrow x = \pm 3$

and $f''(-3) < 0$

$\Rightarrow$ local maxima at $x = x_0 = -3$

Thus, $\overrightarrow{a} = -3\hat{i} - 2\hat{j} + 3\hat{k}$, $\overrightarrow{b} = -2\hat{i} - 3\hat{j} - \hat{k}$ and $\overrightarrow{c} = 7\hat{i} - 2\hat{j} - 3\hat{k}$

$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 9 - 5 - 26 = -22$

Q. 11 If $(a + \sqrt{2}b\cos x)(a - \sqrt{2}b\cos y) = a^2 - b^2$, where a > b>0, then $\frac{dx}{dy}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ is:

Option 1:

$\frac{a - 2b}{a + 2b}$

Option 2:

$\frac{a - b}{a + b}$

Option 3:

$\frac{a + b}{a - b}$

Option 4:

$\frac{2a + b}{2a - b}$

Correct Answer:

$\frac{a + b}{a - b}$

Solution:

$(a + \sqrt{2}b\cos x)(a - \sqrt{2}b\cos y) = a^2 - b^2$

$\Rightarrow a^2 - \sqrt{2}ab\cos y + \sqrt{2}ab\cos x - 2b^2\cos x\cos y = a^2 - b^2$

Differentiating both sides :

$$0 - \sqrt{2}\text{ab}\left(-\sin \text{y}\frac{\text{dy}}{\text{dx}}\right) + \sqrt{2}\text{ab}(-\sin \text{x})$$

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$$-2b^2\left[\cos x\left(-\sin y\frac{dy}{dx}\right)+\cos y(-\sin x)\right]=0$$

$$\text{At } \left(\frac{\pi}{4},\frac{\pi}{4}\right):$$

$$\text{ab } \frac{dy}{dx}-ab-2b^2\left(-\frac{1}{2}\frac{dy}{dx}-\frac{1}{2}\right)=0$$

$$\Rightarrow \frac{dx}{dy}=\frac{ab+b^2}{ab-b^2}=\frac{a+b}{a-b};a,b>0$$

Q. 12 Given the following two statements:

$(S_1):(q\wedge p)\rightarrow(p\leftrightarrow\sim q)$ is tautology.

$(S_2):\sim q\wedge(\sim p\leftrightarrow q)$ is fallacy. Then :

Option 1:
both (S_1) and (S_2) are correct

Option 2:
only (S_1) is correct

Option 3:
only (S_2) is correct

Option 4:
both (S_1) and (S_2) are not correct

Correct Answer:
both (S_1) and (S_2) are not correct

Solution:
Let TV of (r) denotes truth value of a statement r.

$$\text{Now, if } TV(p)=TV(q)=T$$
$$\Rightarrow TV\ (S_1)=F$$
$$\text{Also, if } TV(p)=T\&TV(q)=F$$
$$\Rightarrow TV\ (S_2)=T$$

Q. 13 Let $\int(x)=|x-2|$ and $g(x)=f(f(x)),x\epsilon[0,4]$. Then $\int_0^3(g(x)-f(x))dx$ is equal to:

Correct Answer:
1

Solution:

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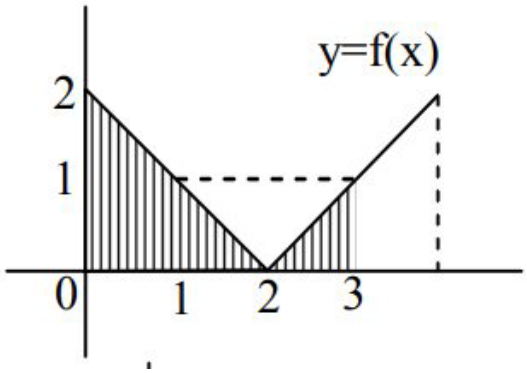
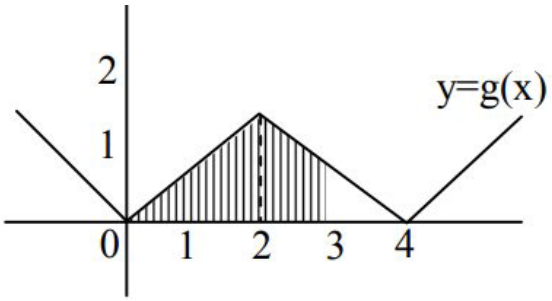
$$\int_0^3 g(x) - f(x) = \int_0^3 ||x-2|-2|dx - \int_0^3 |x-2 \mid dx$$

$$= \left(\frac{1}{2} \times 2 \times 2 + 1 + \frac{1}{2} \times 1 \times 1\right) - \left(\frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 1 \times 1\right)$$

$$= \left(2 + 1 + \frac{1}{2}\right) - \left(2 + \frac{1}{2}\right) = 1$$





Q. 14 Let $u = \frac{2z+i}{z-ki}, z = xiy$ and k>0. If the curve represented by $Re(u) + Im(u) = 1$ intersects the y-axis at the point P and Q where PQ=5, then the value of k is:

Option 1:
3/2

Option 2:
1/2

Option 3:
4

Option 4:
2

Correct Answer:
2

Solution:

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$$u = \frac{2z + i}{z - ki}$$
$$= \frac{2x^2 + (2y+1)(y-k)}{x^2 + (y-k)^2} + i\frac{(x(2y+1) - 2x(y-k)}{x^2 + (y-k)^2}$$

Since Re(u) + Im(u) = 1

$$\Rightarrow 2x^2 + (2y+1)(y-k) + x(2y+1) - 2x(y-k) = x^2 + (y-k)^2$$

Since, the curve intersects the y-axis at the points P and Q

Let P = (0, y_1) and Q = (0, y_2)

$$\Rightarrow y^2 + y - k - k^2 = 0$$
$$y_1 + y_2 = -1$$
$$y_1 y_2 = -k - k^2$$

Given, PQ = 5

$$\Rightarrow |y_1 - y_2| = 5 \Rightarrow k^2 + k - 6 = 0$$
$$\Rightarrow k = -3, 2$$
$$\text{So}, k = 2(k > 0)$$

Q. 15 A survey shows thet 63% of the people in a city read newspaper A whereas 76% read newspaper. If x% of the people read both the newspapers, then a possible value of x cab be:

Option 1:
29

Option 2:
37

Option 3:
65

Option 4:
55

Correct Answer:
55

Solution:
$$n(B) \leq n(A \cup B) \leq n(U)$$
$$\Rightarrow 76 \leq 76 + 63 - x \leq 100$$
$$\Rightarrow -63 \leq -x \leq -39$$
$$\Rightarrow 63 \geq x \geq 39$$

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Q. 16 Let f be a twice differentiable functin on (1,6). If $(2) = 8, f'(2) = 5, f'(x) \geq 1$ and $f''(x) \geq 4$, for all $x\epsilon(1,6)$ then:

Option 1:
$f(5) + f'(5) \leq 26$

Option 2:
$f(5) + f'(5) \geq 28$

Option 3:
$f(5) + f''(5) \leq 20$

Option 4:
$f(5) \leq 10$

Correct Answer:
$f(5) + f'(5) \geq 28$

Solution:
$\mathrm{f}(2) = 8, \mathrm{f}'(2) = 5, \mathrm{f}'(\mathrm{x}) \geq 1, \mathrm{f}''(\mathrm{x}) \geq 4, \forall \mathrm{x} \in (1,6)$

$$f''(x) = \frac{f'(5) - f'(2)}{5-2} \geq 4 \Rightarrow f'(5) \geq 17$$

$$f'(x) = \frac{f(5) - f(2)}{5-2} \geq 1 \Rightarrow f(5) \geq 11$$

$$f'(5) + f(5) \geq 28$$

Q. 17 Let $f(x) = \int \frac{\sqrt{x}}{(1+x)^2} dx (x \geqslant 0)$. Then $f(3) - f(1)$ is equal to:

Option 1:
$-\frac{\pi}{12} + \frac{1}{2} + \frac{\sqrt{3}}{4}$

Option 2:
$\frac{\pi}{6} + \frac{1}{2} + \frac{\sqrt{3}}{4}$

Option 3:
$-\frac{\pi}{6} + \frac{1}{2} + \frac{\sqrt{3}}{4}$

Option 4:
$\frac{\pi}{12} + \frac{1}{2} - \frac{\sqrt{3}}{4}$

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Correct Answer:

$$\frac{\pi}{12}+\frac{1}{2}-\frac{\sqrt{3}}{4}$$

Solution:

$$f(x)=\int_1^3 \frac{\sqrt{x}dx}{(1+x)^2}=\int_1^{\sqrt{3}} \frac{t\cdot 2tdt}{(1+t^2)^2}(\text{ put } \sqrt{x}=t)$$

$$=\left(-\frac{t}{1+t^2}\right)_t^{\sqrt{3}}+\left(\tan^{-1}t\right)_1^{\sqrt{3}}[\text{ Applying by parts }]$$

$$=-\left(\frac{\sqrt{3}}{4}-\frac{1}{2}\right)+\frac{\pi}{3}-\frac{\pi}{4}$$

$$=\frac{1}{2}-\frac{\sqrt{3}}{4}+\frac{\pi}{12}$$

Q. 18 Let $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ be a given ellipse length of whose latus rectum is 10. If its eccentricity is the maximum value of the function, $\phi(t)=\frac{5}{12}+t-t^2$, then a^2+b^2 is equal to:

Option 1:
145

Option 2:
116

Option 3:
126

Option 4:
135

Correct Answer:
126

Solution:

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b);\frac{2b^2}{a}=10\Rightarrow b^2=5a \qquad \ldots(1)$$

$$\text{Now, } \phi(\mathrm{t})=\frac{5}{12}+\mathrm{t}-\mathrm{t}^2=\frac{8}{12}-\left(\mathrm{t}-\frac{1}{2}\right)^2$$

$$\phi(\mathrm{t})_{\max}=\frac{8}{12}=\frac{2}{3}=\mathrm{e}\Rightarrow \mathrm{e}^2=1-\frac{\mathrm{b}^2}{\mathrm{a}^2}=\frac{4}{9} \qquad \ldots(2)$$

from (1) and (2)

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$\Rightarrow a^2 = 81$

So , $a^2 + b^2 = 81 + 45 = 126$

Q. 19 Let P(3,3) be a point on the hyperbola, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal to it at P intersects the x-axis at (9,0) and e is its eccentricity then the ordered pair (a^2, e^2) is equal to:

Option 1:

$\left(\frac{9}{2}, 3\right)$

Option 2:

$\left(\frac{3}{2}, 2\right)$

Option 3:

$\left(\frac{9}{2}, 2\right)$

Option 4:

$(9, 3)$

Correct Answer:

$\left(\frac{9}{2}, 3\right)$

Solution:

Since, $(3, 3)$ lies on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\frac{9}{a^2} - \frac{9}{b^2} = 1 \quad \ldots(1)$

Now, normal at (3, 3) is

$y - 3 = -\frac{a^2}{b^2}(x - 3)$

which passes through (9, 0)

$\Rightarrow b^2 = 2a^2 \quad \ldots(2)$

So, $e^2 = 1 + \frac{b^2}{a^2} = 3$

Also, $a^2 = \frac{9}{2}$(from (1)& (2))

Thus, $(a^2, e^2) = \left(\frac{9}{2}, 3\right)$

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Q. 20 Let α and β be the roots of $x^2 - 3x + p = 0$ and $\gamma \ and \ \delta$ be the roots of $x^2 - 6x + q = 0$. If $\alpha, \beta, \gamma, \delta$ forms a geometric progression. Then the ratio $(2q + p) : (2q - p)$ is :

Option 1:
3:1

Option 2:
9:7

Option 3:
5:3

Option 4:
33:31

Correct Answer:
9:7

Solution:
The general term or n[th] term of a geometric progression is $a_n = ar^{n-1}$

Now,

$x^2 - 3x + p = 0$
roots are α & β
$\alpha + \beta = 3$
$\alpha\beta = p$

$x^2 - 6x + q = 0$
roots are γ & δ
$\gamma + \delta = 6$
$\gamma\delta = q$

α, β, γ & δ are in GP
Let, $\alpha = a, \ \beta = ar, \ \gamma = ar^2$ and $\delta = ar^3$

$a + ar = 3 \qquad \ldots(1)$
$ar^2 + ar^3 = 6 \qquad \ldots(2)$
$(2) \div (1)$
$r^2 = 2$

So, $\dfrac{2q+p}{2q-p} = \dfrac{2r^5 + r}{2r^5 - r} = \dfrac{2r^4 + 1}{2r^4 - 1} = \dfrac{9}{7}$

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Q. 21 Suppose differntial function f(x) satisfies the identity $f(x+y) = f(x) + f(y) + xy^2 + x^2y$, for all real x any y. If $\lim_{x\to 0} \frac{(x)}{x} = 1$ then f'(3) is equal to _____

Correct Answer:
10

Solution:

$$\text{Since, } \lim_{x\to 0} \frac{f(x)}{x} \text{ exist } \Rightarrow f(0) = 0$$

$$\text{Now, } f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(h) + xh^2 + x^2h}{h} (\text{ take } y = h)$$

$$= \lim_{h\to 0} \frac{f(h)}{h} + \lim_{h\to 0}(xh) + x^2$$

$$\Rightarrow f'(x) = 1 + 0 + x^2$$

$$\Rightarrow f'(3) = 10$$

Q. 22 If the system of equations

$$x - 2y + 3z = 0$$

$$2x + y + z = b$$

$$x - 7y + az = 24,$$ has infinitely many solutions, then a-b is equal to_____

Correct Answer:
5

Solution:

$$\text{D} = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & 1 \\ 1 & -7 & 8 \end{vmatrix} = 0 \Rightarrow \text{a} = 8$$

$$\text{also, } \text{D}_1 = \begin{vmatrix} 9 & -2 & 3 \\ \text{b} & 1 & 1 \\ 24 & -7 & 8 \end{vmatrix} = 0 \Rightarrow \text{b} = 3$$

$$\text{hence, a} - \text{b} = 8 - 3 = 5$$

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Q. 23 The probability of a man hitting a target is $\frac{1}{10}$. The least number of shots required, so that the probability of his hitting the target at least once in greater than $\frac{1}{4}$ is_____

Correct Answer:
3

Solution:

$$\text{We have, } 1 - \text{ (probability of all shots result in failure) } > \frac{1}{4}$$

$$\Rightarrow 1 - \left(\frac{9}{10}\right)^{n} > \frac{1}{4}$$

$$\Rightarrow \frac{3}{4} > \left(\frac{9}{10}\right)^{n} \Rightarrow n \geq 3$$

Q. 24 If the equation of a Plane P, passing through the intersection of the planes, $x - 4y - z + 7 = 0$and $3x + y + 5z = 8$ is $ax + by + 6z = 15$ for some $a, b \epsilon R$, then the distance of the point (3,2-1) from the plane P is_____

Correct Answer:
3

Solution:

$$\mathrm{D}_1 = \begin{vmatrix} -7 & 4 & -1 \\ 8 & 1 & 5 \\ 15 & \mathrm{b} & 6 \end{vmatrix} = 0 \Rightarrow \mathrm{b} = -3$$

$$\mathrm{D} = \begin{vmatrix} 1 & 4 & -1 \\ 3 & 1 & 5 \\ \mathrm{a} & \mathrm{b} & 6 \end{vmatrix} = 0 \Rightarrow 21\mathrm{a} - 8\ \mathrm{b} - 66 = 0$$

$$P : 2x - 3y + 6z = 15$$

$$\text{so required distance } = \frac{21}{7} = 3$$

Q. 25 Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to _____

Correct Answer:
8

Solution:

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Given $\left(2x^2+3x+4\right)^{10}=\sum_{r=0}^{20} a_r x^r \quad \ldots(1)$

replace x by $\frac{2}{x}$ in above identity :

$$\frac{2^{10}\left(2x^2+3x+4\right)^{10}}{x^{20}}=\sum_{r=0}^{20}\frac{a_r 2^r}{x^r}$$

$$\Rightarrow 2^{10}\sum^{20} a_r x^r=\sum^{20} a_r 2^r x^{(20-r)} \text{ (from (1))}$$

now, comparing coefficient of $\mathbf{x}^7$ from both sides

(take r = 7 in L.H.S. & r = 13 in R.H.S.)

$$2^{10}a_7=a_{13}2^{13}\Rightarrow\frac{a_7}{a_{13}}=2^3=8$$

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4th September Shift 2

Physics

Q. 1 A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x - direction, where a and E_0 are constants. Initially the particle was at rest at x=0. Other than the initial position the kinetic energy of the particle becomes zero when the distance of the particle from the origin is:

Option 1:
a

Option 2:
$\sqrt{\frac{2}{a}}$

Option 3:
$\sqrt{\frac{3}{a}}$

Option 4:
$\sqrt{\frac{1}{a}}$

Correct Answer:
$\sqrt{\frac{3}{a}}$

Solution:
$E = E_0\left(1 - ax^2\right)$

As we know that force acting on a charge in an electric field is qE and the acceleration will be -

$$a = \frac{qE}{m}$$

$$\begin{aligned} &\frac{vdv}{dx} = \frac{qE_0}{m}\left(1 - ax^2\right) \\ \Rightarrow \quad &\frac{v^2}{2} = \frac{qE_0}{m}\left[x - \frac{ax^3}{3}\right] = 0 \\ \Rightarrow \quad &x = \sqrt{\frac{3}{a}} \end{aligned}$$

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Q. 2 A series L-R circuit is connected to a battery of emf V. If the circuit is switched on at t=0, then the time at which the energy stored in the inductor reaches $\left(\frac{1}{n}\right)$ times of its maximum value, is:

Option 1:
$\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$

Option 2:
$\frac{L}{R}\ln\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)$

Option 3:
$\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}+1}\right)$

Option 4:
$\frac{L}{R}\ln\left(\frac{\sqrt{n}-1}{\sqrt{n}}\right)$

Correct Answer:
$\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$

Solution:

$$\frac{1}{2}\mathrm{Li}^2 = \frac{1}{n}\frac{1}{2}\mathrm{Li}_0^2$$

$$\Rightarrow \quad \mathbf{i} = \frac{\mathbf{i}_0}{\sqrt{\mathbf{n}}}$$

During growth of current,

$$\mathbf{i} = \mathbf{i}_0\left(1-\mathbf{e}^{-t/\tau}\right) \quad \left(\tau = \frac{\mathbf{L}}{\mathbf{R}}\right)$$

$$e^{t/\tau} = \frac{\sqrt{n}}{\sqrt{n}-1}$$

$$t = \frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$$

Q. 3 Consider two uniform discs of the same thickness and different radii R_1=R and $R_2 = \alpha R$ made of the same material. If the ratio of their moments of the interia I_1 and I_2, respectively, about their axes is $I_1 : I_2 = 1 : 16$ then the value of α is:

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Option 1:
$2\sqrt{2}$

Option 2:
$\sqrt{2}$

Option 3:
2

Option 4:
4

Correct Answer:
2

Solution:

Moment of inertia of disc is given by $1 = \frac{MR^2}{2} = \frac{[\rho(\pi R^2)\,t]\,R^2}{2}$

$$\mathrm{I} \propto \mathrm{R}^4$$

$$\frac{\mathrm{I}_2}{\mathrm{I}_1} = \left(\frac{\mathrm{R}_2}{\mathrm{R}_1}\right)^4$$

$$\frac{16}{1} = \alpha^4$$

$$\alpha = 2$$

Q. 4 The electric field of a plane electromagnetic wave is given by

$$\overrightarrow{E} = E_0\,(\hat{x} + \hat{y})\sin(kz - \omega t)$$

Its magnetic field will be given by:

Option 1:
$\frac{E_0}{c}(-\hat{x} + \hat{y})\sin(kz - \omega t)$

Option 2:
$\frac{E_0}{c}(\hat{x} + \hat{y})\sin(kz - \omega t)$

Option 3:
$\frac{E_0}{c}(\hat{x} - \hat{y})\sin(kz - \omega t)$

Option 4:
$\frac{E_0}{c}(\hat{x} - \hat{y})\cos(kz - \omega t)$

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Correct Answer:

$\frac{E_0}{c}(-\hat{x}+\hat{y})\sin(kz-\omega t)$

Solution:

$\vec{\mathbf{E}} \times \vec{\mathbf{B}}$ is along $\vec{\mathbf{C}}$

$\hat{\mathbf{E}} \times \hat{\mathbf{B}} = \hat{\mathbf{C}}$

Since the electric field is given in the question is -

$\vec{E} = E_0\,(\hat{x}+\hat{y})\sin(kz-\omega t)$

So by the cross product we can say that -

$\Rightarrow \vec{\mathbf{B}} = \frac{\mathbf{E}_0}{\mathbf{C}}(-\hat{\mathbf{x}}+\hat{\mathbf{y}})\sin(\mathbf{kz}-\omega\mathbf{t})$

Q. 5 The driver of a bus approching a big wall notices that the frequency of his bus's horn changes from 420Hz to 490Hz when he hears if after it gets reflected from the wall. Find the speed of the bus if speed of the sound is 330ms^{-1}

Option 1:

91 kmh^{-1}

Option 2:

81 kmh^{-1}

Option 3:

61 kmh^{-1}

Option 4:

71 kmh^{-1}

Correct Answer:

91 kmh^{-1}

Solution:

Let the velocity of car be u

Then $490 = \frac{v+u}{v-u}420$ or

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$\frac{330+u}{330-u} \times 420 = 490$

$\Rightarrow \quad \frac{330+u}{330-u} = \frac{7}{6}$

$\Rightarrow \quad 13u = 330$

$\Rightarrow \quad u = \frac{330}{13} = 25.38 \text{ m/s}$

$= 91 \text{ km/h}$

Q. 6 Find the binding energy (in MeV) per neuclon for $^{120}_{50}Sn$. Mass of proton $m_p = 1.00783U$, mass of neutron $m_n = 1.00867U$ and mass of tin nucleus $m_{Sn} = 119.902199U$.

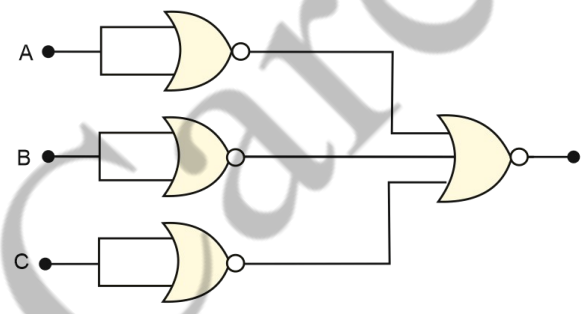
(take 1U = 931 MeV)

Correct Answer:
8.5

Solution:

$$\text{Binding energy per nucleon } = \frac{1020.563131}{120} = 8.504\text{MeV} \approx 8.5\text{MeV}$$

Q. 7 Identify the operation performed by the circuit given below:



Option 1:
NAND

Option 2:
OR

Option 3:
AND

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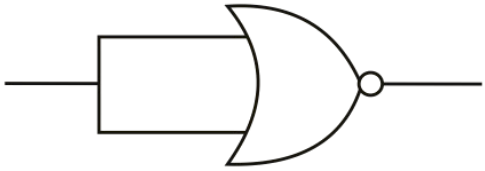
Option 4:
NOT

Correct Answer:
AND

Solution:



Behaves like a not gate so boolen equation will be
$y = \overline{\bar{A} + \bar{B} + \bar{C}}$
$y = A \cdot B \cdot C$
whole arrangement behaves like a AND gate

Q. 8 A capacitor C is fully charged with voltage V_0. After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor $\frac{C}{2}$. The energy loss in the process after the charge is distributed between the two capacitors is:

Option 1:
$\frac{1}{2}CV_0^2$

Option 2:
$\frac{1}{3}CV_0^2$

Option 3:
$\frac{1}{4}CV_0^2$

Option 4:
$\frac{1}{6}CV_0^2$

Correct Answer:
$\frac{1}{6}CV_0^2$

Solution:

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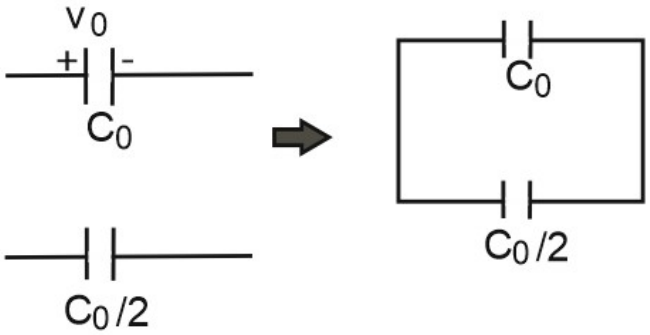


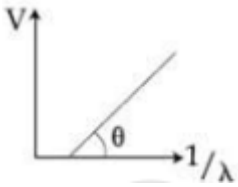
$$\text{Heat loss} =$$

$$H = \frac{C_1 C_2}{2\left(C_1 + C_2\right)}\left(V_1 - V_2\right)^2$$

$$= \frac{C \times \frac{C}{2}}{2\left(C + \frac{C}{2}\right)}\left(V_0 - 0\right)^2 = \frac{C}{6}V_0^2$$

$$H = \frac{1}{6}CV_0^2$$

Q. 9 A photoelectric effect experiment, the graph stopping potential V versus reciprocal of wavelength obtained is shown in the figure. As the intensity of incident radiation is increased:



Option 1:
straight line shifts to right

Option 2:
Slope of the straight line get more steep

Option 3:
straight line shifts to left

Option 4:
graph does not change

Correct Answer:
graph does not change

Solution:

$$\mathrm{eV_s} = \mathrm{hv} - \mathrm{w}$$

$$V_s = \frac{hv}{e} - \frac{w}{e}$$

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Frequency and work function are constant therefore graph does not change.

Q. 10 A circular coil has moment of interia 0.8kg m^2 around any diameter and is carrying current to produce a magnetic moment of 20 Am^2. The coil is kept initially in a vertical position and it can rotate freely around horizontal diameter. When a uniform magnetic field of 4T is applied along the vertical, it starts rotating around its horizontal diameter. The angular speed the coil acquires after rotating by 60° will be:

Option 1:
10 rad s^{-1}

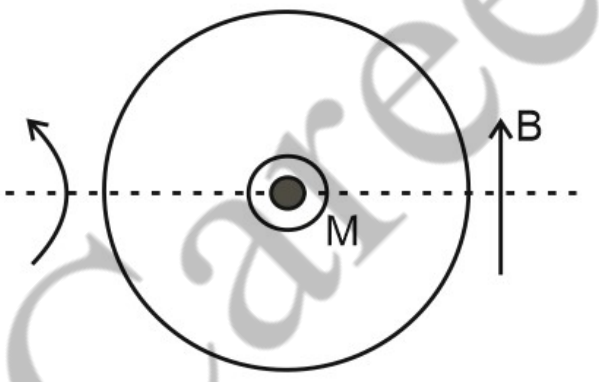
Option 2:
10 πrad s^{-1}

Option 3:
20 πrad s^{-1}

Option 4:
20 rad s^{-1}

Correct Answer:
10 rad s^{-1}

Solution:



$$U_i + K_i = U_f + K_f$$
$$0 + 0 = -MB\cos 30^\circ + \frac{1}{2}I\omega^2$$
$$20 \times 4 \times \frac{\sqrt{3}}{2} = \frac{1}{2}(0.8)\omega^2$$
$$\omega = \sqrt{100\sqrt{3}} = 10(3)^{1/4}$$

So the nearest answer available in the option is the 10 rad/s

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Q. 11 A small ball of mass m is thrown upward with velocity u from the ground. the ball experiences a resistive force mkv^2 where v is its speed. The maximum height attained by the balls is:

Option 1:

$\frac{1}{2k}\tan^{-1}\frac{ku^2}{g}$

Option 2:

$\frac{1}{2}\ln\left(1+\frac{ku^2}{2g}\right)$

Option 3:

$\frac{1}{k}\tan^{-1}\frac{ku^2}{2g}$

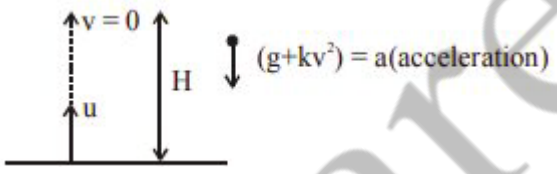
Option 4:

$\frac{1}{2k}\ln\left(1+\frac{ku^2}{g}\right)$

Correct Answer:

$\frac{1}{2k}\ln\left(1+\frac{ku^2}{g}\right)$

Solution:



$\vec{F} = mkv^2 - mg$

$\vec{a} = \frac{\vec{F}}{m} = -\left[kv^2 + g\right]$

$\Rightarrow \quad v \cdot \frac{dv}{dh} = -\left[kv^2 + g\right]$

$\Rightarrow \int_u^0 \frac{v\cdot dv}{kv^2+g} = -\int_0^H dh$

$\frac{1}{2\,K}\ln\left[kv^2+g\right]_u^0 = -H$

$\Rightarrow \frac{1}{2\,K}\ln\left[\frac{ku^2+g}{g}\right] = H$

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Q. 12 Two identical cylindrical vessels are kept on the ground and each contains the same liquid of density d. The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other, x_2. When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is

Option 1:
$gdS\left(x_2^2+x_1^2\right)$

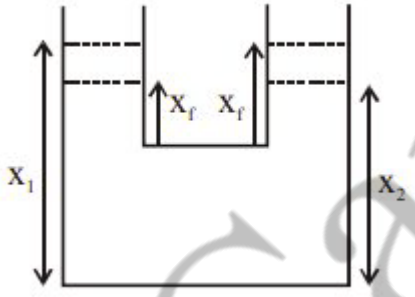
Option 2:
$gdS\left(x_2+x_1\right)^2$

Option 3:
$\frac{3}{4}gdS\left(x_2-x_1\right)^2$

Option 4:
$\frac{1}{4}gdS\left(x_2-x_1\right)^2$

Correct Answer:
$\frac{1}{4}gdS\left(x_2-x_1\right)^2$

Solution:



$$U_i=(\rho S x_1)\,g\cdot\frac{x_1}{2}+(\rho S x_2)\,g\cdot\frac{x_2}{2}$$

$$U_f=(\rho S x_f)\,g\cdot\frac{x_f}{2}\times 2$$

By volume conservation

$$Sx_1+Sx_2=S\,(2x_f)$$

$$x_f=\frac{x_1+x_2}{2}$$

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$$\Delta U = \rho \mathrm{Sg}\left[\left(\frac{x_1^2}{2}+\frac{x_2^2}{2}\right)-x_f^2\right]$$
$$= \rho \mathrm{Sg}\left[\frac{x_1^2}{2}+\frac{x_2^2}{2}-\left(\frac{x_1+x_2}{2}\right)^2\right]$$
$$= \frac{\rho Sg}{2}\left[\frac{x_1^2}{2}+\frac{x_2^2}{2}-x_1x_2\right]$$
$$= \frac{\rho Sg}{4}(x_1-x_2)^2$$
$$= \frac{1}{4}gdS(x_2-x_1)^2$$

Here density is represented as - 'd'

Q. 13 A body is moving in a low circular orbit about a planet of mass M and radius R. The radius of the orbit can be taken to be R itself. Then the ratio of the speed of this body in the orbit to escape velocity from the planet is:

Option 1:
$\frac{1}{\sqrt{2}}$

Option 2:
2

Option 3:
1

Option 4:
$\sqrt{2}$

Correct Answer:
$\frac{1}{\sqrt{2}}$

Solution:
$\mathrm{V_{orbit}} = \sqrt{\frac{\mathrm{GM}}{\mathrm{R}}}$
$\mathrm{V_{escape}} = \sqrt{\frac{2\mathrm{GM}}{\mathrm{R}}}$
$\frac{\mathrm{V_{orbit}}}{\mathrm{V_{escape}}} = \frac{1}{\sqrt{2}}$

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Q. 14 A cube of metal is subjected to a hydrostatic pressure of 4 GPa. The percentage change in the length of the side of the cube is close to:

(Given bulk moodulus of metal, $B = 8 \times 10^{10} Pa$

Option 1:
5

Option 2:
0.6

Option 3:
20

Option 4:
1.67

Correct Answer:
1.67

Solution:

$$\mathrm{B} = \frac{-\frac{\Delta \mathrm{P}}{\Delta \mathrm{V}}}{\mathrm{V}}$$

$$\left|\frac{\Delta \mathrm{V}}{\mathrm{V}}\right| = \frac{\Delta \mathrm{P}}{\mathrm{B}}$$

$$= \frac{4 \times 10^9}{8 \times 10^{10}} = \frac{1}{20}$$

$$\frac{\Delta \ell}{\ell} = \frac{1}{3} \times \frac{\Delta \mathrm{V}}{\mathrm{V}} = \frac{1}{60}$$

$$\text{Percentage change}$$

$$= \frac{\Delta \ell}{\ell} \times 100\%$$

$$= \frac{100}{60}\% = 1.67\%$$

Q. 15 A quantity x is given by (IFv^2/WL^4) in terms of a moment of interia I, force F, velocity v, work W and length L. the dimensional formula for the x is same as that of:

Option 1:
planck's constant

Option 2:
force constant

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Option 3:
Energy density

Option 4:
coefficient of viscosity

Correct Answer:
Energy density

Solution:

$$x = \frac{IFV^2}{WL^4}$$

$$[x] = \frac{[ML^2][MLT^{-2}][LT^{-1}]^2}{[ML^2\ T^{-2}][L]^4}$$

$$[x] = [ML^{-1}\ T^{-2}]$$

$$[\text{Energy density}] = \left[\frac{E}{V}\right]$$

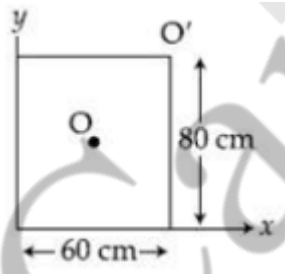
$$= \left[\frac{ML^2\ T^{-2}}{L^3}\right]$$

$$= [ML^{-1}\ T^{-2}]$$

$$\text{Same as } x$$

So answer is Energy density

Q. 16



For a uniform rectangular sheet shown in the figure, the ratio of moments of interia about the axes perpendicular to the sheet and passing through O (the centre of mass) and O' (corner point) is:

Option 1:
$\frac{2}{3}$

Option 2:
$\frac{1}{4}$

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Option 3:

$\frac{1}{8}$

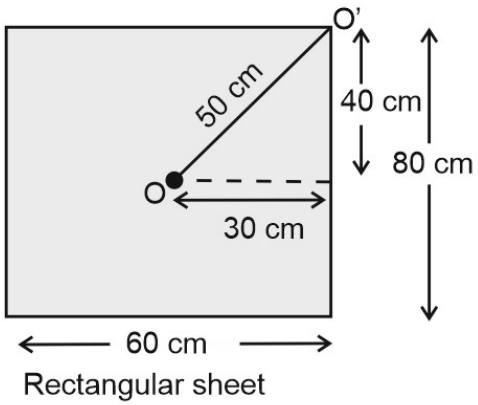
Option 4:

$\frac{1}{2}$

Correct Answer:

$\frac{1}{4}$

Solution:



$$I_O = \frac{M}{12}\left[L^2 + B^2\right] = \frac{M}{12}\left[80^2 + 60^2\right]$$

$$I_{O'} = I_0 + Md^2\{ \text{ parallel axis theorem}$$

$$= \frac{M}{12}\left[80^2 + 60^2\right] + M[50]^2$$

$$\frac{I_O}{I_{O'}} = \frac{M/12\left[80^2+60^2\right]}{\frac{M}{12}[80^2+60^2]+M[50]^2} = \frac{1}{4}$$

Q. 17 Match the thermodynamic process taking place in a system with the correct conditions. In the table: ΔQ is the heat supplied , ΔW is the work done and ΔU is change in internal energy of the system.

Process	Condition
(I) Adiabatic	(A) $\Delta W = 0$
(II) Isothermal	(B) $\Delta Q = 0$
(III) Isochoric	(C) $\Delta U \neq 0, \Delta Q \neq 0, \Delta W \neq 0$
(IV) Isobaric	(D) $\Delta U = 0$

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Option 1:
(I)- (A), (II)-(B), (III) - (D), (IV)-(D)

Option 2:
(I)- (B), (II)-(A), (III) - (D), (IV)-(C)

Option 3:
(I)- (A), (II)-(A), (III) - (C), (IV)-(C)

Option 4:
(I)- (B), (II)-(D), (III) - (A), (IV)-(C)

Correct Answer:
(I)- (B), (II)-(D), (III) - (A), (IV)-(C)

Solution:
(I) Adiabatic process $\Rightarrow \Delta Q = 0$
No exchange of heat takes place with surroundings
(II) Isothermal proess $\Rightarrow$ Temperature remains same

$$\Delta \mathrm{U} = \frac{\mathrm{f}}{2}\mathrm{nR}\Delta \mathrm{T} \Rightarrow \Delta \mathrm{U} = 0$$

No change in intemal energy $[\Delta \mathrm{U} = 0]$

(III) Isochoric process Volume remains constant

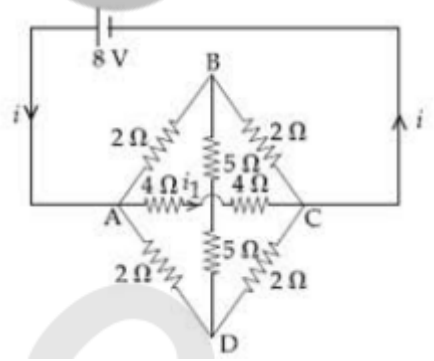
$$\Delta \mathrm{V} = 0$$
$$\mathrm{W} = \int \mathrm{PdV} = 0$$

Hence work done is zero.

(IV) Isobaric process $\Rightarrow$ Pressure remains constant

$$\mathbf{W} = \mathrm{P}\Delta \mathrm{V} \neq 0$$
$$\Delta \mathrm{U} = \frac{\mathrm{f}}{2}\mathrm{nR}\Delta \mathrm{T} = \frac{\mathrm{f}}{2}[\mathrm{P}\Delta \mathrm{V}] \neq 0$$
$$\Delta Q = nC_p\Delta T \neq 0$$

Q. 18 The value of the current i_1 flowing from A to C in the circuit diagram is:



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Option 1:
2 A

Option 2:
4 A

Option 3:
1 A

Option 4:
5 A

Correct Answer:
1 A

Solution:

$$\text{Voltage across } AC = 8V$$

$$\mathrm{R_{AC}} = 4 + 4 = 8\Omega$$

$$i_1 = \frac{V}{R_{AC}} = \frac{8}{8} = 1 \text{ A}$$

Q. 19 A person pushes a box on a rough horizontal surface. He applies a force of 200N over a distance of 15m. Thereafter, he gets progressively tired and his applied force reduces linearly with distance to 100N. The total distance through which the box has been moved is 30m. What is the work done by the person during the total movement of the box?

Option 1:
3280 J

Option 2:
2780 J

Option 3:
5960 J

Option 4:
5250 J

Correct Answer:
5250 J

Solution:

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$F = 200\ \text{N} \quad \text{for } 0 \leq x \leq 15$

$= 200 - \frac{100}{15}(x - 15) \text{ for } \quad 15 \leq x < 30$

$W = \int F dx$

$= \int_0^{15} 200\ dx + \int_{15}^{30} \left(300 - \frac{100}{15}x\right) dx$

$= 200 \times 15 + 300 \times 15 - \frac{100}{15} \times \frac{\left(30^2 - 15^2\right)}{2}$

$= 3000 + 4500 - 2250$

$= 5250\ \text{J}$

Q. 20 A paramagnetic sample shows a net magnetism of 6 A/m when it is placed in an external magnetic field of 0.4T at a temperature of 4K. When the sample is placed in a external magnetic field of 0.3T at a temperature of 24K, then the magnetism will be:

Option 1:
1 A/m

Option 2:
4 A/m

Option 3:
2.25 A/m

Option 4:
0.75 A/m

Correct Answer:
0.75 A/m

Solution:
For paramagnetic material According to curies law

$\chi \propto \frac{1}{T}$

$\chi \propto \frac{1}{T}$

$\Rightarrow \chi_1 T_1 = \chi_2 T_2$

$\Rightarrow \frac{6}{0.4} \times 4 = \frac{I}{0.3} \times 24$

$I = \frac{0.3}{0.4} = 0.75\ \text{A/m}$

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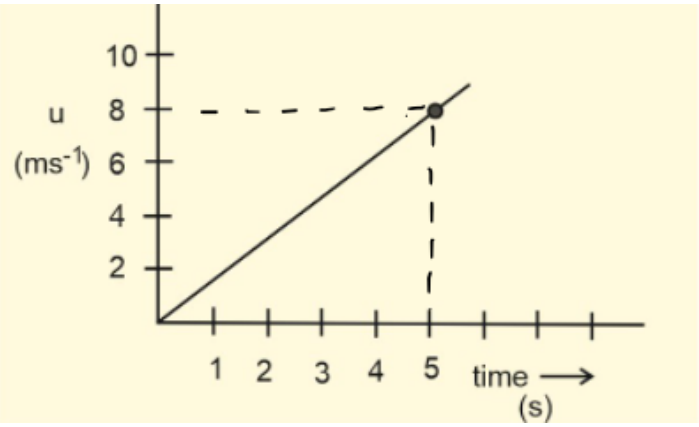
Q. 21 The speed versus time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval t=0 to t=5s will be_____



Option 1:
20

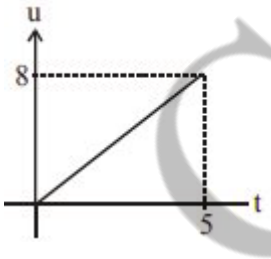
Option 2:
10

Option 3:
30

Option 4:
40

Correct Answer:
20

Solution:



$$\text{Distance} = \int \text{vdt}$$
$$\text{Area under graph} = \frac{1}{2} \times 5 \times 8 = 20$$

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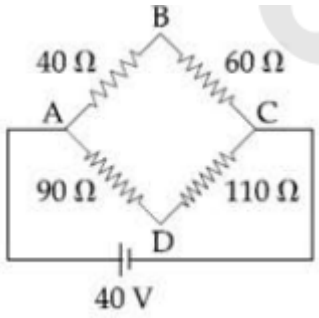
Q. 22

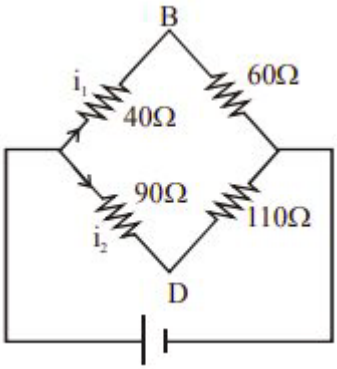


Four resistance 40Ω, 60 Ω, 90Ω and 110Ω make the arms of a quadrilateral ABCD. Across AC is battery of emf 40V and internal resistance negligible. The potential difference across BD in V is_____.

Correct Answer:

2

Solution:



$i_1 = \frac{40}{40+60} = 0.4$

$i_2 = \frac{40}{90+110} = \frac{1}{5}$

$v_B + i_1(40) - i_2(90) = v_D$

$v_B - v_D = \frac{1}{5}(90) - \frac{4}{10} \times 40$

$v_B - v_D = 18 - 16 = 2$

Q. 23 The distance between an object and a screen is 100cm. A lens can be produce real image of the object on the screen for two different positions between the screen and the object. The distance between these two positions is 40cm. if the power of the lens is close to $\left(\frac{N}{100}\right) D$ where N is an integer, the value of N is_____-

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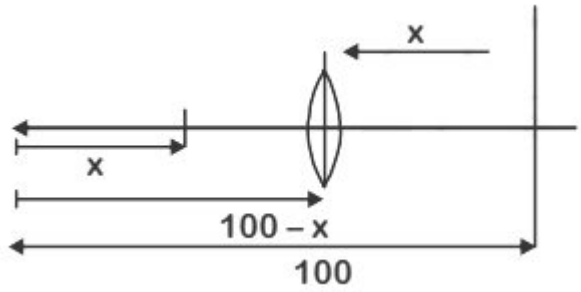
Correct Answer:
476

Solution:



$$\text{Clearly } 100 - x - x = 40$$
$$\Rightarrow \quad 60 = 2x$$
$$\therefore \quad x = 30, \text{ And } 100 - x = 70$$
$$\Rightarrow \quad \tfrac{1}{70} + \tfrac{1}{30} = \tfrac{1}{f}$$
$$\therefore \quad f = \left(\tfrac{21}{100}\right)$$
$$\therefore \quad \text{Power } = \tfrac{1}{f} = \left(\tfrac{100}{21}\right) \approx 4.76$$
$$\text{N/ } 100 = 4.76$$
$$\therefore N = 476$$

Q. 24 The change in the magnitude of the volume of an ideal gas when a small additional pressure ΔP is applied at a constant temperature, is the same as the change when the tempertaure is reduced by a small quanity ΔT at a constant pressure. The initial tempertaure and pressure of the gas were 300K and 2atm, respectively. If $\Delta T = C\,|\Delta P|$ then the value of C in (K/atm)is____

Correct Answer:
150

Solution:
$$\mathrm{PV = nRT}$$
$$\mathrm{P\Delta V + V\Delta P = 0} \quad \text{(for constant temp.)}$$
$$\mathrm{P\Delta V = nR\Delta T} \quad \text{(for constant pressure)}$$
$$\Delta \mathrm{T} = \frac{\mathrm{P\Delta V}}{\mathrm{nR}}$$

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$$\Delta P = -\frac{P\Delta V}{V} \quad (\Delta V \text{ is same in both cases })$$

$$\frac{\Delta T}{\Delta P} = \frac{P\Delta V}{nR}\frac{V}{-P\Delta V} = \frac{-V}{nR} = -\frac{T}{P}$$

$$(PV = nRT)$$

$$\left(\frac{V}{nR} = \frac{T}{P}\right)$$

$$\left|\frac{\Delta T}{\Delta P}\right| = \left|\frac{-300}{2}\right| = 150$$

Q. 25 Orange light of wavelength 6000×10^{-10}m illuminates a single slit of width $0.64 \times 10^{-4} m$. The maximum possible number of diffraction minima produced on both the sides of the central maximum is_____

Correct Answer:
198

Solution:
Condition for minimum,

$$d \sin\theta = n\lambda$$

$$\therefore \sin\theta = \frac{n\lambda}{d} < 1$$

$$n < \frac{d}{\lambda} = \frac{6 \times 10^{-5}}{6 \times 10^{-7}} = 100$$

$\therefore$ Total number of minima on one side $= 99$

Total number of minima $= 198$

Correct Answer is 198

Chemistry

Q. 1 The reaction in which the hybridization of the underlined atom is affected is:

Option 1:
$H_3\underline{P}O_2 \xrightarrow{\text{Disproportionation}}$

Option 2:
$H_2\underline{S}O_4 + NaCl \xrightarrow{420\ K}$

Option 3:
$\underline{N}H_3 \xrightarrow{H^+}$

Option 4:
$\underline{Xe}F_4 + SbF_5 \longrightarrow$

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Correct Answer:
$\underline{Xe}F_4 + SbF_5 \longrightarrow$

Solution:
The reactions will be -

1 $H_2SO_4 + NaCl \xrightarrow{420k} NaHSO_4 + HCl$
$sp^3 \qquad\qquad\qquad sp^3$

2 $XeF_4 + SbF_5 \rightarrow [XeF_3]^+ [SbF_6]^-$
$sp^3d^2 \quad sp^3d \quad sp^3d \quad sp^3d^2$

3 $NH_3 \xrightarrow{H^+} NH_4^+$
$sp^3 \qquad sp^3$

4 $H_3PO_2 \xrightarrow[\text{Reaction}]{\text{Disproportionation}} PH_3 + H_3PO_4$
$sp^3 \qquad\qquad sp^3 \quad sp^3$

So, the hybridization of the underlined atom Xe is affected.

Therefore, the correct option is (4).

Q. 2 250 ml of a waste solution obtained from the workshop of goldsmith contains 0.1 M $AgNO_3$ and 0.1 M AuCl. The solution was electrolysed at 2 V by passing a current of 1A for 15 minutes. The metal/ metals electrodeposited will be:

$\left(E^\circ_{Ag^+/Ag} = 0.80V, E^\circ_{Au^+/Au} = 1.69V\right)$

Option 1:
only gold

Option 2:
sliver and gold in proportion to their atmoic weights

Option 3:
only sliver

Option 4:
sliver and gold in equal mass proportion

Correct Answer:
sliver and gold in proportion to their atmoic weights

Solution:
Given

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$\left(\mathrm{E}^{\circ}_{\mathrm{Ag^{+}/Ag}} = 0.80\mathrm{V}, \mathrm{E}^{\circ}_{\mathrm{Au^{+}/Au}} = 1.69\mathrm{V}\right)$

The solution was electrolyzed at 2 V, and the Electrode Potential of both metal ions is less than 2 V.

So both Ag^{+} & Au^{+} will reduce and their equal gm equivalent will reduce because molarity is also the same.

So, gmeq Ag = gmeq of Au

$$\frac{\mathrm{Wt_{Ag}}}{\mathrm{E_{qwt_{Ag}}}} = \frac{\mathrm{Wt_{Au}}}{\mathrm{E_{qwt_{Au}}}}$$

$$\frac{\mathrm{wt_{Ag}}}{\mathrm{wt_{Au}}} = \frac{\mathrm{Eq_{wt_{Ag}}}}{\mathrm{Eq_{wt_{Au}}}} = \frac{\mathrm{At.wt_{Ag}}}{\mathrm{At.wt_{Au}}}$$

So, Silver and gold in proportion to their atomic weights.

Therefore, the correct option is (2).

Q. 3 An alkaline earth metal 'M' readily forms water soluble sulphate and water insoluble hydroxide. Its oxide MO is very stable to heat and does not have rock-salt structure M is:

Option 1:
Sr

Option 2:
Ca

Option 3:
Mg

Option 4:
Be

Correct Answer:
Be

Solution:
M is "Be" alkaline earth metal.

Be readily forms water-soluble sulphate $BaSO_4$

$Be(OH)_2$ is water-insoluble hydroxide.

BeO is very stable to heat and does not have a rock-salt structure.

Therefore, the correct option is (4).

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Q. 4 The incorrect statement(s) among (a)-(c) is (are):

(a) W(VI) is more stable than Cr(VI)

(b) in the presence of HCl, permaganate titrations provide satisfactory results.

(c) some lanthnoid oxides can be used as phosphorous

Option 1:
(b) and (c) only

Option 2:
(a) and (b) only

Option 3:
(b) only

Option 4:
(a) only

Correct Answer:
(b) only

Solution:
(a) W(VI) is more stable than Cr(VI). It is true and given in NCERT.

(b) in the presence of HCl permanganate titrations provide satisfactory results. It is incorrect.

Permanganate titrations in presence of hydrochloric acid are unsatisfactory since hydrochloric acid is oxidised to chlorine.

(c) Some individual Ln oxides are used as phosphors in television screens and similar fluorescing surfaces.

So,(b) is incorrect.

Therefore, the correct option is (3).

Q. 5 The shortest wavelength of H atom in the Lyman series is λ_1. The longest wavelength in the Balmer series He^+ is:

Option 1:
$\frac{36\lambda_1}{5}$

Option 2:
$\frac{5\lambda_1}{9}$

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Option 3:

$\frac{9\lambda_1}{5}$

Option 4:

$\frac{27\lambda_1}{5}$

Correct Answer:

$\frac{9\lambda_1}{5}$

Solution:

We know,

$$\Delta E = \frac{hc}{\lambda}$$

So, $\lambda = \frac{hc}{\Delta E}$ for λ minimum(Shortest)

$$\Delta E = \text{maximum}$$

for f H atoms in the Lyman series n = 1 & for $\Delta E_{\text{maximum}}$ Transition must be form n = ∞ to n = 1,

Z = 1

$$\text{So } \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda_1} = R_H (1)^2 (1 - 0)$$

$$\frac{1}{\lambda_1} = R_H \times (1)^2 \Rightarrow \lambda_1 = \frac{1}{R_H}$$

For longest wavelength $\Delta E_{\text{minimum}}$ for Balmer series n = 3 to n = 2 will have $\Delta E_{\text{minimum}}$ for He^+ , Z = 2

$$\text{So } \frac{1}{\lambda_2} = R_H \times Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda_2} = R_H \times (2)^2 \left(\frac{1}{4} - \frac{1}{9}\right)$$

$$\frac{1}{\lambda_2} = R_H \times \frac{5}{9}$$

$$\lambda_2 = \frac{1}{R_H} \times \frac{9}{5}$$

$$\lambda_2 = \lambda_1 \times \frac{9}{5}$$

Therefore, the correct option is (3).

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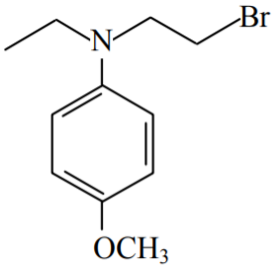
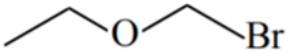
Q. 6 Which of the following compunds will form precipitate with aq. $AgNO_3$ solution most readily?

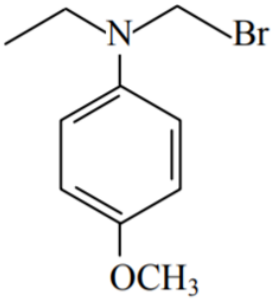
Option 1:



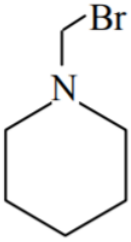
Option 2:



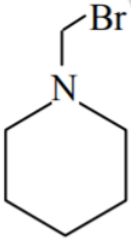
Option 3:



Option 4:



Correct Answer:



Solution:
Reaction of an Alkyl halide with $\mathrm{AgNO_3}$ occurs as

$$\mathrm{R-X + aq\cdot AgNO_3 \xrightarrow{R.D.S} R^{\oplus} + \underset{(PPT)}{AgX}}$$

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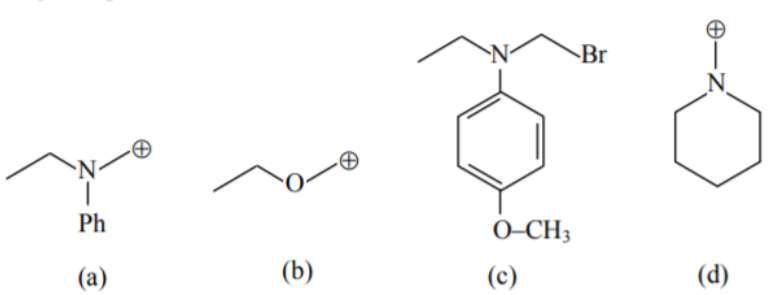
So rate of formation of AgX depend's on stability of carbocation (R^+). In given question formed carbocation will be



(a) (b) (c) (d)

The most stable carbocation is (d) so,

(d) will form the precipitate with aq. $AgNO_3$ solution most readily.

Therefore, the correct option is (4).

Q. 7 In the equilibrium constant for $A \rightleftharpoons B + C$ is $K_{eq}^{(1)}$ and that of $B + C \rightleftharpoons P$ is $K_{eq}^{(2)}$, then equilibrium constant for $A \rightleftharpoons P$ is:

Option 1:
$K_{eq}^{(1)} / K_{eq}^{(2)}$

Option 2:
$K_{eq}^{(2)} - K_{eq}^{(1)}$

Option 3:
$K_{eq}^{(2)} + K_{eq}^{(1)}$

Option 4:
$K_{eq}^{(2)} K_{eq}^{(1)}$

Correct Answer:
$K_{eq}^{(2)} K_{eq}^{(1)}$

Solution:

$$A \rightleftharpoons B + C \quad K_{eq}^{(1)} = \frac{[B][C]}{[A]} \ldots\ldots (1)$$

$$B + C \rightleftharpoons P \quad K_{eq}^{(2)} = \frac{[P]}{[B][C]} \quad \ldots..(2)$$

For,

$$A \rightleftharpoons P \quad K_{eq} = \frac{[P]}{[A]}$$

Multiplying equations (1) & (2)

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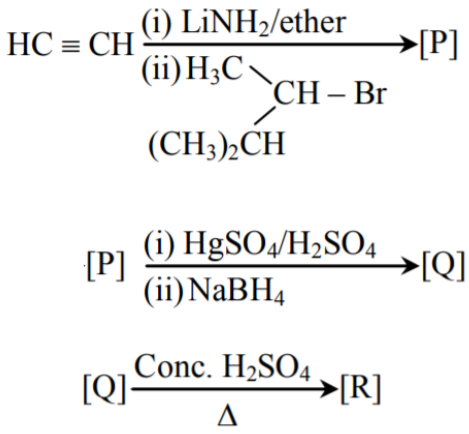
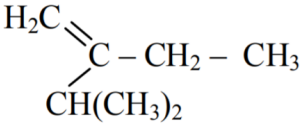
$$K_{eq}^{(1)} \times K_{eq}^{(2)} = \frac{[P]}{[A]} = K_{eq}$$

Therefore, the correct option is (4).

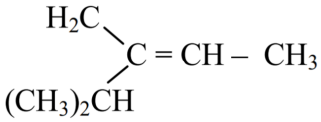
Q. 8 The major product [R] in the following sequence reaction is:



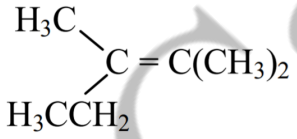
Option 1:



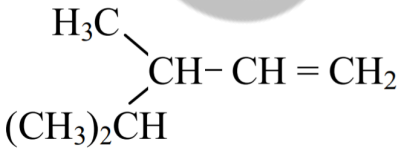
Option 2:



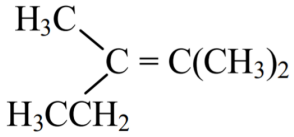
Option 3:



Option 4:



Correct Answer:



Solution:
The reactions will be-

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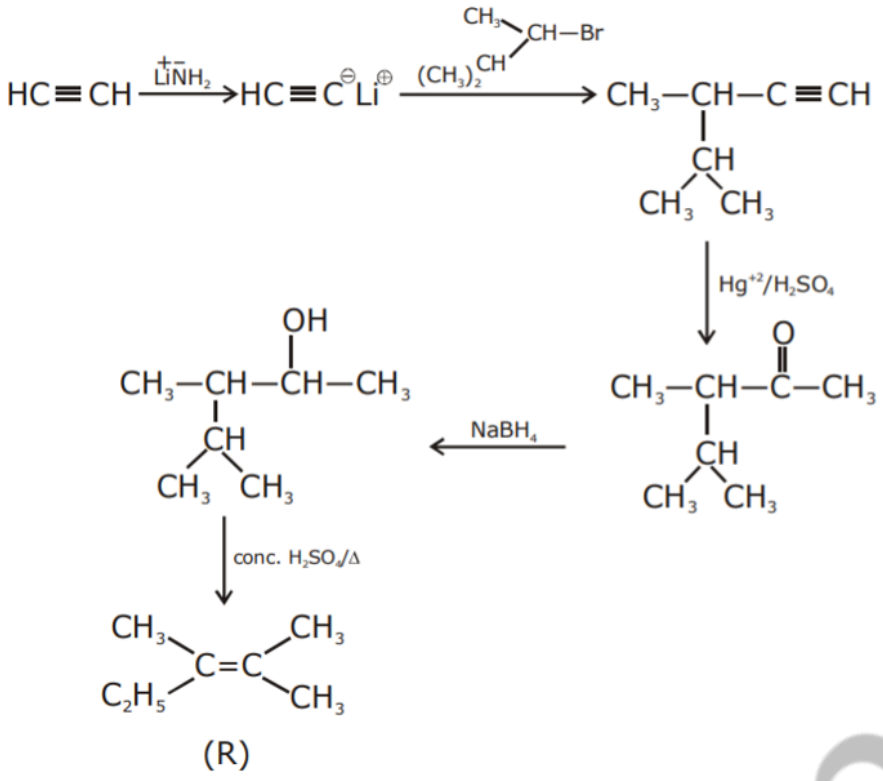

Therefore, the correct option is (3).

Q. 9 The Crystal Field Stabilization Energy (CFSE) of $[CoF_3(H_2O)_3]\ (\Delta_0 < P)$ is:

Option 1:
-0.8 Δ_0 + 2P

Option 2:
-0.4 Δ_0

Option 3:
- 0.8 Δ_0

Option 4:
-0.4 Δ_0 + P

Correct Answer:
-0.4 Δ_0

Solution:
$[CoF_3(H_2O)_3]\ (\Delta_0 < P)$ Means all ligands behaves as weak field ligands

So, $Co^{3+}\ (3d^6) = t_2g^4eg^2$

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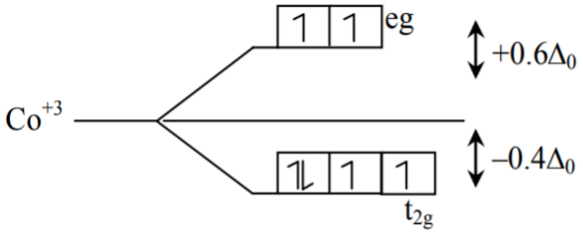

$$\text{CFSE} = [-0.4 \times 4 + 0.6 \times 2]\Delta_0$$

$$\text{CFSE} = [-1.6 + 1.2]\Delta_0$$

$$\text{CFSE} = [-0.4\Delta_0]$$

Therefore, the correct option is (2).

Q. 10 The process of the clacination and roasting in metallurigical industries, respectively, can lead to:

Option 1:
Global warming and photochemical smog

Option 2:
Global warming and acid rain

Option 3:
Photochemical smog and ozone layer depletion

Option 4:
Photochemical somg and Global warming

Correct Answer:
Global warming and acid rain

Solution:
We know

Calcination Releases → CO_2 → Global warming

Roasting Releases → SO_2 → Acid Rain

The processes of calcination and roasting in metallurgical industries, respectively, can lead to Global warming and acid rain Due to the industrial processes SO_2 gas is released which is responsible for acid rain & CO_2 gas for global warming.

Therefore, the correct option is (2).

Q. 11 Five moles of an ideal gas at 1 bar and 298K is expanded into vaccum to double the volume. The work done is:

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Option 1:
$C_V(T_2 - T_1)$

Option 2:
$-RT(V_2 - V_1)$

Option 3:
$-RT \ln \frac{V_2}{V_1}$

Option 4:
zero

Correct Answer:
zero

Solution:
As the expansion is done in a vacuum that is in absence of ext. pressure.

If ext. pressure = 0

So, Work Done = zero

Therefore, the correct option is (4).

Q. 12 The one that can exhibit the highest paramagnetic behavior among the following is:

gly- glycinato, bpy- 2,2'-bipyridine

Option 1:
$[\mathrm{Pd(gly)_2}]$

Option 2:
$[\mathrm{Fe(en)(bpy)\,(NH_3)_2}]^{2+}$

Option 3:
$[\mathrm{Co(OX)_2(OH)_2}]^{-}\ (\Delta_0 > \mathrm{P})$

Option 4:
$[\mathrm{Ti\,(NH_3)_6}]^{3+}$

Correct Answer:
$[\mathrm{Co(OX)_2(OH)_2}]^{-}\ (\Delta_0 > \mathrm{P})$

Solution:

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1. $(Pd(gly)_2) \Rightarrow pd^{2+}(4d^8) \rightarrow$ Square planar
   $n = 0, \mu = 0$ diamagentic

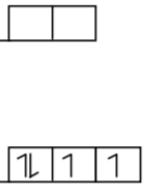
2. $(Fe(en)(bpy)(NH_3)_2)^{2+}$
   $Fe^{2+} \Rightarrow 3d^6 (t_2g^6 eg^o) \Rightarrow n = 0, \mu = 0$

3. $[Co(OX_2)(OH_2)]^- (\Delta_0 > P) \Rightarrow Co^{+5}(3d^4) \Rightarrow t_2g^4\, eg^0$

$[Co(OX)_2(OH)_2]^-$ $\Delta_0 > P$ [S.F.L]

$Co = 3d^7\, 4s^2$

$Co^{+5} = 3d^4\, 4s^0$



So, $[Co(OX)_2(OH)_2]^-$ $\Delta_0 > P$ has

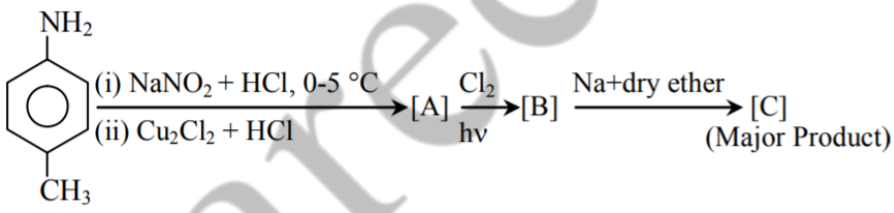
$n = 2, \mu = \sqrt{8}$

4. $(Ti(NH_3)_6)^{3+} \Rightarrow Ti^{3+}(3d^1) \Rightarrow \mu = \sqrt{3}$

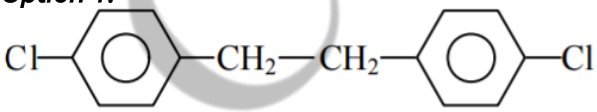
the highest number of unpaired e^- s. so it is most paramagnetic.

Therefore, the correct option is (3).

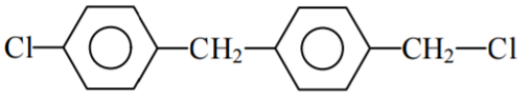
Q. 13 In the following reaction sequence, [C] is:



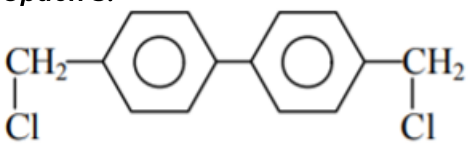
Option 1:



Option 2:



Option 3:



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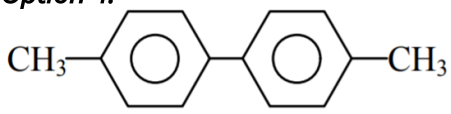
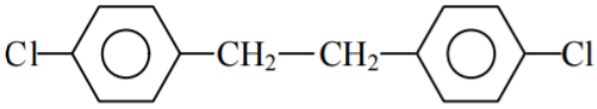
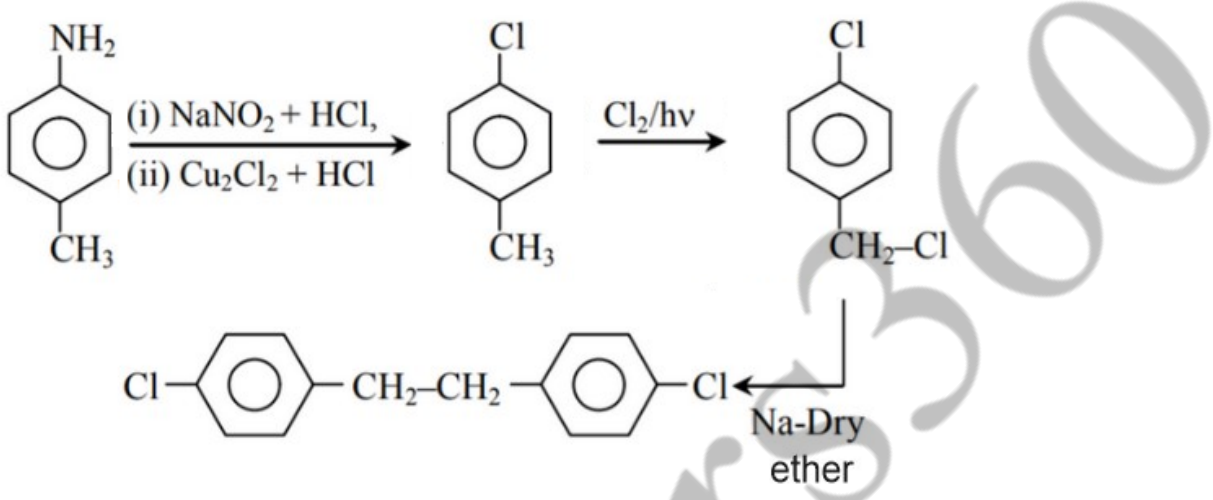
Option 4:



Correct Answer:



Solution:
The reaction will be-



Therefore, the correct option is (1).

Q. 14 The process that is NOT endothermic in nature is:

Option 1:
$Ar(g) + e^- \rightarrow Ar^-(g)$

Option 2:
$H(g) + e^- \rightarrow H^-(g)$

Option 3:
$O^-(g) + e^- \rightarrow O^{2-}(g)$

Option 4:
$Na(g) \rightarrow Na^+(g) + e^-$

Correct Answer:
$H(g) + e^- \rightarrow H^-(g)$

Solution:
$\mathrm{H_{(g)} + e^- \rightarrow H^-}$ is exothermic

Rest of all endothermic processes.

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Therefore, the correct option is (2).

Q. 15 The mechanism of action of "Terfenadine" (Seldane) is:

Option 1:
Activates the histamine receptor

Option 2:
Inhibits the secretion of histamine

Option 3:
helps in the secretion of histamine

Option 4:
inhibits the action of histamine receptor

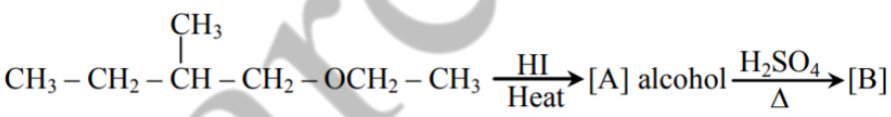
Correct Answer:
inhibits the action of histamine receptor

Solution:
Seldane is an antihistamine drug.

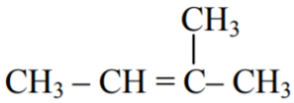
The mechanism of action of “Terfenadine” (Seldane) is to inhibit the action of histamine receptor.

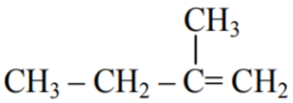
Therefore, the correct option is (4).

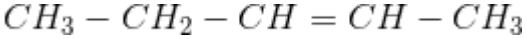
Q. 16 The major product [B] in the following reaction is:



Option 1:

Option 2:


Option 3:


Option 4:


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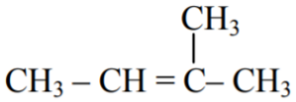
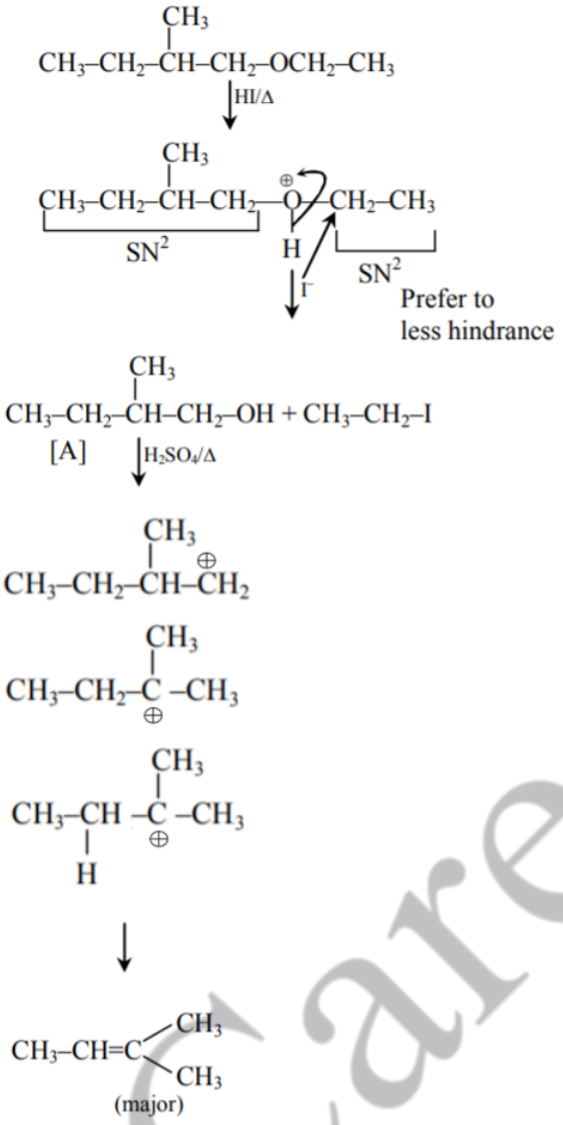
Correct Answer:



Solution:
The reaction will be -



Therefore, the correct option is (2).

Q. 17 A sample of red ink (a colloidal suspension) is prepared by mixing eosin dye, egg white, HCHO and water. The component which ensures stability of the ink sample is:

Option 1:
Egg white

Option 2:
Water

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Option 3:
HCHO

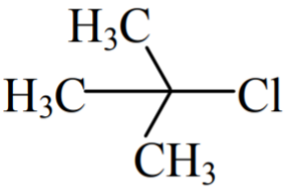
Option 4:
Eosin dye

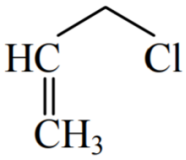
Correct Answer:
Egg white

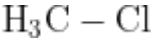
Solution:
The component which ensures the stability of the ink sample is Egg White.

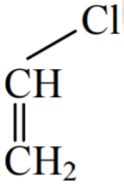
Therefore, the correct option is (1).

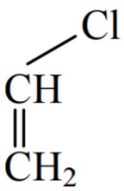
Q. 18 Among the following compunds, which one has the shortest C-Cl bond?

Option 1:


Option 2:


Option 3:


Option 4:


Correct Answer:


Solution:

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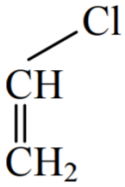
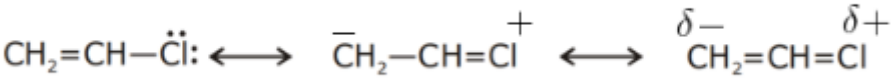
In the structure of  , C—Cl bond is shortest due to resonance of lone pair of –Cl.

The resonating structures are given below :



Due to resonance C—Cl bond acquires partial double bond character. Hence C—Cl bond length is the least.

Therefore, the correct option is (4).

Q. 19 The molecule in which hybrid Mos involve only one d-orbital of the central atom is:

Option 1:
$[Ni(CN)_4]^{2-}$

Option 2:
BrF_5

Option 3:
XeF_4

Option 4:
$[CrF_6]^{3-}$

Correct Answer:
$[Ni(CN)_4]^{2-}$

Solution:
Hybridisation of the given molecules-

(1) $[Ni(CN)_4]^{2-} \rightarrow dsp^2$
(2) BrF_5 – Sp^3d^2
(3) XeF_4 – Sp^3d^2
(4) $(CrF_6)^{3-}$ – d^2Sp^3

So, in $[Ni(CN)_4]^{2-}$ moleculehybrid MOs involve only **one** d-orbital of the central atom.

Therefore, the correct option is (1).

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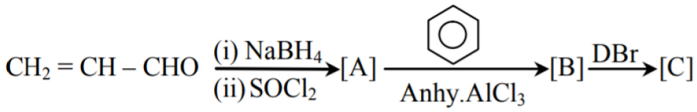
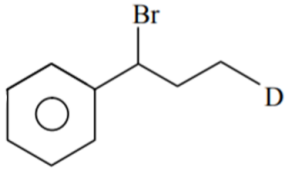
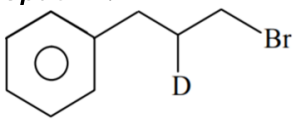
Q. 20 The major product [C] of the following reaction sequence will be:



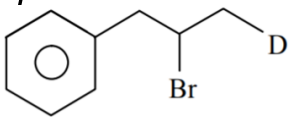
Option 1:



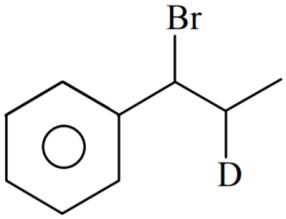
Option 2:



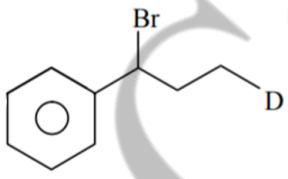
Option 3:



Option 4:



Correct Answer:



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Solution:
The reaction will be-

$$[Fe^{2+} \rightarrow Fe^{3+} + e^-] \times 2$$

$$\frac{H_2O_2 + 2e^- \rightarrow 2HO^{\ominus}}{2Fe^{2+} + H_2O_2 \rightarrow 2Fe^{3+} + 2HO^{\ominus}}$$

$x = 2 \quad y = 2$

$$[8H^+ + MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O] \times 2$$

$$[H_2O_2 \rightarrow O_{2(g)} + 2H^+ + 2e^-] \times 5$$

$$\Rightarrow 16H^+ + 2MnO_4^- + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_{2(g)} + 10H^+$$

$$\Rightarrow 6H^+ + 2MnO_4^- + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_{2(g)}$$

So $x' = 2 \quad y' = 8 \quad z' = 5$

So $x + y + x' + y' + z'$

$\Rightarrow 2 + 2 + 2 + 8 + 5$

$\Rightarrow 19$

Therefore, the correct option is (1).

Q. 21 The osmotic pressure of a solution of NaCl is 0.10 atm and that of a glucose solution is 0.20 atm. The osmotic pressure of a solution formed by mixing 1L of the sodium chloride solution 2L of the glucose solution is $x \times 10^{-3}$atm. x is _____ (nearest interger)

Correct Answer:
167

Solution:
We know this formula.

$$\pi_f V_f = \pi_1 V_1 + \pi_2 V_2$$

$$\pi_f = \frac{\pi_1 V_1 + \pi_2 V_2}{V_f}$$

$$\pi_f = \frac{0.1 \times 1 + 0.2 \times 2}{3}$$

$$\pi_f = \frac{0.5}{3} = \frac{500}{3} \times 10^{-3}$$

$$x = 167$$

Ans = 167

Alternate Solution

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Osmotic pressure = $\pi = i \times C \times RT$

For NaCl i = 2 so

$\pi_{NaCl} = i \times C_{NaCl} \times RT$ $\quad C_{NaCl}$ = conc. of NaCl

$0.1 = 2 \times C_{NaCl} \times RT$

$C_{NaCl} = \dfrac{0.05}{RT}$ $\quad C_{glucose}$ = conc. of glucose

For glucose i = 1 so

$\pi_{Glucose} = i \times C_{glucose} \times RT$

$0.2 = 1 \times C_{glucose} \times RT$

$C_{glucose} = \dfrac{0.2}{RT}$

η_{NaCl} = No. of moles NaCl

η_{NaCl} in 1 L = $C_{NaCl} \times V_{Litre} = \dfrac{0.05}{RT}$

$\eta_{glucose}$ = No. of moles glucose

$\eta_{glucose}$ in 2 L = $C_{glucose} \times V_{Litre} = \dfrac{0.4}{RT}$

Concentration = mole/volume

$V_{Total} = 1 + 2 = 3L$

so final conc. NaCl = $\dfrac{0.05}{3RT}$

Final conc. glucose = $\dfrac{0.4}{3RT}$

$\pi_{Total} = \pi_{NaCl} + \pi_{glucose}$

$= [i \times C_{NaCl} + C_{glucose}] \times RT$

$= \left(\dfrac{2 \times 0.05}{3RT} + \dfrac{0.4}{3RT}\right) \times RT$

$= \dfrac{0.5}{3}$ atm

$= 0.1666$ atm

$= 166.6 \times 10^{-3}$ atm

$\Rightarrow 167.00 \times 10^{-3}$ atm

$x = 167$

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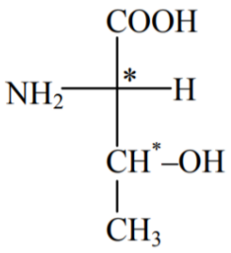
Ans = 167

Q. 22 The number of chiral centres present in threonine is____

Correct Answer:
2

Solution:
The structure of Threonine is :



There are 2-chiral centers present.

Ans =2

Q. 23 A 100mL solution was made by adding 1.43g of $\mathrm{Na_2CO_3.xH_2O}$. The normality of the solution is 0.1N. The value of x is_____.

(The atomic mass of Na is 23 g/mol)

Correct Answer:
10

Solution:
Molar mass(M) of $Na_2CO_3.xH2O$
M = 23 × 2 + 12 + 48 + 18x
M = 46 + 12 + 48 + 18x
M = (106 + 18x)

As n_{factor} in dissolution will be determined from net cationic or anionic charge; which is 2 so.

Eq.wt. = $\frac{M}{2}$ = (53 + 9x)

$$\mathrm{Gm_{eq}} = \frac{\mathrm{wt}}{\mathrm{Eq.wt.}} = \frac{1.43}{53 + 9x}$$

$$\mathrm{Normality} = \frac{\mathrm{Gm_{eq}}}{\mathrm{V_{litre}}}$$

As volume = 100 ml = 0.1 Litre

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$$\text{Normality } = 0.1 = \frac{\frac{1.43}{53+9x}}{0.1}$$

$$10^{-2} = \frac{1.43}{53+9x}$$

53 + 9x = 143
9x = 90
x = 10.00

Ans = 10

Q. 24 The number of molecules with energy greater than the threshold energy for a reaction increases five fold by a rise of temperature from 27°C to 42°C. Its energy of activation in J/mol is _____. (Take ln 5 =1.6094 ; R=8.314 J mol^{-1} K^{-1})

Correct Answer:
84297

Solution:
Given,

T_1 = 300K

T_2 = 315K

As per question $K_{T2} = 5K_{T2}$ as molecules activated are increased five times so k will increases five time.

Now

$$\ln\left(\frac{K_{T_2}}{K_{T_1}}\right) = \frac{Ea}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln 5 = \frac{Ea}{R}\left(\frac{1}{300} - \frac{1}{315}\right)$$

$$\ln 5 = \frac{Ea}{R}\left(\frac{15}{300 \times 315}\right)$$

So,

$$Ea = \frac{1.6094 \times 8.314 \times 300 \times 315}{15}$$

$$Ea = 84297.47 \text{ Joules/mole}$$

Ans = 84297

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Q. 25 Consider the following equations:

$2Fe^{2+} + H_2O_2 \rightarrow xA + yB$ (in basic medium)

$2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow x'C + y'D + z'E$ (in acidic medium)

The sum of stoichiometric coeffcients x, y, x', y' and z' for the products A, B, C, D and E, respectively, is_____.

Correct Answer:
19

Solution:

$$[Fe^{2+} \rightarrow Fe^{3+} + e^-] \times 2$$

$$\frac{H_2O_2 + 2e^- \rightarrow 2HO^{\ominus}}{2Fe^{2+} + H_2O_2 \rightarrow 2Fe^{3+} + 2HO^{\ominus}}$$

$x = 2 \quad y = 2$

$$[8H^+ + MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O] \times 2$$

$$[H_2O_2 \rightarrow O_{2(g)} + 2H^+ + 2e^-] \times 5$$

$$\Rightarrow 16H^+ + 2MnO_4^- + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_{2(g)} + 10H^+$$

$$\Rightarrow 6H^+ + 2MnO_4^- + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_{2(g)}$$

So $x' = 2 \quad y' = 8 \quad z' = 5$

So $x + y + x' + y' + z'$

$\Rightarrow 2 + 2 + 2 + 8 + 5$

$\Rightarrow 19$

Maths

Q. 1 The circle is passing through the intersection of the circle, $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 - 4y = 0$, having its centre on the line, $2x - 3y + 12 = 0$, also passes through the point:

Option 1:
(-1,3)

Option 2:
(-3,6)

Option 3:
(-3,1)

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Option 4:
(1,-3)

Correct Answer:
(-3,6)

Solution:
Let S be the circle passing through point of intersection of S_1 & S_2

$$\therefore S = S_1 + \lambda S_2 = 0$$
$$\Rightarrow S : \left(x^2 + y^2 - 6x\right) + \lambda\left(x^2 + y^2 - 4y\right) = 0$$
$$\Rightarrow S : x^2 + y^2 - \left(\frac{6}{1+\lambda}\right)x - \left(\frac{4\lambda}{1+\lambda}\right)y = 0 \qquad \ldots(1)$$

$$\text{Centre}\left(\frac{3}{1+\lambda}, \frac{2\lambda}{1+\lambda}\right) \text{ lies on}$$
$$2x - 3y + 12 = 0 \Rightarrow \lambda = -3$$

Put in (1)

$$\Rightarrow S : x^2 + y^2 + 3x - 6y = 0$$

Now check options point (–3, 6)

Q. 2 Let x=4 be a directix to an ellipse whose centre is at the origin and its eccentricity is $\frac{1}{2}$. If P $(1, \beta), \beta > 0$ is a point on this ellipse, then the equation of the normal to it at P is:

Option 1:
$4x - 3y = 2$

Option 2:
$8x - 2y = 5$

Option 3:
$7x - 4y = 1$

Option 4:
$4x - 2y = 1$

Correct Answer:
$4x - 2y = 1$

Solution:
$$\text{Ellipse} \ : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$
$$\text{directrix} \ : x = \frac{a}{e} = 4 \ \ \& \ \ e = \frac{1}{2}$$

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$\Rightarrow a = 2 \& \ b^2 = a^2\left(1 - e^2\right) = 3$

$\Rightarrow$ Ellipse is $\frac{x^2}{4} + \frac{y^2}{3} = 1$

p is $\left(1, \frac{3}{2}\right)$

4x - 2y = 1 satisies this point

Q. 3 The minimum value of $2^{\sin x} + 2^{\cos x}$ is:

Option 1:
$2^{-1+\frac{1}{\sqrt{2}}}$

Option 2:
$2^{-1+\sqrt{2}}$

Option 3:
$2^{1-\sqrt{2}}$

Option 4:
$2^{1-\frac{1}{\sqrt{2}}}$

Correct Answer:
$2^{1-\frac{1}{\sqrt{2}}}$

Solution:
Apply AM $\geq$ GM

$\Rightarrow \frac{2^{\sin x} + 2^{\cos x}}{2} \geq \sqrt{2^{\sin x} \cdot 2^{\cos x}}$

$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2\sqrt{2^{\sin x + \cos x}}$

$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2^{1+\left(\frac{\sin x + \cos x}{2}\right)}$

$\Rightarrow \min\left(2^{\sin x} + 2^{\cos x}\right) = 2^{1-\frac{1}{\sqrt{2}}}$

Q. 4 The angle of elevation of a cloud C from a point P, 200 m above a still lake is 30°. If the angle of depression of the images of C in the lake from the point P is 60°, then PC (in m) is equal to:

Option 1:
100

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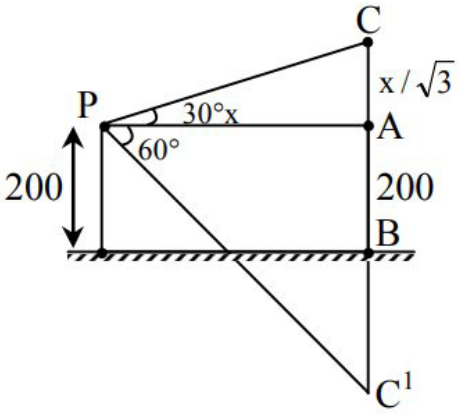
Option 2:
$200\sqrt{3}$

Option 3:
400

Option 4:
$400\sqrt{3}$

Correct Answer:
400

Solution:



Let PA = x

For triangle APC

$$AC = \frac{PA}{\sqrt{3}} = \frac{x}{\sqrt{3}}$$
$$AC^1 = AB + BC^1$$
$$AC^1 = 400 + \frac{x}{\sqrt{3}}$$

$$\text{From } \Delta C^1PA : AC^1 = \sqrt{3}PA$$
$$\Rightarrow \left(400 + \frac{x}{\sqrt{3}}\right) = \sqrt{3}x \Rightarrow x = (200)(\sqrt{3})$$
$$\text{from } \Delta APC : PC = \frac{2x}{\sqrt{3}} \Rightarrow PC = 400$$

Q. 5 If the system of the equations

$$x + y + z = 2$$

$$2x + 4y - z = 6$$

$$3x + 2y - \lambda z = \mu$$

has infinitely many solutions, then:

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Option 1:
$\lambda + 2\mu = 14$

Option 2:
$2\lambda - \mu = 5$

Option 3:
$\lambda - 2\mu = -5$

Option 4:
$2\lambda + \mu = 14$

Correct Answer:
$2\lambda + \mu = 14$

Solution:
$\Delta = \Delta_x = \Delta_y = \Delta_z = 0$

$$\text{Now } \Delta = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & -1 \\ 3 & 2 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda = \frac{9}{2}$$

$$\Delta_{x=0} \Rightarrow \begin{vmatrix} 2 & 1 & 1 \\ 6 & 4 & -1 \\ \mu & 2 & -\frac{9}{2} \end{vmatrix} = 0$$

$$\Rightarrow \mu = 5$$

$$\text{for } \lambda = \frac{9}{2} \ \& \ \mu = 5, \Delta_y = \Delta_z = 0$$

Now check option

Q. 6 The area (in sq. units) of the largest rectangle ABCD whose vertices A and B lie on the x-axis and vertices C and D lie on the probola, $y = x^2 - 1$ below the x-axis is:

Option 1:
$\frac{2}{3\sqrt{3}}$

Option 2:
$\frac{1}{3\sqrt{3}}$

Option 3:
$\frac{4}{3}$

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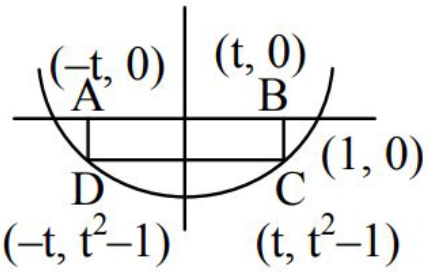
Option 4:

$\frac{4}{3\sqrt{3}}$

Correct Answer:

$\frac{4}{3\sqrt{3}}$

Solution:



$$\text{Area } (A) = 2t.\left(1 - t^2\right)$$
$$(0 < t < 1)$$
$$A = 2t - 2t^3$$
$$\frac{dA}{dt} = 2 - 6t^2$$
$$t = \frac{1}{\sqrt{3}}$$

$$\Rightarrow A_{max} = \frac{2}{\sqrt{3}}\left(1 - \frac{1}{3}\right) = \frac{4}{3\sqrt{3}}$$

Q. 7

The function $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, |x| \leq 1 \\ \frac{1}{2}(|x| - 1), |x| > 1 \end{cases}$ is:

Option 1:

continuous on R -{1} and differentiable on R- {-1,1}.

Option 2:

both continous and differentiable on R - {-1}

Option 3:

continous on R - {-1} and differntiable on R - {-1,-1}

Option 4:

both continous and differntiable on R - {-1}

Correct Answer:

continuous on R -{1} and differentiable on R- {-1,1}.

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Solution:

for continuity at x = – 1

$$\text{L.H.L.} = \frac{\pi}{4} - \frac{\pi}{4} = 0$$
$$\text{R.H.L} = \frac{\pi}{4} - \frac{\pi}{4} = 0$$

so, continuous at x = –1

for continuity at x = 1

L.H.L. = 0

$$\text{R.H.L.} = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$$

so, not continuous at x = 1

For differentiability at x = –1

$$\text{L.H.D} = \frac{1}{1+1} = \frac{1}{2}$$
$$\text{R.H.D.} = -\frac{1}{2}$$

so, non differentiable at x = –1 $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, & x \in (-\infty, -1) \cup [1, \infty) \\ -\frac{(x+1)}{2}, & x \in (-1, 0] \\ \frac{x-1}{2}, & x \in (0, 1) \end{cases}$

Q. 8 If for some positive integer n, the coefficients of three consecutive terms in the binomial expansion of $(1+x)^{n+5}$ are in the ratio 5:10:14, then the largest coefficient in this expansion is:

Option 1:
462

Option 2:
330

Option 3:
792

Option 4:
252

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Correct Answer:
462

Solution:
Let n + 5 = N

$$^{N}C_{r-1} : {}^{N}C_{r} : {}^{N}C_{r+1} = 5 : 10 : 14$$
$$\Rightarrow \frac{{}^{N}C_{r}}{{}^{N}C_{r-1}} = \frac{N+1-r}{r} = 2$$
$$\frac{N_{C_{r+1}}}{N_{C_{r}}} = \frac{N-r}{r+1} = \frac{7}{5}$$
$$\Rightarrow r = 4,\ N = 11$$
$$\Rightarrow (1+x)^{11}$$
$$\text{Largest coefficient } = {}^{11}C_{6} = 462$$

Q. 9 Supportive the vectors x_1, x_2 and x_3 solutions of the system of linear equations, Ax = b when the vector b on the right side is equal to b_1, b_2 and b_3 respectively. If

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, b_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \& \ b_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

then the determinant A is equal to:

Option 1:
4

Option 2:
2

Option 3:
$\frac{1}{2}$

Option 4:
$\frac{3}{2}$

Correct Answer:
2

Solution:
$$Ax_1 = b_1$$
$$Ax_2 = b_2$$
$$Ax_3 = b_3$$
$$\Rightarrow |A| \begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{vmatrix}$$

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$\Rightarrow |A| = \frac{4}{2} = 2$

Q. 10 The contrapositive of the statement:

'If a function f is differentiable at a, then it is also continuous at a', is:

Option 1:
If function f is continous at a, then it is not differentiable at a

Option 2:
If function f is not continous at a, then it is not differentiable at a

Option 3:
If function f is not continous at a, then it is differentiable at a.

Option 4:
If function f is continous at a, then it is differentiable at a

Correct Answer:
If function f is not continous at a, then it is not differentiable at a

Solution:
p = function is differentiable at a

q = function is continuous at a

contrapositive of statements p → q is

$\sim q \rightarrow \sim p$

Q. 11 The solution of the differntial equation $\frac{dy}{dx} - \frac{y+3x}{\log_e(y+3x)} + 3 = 0$ is:

(where C is a constant of intergration)

Option 1:
$x - \frac{1}{2}(\log_e(y+3x))^2 = C$

Option 2:
$x - \log_e(y+3x) = C$

Option 3:
$y + 3x - \frac{1}{2}(\log_e x^2) = C$

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Option 4:

$x - 2\log_e(y + 3x) = C$

Correct Answer:

$x - \frac{1}{2}(\log_e(y + 3x))^2 = C$

Solution:

Let

ln (y + 3x) = z

$$\frac{1}{y+3x}\left(\frac{dy}{dx}+3\right)=\frac{dz}{dx}$$

$$\frac{dy}{dx}+3=\frac{y+3x}{\ell n(y+3x)}(\text{ given })$$

$$\frac{\mathrm{dz}}{\mathrm{dx}}=\frac{1}{\mathrm{z}}$$

$$\Rightarrow \mathrm{zdz}=\mathrm{dx} \Rightarrow \frac{\mathrm{z}^2}{2}=\mathrm{x}+\mathrm{C}$$

$$\Rightarrow \mathrm{zdz}=\mathrm{dx} \Rightarrow \frac{\mathrm{z}^2}{2}=\mathrm{x}+\mathrm{C}$$

$$\Rightarrow \frac{1}{2}\ln^2(\mathrm{y}+3\mathrm{x})=\mathrm{x}+\mathrm{C}$$

$$\Rightarrow \mathrm{x}-\frac{1}{2}(\ln(\mathrm{y}+3\mathrm{x}))^2=\mathrm{C}$$

Q. 12 In a game two players, A and B take turns in throwing a pair of fair dice starting with player A and a total of scores on the two dice, in each throw is noted. A wins the game if he throws total of 6 before B throws a total of 7 and B wins the game if he throws a total 7 before A throws a total six. The game stops as soon as either of the players wins. The probability of A winning the game is:

Option 1:

$\frac{5}{31}$

Option 2:

$\frac{31}{61}$

Option 3:

$\frac{5}{6}$

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Option 4:
$\frac{30}{61}$

Correct Answer:
$\frac{30}{61}$

Solution:
$P(\text{Sum } 6) = \frac{5}{36}$
$(1,5),(5,1),(2,4),(4,2),(3,3)$
$P(\text{Sum } 7) = \frac{6}{36} = \frac{1}{6}$
$(1,6),(6,1),(2,5),(5,2),(3,4),(4,3)$

If A represents total of 6 for A, A' represents total which is other than 6, B represents total of 7 for B, and B' represents when it is not 7

$A \text{ wins when } A,\ A'B'A,\ A'B'A'B'A, \ldots\ldots$

$\therefore P(\text{ A wins }) = \frac{5}{36} + \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36} + \frac{31}{36} \times \frac{5}{6} \times \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36} + \ldots$
$= \frac{\frac{5}{36}}{1 - \frac{31}{36} \times \frac{5}{6}} = \frac{30}{61}$

Q. 13 The distance of the point (1, -2, 3) from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is:

Option 1:
$\frac{7}{5}$

Option 2:
1

Option 3:
$\frac{1}{7}$

Option 4:
7

Correct Answer:
1

Solution:
equation of line parallel to $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ passes through $(1, -2, 3)$ is

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$$\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{-6}=r$$
$$x=2r+1$$
$$y=3r-2$$
$$z=-6r+3$$
$$\text{so } 2r+1-3r+2-6r+3=5$$
$$r=\frac{1}{7}$$

$$\text{x}=\frac{9}{7},\text{y}=\frac{-11}{7},\text{z}=\frac{15}{7}$$

$$\text{Distance is } =\sqrt{\left(\frac{9}{7}-1\right)^2+\left(2-\frac{11}{7}\right)^2+\left(3-\frac{15}{7}\right)^2}$$
$$=\sqrt{\left(\frac{2}{7}\right)^2+\left(\frac{3}{7}\right)^2+\left(\frac{6}{7}\right)^2}$$
$$=\frac{1}{7}\sqrt{4+9+36}=1$$

Q. 14 Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation, $x^2-x+2\lambda=0$ and α and γ are the roots of the equation, $3x^2-10x+27\lambda=0$, then $\frac{\beta\gamma}{\lambda}$ is equal to:

Option 1:
27

Option 2:
18

Option 3:
9

Option 4:
36

Correct Answer:
18

Solution:
$$\alpha+\beta=1,\alpha\beta=2\lambda$$
$$\alpha+\beta=\frac{10}{3},\quad \alpha\gamma=\frac{27\lambda}{3}=9\lambda$$
$$\gamma-\beta=\frac{7}{3}$$

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$$\frac{\gamma}{\beta}=\frac{9}{2}\Rightarrow\gamma=\frac{9}{2}\beta=\frac{9}{2}\times\frac{2}{3}\Rightarrow\gamma=3$$
$$\frac{9}{2}\beta-\beta=\frac{7}{3}$$
$$\frac{9}{2}\beta=\frac{7}{3}\Rightarrow\beta=\frac{2}{3}$$
$$\alpha=1-\frac{2}{3}=\frac{1}{3}$$
$$2\lambda=\frac{2}{9}\Rightarrow\lambda=\frac{1}{9}$$
$$\frac{\beta\gamma}{\lambda}=\frac{\frac{2}{3}\times 3}{\frac{1}{9}}=18$$

Q. 15 The integral $\int_{\pi/6}^{\pi/3}\tan^3 x\sin^2 3x(2\sec^2\sin^2 3x+3\tan x\sin 6x)dx$ is equal to:

Option 1:
$\frac{7}{18}$

Option 2:
$-\frac{1}{9}$

Option 3:
$-\frac{1}{18}$

Option 4:
$\frac{9}{2}$

Correct Answer:
$-\frac{1}{18}$

Solution:
$$\mathrm{I}=\int_{\pi/6}^{\pi/3}\left(\left(2\tan^3\mathrm{x}\cdot\sec^2\mathrm{x}\cdot\sin^4 3\mathrm{x}\right)+\left(3\tan^4\mathrm{x}\cdot\sin^3 3\mathrm{x}\cdot\cos 3\mathrm{x}\right)\right)\mathrm{dx}$$
$$\Rightarrow \mathrm{I}=\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\mathrm{d}\left((\sin 3\mathrm{x})^4(\tan\mathrm{x})^4\right)\qquad\Rightarrow I=-\frac{1}{18}$$
$$\Rightarrow \mathrm{I}=\left((\sin 3\mathrm{x})^4(\tan\mathrm{x})^4\right)_{\pi/6}^{\pi/3}$$

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Q. 16 Let $a_1, a_2, \ldots\ldots\ldots\ldots a_n$ be given A.P, whose common difference is an integer and $S_n = a_1 + a_2 + \ldots\ldots\ldots a_n$. If $a_1 = 1, a_n = 300$ and $15 \leq n \leq 50$, then the ordered pair (S_{n-4}, a_{n-4}) is equal to:

Option 1:
(2490, 249)

Option 2:
(2480, 249)

Option 3:
(2480, 248)

Option 4:
(2490, 248)

Correct Answer:
(2490, 248)

Solution:

$a_n = a_1 + (n-1)d$

$\Rightarrow 300 = 1 + (n-1)d$

$\Rightarrow (n-1)d = 299 = 13 \times 23$

since, $n \in [15, 50]$

$\therefore n = 24$ and $d = 13$

$a_{n-4} = a_{20} = 1 + 19 \times 13 = 248$

$\Rightarrow a_{n-4} = 248$

$S_{n-4} = \frac{20}{2}\{1 + 248\} = 2490$

Q. 17 If a and b are real numbers such that $(2+\alpha)^4 = a + b\alpha$, where $\alpha = \frac{-1+i\sqrt{3}}{2}$, then a+b is equal to:

Option 1:
9

Option 2:
24

Option 3:
33

Option 4:
57

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Correct Answer:
9

Solution:

$\alpha = \omega$

$\Rightarrow (2+\omega)^4 = \mathrm{a} + \mathrm{b}\omega \quad (\omega^3 = 1)$

$\Rightarrow 2^4 + 4 \cdot 2^3\omega + 6 \cdot 2^2\omega^3 + 4 \cdot 2 \cdot \omega^3 + \omega^4 = \mathrm{a} + \mathrm{b}\omega$

$\Rightarrow 16 + 32\omega + 24\omega^2 + 8 + \omega = \mathrm{a} + \mathrm{b}\omega$

$\Rightarrow 24 + 24\omega^2 + 33\omega = \mathrm{a} + \mathrm{b}\omega$

$\Rightarrow 24(1 + \omega + \omega^2) + 9\omega = \mathrm{a} + \mathrm{b}\omega$

$\Rightarrow \mathrm{a} = 0, \ \mathrm{b} = 9$

Note that

$1 + \omega + \omega^2 = 0$

Q. 18 If the perpendicular bisector of the line segment joining the points P(1,4) and Q (k,3) has y-intercept equal to -4 then a value of k is:

Option 1:
-2

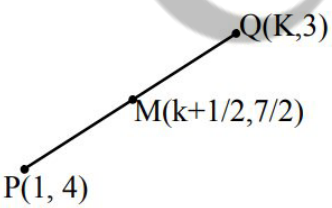
Option 2:
-4

Option 3:
$\sqrt{14}$

Option 4:
$\sqrt{15}$

Correct Answer:
-4

Solution:



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$$\text{Slope } = \mathrm{m} = \frac{1}{1-\mathrm{k}}$$

equation of perpendicular bisector is

$$y+4=(k-1)(x-0)$$

$$\Rightarrow y+4=x(k-1)$$

$$\Rightarrow \frac{7}{2}+4=\frac{k+1}{2}(k-1)$$

$$\Rightarrow \frac{15}{2}=\frac{\mathrm{k}^2-1}{2} \Rightarrow \mathrm{k}^2=16 \Rightarrow \mathrm{k}=4,-4$$

Q. 19 Let $f:(0,\infty)\to(0,\infty)$ be a differentiable function such that f(1)=e and $\lim\limits_{t\to x}\frac{t^2f^2(x)-x^2f^2(t)}{t-x}=0$. If f(x)=1, then x is equal to:

Option 1:

$\frac{1}{e}$

Option 2:

2e

Option 3:

$\frac{1}{2e}$

Option 4:

e

Correct Answer:

$\frac{1}{e}$

Solution:

$$L=\mathrm{Lim}_{t\to x}\frac{t^2f^2(x)-x^2f^2(t)}{t-x}$$

using L H'opital. rule

$$\mathrm{L}=\mathrm{Lim}_{\mathrm{t}\to\mathrm{x}}\frac{2\mathrm{tf}^2(\mathrm{x})-\mathrm{x}^2\cdot 2\mathrm{f}'(\mathrm{t})\cdot\mathrm{f}(\mathrm{t})}{1}$$

$$\Rightarrow \mathrm{L}=2\mathrm{xf}(\mathrm{x})(\mathrm{f}(\mathrm{x})-\mathrm{xf}'(\mathrm{x}))=0(\text{ given })$$

$$\Rightarrow \mathrm{f}(\mathrm{x})=\mathrm{xf}'(\mathrm{x}) \Rightarrow \int\frac{\mathrm{f}'(\mathrm{x})\mathrm{dx}}{\mathrm{f}(\mathrm{x})}=\int\frac{\mathrm{dx}}{\mathrm{x}}$$

$$\Rightarrow \ell\mathrm{n}|\mathrm{f}(\mathrm{x})|=\ell\mathrm{n}|\mathrm{x}|+\mathrm{C}$$

$$\because \mathrm{f}(1)=\mathrm{e},\mathrm{x}>0,\mathrm{f}(\mathrm{x})>0$$

$$\Rightarrow \mathrm{f}(\mathrm{x})=\mathrm{ex},\text{ if } \mathrm{f}(\mathrm{x})=1 \Rightarrow \mathrm{x}=\frac{1}{\mathrm{e}}$$

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Q. 20 Let $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^{n} Y_i = T$, where each X_i contains 10 elements and each Y_i contains 5 elements. If each element of the set T is an element of exactly 20 sets $X_i's$ and exactly 6 sets $Y_i's$ then n is equal to:

Option 1:
15

Option 2:
50

Option 3:
45

Option 4:
30

Correct Answer:
30

Solution:

$n(X_i) = 10$

$\bigcup_{i=1}^{50} X_i = T, \Rightarrow n(T) = 500$

each element of T belongs to exactly 20 elements of X_i

$\Rightarrow \frac{500}{20} = 25 \text{ distinct element}$

$\text{so } \frac{5n}{6} = 25 \Rightarrow n = 30$

Q. 21 A test consists of 6 multiple choice questions, each having 4 alternative answer of which only one is correct. The number of ways, in which a candidate answers all six questions such that exactly four of the answers are correct, is_____

Correct Answer:
135

Solution:
First choose 4 questions that will be answered correctly: ${}^6C_4$

Now, there is only one way to answer these questions

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Now to answer the remaining 2 questions, we can mark any of the three incorrect options, so 3 choices for each

$$\text{Number of Ways } = {}^6C_4 \cdot 1^4 \cdot 3^2$$
$$= 15 \times 9$$
$$= 135$$

Q. 22 If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of $\left|\hat{i} \times \left(\vec{a} \times \hat{i}\right)\right|^2 + \left|\hat{j} \times \left(\vec{a} \times \hat{j}\right)\right|^2 + \left|\hat{k} \times \left(\vec{a} \times \hat{k}\right)\right|^2$ is equal to _____.

Correct Answer:
18

Solution:
$\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$
$|\vec{a}| = \sqrt{2^2 + 1^2 + 2^2} = 3$

$\sum |\vec{a} - (\vec{a} \cdot \hat{i})i|^2$
$\Rightarrow \sum \left(|\vec{a}|^2 + (\vec{a} \cdot i)^2 - 2(\vec{a} \cdot i)^2\right)$
$\Rightarrow 3|\vec{a}|^2 - \Sigma(\vec{a}.\hat{i})^2$
$\Rightarrow 2|\vec{a}|^2$
$\Rightarrow 18$

Q. 23 Let {x} and [x] denote the fractional part of x and the greatest integer $\leq x$ respectively of a real number x. If $\int_0^n \{x\}dx, \int_0^n [x]dx$ and $10(n^2 - n), (n \epsilon N, n > 1)$ are three consecutive terms of a G.P, then n is equal to_____

Correct Answer:
21

Solution:
$$\int_0^n \{x\}dx = n\int_0^1 \{x\}dx = n\int_0^1 xdx = \frac{n}{2}$$
$$\int_0^n [x]dx = \int_0^n (x - \{x\})dx = \frac{n^2}{2} - \frac{n}{2}$$
$$\Rightarrow \left(\frac{n^2 - n}{2}\right)^2 = \frac{n}{2} \cdot 10 \cdot n(n-1)(\text{ where } n > 1)$$
$$\Rightarrow \frac{n-1}{4} = 5 \Rightarrow n = 21$$

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Q. 24 If the variance of the following frequency distribution:

Class	$10-20$	$20-30$	$30-40$
Frequency	2	x	2

is 50, then x is equal to _____

Correct Answer:
4

Solution:
Variance is independent of shifting of origin

Class	$10-20$	$20-30$	$30-40$
Frequency	2	x	2

$\Rightarrow$

Class (x_i)	15	25	35
Frequency (f_i)	2	x	2

or

Class (x_i)	-10	0	10
Frequency (f_i)	2	x	2

$$\Rightarrow \text{ Variance } (\sigma^2) = \frac{\sum x_i^2 f_i}{\sum f_i} - (\bar{x})^2$$

$$\Rightarrow 50 = \frac{200+0+200}{x+4} - 0$$

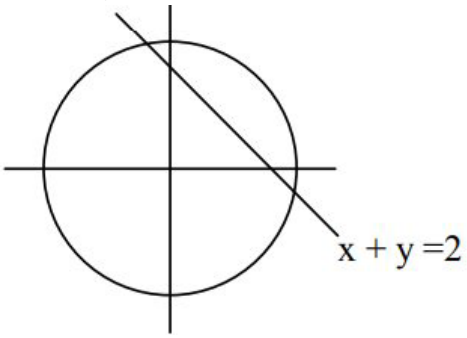
$$\Rightarrow 200 + 50x = 200 + 200$$

$$\Rightarrow x = 4$$

Q. 25 Let PQ be a diameter of the circle $x^2 + y^2 = 9$. If α and β are the lengths of the perpendiculars from P and Q on the straight line , x+y =2 respectively, then the maximum value of $\alpha\beta$ is_____

Correct Answer:
11

Solution:



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Let $P(3\cos\theta, 3\sin\theta)$

$Q(-3\cos\theta, -3\sin\theta)$

$$\alpha = \left|\frac{3\cos\theta + 3\sin\theta - 2}{\sqrt{1+1}}\right|$$

$$\beta = \left|\frac{-3\cos\theta - 3\sin\theta - 2}{\sqrt{1+1}}\right|$$

$$\Rightarrow \alpha\beta = \frac{|(3\cos\theta + 3\sin\theta)^2 - 4|}{2}$$

$$\Rightarrow \alpha\beta = \frac{5 + 9\sin 2\theta}{2} \leq 7$$

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5th September Shift 1

Physics

Q. 1 Assume that the displacement (s) of air is proportional to the pressure difference (Δp) created by a sound wave. Displacement (s) further depends on the speed of sound (ϑ), density of air (ρ) and the frequency (f). If $\Delta p \sim 10Pa,\ \vartheta \sim 300m/s,\ \rho \sim 1\ kg/m^3$ and $f - 1000\ Hz$, then s will be of the order of (take the multiplicative constant to be 1)

Option 1:
$\frac{3}{100}mm$

Option 2:
$10\ mm$

Option 3:
$\frac{1}{10}\ mm$

Option 4:
$1\ mm$

Correct Answer:
$\frac{3}{100}mm$

Solution:

$\because \Delta P_0 = \mathbf{B}\frac{\omega}{\mathbf{v}} \times \mathbf{S}_0$

$\Rightarrow \mathbf{S} = \frac{\Delta \mathbf{P} \times \mathbf{v}}{\mathbf{B}\omega} = \frac{\Delta \mathbf{P} \times \mathbf{v}}{\rho \mathbf{v}^2 \times \mathbf{2}\pi v}$

$\Rightarrow \mathbf{S} \propto \frac{\Delta \mathbf{P}}{\rho \mathbf{v} \mathbf{f}}$

$\Rightarrow \quad \mathbf{S} = \frac{\Delta P}{\rho \mathbf{P}}$

$= \frac{10}{1 \times 300 \times 1000}$

$= \frac{1}{30}\ \mathrm{mm}$

$\approx \frac{3}{100}\ \mathrm{mm}$

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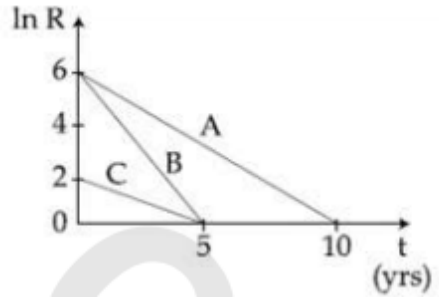
Q. 2 Activities of three radioactive substances A, B and C are represented by the curves A, B and C, in the figure. Then their half-lives $T_{\frac{1}{2}}(A) : T_{\frac{1}{2}}(B) : T_{\frac{1}{2}}(C)$ are in the ratio :



Option 1:
$2 : 1 : 1$

Option 2:
$3 : 2 : 1$

Option 3:
$2 : 1 : 3$

Option 4:
$4 : 3 : 1$

Correct Answer:
$2 : 1 : 3$

Solution:
$\mathrm{R} = \mathrm{R_0 e^{-\lambda t}}$

$\ln \mathrm{R} = \ln \mathrm{R_0} - \lambda \mathrm{t}$

$\lambda_{\mathrm{A}} = \frac{6}{10} \Rightarrow \mathrm{T_A} = \frac{10}{6} \ln 2$

$\lambda_{\mathrm{B}} = \frac{6}{5} \Rightarrow \mathrm{T_B} = \frac{5 \ln 2}{6}$

$\lambda_{\mathrm{C}} = \frac{2}{5} \Rightarrow \mathrm{T_C} = \frac{5 \ln 2}{2}$

$\frac{10}{6} : \frac{5}{6} : \frac{15}{6} :: 2 : 1 : 3$

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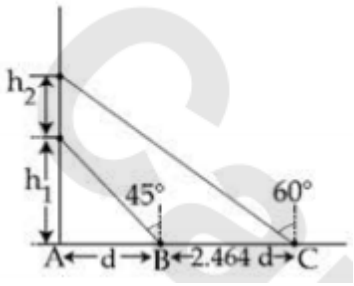
Q. 3 A baloon is moving up in air vertically above a point A on the ground. When it is at a height h_1, a girl standing at a distanced (point B) from A (see figure) sees it at an angle 45° with respect to the vertical. When the balloon climbs up a further height h_2, it is seen at an angle 60° with respect to the vertical if the gitl moves further by a distance $2.464\ d$ (point C). Then the height h_2 is (given $\tan 30^\circ = 0.5774$) :



Option 1:
$1.464\ d$

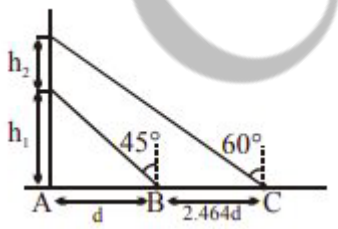
Option 2:
$0.732\ d$

Option 3:
$0.464\ d$

Option 4:
d

Correct Answer:
d

Solution:



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$$\frac{h_1}{d} = \tan 45^\circ \Rightarrow h_1 = d \ldots$$

$$\frac{h_1 + h_2}{d + 2.464d} = \tan 30^\circ$$

$$\Rightarrow (h_1 + h_2) \times \sqrt{3} = 3.46\ d$$

$$(h_1 + h_2) = \frac{3.46\ d}{\sqrt{3}}$$

$$\Rightarrow d + h_2 = \frac{3.46\ d}{\sqrt{3}}$$

$$h_2 = d$$

Q. 4 An electron is constrained to move along the y-axis with a speed of $0.1\ c$ (c is the speed of light) in the presence of electromagnetic wave, whose electric field is $\vec{E} = 30\ \hat{j} \sin(1.5 \times 10^7 t - 5 \times 10^{-2} x) V/m.$

The maximum magnetic force experienced by the electron will be :

(given $c = 3 \times 10^8 ms^{-1}$ and electron charge $= 1.6 \times 10^{-19} C$)

Option 1:

$3.2 \times 10^{-18} N$

Option 2:

$2.4 \times 10^{-18} N$

Option 3:

$4.8 \times 10^{-19} N$

Option 4:

$1.6 \times 10^{-19} N$

Correct Answer:

$4.8 \times 10^{-19} N$

Solution:

$\Rightarrow \mathrm{E} = \overrightarrow{\mathrm{E}} = 30\hat{\mathrm{j}} \sin(1.5 \times 10^7 \mathrm{t} - 5 \times 10^{-2} \mathrm{x})\ \mathrm{V/m}$

$\Rightarrow \mathrm{B} \Rightarrow \mathrm{E/V} \Rightarrow \frac{30}{1.5 \times 10^7} \times 5 \times 10^{-2}$

$\Rightarrow 10^{-7} \mathrm{Tesla}$

$\Rightarrow \mathrm{F_{mag}} = \mathrm{q}(\overrightarrow{\mathrm{V}} \times \overrightarrow{\mathrm{B}}) = |\mathrm{qVB}|$

$= 1.6 \times 10^{-19} \times 0.1 \times 3 \times 10^8 \times 10^{-7}$

$= 4.8 \times 10^{-19}\ \mathrm{N}$

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Q. 5 A helicopter rises from rest on the ground vertically upwards with a constant acceleration g. A food packet is dropped from the helicopter when it is at a height h. The time taken by the packet to reach the ground is close to [g is the acceleration due to gravity]:

Option 1:

$t = \frac{2}{3}\sqrt{\left(\frac{h}{g}\right)}$

Option 2:

$t = 1.8\sqrt{\frac{h}{g}}$

Option 3:

$t = 3.4\sqrt{\left(\frac{h}{g}\right)}$

Option 4:

$t = \sqrt{\frac{2h}{3g}}$

Correct Answer:

$t = 3.4\sqrt{\left(\frac{h}{g}\right)}$

Solution:

After time t_o the food packet is released

$$t_0 = \sqrt{\frac{2\ h}{g}}$$

Let V_o be the speed attained by helicopter after height h

$$\Rightarrow V_0 = \sqrt{\frac{2\ h}{g}} \times g = \sqrt{2gh}$$

$$\therefore t_1 = \text{ time to reach top } = \frac{v_0}{g} = \sqrt{\frac{2\ h}{g}}$$

Let h ' be the further height attained by the food packet when released

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$$\therefore \quad H = h + h' = 2\ h$$

$$\therefore \quad t_2 = \text{ time of fall } = \sqrt{\frac{2 \times (2\ h)}{g}}$$

$$= 2\sqrt{\frac{h}{g}}$$

$$\therefore \quad \text{Total time } = t_1 + t_2$$

$$= (2 + \sqrt{2})\sqrt{\frac{h}{g}}$$

$$= 3.4\sqrt{\frac{h}{g}}$$

Q. 6 A galvanometer of resistance G is converted into a voltmeter of range $0 - 1V$ by connecting a resistance R_1 in series with it, The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to $0 - 2V$ will be :

Option 1:
g

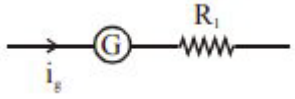
Option 2:
r_1

Option 3:
$R_1 - G$

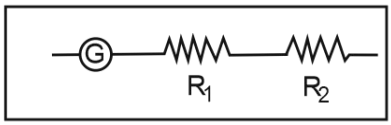
Option 4:
$R_1 + G$

Correct Answer:
$R_1 + G$

Solution:



$$\Rightarrow 1 = i_g\,(G + R_1)$$



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$\Rightarrow 2 = i_g\,(R_1 + R_2 + G)$

$Divide\ (1)\ by\ (2)$

$\Rightarrow \frac{1}{2} = \frac{G + R_1}{G + R_1 + R_2}$

$G + R_1 + R_2 = 2G + 2\ h$

$(R_2 = G + R_1)$

Q. 7 Two capacitors of capacitances C and $2C$ are charged to potential differences V and $2V$, respectively. These are then connected in parallel in such a manner that the positive terminal of one is connected to the negative terminal of the other. The final energy of this configuration is :

Option 1:

$\frac{25}{6}CV^2$

Option 2:

$\frac{3}{2}CV^2$

Option 3:

zero

Option 4:

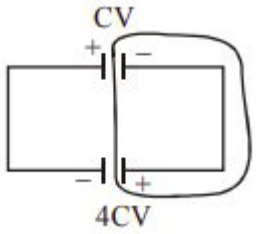
$\frac{9}{2}CV^2$

Correct Answer:

$\frac{3}{2}CV^2$

Solution:

$Q_1 = CV \quad Q_2 = 2C \times 2\ V = 4CV$



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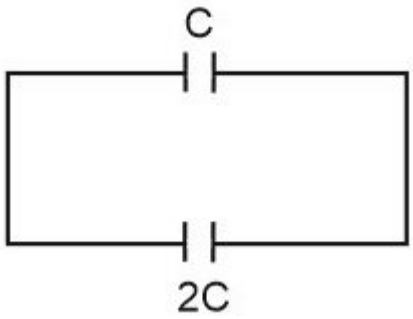
$\Rightarrow$ By conservation of charge

$q_i = q_f$

$Q_1 + Q_2 = q_1 + q_2$

$4CV - CV = (C + 2C)V_C$

$V_C = \frac{3CV}{3C} \Rightarrow V$



$\Rightarrow \frac{1}{2} \times (3C) \times V_c^2$

$= \frac{1}{2} \times 3C \times V^2 = \frac{3}{2}CV^2$

Q. 8 With increasing biasing voltage of a photodiode, the photocurrent magnitude :

Option 1:
remains constant

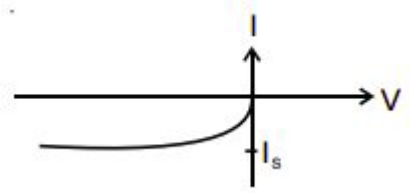
Option 2:
increases initially and after attaining certain value, it decreases

Option 3:
increases linearly

Option 4:
increases initially and saturates finally

Correct Answer:
increases initially and saturates finally

Solution:
I-V characteristic of a photodiode is as follows:



On increasing the potential difference the current first increases and then attains a saturation.

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Q. 9 The value of the acceleration due to gravity is g_1 at a height $h = \frac{R}{2}$ ($R =$ radius of the earth) from the surface of the earth. It is again equal to g_1 at a depth d below the suface of the earth. The ratio $\left(\frac{d}{R}\right)$ equals :

Option 1:
$\frac{4}{9}$

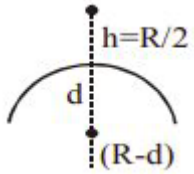
Option 2:
$\frac{5}{9}$

Option 3:
$\frac{1}{3}$

Option 4:
$\frac{7}{9}$

Correct Answer:
$\frac{5}{9}$

Solution:



$$g_1 = \frac{GM}{\left(R+\frac{R}{2}\right)^2} \ldots (1)$$

$$g_2 = \frac{GM(R-d)}{R^3} \ldots (2)$$

$$g_1 = g_2$$

$$\frac{\mathrm{GM}}{\left(\frac{3\mathrm{R}}{2}\right)^2} = \frac{\mathrm{GM(R-d)}}{\mathrm{R}^3}$$

$$\Rightarrow \frac{4}{9} = \frac{(\mathrm{R-d})}{\mathrm{R}}$$

$$4R = 9R - 9d$$

$$5\mathrm{R} = 9\ \mathrm{d} \Rightarrow \frac{\mathrm{d}}{\mathrm{R}} = \frac{5}{9}$$

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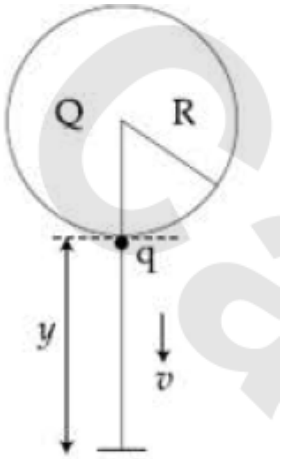
Q. 10 A solid sphere of radius R carries a charge $Q+q$ distributed uniformy over its volume. A very small point like piece of it of mass m gets detached from the bottom of the sphere and falls down vertically under gravity. This piece carries charge q. If it acquires a speed ν when it has fallen through a vertical height y (see figure), then :(assume the remaining portion to be spherical).



Option 1:

$$\nu^2 = y\left[\frac{qQ}{4\pi\epsilon_0 R^2 ym} + g\right]$$

Option 2:

$$\nu^2 = y\left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g\right]$$

Option 3:

$$\nu^2 = 2y\left[\frac{Qq\,R}{4\pi\epsilon_0 (R+y)^3 m} + g\right]$$

Option 4:

$$\nu^2 = 2y\left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g\right]$$

Correct Answer:

$$\nu^2 = 2y\left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g\right]$$

Solution:

$\frac{1}{2}mv^2 = mgy + \Delta U_{elec}$

$\Delta U_{elec.} = kQq\left[\frac{1}{R} - \frac{1}{(R+y)}\right]$

$\Rightarrow \quad v^2 = 2y\left[\frac{qQ}{4\pi\varepsilon_0 R(R+y)m} + g\right]$

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Q. 11 A wheel is rotating freely with an angular speed ω on a shaft. The moment of inertia of the wheel is I and the moment of inertia of the shaft is negligible . Another wheel of moment of inertia $3I$ initially at rest is suddenly coupled to the same shaft. The resultant fractional loss in the kinetic energy of the system is :

Option 1:
$\frac{5}{6}$

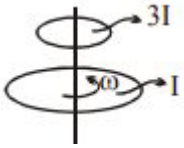
Option 2:
$\frac{1}{4}$

Option 3:
0

Option 4:
$\frac{3}{4}$

Correct Answer:
$\frac{3}{4}$

Solution:



By anglar momentum conservation

$$\omega I + 3I \times 0 = 4I\omega' \Rightarrow \omega' = \frac{\omega}{4}$$

$$(\mathrm{KE})_{\mathrm{i}} = \frac{1}{2}\mathrm{I}\omega^2$$

$$(\mathrm{KE})_{\mathrm{f}} = \frac{1}{2} \times (4\mathrm{I}) \times \left(\frac{\omega}{4}\right)^2 = \frac{\mathrm{I}\omega^2}{8}$$

$$\Delta\mathrm{KE} = \frac{3}{8}\mathrm{I}\omega^2$$

$$\text{fractional loss } = \frac{\Delta\mathrm{KE}}{\mathrm{KE}_1} = \frac{\frac{3}{8}\mathrm{I}\omega^2}{\frac{1}{2}\mathrm{I}\omega^2} = \frac{3}{4}$$

Q. 12 For a concave lens of focal length f, the relation between object and image distances u and ν, respectively, from its pole can best be represented by $(u = \nu)$ is the reference line) :

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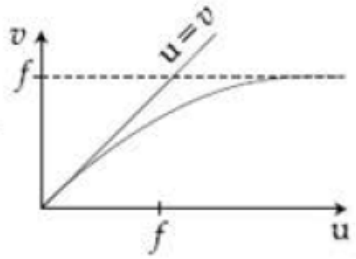
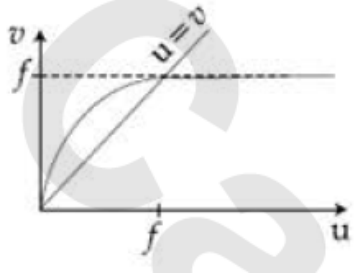
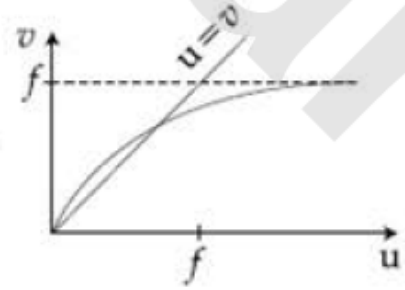
Option 1:



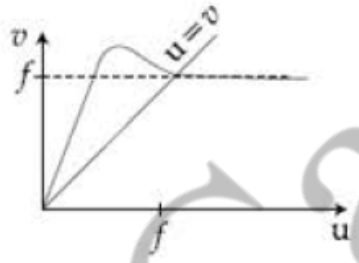
Option 2:



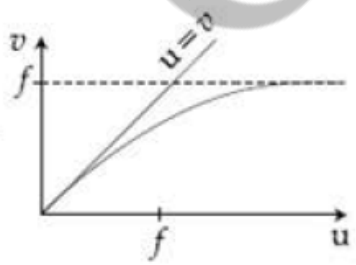
Option 3:



Option 4:



Correct Answer:



Solution:

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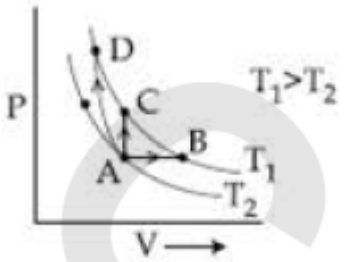
$$v = \frac{uf}{u+f}$$

Case-I
If $v = u$
$\Rightarrow f + u = f$
$\Rightarrow u = 0$

Case-II
If $u = \infty$
$then v = f$
Only option (4) satisfies this condition.

Q. 13 Three different processes that can occur in an ideal monocromatic gas are shown in the $P\ vs\ V$ diagram.The paths are labelled as $A \to B, A \to C$ and $A \to D$. The change in internal energies during these process are taken as E_{AB}, E_{AC} and E_{AD} and the workdone as $W_{AB},\ W_{AC}$ and W_{AD}. The correct relation between these parameters are :



Option 1:
$E_{AB} = E_{AC} < E_{AD},\ W_{AB} > 0,\ W_{AC} = 0,\ W_{AD} < 0$

Option 2:
$E_{AB} = E_{AC} = E_{AD},\ W_{AB} > 0,\ W_{AC} = 0,\ W_{AD} < 0$

Option 3:
$E_{AB} < E_{AC} < E_{AD},\ W_{AB} > 0,\ W_{AC} > W_{AD}$

Option 4:
$E_{AB} > E_{AC} > E_{AD},\ W_{AB} < W_{AC} < W_{AD}$

Correct Answer:
$E_{AB} = E_{AC} < E_{AD},\ W_{AB} > 0,\ W_{AC} = 0,\ W_{AD} < 0$

Solution:
$\Delta U = nC_v\Delta T = \text{ same}$
$AB \to \text{ volume is increasing } \Rightarrow W > 0$
$AD \to \text{ volume is decreasing } \Rightarrow W < 0$
$AC \to \text{ volume is constant } \Rightarrow W = 0$

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Q. 14 Number of molecules in a volume of $4\ cm^3$ of a perfect monoatomic gas at some temperature T and at pressure of $2\ cm$ of mercury is close to ?(Given, mean kinetic energy of a molecule (at T) is $4 \times 10^{-14} erg,\ g = 980 cm/s^2$, density of mercury $= 13.6\ g/cm^3$)

Option 1:
4.0×10^{18}

Option 2:
4.0×10^{16}

Option 3:
5.8×10^{16}

Option 4:
5.8×10^{18}

Correct Answer:
4.0×10^{18}

Solution:
$\mathrm{n} = \frac{\mathrm{PV}}{\mathrm{RT}}, \frac{3}{2}\mathrm{kT} = 4 \times 10^{-14}$

$\mathrm{N} = \frac{\mathrm{PV}}{\mathrm{RT}} \times \mathrm{Na}$

$= \frac{2 \times 13.6 \times 980 \times 4}{\frac{8}{3} \times 10^{-14}} = 3.99 \times 10^{18} \approx 4 \times 10^{18}$

Q. 15 A square loop of side $2a$, and carrying current I, is kept in XZ plane with its centre at origin. A long wire carrying the same current I is placed parallel to the z-axis and passing through the point $(0, b, 0), (b >> a)$. The magnitude of the torque on the loop about z-axis is given by :

Option 1:
$\frac{\mu_0 I^2 a^2}{2\pi b}$

Option 2:
$\frac{\mu_0 I^2 a^3}{2\pi b^2}$

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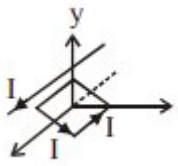
Option 3:
$\frac{2\mu_0 I^2 a^2}{\pi b}$

Option 4:
$\frac{2\mu_0 I^2 a^3}{\pi b^2}$

Correct Answer:
$\frac{2\mu_0 I^2 a^2}{\pi b}$

Solution:



$\vec{\tau} = \vec{\mathrm{M}} \times \vec{\mathrm{B}}$

$= 4\mathrm{a}^2\mathrm{I} \times \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{b}}$

$= \frac{2\mu_0 I^2 a^2}{\pi b}$

Q. 16

A physical quantity z depends on four observables a, b, c and d as $z = \frac{a^2 b^{\frac{2}{3}}}{\sqrt{c}\, d^3}$. The percentages of error in the measurment of a, b, c and d are $2\ ^o/_o, 1.5\ ^o/_o, 4\ ^o/_o$ and $2.5\ ^o/_o$ respectively. The percentage of error in z is :

Option 1:
$12.25\ ^o/_o$

Option 2:
$16.5\ ^o/_o$

Option 3:
$13.5\ ^o/_o$

Option 4:
$14.5\ ^o/_o$

Correct Answer:
$14.5\ ^o/_o$

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Solution:

$$\frac{\Delta Z}{Z}=\frac{2\Delta \mathrm{a}}{\mathrm{a}}+\frac{2}{3}\frac{\Delta \mathrm{b}}{\mathrm{b}}+\frac{1}{2}\frac{\Delta \mathrm{c}}{\mathrm{c}}+\frac{3\Delta \mathrm{d}}{\mathrm{d}}=14.5\%$$

Q. 17 A hollow spherical shell at outer radius R floats just submerged under the water surface. The inner radius of the shell is r. If the sphecific gravity of the shell material is $\frac{27}{8}$ w.r.t water, the value of r is :

Option 1:

$\frac{8}{9}R$

Option 2:

$\frac{4}{9}R$

Option 3:

$\frac{2}{3}R$

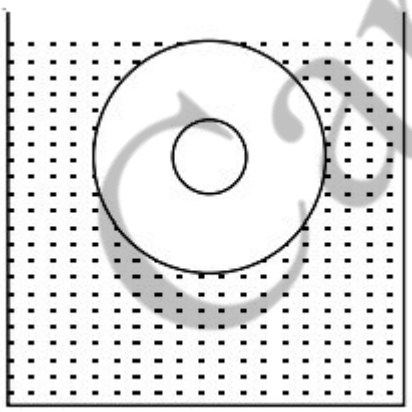
Option 4:

$\frac{1}{3}R$

Correct Answer:

$\frac{8}{9}R$

Solution:



$\frac{4}{3}\pi\left(\mathrm{R}^3-\mathrm{r}^3\right)\rho_{\mathrm{m}}\mathrm{g}=\frac{4}{3}\pi\mathrm{R}^3\rho_{\mathrm{w}}\mathrm{g}$

$1-\left(\frac{\mathrm{r}}{\mathrm{R}}\right)^3=\frac{8}{27}$

$\Rightarrow \frac{\mathrm{r}}{\mathrm{R}}=\left(\frac{19}{27}\right)^{1/3}=\frac{19^{1/3}}{3}$

$=0.88\simeq\frac{8}{9}$

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$$\Rightarrow r = \frac{8R}{9}$$

Q. 18 In a resonance tube experiment when the tube is filled with water up to a height of $17.0\ cm$ from bottom, it resonates with a given tuning fork. When the water level is raised the next resonance with the same tuning fork occurs at a height of $24.5\ cm$. If the velocity of sound in air is $330\ m/s$, the tuning fork frequency is :

Option 1:
$2200\ Hz$

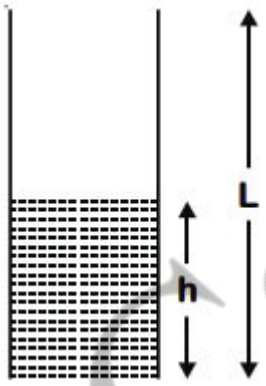
Option 2:
$550\ Hz$

Option 3:
$1100\ Hz$

Option 4:
$3300\ Hz$

Correct Answer:
$2200\ Hz$

Solution:



$\mathrm{L} - \mathrm{h}_1 = \mathrm{n}\frac{\lambda}{2} + \frac{\lambda}{4}$

$\mathrm{L} - \mathrm{h}_2 = (\mathrm{n}-1)\frac{\lambda}{2} + \frac{\lambda}{4}$

$\Rightarrow \quad \frac{\lambda}{2} = \mathrm{h}_2 - \mathrm{h}_1 = 24.5 - 17.0 = 7.5\ \mathrm{cm}$

$\Rightarrow \quad \lambda = 15\ \mathrm{cm}$

$\mathrm{u} = \mathrm{f}\lambda$

$\mathrm{f} = 2200\ \mathrm{Hz}$

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Q. 19 An electrical power line, having a total resistance of $2\ \Omega$, delivers $1\ kW$ at $220\ V$. The efficiency of the transmission line is approximately :

Option 1:
$72\ ^{o}/_{o}$

Option 2:
$91\ ^{o}/_{o}$

Option 3:
$85\ ^{o}/_{o}$

Option 4:
$96\ ^{o}/_{o}$

Correct Answer:
$96\ ^{o}/_{o}$

Solution:
$\mathrm{vi} = 10^3$

$\mathrm{i} = \frac{1000}{220}$

$\text{loss } = \mathrm{i}^2\mathrm{R} = \left(\frac{50}{11}\right)^2 \times 2$

$\text{efficiency } = \frac{1000}{1000+\mathrm{i}^2\mathrm{R}} \times 100 = 96\%$

Q. 20 A bullet of mass $5g$, travelling with a speed of $210\ m/s$, strikes a fixed wooden target. One half of its kinetic energy is converted into heat in the bullet while the other half is converted into heat in the wood. The rise of temperature of the bullet if the specific heat of its material is $0.030\ cal(g - ^{\circ} C)\ (1\ cal = 4.2 \times 10^7\ ergs)$

Option 1:
$87.5^{\circ}C$

Option 2:
$83.3^{\circ}C$

Option 3:
$119.2^{\circ}C$

Option 4:
$38.4^{\circ}C$

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Correct Answer:
$87.5°C$

Solution:
$\frac{1}{2}mv^2 \times \frac{1}{2} = ms\Delta T$

$\Delta T = \frac{v^2}{4\times5} = \frac{210^2}{4\times30\times4.200}$

$= 87.5°\text{C}$

Q. 21 A force $\vec{F} = (\hat{i} + 2\hat{j} + 3\hat{k})N$ acts at a point $(4\hat{i} + 3\hat{j} - \hat{k})m$. Then the magnitude of torque about the point $(\hat{i} + 2\hat{j} + \hat{k})m$ will be $\sqrt{x}\ N - m$. The value of x is _____.

Correct Answer:
195

Solution:
$\vec{\tau} = (\vec{r}_2 - \vec{r}_1) \times \vec{F}$
$= [(4\hat{i} + 3\hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + \hat{k})] \times \vec{F}$
$= (3\hat{i} + \hat{j} - 2\hat{k}) \times (\hat{i} + 2\hat{j} + 3\hat{k})$
$$\tau = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & 2 & 3 \end{vmatrix}$$
$= 7\hat{i} - 11\hat{j} + 5\hat{k}$
$|\vec{\tau}| = \sqrt{195}$

Q. 22 A beam of electrons of energy E scatters from a target having atomic spacing of $1\,\mathring{A}$. The first maximum intensity occurs at $\theta = 60°$. Then E (in eV) is _____.

(Plank constant $h = 6.64 \times 10^{-34} Js$,

$1\ eV = 1.6 \times 10^{-19} J$,

electron mass $m = 9.1 \times 10^{-31} kg$)

Correct Answer:
50.47

Solution:

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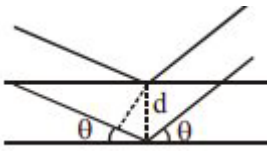

$2\ \mathrm{d}\sin\theta = \lambda = \frac{\mathrm{h}}{\sqrt{2\mathrm{mE}}}$

$2 \times 10^{-10} \times \frac{\sqrt{3}}{2} = \frac{6.6\times10^{-34}}{\sqrt{2\mathrm{mE}}}$

$\mathrm{E} = \frac{1}{2} \times \frac{6.64^2\times10^{-48}}{9.1\times10^{-31}\times3\times1.6\times10^{-19}} = 50.47$

Q. 23 A particle of mass $200\ MeV/c^2$ collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is $\frac{N}{4}$. The value of N is :

(Given the mass of the hydrogen atom to be $1\ GeV/c^2$)_____.

Correct Answer:
51

Solution:
$M_0 = 200\mathrm{MeV/c^2}; \mathrm{m} = 1\mathrm{GeV/c^2}$
$\text{Initial velocity } = v_0$

$\mathrm{Mv_0} = \mathrm{mv}$

$\therefore \mathrm{v} = \left(\frac{M}{m}\right)\mathrm{v_0}$

$\frac{1}{2}\mathrm{Mv_0^2} = \frac{1}{2}\mathrm{mv}^2 + \frac{\Delta\mathrm{E_0}\times3}{4} \quad (\Delta\mathrm{E_0} = 13.6\mathrm{eV})$

$\Rightarrow \quad \frac{1}{2}\mathrm{Mv_0^2}\left(1 - \frac{\mathrm{M}}{\mathrm{m}}\right) = \frac{3}{4}\Delta\mathrm{E_0}$

$\Rightarrow \frac{1}{2}\mathrm{Mv_0^2} = \frac{3}{4}\times\Delta\mathrm{E_0}\times\frac{10}{8} = \frac{1}{4}\left(3\Delta\mathrm{E_0}\times\frac{10}{8}\right)$

$= \frac{51}{4} = \frac{51}{4}\mathrm{eV}$

$\therefore \mathrm{N} = 51$

Q. 24 A compound microscope consists of an objective lens of focal length $1\ cm$ and an eye piece of focal length $5\ cm$ with a seperation of $10\ cm$.The distance between an object and the objective lens, at which the strain on the eye is minimum is $\frac{n}{40}cm$ The value of n is _____.

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Correct Answer:
50

Solution:

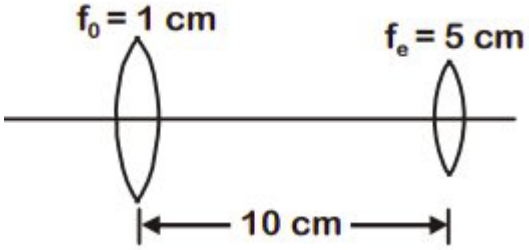

Image by objective is formed at the focus of eye-piece

$\therefore For \;\; objective, v = 5, u, f = 1 \text{ cm}$

$$\frac{1}{5} - \frac{1}{u} = \frac{1}{1} \Rightarrow \frac{1}{5} - 1 = \frac{1}{u}$$

$$\therefore \quad |u| = \frac{5}{4} cm \Rightarrow |u| = \frac{50}{40} cm$$

$$\therefore \quad n = 50$$

Q. 25 Two concentric circular coils, C_1 and C_2 , are placed in the XY plane. C_1 has 500 turns, and a radius of $1\ cm$. C_2 has 200 turns and radius of $20\ cm$. C_2 carries a time dependent current $I(t) = \left(5t^2 - 2t + 3\right) A$ where t is in s. The emf indiuced in C_1 (in mV), at the instant $t = 1s$ is $\frac{4}{x}$. The value of x is _____.

Correct Answer:
5

Solution:

$.1 = 5t^2 - 2t + 3$

$\frac{dI}{dt} = 10t - 2$

$$\text{At } (t = 1 \text{ sec }) \left(\frac{dI}{dt}\right) = 8 \text{ A/s}$$

$$\phi = \left(\frac{\mu_0 \times 200 \times 1 \times 100}{2 \times 20}\right) \times \frac{\pi \times 500}{100 \times 100}$$

$$\therefore \quad \mathbf{e} = \left|\frac{d\phi}{dt}\right| = \frac{\mu_0 \times 200 \times 100 \times 500 \times \pi}{2 \times 20 \times 100 \times 100} \left(\frac{dl}{dt}\right)$$

$$\Rightarrow \quad e = \left|\frac{d\phi}{dt}\right| = \left(\frac{8}{10}\right) \text{m volt } = \frac{4}{5} \text{ m volt}$$

$$x = 5$$

Chemistry

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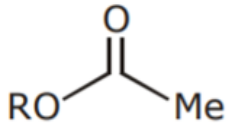
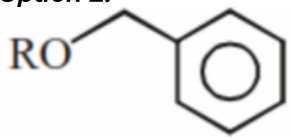
Q. 1 Which of the following derivatives of alcohols is unstable in an aqueous base?

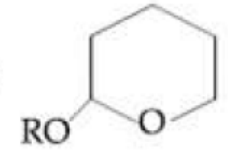
Option 1:



Option 2:



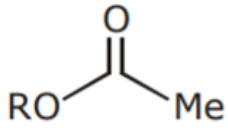
Option 3:



Option 4:

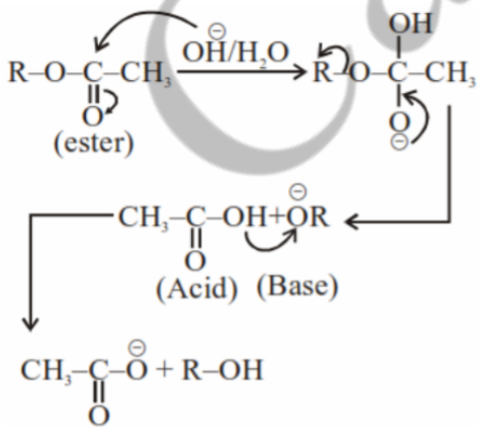
$RO - CMe_3$

Correct Answer:



Solution:

Hydrolysis of ester occurs in a basic medium due to it is unstable in an aqueous base.



Therefore, the correct option is (1).

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Q. 2 The values of the crystal field stabilization energies for high spin d^6 metal ion in octahedral and tetrahedral fields, respectively, are :

Option 1:
$-0.4\ \Delta_0 \text{ and } -0.6\ \Delta_t$

Option 2:
$-2.4\ \Delta_0 \text{ and } -0.6\ \Delta_t$

Option 3:
$-1.6\ \Delta_0 \text{ and } -0.4\ \Delta_t$

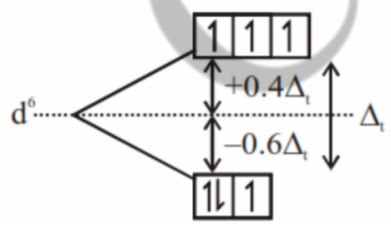
Option 4:
$-0.4\ \Delta_0 \text{ and } -0.27\ \Delta_t$

Correct Answer:
$-0.4\ \Delta_0 \text{ and } -0.6\ \Delta_t$

Solution:
For high spin octahedral field

$$\text{CFSE} = (4)\,(-0.4\Delta_0) + 2\,(0.6\Delta_0) = -0.4\Delta_0$$

For high spin tetrahedral field



$$\text{CFSE} = 3\,(-0.6\Delta_t) + 3\,(0.4\Delta_t) = -0.6\Delta_t$$

The values of the crystal field stabilization energies for a high spin d^6 metal ion in octahedral and tetrahedral fields, respectively, are $-0.4\Delta_0 \text{ and } -0.6\Delta_t$

Therefore, the correct option is (1).

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Q. 3 Consider the following reaction :

$$N_2O_4(g) \rightleftharpoons 2NO_2(g);\ \Delta H^0 = +58\ kJ$$

For each of the following cases (a, b), the direction in which the equilibrium shifts is :

(a) Temperature is decreased.

(b) Pressure is increased by adding N_2 at constant T.

Option 1:
(a) towards product, (b) towards reactant

Option 2:
(a) towards reactant, (b) towards product

Option 3:
(a) towards reactant, (b) no change

Option 4:
(a) towards product, (b) no change

Correct Answer:
(a) towards reactant, (b) no change

Solution:
The reaction is -

$$N_2O_4(\ g) \rightleftharpoons 2NO_2(\ g); \Delta H^\circ = +58\ kJ$$

Here, $\Delta H^\circ = +58\ kJ > 0$

Enthalpy is greater than zero as it is an endothermic reaction, so for forwarding reaction needs energy,

(a) Temperature is decreased, then it will go backward or towards reactant.

(b) Pressure is increased by adding N_2 at constant T, Here T is constant and N_2 is treated as an inert gas, As 'N2 ' will not react with both N_2O_4 & NO_2, as moles increases in reactants, as much as in products. Hence no effect on the equilibrium.

Option : (a) towards reactant, (b) no change

Therefore, the correct option is (3).

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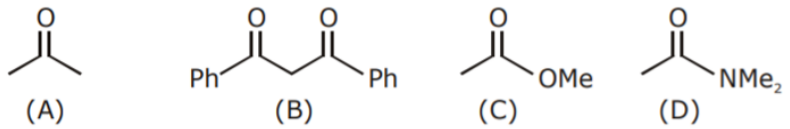
Q. 4 The increasing order of the acidity of the **α**—hydrogen of the following compounds is :



Option 1:
(D) < (C) < (A) < (B)

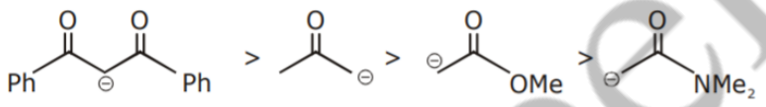
Option 2:
(B) < (C) < (A) < (D)

Option 3:
(A) < (C) < (D) < (B)

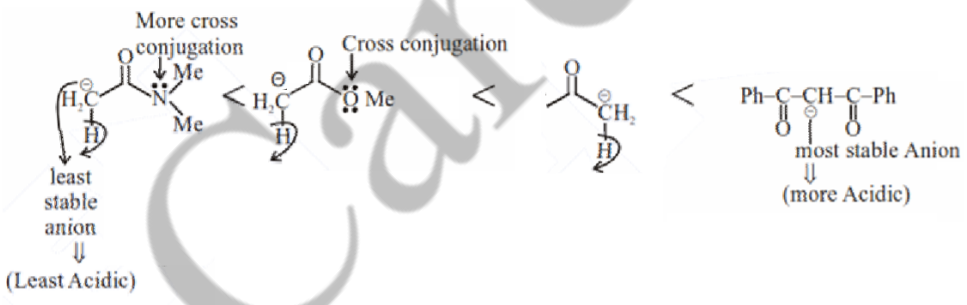
Option 4:
(C) < (A) < (B) < (D)

Correct Answer:
(D) < (C) < (A) < (B)

Solution:
Stability order



Reason:



Option: (D) < (C) < (A) < (B)

Therefore, the correct option is (1).

Q. 5 A diatomic molecule X_2 has a body-centred cubic (bcc) structure with a cell edge of 300 pm. The density of the molecule is $6.17\ g\ cm^{-3}$. The number of molecules present in 200 g of X_2 is :

(Avogadro constant $(N_A) = 6 \times 10^{23} mol^{-1}$)

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Option 1:
$40\ N_A$

Option 2:
$8\ N_A$

Option 3:
$4\ N_A$

Option 4:
$2\ N_A$

Correct Answer:
$4\ N_A$

Solution:
Asking : The number of molecules present in 200 g of X_2 ?

1 mole = 6 X 10^{23} Molecules. = N_A Molecules

Mole = mass/ Molar mass

Given mass = 200 g

Now need to find Molar mass.

We know this formula For **Molar mass**,

$$\text{Density of unit cell } = \frac{\text{Z} \times \text{M}}{\text{N}_\text{A} \times \text{a}^3}$$

Z = Number of atoms

M = Molar mass

a = Cell edge length

N_A = Avogadro constant (N_A) = 6 X 10^{23} mol^{-1}

A diatomic molecule X_2 has a body-centered cubic (bcc) structure with a cell edge of 300 pm. The density of the molecule is 6.17 g cm^{-3}.

So,

Z = 2 for BCC

a = 300 pm = 300 X 10^{-10} cm

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d = 6.17 g cm^{-3}

Now,

$$d = 6.17\ g/cm^3 = \frac{2 \times M}{6 \times 10^{23} \times (300 \times 10^{-10})^3}$$

$$M = \frac{6.17 \times 6 \times 9 \times 3 \times 10^{-1}}{2}$$

$$M = 81 \times 6.17 \times 10^{-1}$$

$$M = 49.97\ g/mol$$

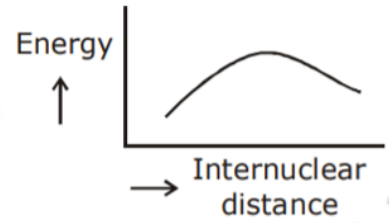
Hence ,$\text{Mole} = \frac{200g}{49.97g/mol} \approx 4\ mol$

So, 4 mole = 4 X 6 X 10^{23} Molecules. = **$4N_A$ Molecules**

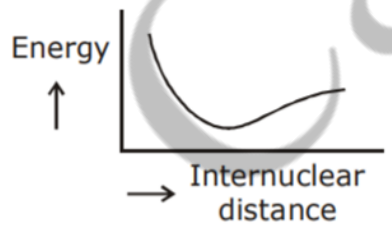
Therefore, the correct option is (3).

Q. 6 The potential energy curve for the H_2 molecule as a function of internuclear distance is :

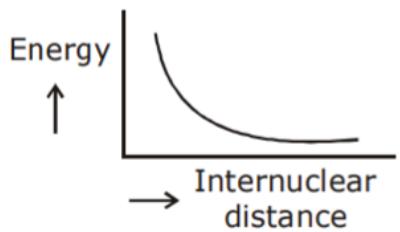
Option 1:



Option 2:



Option 3:



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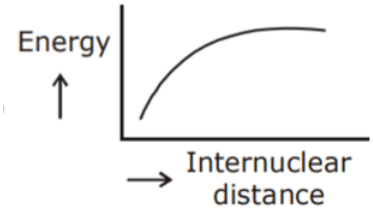
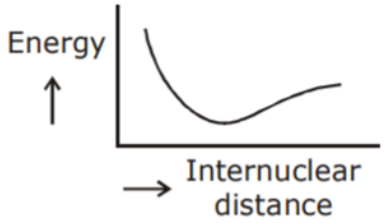
Option 4:

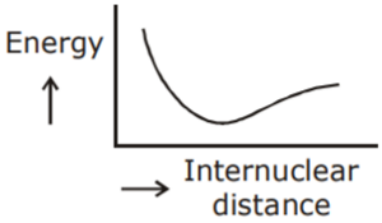


Correct Answer:

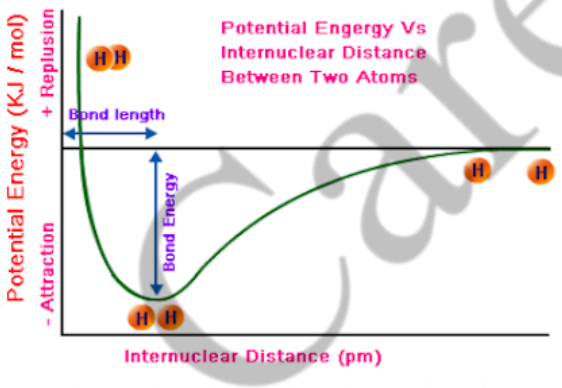


Solution:

The potential energy curve for H_2 molecules is.



It will follow the same as



Therefore, the correct option is (2).

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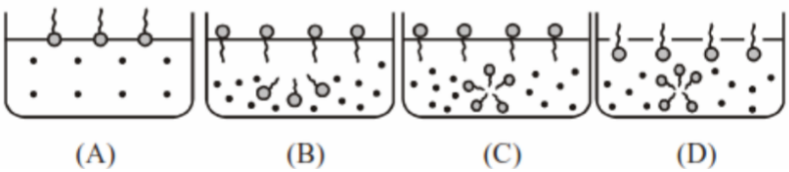
Q. 7 Identify the correct molecular picture showing what happens at the critical micellar concentration (CMC) of an aqueous solution of a surfactant

(○ polar head; ~ non-polar tail; • water).



Option 1:
(D)

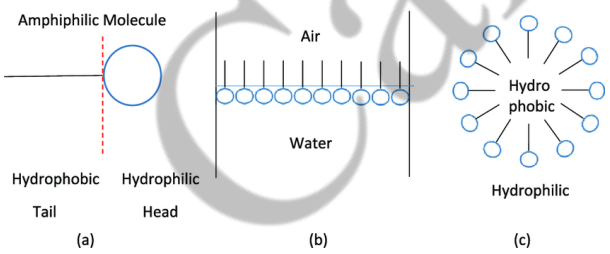
Option 2:
(B)

Option 3:
(A)

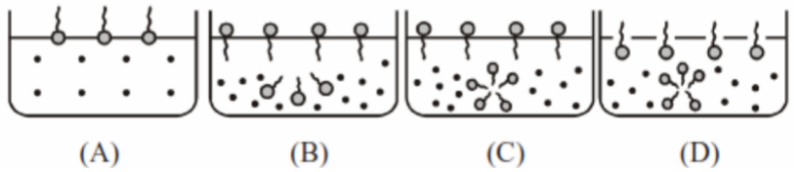
Option 4:
(C)

Correct Answer:
(D)

Solution:
The mechanism of micellar concentration at the critical micellar concentration.



Only Image (D) is following the same pattern among them.



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Therefore, the correct option is (1).

Q. 8 The difference between the radii of 3rd and 4th orbits of Li^{2+} is ΔR_1. The difference between the radii of 3rd and 4th orbits of He^{+} is ΔR_2. Ratio $\Delta R_1 : \Delta R_2$ is :

Option 1:
8 : 3

Option 2:
3 : 8

Option 3:
2 : 3

Option 4:
3 : 2

Correct Answer:
2 : 3

Solution:
We know the formula,

The radius of nth Bohr's Orbit

$$\mathrm{r} = 0.529 \times \frac{\mathrm{n}^2}{\mathrm{Z}}\ \mathrm{A}^{\circ}$$

Where
n = principal quantum number of orbit.
Z = Atomic number

Now,

The difference between the radii of 3rd and 4th orbits of Li^{2+} is ΔR_1.

$$(\mathrm{R}_4 - \mathrm{R}_3)_{\mathrm{Li}^{+2}} = \frac{0.529}{3}\left\{4^2 - 3^2\right\} = \Delta \mathrm{R}_1$$

The difference between the radii of 3rd and 4th orbits of He^{+} is ΔR_2.

$$(\mathrm{R}_4 - \mathrm{R}_3)_{\mathrm{He}^{+2}} = \frac{0.529}{2}\left\{4^2 - 3^2\right\} = \Delta \mathrm{R}_2$$

Ratio $\Delta R_1 : \Delta R_2$ is

$$\frac{\Delta \mathrm{R}_1}{\Delta \mathrm{R}_2} = \frac{(\mathrm{R}_4 - \mathrm{R}_3)_{\mathrm{Li}^{2+}}}{(\mathrm{R}_4 - \mathrm{R}_3)_{\mathrm{He}^{+}}} = \frac{\frac{4^2}{3} - \frac{3^2}{3}}{\frac{4^2}{2} - \frac{3^2}{2}} = \frac{7/3}{7/2} = \frac{2}{3}$$

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Therefore, the correct option is (3).

Q. 9 In the sixth period, the orbitals that are filled are :

Option 1:
6s, 4f, 5d, 6p

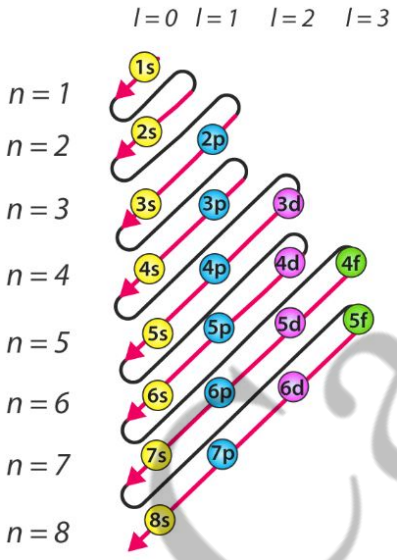
Option 2:
6s, 5d, 5f, 6p

Option 3:
6s, 5f, 6d, 6p

Option 4:
6s, 6p, 6d, 6f

Correct Answer:
6s, 4f, 5d, 6p

Solution:
Energy order of orbital's according to Aufbau principle-



The order of orbitals filling is 6s, 4f, 5d, 6p.

Therefore, the correct option is (1).

Q. 10 The most appropriate reagent for conversion of C_2H_5CN into $CH_3CH_2CH_2NH_2$ is :

Option 1:
$NaBH_4$

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Option 2:
CaH_2

Option 3:
$LiAlH_4$

Option 4:
$Na(CN)BH_3$

Correct Answer:
$LiAlH_4$

Solution:
The given conversion can be carried out by $LiAlH_4$

$$CH_3 - CH_2 - C \equiv N \xrightarrow{LiAlH_4} CH_3 - CH_2 - CH_2 - NH_2$$

Therefore, the correct option is (3).

Q. 11 If a person is suffering from the deficiency if nor-adrenaline, what kind of drug can be suggested ?

Option 1:
Anti-inflammatory

Option 2:
Antidepressant

Option 3:
Antihistamine

Option 4:
Analgesic

Correct Answer:
Antidepressant

Solution:
Non-adrenaline is a neurotransmitter. If nor-adrenaline is low, a person may suffer from depression. Hence, an antidepressant drug is suggested.

Anti-depressant → drug, which enhances the mood.

Therefore, the correct option is (2).

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Q. 12 Which of the following is not an essential amino acid ?

Option 1:
Tyrosine

Option 2:
Leucine

Option 3:
Valine

Option 4:
Lysine

Correct Answer:
Tyrosine

Solution:
Tyrosine is not an essential amino acid.

See the list below.

Essential AA	Nonessential AA	Conditionally essential AA
Arginine	Alanine	Cysteine
Histidine	Asparagine	Glutamine
Isoleucine	Aspartate	Hydroxyproline
Leucine	Glutamate	Proline
Lysine	Glycine	Taurine
Methionine	Serine	
Phenylalanine	Tyrosine	
Threonine		
Tryptophan		
Valine		

Therefore, the correct option is (1).

Q. 13 The correct electronic configuration and spin-only magnetic moment (BM) of $\mathrm{Gd}^{3+}(\mathrm{Z}=64)$, respectively, are :

Option 1:
$[\mathrm{Xe}]\ 4\mathrm{f}^{7}$ and 8.9

Option 2:
$[\mathrm{Xe}]\ 4\mathrm{f}^{7}$ and 7.9

Option 3:
$[\mathrm{Xe}]\ 5\mathrm{f}^{7}$ and 8.9

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Option 4:
$[Xe]\ 5f^7$ and 7.9

Correct Answer:
$[Xe]\ 4f^7$ and 7.9

Solution:
Electronic configuration of Gd^{3+} is ${}_{64}Gd^{3+} = [Xe]4f^7$ [Xe] 1 1 1 1 1 1 1

Gd^{3+} having 7 unpaired electrons.

Magnetic moment $(\mu) = \sqrt{n(n+2)}B.M.$

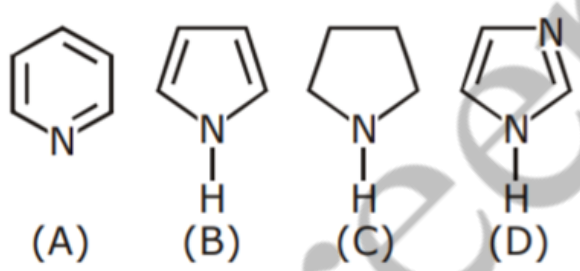
Where n = number of unpaired electrons.

So,

$\mu = \sqrt{7(7+2)} = 7.9\ B.M.$

Therefore, the correct option is (2).

Q. 14 The increasing order of basicity of the following compounds is :



Option 1:
(A) < (B) < (C) < (D)

Option 2:
(B) < (A) < (D) < (C)

Option 3:
(D) < (A) < (B) < (C)

Option 4:
(B) < (A) < (C) < (D)

Correct Answer:
(B) < (A) < (D) < (C)

Solution:

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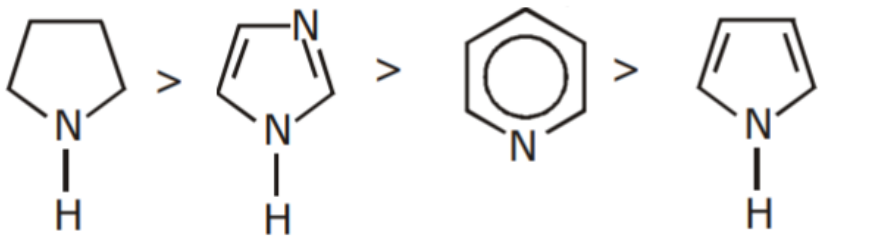
Correct order of basic strength is:



$pK_b = 2.7$ $pK_b = 7.05$ $pK_b = 8.75$ $pK_b = 13.6$

Therefore, the correct order of Basic strength is (B) < (A) < (D) < (C).

Therefore, the correct option is (2).

Q. 15 A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives for A and B are $300\ s$ and $180\ s$, respectively. If the concentrations of A and B are equal initially, the time required for the concentration of A to be four times that of B (in s) is : (Use e ln 2 = 0.693)

Option 1:
180

Option 2:
900

Option 3:
300

Option 4:
120

Correct Answer:
900

Solution:
We know this for first-order reactions-

$$A_t = A_0 \cdot e^{-k_1 t}$$

$$B_t = B_0 \cdot e^{-k_1 t}$$

So, half-lives

$$k_1 = \frac{\ln 2}{300}$$

$$k_2 = \frac{\ln 2}{180}$$

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The concentration of A to be four times that of B

So, A_t and B_t are related as [A] = 4[B]

$$A_0 \cdot e^{-k_1 t} = 4 \times B_0 \cdot e^{-k_2 t}$$

If the concentrations of A and B are equal initially, $A_0 = B_0$

Then,

$$e^{-k_1 t} = 4e^{-k_2 t}$$

$$e^{(k_2 - k_1)t} = 4$$

$$(k_2 - k_1)t = \ln 4$$

$$\left(\frac{\ln 2}{180} - \frac{\ln 2}{300}\right) t = 2\ln 2$$

$$\frac{t}{180} - \frac{t}{300} = 2$$

$$\frac{t}{3} - \frac{t}{5} = 120$$

$$\frac{2t}{15} = 120$$

t = 900 sec

Therefore, the correct option is (2).

Q. 16 The structure of PCl_5 in the solid state is :

Option 1:
tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$

Option 2:
square planar $[PCl_4]^+$ and octahedral $[PCl_6]^-$

Option 3:
square pyramidal

Option 4:
trigonal bipyramidal

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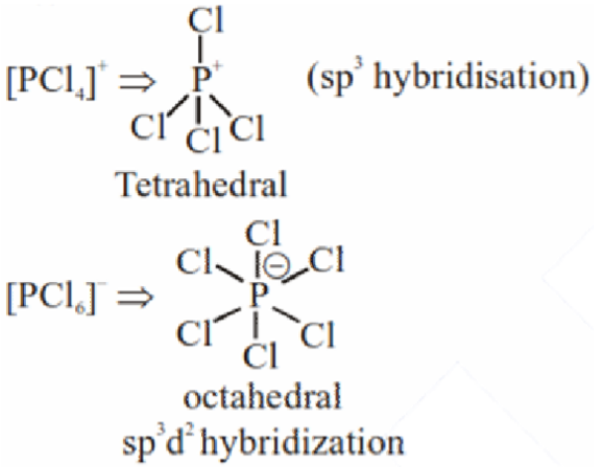
Correct Answer:
tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$

Solution:
In solid-state PCl_5 exist in ion-pair as $[PCl_4]^+$ and $[PCl_6]^-$



So, tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$

Therefore, the correct option is (1).

Q. 17 An Ellingham diagram provides information about :

Option 1:
The conditions of pH and potential under which a species is thermodynamically stable

Option 2:
The temperature dependence of the standard Gibbs energies of formation of some metal oxides.

Option 3:
The pressure dependence of the standard electrode potentials of the extraction of metals.

Option 4:
The kinetics of the reduction process.

Correct Answer:
The temperature dependence of the standard Gibbs energies of formation of some metal oxides.

Solution:
Ellingham diagram provides information about temperature dependence of the standard Gibbs energies of **formation of some metal oxides.**

See below Ellingham diagram-

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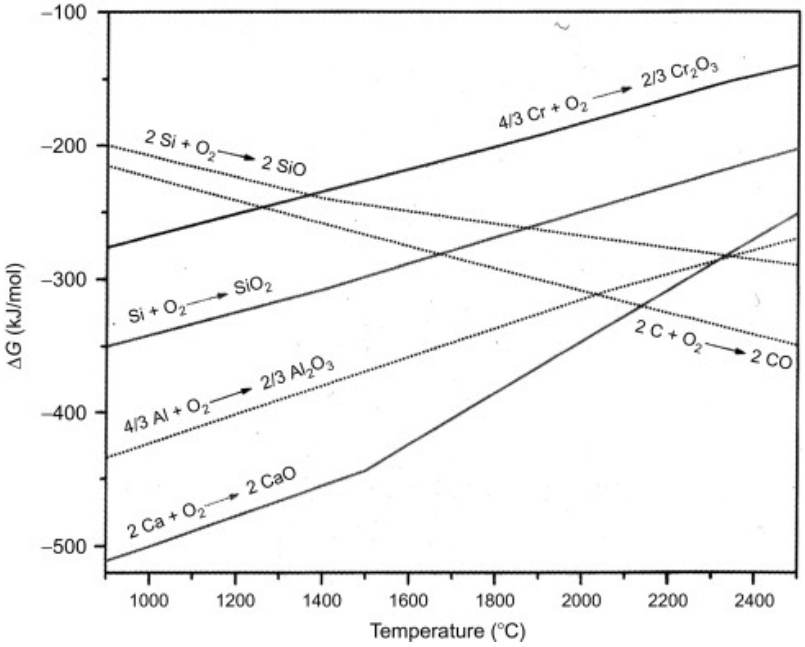
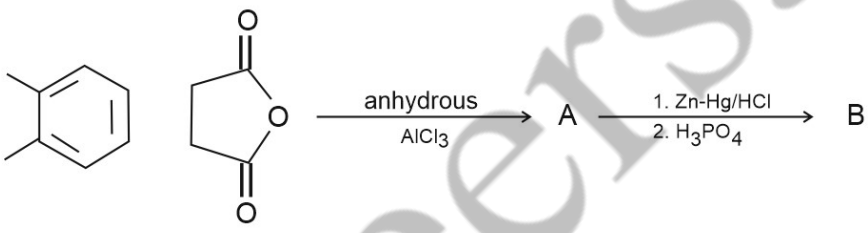
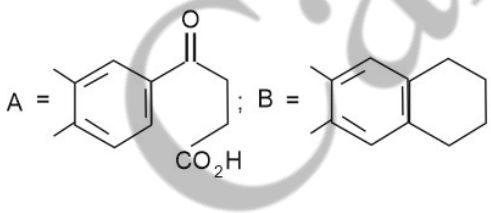


Therefore, the correct option is (2).

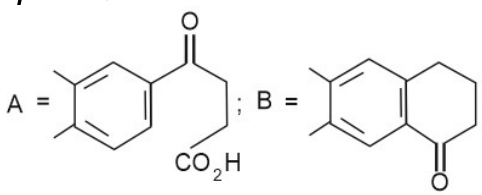
Q. 18 In the following reaction sequence the major products A and B are :



Option 1:



Option 2:



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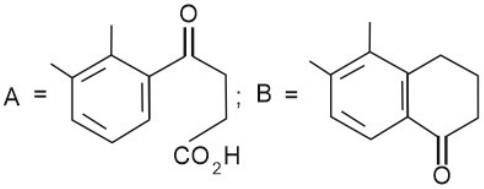
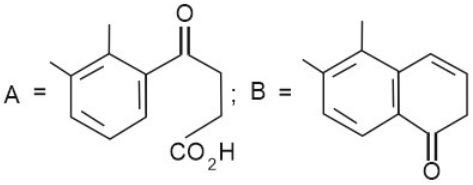
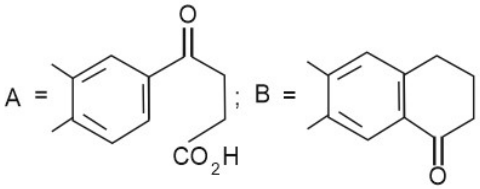
Option 3:



Option 4:

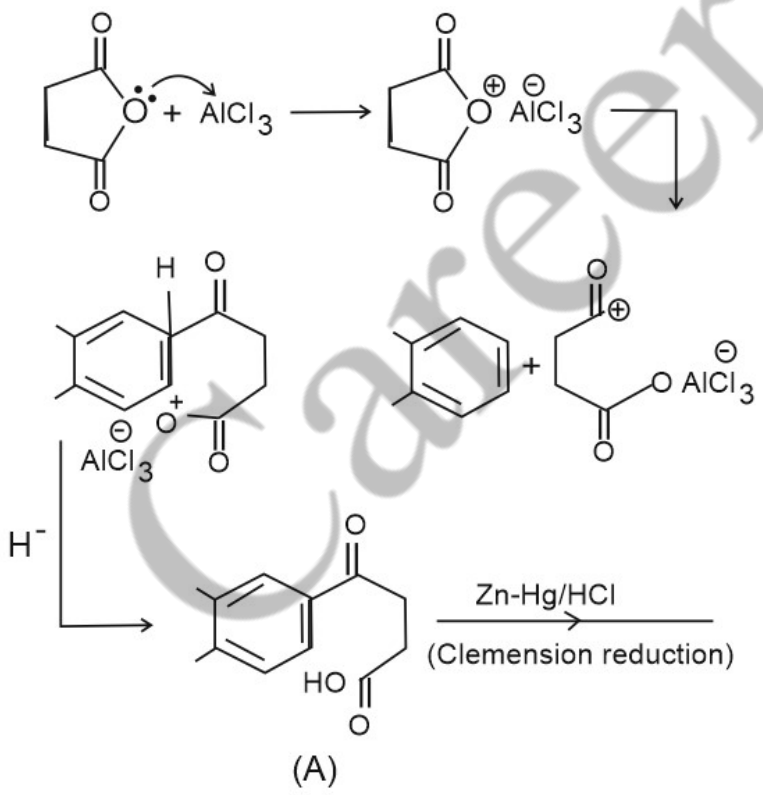


Correct Answer:



Solution:

The reaction will be -



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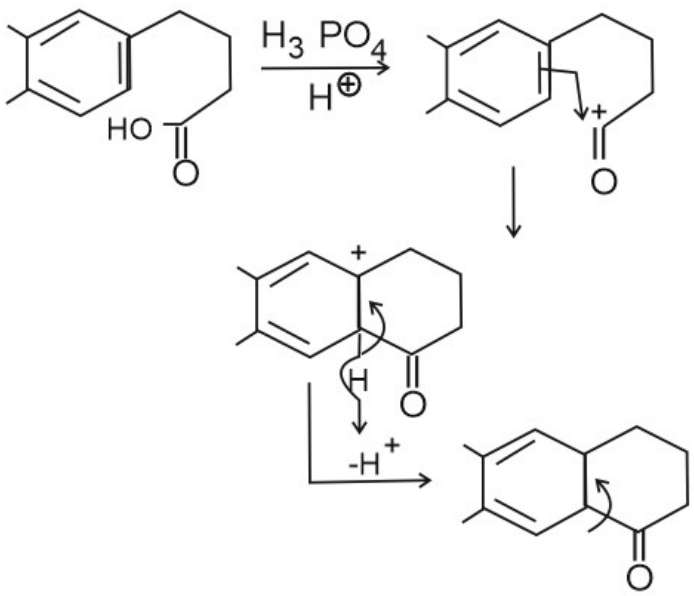

Therefore, the correct option is (2).

Q. 19 The equation that represents the water-gas shift reaction is :

Option 1:

$$CH_4(g) + H_2O(g) \xrightarrow[Ni]{1270K} CO(g) + 3H_2(g)$$

Option 2:

$$2C(s) + O_2(g) + 4N_2 \xrightarrow{1273K} 2CO(g) + 4N_2(g)$$

Option 3:

$$C(s) + H_2O(g) \xrightarrow{1270K} CO(g) + H_2(g)$$

Option 4:

$$CO(g) + H_2O(g) \xrightarrow[catalyst]{673\ K} CO_2(g) + H_2(g)$$

Correct Answer:

$$CO(g) + H_2O(g) \xrightarrow[catalyst]{673\ K} CO_2(g) + H_2(g)$$

Solution:

Let check given equations one by one-

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(1) Water gas shift reaction

$$CO_{(g)} + H_2O_{(g)} \xrightarrow[\text{Catalyst}]{673K} CO_{2(g)} + H_{2(g)}$$

(2) Water gas is produced by this reaction.

$$CH_{4(g)} + H_2O_{(g)} \xrightarrow[Ni]{1270K} CO_{(g)} + 3\ H_{2(g)}$$

(3) Water gas is produced by this reaction

$$C_{(s)} + H_2O_{(g)} \xrightarrow{1270K} CO_{(g)} + H_{2(g)}$$

(4) producer gas is produced by this reaction.

$$2C_{(s)} + O_{2(g)} + 4N_{2(g)} \xrightarrow{1270K} 2CO_{(g)} + 4N_{2(g)}$$

Therefore, the correct option is (4).

Q. 20 The condition that indicates a polluted environment is :

Option 1:
eutrophication

Option 2:
0.03% of CO_2 in the atmosphere

Option 3:
BOD value of 5 ppm

Option 4:
pH of rain water to be 5.6

Correct Answer:
eutrophication

Solution:
In Eutrophication nutrient-enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity. It indicates a polluted environment.

Therefore, the correct option is (1).

Q. 21 The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is _____

Correct Answer:
18

Solution:

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Combustion reaction of 1 mole of propane and 2 moles of butane-

$$\underset{1\,\text{mol}}{C_3H_8} + \underset{5\,\text{mol}}{5O_2} \rightarrow 3CO_2 + 4H_2O$$

$$\underset{2\,\text{mol}}{C_4H_{10}} + \underset{13\,\text{mol}}{\frac{13}{2}O_2} \rightarrow 4CO_2 + 5H_2O$$

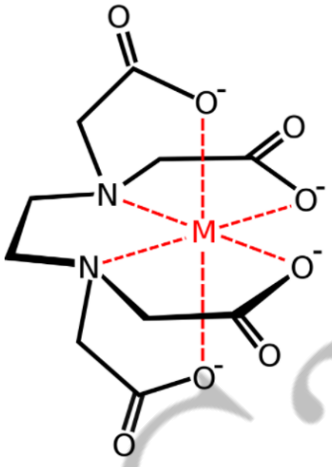
So, Total required mol of O_2 = 5 + 13 = 18.

Ans = 18

Q. 22 The total number of coordination sites in ethylenediaminetetraacetate $(EDTA^{4-})$ is ____.

Correct Answer:
6

Solution:
$EDTA^{4-}$ is a hexadentate ligand, so its donation sites are six.



Ans = 6

Q. 23 The number of chiral carbon(s) present in peptide, Ile-Arg-Pro, is _____.

Correct Answer:
4

Solution:
We know

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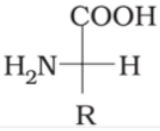
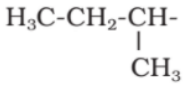
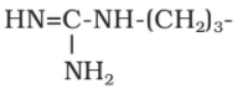
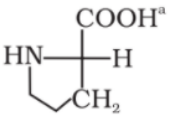
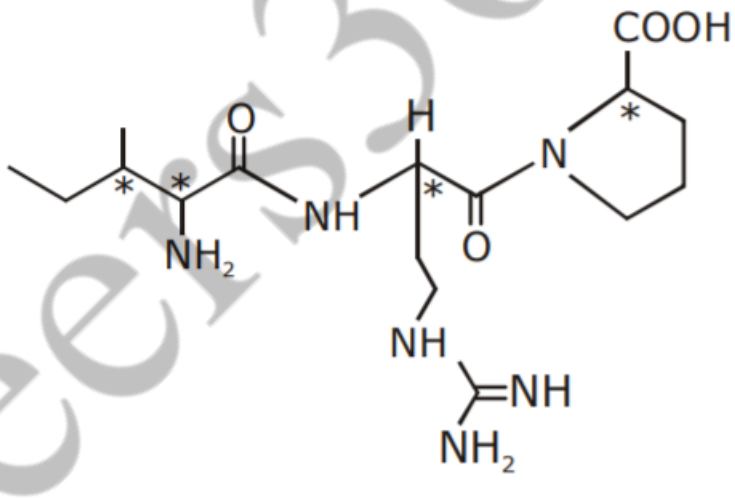
Natural Amino Acids 

Where R is given below-

Name of the amino acids	Characteristic feature of side chain, R	Three letter symbol	One letter code
Isoleucine*		Ile	I
Arginine*		Arg	R
Proline		Pro	P

Now, Structure of $\mathrm{Ile\text{-}Arg\text{-}Pro}$ will be -



The number of chiral carbon(s) present in peptide, $\mathrm{Ile\text{-}Arg\text{-}Pro}$, is **4.**

Ans = 4

Q. 24 A soft drink was bottled with a partial pressure of CO_2 of 3 bar over the liquid at room temperature. The partial pressure of CO_2 over the solution approaches a value of 30 bar when $44\ g$ of CO_2 is dissolved in $1\ kg$ of water at room temperature, The approximate pH of the soft drink is _____$\times 10^{-1}$.

(First dissociation constant of $H_2CO_3 = 4.0 \times 10^{-7};\ log\ 2 = 0.3;$ density of the soft drink $= 1\ g\ mL^{-1}$)

Correct Answer:
37

Solution:

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The partial pressure of CO_2 over the solution approaches a value of 30 bar when $44\ g$ of CO_2 is dissolved in $1\ kg$ of water at room temperature.

Then, $\text{Molarity } = \dfrac{\text{mole}}{\text{Volume(L)}}$

$\text{Molarity } = \dfrac{\frac{\text{Mass}}{\text{Molar mass}}}{\text{Volume(L)}}$

and density = Mass/volume

so, Volume = density X Mass = 1 x 1 = 1 litre

Then, $\text{Molarity } = \dfrac{\frac{44}{44}}{1} = 1 \text{ mole /litre}$

Now,

A soft drink was bottled with a partial pressure of CO_2 of 3 bar over the liquid at room temperature.

Reaction : $\mathrm{CO_2 + H_2O \rightarrow H_2CO_3}$

For, 30 Bar → 1 mol/litre

Then, 3 Bar → 0.1 mol/litre

Now,

	$H_2CO_3 \rightleftharpoons$	$H^{\oplus}$ +	HCO_3^-
t = 0	0.1	0	0
Eq.	$0.1(1-\alpha)$	0.1α	0.1α

First dissociation constant :

$$4 \times 10^{-7} = \frac{0.1\alpha^2}{1-\alpha}$$

$$4 \times 10^{-7} = 0.1\alpha^2 \quad \because (1-\alpha \approx 1)$$

$$\alpha^2 = 4 \times 10^{-6}$$

$$\alpha = 2 \times 10^{-3}$$

So,

$$\left[\mathrm{H^+}\right] = 0.1\alpha = 2 \times 10^{-4}\mathrm{M}$$

Now, we pH = - log([H^+])

$$\mathrm{pH} = -\log[2 \times 10^{-4}]$$

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$$\mathrm{pH} = -[\log(2) + (-4)\log(10)]$$

$$\mathrm{pH} = -[0.3 + (-4)]$$

$$\mathrm{pH} = 3.7 = 37 \times 10^{-1}$$

Ans = 37

Q. 25 An oxidation-reduction reaction in which 3 electrons are transferred has a ΔG^0 of $17.37\ kJ\ mol^{-1}$ at $25^{\circ}C$. The value of E^0_{cell} (in V) is _____$\times 10^{-2}$.

$(1\ F = 96,500\ C\ mol^{-1})$

Correct Answer:
-6

Solution:
We know the formula

$$\Delta \mathrm{G}^{\circ} = -\mathrm{nFE}^{\circ}_{\mathrm{cell}}$$

Given

n = 3 electrons

F =96500 C/mol

$$\Delta \mathrm{G}^{\circ} = 17.37\ \mathrm{kJ\ mol}^{-1}$$

Now,

$$17.37 \times 1000 = -3 \times 96500 \times \mathrm{E}^{\circ}_{\mathrm{cell}}$$

$$\Rightarrow \mathrm{E}^{\circ} = -6 \times 10^{-2}\ \mathrm{V}$$

Ans = (– 6)

Maths

Q. 1 If $y = y(x)$ is the solution of the differential equation $\frac{5+e^x}{2+y}.\frac{dy}{dx} + e^x = 0$ satisfying $y(0) = 1$, then a value of $y\ (log_e 13)$ is :

Correct Answer:
-1

Solution:

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$$\frac{1}{2+y}\frac{dy}{dx} = \frac{-e^x}{5+e^x}$$
$$\Rightarrow \frac{dy}{2+y} = \frac{-e^x}{5+e^x}\cdot dx$$
$$\Rightarrow \int \frac{dy}{2+y} = \int \frac{-e^x}{5+e^x}\cdot dx$$

$$\ln(y+2) = -\ln(e^x+5) + \ln C$$
$$= \ln\left(\frac{C}{e^x+5}\right)$$
$$\Rightarrow 2+y = \frac{C}{e^x+5}$$
$$\Rightarrow y = \frac{C}{e^x+5} - 2$$

y(0) = 1 (Given)

$$\Rightarrow 1 = \frac{C}{e^0+5} - 2$$
$$\Rightarrow C = 18$$

$$\Rightarrow y = \frac{18}{e^{\log_e 13}+5} - 2$$
$$\Rightarrow y = -1$$

Q. 2 The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$, is :

Option 1:
$\frac{5}{9}$

Option 2:
$\frac{25}{81}$

Option 3:
$\frac{5}{27}$

Option 4:
$\frac{25}{9}$

Correct Answer:
$\frac{25}{81}$

Solution:

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$9x^2 - 18|x| + 5 = 0$
$9|x|^2 - 15|x| - 3|x| + 5 = 0 \left(\because x^2 = |x|^2\right)$
$3|x|(3|x| - 5) - (3|x| - 5) = 0$
$|x| = \frac{1}{3}, \frac{5}{3}$
$x = \pm\frac{1}{3}, \pm\frac{5}{3}$
$\text{Product of roots } = \frac{25}{81}$

Q. 3 The negation of the Boolean expression $x \leftrightarrow\sim y$ is equivalent to :

Option 1:
$(x \wedge y) \vee (\sim x \wedge \sim y)$

Option 2:
$(x \wedge y) \wedge (\sim x \vee \sim y)$

Option 3:
$(x \wedge \sim y) \vee (\sim x \wedge y)$

Option 4:
$(\sim x \wedge y) \vee (\sim x \wedge \sim y)$

Correct Answer:
$(x \wedge y) \vee (\sim x \wedge \sim y)$

Solution:
$\mathrm{p} \leftrightarrow \mathrm{q} \equiv (\mathrm{p} \rightarrow \mathrm{q}) \wedge (\mathrm{q} \rightarrow \mathrm{p})$
$\mathrm{x} \leftrightarrow\sim \mathrm{y} \equiv (\mathrm{x} \rightarrow\sim \mathrm{y}) \wedge (\sim \mathrm{y} \rightarrow \mathrm{x})$
$\therefore \mathrm{p} \rightarrow \mathrm{q} \equiv\sim \mathrm{p} \vee \mathrm{q}$
$\mathrm{x} \leftrightarrow\sim \mathrm{y} \equiv (\sim \mathrm{x} \vee \sim \mathrm{y}) \vee (\mathrm{y} \vee \mathrm{x})$
$\sim (\mathrm{x} \leftrightarrow\sim \mathrm{y}) \equiv (\mathrm{x} \wedge \mathrm{y}) \vee (\sim \mathrm{x} \wedge \sim \mathrm{y})$

Q. 4 The mean and variance of 7 observations are 8 and 16, respectively .If five observations are $2, 4, 10, 12, 14$, then the absolute difference of the remaining two observations is :

Option 1:
1

Option 2:
4

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Option 3:
2

Option 4:
3

Correct Answer:
2

Solution:

$$\bar{x} = \frac{2+4++10+12+14+x+y}{7} = 8$$

$$x + y = 14 \qquad \ldots(1)$$

$$(\sigma)^2 = \frac{\sum (\mathrm{x_i})^2}{\mathrm{n}} - \left(\frac{\sum \mathrm{x_i}}{\mathrm{n}}\right)^2$$

$$16 = \frac{4+16+100+144+196+\mathrm{x}^2+\mathrm{y}^2}{7} - 8^2$$

$$16 + 64 = \frac{460+\mathrm{x}^2+\mathrm{y}^2}{7}$$

$$560 = 460 + x^2 + y^2$$

$$\mathrm{x}^2 + \mathrm{y}^2 = 100 \qquad \ldots(2)$$

Clearly by (i) and (ii), $|x - y| = 2$

Q. 5 If $2^{10} + 2^9 \cdot 3^1 + 2^8 \cdot 3^2 + \ldots\ldots + 2.3^9 + 3^{10} = S - 2^{11}$, then S is equal to

Option 1:
$3^{11} - 2^{11}$

Option 2:
3^{11}

Option 3:
$\frac{3^{11}}{2} + 2^{10}$

Option 4:
2.3^{11}

Correct Answer:
3^{11}

Solution:

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$$a = 2^{10}; r = \frac{3}{2}; n = 11(G.P)$$
$$S = \left(2^{10}\right) \frac{\left(\left(\frac{3}{2}\right)^{11} - 1\right)}{\frac{3}{2} - 1} = 2^{11}\left(\frac{3^{11}}{2^{11}} - 1\right)$$
$$S = 3^{11} - 2^{11} = S - 2^{11}(\text{ Given })$$
$$\therefore S = 3^{11}$$

Q. 6 If $3^{2\sin 2\alpha - 1}, 14$ and $3^{4-2\sin 2\alpha}$ are the first three terms of an A.P. for some α, then the sixth term of this A.P. is :

Option 1:
66

Option 2:
81

Option 3:
65

Option 4:
78

Correct Answer:
66

Solution:
Three-term are in AP

$$3^{2\sin 2\alpha - 1}, 14, 3^{4-2\sin\alpha}$$

Then

$$3^{4-\sin 2\alpha} + 3^{2\sin 2\alpha - 1} = 28$$
$$\text{Let } 3^{2\sin 2\alpha} = t$$
$$\frac{81}{t} + \frac{t}{3} = 28$$
$$t = 81, 3$$
$$3^{2\sin 2\alpha} = 3^1, 3^4$$

$$\sin 2\alpha = \frac{1}{2}, 2(\text{ rejected })$$

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$$\text{First term a} = 3^{2\sin 2\alpha - 1}$$
$$\text{a} = 1$$
$$\text{Second term} = 14$$
$$\therefore \text{ common difference d} = 13$$
$$T_6 = \text{a} + 5\text{ d}$$
$$T_6 = 1 + 5 \times 13$$
$$T_6 = 66$$

Q. 7 If the volume of a parallelopiped, whose coterminous edges are given by the vectors

$\vec{a} = \hat{i} + \hat{j} + n\hat{k},\ \vec{b} = 2\hat{i} + 4\hat{j} - n\hat{k}$ and

$\vec{c} = \hat{i} + n\hat{j} + 3\hat{k}\ (n \geqslant 0)$, is $158\ cu.units$, then :

Option 1:
$\vec{a} \cdot \vec{c} = 17$

Option 2:
$\vec{b} \cdot \vec{c} = 10$

Option 3:
$n = 7$

Option 4:
$n = 9$

Correct Answer:
$\vec{b} \cdot \vec{c} = 10$

Solution:

$$\text{v} = [\vec{\text{a}}\,\vec{\text{b}}\,\vec{c}]$$
$$158 = \begin{vmatrix} 1 & 1 & \text{n} \\ 2 & 4 & -\text{n} \\ 1 & \text{n} & 3 \end{vmatrix}, \text{n} \geq 0$$
$$158 = 1\left(12 + \text{n}^2\right) - (6 + \text{n}) + \text{n}(2\text{n} - 4)$$
$$158 = \text{n}^2 + 12 - 6 - \text{n} + 2\text{n}^2 - 4\text{n}$$
$$3\text{n}^2 - 5\text{n} - 152 = 0$$
$$\text{n} = 8, -\frac{38}{6}\ (\text{rejected})$$
$$\vec{\text{a}} \cdot \vec{c} = 1 + \text{n} + 3\text{n} = 1 + 4\text{n} = 33$$
$$\vec{\text{b}} \cdot \vec{c} = 2 + 4\text{n} - 3\text{n} = 2 + \text{n} = 10$$

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Q. 8 If S is the sum of the first 10 terms of the series

$$\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+\tan^{-1}\left(\frac{1}{21}\right)+.....,$$

then $\tan(S)$ is equal to :

Option 1:

$\frac{5}{6}$

Option 2:

$\frac{5}{11}$

Option 3:

$-\frac{6}{5}$

Option 4:

$\frac{10}{11}$

Correct Answer:

$\frac{5}{6}$

Solution:

$$\mathrm{S}=\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+\ldots$$

$$\mathrm{S}=\tan^{-1}\left(\frac{2-1}{1+1.2}\right)+\tan^{-1}\left(\frac{3-2}{1+2\times 3}\right)+\tan^{-1}\left(\frac{4-3}{1+3\times 4}\right)+\ldots+\tan^{-1}\left(\frac{11-10}{1+10\times 11}\right)$$

$$\mathrm{S}=\left(\tan^{-1}2-\tan^{-1}1\right)+\left(\tan^{-1}3-\tan^{-1}2\right)+\left(\tan^{-1}4-\tan^{-1}3\right)+\ldots\ldots+\left(\tan^{-1}(11)-\tan^{-1}(10)\right)$$

$$\mathrm{S}=\tan^{-1}11-\tan^{-1}1=\tan^{-1}\left(\frac{11-1}{1+11}\right)$$

$$\tan(\mathrm{S})=\frac{11-1}{1+11\times 1}=\frac{10}{12}=\frac{5}{6}$$

Q. 9 If the four complex numbers $z,\overline{z},\overline{z}-2Re(\overline{z})$ and $z-2Re(z)$ represent the vertices of a square of side 4 units in the Argand plane, the $|z|$ is equal to :

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Option 1:
$4\sqrt{2}$

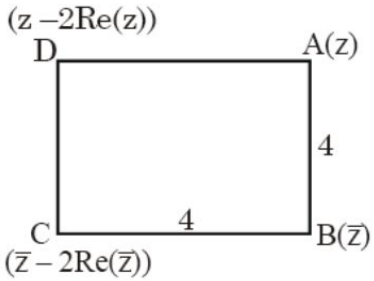
Option 2:
4

Option 3:
$2\sqrt{2}$

Option 4:
2

Correct Answer:
$2\sqrt{2}$

Solution:



Let z = x + iy

Length of side = 4

AB = 4

$|z - \bar{z}| = 4$
$|2y| = 4; |y| = 2$

BC = 4

$| \bar{z} - (\bar{z} - 2\,\mathrm{Re}(\bar{z}) | = 4$
$|2x| = 4; |x| = 2$
$|z| = \sqrt{x^2 + y^2} = \sqrt{4+4} = 2\sqrt{2}$

Q. 10 A survey shows that $73\ ^{o}/_{o}$ of the persons working in an office like coffee, whereas $65\ ^{o}/_{o}$ like tea. If x denotes the percentage of them, who like both coffee and tea, then x cannot be :

Option 1:
63

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Option 2:
36

Option 3:
54

Option 4:
38

Correct Answer:
36

Solution:
$\mathrm{C} \rightarrow$ person like coffee

$\mathrm{T} \rightarrow$ person like Tea

$n(C) = 73$

$\mathrm{n(T)} = 65$

$n(C \cup T) \leq 100$

$\mathrm{n(C) + n(T) - n(C \cap T) \leq 100}$

$73 + 65 - x \leq 100$

$x \geq 38$

$73 - x \geq 0 \Rightarrow x \leq 73$

$65 - x \geq 0 \Rightarrow x \leq 65$

$38 \leq x \leq 65$

Q. 11 If the co-ordinates of two points A and B are $(\sqrt{7}, 0)$ and $(-\sqrt{7}, 0)$ respectively and P is any point on the conic, $9x^2 + 16y^2 = 144,$ then $PA + PB$ is equal to :

Option 1:
16

Option 2:
8

Option 3:
6

Option 4:
9

Correct Answer:
8

Solution:

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$$\frac{x^2}{16}+\frac{y^2}{9}=1$$

$$a=4; b=3; e=\sqrt{\frac{16-9}{16}}=\frac{\sqrt{7}}{4}$$

A and B are foci

$$\Rightarrow \mathrm{PA}+\mathrm{PB}=2\mathrm{a}=2\times 4=8$$

Q. 12 If the point P on the curve, $4x^2+5y^2=20$ is farthest from the point $Q(0,-4)$, then PQ^2 is equal to :

Option 1:
36

Option 2:
48

Option 3:
21

Option 4:
29

Correct Answer:
36

Solution:

Given ellipse is $\frac{x^2}{5}+\frac{y^2}{4}=1$

Let point P is $(\sqrt{5}\cos\theta, 2\sin\theta)$

$$(\mathrm{PQ})^2=5\cos^2\theta+4(\sin\theta+2)^2$$
$$(\mathrm{PQ})^2=\cos^2\theta+16\sin\theta+20$$
$$(\mathrm{PQ})^2=-\sin^2\theta+16\sin\theta+21$$
$$=85-(\sin\theta-8)^2$$

will be maximum when $\sin\theta=1$

$$\Rightarrow (\mathrm{PQ})^2\max=85-49=36$$

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Q. 13 Let $\lambda \in R$. the system of linear equations

$$2x_1 - 4x_2 + \lambda x_3 = 1$$

$$x_1 - 6x_2 + x_3 = 2$$

$$\lambda x_1 - 10x_2 + 4x_3 = 3$$

is inconsistent for :

Option 1:
exactly one negative value of λ.

Option 2:
exactly one positive value of λ.

Option 3:
every value of λ.

Option 4:
exactly two value of λ.

Correct Answer:
exactly one negative value of λ.

Solution:

$$\mathrm{D} = \begin{vmatrix} 2 & -4 & \lambda \\ 1 & -6 & 1 \\ \lambda & -10 & 4 \end{vmatrix} = 2(3\lambda + 2)(\lambda - 3)$$

$$\mathrm{D}_1 = -2(\lambda - 3)$$
$$\mathrm{D}_2 = -2(\lambda + 1)(\lambda - 3)$$
$$\mathrm{D}_3 = -2(\lambda - 3)$$

$$\text{When } \lambda = 3, \text{ then}$$
$$\mathrm{D} = \mathrm{D}_1 = \mathrm{D}_2 = \mathrm{D}_3 = 0$$
$$\Rightarrow \text{ Infinite many solution}$$

$$\text{when } \lambda = -\frac{2}{3} \text{ then } \mathrm{D}_1, \mathrm{D}_2, \mathrm{D}_3. \text{ none of them is zero so equations are inconsistant}$$

$$\therefore \lambda = -\frac{2}{3}$$

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Q. 14 If the minimum and the maximum values of the function

$f:\left[\frac{\pi}{2},\frac{\pi}{2}\right]\to R,$ defined by

$$f(\theta)=\begin{vmatrix}-\sin^2\theta & -1-\sin^2\theta & 1\\ -\cos^2\theta & -1-\cos^2\theta & 1\\ 12 & 10 & -2\end{vmatrix}$$

are m and M respectively, then the ordered pair (m,M) is equal to :

Option 1:
$(0,2\sqrt{2})$

Option 2:
$(-4,0)$

Option 3:
$(-4,4)$

Option 4:
$(0,4)$

Correct Answer:
$(-4,0)$

Solution:

$C_3\to C_3-(C_1-C_2)$

$$f(\theta)=\begin{vmatrix}\sin^2\theta & -1-\sin^2\theta & 0\\ -\cos^2\theta & -1-\cos^2\theta & 0\\ 12 & 10 & -4\end{vmatrix}$$

$$=-4\left[\left(1+\cos^2\theta\right)\sin^2\theta-\cos^2\theta\left(1+\sin^2\theta\right)\right]$$

$$=-4\left[\sin^2\theta+\cos^2\theta-\cos^2\theta-\cos^2\theta\sin^2\theta\right]$$

$f(\theta)=4\cos 2\theta$

$\theta\in\left[\frac{\pi}{4},\frac{\pi}{2}\right]$

$2\theta\in\left[\frac{\pi}{2},\pi\right]$

$f(\theta)\in[-4,0]$

$(m,M)=(-4,0)$

Q. 15 If (a,b,c) is the image of the point $(1,2,-3)$ in the line,

$\frac{x+1}{2}=\frac{y-3}{-2}=\frac{z}{-1}$, then $a+b+c$ is equal to :

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Option 1:
2

Option 2:
-1

Option 3:
3

Option 4:
1

Correct Answer:
2

Solution:
Line is $\frac{x+1}{2}=\frac{y-3}{-2}=\frac{z}{-1}=\lambda$

Let point R is $(2\lambda-1,-2\lambda+3,-\lambda)$

Direction ratio of PQ $=(2\lambda-2,-2\lambda+1,3-\lambda)$PQ is perpendicular to line

$\Rightarrow 2(2\lambda-2)-2(-2\lambda+1)-1(3-\lambda)=0$

$4\lambda-4+4\lambda-2-3+\lambda=0$

$9\lambda=9\Rightarrow\lambda=1$

$\Rightarrow$ point R is $(1,1,-1)$

$\frac{a+1}{2}=1\Rightarrow a=1$

$\frac{b+2}{2}=1\Rightarrow b=0$

$\frac{c-3}{2}=-1\Rightarrow c=1$

$\Rightarrow a+b+c=2$

Q. 16 If the function

$$f(x)=\begin{cases}k_1(x-\pi)^2-1 & x\le\pi\\ k_2\cos x, & x>\pi\end{cases} \text{ is twice}$$

differentiable, then the ordered pair (k_1,k_2) is equal to :

Option 1:
$\left(\frac{1}{2},1\right)$

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Option 2:
$(1, 0)$

Option 3:
$\left(\frac{1}{2}, -1\right)$

Option 4:
$(1, 1)$

Correct Answer:
$\left(\frac{1}{2}, 1\right)$

Solution:
f(x) is continuous and differentiable $\mathrm{f}\left(\pi^{-}\right) = \mathrm{f}(\pi) = \mathrm{f}\left(\pi^{+}\right)$

$k_1((\pi) - \pi)^2 - 1 = k_2 \cos(\pi)$
$-1 = k_2$

$$f(x) = \begin{cases} 2k_1(x - \pi); x \leq \pi \\ -k_2 \sin x; x > \pi \end{cases}$$

$\mathrm{f}'\left(\pi^{-}\right) = \mathrm{f}'\left(\pi^{+}\right)$
$0 = 0$
so, differentiable at $x = 0$

$$f'(x) = \begin{cases} 2k_1 & ; x \leq \pi \\ -k_2 \cos x; x > \pi \end{cases}$$

$f'\left(\pi^{-}\right) = f''\left(\pi^{+}\right)$
$2k_1 = k_2$
$k_1 = \frac{k_2}{2}$

$\Rightarrow (k_1, k_2) = \left(\frac{1}{2}, 1\right)$

Q. 17 If the common tangent to the parabolas, $y^2 = 4x$ and $x^2 = 4y$ also touches the circle, $x^2 + y^2 = c^2$, then c is equal to :

Option 1:
$\frac{1}{2\sqrt{2}}$

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Option 2:

$\frac{1}{\sqrt{2}}$

Option 3:

$\frac{1}{4}$

Option 4:

$\frac{1}{2}$

Correct Answer:

$\frac{1}{\sqrt{2}}$

Solution:

$y = mx + \frac{1}{m}$ (tangent at $y^2 = 4x$)

$y = mx - m^2$ (tangent at $x^2 = 4y$)

$\frac{1}{m} = -m^2$ (for common tangent)

$m^3 = -1$

$m = -1$

$y = -x - 1$

$x + y + 1 = 0$

This line touches circle

$\therefore$ apply $p = r$

$c = \left|\frac{0+0+1}{\sqrt{2}}\right| = \frac{1}{\sqrt{2}}$

Q. 18 If α is the positive root of the equation,

$p(x) = x^2 - x - 2 = 0$, then

$\lim\limits_{x \to \alpha^+} \frac{\sqrt{1 - \cos(p(x))}}{x + \alpha - 4}$ is equal to :

Option 1:

$\frac{3}{2}$

Option 2:

$\frac{3}{\sqrt{2}}$

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Option 3:

$\frac{1}{\sqrt{2}}$

Option 4:

$\frac{1}{2}$

Correct Answer:

$\frac{3}{\sqrt{2}}$

Solution:

$x^2 - x - 2 = 0$

$\text{roots are } 2\& -1$

$\Rightarrow \lim\limits_{x\to 2^+} \frac{\sqrt{1-\cos(x^2-x-2)}}{(x-2)}$

$= \lim\limits_{x\to 2^+} \frac{\sqrt{2\sin^2\frac{(x^2-x-2)}{2}}}{(x-2)}$

$= \lim\limits_{x\to 2^+} \frac{\sqrt{2}\sin\left(\frac{(x-2)(x+1)}{2}\right)}{(x-2)}$

$= \frac{3}{\sqrt{2}}$

Q. 19 If $\int (e^{2x} + 2e^x - e^{-x} - 1)e^{(e^x+e^{-x})}dx = g(x)e^{(e^x+e^{-x})} + c,$

where c is a constant of integration, then $g(0)$ is equal to :

Option 1:

e

Option 2:

e^2

Option 3:

1

Option 4:

2

Correct Answer:

2

Solution:

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$$e^{2x}+2e^{x}-e^{-x}-1$$
$$=e^{x}\left(e^{x}+1\right)-e^{-x}\left(e^{x}+1\right)+e^{x}$$
$$=\left[\left(e^{x}+1\right)\right]\left(e^{x}-e^{-x}\right)+e^{x}]$$

$$\text{so I}=\int\left(e^{x}+1\right)\left(e^{x}-e^{-x}\right)e^{e^{x}+e^{-x}}+\int e^{x}\cdot e^{e^{x}+e^{-x}}dx$$
$$=\left(e^{x}+1\right)e^{e^{x}+e^{-x}}-\int e^{x}\cdot e^{e^{x}+e^{-x}}dx+\int e^{x}\cdot e^{e^{x}+e^{-x}}dx$$
$$=\left(e^{x}+1\right)e^{e^{x}+e^{-x}}+C$$
$$\therefore g(x)=e^{x}+1\Rightarrow g(0)=2$$

Q. 20 The value of $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}}\frac{1}{1+e^{sinx}}dx$ is :

Option 1:
$\frac{\pi}{4}$

Option 2:
π

Option 3:
$\frac{\pi}{2}$

Option 4:
$\frac{3\pi}{2}$

Correct Answer:
$\frac{\pi}{2}$

Solution:
Given,

$$I=\int_{-\pi/2}^{\pi/2}\frac{1}{1+e^{\sin x}}dx \qquad \ldots(1)$$

Apply Kings property

$$I=\int_{-\pi/2}^{\pi/2}\frac{1}{1+e^{\sin\left(\frac{\pi}{2}+\left(-\frac{\pi}{2}\right)-x\right)}}\,dx=\int_{-\pi/2}^{\pi/2}\frac{1}{1+e^{-\sin x}}\,dx$$
$$=\int_{-\pi/2}^{\pi/2}\frac{e^{\sin x}}{1+e^{\sin x}}dx \qquad \ldots(2)$$

Add (1) and (2)

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$$2I = \int_{-\pi/2}^{\pi/2} dx = \pi$$
$$I = \frac{\pi}{2}$$

Q. 21 Let $f(x) = x \cdot \left[\frac{x}{2}\right]$, for $-10 < x < 10$, where $[t]$ denotes the greatest integer function. Then the number of points of discontinuity of f is equal to _____.

Correct Answer:
8

Solution:
$x \in (-10, 10)$

$\frac{x}{2} \in (-5, 5) \rightarrow 9$ integer

check continuity at $x = 0$

$$\left.\begin{array}{l} f(0) = 0 \\ f(0^+) = 0 \\ f(0^-) = 0 \end{array}\right\} \text{ continuous at } x = 0$$

the function will be discontinuous when

$$\frac{x}{2} = \pm 4, \pm 3, \pm 2, \pm 1$$

8 points of discontinuity

Q. 22 If the line, $2x - y + 3 = 0$ is at a distance $\frac{1}{\sqrt{5}}$ and $\frac{2}{\sqrt{5}}$ from the lines $4x - 2y + \alpha = 0$ and $6x - 3y + \beta = 0$, respectively, then the sum of all possible values of α and β is _____.

Correct Answer:
30

Solution:
Apply distance between parallel line formula

Distance between the parallel line

$$2x - y + 3 = 0 \qquad \ldots(1)$$
$$4x - 2y + \alpha = 0 \qquad \ldots(2)$$

Divide equation (2) by 2, to make coefficient of x and y same

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$\Rightarrow 2x - y + \frac{\alpha}{2} = 0$

$\Rightarrow \left|\frac{\frac{\alpha}{2} - 3}{\sqrt{2^2 + 1^2}}\right| = \frac{1}{\sqrt{5}}$

$\Rightarrow \left|\frac{\frac{\alpha-6}{2}}{\sqrt{5}}\right| = \frac{1}{\sqrt{5}}$

$\Rightarrow |\alpha - 6| = 2$

$\Rightarrow \alpha = 8 \text{ and } 4$

Sum = 12

Again, Multiply equation (1) by 3

$6x - 3y + \beta = 0$

$6x - 3y + 9 = 0$

$\left|\frac{\beta - 9}{3\sqrt{5}}\right| = \frac{2}{\sqrt{5}}$

$|\beta - 9| = 6 \Rightarrow \beta = 15, 3$

$\text{sum } = 18$

Sum = 12 + 18 = 30

Q. 23 The number of words, with or without meaning, that can be formed by taking 4 letters at a time from the letters of the word 'SYLLABUS' such that two letters are distinct and two letters are alike, is _____.

Correct Answer:
240

Solution:
SYLLABUS

2S, 2L, 1A, 1B, 1U, 1Y

We need to form word of the type XYZZ

$\text{selection of 2 alike letters } = {}^2C_1$

$\text{selection of 2 distinct letters } = {}^5C_2$

Ways of arranging these 4 selected words: 4! / 2!

Total number of ways

$${}^2C_1 \times {}^5C_2 \times \frac{4!}{2!} = 240$$

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Q. 24 The natural number m, for which the coefficient of x in the binomial expansion of $\left(x^m + \frac{1}{x^2}\right)^{22}$ is 1540, is _____.

Correct Answer:
13

Solution:

$$T_{r+1} = {}^{22}C_r (x^m)^{22-r} \left(\frac{1}{x^2}\right)^r = {}^{22}C_r x^{22m-mr-2r}$$

$\because {}^{22}C_3 = {}^{22}C_{19} = 1540$

$r = 3 \text{ or } 19$

$22\ m - mr - 2r = 1$

$$m = \frac{2r+1}{22-r}$$

$$r = 3,\ m = \frac{7}{19} \notin \mathbb{N}$$

$$r = 19,\ m = \frac{38+1}{22-19} = \frac{39}{3} = 13$$

$m = 13$

Q. 25 Four fair dice are thrown independently 27 times. Then the expected number of times, at least two dice show up a three or a five, is _____.

Correct Answer:
11

Solution:
4 dice are independently thrown. Each die has the probability to show 3 or 5 is

$$p = \frac{2}{6} = \frac{1}{3}$$

$$\therefore \quad q = 1 - \frac{1}{3} = \frac{2}{3} (\text{ not showing 3 or 5})$$

The experiment is performed with 4 dices independently

Therefore, Their binomial distribution is

$$(q+p)^4 = (q)^4 + {}^4C_1 q^3 p + {}^4C_2 q^2 p^2 + {}^4C_3 q p^3 + {}^4C_4 P^4$$

$\therefore$ In one throw of each dice probability of showing 3 or 5 at least twice is

$$= p^4 + {}^4C_3 q p^3 + {}^4C_2 q^2 p^2$$

$$= \frac{33}{81}$$

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$\therefore$ Such experiment performed 27 times

$\therefore$ so expected out comes $= np$

$= \frac{33}{81} \times 27 = 11$

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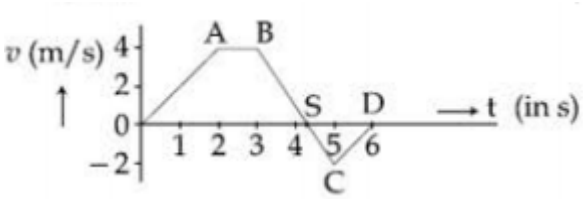
5th September Shift 2

Physics

Q. 1 The velocity (v) and time (t) graph of the body in straight line motion is shown in the figure. The point S is at 4.33 seconds. The total distance covered by the body in 6 second is:



Option 1:
$\frac{11}{4}m$

Option 2:
$\frac{49}{4}m$

Option 3:
$\frac{37}{3}m$

Option 4:
$12m$

Correct Answer:
$\frac{37}{3}m$

Solution:
Total distance=area under the v-t graph=area of trapezium(OABS)+area of triangle=
$\frac{1}{2}(4.333+1)\times 4+\frac{1}{2}\times 1.667\times 2=\frac{37}{3}m$

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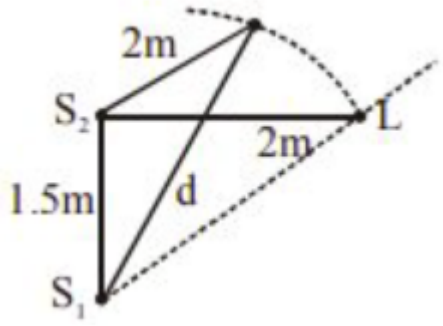
Q. 2 Two coherent sources of sound S_1 and S_2 produce sound waves of the same wavelength $\lambda = 1m$ in the phase. S_1 and S_2 are placed 1.5m apart. A listener located at L, directly in front of S_2 finds that the intensity is at a min. When hs is 2m away from S_2. The listener moves away from S_1 keeping his distance from S_2 fixed. The adjacent maximum of intensity is observed when the listener is at a distance d from S_1. Then d is:



Option 1:
5m

Option 2:
12m

Option 3:
2m

Option 4:
3m

Correct Answer:
3m

Solution:

Initially
$S_2\ L = 2\ m$
$S_1\ L = \sqrt{2^2 + (3/2)^2}$
$S_1\ L = \frac{5}{2} = 2.5\ m$
$\Delta x = S_1\ L - S_2\ L = 0.5\ m$
$Sosince \lambda = 1\ m \quad \therefore \Delta x = \frac{\lambda}{2}$
So while listener moves away from S_1
Then, $\Delta x\ (= S_1L - S_2L)$ increases and hence, at $\Delta x = \lambda$ first maxima will appear.
$\Delta x = \lambda = S_1L - S_2L$
$1 = d - 2$
$\Rightarrow d = 3m$

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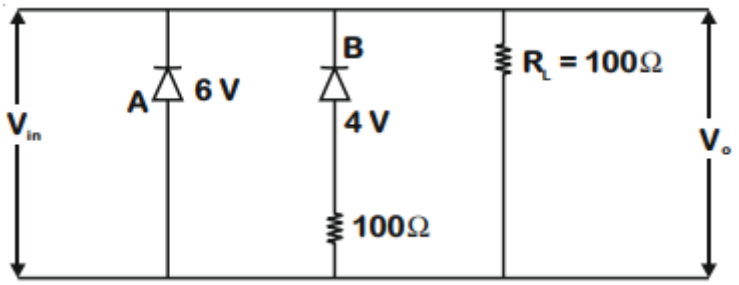
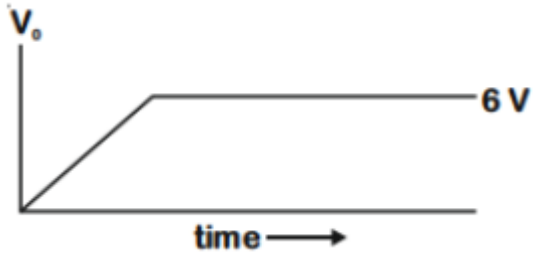
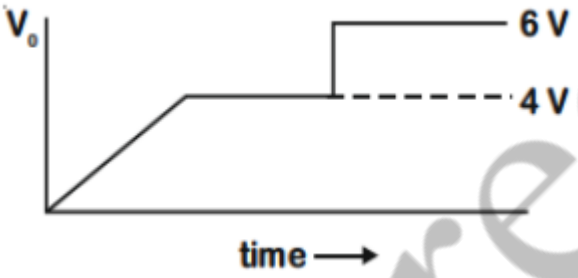
Q. 3 Two Zener diodes (A and B) having a breakdown voltage of 6V and 4V respectively, are connected. The voltage V_0 variation with i/p voltage linearly increasing with time is given by: $(V_{input} = 0)$ (V at t= 0)



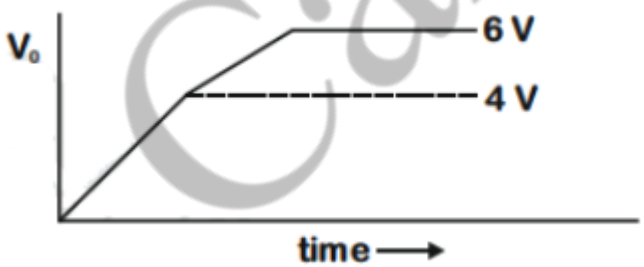
Option 1:



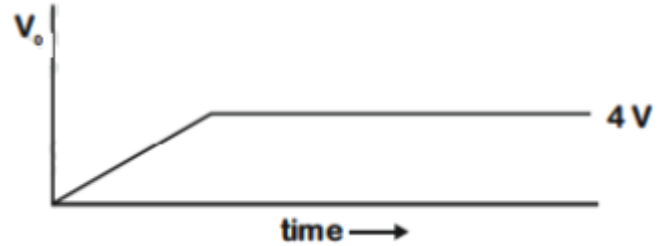
Option 2:



Option 3:



Option 4:



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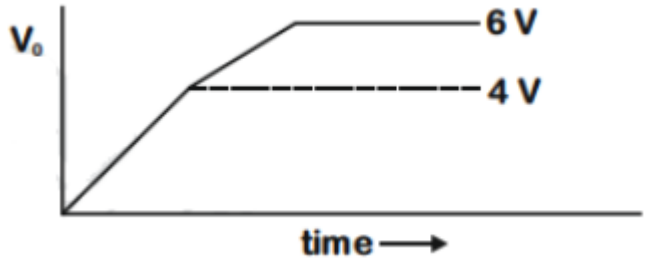
Correct Answer:

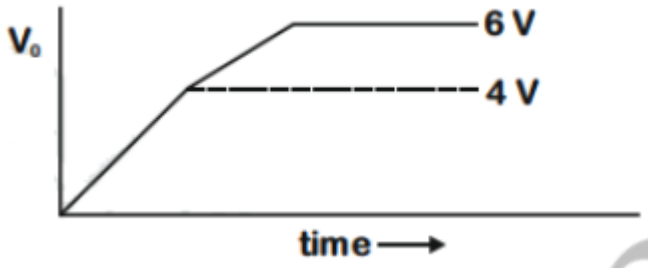


Solution:

Till input voltage Reaches $4V$ No zener is in Breakdown Region So $\mathrm{V_0 = V_i}$ Then Now when $\mathrm{V_i}$ changes between $4V\ to\ 6V$ One Zener with $4V$ will Breakdown are P.D. across This Zener will become constant and Remaining Potential will drop. acro Resistance in series with $4V$ Zener.

Now, the current in the circuit increases Abruptly and the source must have an internal resistance. Some potential will get drop across the source also so the correct graph between $\mathrm{V_0}$ and t. will be

We have to Assume some resistance in series with the source



Q. 4 In a experiment verify stokes law, a small spherical ball of radius r and density ρ falls under gravity through distance h in air before entering a tank of water. If the terminal velocity of ball inside water is same as its velocity just before entering the water surface, then the value of h is proportional to :(ignore viscosity of air)

Option 1:

r^4

Option 2:

r^3

Option 3:

r

Option 4:

r^2

Correct Answer:

r^4

Solution:

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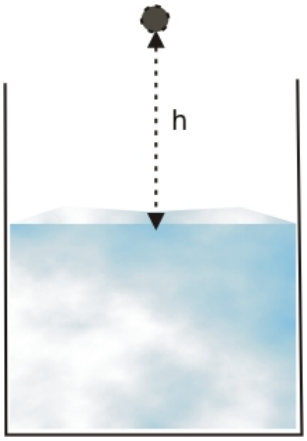
Correct answer r^4



.As,

$$v = \sqrt{2gh}$$

So.

$$\rho \times \frac{4}{3} \times \pi \times r^3 \times g = 6 \times \pi \times \eta \times r \times \sqrt{2gh}$$

From here

$$r^2 = k\sqrt{h}$$

where k = constant

So,

$$r^4 \propto h$$

Q. 5 A spaceship in space sweeps stationary interplanteray dust. As a result, its mass increases at a rate $\frac{dM(t)}{dt} = bv^2(t)$ where v(t) is its instantaeous velocity. The instantaeous accelartion of the satellite is:

Option 1:
$\frac{-2bv^3}{M(t)}$

Option 2:
$\frac{-bv^3}{M(t)}$

Option 3:
$-bv^3(t)$

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Option 4:

$$\frac{-bv^3}{2M(t)}$$

Correct Answer:

$$\frac{-bv^3}{M(t)}$$

Solution:

As given

$$\frac{dM(t)}{dt} = bv^2(t)$$

As we know,

$$F = \frac{d(Mv)}{dt}$$

$So,$

$$F = M\frac{dv}{dt} + v\frac{dM}{dt}$$

$$F = M\left(\frac{dv}{dt}\right) + v\left(bv^2\right)$$

$\text{We know that the net force is zero. } F = 0$

$$\frac{dv}{dt} = a = \left(\frac{-bv^3}{M(t)}\right)$$

Where M(t) represents Mass as a function of times.

The correct option is 2nd option

Q. 6 Two different wires having length L_1 and L_2 and respective temperature coefficient of linear expression$\alpha_1 and \alpha_2$ are joined end to end. then the effective temperature coefficent of linear expansin is:

Option 1:

$$\frac{\alpha_1 L_1 + \alpha_2 L_2}{L_1 + L_2}$$

Option 2:

$$2\sqrt{\alpha_1 \alpha_2}$$

Option 3:

$$\frac{4\alpha_1\alpha_2}{(\alpha_1 + \alpha_2)} \times \frac{L_2 L_1}{(L_1 + L_2)^2}$$

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Option 4:

$\frac{\alpha_1 + \alpha_2}{2}$

Correct Answer:

$\frac{\alpha_1 L_1 + \alpha_2 L_2}{L_1 + L_2}$

Solution:

Correct Answer: a) $\frac{\alpha_1 L_1 + \alpha_2 L_2}{L_1 + L_2}$

The coefficient of linear expansion of the first wire of length L_1 is

$$\Delta L_1 = \alpha, L_1 \Delta t$$

and,

The coefficient of linear expansion of first wire of length L_2 is

$$\Delta L_2 = \alpha_2 L_2 \Delta t$$

Now, if a single wire of linear expansion α is taken instead of two wires,

$$\alpha = \frac{\Delta \mathrm{L}_1 + \Delta \mathrm{L}_2}{(L_1 + \mathrm{L}_2) \times \Delta \mathrm{t}}$$

Putting the value of $\Delta L_1 \text{and} \Delta L_2$

We get,

$$\alpha = \frac{L_1 \alpha_1 + L_2 \alpha_2}{L_1 + L_2}$$

Q. 7 In an adiabatic process, the density of a diatomic gas becomes 32 times its initial value. The final pressure of the gas is found to be n times the intial pressure. The value of n is:

Option 1:

128

Option 2:

$\frac{1}{3^2}$

Option 3:

326

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Option 4:
32

Correct Answer:
128

Solution:
Correct Answer: a)128

$$\left(\frac{V_1}{V_2}\right) = \left(\frac{\rho_2}{\rho_1}\right)$$

Because

$$\left(\rho \propto \frac{1}{V}\right)$$

$$P_1V_1^{\gamma} = P_2V_2^{\gamma}$$

$$\Rightarrow \quad \left(\frac{V_1}{V_2}\right)^{\gamma} = \left(\frac{P_2}{P_1}\right)$$

$$\left(\frac{\rho_2}{\rho_1}\right)^{\gamma} = \frac{P_2}{P_1}$$

$$\Rightarrow (32)^{\frac{7}{5}} = \frac{P_2}{P_1}$$

So,

$$\frac{P_2}{P_1} = 128$$

Q. 8 An iron rod of volume $10^{-3}m^3$ and relative permeability 1000 is placed as core in a solenoid with 10 turns/cm . If a current of 0.5A is passed through the solenoid, then the magentic moment of rod will be:

Option 1:
$500 \times 10^2 Am^2$

Option 2:
$0.5 \times 10^2 Am^2$

Option 3:
$50 \times 10^2 Am^2$

Option 4:
$5 \times 10^2 Am^2$

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Correct Answer:

$5 \times 10^2 Am^2$

Solution:

Correct Answer: c) $50 \times 10^2 Am^2$

$$\text{Here, } n = 10 \text{ turns /cm} = 1000 \text{ turns /m}$$

$$V = 10^{-3}m^3, \mu_r = 1000, i = 0 \cdot 5A$$

$$\text{We know that } B = \mu_0(H + I)$$

$$, I = \frac{B}{\mu_0} - H = \frac{\mu H}{\mu_0} - H$$

$$= \mu_r H - H = (\mu_r - 1)\, H$$

$$5 \times 10^2 A - m^2$$

Q. 9 Ten charges are placed on the circumference of a circle of radius R with constant angular sepration between successive charges. Alternate charges 1.3.5.7.9 have charge (+q) each while 2,4,6,8,10 have charge (-q) each. The potential V and the electric field E at the centre of circle are respectively (Take V =0 at w):

Option 1:

V=0, E=0

Option 2:

$$V = \frac{10q}{4\pi E_0 R}, E = \frac{10q}{4\pi E_0 R^2}$$

Option 3:

$$V = 0, E = \frac{10q}{4\pi E_0 R^2}$$

Option 4:

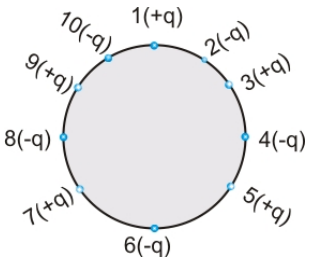
$$V = \frac{10q}{4\pi E_0 R}, E = 0$$

Correct Answer:

V=0, E=0

Solution:

Correct Answer: a) V=0, E=0



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As electric field is a vector quantity, so the electric due to each charge will cancel out each other due to symmetry as equal number of positive and negative charges are present at equal distances from the centre. And potential is a scalar quantity, it will also be zero as potential due to 5 positive charges will cancel the potential due to 5 negative charges

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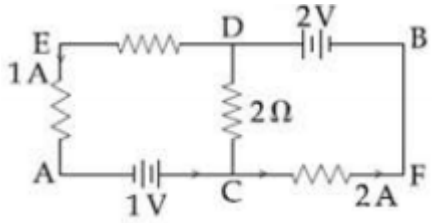
Q. 10 In the circuit given in figure current is different branches and the value of the one resistor are shown. The potential at point B with respect to A is?



Option 1:
-1 V

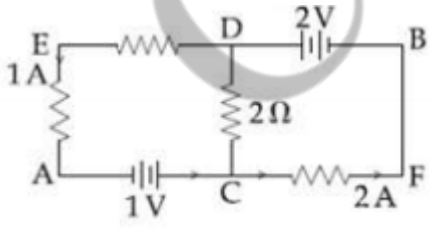
Option 2:
+ 1V

Option 3:
+2V

Option 4:
2V

Correct Answer:
+ 1V

Solution:
Correct Answer: 2) + 1V



At C,

i+1=2

Or. i=1 A

Let V_A=0, then V_C=1V

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$V_D=V_C+2 \times i=3V$

$V_B=V_D-2=1V$

So, $|V_B - V_A| = 1V$

Q. 11 A driver in a car approaching a vertical wall notices that the frequency of his ear horn has changed from 440 Hz to 480 Hz when it gets reflected from the wall. If the speed of sound in air is 345 m/s, then the speed of the car is:

Option 1:
36km/hr

Option 2:
18 km/hr

Option 3:
24 km/hr

Option 4:
54 km/hr

Correct Answer:
54 km/hr

Solution:
Correct Answer: 4) 54 km/hr

$f_0 = \left(\frac{v+u}{v-u}\right) \times f_s$

$Or, 480 = \left(\frac{v+u}{v-u}\right) \times 440$

$Or, \left(\frac{48}{44}\right) = \frac{v+u}{v-u} = \frac{345+u}{345-u}$

$Or, 1.0909 \times 345 - 1.0909u = 345 + u$

$Or, u = 15m/s = 54km/hr$

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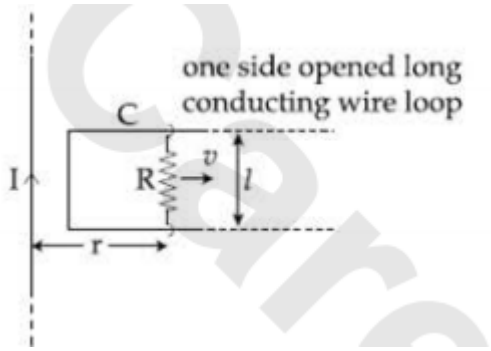
Q. 12 An infinitely long straight wire carrying current I, one side open rectangular loop & a conductor C with a sliding connector are located in the same place. The conductor has the length l and resistance R. If slides to the right with a velocity c, The resistance of conductor & and self-inductance of the loop are negligible. The induced current in the loop as a function of separation r, between the connector & straight wire, is:



Option 1:

$2\frac{\mu_o}{\pi}\cdot\frac{Ivl}{Rr}$

Option 2:

$\frac{\mu_o}{2\pi}\cdot\frac{Ivl}{Rr}$

Option 3:

$\frac{\mu_o}{\pi}\cdot\frac{Ivl}{Rr}$

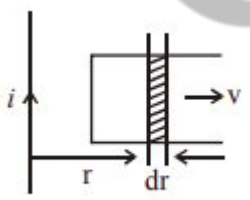
Option 4:

$\frac{\mu_o}{4\pi}\cdot\frac{Ivl}{Rr}$

Correct Answer:

$\frac{\mu_o}{2\pi}\cdot\frac{Ivl}{Rr}$

Solution:



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$B = \frac{\mu_0 i}{2\pi r}$

$\phi = \frac{\mu_0 i}{2\pi r} \ell dr$

$\Rightarrow \frac{d\phi}{dt} = \frac{\mu_0 i \ell}{2\pi r} \cdot \frac{dr}{dt}$

$\Rightarrow e = \frac{\mu_0}{2\pi} \cdot \frac{iv\ell}{r}$

$i = \frac{e}{R} = \frac{\mu_0}{2\pi} \cdot \frac{iv\ell}{Rr}$

Q. 13 A radioactive nucleus decays by two different processes. The half-life for the 1st process is 10s and that for the 2nd process 100s. The effective half-life (in seconds) of the nucleus is close to:

Correct Answer:
9

Solution:
Correct Answer: 2) 9 sec

Radioactive decay rate is directly proportional to number of nucleus present at any time

$\frac{dN}{dt} = -\lambda N$ _ (i)

where λ is a constant related to half life as $\lambda = \frac{\ln 2}{T_{\frac{1}{2}}}$ where $T_{\frac{1}{2}}$ is half life time of radioactive decay

The total rate of decay is $\frac{dN}{dt} = -(\lambda_1 + \lambda_2) N = \lambda_{effective} N$ ...(ii)

$\lambda_1 = \frac{\ln 2}{T_1}$

$\lambda_2 = \frac{\ln 2}{T_2}$

$\lambda_{effective} = \frac{\ln 2}{T_{effective}}$

substituting in equation (ii)

$\left(\frac{\ln 2}{T_1} + \frac{\ln 2}{T_2}\right) N = \frac{\ln 2}{T_{effective}} N$

$\frac{1}{T} = \frac{1}{T_1} + \frac{1}{T_2}$

$\frac{1}{T} = \frac{1}{10} + \frac{1}{100} = \frac{11}{100}$

$\Rightarrow T = \frac{100}{11} = 9 sec$

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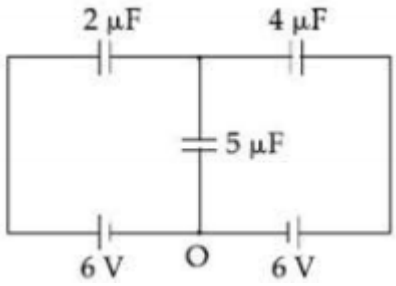
Q. 14 In the electrical circuit shown, the charge on $5\mu F$ the capacitor is:



Option 1:
$5.45\mu c$

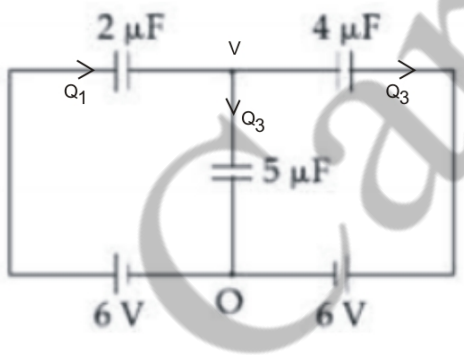
Option 2:
$10.90\mu c$

Option 3:
$16.36\mu c$

Option 4:
$18.00\mu c$

Correct Answer:
$16.36\mu c$

Solution:
Correct Answer: 3) $16.36\mu c$



Let the potential difference between $2\mu F$ and $4\mu F$ be:

$$Q_1 = Q_2 + Q_3$$
$$\Rightarrow (6 - V)2 = 4(V - 6) + 5(V)$$
$$\Rightarrow 12 - 2V = 4V - 24 + 5V$$
$$\Rightarrow V = \frac{36}{11}$$
$$\Rightarrow Q_3 = 5 \times \frac{36}{11} = 16.36\mu C$$

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Q. 15 If acceleration due to gravity on the earth's surface at the poles is g and angular velocity of the earth about the axis passing through the poles is ω. An object is weighted at the equator & at the height of h above the poles by using a spring balance. If the weights are found to be the same, then h is (h <<R where R = radius of the earth)

Option 1:

$\frac{R^2\omega^2}{4g}$

Option 2:

$\frac{R^2\omega^2}{g}$

Option 3:

$\frac{R^2\omega^2}{2g}$

Option 4:

$\frac{R^2\omega^2}{8g}$

Correct Answer:

$\frac{R^2\omega^2}{2g}$

Solution:

Correct Answer: 3) $\frac{R^2\omega^2}{2g}$

$g_{eff} = g - Rw^2\cos^2\theta$

At equator, $\theta = 0^\circ$

So, $g_{eff} = g - Rw^2$.....(1)

Acceleration due to gravity at height h above earth's surface,

$g' = g\left(1 - \frac{2h}{R}\right)$....(2)

According to the question

$$g_{eff} = g'$$

$$\Rightarrow g\left(1 - \frac{2h}{R}\right) = g - Rw^2$$

$$\Rightarrow \frac{2hg}{R} = w^2R \Rightarrow h = \frac{R^2w^2}{2g}$$

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Q. 16 An galvanometer is used in the laboratory for detecting the null point in the electrical experiments. If on passing a current of 5 mA it produces a deflection of 2°, its figure of merit is close to:-

Option 1:
$6 \times 10^{-3} A/div$

Option 2:
$333^{\circ} A/div$

Option 3:
$3 \times 10^{-3} A/div$

Option 4:
$666^{\circ} A/div$

Correct Answer:
$3 \times 10^{-3} A/div$

Solution:
Correct Answer: c) $3 \times 10^{-3} A/div$

$$\text{figure of merit}$$

$$G = \frac{K}{NBA} = \frac{I}{\theta}$$

$$\Rightarrow G = \frac{6 \times 10^{-3}}{2} = 3 \times 10^{-3} \text{Am}^2$$

or,

$3 \times 10^{-3} \text{A/div}$

Q. 17 The quantities $x = \frac{1}{\sqrt{\mu_o \epsilon_o}}, y = \frac{E}{B}$ and $z = \frac{l}{CR}$ are defined where C - capacitance, R - resistance, l - length, E - Electric field, B - magnetic field $\epsilon_o \mu_o$ - free space permittivity and permeability respectively, then:

Option 1:
only y and z have same dimension

Option 2:
only x and z have same dimension

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Option 3:
only x and y have same dimension

Option 4:
x, y, and z have same dimension

Correct Answer:
x, y, and z have same dimension

Solution:
Correct Answer: 4) x, y, and z have same dimension

$$x = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

As we know, speed of light,

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\therefore [c] = [x] = \left[LT^{-1}\right]$$

$$y = \frac{E}{B}$$

And c is also equal to $\frac{E}{B}$

$$\therefore [c] = [y] = \left[LT^{-1}\right]$$

$$z = \frac{l}{CR} \Rightarrow [z] = \frac{[L]}{[C][R]} = \frac{(L]}{[M^{-1}L^{-2}T^{4}A^{2}]\,[ML^{2-3}A^{-2}]} = [LT^{-1}]$$

Q. 18 Match the columns:

I **Radiation**	**II** **Wavelength**
a. Microwave	i. 100m
b. Gamma rays	ii. 10^{-15} m
c. All radio waves	iii. 10^{-10} m
d. X-rays	iv. 10^{-3} m

Option 1:
a - (i), b - (iii), c - (iv), d - (ii)

Option 2:
a - (ii), b - (i), c - (iv), d - (ii)

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Option 3:
a - (iii), b - (i), c - (i), d - (iv)

Option 4:
a - (iv) , b - (ii), c - (i), d - (iii)

Correct Answer:
a - (iv) , b - (ii), c - (i), d - (iii)

Solution:
Correct Answer: 4) a - (iv) , b - (ii), c - (i), d - (iii)

The radiation ad corresponding wavelengths are given below

Microwave-$10^{-3}m$

AM Radio waves-$100m$

Gamma rays-$10^{-15}m$

X-rays- $10^{-10}m$

Q. 19 A parallel plate capacitor has a plate of length l, width w, & separation of plates is d. If it is connected to a battery of emf V, A dielectric slab of the same thickness d & of dielectric constant K = 4 is being inserted between the plates of the capacitor. At what length of the slab inside states, will the energy stored in the capacitor be two times its initial energy stored?

Option 1:
$\frac{2l}{3}$

Option 2:
$\frac{l}{2}$

Option 3:
$\frac{l}{3}$

Option 4:
$\frac{l}{4}$

Correct Answer:
$\frac{l}{3}$

Solution:

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Correct Answer: 3) $\frac{l}{3}$

Let the dielectric insert up to length x

For dielectric inserted part

$$C_1 = \frac{K\epsilon_0 * w * x}{d}$$

and for the remaining part

$$C_2 = \frac{\epsilon_0 * w * (l - x)}{d}$$

Now,

$$C_{eq} = C_1 + C_2 = \frac{\epsilon_0 * w}{d} * (l + x(k - 1))$$

$$U_{final} = \frac{1}{2}C_{eq}V^2 = \frac{\epsilon_0 * w * V^2}{2d} * (l + x(k - 1))$$

$$U_{initial} = \frac{1}{2}CV^2 = \frac{\epsilon_0 * w * l * V^2}{2d}$$

But, $U_{fial} = 2U_{initial}$

On solving we get $x = \frac{l}{3}$

Q. 20 A ring is hung on a nail. It can oscillate without slipping or sliding (i) in its plane with a time period T_1 and (ii) back and forth in a direction perpendicular to its plane with a period T_2 The ratio $\frac{T_1}{T_2}$ will be

Option 1:
$\frac{\sqrt{2}}{3}$

Option 2:
$\frac{2}{3}$

Option 3:
$\frac{2}{\sqrt{3}}$

Option 4:
$\frac{3}{\sqrt{2}}$

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Correct Answer:

$\frac{2}{\sqrt{3}}$

Solution:

Correct Answer: c) $\frac{2}{\sqrt{3}}$

$$\frac{T_1}{T_2} = \sqrt{\frac{I_1}{I_2}}$$

$$I_1 = MR^2 + MR^2 = 2MR^2$$

$$I_2 = \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2$$

$$\frac{T_1}{T_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{2*2}{3}} = \frac{2}{\sqrt{3}}$$

Q. 21 A body of mass 2kg is driven by an engine delivering a constant power of 1J/s. The body starts from rest and moves in a straight line. After 9s, the body has moved a distance (in m) .....

Correct Answer:

18

Solution:

Correct Answer: 18

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$$P = \frac{dW}{dt} = 1$$

$$W = t$$
$$W = \frac{1}{2}m(V_2^2 - V_1^2) = t$$

$$W = \frac{1}{2}mV_2^2 = t$$

$$V_2 = V = \sqrt{t}$$

$$\frac{dx}{dt} = \sqrt{t}$$
$$\int dx = \int \sqrt{t}dt$$

$$x = \frac{t^{\frac{3}{2}}}{\frac{3}{2}}$$

$$for\ t = 9\ sec, x = 18m$$

Q. 22 Nitrogen gas at $300^o\ C$. The temp(in K) at which the rms speed of H_2 molecule would be equal to the rms speed of a nitrogen molecule is (Molar mass of N_2 gas is 28gm).
Up to 2 digits after decimal

Correct Answer:
40.94

Solution:
Correct Answer: 41

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

For N_2 $T_1 = 300 + 273.15 = 573.15K\ and\ \ M_1 = 28$

For H_2 $$\frac{V_{N_2}}{V_{H_2}} = 1 = \sqrt{\frac{T_1}{T_2} * \frac{M_2}{M_1}} = \sqrt{\frac{573.15}{T} * \frac{2}{28}}$$
$$T = 40.94K$$

Q. 23 A prism of angle $A = 1^\circ$ has a refractive index $\mu = 1.5$. A good estimate for minimum angle of deviation(in degrees) is close to N/10. The value of N is...

Correct Answer:
5

Solution:
Correct Answer: 5

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$$\mu = \frac{Sin(\frac{(A+\delta_m)}{2})}{Sin(\frac{A}{2})}$$

Since A and δ_m is very small

$$\mu = \frac{(\frac{(A+\delta_m)}{2})}{\frac{A}{2}}$$

$$1.5 = \frac{A+\delta_m}{A}$$

$$\delta_m = \frac{A}{2} = \frac{N}{10}$$

$$N = 5$$

Q. 24 The surface of a metal is illuminated alternatively with photons of energies E=4eV and E=2.5eV resp. The ratio of maximum speeds of the photoelectrons emitted is the two cases is 2. The work function of the metal in eV is .....

Correct Answer:
2

Solution:

$$\mathrm{E}_1 = \phi + \mathrm{K}_1 \ldots (1)$$

$$\mathrm{E}_2 = \phi + \mathrm{K}_2 \ldots (2)$$

$$E_1 - E_2 = K_1 - K_2 \ \ldots (3)$$

$$Now \ \ \frac{V_1}{V_2} = 2$$

$$\frac{\mathrm{K}_1}{\mathrm{K}_2} = 4$$

$$\mathrm{K}_1 = 4\ \mathrm{K}_2$$

$$\text{Now from equation } (3) \Rightarrow 4 - 2.5 = 4\ \mathrm{K}_2 - \mathrm{K}_2$$

$$1.5 = 3\ \mathrm{K}_2$$

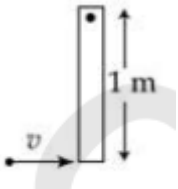
$$\mathrm{K}_2 = 0.5\mathrm{eV}$$

$$\text{Now putting This Value in equation } (2)$$

$$2.5 = \phi + 0.5\mathrm{eV}$$

$$\phi = 2\mathrm{ev}$$

Q. 25 A thin rod of mass 0.9kg length 1m is suspended at rest from one end so that it can oscillate in the vertical plane. A particle of mass 0.1 kg moves in a straight line with a velocity 80m/s hits the rod at its bottom most point & sticks to it. The angular speed in rad/sec of it immediately after the collision will be



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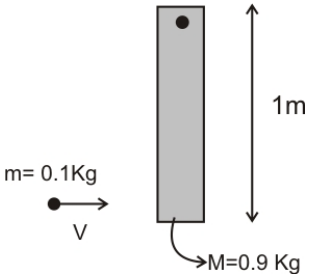
Correct Answer:
20

Solution:
Correct Answer: 20



$$mvR = I\omega$$
$$\Rightarrow 0.1 \times 80 \times 1 = \left\{\frac{ML^2}{3} + mL^2\right\}\omega$$
$$\Rightarrow 8 = \left\{\frac{0.9}{3} + 0.1\right\}\omega$$
$$\Rightarrow \omega = 20 rad/s$$

Chemistry

Q. 1 Lattice enthalpy and enthalpy of solution of NaCl are $788\ \text{kJ mol}^{-1}$ and $4\ \text{kJ mol}^{-1}$, respectively. The hydration enthalpy of NaCl is:

Option 1:
-780 $kJmol^{-1}$

Option 2:
780 $kJmol^{-1}$

Option 3:
784 $kJmol^{-1}$

Option 4:
-784 $kJmol^{-1}$

Correct Answer:
-784 $kJmol^{-1}$

Solution:
Lattice enthalpy = 788kJ/mol

Enthalpy of solution = 4kJ/mol

The reaction occurs as follows: $\text{NaCl(s)} \xrightarrow{788\text{kj/mol}} \text{NaCl(g)} \xrightarrow{x} \text{Na}^+\text{Cl}^-\text{(aq)}$

Enthalpy of solution = enthalpy of hydration + lattice energy

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Hydration enthalpy of NaCl

= -788 + 4

= -784 kJ/mol

Therefore, Option(4) is correct.

Q. 2 The correct order of ionic radii of $O^{2-}, N^{3-}, F^{-}, Mg^{2+}, Na^{+}, Al^{3+}$

Option 1:
$N^{3-} < F^{-} < O^{2-} < Mg^{2+} < Na^{+} < Al^{3+}$

Option 2:
$Al^{3+} < Na^{+} < Mg^{2+} < O^{2-} < F^{-} < N^{3-}$

Option 3:
$N^{3-} < O^{2-} < F^{-} < Na^{+} < Mg^{2+} < Al^{3+}$

Option 4:
$Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

Correct Answer:
$Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

Solution:
The charge to mass ratio(z/e) of the given species is given as follows:

O^{2-} = z/e = 8/10 = 0.8

N^{3-} = 7/10 = 0.7

F^{-} = 9/10 = 0.9

Mg^{2+} = 12/10 = 1.2

Na^{+} = 11/10 = 1.2

Al^{3+} = 13/10 = 1.3

Now, higher the z/e, smaller is the atomic radii.

Thus, the atomic radii follows the order given below:

$Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

Therefore, Optioin(4) is correct.

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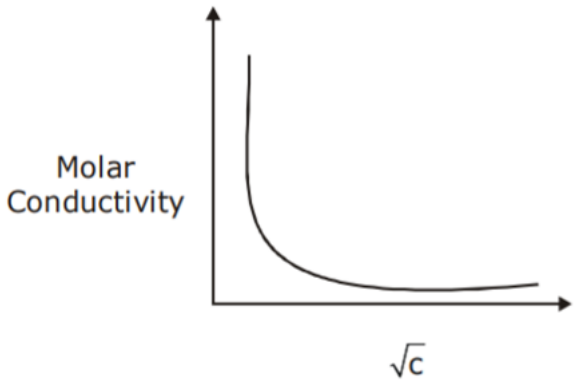
Q. 3 The variation of molar conductivity with concentration of an electrolyte (X) in aqueous solution is shown in the given figure.



The electrolyte (X) is:

Option 1:
CH_3COOH

Option 2:
HCl

Option 3:
NaCl

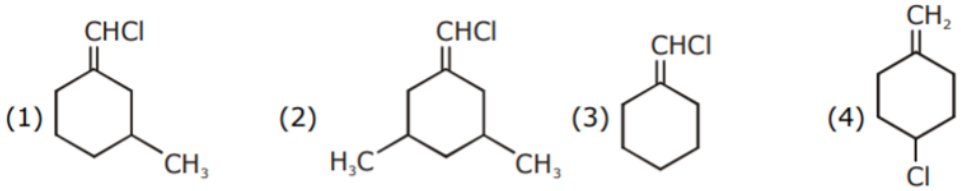
Option 4:
KNO_3

Correct Answer:
CH_3COOH

Solution:
The electrolyte X is CH_3COOH. Because it is a weak acid and on dilution it's molar conductivity increases.

Therefore, Option(1) is correct**.**

Q. 4 Among the following compound geometrical isomerism to exhibited by :



Option 1:
1 and 4

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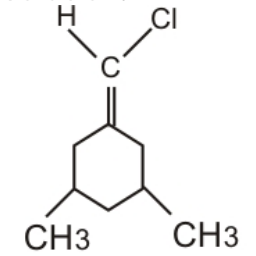
Option 2:
2 and 4

Option 3:
Only 3

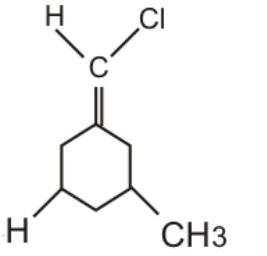
Option 4:
1 and 2

Correct Answer:
1 and 2

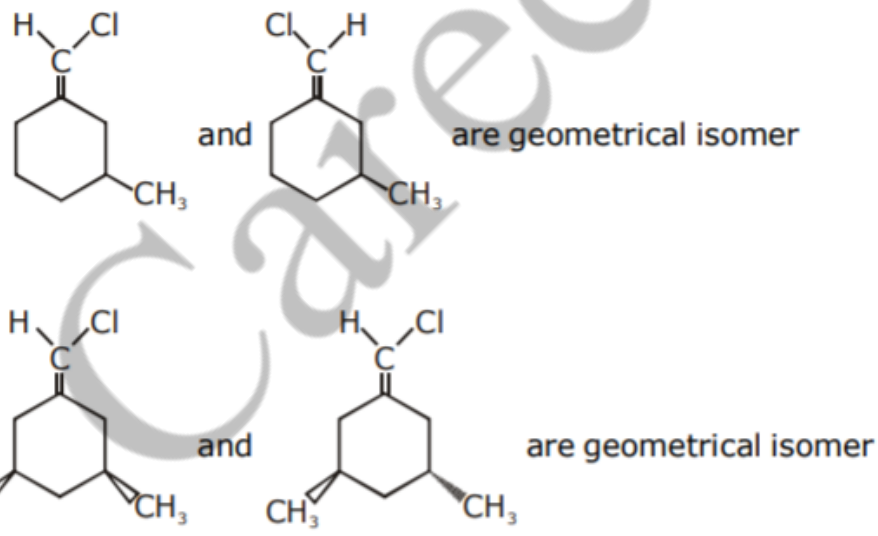
Solution:



Geometrical isomer due to ring



Geometrical isomer due to C = C



Both (1) and (2) will show geometrical isomerism.

Therefore, Option(4) is correct.

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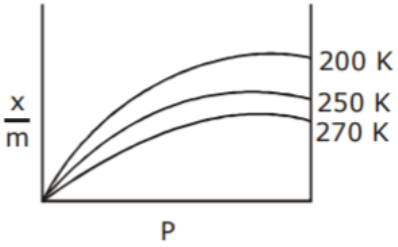
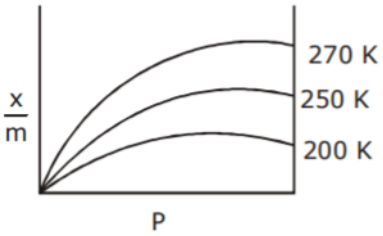
Q. 5 Adsorption of a gas follows frenndlich absorption isotherm. If x i s the man of the gas absorbed on mass m of the adsorbent, the correct plot of $\frac{x}{m}$ v/s p is:

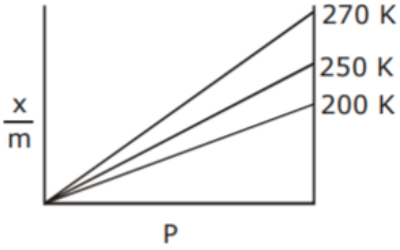
Option 1:



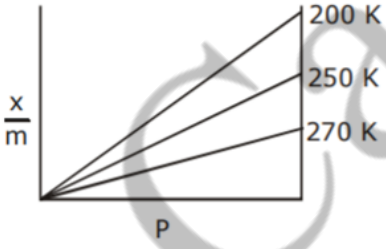
Option 2:



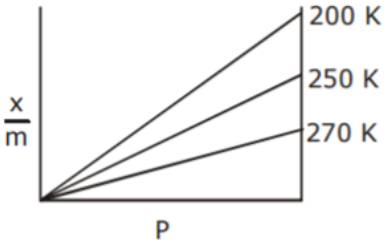
Option 3:



Option 4:



Correct Answer:



Solution:
The correct plot of x/m vs P is given as:

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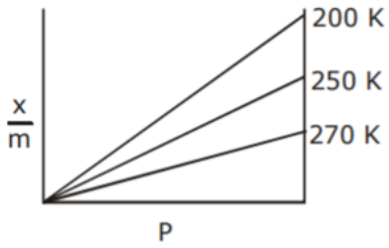

As temperature increases, adsorption decreases.

Therefore, Option(4) is correct.

Q. 6 Which one of the following polymers is not obtained by condensation polymerisation ?

Option 1:
$Nylon\ 6,6$

Option 2:
Buna N

Option 3:
Nylon 6

Option 4:
Bakelite

Correct Answer:
Buna N

Solution:
Buna-N is a condensation polymer.

Therefore, Option(2) is correct.

Q. 7 The correct statement about parobability density (except at a distance from nucleus) is :-

Option 1:
it can be zero for 1s orbital

Option 2:
It can never be zero for 2s orbital

Option 3:
It can be negative for 2p orbital

Option 4:
it can be zero for 3p orbital

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Correct Answer:
it can be zero for 3p orbital

Solution:
Probability density can be zero for 3*p* orbital.

Therefore, Option(4) is correct.

Q. 8 Boron & Silicon of very high priority can be obtained through:

Option 1:
Zone refining

Option 2:
Liquation

Option 3:
electrolytic refining

Option 4:
Vapour Phase refining

Correct Answer:
Zone refining

Solution:
Boron and silicon of very high purity can be obtained through zone refining.

Therefore, Option(1) is correct.

Q. 9 The one that is NOT suitable for the removal of permenent hardness of water :-

Option 1:
Clark's method

Option 2:
Ion-exchange method

Option 3:
Calgon's method

Option 4:
Treatment with sodium carbonate.

Correct Answer:
Clark's method

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Solution:
Clark's method is not suitable for the removal of permanent hardness of the water. Rather it is used for the removal of bicarbonates.

Therefore, Option(1) is correct.

Q. 10 The major product form in the following reaction

$$CH_3CH = CHCH(CH_3)_2 \overset{HBr}{\rightarrow}$$

Option 1:
$CH_3CH_2CH(Br)CH(CH_3)_2$

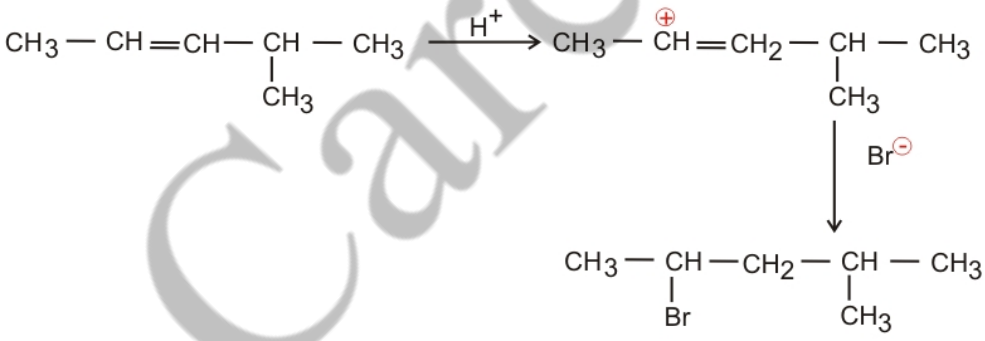
Option 2:
$Br(CH_2)_3CH(CH_3)_2$

Option 3:
$CH_3CH(Br)CH_2CH(CH_3)_2$

Option 4:
$CH_3CH_2CH_2C(Br)(CH_3)_2$

Correct Answer:
$CH_3CH(Br)CH_2CH(CH_3)_2$

Solution:
It follows Markonikov rule. The mechanism is given below:



Carbocation will be formed on the left side due to more stability due to a greater number of alpha hydrogens

Therefore, Option(3) is correct.

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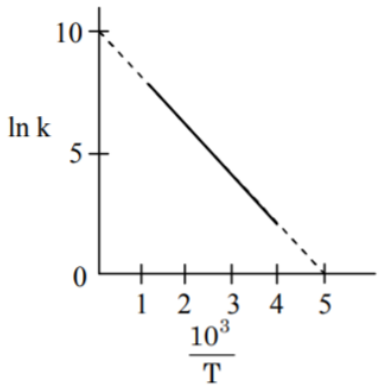
Q. 11 The rate constant (k) of a reaction is measured at different temperature (t) and data are plotted in given fig. the activation energy of the reaction in KJ/mol is (R--gas constant)



Option 1:
2R

Option 2:
$\frac{2}{R}$

Option 3:
R

Option 4:
$\frac{1}{R}$

Correct Answer:
2R

Solution:
We know that

$$\ln k = \frac{-E_a}{RT} + \ln A \text{ is a line, i.e, } y = mx + c$$

Now, y = lnk, m = $-E_a/R$, x = 1/T, C = lnA

Now, we have:

$$m = \frac{-E_a}{R} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 10}{5 - 0} = -2$$

Thus, E_a = 2R

Therefore, option(1) is correct.

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Q. 12 An element crystallites in a face centered cubic (fcc) unit cell with cell edge a. The distance between the centre of two nearest octahedral voids in the crystal lattice is:

Option 1:
$\sqrt{2a}$

Option 2:
$\frac{a}{2}$

Option 3:
$\frac{a}{\sqrt{2}}$

Option 4:
a

Correct Answer:
$\frac{a}{\sqrt{2}}$

Solution:
The octahedral void is at the middle of the edge in FCC.

The equation is given as:

$$x = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \frac{a}{\sqrt{2}}$$

Therefore, Option(3) is correct.

Q. 13 Reaction of ammonia with exam of Cl_2 gives:

Option 1:
NCl_3 and HCl

Option 2:
NH_4Cl and HCl

Option 3:
NH_4Cl and N_2

Option 4:
NCl_3 and NH_4Cl

Correct Answer:
NCl_3 and HCl

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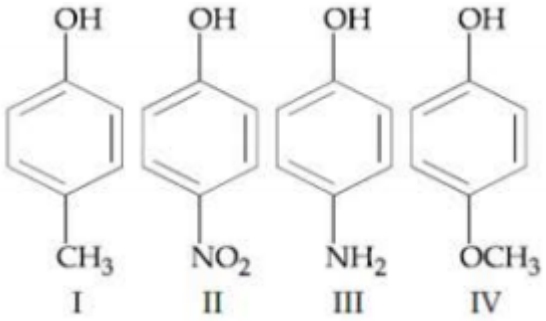
Solution:
The reaction occurs as follows:

$$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$$

(excess)

Therefore, Option(1) is correct.

Q. 14 The increasing order of boiling pts of the following compound is :



Option 1:
I<III<IV<II

Option 2:
I<IV<II<III

Option 3:
IV<I<II<III

Option 4:
III<I<II<IV

Correct Answer:
I<IV<II<III

Solution:
The value of boiling point of given structures is:

I = 202^oC

II = 279^oC

III = 284^oC

IV = 243^oC

Experimental data suggest that boiling point order is:

III > II > IV > I

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Therefore, Option(2) is correct.

Q. 15 Hydrogen peroxide in pure state is:

Option 1:
non-planar and almost colorless

Option 2:
linear and almost colorless

Option 3:
planar and blue in color

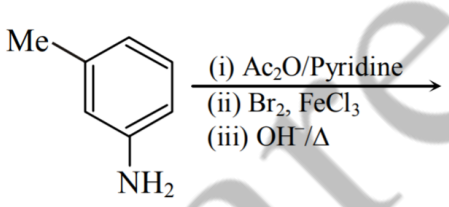
Option 4:
linear and blue in color

Correct Answer:
non-planar and almost colorless

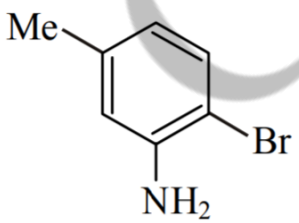
Solution:
Hydrogen peroxide is non-planar and almost colorless (very pale blue) liquid.

Therefore, Option(1) is correct.

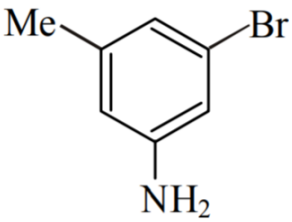
Q. 16 The final major product of the following reaction is :



Option 1:



Option 2:



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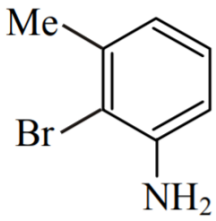
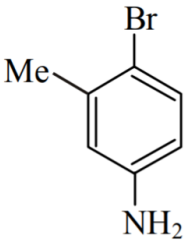
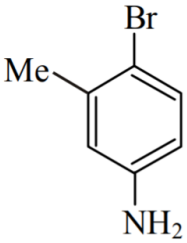
Option 3:



Option 4:

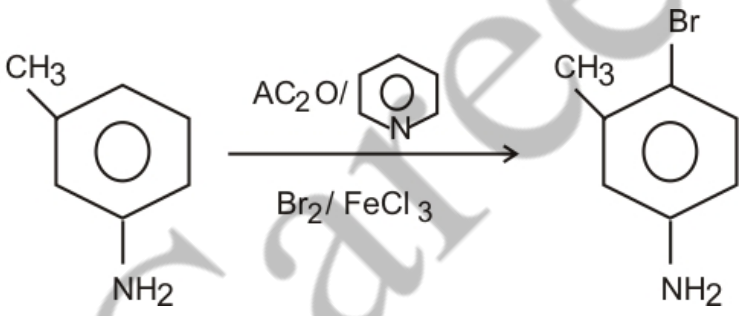


Correct Answer:



Solution:

The reaction occurs as follows:



Mechanism

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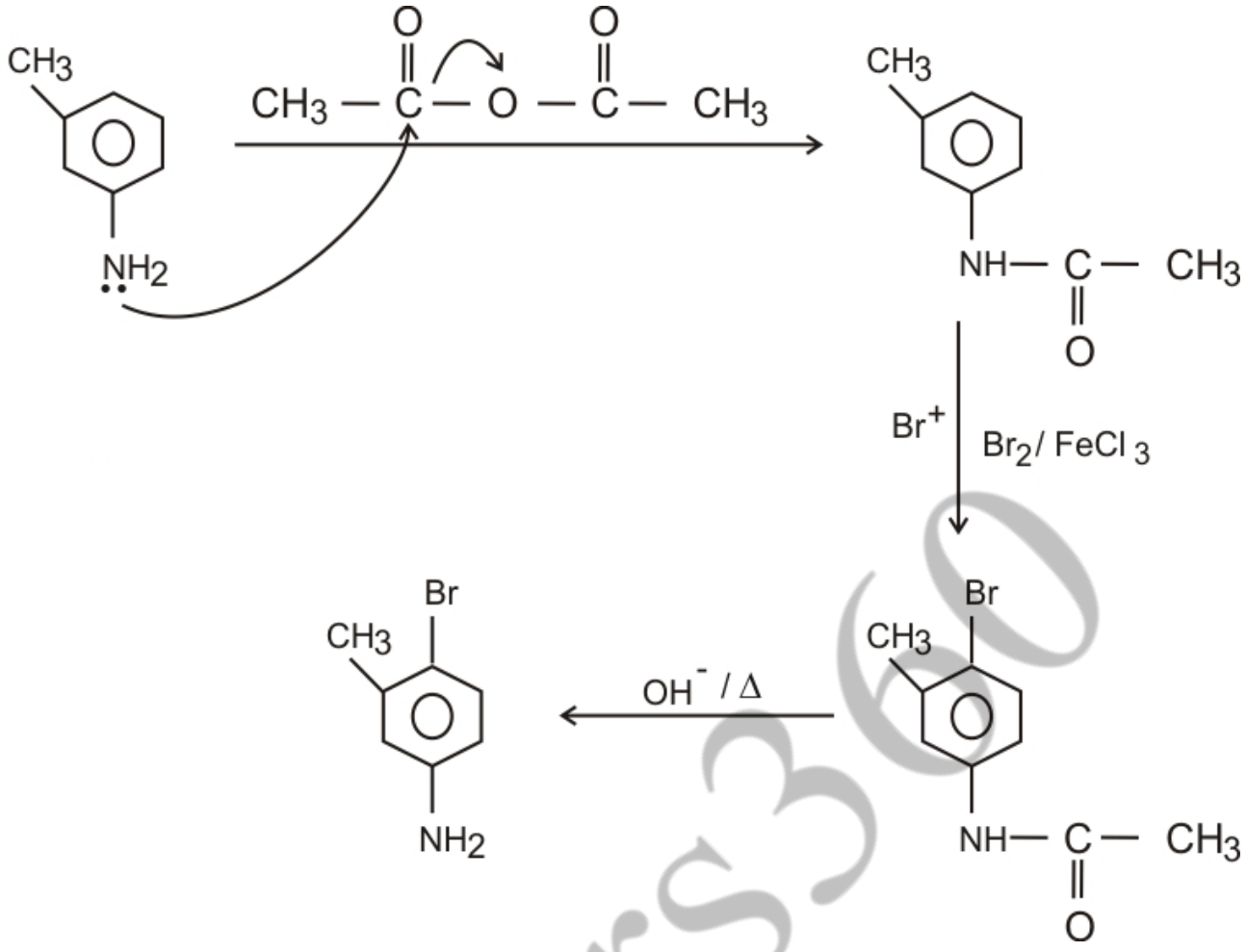

Therefore, Option(4) is correct.

Q. 17 The compound that has the largest H-M-H bond angle (M=N,O,S,C)is:

Option 1:
CH_4

Option 2:
H_2O

Option 3:
H_2S

Option 4:
NH_3

Correct Answer:
CH_4

Solution:
CH_4 has the largest H-M-H bond angle which is 109.5^o.

The bond angle of other species are given below:

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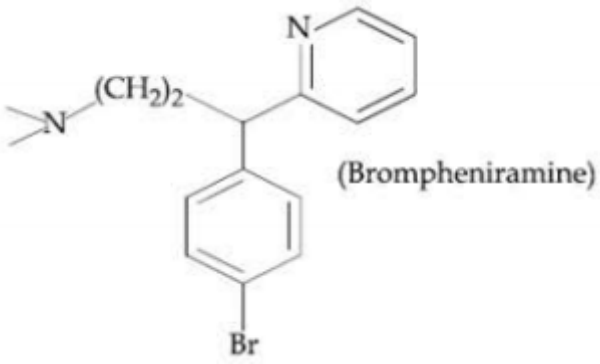
$H_2O = 101.9^o$

$NH_3 = 107^o$

$H_2S = 90^o$

Therefore, Option(1) is correct.

Q. 18 The following molecules act as an :



(Brompheniramine)

Option 1:
Antidepranant

Option 2:
Anti-bacterial

Option 3:
Antiseptic

Option 4:
Anti-histamine

Correct Answer:
Anti-histamine

Solution:
Brompheniramine works by blocking the action of histamine, which causes allergic symptoms.

Hence, it acts as an antihistamine.

Therefore, Option(3) is correct.

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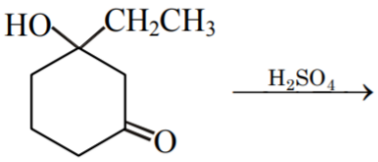
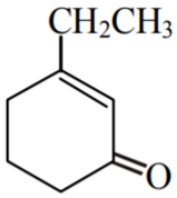
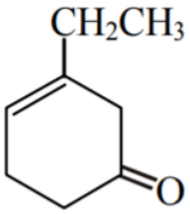
Q. 19 The major product of the following reaction is -



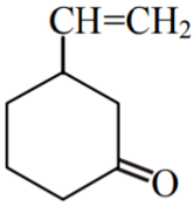
Option 1:



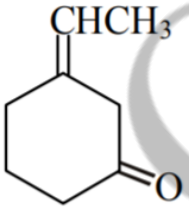
Option 2:



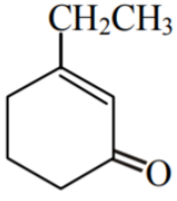
Option 3:



Option 4:



Correct Answer:



Solution:

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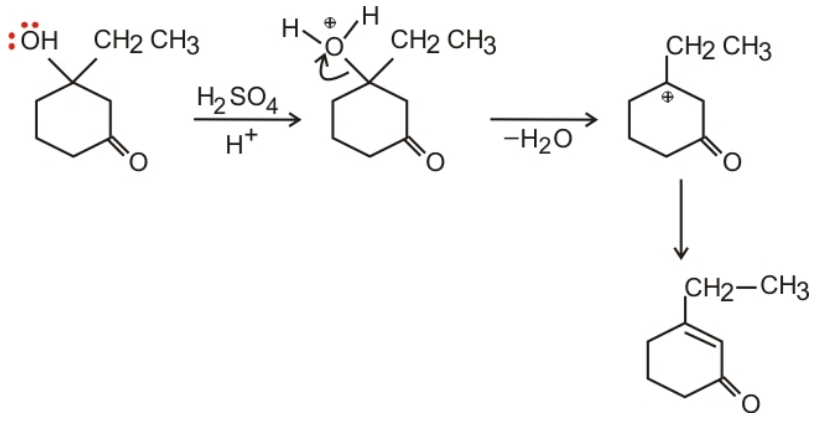

Therefore, Option(1) is correct.

Q. 20 Consider the complex ions -

trans-$[Co(en)_2Cl_2]^+$ (A) and cis -$[Co(en)_2Cl_2]^+$ (B)

The correct statement regarding them is :-

Option 1:
Both (A) and (B) cannot be optically active.

Option 2:
Both (A) and (B) can be optically active.

Option 3:
(A) cannot be optically active, but (B) can be optically active.

Option 4:
(A) can be optically active, but (B) cannot be optically active.

Correct Answer:
(A) cannot be optically active, but (B) can be optically active.

Solution:
Trans - $[Co(en)_2Cl_2]^+$ has plane of symmetry. So it is not optically active.

Cis - $[Co(en)_2Cl_2]^+$ does not have any plane of symmetry, so it can be optically active.

Therefore, Option(3) is correct.

Q. 21 The volume in ml of $0.02\ M K_2\ Cr_2\ O_7$ soln. required to react with 0.288gm of ferrous oxalate in acidic medium is (Molar mass of $Fe = 56\ g/mol$)

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Correct Answer:
50

Solution:
The reaction occurs as follows:

$$\mathrm{Fe(C_2O_4)} + \mathrm{K_2Cr_2O_7} \rightarrow \mathrm{Fe^{3+}} + \mathrm{CO_2} + \mathrm{Cr^{3+}}$$

From law of equivalence, we know:

meq of $Fe(C_2O_4)$ = meq of $K_2Cr_2O_7$

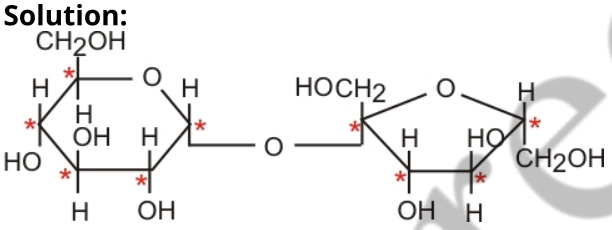
$$\frac{0.288}{144} \mathrm{x}\, 3 = \frac{6 \mathrm{x}\, 0.02 \mathrm{x}\, \mathrm{V}}{1000}$$

Thus, V = 50

Thus, volume in ml required of $K_2Cr_2O_7$ is 50.

Q. 22 The number of chiral carbons present in sucrose is _____ .

Correct Answer:
9

Solution:


Thus, the total number of chiral carbons present in sucrose is 9.

Q. 23 Considering that $\Delta_0 > P$ the magnetic moment in BM of $[Ru(H_2O)_6]^{2+}$ would be ____?

Correct Answer:
0

Solution:
$[Ru(H_2O)]^{2+}$

Ru^{2+} has d^6 configuration and since $\Delta_0 > P$, hence pairing will occur.

Thus, there will be no unpaired electrons.

Hence, the magnetic moment of the complex will be 0 BM.

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Q. 24 For a reaction $X + Y \rightleftharpoons Z$ 1.0 mol of X 1.5mol of Y and 0.5mol of Z were taken in a 1L vessel and allowed to react. At equilibrium, concentration of Z was 1mol/L . The equilibrium constant of reaction is x/16. What is x?

Correct Answer:
16

Solution:
The reaction occurs as follows:

$$\mathrm{X} + \mathrm{Y} \rightleftharpoons 2\mathrm{Z}$$

t = 0	1	1.5	0.5
t = teq	(1-x)	(1.5-x)	(0.5 + 2x)

Given, 0.5 + 2x = 1

Thus, x = 0.25

Now, K_{eq} is given as:

$$\mathrm{K_{eq}} = \frac{1^2}{(1.25)(0.75)} = \frac{16}{15}$$

Hence, x = 16.

Q. 25 For a dimerization reaction,

$$2\ \mathrm{A(\ g)} \longrightarrow \mathrm{A_2(\ g)} \text{ at } 298\ \mathrm{K}, \Delta\mathrm{U}^{\ominus} = -20\ \mathrm{kJ\ mol^{-1}}$$

$$\Delta\mathrm{S}^{\ominus} = -30\ \mathrm{J\ K^{-1}\ mol^{-1}} \text{ then the } \Delta\mathrm{G}^{\ominus} \text{ will be } ____ \text{ J.}$$

Correct Answer:
-13538.00

Solution:
The reaction occurs as follows:

$$2\mathrm{A(g)} \rightarrow \mathrm{A_2(g)}$$

$$\Delta\mathrm{U} = -20\mathrm{kJ/mol}$$
$$\Delta\mathrm{n_g} = -1$$

$$\Delta\mathrm{H} = \Delta\mathrm{U} + \Delta\mathrm{nRT}$$

$$\Delta\mathrm{H} = -20 + \frac{(-1)\ \mathrm{x}\ 8.314\ \mathrm{x}\ 298}{1000}$$

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$$\Delta H = -22.48$$

Now we have,

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = -22.48 - \frac{298 \text{ x } (-30)}{1000}$$

$$\Delta G = -13538 \text{ J/mol}$$

Maths

Q. 1 $\lim_{x\to 0} \frac{x\left(e^{\left(\sqrt{1+x^2+x^4}-1/x\right)}-1\right)}{\sqrt{1+x^2+x^4}-1}$

Option 1:
is equal to 1

Option 2:
is equal to 0

Option 3:
doesn't exist

Option 4:
is equal to $\sqrt{e}$

Correct Answer:
is equal to 1

Solution:

$$\lim_{x\to 0} \frac{\left(e^{\left(\sqrt{1+x^2+x^4}-1\right)/x}-1\right)}{\frac{\sqrt{1+x^2+x^4}-1}{x}}$$

$$\lim_{x\to 0} \frac{\sqrt{1+x^2+x^4}-1}{x} \qquad \left(\frac{0}{0} \text{ form}\right)$$

$$\lim_{x\to 0} \left(\frac{1\,(2x+4x^3)}{2x\sqrt{1+x^2+x^4}}\right) = 0$$

The limit is of the form $\lim_{h\to 0} \frac{e^h-1}{h}$

It is equal to 1

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Q. 2 If the sum of first 20 terms of the series $\log_{\left(7^{\frac{1}{2}}\right)} x+\log_{\left(7^{\frac{1}{3}}\right)} x+\log_{\left(7^{\frac{1}{4}}\right)} x+\cdots$ is 460 then x=?

Option 1:
e^2

Option 2:
$7^{\frac{1}{2}}$

Option 3:
$7^{\frac{46}{21}}$

Option 4:
7^2

Correct Answer:
7^2

Solution:
$\log_{7^{1/2}} x+\log_{7^{1/3}} x+\log_{7^{1/4}} x+\ldots$

$2 \log_7 x+3 \log_7 x+4 \log_7 x+\ldots$

$$\begin{aligned} S_{20} &= (\log_7 x)(2+3+4+\ldots 21) \\ &= (\log_7 x)(1+2+3+4+\ldots 21-1) \\ &= (\log_7 x)\left(\frac{21 \times 22}{2}-1\right) \\ &= (\log_7 x)(230) \\ &= 230 \log_7 x \end{aligned}$$

$$\therefore \quad \log_7 x=\frac{460}{230}$$

$\Rightarrow \log_7 x=2$

$\Rightarrow x=49$

Q. 3 If $L=\sin^2\left(\frac{\pi}{16}\right)-\sin^2\left(\frac{\pi}{8}\right)$ and $M=\cos^2\left(\frac{\pi}{16}\right)-\sin^2\left(\frac{\pi}{8}\right)$ then:

Option 1:
$L=\frac{1}{2\sqrt{2}}+\frac{1}{2}\cos\left(\frac{\pi}{8}\right)$

Option 2:
$L=\frac{1}{2\sqrt{2}}-\frac{1}{4}\cos\left(\frac{\pi}{8}\right)$

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Option 3:

$$M = \frac{1}{2\sqrt{2}} + \frac{1}{2}\cos\left(\frac{\pi}{8}\right)$$

Option 4:

$$M = \frac{1}{4\sqrt{2}} + \frac{1}{4}\cos\left(\frac{\pi}{8}\right)$$

Correct Answer:

$$M = \frac{1}{2\sqrt{2}} + \frac{1}{2}\cos\left(\frac{\pi}{8}\right)$$

Solution:

$$L = \sin^2\left(\frac{\pi}{16}\right) - \sin^2\left(\frac{\pi}{8}\right)$$
$$= \frac{1-\cos(x/8)}{2} - \left(\frac{1-\cos(\pi/4)}{2}\right)$$
$$= \frac{\cos\pi/4 - \cos(x/8)}{2} - \frac{1}{2}$$
$$= \frac{1}{26} - \frac{1}{3}\cos(x/8)$$

$$M = \cos^2(\pi/16) - \sin^2(\pi/8)$$
$$= \left(\frac{\cos(x/8)+1)}{2}\right) - \left(\frac{1-\cos(\pi/4)}{2}\right)$$
$$= \frac{1}{2}\cos(\pi/8) + \frac{1}{2\sqrt{2}}$$

Q. 4 If the mean and standard deviation of the data 3,5,7, a,b are 5&2 respectively. Roots of the equation:-

Option 1:

$$x^2 - 10x + 19 = 0$$

Option 2:

$$x^2 - 20x + 18 = 0$$

Option 3:

$$2x^2 - 20x + 19 = 0$$

Option 4:

$$x^2 - 10x + 18 = 0$$

Correct Answer:

$$x^2 - 10x + 19 = 0$$

Solution:

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$3, 5, 7, a, b$

$$\frac{3+5+7+a+b}{5} = 5$$

$$a + b = 10 - - - 0 - (1)$$

$$\frac{4+0+4+(a-5)^2+(b-5)^2}{5} = 4$$

$$a^2 + b^2 - 10(a+b) + 50 = 12$$

$$a^2 + b^2 = 62$$

$$(a+b)^2 - 2ab - 100 + 50 = 12$$
$$ab = 19$$
$$\text{Equation is } x^2 - 10x + 19 = 0$$

Q. 5 The area in sq units of the region $A = \left\{(x, y) = (x-1)\,[x] \leq y \leq 2\sqrt{x},\ \ 0 \leq x \leq 2\right\}$ where [t] denotes the greatest integer function is

Option 1:
$\frac{8}{3}\sqrt{2} - \frac{1}{2}$

Option 2:
$\frac{4}{3}\sqrt{2} + 1$

Option 3:
$\frac{4}{3}\sqrt{2} - \frac{1}{2}$

Option 4:
$\frac{8}{3}\sqrt{2} - 1$

Correct Answer:
$\frac{8}{3}\sqrt{2} - \frac{1}{2}$

Solution:
$A = \{(x, y) : (x-1)[x] \leqslant y \leqslant 2\sqrt{x},\ \ 0 \leq x \leq 2\}$

$= \{(x, y) : 0 \leq y \leq 2\sqrt{x},\quad 0 \leq x < 1;\ \ (x-1) \leq y \leq 2\sqrt{x};\ \ 1 \leq x < 2\}$

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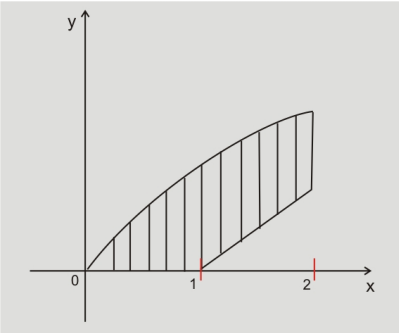

$$= \int_0^1 2\sqrt{x}dx + \int_1^2 (2\sqrt{x} - (x-1))dx$$
$$= \frac{4}{3} + \left(\frac{8\sqrt{2}}{5} - \frac{4}{3} - \frac{1}{2}\right)$$
$$= \frac{8\sqrt{2}}{3} - \frac{1}{2}$$

Q. 6 If the sum of the second , 3rd, 4th terms of a positive term GP is 3 & the sum of 6th , 7th & 8th terms is 243 then the sum of first 50 terms of GP is

Option 1:
$\frac{1}{13}\left(3^{50}-1\right)$

Option 2:
$\frac{1}{26}\left(3^{50}-1\right)$

Option 3:
$\frac{1}{26}\left(3^{49}-1\right)$

Option 4:
$\frac{2}{13}\left(3^{50}-1\right)$

Correct Answer:
$\frac{1}{26}\left(3^{50}-1\right)$

Solution:
General Term of a GP

So, the general term or n^{th} term of a geometric progression is $a_n = ar^{n-1}$

Sum of n-term of a GP

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Let S_n be the sum of n terms of the G.P. with the first term 'a' and common ratio 'r'. Then

$$S_n = a\left(\frac{r^n - 1}{r - 1}\right)$$

Now,

$$ar + ar^2 + ar^3 = 3$$
$$ar^5 + ar^6 + ar^7 = 243$$
$$Dividing\ we\ get$$
$$r^4 = 81 \Rightarrow r = 3$$

So,

$$a(3 + 9 + 27) = 3$$
$$\Rightarrow a = \frac{1}{13}$$

$$\therefore S_{50} = \frac{a\left(r^{50} - 1\right)}{(r - 1)}$$
$$= \frac{1}{13}\left(\frac{3^{50} - 1}{2}\right)$$

Q. 7 If $\int \frac{\cos\theta}{5 + 7\sin\theta - 2\cos\theta} d\theta = A\log_e(B(\theta)) + C$ where C=constant of integration, then $B(\theta) =$

Option 1:
$$\frac{2\sin\theta + 1}{5(\sin\theta + 3)}$$

Option 2:
$$\frac{2\sin\theta + 1}{\sin\theta + 3}$$

Option 3:
$$\frac{5(\sin\theta + 3)}{2\sin\theta + 1}$$

Option 4:
$$\frac{5(2\sin\theta + 1)}{\sin\theta + 3}$$

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Correct Answer:

$$\frac{5(2\sin\theta+1)}{\sin\theta+3}$$

Solution:

$$\int \frac{\cos\theta}{5+7\sin\theta-2\left(1-\sin^2\theta\right)}d\theta$$

$$\int \frac{\cos\theta}{2\sin^2\theta+7\sin\theta+3}d\theta$$

$$\int \frac{\cos\theta}{(2\sin\theta+3)(\sin\theta+1)}d\theta$$

$$\text{Let}\ \ \sin\theta=t \qquad \Rightarrow \qquad \cos\theta d\theta=dt$$

$$\int \frac{dt}{(2t+1)}(t+3)$$

$$\left[\frac{2}{5}\int\frac{dt}{2t+1}-\frac{1}{5}\int\frac{dt}{2t+3}\right]$$

$$\frac{2}{5}\ln(2\sin\theta+1)-\frac{1}{5}\ln(\sin\theta+3)+C$$

$$\frac{1}{5}\ln\left(\frac{(2\sin\theta+1)}{\sin\theta+3}\right)+C$$

Q. 8 The derivative of $\tan^{-1}\left\{\frac{\sqrt{1+x^2}-1}{x}\right\}$ with respect to $\tan^{-1}\left\{\frac{2x\sqrt{1-x^2}-1}{1-2x^2}\right\}$ at $x=\frac{1}{2}$ is

Option 1:

$$\frac{2\sqrt{3}}{5}$$

Option 2:

$$\frac{\sqrt{3}}{10}$$

Option 3:

$$\frac{2\sqrt{3}}{3}$$

Option 4:

$$\frac{\sqrt{3}}{12}$$

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Correct Answer:

$\frac{\sqrt{3}}{10}$

Solution:

$\tan^{-1}\left\{\frac{\sqrt{1+x^2}-1}{x}\right\}$

Let $x=\tan\theta$

$\tan^{-1}\left\{\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right\}=\frac{\theta}{2}=\frac{\tan^{-1}x}{2}$

$\tan^{-1}\left\{\frac{2x\sqrt{1-x^2}}{1-2x^2}\right\}$ let $x=\sin\alpha$

$=\tan^{-1}\left\{\frac{2\sin\alpha\sqrt{1-\sin^2\alpha}}{1-2\sin^2\alpha}\right\}$

$=2\alpha=2\sin^{-1}x$

$\frac{dy}{dx}=\frac{\frac{1}{2}+\frac{1}{1+x^2}}{2\times\frac{1}{\sqrt{1-x^2}}}$

$=\frac{1}{4}\frac{\sqrt{1-x^2}}{1+x^2}$

at $x=\frac{1}{2}$

$\frac{1}{4}\times\frac{\sqrt{1-\frac{1}{4}}}{1+\frac{1}{4}}=\frac{\frac{\sqrt{3}}{2}}{5}=\frac{\sqrt{3}}{10}$

Q. 9 Let $y=y(x)$ be the solution of the differential equation $\cos x\frac{dy}{dx}+2y\sin x=\sin 2x,\ x\in\left(0,\frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{3}\right)=0$ then $y\left(\frac{\pi}{4}\right)$ is equal to:

Option 1:

$2+\sqrt{2}$

Option 2:

$\frac{1}{\sqrt{2}}-1$

Option 3:

$2-\sqrt{2}$

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Option 4:

$\sqrt{2}-2$

Correct Answer:

$\sqrt{2}-2$

Solution:

$\cos x\frac{dy}{dx}+2y\sin x=\sin 2x$

$\Rightarrow \frac{dy}{dx}+2y\tan x=2\sin x$

$I.F.=e^{2\int \frac{\sin x}{\cos x}dx}$

$=e^{\ln \sec^2 x}=\sec^2 x$

$\Rightarrow y\times \sec^2 x=2\int \sin x\times \sec^2 x dx$

$\Rightarrow y\sec^2 x=\frac{2}{\cos x}+c$

$0=4+c\Rightarrow c=-4$

$\text{at } x=\pi/4$

$y\times 2=2\sqrt{2}-4$

$\Rightarrow y=\sqrt{2}-2$

Q. 10 There are 3 sections in a question paper & each section contains 5 questions. A candidate has to answer a total of 5 questions, choosing at least one question from each section. Then the number of ways in which the candidate can choose questions is:

Option 1:

3000

Option 2:

2250

Option 3:

2255

Option 4:

1500

Correct Answer:

2250

Solution:

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He can attempt (3,1,1) questions from sections A, B and C, OR

He can attempt (2,2,1) questions from sections A, B and C

Case I: 3, 1, 1

The number of ways is ${}^3C_1 \times \left({}^5C_3\ {}^5C_1\ {}^5C_1\right) = 750$

Case II: 2, 2, 1

The number of ways is ${}^3C_2 \times \left({}^5C_2\ {}^5C_2\ {}^5C_1\right) = 1500$

The total number of ways 2250.

Q. 11 If x=1 is a critical point of a function $f(x) = (3x^2 + ax - 2 - a)e^x$ then

Option 1:

$x = 1$ is a local maxima and $x = \dfrac{-2}{3}$ is a local minima of f.

Option 2:

$x = \dfrac{-2}{3}$ is a local maxima and $x = 1$ is a local minima of f.

Option 3:

$x = \dfrac{-2}{3}$ and $x = 1$ is a local minima of f.

Option 4:

$x = \dfrac{-2}{3}$ and $x = 1$ is a local maxima of f.

Correct Answer:

$x = \dfrac{-2}{3}$ is a local maxima and $x = 1$ is a local minima of f.

Solution:

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$f(x) = (3x^2 + ax - 2 - a)e^x$

$f'(x) = (6x + a)e^x + (3x^2 + ax - 2 - a)e^x$

$f'(x)|_{atx=1} = (6 + a)e^1 + (3 + a - 2 - a)e^1 = 0$
$a + 7 = 0$
$a = -7$

$f'(x) = (3x^2 - x - 2)e^x$
for minima and maxima f'(x) = 0

$f'(x) = (3x^2 - x - 2)e^x = 0$
$3x^2 - x - 2 = 0$
$3x^2 - 3x + 2x - 2 = 0$
$3x(x - 1) + 2(x - 1) = 0$
$(x - 1)(3x + 2) = 0$
$x = 1 \; or \; x = \dfrac{-2}{3}$

$f''(x) = (6x - 1)e^x + (3x^2 - x - 2)e^x$

$f''(x)|_{atx=1} = 5e > 0$

$\Rightarrow$ local minima at $x = 1$

$f''(x)|_{atx=-2/3} = -5e^{\frac{-2}{3}} < 0$

$\Rightarrow$ local maxima at $x = -2/3$

Q. 12 If for some $\alpha \in R$, the lines $L_1 : \dfrac{x+1}{2} = \dfrac{y-2}{-1} = \dfrac{z-1}{1}$ and $L_2 : \dfrac{x+2}{\alpha} = \dfrac{y+1}{5-\alpha} = \dfrac{z+1}{1}$ are coplanar then the line L_2 passes through the point:

Option 1:
(2,-10, -2)

Option 2:
(10,2,2)

Option 3:
10,-2,-2)

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Option 4:
(-2,10,2)

Correct Answer:
(2,-10, -2)

Solution:
Given equations of line

$$L_1 : \frac{x+1}{2} = \frac{y-2}{-1} = \frac{z-1}{1}$$

$$L_2 : \frac{x+2}{\alpha} = \frac{y+1}{5-\alpha} = \frac{z+1}{1}$$

For the coplanarity of two lines

$$\begin{vmatrix} -1+2 & 2+1 & 1+1 \\ 2 & -1 & 1 \\ \alpha & 5-\alpha & 1 \end{vmatrix} = 0$$

$$1(-1-5+\alpha) - 3(2-\alpha) + 2(10-2\alpha+\alpha) = 0$$

$$\alpha = -4$$

Now

$$L_2 : \frac{x+2}{-4} = \frac{y+1}{9} = \frac{z+1}{1}$$

$$\text{point } (2,-10,-2) \text{ satisfy the equation.}$$

Q. 13 If the length of the chord of the circle $x^2 + y^2 = r^2 \ (r > 0)$ along the line $y - 2x = 3$ is r then r^2 is equal to

Option 1:
$\frac{9}{5}$

Option 2:
12

Option 3:
$\frac{12}{5}$

Option 4:
$\frac{24}{5}$

Correct Answer:
$\frac{12}{5}$

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Solution:

Length of the chord by the line y = mx + c on the circle $x^2 + y^2 = a^2$ is

$$= 2\sqrt{\frac{a^2(1+m^2) - c^2}{1+m^2}}$$

Given equation of line, y = 2x +3

and circle $x^2 + y^2 = r^2$

according to question

$$r = 2\sqrt{\frac{r^2(1+4) - 9}{1+4}}$$
$$\frac{r^2}{4} = \frac{5r^2 - 9}{5}$$
$$r^2 = \frac{12}{5}$$

Q. 14 Which of the following points lies on the tangent to the curve $x^4e^y + 2\sqrt{y+1} = 3$ at the point (1,0)?

Option 1:
(2,2)

Option 2:
(-2,6)

Option 3:
(-2,4)

Option 4:
(2,6)

Correct Answer:
(-2,6)

Solution:

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$x^4e^y + 2\sqrt{y+1} = 3$

$4x^3e^y + x^4e^y\frac{dy}{dx} + \frac{1}{\sqrt{y+1}}\frac{dy}{dx} = 0$

$\frac{dy}{dx}_{at(1,0)} = -2$

Equation of tangent

$(y-0) = -2(x-1)$

$y = -2x + 2$

$(-2, 6)$ satisfy this equation

Q. 15 If α & β are the roots of the equation $7x^2 - 3x - 2 = 0$ then value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ equals?

Option 1:
1/24

Option 2:
3/8

Option 3:
27/16

Option 4:
27/32

Correct Answer:
27/16

Solution:
Given quad equation $7x^2 - 3x - 2 = 0$

$\alpha + \beta = \frac{3}{7}, \quad \alpha\beta = -\frac{2}{7}$

$\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2} = \frac{\alpha(1-\beta^2) + \beta(1-\alpha^2)}{(1-\alpha^2)(1-\beta^2)}$

$\Rightarrow \frac{(\alpha+\beta)(1-\alpha\beta)}{1-[(\alpha+\beta)^2 - 2\alpha\beta] + (\alpha\beta)^2}$

Put the value of $\alpha + \beta$ and $\alpha\beta$

$\Rightarrow \frac{27}{16}$

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Q. 16 The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is

Option 1:
$-2^{15}i$

Option 2:
$2^{15}i$

Option 3:
-2^{15}

Option 4:
6^5

Correct Answer:
$-2^{15}i$

Solution:

$$\left[\frac{-1+i\sqrt{3}}{1-i}\right]^{30} = \left(\frac{2\omega}{1-i}\right)^{30}$$

$$= \frac{2^{30}\cdot\omega^{30}}{((1-i)^2)^{30}}$$

$$= \frac{2^{30}\cdot 1}{(1+i^2-2i)^{15}}$$

$$= \frac{2^{30}}{-2^{15}\cdot i^{15}}$$

$$= -2^{15}i$$

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Q. 17 If $a+x=b+y=c+z+1$ wherea,b,c,x,y,z are non zero distinct real nos then $\begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix}$ is equal to:

Option 1:
$y(a-c)$

Option 2:

$y(a-b)$

Option 3:

$y(b-a)$

Option 4:

0

Correct Answer:

$y(a-b)$

Solution:

Given that a + x = b + y = c + z + 1 = 0

$$\begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix} = \begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} + \begin{vmatrix} x & y & x+a \\ y & y & y+b \\ z & y & z+c \end{vmatrix}$$

$$\text{Since, } \begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} = 0$$

$$\because \begin{vmatrix} x & a & x+a \\ y & b & y+b \\ z & c & z+c \end{vmatrix} = \begin{vmatrix} x & a & a \\ y & b & b \\ z & c & c \end{vmatrix} + \begin{vmatrix} x & a & x \\ y & b & y \\ z & c & z \end{vmatrix} = 0+0$$

Now

$$\begin{vmatrix} x & y & x+a \\ y & y & y+b \\ z & y & z+c \end{vmatrix} = y \begin{vmatrix} x & 1 & x+a \\ y & 1 & y+b \\ z & 1 & z+c \end{vmatrix}$$

$$y \begin{vmatrix} x & 1 & x+a \\ y & 1 & y+b \\ z & 1 & z+c \end{vmatrix} = y \begin{vmatrix} x & 1 & x \\ y & 1 & y \\ z & 1 & z \end{vmatrix} + y \begin{vmatrix} x & 1 & a \\ y & 1 & b \\ z & 1 & c \end{vmatrix}$$

$$\Rightarrow y \begin{vmatrix} x & 1 & a \\ y & 1 & b \\ z & 1 & c \end{vmatrix}$$

$C_1 \to C_1 + C_3$

$$y \begin{vmatrix} x+a & 1 & a \\ y+b & 1 & b \\ z+c & 1 & c \end{vmatrix} = y \begin{vmatrix} x+a & 1 & a \\ x+a & 1 & b \\ x+a-1 & 1 & c \end{vmatrix}$$

$R_1 \to R_1 - R_2$

$R_2 \to R_2 - R_3$

$$\Rightarrow y \begin{vmatrix} 0 & 0 & a-b \\ 1 & 0 & b-c \\ x+a-1 & 1 & c \end{vmatrix} = y((a-b))$$

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Q. 18 If the line y=mx+c is a common tangent to the hyperbola $\frac{x^2}{100}-\frac{y^2}{64}=1$ and the circle $x^2+y^2=36$ then which of the following is true?

Option 1:
$4c^2=369$

Option 2:
$5m=4$

Option 3:
$8m+5=0$

Option 4:
$c^2=369$

Correct Answer:
$4c^2=369$

Solution:
Equation of the circle $x^2+y^2=36$

Equation of a tangent to the circle $y=mx\pm 6\sqrt{1+m^2}$

Equation of Hyperbola $y=mx\pm\sqrt{100m^2-64}$

For the common tangent

$36(1+m^2)=(100m^2-64)$
$64m^2=100$
$m^2=\frac{100}{64}$
$Now, c=6\sqrt{1+m^2}=$
$c^2=36(1+m^2)=\frac{369}{4}$
$4c^2=369$

Q. 19 The statement $(p\rightarrow(q\rightarrow p))\rightarrow(p\rightarrow(p\vee q))$ is

Option 1:
a contadiction

Option 2:
is equivalent to $(p\vee q)\wedge(\sim p)$

Option 3:
a tautology

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Option 4:
is equivalent to $(p \wedge q) \vee (\sim q)$

Correct Answer:
a tautology

Solution:

p	q	$q \rightarrow p$	$p \rightarrow (q \rightarrow p)$	$p \vee q$	$p \rightarrow p \vee q$	$(p \rightarrow (q \rightarrow P)) \rightarrow (p \rightarrow (p \vee q))$
T	T	T	T	T	T	T
T	F	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	T	F	T	T

$(p \rightarrow (q \rightarrow P)) \rightarrow (p \rightarrow (p \vee q))$

The statement is a tautology

Q. 20 If the system of linear equations

$$x + y + 3z = 0$$

$$x + 3y + k^2 z = 0$$

$$3x + y + 3z = 0$$

has non zero solution (x, y, z) for $some$ $k \in R$ then $x + \left(\frac{y}{z}\right) = ?$

Option 1:
-9

Option 2:
9

Option 3:
3

Option 4:
-3

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Correct Answer:
-3

Solution:
For non-zero solutions to exist,

$$\begin{vmatrix} 1 & 1 & 3 \\ 1 & 3 & k^2 \\ 3 & 1 & 3 \end{vmatrix} = 0$$
$$\Rightarrow 1(9-k^2) - 1(3-3k^2) + 3(1-9) = 0$$
$$k^2 = 9$$

Now the system of equation become

x + y + 3z = 0

x + 3y + 9z = 0

3x + y + 3z = 0

we, get x = 0

3y = -9z

y/z = -3

x+ (y/z) = -3

Q. 21 If line x+y=a and x-y=b touch the curve $y = x^2 - 3x + 2$ at the pts where the curve intersects the x-axis then a/b=?

Option 1:
$\frac{1}{2}$

Option 2:
$\frac{3}{2}$

Option 3:
2

Option 4:
None of the above

Correct Answer:
$\frac{1}{2}$

Solution:

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Given equation of the Curve $y = x^2 - 3x + 2$

The curve intersects at X-axis (1,0) and (2,0)

Now the curve x + y = a and x - y = b touch the curve $y = x^2 - 3x + 2$

at (1,0) and (2,0)

a = 1 and y = 2

a/b = 1/2

Q. 22 If $A = \{a, b, c\}$ and $B = \{(1, 2, 3, 4)\}$ then the no. of elements in the set $C = \{f : A \to B | 2 \in f(A)\}$ and f is not one-one is.....

Option 1:
15

Option 2:
16

Option 3:
18

Option 4:
19

Correct Answer:
19

Solution:
Let say a has image 2 then b and c both should have single image 1/3/4, Total selection will be 3

If a has option to choose 1/3/5 then b and c have only one option 2 Total selection will be 3

Here if we rearrange a,b c it has 3 option so total number of ways is 3 X (3 + 3) = 18

Now if a b c has one option as 2

Total selection is 18 +1 = 19.

Q. 23 The coefficient of x^4 in the expansion of $\left(1 + x + x^2 + x^3\right)^6$ in powers of x is.....

Option 1:
60

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Option 2:
90

Option 3:
120

Option 4:
150

Correct Answer:
120

Solution:

$\left[1+x+x^2+x^3\right]^6$

$=\left[1+x+x^2(1+x)\right]^6$

$=\left[(1+x^2)(1+x)\right]^6$

$=(1+x^2)^6(1+x)^6$

To form x^4 term by multiplying these 2 brackets, following combinations are possible

$\text{Coefficient of } x^0 \text{ in } (1+x^2)^6 \times \text{Coefficient of } x^4 \text{ in } (1+x)^6$
$= {}^6C_0 \cdot {}^6C_4 = 15$

$\text{Coefficient of } x^2 \text{ in } (1+x^2)^6 \times \text{Coefficient of } x^2 \text{ in } (1+x)^6$
$= {}^6C_1 \cdot {}^6C_2 = 90$

$\text{Coefficient of } x^4 \text{ in } (1+x^2)^6 \times \text{Coefficient of } x^0 \text{ in } (1+x)^6$
$= {}^6C_2 \cdot {}^6C_0 = 15$

$\text{Total } = 120$

Q. 24 Let the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ such that $|\vec{a}| = 2, \left|\vec{b}\right| = 4, |\vec{c}| = 4$. If the projection of $\vec{b}$ on $\vec{a}$ is equal to the projection of $\vec{c}$ on $\vec{a}$ and $\vec{b}$ is $\perp$ $\vec{c}$ then the value of $\left|\vec{a}+\vec{b}-\vec{c}\right|$ = ?

Option 1:
$\sqrt{6}$

Option 2:
6

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Option 3:
3

Option 4:
$\sqrt{3}$

Correct Answer:
6

Solution:

$$\text{Given } |\vec{a}| = 2, |\vec{b}| = 4, |\vec{c}| = 5$$
$$\vec{b} \cdot \hat{a} = \vec{c} \cdot \hat{a} \Rightarrow \vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$$
$$\vec{b} \cdot \vec{c} = 0$$
$$|\vec{a} + \vec{b} - \vec{c}|^2 = (\vec{a} + \vec{b} - \vec{c}) \cdot (\vec{a} + \vec{b} - \vec{c})$$
$$= |\vec{a}|^2 + \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} + \vec{a} \cdot \vec{b} + |\vec{b}|^2 - \vec{b} \cdot \vec{c} - \vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c} + |\vec{c}|^2$$
$$= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2$$
$$|\vec{a} + \vec{b} - \vec{c}| = \sqrt{|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2}$$
$$= \sqrt{2^2 + 4^2 + 4^2}$$
$$= 6$$

Q. 25 In a bombing attack there is 50% chance that a bomb will hit the target Atleast two independent hits are required to destroy the target completely. Then the minimum no of bombs that must be dropped to ensure that there is atleast 99% chance of completely destroying the target is ..

Option 1:
9

Option 2:
10

Option 3:
11

Option 4:
12

Correct Answer:
11

Solution:
The probability of success in one strike is p = 1/2

probability of failure = q = 1/2

Now, the probability of r success

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$$P(x = r) = {}^{n}C_r \cdot \left(\frac{1}{2}\right)^{r} \cdot \left(\frac{1}{2}\right)^{n-r} = {}^{n}C_r \left(\frac{1}{2}\right)^{n}$$

From the given question $P(x \geq 2) \geq 0.99$

$\Rightarrow 1 - P(x < 2) \geq 0.99$
$\Rightarrow 1 - P(x = 0) - P(x = 1) \geq 0.99$
$\Rightarrow 1 - 0.99 \geq \frac{1+n}{2^n}$
$\Rightarrow 2^n \geq 100 + 100n$
For $n = 10, 2^n < 100 + 100n$
For $n = 11, 12, 13$
$2^n > 100 + 100n$

n = 11 is the least number.

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6th September Shift 1

Physics

Q. 1 You are given that mass of $^{7}_{3}Li = 7.0160u$

mass of $^{4}_{2}He = 4.0020u$

mass of $^{1}_{1}H = 1.0079u$

When 20g of $^{7}_{3}Li$ is converted into $^{4}_{2}He$ by proton capture, the energy liberated (in kWh), is:- (mass of nucleon $= 1\frac{GeV}{C^2}$)

Option 1:
8×10^6

Option 2:
1.33×10^6

Option 3:
4.5×10^5

Option 4:
6.82×10^5

Correct Answer:
1.33×10^6

Solution:
The nuclear reaction is
$_3Li^7 + \ _1H^1 \rightarrow 2\ _2\text{He}^4 + Q$
Mass defect, $\Delta m = m_{Li} + m_p + 2m_{He}$
$\Rightarrow \Delta m = 7.0160 + 1.0079 - 2 \times 4.0026 = 0.0187 a.m.u$
$Q = 0.0187 \times 931\text{MeV} = 17.40\ \text{MeV}$
Number of nuclei in 20g of $3\text{Li}^7 = \frac{6.023 \times 10^{23}}{7} \times 20 = 1.72 \times 10^{24}$
Total energy released when 20g of $_3\text{Li}^7$ is converted into Helium,
$\text{Q} = 17.40 \times 1.72 \times 10^{24}\text{MeV} = 17.40 \times 1.72 \times 1.6 \times 10^{-13} \times 10^{24} J$
As $1kWh = 3.6 \times 10^6 J$

$\therefore E = \frac{17.40 \times 1.72 \times 1.6 \times 10^{11}}{3.6 \times 10^6} kWh$

$E = 1.33 \times 10^6 \text{kWh}$

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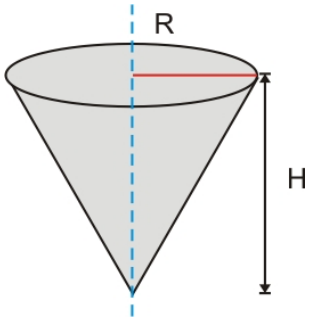
Q. 2 Shown in the figure is a hollow ice cream cone (it is open at the top). If its mass is 'M', radius of its top, R and height H, then its M.O.I about its axis is :



Option 1:
$\frac{MH^2}{3}$

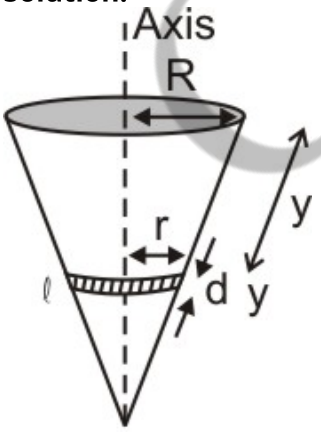
Option 2:
$\frac{MR^2}{3}$

Option 3:
$\frac{M(R^2+H^2)}{4}$

Option 4:
$\frac{MR^2}{2}$

Correct Answer:
$\frac{MR^2}{2}$

Solution:



We will take an elemental ring of thickness dy and radius r at a slant height y as show in the figure

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$given,\ \text{Mass} = M$
$\text{Radius} = R$
$I_{HC} = \int dI\ \text{ring} = \int (dm) r^2$

$dm = \frac{M}{\pi R l}(2\pi r dy)$

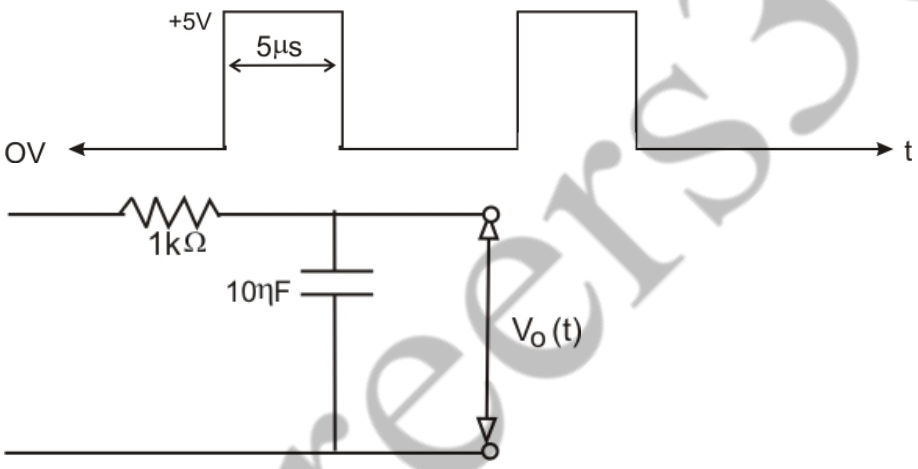
where dm is the mass of the elemental ring

$$I_{HC} = \int \frac{2M}{R\ell} r^3 dy$$

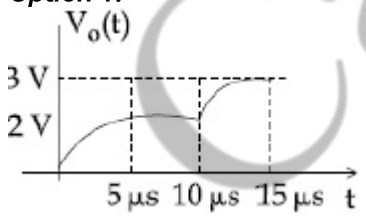
$$\frac{r}{R} = \frac{y}{l} \quad \Rightarrow \quad r = \frac{R}{l} y$$

$$I_{HC} = \frac{2M}{Rl} \frac{R^3}{l^3} \int_0^l y^3 dy = \frac{MR^2}{2}$$

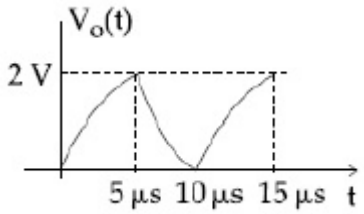
Q. 3 For the given i/p voltage waveform $V_{in}(t)$ the o/p voltage waveform $V_0(t)$ across the capacitor is correctly depicted by:-



Option 1:



Option 2:



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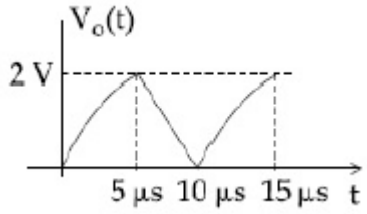
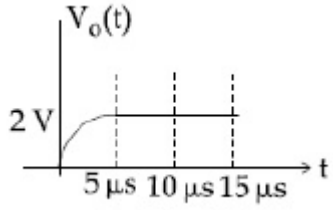
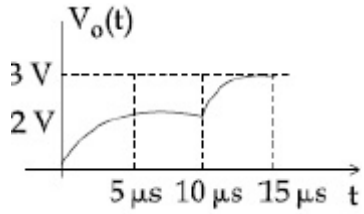
Option 3:



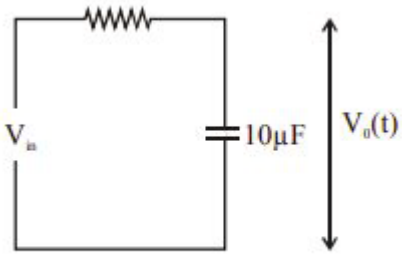
Option 4:



Correct Answer:



Solution:



Now for next $5\mu s$ $V_0(t) = 5 - 3.79e^{-\frac{t}{RC}}$
after $5\mu s$ again $V_0(t) = 2.79$ Volt $\approx 3V$

Most approperiate Ans.(1)

Q. 4 A point-like object is placed at a distance of 1m in front of convex lens of focal length 0.5m. A plane mirror is placed at a distance of 2m behind the lens. The position and nature of the final image formed by the system is:-

Option 1:
1m from the mirror, real

Option 2:
1m from the mirror, virtual

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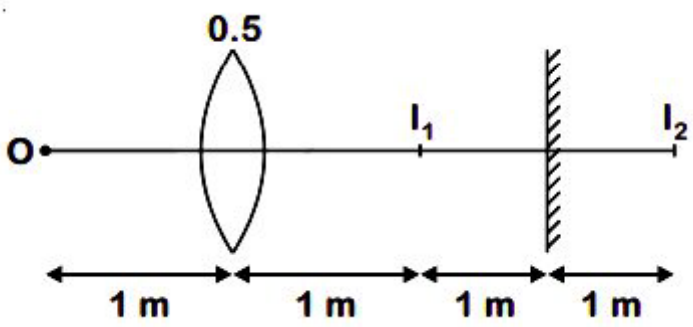
Option 3:
2.6m from the mirror, real

Option 4:
2.6m from the mirror, virtual

Correct Answer:
2.6m from the mirror, real

Solution:



Object is at 2f. So image will also be at ' 2f. (I_1)

Image of I_1 will be 1 m *behind mirror.*

i.e. $\Rightarrow I_2$
Now I_2 will be object for lens.
$\therefore$ u $\Rightarrow -3$ m
f $\Rightarrow +0.5$ m
$\frac{1}{v} \Rightarrow \frac{1}{f} + \frac{1}{u}$
$\Rightarrow \frac{1}{+0.5} + \frac{1}{-3}$
v $\Rightarrow \frac{3}{5} \Rightarrow 0.6$ m
So total distance from mirror $\Rightarrow 2 + 0.6 \Rightarrow 2.6$ m *and real image*

Q. 5 An AC circuit has $R = 100\Omega, C = 2\mu F$ and L = 800 mH, connected in series. The quality factor of the circuit is:-

Correct Answer:
2

Solution:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{100}\sqrt{\frac{80 \times 10^{-3}}{2 \times 10^{-6}}} = 2$$

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Q. 6 An e^- , a doubly ionized. He ion (He^{2+}) and a proton are having the same K.E. The relation between their respective De- Broglie wavelength $\lambda_e, \lambda_{He^{++}} \; and \; \lambda_p$

Option 1:
$\lambda_e < \lambda_{He^{++}} = \lambda_p$

Option 2:
$\lambda_e < \lambda_p < \lambda_{He^{++}}$

Option 3:
$\lambda_e > \lambda_{He^{++}} > \lambda_p$

Option 4:
$\lambda_e > \lambda_p > \lambda_{He^{++}}$

Correct Answer:
$\lambda_e > \lambda_p > \lambda_{He^{++}}$

Solution:

$$\lambda = \frac{h}{\sqrt{2mK}}$$

$m_{He} = 4m_p$

$\lambda_e = \frac{h}{\sqrt{2m_e K}}$

$\lambda_p = \frac{h}{\sqrt{2m_p K}}$

$$\lambda_{H_e} = \frac{h}{\sqrt{2\left(m_{He}\right)K}} = \frac{h}{\sqrt{2(4m_p)K}}$$

$m_{He} > m_p > m_e$

$\therefore \lambda_e > \lambda_p > \lambda_{He}$

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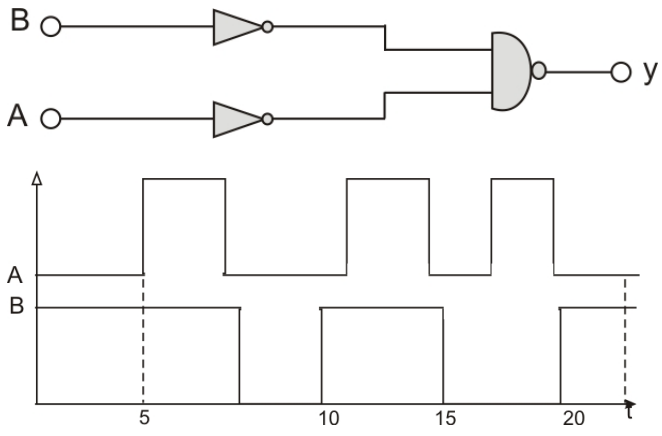
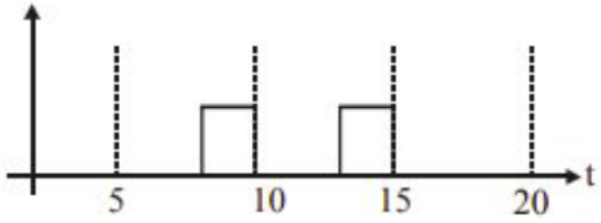
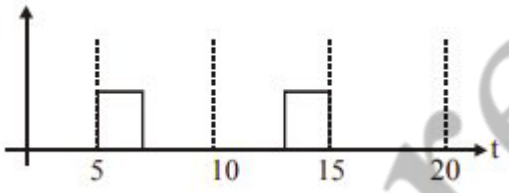
Q. 7 Identify the correct o/p signal Y in the given combination of gates (as shown) for the given I/P A and B



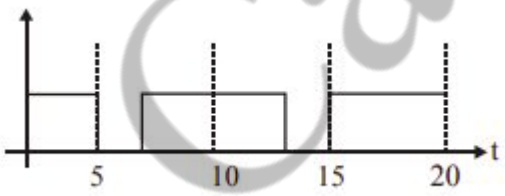
Option 1:



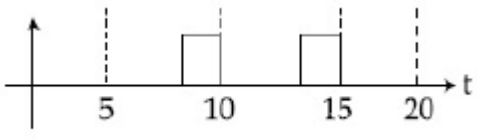
Option 2:



Option 3:



Option 4:



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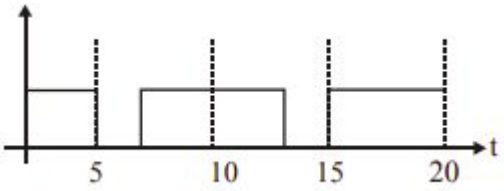
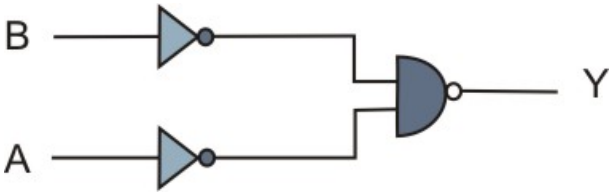
Correct Answer:

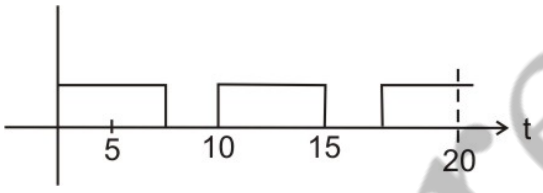


Solution:



$Y = \overline{\bar{A}\bar{B}} = A + B$

Inputs		Output
A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1



Q. 8 An e is moving along the +x direction with velocity $6 \times 10^6 m/s$. It enters a region of the uniform $\overrightarrow{E}$ of 300 V/cm pointing along +y direction. The magnitude and direction of the magnetic field set up in the region such that e^- keeps moving along the x- direction will be:

Option 1:
$5 \times 10^{-3} T$ along -z direction

Option 2:
$3 \times 10^{-4} T$ along -z direction

Option 3:
$3 \times 10^{-4} T$ along +z direction

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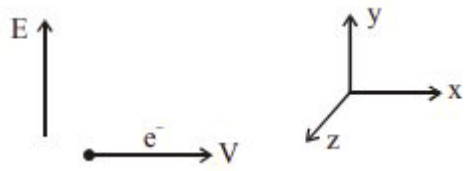
Option 4:

$5 \times 10^{-3} T$ along +z direction

Correct Answer:

$5 \times 10^{-3} T$ along +z direction

Solution:



$\vec{B}$ must be in $+z$ axis.

$\vec{V} = 6 \times 10^{6}\hat{i}$

$\vec{E} = 300\hat{j}\text{V/cm} = 3 \times 10^{4}\ \text{V/m}$

$q\vec{E} + q\vec{V} \times \vec{B} = 0$

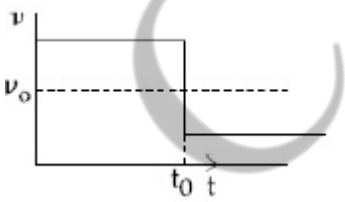
$E = VB$

$B = \frac{E}{V} = \frac{3\times10^{4}}{6\times10^{6}} = 5 \times 10^{-3}\ \text{T}$

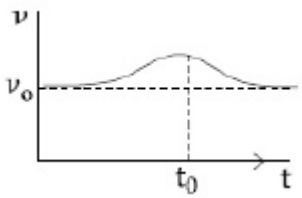
Q. 9 A sound source 'S' is moving along a straight track with the speed v, and is emitting sound of frequency ν_0 (see figure). An observer is standing at a finite distance, at the point O, from the track. The time variation of frequency heard by observer is best represented.

(E_0 represents the instant when the distance between the source and the observer is minimum:-

Option 1:



Option 2:



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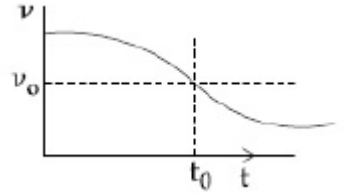
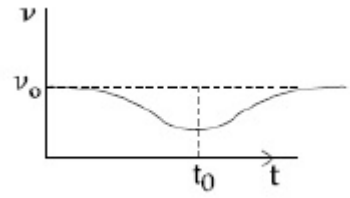
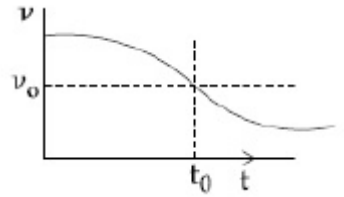
Option 3:



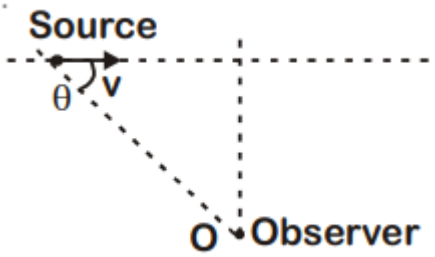
Option 4:



Correct Answer:



Solution:



ν- Apparent frequency

ν_0-Original frequency

While approaching the observer

$$\nu = \nu_0 \left(\frac{c}{\mathbf{c} - v\cos\theta} \right)$$

While receding from the observer

$$\nu = \nu_0 \left(\frac{\mathbf{c}}{\mathbf{c} + v\cos\theta} \right)$$

When the observer approaches the observer its apparent frequency is more than the original frequency as it just passes the observer it is equal to the original frequency for a moment subsequently when it recedes from the observer the apparent frequency is less than the original frequency

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Q. 10 An object of mass 'm' is suspended at the end of a mass less wire of length L and area of cross-section A. Young's modules of the maternal of the wire is Y. if the mass is pulled down slightly its freq. of oscillation along the vertical direction is :

Option 1:

$$f=\frac{1}{2\pi}\sqrt{\frac{mA}{YL}}$$

Option 2:

$$f=\frac{1}{2\pi}\sqrt{\frac{YL}{mA}}$$

Option 3:

$$f=\frac{1}{2\pi}\sqrt{\frac{YA}{mL}}$$

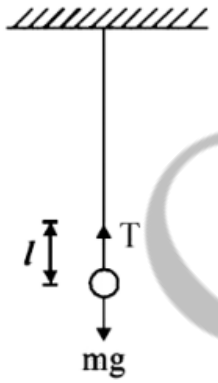
Option 4:

$$f=\frac{1}{2\pi}\sqrt{\frac{mL}{YA}}$$

Correct Answer:

$$f=\frac{1}{2\pi}\sqrt{\frac{YA}{mL}}$$

Solution:
At equilibrium



$T=mg$ .....(i)

$$Y=\frac{T/A}{l/L}$$

$$\Rightarrow T=\frac{YAl}{L}\ldots(ii)$$

From
(i) and (ii)

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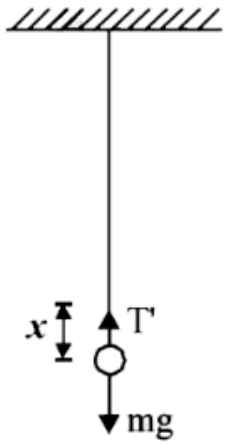
$$mg = \frac{YAl}{L} \ldots (iii)$$

Now for a small displacement x



$$\text{Restoring force} \quad = -[T' - mg] = -\left[\frac{YA(l+x)}{L} - \frac{YAl}{L}\right] = \frac{-YAx}{L}$$ Comparing with

$F = -m\omega^2 x$

$m\omega^2 = \frac{YA}{L}$

$\Rightarrow \omega = \sqrt{\frac{YA}{mL}} \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{YA}{mL}}$

$\text{Frequency} \quad = f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{YA}{mL}}$

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Q. 11 A particle of change 'q' and mass 'm' is moving with a velocit $-v\hat{i}\ (v \neq 0)$ towards a large screen in the Y-Z plane at a distance 'd'. if there is a magnetic field $\overrightarrow{B} = Bo\hat{k}$, the maximum value of v for which the particle will not hit the screen is:

Option 1:

$\frac{qdBo}{m}$

Option 2:

$\frac{qdBo}{3m}$

Option 3:
$\frac{qdBo}{2m}$

Option 4:
$\frac{2qdBo}{m}$

Correct Answer:
$\frac{qdBo}{m}$

Solution:
As velocity and the magnetic field are perpendicular to each other So particle will perform uniform circular motion in the x-y plane

And its radius is given as

$$r = \frac{mv}{qB_0}$$

So if the particle will not hit the screen then

$$r \leq d$$

$$\frac{mv}{qB_0} \leq d$$

$$v \leq \frac{dqB_0}{m}$$

$$\Rightarrow V_{max} = \frac{qdB_0}{m}$$

Q. 12 Molecules of an ideal gas are known to have 3 translatinal D.O.F and two rotational D.O.F. The gas is maintained at temperature T. The total internal energy, U of a mole of this gas and the value of $\gamma \left(= \frac{cp}{cv}\right)$ are given respectively, by:

Option 1:
$U = \frac{5}{2}RT$ and $\gamma = \frac{6}{5}$

Option 2:
$U = 5\ RT$ and $\gamma = \frac{6}{5}$

Option 3:
$U = \frac{5}{2}RT$ and $\gamma = \frac{7}{5}$

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Option 4:

$U = 5\ RT$ and $\gamma = \frac{7}{5}$

Correct Answer:

$U = \frac{5}{2}RT$ and $\gamma = \frac{7}{5}$

Solution:

For 1 mole $U = \frac{fRT}{2}$ and $\gamma = \frac{C_p}{C_v} = \frac{f+2}{f}$

So if $U = 5RT$ then we get $f = 10$ (which is not possible)

and $U = \frac{5}{2}RT$ then we get $f = 5$ and $\gamma = \frac{C_p}{C_v} = \frac{f+2}{f} = \frac{7}{5}$

So option C is correct

Q. 13 An insect is at the bottom of a hemisphere ditch of radius 1m. It crawls up the ditch but starts slipping after it is at height 'h' from the bottom. If the coefficient of friction b/w the ground and insect is 0.75. Then h is : $(g = 10m/s^2)$

Option 1:

0.6 m

Option 2:

0.45 m

Option 3:

0.2 m

Option 4:

0.8 m

Correct Answer:

0.2 m

Solution:

Till the component of its weight along with the bowl is balanced by limiting frictional force, then only the insect crawl up the bowl, up to a certain height h

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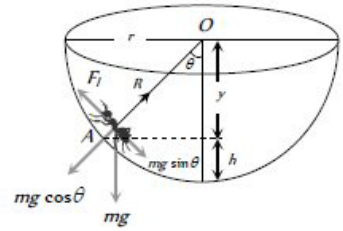

Let m=mass of the insect, r=radius of the bowl, μ= coefficient of friction for limiting condition at point

$$R = mg\cos\theta \quad \ldots\ldots(\text{i}) \quad \text{and} \quad F_l = mg\sin\theta \quad \ldots\ldots (\text{ii})$$

Dividing (ii) by (i)

$$\tan\theta = \frac{F_l}{R} = \mu \quad [\text{ using } F_l = \mu R]$$

$$\therefore \quad \frac{\sqrt{r^2-y^2}}{y} = \mu \quad \text{or } y = \frac{r}{\sqrt{1+\mu^2}}$$

$$\text{So} \quad h = r - y = r\left[1 - \frac{1}{\sqrt{1+\mu^2}}\right], \therefore h = r\left[1 - \frac{1}{\sqrt{1+\mu^2}}\right]$$

where

h = height up to which insect can climb

m = mass of insect

r = radius of the bowl

$\mu =$ coefficient of friction

$$h = r\left[1 - \frac{1}{\sqrt{1+\mu^2}}\right] = 1\left[1 - \frac{1}{\sqrt{1+(\frac{3}{4})^2}}\right] = 1(1-\frac{4}{5}) = \frac{1}{5} = 0.2\ m$$

Q. 14 A screw gauge has 50 division on its circular scale. The circular scale is 4 units ahead of the pitch scale marking, prior to use. Upon one complete rotation of the circular scale, a displacement of 0.5 mm is noticed on the pitch scale. The nature of zero error involved, and the least count of the screw gauge are respectively.

Option 1:
positive, $0.1\mu m$

Option 2:
positive, $0.1mm$

Option 3:
negative, $2\ \mu m$

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Option 4:
positive, $10\ \mu m$

Correct Answer:
positive, $10\ \mu m$

Solution:
A screw gauge has 50 division on its circular scale

i.e N=50

Upon one complete rotation of the circular scale, a displacement of 0.5 mm is noticed on the pitch scale

i.e Pitch of the circular scale P=0.5 mm

So, So least count of screw gauge $\mathrm{LC}=\frac{\mathrm{P}}{\mathrm{N}}=\frac{0.5}{50}\mathrm{mm}=0.01\mathrm{mm}=10\mu\ m$

And The circular scale is 4 units ahead of the pitch scale marking, prior to use.

i.e nature of zero error involved is positive.

Q. 15 A clock has a continously moving second's hand of $0.1m$ length. The average acceleration of the tip of the hand (in m/s^2 is of order of :

Option 1:
10^{-4}

Option 2:
10^{-1}

Option 3:
10^{-3}

Option 4:
10^{-2}

Correct Answer:
10^{-3}

Solution:

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$\mathrm{R} = 0.1\ \mathrm{m}$

$\omega = \frac{2\pi}{\mathrm{T}} = \frac{2\pi}{60} = 0.105\mathrm{rad/sec}$

$a = \omega^2 R$

$= (0.105)^2(0.1)$

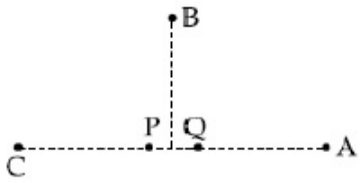
$= 0.0011$

$= 1.1 \times 10^{-3}$

Average acceleration is of the order of 10^{-3}

Q. 16 In the figure below, P and Q are two equally intense coherent sources emitting radiation of wavelength 20 m. The separation b/w P and Q is 5 m and the phase of P is ahead of that of Q by $90°C$. A, B and C are 3 distinct points of observation, each equidistant from the mid-point of PQ

The transition of radiation at A, B and C will be in ratio:-



Option 1:

$2 : 1 : 0$

Option 2:

$4 : 1 : 0$

Option 3:

$0 : 1 : 4$

Option 4:

$0 : 1 : 2$

Correct Answer:

$2 : 1 : 0$

Solution:

$$PA - QA = PQ = 5m = \frac{20}{4} = \frac{\lambda}{4} = \Delta x$$

$$\phi' = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$$

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$Phase\ of\ Q\ is\ greater\ by\ \left(90^\circ = \frac{\pi}{2}\right)\ than\ P.$

$Net\ phase\ difference\ \phi = \pi/2 - \pi/2 = 0$

$$I_{net} = I_0 + I_0 + 2\sqrt{I_0^2}\cos\phi$$
$$= 2I_0 + 2I_0\cos\phi$$
$$= 2I_0(1 + \cos\phi)$$

$$I_A = 4I_0$$

At $\phi = 90^\circ = \pi/2$

$$I_B = 2I_0$$

At C:

$$PC - QC = PQ = -5m = -\frac{20}{4} = -\frac{\lambda}{4} = \Delta x$$

$$\phi' = -\pi/2$$

$$I_A : I_B : I_C = 4I_0 : 2I_0 : 0 = 2 : 1 : 0$$

Q. 17 A satelete is in an all optical orbit around planet P. It is observed that the velocity of the satelite when it is farthest from the planet is 6 times less than that when it is closest to the planet. The ratio of distances b/w the satelite and the planet at closest and farthest point is :-

Option 1:
$3 : 4$

Option 2:
$1 : 6$

Option 3:
$1 : 2$

Option 4:
$1 : 3$

Correct Answer:
$1 : 6$

Solution:
As the angular momentum remains conserved. So,

$$L_{Nearest} = L_{Farthest}$$

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$$m \times v_n \times r_n = m \times v_f \times r_f$$

Also,

$$V_f = \frac{V_n}{6}$$

As,

$$V_n \times r_n = V_f \times r_f$$

$$V_n \times r_n = \frac{V_n}{6} \times r_f$$

So,

$$\frac{r_n}{r_f} = \frac{1}{6}$$

Q. 18 If the $P.E.$ b/w two molecules is given by

$$U = \frac{-A}{r^6} + \frac{B}{r^{12}}$$

then at equilibrium, seperation b/w the molecules and the $P.E.$ are :-

Option 1:

$$\left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{2B}$$

Option 2:

$$\left(\frac{B}{A}\right)^{\frac{1}{6}}, 0$$

Option 3:

$$\left(\frac{B}{2A}\right)^{\frac{1}{6}}, -\frac{A^2}{2B}$$

Option 4:

$$\left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{4B}$$

Correct Answer:

$$\left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{4B}$$

Solution:

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As,

$$U = \frac{-A}{r^6} + \frac{B}{r^{12}}$$

$$\text{For } eq^m : -\frac{dU}{dr} = 0$$

$$\Rightarrow \frac{dU}{dr} = \frac{-(-6)A}{r^7} + \frac{(-) \times B \times 12}{r^{13}} = 0$$

$$\Rightarrow \frac{6A}{r^7} = \frac{12B}{r^{13}}$$

$$\Rightarrow \quad r = \left(\frac{2B}{A}\right)^{1/6}$$

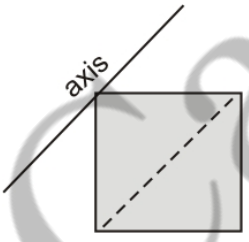
At this r :-

$$U = \frac{-A}{(\frac{2B}{A})^{\frac{1}{6}\times 6}} + \frac{B}{(\frac{2B}{A})^{\frac{1}{6}\times 12}}$$

$$U = \frac{-A^2}{2B} + \frac{A^2}{4B}$$

$$U = \frac{-A^2}{4B}$$

Q. 19 Four point masses each of mass $'m'$, are placed at the corners of a square of side $'l'$. The square is rotating with angular frequency $'\omega'$, about an axis passing through one of the corner of the square and parallel to its diagonal , as shown in figure. The angular momentum of the square about this axis is :-



Option 1:

ml^2w

Option 2:

$2ml^2w$

Option 3:

$3ml^2w$

Option 4:

$4ml^2w$

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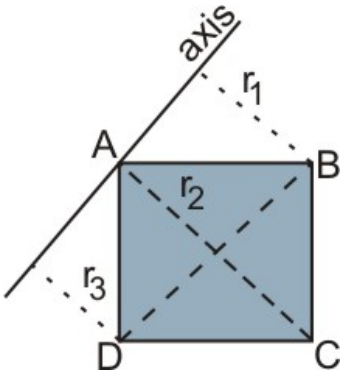
Correct Answer:

$3ml^2w$

Solution:



Since, BD is parallel to axis.

$r_1 = r_3$ Half of the diagonal square $\frac{l}{\sqrt{2}}$

Also,

r_2 = Equal to the diagonal of square = $\sqrt{2}l$

$$So,$$
$$(MOI) = \sum M_i r_i^2 \Rightarrow mr_1^2 + mr_2^2 + mr_3^2$$

Because all the masses are same = m

By putting the value of r_1, r_2 , r_3

We get

$$I = 3ml^2$$

So, Angular momentum = $I\omega$

Angular momentum = $3ml^2\omega$

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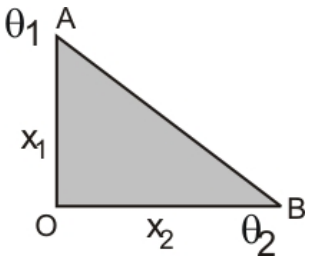
Q. 20 Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point 'O' is perpendicular to the hypotenuse, the $\frac{Q_1}{Q_2}$ is proportional to



Option 1:

$\frac{x_2}{x_1}$

Option 2:

$\frac{x_1^3}{x_2^3}$

Option 3:

$\frac{x_1}{x_2}$

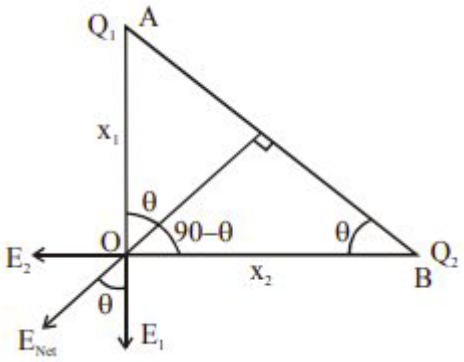
Option 4:

$\frac{x_2^2}{x_1^2}$

Correct Answer:

$\frac{x_1}{x_2}$

Solution:



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$E_2 =$ electric field due to Q_2

$= \frac{kQ_2}{x_2^2}$

$E_1 = \frac{kQ_1}{x_1^2}$

From diagram

$\tan\theta = \frac{E_2}{E_1} = \frac{x_1}{x_2}$

$\frac{kQ_2}{x_2^2 \times \frac{kQ_1}{x_1^2}} = \frac{x_1}{x_2}$

$\frac{Q_2 x_1^2}{Q_1 x_2^2} = \frac{x_1}{x_2}$

$\frac{Q_2}{Q_1} = \frac{x_2}{x_1}$

$\frac{Q_1}{Q_2} = \frac{x_1}{x_2}$

Q. 21

Suppose that the intensity of laser is $\left(\frac{315}{\pi}\right)\frac{W}{m^2}$. The rms electric field, in units of $\frac{V}{m}$ associated with this source is close to the nearest integer is :-_____. Take

$\epsilon_0 = 8.85 \times 10^{-12} C^2 N m^{-2}, C = 3 \times 10^2 m/s$

Correct Answer:

194

Solution:

$\because I = \frac{1}{2}\varepsilon_0 E_0^2 c$

$\Rightarrow E_0 = \sqrt{\frac{21}{\varepsilon_0 c}}$

$\therefore E_{rms} = \frac{E_0}{\sqrt{2}} = \sqrt{\frac{1}{\varepsilon_0 c}}$

$= \sqrt{\frac{315}{\pi} \times \frac{1}{8.86 \times 10^{-12} \times 3 \times 10^8}}$

$= 194$

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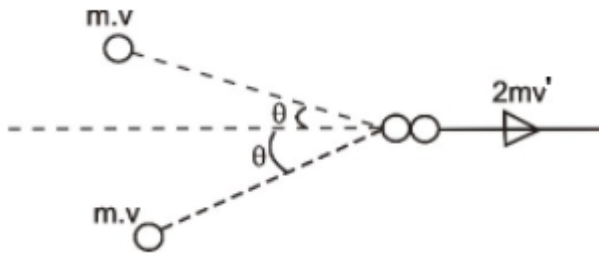
Q. 22 Two bodies of same mass are moving with the same speed , but in different direction. in a plane. They have a completely inelastic collision and move together thereafter with a final speed which is half of their initial speed. Thw angle b/w the intial velocities of the two bodies (in degree) is _____.

Correct Answer:
120

Solution:



Let us apply the conservation of momentum in x - direction. So,

$$mv\cos\theta + mv\cos\theta = 2mv'$$

Also from the question -

$$v' = v/2$$

So,

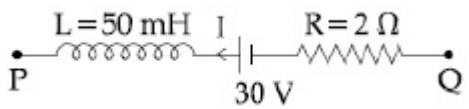
$$2mv\cos\theta = 2m\frac{v}{2}$$

So, the angle between initial velocities of the masses is = $2\theta = 120^o$

Q. 23 A part of the complete circuit is shown in the figure. At some instant, the value of current $I = 1A$ and it is decreasing at a rate of

$10^2 A/s$. The value of the potential difference $V_P - V_Q$ (in volts) at that instant, is

_____



Correct Answer:
33

Solution:

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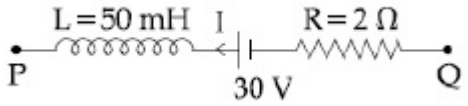

$\frac{Ldi}{dt} = 5$
$V_P - 5 - 30 + 2 \times 1 = VQ$
$V_P - V_Q = 33 \text{ volt}$

Q. 24 Initially, a gas of diatomic molecules is contained in a cylinder of volume V, at a pressure P_1 and temperature 250k. Assume that 25% of the molecules gets dissociated causing a change in no.of moles. The pressure of the resulting gas at temperature 2000 k , when contained in the volume $2V_1$ is given by p_2 The ratio of $\frac{p_2}{p_1} =$ _____

Correct Answer:
5

Solution:

$\mathrm{PV} = \mathrm{nRT}$
$\mathrm{P_1\ V_1} = \mathrm{nR250}$
$\mathrm{P_2\ (2\ V_1)} = \frac{5\mathrm{n}}{4}\mathrm{R} \times 2000$
Divide
$\frac{P_1}{2P_2} = \frac{4 \times 250}{5 \times 2000}$
$\frac{P_1}{P_2} = \frac{1}{5}$
$\frac{P_2}{P_1} = 5$

Q. 25 The density of a solid metal sphere is determined by measuring its mass and its diameter. The maximum error in the density of the sphere is $\left(\frac{x}{100}\right)\%$. If the relative errors in measuring the mass and diameter are 6% and 1.5 % respectively. The value of 'x' is

Correct Answer:
1050

Solution:
As we know-

$\rho = \frac{mass}{Volume} = \frac{m}{\frac{4}{3}\pi R^3} = \frac{m}{\frac{\pi}{6}D^3}$

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$$\Rightarrow \frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 3\frac{\Delta D}{D}$$

$$\Rightarrow \frac{\Delta\rho}{\rho} = \frac{6}{100} + \frac{1.5 \times 3}{100}$$

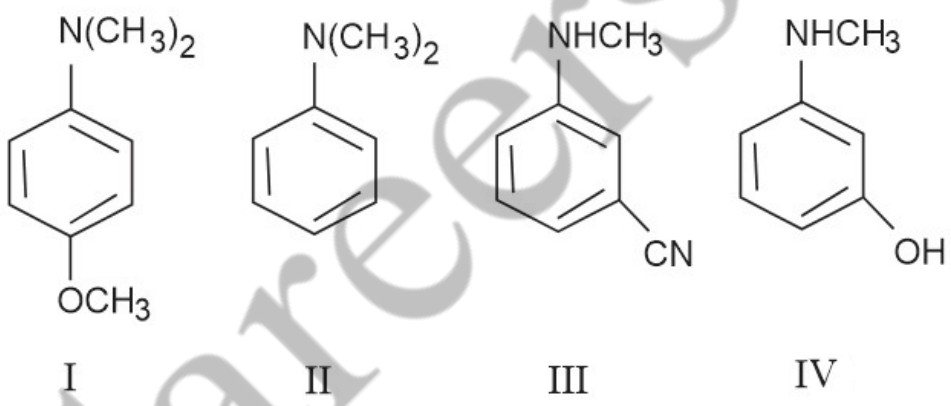
$$\Rightarrow \left(\frac{\Delta\rho}{\rho} \times 100\right)\% = 10.5\%$$

So, $\left(\frac{x}{100}\right)\% = \left(\frac{\Delta\rho}{\rho} \times 100\right)\% = 10.5\%$

$\therefore x = 1050$

Chemistry

Q. 1 The increasing order of $\mathrm{pK_b}$ values of the following compounds is :



Option 1:
II<IV<III<I

Option 2:
II<I<III<IV

Option 3:
I<II<III<IV

Option 4:
I<II<IV<III

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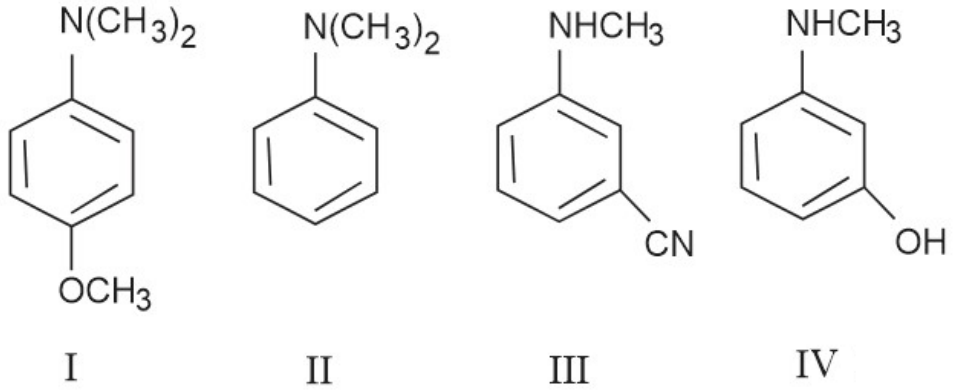
Correct Answer:
I<II<IV<III

Solution:



I II III IV

III contains -M and -I group, hence it is the least basic.

IV contains -I group

I contains +M group and is more basic than II.

Thus order of strength: I > II > IV > III.

Order of $\mathrm{pK_b}$: I < II < IV < III.

Therefore, **Option(4) is correct.**

Q. 2 The presence of soluble fluoride ion up to 1ppm concentration in drinking water is

Option 1:
Harmful to skin

Option 2:
Harmful to bone

Option 3:
Harmful to teeth

Option 4:
Safe for teeth

Correct Answer:
Safe for teeth

Solution:

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Fluoride ion concentration upto 1 ppm is safe for teeth.

Therefore, **Option(4) is correct.**

Q. 3 Kraft temperature is the temperature

Option 1:
Above which the aqueous solution of detergents starts boiling

Option 2:
Below which the formation of micelles takes place.

Option 3:
Above which the formation of micelles takes place.

Option 4:
Below which the aqueous solution of detergents starts freezing.

Correct Answer:
Above which the formation of micelles takes place.

Solution:
Micelle formation takes place above the kraft's temperature.

Therefore, **Option(3) is correct.**

Q. 4 For the reaction:

$$\mathrm{Fe_2N(s)} + \frac{3}{2}\mathrm{H_2(g)} \rightleftharpoons 2\mathrm{Fe(s)} + \mathrm{NH_3(g)}$$

Option 1:
$K_c = K_p(RT)^{-\frac{1}{2}}$

Option 2:
$K_c = K_p(RT)^{\frac{1}{2}}$

Option 3:
$K_c = K_p(RT)$

Option 4:
$K_c = K_p(RT)^{\frac{3}{2}}$

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Correct Answer:

$K_c = K_p(RT)^{\frac{1}{2}}$

Solution:

The reaction occurs as follows:

$$Fe_2N(s) + \frac{3}{2}H_2(g) \rightleftharpoons 2Fe(s) + NH_3(g)$$

$$\Delta n = -\frac{1}{2}$$

We know that:

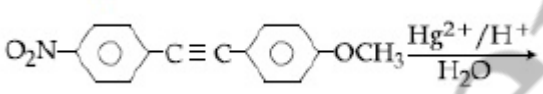
$$k_p = k_c(RT)^{\Delta n}$$

$$\text{Thus, } k_p = k_c(RT)^{-\frac{1}{2}}$$

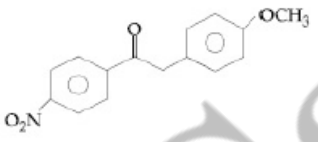
$$k_c = k_p(RT)^{\frac{1}{2}}$$

Therefore, **Option(2) is correct.**

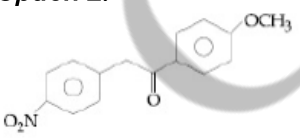
Q. 5 The major product obtained from the following reaction is :



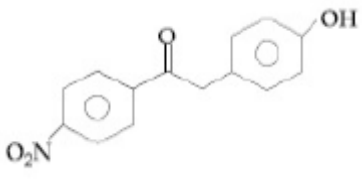
Option 1:



Option 2:



Option 3:



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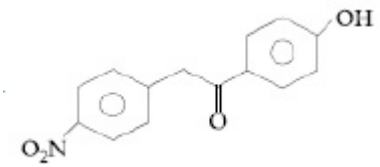
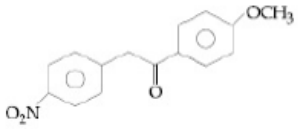
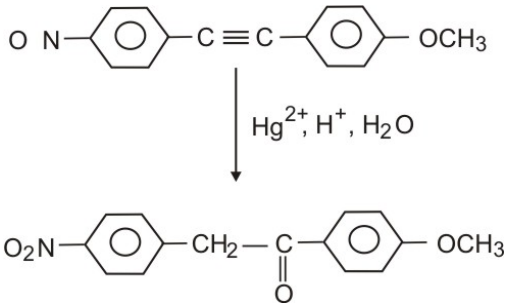
Option 4:



Correct Answer:



Solution:
The reaction occurs as follows:



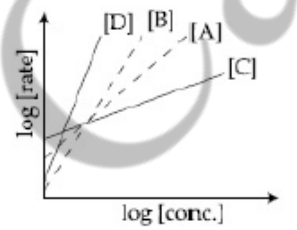
Addition follows Markovnikov's rule.

Therefore, **Option(2) is correct.**

Q. 6 Consider the following reaction

$$A \rightarrow P1 : B \rightarrow P2 \rightarrow C \rightarrow P3, D \rightarrow P4$$

The order of above reaction one a, b, c and d respectively. The following graph is obtained when log[rate] vs log[conc.] are plotted.



Among the following the correct sequence of the order of reaction are

Option 1:

$d > b > a > c$

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Option 2:
$d > a > b > c$

Option 3:
$c > a > b > d$

Option 4:
$a > b > c > d$

Correct Answer:
$d > b > a > c$

Solution:
Rate is given as:

$$\text{Rate} = \text{k}[\text{Reactant}]\text{n}$$

Thus, $\log\ \text{rate} = \text{n}[\text{Reactant}] + \text{C}$

Thus, slope of log(rate) vs log[reactant] gives the order of the reaction.

Hence, from the given graph,

Order: d > b > a > c

Therefore, **Option(1) is correct.**

Q. 7 **Assertion (A):** Ethane polymerized in the presence of Ziegler Natta catalyst at high temperature and pressure is used to make bucket and dustbin.

Reason (R): High-density polymers are closely packed and are chemically inert.

Choose the correct answer:

Option 1:
(A) and (R) both are wrong

Option 2:
(A) is correct but (R) is wrong

Option 3:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Option 4:
Both (A) and (R) are correct and (R) is the correct explanation of (A)

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Correct Answer:
Both (A) and (R) are correct and (R) is the correct explanation of (A)

Solution:
The polymerization of ethene with Ziegler Natta catalyst under high temperature and pressure produces high-density polyethylene which has density due to close packing and is chemically inert. It is used to prepare buckets, dustbin bottles, pipes, etc.

Hence, (A) and (R) are both correct and (R) is the correct explanation of (A).

Therefore, **Option(4) is correct.**

Q. 8 Which of the following compounds show geometrical isomerisation?

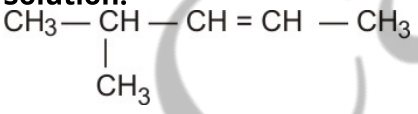
Option 1:
4-Methylpent-1-ene

Option 2:
2-Methylpent-1-ene

Option 3:
2-Methylpent-2-ene

Option 4:
4-Methylpent-2-ene

Correct Answer:
4-Methylpent-2-ene

Solution:


4-Methylpent-2-ene shows geometrical isomerism.

Therefore, **Option(4) is correct.**

Q. 9 The correct statement with respect to dinitrogen:

Option 1:
N_2 is paramagnetic in nature.

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Option 2:
It can combine with dioxygen at 25^oC

Option 3:
It can be used as an inert diluent for reactive chemicals.

Option 4:
Liquid dinitrogen is not used in cryosurgery.

Correct Answer:
It can be used as an inert diluent for reactive chemicals.

Solution:
Nitrogen is used as an inert diluent for reactive chemicals.

Therefore, **Option(3) is correct.**

Q. 10 The set that contains the atomic numbers of only transition elements.

Option 1:
$21, 25, 42, 72$

Option 2:
$21, 32, 53, 64$

Option 3:
$9, 17, 34, 38$

Option 4:
$37, 42, 50, 64$

Correct Answer:
$21, 25, 42, 72$

Solution:
The set containing transition elements is:

$21, 25, 42, 72$

Therefore, **Option(1) is correct.**

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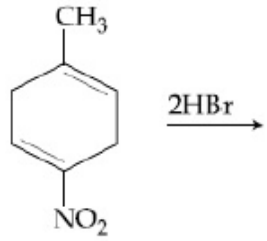
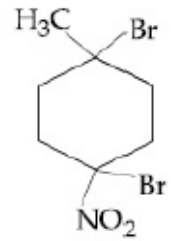
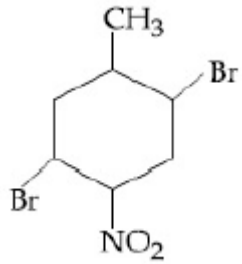
Q. 11 The major product of the following reaction is :



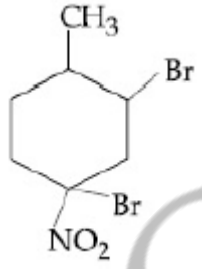
Option 1:



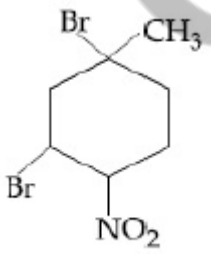
Option 2:



Option 3:



Option 4:



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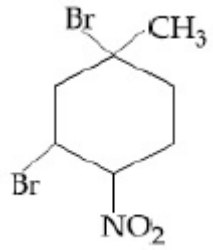
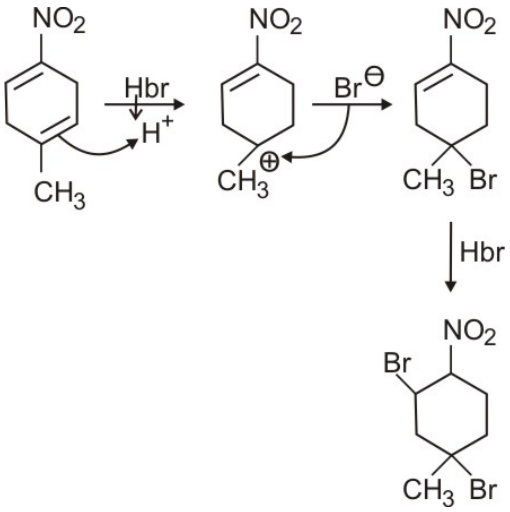
Correct Answer:



Solution:
The reaction occurs as follows:



Therefore, Option(4) is correct.

Q. 12 The incorrect statement is:

Option 1:
Brass is an alloy of copper and nickel.

Option 2:
Dust iron is used to manufacture brought iron.

Option 3:
German silver is an alloy of zinc, copper and nickel.

Option 4:
Bronze is an alloy of copper and tin

Correct Answer:
Brass is an alloy of copper and nickel.

Solution:
Brass is an alloy of copper and zinc, but not of copper and nickel. Thus, other statements are correct.

Therefore, **Option(1) is correct.**

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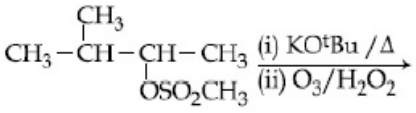
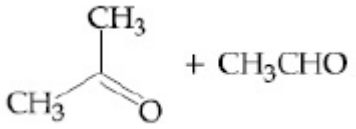
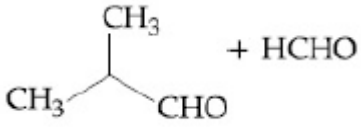
Q. 13 The major products of the following reaction are :



Option 1:

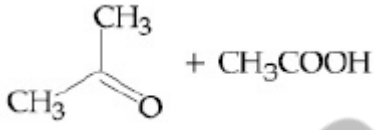


Option 2:

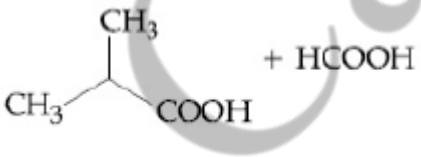


Option 3:

Option 4:

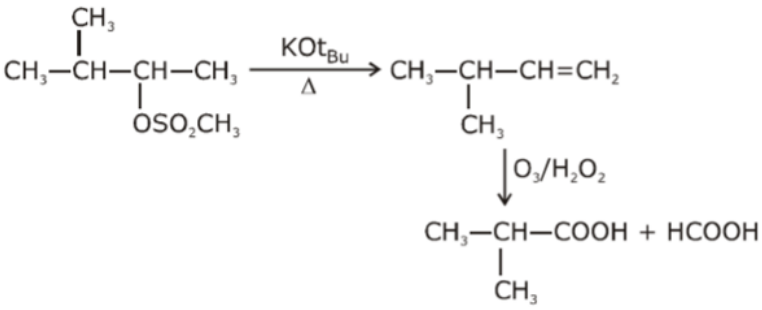


Correct Answer:



Solution:

The reaction occurs as follows:



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Therefore, **Option(3) is correct.**

Q. 14 The species that has spin only magnetic moment $5.9\ BM$ (Td-Tetraherdral)

Option 1:
$[NiCl_4]^{2-}\ (Td)$

Option 2:
$[MnBr_4]^{2-}\ (Td)$

Option 3:
$Ni(CO)_4\,(Td)$

Option 4:
$\left[CN\,[Ni(CN)_4]^{2-}\right]$ Square planner

Correct Answer:
$[MnBr_4]^{2-}\ (Td)$

Solution:
We have :

$$\sqrt{\mathrm{n(n+2)}}\ =\ 5.9\mathrm{BM}$$

Thus, n = 5(unpaired electrons)

In $[Mn(Br)_4]^{2-}, Mn^{2+}$ has 5 unpaired electrons due to weak (Br^-) ligand.

Therefore, **Option(2) is correct.**

Q. 15 The variation of the equilibrium constant with temperature is given below

Temp	eq. constant
$T_1 = 25°C$	$K_1 = 10$
$T_2 = 100°C$	$K_2 = 100$

The value of $\Delta H_o, \Delta G_o$ at T_1 and ΔG_o at T_2 in kJ/mol respectively are close to:

Option 1:
28.4, -7.14, -5.71

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Option 2:
0.64, -7.14, -5.71

Option 3:
0.64, -5.71, -14.09

Option 4:
28.4, -5.71. -14.29

Correct Answer:
28.4, -5.71. -14.29

Solution:

$$(\Delta G^0)_1 = -RT_1 \ln k$$

$$= -8.314 \times 298 \times \ln(10)$$

$$= -5.7 kJ/mol$$

$$(\Delta G^o)_2 = -RT_2 \ln K$$

$$= (-8 \cdot 314) \times 373 \times \ln(100)$$

$$= -14.28 kJ/mol$$

Now, from van't Hoff's Equation -

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{\Delta H^0}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\Rightarrow \ln 10 = \frac{\Delta H^0}{R}\left[\frac{1}{298} - \frac{1}{373}\right]$$

$$\Rightarrow \quad \Delta H^0 = \frac{(\ln 10) \times 298 \times 373 \times R}{75 \times 1000}$$

$$= 28.4 kJ/mol$$

Therefore, **option (4) is correct.**

Q. 16 Arrange the following solution in the decreasing order of pOH

A) $0.01M\ HCl$

B) $0.01M\ NaOH$

C) $0.01M\ CH_3COONa$

D) $0.01M\ NaCl$

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Option 1:
$B > C > D > A$

Option 2:
$B > D > C > A$

Option 3:
$A > D > C > B$

Option 4:
$A > C > D > B$

Correct Answer:
$A > D > C > B$

Solution:
$pOH + pH = 14$

If pH is low(means more acidic) then pOH is high.

Thus, acidic order: $HCl > NaCl > CH_3COONa > NaOH.$

Then the order of pOH will be:

$A > D > C > B.$

Therefore, **Option(3) is correct.**

Q. 17 The lanthanide does not show +4 oxidation state is:

Option 1:
Tb

Option 2:
Ce

Option 3:
Eu

Option 4:
Dy

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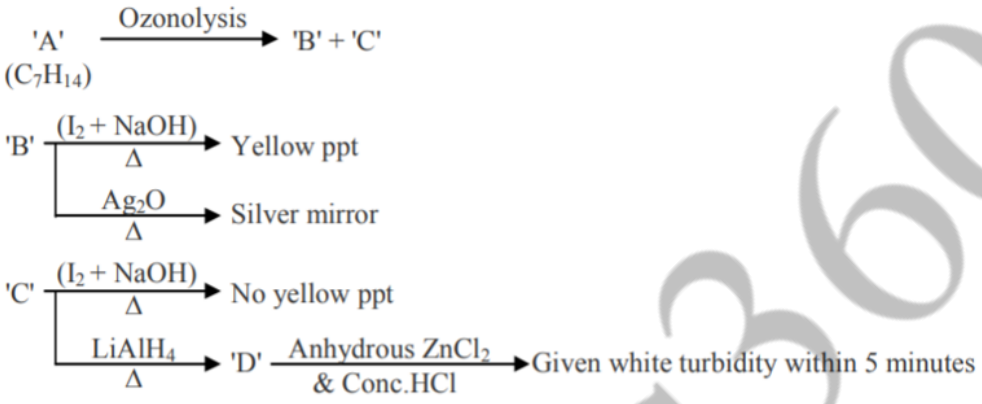
Correct Answer:
Eu

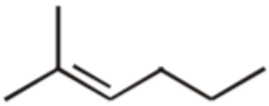
Solution:
Eu does not show +4 oxidation state.

Therefore, **Option(3) is correct.**

Q. 18 Consider the following reactions :

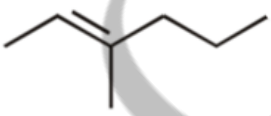


Option 1:

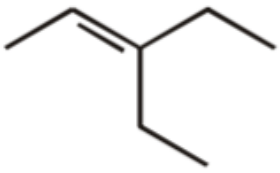


Option 2:

Option 3:



Option 4:



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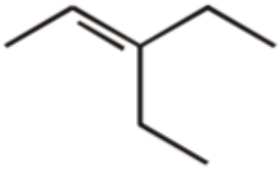
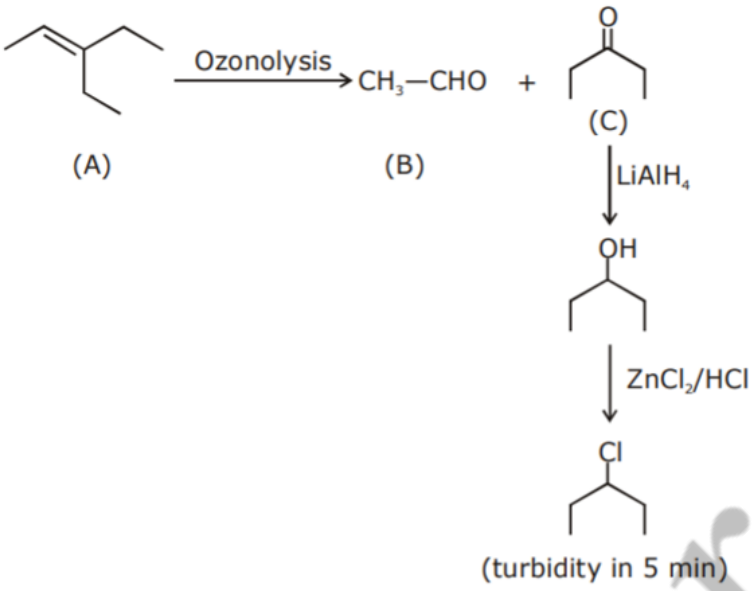
Correct Answer:



Solution:
The reactions occur as follows:



Therefore, Option(4) is correct.

Q. 19 Among the sulphate of alkaline earth metals the solubilities of $BeSO_4$ and $MgSO_4$ in water, respectively, are -

Option 1:
high and poor

Option 2:
poor and poor

Option 3:
high and high

Option 4:
poor and high

Correct Answer:
high and high

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Solution:
$BeSO_4$ and $MgSO_4$ are readily soluble in water. Thus, the solubility of both these sulphates is high in water.

Therefore, **Option(3) is correct.**

Q. 20 Solution of two components containing n_1 moles of 1st component and n_2 moles of 2nd component is prepared, M1 and M2 are molecular weight of component 1 and 2 respectively. If d is the density of the solution in g mL^{-1}, C_2 is the molarity and x_2 is the mole fraction of 2nd component, C_2 can be expressed as:

Option 1:

$$c_2 = \frac{dx_2}{M_2 + x_2(M_2 - M_1)}$$

Option 2:

$$c_2 = \frac{dx_1}{M_2 + x_2(M_2 - M_1)}$$

Option 3:

$$c_2 = \frac{1000dx_2}{M_1 + x_2(M_2 - M_1)}$$

Option 4:

$$c_2 = \frac{1000x_2}{M_1 + x_2(M_2 - M_1)}$$

Correct Answer:

$$c_2 = \frac{1000dx_2}{M_1 + x_2(M_2 - M_1)}$$

Solution:

$$\text{Total mass of Component 1} = n_1M_1$$
$$\text{Total mass of Component 2} = n_2M_2$$

$$\therefore \text{ Total mass of Solution } = n_1M_1 + n_2M_2$$

$$\therefore \text{ Total mass of Solution } = \frac{n_1M_1 + n_2M_2}{d} \text{ ml } = \frac{n_1M_1 + n_2M_2}{1000d} \text{ litre}$$

$$\therefore C_2 = \frac{n_2}{\frac{n_1M_1 + n_2M_2}{1000d}}$$

$$C_2 = \frac{1000dn_2}{n_1M_1 + n_2M_2}$$

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dividing by $(n_1 + n_2)$, we get :

$$C_2 = \frac{1000 d x_2}{x_1 M_1 + x_2 M_2}$$

$$C_2 = \frac{1000 d x_2}{M_1 + (M_2 - M_1) x_2}$$

Therefore, **Option(3) is correct.**

Q. 21 The elevation of boiling point of 0.1 m aq $CrCl_3.xNH_3$ solution is two times that of 0.05 aq. $CaCl_2$ solution. The value of x ____

(Assume 100% ionization of complex and $CaCl_2$, coordination number of Cr is 6 and that all of NH_3 molecules are present inside the coordination sphere.)

Correct Answer:
5

Solution:
ΔT_b of $CrCl_3.xNH_3 = 2(\Delta T_b$ of $CaCl_2)$

$i \times 0.1 = 2 \times 3 \times 0.05$

Thus, i = 3

Therefore, the complex is dissociated into 3 ions.

Hence, complex is $[Cr(NH_3)_5Cl]Cl_2$

Thus, the value is 5.

Q. 22 Potassium chlorate is prepared by the electrolysis of KCl in basic solution
$Cl^- + 6OH^- \rightarrow ClO_3^- + 3H_2O + 6e^-$

If only 60% of current utilized in the reaction, the time (hr) required to produce 10 g of $KClO_3$ using a current of 2A is ___

$(F = 96500 C mol^-,\ KClO_3 = 122 g \cdot mol^-)$

Correct Answer:
11

Solution:
The reaction occurs as follows:

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$$Cl^- + 6OH^- \rightarrow ClO_3^- + 3H_2O + 6e^-$$

Current efficiency = 60%

Now, moles of $KClO_3 = \frac{10}{122.5}$

Thus, moles of $e^- = \frac{60}{122.5}$ (assuming 100% efficiency)

Thus, actual moles of $e^- = \frac{100}{122.5}$

$$\therefore \frac{2 \text{ x t}}{96500} = \frac{100}{122.5}$$

Thus, t = 39387.76 sec. = 10.94 hours.

Thus, time required is approximately 11 hours.

Q. 23 A spherical balloon of radius 3 cm contain He gas has a pressure of 48×10^{-3} bar. At the same temp. the pressure of spherical balloon of radius 12 cm containing the same amount of gas will be ____ $\times 10^{-6}$ bar.

Correct Answer:
750

Solution:
We have:

$r_1 = 3cm$

$P_1 = 48 \times 10^{-3}$ bar

Thus, according to Boyle's law, we have:

$P_1V_1 = P_2V_2$

On putting the values we get:

$$(48 \text{ x } 10^{-3}\text{bar})(4/3\pi(3)^3) = \text{P}_2(4/3\pi(12)^3)$$

$$\text{P}_2 = \frac{48 \text{ x } 10^{-3} \text{ x } (3)^3}{(12)^3}$$

$$\text{P}_2 = 750 \text{ x } 10^{-6} \text{ bar}$$

Thus, 750 is the answer.

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Q. 24 In an estimation of bromine by Carius method, 1.6 g of an organic compound gave 1.88 g of AgBr. The mass % of Bromine in the compound is:

Correct Answer:
50

Solution:
Moles of $AgBr = \frac{1.88}{188}$

Thus, the mole of AgBr = 0.01

Thus, moles of bromine in organic compound = 0.8 g

Now, % of bromine is given as:

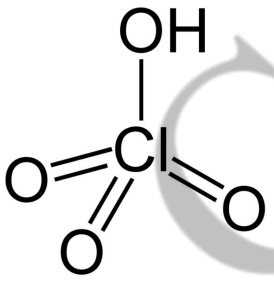
$$\% \text{ of bromine } = \frac{0.8}{1.6} \text{ x } 100 \ = \ 50\%$$

Thus, the correct answer is 50.

Q. 25 The number of $Cl = O$ bond in perchloric acid is _____

Correct Answer:
3

Solution:
$Cl = O$ bond in perchloric acid:



Thus, number of $Cl = O$ bond is 3.

Thus, the correct answer is 3.

Maths

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Q. 1 Two families with 3 members each and one family with 4 members are to be seated in a row. In how many ways can they be seated so that the same family member are not separated?

Option 1:
$2!3!4!$

Option 2:
$(3!)^2\,(4!)$

Option 3:
$(3!)\,(4!)^3$

Option 4:
$(3!)^3\,(4!)$

Correct Answer:
$(3!)^3\,(4!)$

Solution:
Family A has 3 members they can be arranged in $3!$ ways

Family B has 3 members they can be arranged in $3!$ ways

Family C has 4 members they can be arranged in $4!$ ways

Now A, B, C can be arranged in $3!$ ways

The total arrangement is $(3!)^3\,(4!)$

Correct Answer: Option D

Q. 2 The shortest distance b/w the line $\frac{x-1}{0}=\frac{y+1}{-1}=\frac{z}{1}$ and $x+y+z=0, 2x-y+z+3=0$ is

Option 1:
$\frac{1}{\sqrt{2}}$

Option 2:
$\frac{1}{2}$

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Option 3:

$\frac{1}{\sqrt{3}}$

Option 4:

1

Correct Answer:

$\frac{1}{\sqrt{3}}$

Solution:

The shortest distance between the lines

$\frac{x-1}{0}=\frac{y+1}{-1}=\frac{z}{1}$ and $x+y+z+1=0=2x-y+z+3=0$

Line of intersection of planes $\mathrm{x+y+z+1=0=2x-y+z+3}$ is

Eliminating y gives $3x+2z+4=0$

$\Rightarrow \mathrm{x}=\frac{-2\mathrm{z}-4}{3}$

Substituting above x in $\mathrm{x+y+z+1=0}$ gives $3\mathrm{y}+\mathrm{z}-1=0$

$\Rightarrow \mathrm{z}=-3\mathrm{y}+1-(2)$

From (1) and (2), line equation is $\frac{3x+4}{-2}=-3y+1=z$

$\Rightarrow \frac{\mathrm{x}-(-4/3)}{-2/3}=\frac{\mathrm{y}-(1/3)}{-1/3}=\frac{\mathrm{z}}{1}$

Given line is $\frac{x-1}{0}=\frac{y+1}{-1}=\frac{z}{1}$

Shortest distance between above two skew lines is $\frac{(\vec{\mathrm{b}}-\vec{\mathrm{a}})\cdot(\vec{\mathrm{c}}\times\vec{\mathrm{d}})}{|\vec{\mathrm{c}}\times\vec{\mathrm{d}}|}$

$\vec{\mathrm{a}}=(1,-1,0)$

$\vec{\mathrm{b}}=\left(-\frac{4}{3},\frac{1}{3},0\right)$

$\vec{\mathrm{c}}=(0,-1,1)$

$\vec{\mathrm{d}}=\left(-\frac{2}{3},-\frac{1}{3},1\right)$

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$$\therefore \frac{(\vec{b}-\vec{a})\cdot(\vec{c}\times\vec{d})}{|\vec{c}\times\vec{d}|} = \left\| \begin{array}{ccc} 7/3 & -4/3 & 0 \\ 0 & -1 & 1 \\ -2/3 & -1/3 & 1 \end{array} \right\| \Big/ \left\| \begin{array}{ccc} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & 1 \\ -2/3 & -1/3 & 1 \end{array} \right\| = \frac{1}{\sqrt{3}}$$

Correct Answer: Option C

Q. 3 If {P} denotes the fractional part of number P, then $\left\{\frac{3^{200}}{8}\right\}$ is equal to

Option 1:
$\frac{5}{8}$

Option 2:
$\frac{3}{8}$

Option 3:
$\frac{1}{8}$

Option 4:
$\frac{7}{8}$

Correct Answer:
$\frac{1}{8}$

Solution:
$\left\{\frac{3^{200}}{8}\right\}$

$= \left\{\frac{9^{100}}{8}\right\}$

$= \left\{\frac{(8+1)^{100}}{8}\right\}$

Now, let us first find the remainder when $(8+1)^{100}$ is divided by 8

$$(8+1)^{100} = 8^{100} + {}^{100}C_1.8^{99} + \ldots\ldots\ldots\ldots + {}^{100}C_{100}.1$$
$$= 8k+1$$

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Hence remainder is 1 and fractional part is $\frac{1}{8}$

Correct Answer: Option C

Q. 4 If $I_1 = \int_0^1 \left(1 - x^{50}\right)^{100} dx$ and $I_2 = \int_0^1 \left(1 - x^{50}\right)^{101} dx$ such that $I_2 = \alpha I_1$ then $\alpha = ?$

Option 1:
$\frac{5049}{5050}$

Option 2:
$\frac{5050}{5049}$

Option 3:
$\frac{5050}{5051}$

Option 4:
$\frac{5051}{5050}$

Correct Answer:
$\frac{5050}{5051}$

Solution:

$$I_n = \int_0^1 (1 - x^m)^n \, dx$$

$$I_{n+1} = \int_0^1 (1 - x^m)^{n+1} \, dx$$

$$I_{n+1} = \int_0^1 (1 - x^m)(1 - x^m)^n \, dx$$

$$I_{n+1} = I_n - \int_0^1 x^m (1 - x^m)^n \, dx$$

$$I_{n+1} = I_n + \left[\frac{x(1 - x^m)^{n+1}}{m(n+1)}\right]_0^1 - \int_0^1 \frac{(1 - x^m)^{n+1}}{m(n+1)} dn$$

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$$I_{n+1} = I_n - \frac{1}{m(n+1)} I_{n+1}$$

$$m(n+1)I_{n+1} + I_{n+1} = I_n - m(n+1)$$

$$\frac{5050}{5051}$$

Correct Answer: Option C

Q. 5 Let m and n be respectively the min and max value of $\begin{vmatrix} \cos^2 x & 1+\sin^2 x & \sin 2x \\ 1+\cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1+\sin 2x \end{vmatrix}$ then ordered pair (m,n) = ?

Option 1:
(-4,-1)

Option 2:
(1,3)

Option 3:
(-3,3)

Option 4:
(-3,-1)

Correct Answer:
(-3,-1)

Solution:

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$\begin{vmatrix} \cos^2 x & 1+\sin^2 x & \sin 2x \\ 1 & -1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$$

$$\cos^2 x(-1) - \left(1+\sin^2 x\right)[1] + \sin 2x(-1)$$

$$-\cos^2 x - 1 - \sin^2 x - \sin 2x$$

$$-\sin 2x - 2$$

Minimum = -3 and Maximum = -1

Correct Answer: Option D

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Q. 6 Let L_1 be the tangent to the parabola $y^2 = 4(x+1)$ and L_2 be the tangent to the parabola $y^2 = 8(x+2)$

such that L_1 and L_2 intersect at 90^o. Then at L_1 and L_2 meet on the st. line.

Option 1:
2x+1=0

Option 2:
x+2=0

Option 3:
x+3=0

Option 4:
x+2y=0

Correct Answer:
x+3=0

Solution:
Equation of 1st parabola : $y^2 = 4(x+1)$

Equation of tangent : $y = m_1(x+1) + \frac{1}{m_1}$

Equation of 2nd parabola : $y^2 = 8(x+2)$

Equation of tangent : $y = m_2(x+2) + \frac{2}{m_2}$

Now both tangents intersect at 90 degree

$So,\ m_1m_2 = -1$

$$m_1(x+1) + \frac{1}{m_1} = m_2(x+2) + \frac{2}{m_2}$$
$$x(m_1 - m_2) + m_1 - 2m_2 = \frac{2}{m_2} - \frac{1}{m_1}$$
$$x(m_1 - m_2) + m_1 - 2m_2 = \frac{2m_1 - m_2}{m_1m_2}$$
$$x(m_1 - m_2) + m_1 - 2m_2 = \frac{2m_1 - m_2}{-1}$$
$$x(m_1 - m_2) = 3m_2 - 3m_1$$
$$x + 3 = 0$$

Correct Answer: Option C

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Q. 7 If $f(x+y)=f(x)f(y)$ and $\sum_{x=1}^{\infty} f(x)=2, \quad x,y \in N$. Then the value of $\frac{f(4)}{f(2)}$ is ....

Option 1:
$\frac{2}{3}$

Option 2:
$\frac{1}{9}$

Option 3:
$\frac{1}{3}$

Option 4:
$\frac{4}{9}$

Correct Answer:
$\frac{4}{9}$

Solution:

$f(x+y)=f(x)\cdot f(y)$

$f(2)=f(1)^2$

$f(3)=f(1)f(2)=f(1)^3$

$\sum_{x=1}^{\infty} f(x)=f(1)+f(1)^2+\ldots=2$

$\frac{f(1)}{1-f(1)}=2$

$f(1)=\frac{2}{3}$

$\frac{f(4)}{f(2)}=\frac{4}{9}$

Correct answer: Option D

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Q. 8 The value of $\lambda \ \ and \ \ \mu$ for which the system of linear eq $x+y+z=2, x+2y+3z=5, x+3y+\lambda z=\mu$

has infinitely many solutions are respectively ..

Option 1:
5 & 8

Option 2:
4 & 9

Option 3:
6 & 8

Option 4:
5 & 2

Correct Answer:
5 & 8

Solution:
$x+y+z=2$
$x+2y+3z=5$
$x+3y+\lambda z=\mu$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 5$$

and

$$\begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \\ 1 & 3 & \mu \end{vmatrix} = 0 \Rightarrow \mu = 8$$

Correct Answer : Option A

Q. 9 The negation of $p \vee (\sim p \wedge q)$

Option 1:
$\sim p \wedge \sim q$

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Option 2:
$p \wedge \sim q$

Option 3:
$\sim p \vee q$

Option 4:
$\sim p \vee \sim q$

Correct Answer:
$\sim p \wedge \sim q$

Solution:

$\mathbf{p}$	$\mathbf{q}$	$\neg(\mathbf{p} \vee (\neg\mathbf{p} \wedge \mathbf{q}))$
F	F	$\mathbf{T}$
F	T	F
T	F	F
T	T	F

$\mathbf{p}$	$\mathbf{q}$	$(\neg\mathbf{p}\mathbf{\Lambda}\neg\mathbf{q})$
F	F	$\mathbf{T}$
F	T	F
T	F	F
T	T	F

Correct Answer: Option A

Q. 10 The position of moving car at time t is given as $f(t) = 3t^2 + bt + c, t > 0$ where a , b,c are real no.>1 Then the avg speed of the car over the time interval $[t_1, t_2]$ is attained at the pt. ....

Option 1:
$(t_2 - t_1)/2$

Option 2:
$a(t_2 + t_1) + b$

Option 3:
$a(t_2 + t_1) + b$

Option 4:
$(t_2 + t_1)/2$

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Correct Answer:
$(t_2 + t_1)/2$

Solution:
Using mean value theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(t) = \frac{f(t_2) - f(t_1)}{t_2 - t_1}$$

$$\Rightarrow 2at + b = \frac{f(t_2) - f(t_1)}{t_2 - t_1}$$

$$\Rightarrow 2at + b = \frac{a(t_2^2 - t_1^2) + b(t_2 - t_1)}{t_2 - t_1}$$

$$\Rightarrow 2at + b = a(t_2 + t_1) + b$$

$$\Rightarrow \quad t = \frac{t_1 + t_2}{2}$$

Correct Answer : Option D

Q. 11 A ray of light coming from the general eqn of the pt $\left(1, 2\sqrt{3}\right)$ is incident at an angle 30° on the line x=1 at point A. The ray gets reflected on the line x=1 and meets x-axis at pt.B Then the line AB pass via ...

Option 1:
$\left(4, -\sqrt{3}\right)$

Option 2:
$\left(3, -\sqrt{3}\right)$

Option 3:
$\left(3, -1/\sqrt{3}\right)$

Option 4:
$\left(4, -\sqrt{3}/2\right)$

Correct Answer:
$\left(3, -\sqrt{3}\right)$

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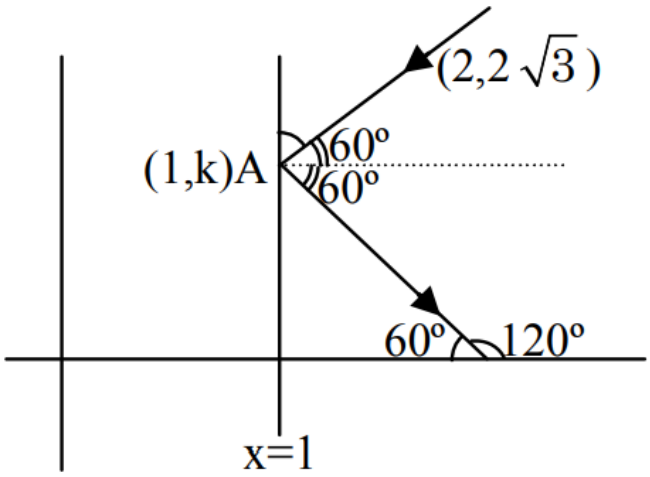
Solution:



$$\tan 60^\circ = \frac{2\sqrt{3} - k}{2 - 1}$$

$$\sqrt{3} = 2\sqrt{3} - k$$

$$\therefore k = \sqrt{3}$$

So point $A(1, \sqrt{3})$

Now slope of line AB in $m_{AB} = \tan 120^\circ$

$$mm_{AB} = -\sqrt{3}$$

Now equation of line AB is

$$y - \sqrt{3} = -\sqrt{3}(x - 1)$$

$$\sqrt{3}x + y = 2\sqrt{3}$$

Now satisfy options

$$\left(3, -\sqrt{3}\right)$$

Correct Answer: Option B

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Q. 12 α, β two roots of the eqn $x^2 - 64x + 256 = 0$ ... Then $\left(\frac{\alpha^3}{\beta^5}\right)^{1/8} + \left(\frac{\beta^3}{\alpha^5}\right)^{1/8} = ?$

Option 1:

3

Option 2:
2

Option 3:
4

Option 4:
1

Correct Answer:
2

Solution:
$x^2 - 64x + 256 = 0$
$\alpha + \beta = 64$
$\alpha\beta = 256$
$\left(\frac{\alpha^3}{\beta^5}\right)^{1/8} + \left(\frac{\beta^3}{\alpha^5}\right)^{1/8}$
$= \frac{\alpha + \beta}{(\alpha\beta)^{5/8}}$
$= \frac{64}{(256)^{5/8}} = 2$

Correct Answer: Option B

Q. 13 $\lim\limits_{x \to 1} \frac{\int_0^{(x-1)^2} t.\cos(t^2)dt}{(x-1)\sin(x-1)}$ is equal to.....

Option 1:
$-\frac{1}{2}$

Option 2:
$\frac{1}{2}$

Option 3:
Does not

Option 4:
0

Correct Answer:
0

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Solution:

$$\lim_{x\to 1} \frac{\int_0^{(x-1)^2} t\cdot\cos(t^2)\,dt}{(x-1)\sin(x-1)}$$

Apply L hospital rule

$$= \lim_{x\to 1} \frac{2(x-1)^3\cos((x-1)^4)}{\sin(x-1)+(x-1)\cos(x-1)}$$

$$= \lim_{x\to 1} \frac{2(x-1)^2\cos(x-1)^4}{\frac{\sin(x-1)}{(x-1)}+\cos(x-1)} = 0$$

Correct Answer: Option D

Q. 14 The region represented by $\{z = x + iy \in C : |z| - Re(z) \leq 1\}$ is also given by

Option 1:
$y^2 \geq 2(x+1)$

Option 2:
$y^2 \leq 2(x+1/2)$

Option 3:
$y^2 \leq (x+1/2)$

Option 4:
$y^2 \geq (x^2+1)$

Correct Answer:
$y^2 \leq 2(x+1/2)$

Solution:

$|z| - \mathrm{Re}(z) \leq 1$
$\Rightarrow \sqrt{x^2+y^2} - x \leq 1$
$\Rightarrow \sqrt{x^2+y^2} \leq 1 + x$

$\Rightarrow x^2 + y^2 \leq 1 + x^2 + 2x$
$\Rightarrow y^2 \leq 2(x+1/2)$

Correct Answer: Option B

Q. 15 Out of 11 consecutive natural no if 3 are selected at random without repeat Then the probability that they are in AP with +ve c.d

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Option 1:

$\frac{5}{33}$

Option 2:

$\frac{10}{99}$

Option 3:

$\frac{3}{101}$

Option 4:

$\frac{15}{101}$

Correct Answer:

$\frac{5}{33}$

Solution:

$$\text{Total no of selections} = {}^{11}C_3$$
$$= \frac{11 \times 10 \times 9}{6}$$
$$= 165$$

No of ways with common difference 1 = 9

No of ways with common difference 2 = 7

No of ways with common difference 3 = 5

No of ways with common difference 4 = 3

No of ways with common difference 5 = 1

$$\text{Required probability} = \frac{1+3+5+7+9}{165}$$
$$= \frac{25}{165} = \frac{5}{33}$$

Correct Answer: Option A

Q. 16 The general eqn of the $DE = \sqrt{1+x^2+y^2+x^2y^2} + xy\frac{dy}{dx}$

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Option 1:

$$\sqrt{1+y^2}-\sqrt{1+x^2}=\frac{1}{2}\log_e\left\{\frac{\sqrt{1+x^2}-1}{\sqrt{1+y^2}+1}\right\}$$

Option 2:

$$\sqrt{1+y^2}+\sqrt{1+x^2}=\frac{1}{2}\log_e\left\{\frac{\sqrt{1+x^2}-1}{\sqrt{1+y^2}+1}\right\}$$

Option 3:

$$\sqrt{1+y^2}+\sqrt{1+x^2}=\frac{1}{2}\log_e\left\{\frac{\sqrt{1+x^2}+1}{\sqrt{1+y^2}-1}\right\}$$

Option 4:

$$\sqrt{1+y^2}-\sqrt{1+x^2}=\frac{1}{2}\log_e\left\{\frac{\sqrt{1+x^2}+1}{\sqrt{1+y^2}-1}\right\}$$

Correct Answer:

$$\sqrt{1+y^2}+\sqrt{1+x^2}=\frac{1}{2}\log_e\left\{\frac{\sqrt{1+x^2}+1}{\sqrt{1+y^2}-1}\right\}$$

Solution:

$$\sqrt{1+x^2+y^2+x^2y^2}+xy\frac{dy}{dx}=0$$

$$\Rightarrow \sqrt{(1+x^2)(1+y^2)}=-xy\frac{dy}{dx}$$

$$\Rightarrow \int\frac{y}{\sqrt{1+y^2}}dy=-\int\frac{\sqrt{1+x^2}}{x}dx$$

$$\Rightarrow \sqrt{1+y^2}=-\sqrt{x^2+1}-\frac{1}{2}\ln\left(\frac{\sqrt{x^2+1}-1}{\sqrt{x^2+1}+1}\right)+C$$

Correct Answer: Option C

Q. 17 Which of the following points lie on the locus of foot of perpendicular drawn upon any tangent to the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$ from any of it's foci?

Option 1:

$(1,2)$

Option 2:

$(-2,\sqrt{3})$

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Option 3:
$(-1, \sqrt{2})$

Option 4:
$(-1, \sqrt{3})$

Correct Answer:
$(-1, \sqrt{3})$

Solution:
$$\frac{x^2}{4} + \frac{y^2}{2} = 1$$

required locus

using the property of the ellipse

$$x^2 + y^2 = 4$$

Correct Answer: Option D

Q. 18 Let a,b,c,d and p be any non zero distinct real numbers such that $(a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) = 0$ then

Option 1:
a,c,p are in A.P

Option 2:
a,b,c,d are in A.P

Option 3:
a,b,c,d are in G.P

Option 4:
a,c,p are in G.P

Correct Answer:
a,b,c,d are in G.P

Solution:
If a, b, c, d are in G P

$$\left(a^2 + a^2r^2 + a^2r^4\right)p^2 - 2\left(a^2r + a^2r^3 + a^2r^5\right)p + \left(a^2r^2 + a^2r^4 + a^2r^6\right) = 0$$
$$\Rightarrow p^2 - 2pr + r^2 = 0$$
$$\Rightarrow (p - r)^2 = 0$$

Correct Answer: Option C

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Q. 19 If $\sum_{i=1}^{n}(x_i - a) = n \ and \ \sum_{i=1}^{n}(x_i - a)^2 = na \ , \ (n, a) > 1$ then the solution of n observation $x_1, x_2, ...x_n$ are

Option 1:
$n\left(\sqrt{a-1}\right)$

Option 2:
$\sqrt{n\,(a-1)}$

Option 3:
$a - 1$

Option 4:
$\sqrt{a-1}$

Correct Answer:
$\sqrt{a-1}$

Solution:

$$\sum_{i=1}^{n}(x_i - a) = n \quad \sum_{i=1}^{n}(x_i - a)^2 = na$$

$$\therefore S \cdot D \cdot \text{ for } (x_i - a)$$

$$= \sqrt{\frac{\sum(x_i - a)^2}{n} - \left(\frac{\sum(x_i - a)}{n}\right)^2}$$

$$= \sqrt{a-1}$$

Correct Answer: Option D

Q. 20 The area in sq unit of the region $A = \left\{x, y : |x| + |y| \leq 1, 2y^2 \geq |x|\right\}$

Option 1:
$\frac{5}{6}$

Option 2:
$\frac{1}{6}$

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Option 3:

$\frac{7}{6}$

Option 4:

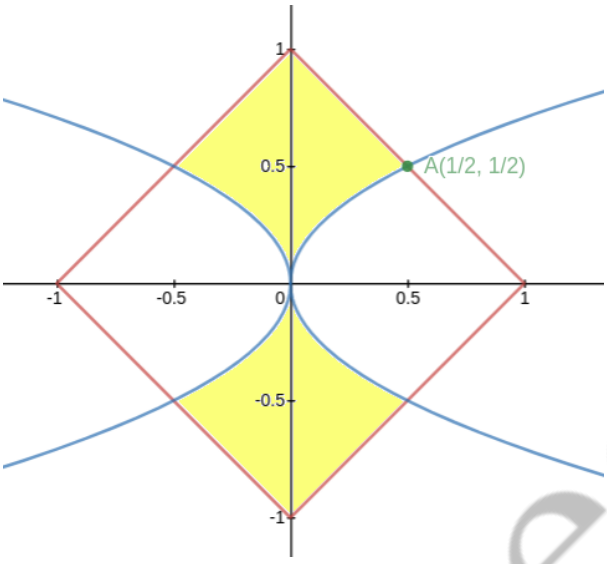
$\frac{1}{3}$

Correct Answer:

$\frac{5}{6}$

Solution:

Area of the shaded region



$$\text{Required area } = 4 \times \int_0^{1/2} \left((1-x) - \sqrt{\frac{x}{2}} \right) dx$$

$$= 4 \times \left[x - \frac{x^2}{2} - \frac{2}{3\sqrt{2}} x^{3/2} \right]_0^{1/2}$$

$$= 4 \times \left[\frac{1}{2} - \frac{1}{8} - \frac{2}{3\sqrt{2}x} 2\sqrt{2} \right]$$

$$= 4 \times \frac{5}{24} = \frac{5}{6}$$

Correct Answer : Option A

Q. 21 If $\vec{a}$ and $\vec{b}$ are unit vectors then the greatest value of $\sqrt{3}\left|\vec{a}+\vec{b}\right| + \left|\vec{a}-\vec{b}\right|$ is

Option 1:

1

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Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
4

Solution:
$|\vec{a}| = 1, |\vec{b}| = 1$
$\sqrt{3}|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|$

$$= \sqrt{3}\sqrt{|\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}|\cos\theta} + \sqrt{|\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos\theta}$$

$$f(\theta) = \sqrt{6(1+\cos\theta)} + \sqrt{2(1-\cos\theta)}$$
$$f'(\theta) = \frac{-6\sin\theta}{2\sqrt{6(1+\cos\theta)}} + \frac{2\sin\theta}{2\sqrt{2(1-\cos\theta)}} = 0$$
$$\frac{6\sin\theta}{2\sqrt{6(1+\cos\theta)}} = \frac{2\sin\theta}{2\sqrt{2(1-\cos\theta)}}$$
$\sin\theta = 0$ *or*
$18(1-\cos\theta) = 6(1+\cos\theta)$
$\cos\theta = \frac{1}{2}$

$$f(\theta) = \sqrt{6 \times (1+\frac{1}{2})} + \sqrt{2 \times (1-\frac{1}{2})}$$
Maximum value $f(\theta) = 4$

Correct Answer :4

Q. 22 Let AD and BC be two verticle pole of A&B on horizontal ground. If AD=8 and BC=11 and AB=10 then the distance of the point M on AB from the pt A st $MD^2 + MC^2$ is min is ....

Option 1:
4

Option 2:
5

Option 3:
6

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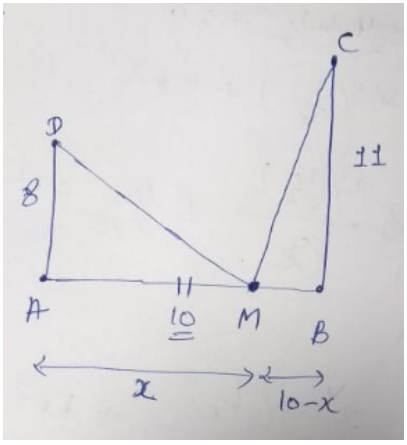
Option 4:
7

Correct Answer:
5

Solution:



$f(x) = MD^2 + MC^2 = 8^2 + x^2 + 11^2 + (10 - x)^2$
$f'(x) = 2x + 2(10 - x)(-1) = 0$
$2x = 2(10 - x)$
$x = 5$
$f''(x) = 4 > 0 \Rightarrow$minima at x = 5
Distance from point A i . e. AM= 5

Correct Answer : 5

Q. 23

Let $f : R \to R$ defined as $f(x) = \begin{cases} x^5 \sin\left(\frac{1}{x} + 5x^2\right), x < 0 \\ 0, x = 0 \\ x^5 \cos\left(\frac{1}{x} + \lambda z^2, x < 0\right) \end{cases}$ The value of λ for which $f''(x)$ exists is.

Option 1:
2

Option 2:
3

Option 3:
5

Option 4:
6

Correct Answer:
5

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Solution:

$$f'(x) = \begin{cases} 5x^4 \sin\left(\frac{1}{x}\right) - x^3 \cos\left(\frac{1}{x}\right) + 10x, x < 0 \\ 0, \quad x = 0 \\ 5x^4 \cos\left(\frac{1}{x}\right) + x^3 \sin\left(\frac{1}{x}\right) + 2\lambda, x > 0 \end{cases}$$

$$f''(x) = \begin{cases} \text{term having x in muliplication} + 10, x < 0 \\ 0, \quad x = 0 \\ \text{term having x in muliplication} + 2\lambda, x > 0 \end{cases}$$

L.H.L = R.H.L.

$2\lambda = 10$
$\lambda = 5$

Correct Answer : 5

Q. 24 Set A has m element and set B has n elements. If the total number of the subset of A is 112 more than the total number of subsets of B then m-n =?

Option 1:
2

Option 2:
3

Option 3:
6

Option 4:
None of the above

Correct Answer:
3

Solution:
$2^m - 2^n = 112$

By hit and trial, m= 7 and n = 4

So, m - n = 3

Q. 25 The angle of elevation of the top of a hill from a point on the horizontal planes passing through the foot of the hill is found to be 45°. After walking a distance of 80 meters towards the top, up a slope inclined at an angle of 30° to the horizontal plane, the angle of elevation of the top of the hill becomes 75°. Then the height of the hill (in meters) is

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Option 1:
28

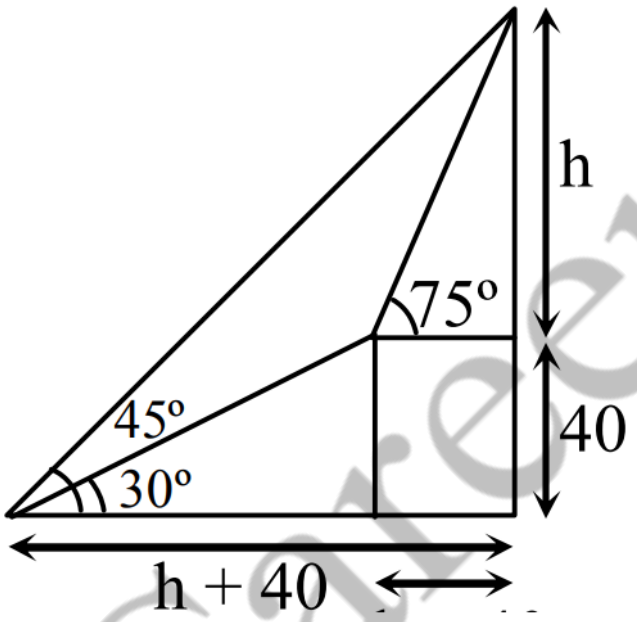
Option 2:
56

Option 3:
40

Option 4:
80

Correct Answer:
80

Solution:



$$\tan 75^\circ = \frac{h}{h + 40 - 40\sqrt{3}}$$

$$\frac{2+\sqrt{3}}{1} = \frac{h}{h + 40 - 40\sqrt{3}}$$

$$\Rightarrow 2h + 80 - 80\sqrt{3} + \sqrt{3}h + 40\sqrt{3} - 120 = h$$

$$\Rightarrow h(\sqrt{3}+1) = 40 + 40\sqrt{3}$$

$$\Rightarrow h = 40$$

Height of hill = 40 + 40 = 80 m

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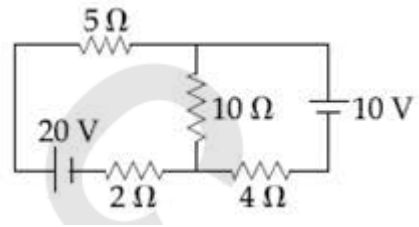
6th September Shift 2

Physics

Q. 1



In the figure shown, the current in the 10 V battery is close to :

Option 1:
0.71 A from positive to negative terminal

Option 2:
0.42 A from positive to negative terminal

Option 3:
0.21 A from positive to negative terminal

Option 4:
0.36 A from negative to positive terminal

Correct Answer:
0.21 A from positive to negative terminal

Solution:

$$E_{eq} = \frac{20 \times 10}{17} = \frac{200}{17}$$

$$\text{and } R_{eq} = \frac{7 \times 10}{17} = \frac{70}{17}$$

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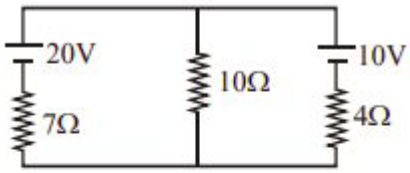

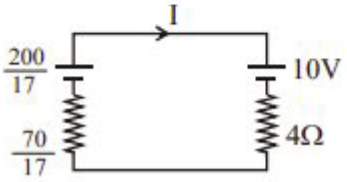

$$\therefore \quad I = \frac{\frac{20}{17} - 10}{4 + \frac{70}{17}} = 0.21 \text{ A}$$

from +ve to -ve terminal

Q. 2 A charged particle going around in a circle can be considered to be a current loop. A particle of mass m carrying charge q is moving in a plane with speed v under the influence of magnetic field $\overrightarrow{B}$. The magnetic moment of this moving particle:

Option 1:

$$-\frac{mv^2\overrightarrow{B}}{2B^2}$$

Option 2:

$$\frac{mv^2\overrightarrow{B}}{2\pi B^2}$$

Option 3:

$$\frac{mv^2\overrightarrow{B}}{B^2}$$

Option 4:

$$\frac{mv^2\overrightarrow{B}}{2B^2}$$

Correct Answer:

$$-\frac{mv^2\overrightarrow{B}}{2B^2}$$

Solution:

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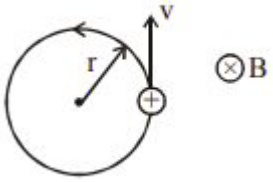


Magnetic moment

$$\mathrm{M} = \mathrm{iA}$$

$$M = \left(\frac{q}{T}\right) \times \pi r^2 = \frac{q\pi r^2}{\left(\frac{2\pi r}{v}\right)} = \frac{qvr}{2}$$

$$M = \frac{qv}{2} \times \frac{vm}{qB}$$

$$\mathrm{M} = \frac{\mathrm{mv}^2}{2\ \mathrm{B}}$$

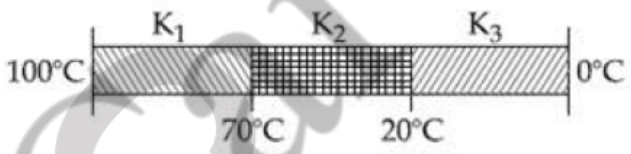
As we can see from the figure, direction of magnetic moment (M) is opposite to magnetic field.

$$\overrightarrow{\mathrm{M}} = -\frac{\mathrm{mv}^2}{2\ \mathrm{B}}\hat{\mathrm{B}}$$

$$= -\frac{\mathrm{mv}^2}{2\ \mathrm{B}^2}\overrightarrow{\mathrm{B}}$$

Q. 3 Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity $K_1, K_2 \ and \ K_3$

respectively. They are joined together at their ends to make a long rod (See figure). One end of the long rod is maintained at 100^oC and the other at 0^oC (see figure). If the joints of the rod are at 70^oC and 20^oC in steady state and there is no loss of energy from the surface of the rod, the correct relationship between $K_1, K_2 \ and \ K_3$ is :



Option 1:

$K_1 : K_3 = 2 : 3,$
$K_2 : K_3 = 2 : 5$

Option 2:

$K_1 < K_3 < K_3$

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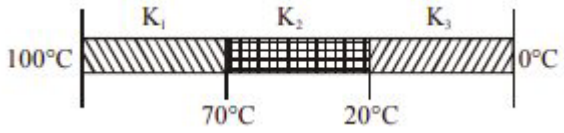
Option 3:
$K_1 : K_2 = 5 : 2,$
$K_1 : K_3 = 3 : 5$

Option 4:
$K_1 < K_2 < K_3$

Correct Answer:
$K_1 : K_3 = 2 : 3,$
$K_2 : K_3 = 2 : 5$

Solution:



Rods are identical have same length (ℓ) and area of cross-section (A)

Combination are in series, so heat current is same for all Rods

$$\left(\frac{\Delta Q}{\Delta t}\right)_{AB} = \left(\frac{\Delta Q}{\Delta t}\right)_{BC} = \left(\frac{\Delta Q}{\Delta t}\right)_{C} = \text{ Heat current}$$

$$\frac{(100-70)K_1\ A}{\ell} = \frac{(70-20)K_2\ A}{\ell} = \frac{(20-0)K_3\ A}{\ell}$$

$$30\ K_1 = 50\ K_2 = 20\ K_3$$
$$3\ K_1 = 2\ K_3$$

$$\frac{K_1}{K_3} = \frac{2}{3} = 2 : 3$$
$$5\ K_2 = 2\ K_3$$

$$\frac{K_2}{K_3} = \frac{2}{5} = 2 : 5$$

Q. 4 Two identical electric point dipoles have dipole moments $\vec{p_1} = p\hat{i}$ *and* $\vec{p_2} = -p\hat{i}$ and are held on the x axis at distance 'a' from each other. When released, they move along the x-axis with the direction of their dipole moments remaining uchanged. If the mass of each dipole is 'm', their speed when they are infinitely far apart is :

Option 1:
$$\frac{P}{a}\sqrt{\frac{1}{\pi\epsilon_0 ma}}$$

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Option 2:

$$\frac{P}{a}\sqrt{\frac{1}{2\pi\epsilon_0 ma}}$$

Option 3:

$$\frac{P}{a}\sqrt{\frac{2}{\pi\epsilon_0 ma}}$$

Option 4:

$$\frac{P}{a}\sqrt{\frac{3}{2\pi\epsilon_0 ma}}$$

Correct Answer:

$$\frac{P}{a}\sqrt{\frac{1}{2\pi\epsilon_0 ma}}$$

Solution:

Using energy conservation:

$$\mathrm{KE_i} + \mathrm{PE_i} = \mathrm{KE_f} + \mathrm{PE_f}$$

$$\mathrm{O} + \frac{2\mathrm{KP}}{\mathrm{a}^3} \times \mathrm{P} = \frac{1}{2}\mathrm{mv}^2 \times 2 + 0$$

$$\mathrm{V} = \sqrt{\frac{2\mathrm{P}^2}{4\pi\varepsilon_0 \mathrm{a}^3\ \mathrm{m}}} = \frac{\mathrm{P}}{\mathrm{a}}\sqrt{\frac{1}{2\pi\varepsilon_0 \mathrm{am}}}$$

Q. 5 For a plane electromagnetic wave, the magnetic field at a point x and time t is :

$$\vec{B}(x,t) = \left[1.2 \times 10^{-7}\ \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)\hat{k}\right] T$$

The instantaneous electric field $\vec{E}$ corresponding to $\vec{B}$ is :

(speed of light $c = 3 \times 10^8\ ms^{-1}$)

Option 1:

$$\vec{E}(x,t) = \left[-36\ \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)\hat{j}\right] \frac{V}{m}$$

Option 2:

$$\vec{E}(x,t) = \left[36\ \sin(1 \times 10^3 x + 0.5 \times 10^{11} t)\hat{j}\right] \frac{V}{m}$$

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Option 3:

$$\vec{E}(x,t) = \left[36\ \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)\hat{k}\right] \frac{V}{m}$$

Option 4:

$$\vec{E}(x,t) = \left[36\ \sin(1 \times 10^3 x + 1.5 \times 10^{11} t)\hat{i}\right] \frac{V}{m}$$

Correct Answer:

$$\vec{E}(x,t) = \left[-36\ \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)\hat{j}\right] \frac{V}{m}$$

Solution:

$$\vec{B} = 1.2 \times 10^{-7} \sin\left(0.5 \times 10^3 x + 1.5 \times 10^{11} t\right) \hat{k}T$$

$$E_0 = cB_0 = 36\frac{v}{m}$$

Wave is travelling along -x axis and $\vec{B}$ is along the +z axis.

$\vec{\mathrm{E}}$ must be along -y-axis, as we know that $\vec{\mathrm{E}} \times \vec{\mathrm{B}}$ points in the direction of propagation.

Q. 6 Two planets have masses M and 16 M and their radii are a and 2a, respectively. The separation etween the centres of the planets is 10 a. A body of mass m is fired from the surface of the larger planet towards the smaller planet along the line joining their centres. For the body to be able to reach at the surface of smaller planet, the minimum firing speed needed is :

Option 1:

$$2\sqrt{\frac{GM}{a}}$$

Option 2:

$$4\sqrt{\frac{GM}{a}}$$

Option 3:

$$\sqrt{\frac{GM^2}{ma}}$$

Option 4:

$$\frac{3}{2}\sqrt{\frac{5\ GM}{a}}$$

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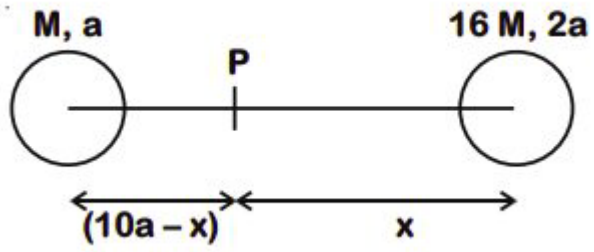
Correct Answer:

$\frac{3}{2}\sqrt{\frac{5\ GM}{a}}$

Solution:



Velocity should be given so as to reach a point where field is zero.

$$\frac{16\mathrm{M}}{x^2} = \frac{\mathrm{M}}{(10\mathrm{a} - \mathrm{x})^2}$$

$$x = 8a$$

By CoE

$$-\frac{\mathrm{GMm}}{8\mathrm{a}} - \frac{16\mathrm{GMm}}{2\mathrm{a}} + \frac{1}{2}\mathrm{mv}^2 = -\frac{16\mathrm{GMm}}{8\mathrm{a}} - \frac{\mathrm{GMm}}{2\mathrm{a}}$$

$$v = \frac{3}{2}\sqrt{\frac{5Gm}{a}}$$

Q. 7 A particle moving in the xy plane experiences a velocity dependent force $\vec{F} = k\left(v_y\hat{i} + v_x\hat{j}\right)$, where $v_x \ \ and \ \ v_y$ are the x and y components of its velocity $\vec{v}$. If $\vec{a}$ is the acceleration of the particle, then which of the following statements is true for the particle ?

Option 1:

quantity $\overrightarrow{v} \times \overrightarrow{a}$ is constant in time

Option 2:

$\overrightarrow{F}$ arises due to a magnetic field

Option 3:

Kinetic energy of particle is constant in time

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Option 4:
quantity $\overrightarrow{v} . \overrightarrow{a}$ is constant in time

Correct Answer:
quantity $\overrightarrow{v} \times \overrightarrow{a}$ is constant in time

Solution:
$\frac{\mathrm{dv_x}}{\mathrm{dt}} = \frac{\mathrm{k}}{\mathrm{m}} \mathrm{v_y}$

$\frac{\mathrm{dv_y}}{\mathrm{dt}} = \frac{\mathrm{k}}{\mathrm{m}} \mathrm{v_x}$

$$\frac{dv_y}{dv_x} = \frac{v_x}{v_y} \Rightarrow \int v_y dv_y = \int v_x dv_x$$

$\mathrm{v_y^2} = \mathrm{v_x^2} + \mathrm{C}$
$\mathrm{v_y^2} - \mathrm{v_x^2} = \text{constant}$

$$\overrightarrow{\mathrm{v}} \times \overrightarrow{\mathrm{a}} = \left(\mathrm{v_x}\hat{\mathrm{i}} + \mathrm{v_y}\hat{\mathrm{j}}\right) \times \frac{\mathrm{k}}{\mathrm{m}} \left(\mathrm{v_y}\hat{\mathrm{i}} + \mathrm{v_x}\hat{\mathrm{j}}\right)$$
$$= \left(\mathrm{v_x^2}\hat{\mathrm{k}} - \mathrm{v_y^2}\hat{\mathrm{k}}\right) \frac{\mathrm{k}}{\mathrm{m}}$$
$$= \left(v_x^2 - v_y^2\right) \frac{k}{m} \hat{k} = Constant$$

Q. 8 Particle A of mass m_1 moving with velocity $\left(\sqrt{3}\hat{i} + \hat{j}\right) ms^{-1}$ collides with another particle B of mass m_2 which is at rest initially. Let $\overrightarrow{V_1}$ and $\overrightarrow{V_2}$ be the velocities of particles A and B after collision respectively. If $m_1 = 2m_2$ and after collision $\overrightarrow{V_1} = \left(\hat{i} + \sqrt{3}\hat{j}\right) ms^{-1}$, the angle between $\overrightarrow{V_1}$ and $\overrightarrow{V_2}$ is :

Option 1:
15^o

Option 2:
60^o

Option 3:
-45^o

Option 4:
105^o

Correct Answer:
105^o

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Solution:

$\vec{v}_{01} = (\sqrt{3}\hat{i} + \hat{j})\text{m/s}$

$\vec{v}_{\infty} = \vec{0}$

$m_1 = 2\ m_2$

After collision, $\vec{v}_1 = (\hat{i} + \sqrt{3}\hat{j})\text{m/s}$

$\vec{v}_2 = ?$

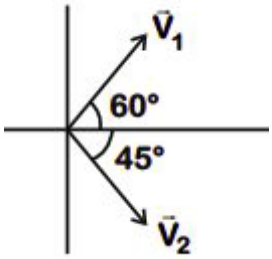
Applying conservation of linear momentum.

$2\ m_2(\sqrt{3}\hat{i} + \hat{j}) + 0 = 2\ m_2(\hat{i} + \sqrt{3}\hat{j}) + m_2\vec{v}_2$

$\vec{v}_2 = 2(\sqrt{3}\hat{i} + \hat{j}) - 2(\hat{i} + \sqrt{3}\hat{j})$

$= 2(\sqrt{3}\hat{i} - \hat{j}) + 2(\hat{i} - \sqrt{3}\hat{j})$

$\vec{v}_2 = 2(\sqrt{3} - 1)(\hat{i} - \hat{j})$



for angle between $\vec{v}_1 \& \vec{v}_2$

$$\cos\theta = \frac{\vec{v}_1, \vec{v}_2}{\vec{v}_1\vec{v}_2} = \frac{2(\sqrt{3}-1)(1-\sqrt{3})}{2 \times 2\sqrt{2}(\sqrt{3}-1)}$$

$$\cos\theta = \frac{1-\sqrt{3}}{2\sqrt{2}} \Rightarrow \theta = 105^\circ$$

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Q. 9 When a car is at rest, its driver sees rain drops falling on it vertically. When driving the car with speed v, he sees that rain drops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to $(1+\beta)v$, this angle changes to 45°. The value of 100β is close to :

Option 1:
50

Option 2:
41

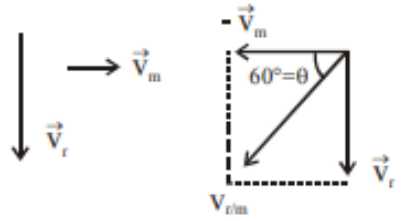
Option 3:
37

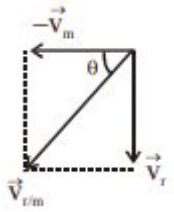
Option 4:
73

Correct Answer:
73

Solution:
Rain is falling vertically downwards.

$\overrightarrow{v}_{r/m} = \overrightarrow{v}_r - \overrightarrow{v_m}$





$\tan 60^\circ = \frac{v_r}{v_m} = \sqrt{3}$
$v_r = v_m\sqrt{3} = v\sqrt{3}$
Now, $v_m = (1 + B)v$
and $\theta = 45^\circ$
$\tan 45 = \frac{v_r}{v_m} = 1$
$v_r = v_m$
$v\sqrt{3} = (1 + \beta)v$
$\sqrt{3} = 1 + \beta$
$\Rightarrow \beta = \sqrt{3} - 1 = 0.73$

$100\beta = 73$

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Q. 10 Given the masses of various atomic particles
$m_p = 1.0072\ u, m_n = 1.0087\ u, m_e = 0.000548\ u, m_v^- = 0, m_d = 2.0141\ u,$
where
$p \equiv proton, n \equiv neutron, e \equiv electron, \bar{v} \equiv antineutrino \quad and \quad d \equiv deuteron.$

Which of the following process is allowed by momentum and energy conservation ?

Option 1:
$n + n \rightarrow$ deuterium atom (electron bound to the nucleus)

Option 2:
$p \rightarrow n + e^{+} + \bar{v}$

Option 3:
$n + p \rightarrow d + \gamma$

Option 4:
$e^{+} + e^{-} \rightarrow \gamma$

Correct Answer:
$n + p \rightarrow d + \gamma$

Solution:
The mass of the product nucleus is always less than the sum of the individual masses of the constituent reactants because energy is removed when the nucleus is formed.

The mass which is removed from the total mass of the original particles is known as the mass defect and it represents the energy released when the nucleus is formed

$\Delta \text{m should be positive}$
$(m_p + m_n) > m_d$

Only in case-III,
$\text{M}_{\text{LHS}} > \text{M}_{\text{RHS}}$

i.e. total mass on the reactant side is greater than that on the product side. Hence it will only be allowed.

Q. 11 A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit :

Option 1:
ammeter is always used in parallel and voltmeter is series

Option 2:
Both ammeter and voltmeter must be connected in parallel

Option 3:
ammeter is always connected in series and voltmeter in parallel

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Option 4:
Both, ammeter and voltmeter must be connected in series

Correct Answer:
ammeter is always connected in series and voltmeter in parallel

Solution:
Ammeter :- In series connection, the same current flows through all the components. It aims at measuring the curent flowing through the circuit and hence, it is connected in series.

Voltmeter :- A voltmeter measures voltage change between two points in a circuit, So we have to place the voltmeter in parallal with the circuit component.

Q. 12 Consider the force F on a charge 'q' due to a uniformly charged a spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F, if 'q' is placed at distance r from the centre of the shell?

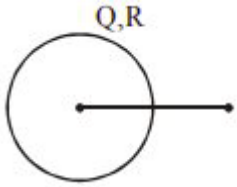
Option 1:
$F = \frac{1}{4\pi\epsilon_0}\frac{Qq}{r^2}\ for\ r > R$

Option 2:
$\frac{1}{4\pi\epsilon_0}\frac{qQ}{R^2} > F > 0\ for\ r < R$

Option 3:
$F = \frac{1}{4\pi\epsilon_0}\frac{Qq}{r^2}\ for\ r < R$

Option 4:
$F = \frac{1}{4\pi\epsilon_0}\frac{Qq}{r^2}$ for all r

Correct Answer:
$F = \frac{1}{4\pi\epsilon_0}\frac{Qq}{r^2}\ for\ r > R$

Solution:


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$\text{Inside the shell E} = 0 \;\; hence \;\; F = 0$

$\text{Oustside the sheell E} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{\text{Q}}{\text{r}^2}$

$\text{E} = \frac{1}{4\pi\varepsilon_0}\frac{\text{Q}}{\text{r}^2}$

$\text{hence F} = \frac{1}{4\pi\varepsilon_0}\frac{\text{Qq}}{\text{r}^2} \text{ for r} > \text{R}$

Q. 13 A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm. 5.55 mm, 5.45 mm ; 5.65 mm. The average of these four readings is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter of the pencil should therefore be recorded as:

Option 1:
$(5.5375 \pm 0.0739)mm$

Option 2:
$(5.5375 \pm 0.0740)mm$

Option 3:
$(5.538 \pm 0.074)mm$

Option 4:
$(5.54 \pm 0.07)mm$

Correct Answer:
$(5.54 \pm 0.07)mm$

Solution:
$\text{d}_{\text{av}} = 5.5375 \text{ mm}$

$\Delta\text{d} = 0.07395 \text{ mm}$

$\therefore \text{ Measured data are up to two digits after decimal}$

$\therefore d = (5.54 \pm 0.07)\text{mm}$

Use significant figures. Answer must be upto three significant figures.

Q. 14 A double convex lens has power P and same radii of curvature R of both the surfaces. The radius of curvature of surface of a plano-convex lens made of the same material with power 1.5 P is :

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Option 1:

$2R$

Option 2:

$\frac{R}{2}$

Option 3:

$\frac{3R}{2}$

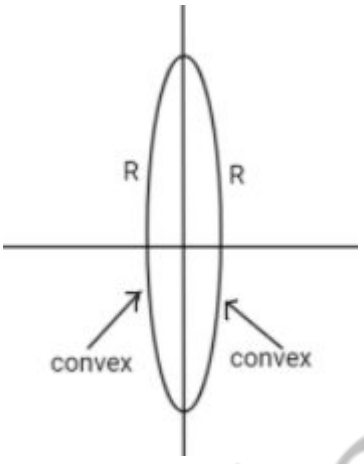
Option 4:

$\frac{R}{3}$

Correct Answer:

$\frac{R}{3}$

Solution:

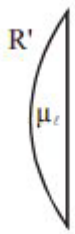


$$R_1 = R_2 = R$$

$Power = P$

Refractive index is assume(μ_l)

$$P = \frac{1}{f} = (\mu_l - 1)\left(\frac{2}{R}\right)$$



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$$P' = \frac{1}{f'} = (\mu_\ell - 1)\left(\frac{1}{R'}\right)$$

$$P' = \frac{3}{2}P$$

$$(\mu_\ell - 1)\left(\frac{1}{R'}\right) = \frac{3}{2}(\mu_\ell - 1)\left(\frac{2}{R}\right)$$

$$\therefore \quad R' = \frac{R}{3}$$

Q. 15 A square loop of side 2 a and carrying current I is kept in xz plane with its centre at the origin. A long wire carrying the same current I is placed parallel to the z-axis and passes through point $(0, b, 0)$. The magnitude of the torque on the loop about the z-axis will be:

Option 1:

$$\frac{2\mu_0 I^2 a^2}{\pi b}$$

Option 2:

$$\frac{2\mu_0 I^2 a^2 b}{\pi(a^2 + b^2)}$$

Option 3:

$$\frac{\mu_0 I^2 a^2 b}{2\pi(a^2 + b^2)}$$

Option 4:

$$\frac{\mu_0 I^2 a^2}{2\pi b}$$

Correct Answer:

$$\frac{2\mu_0 I^2 a^2 b}{\pi(a^2 + b^2)}$$

Solution:

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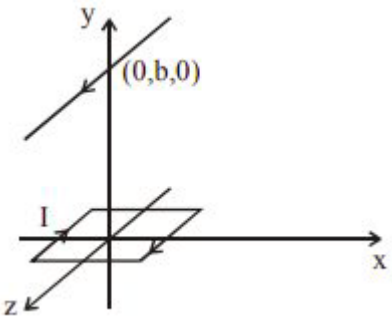

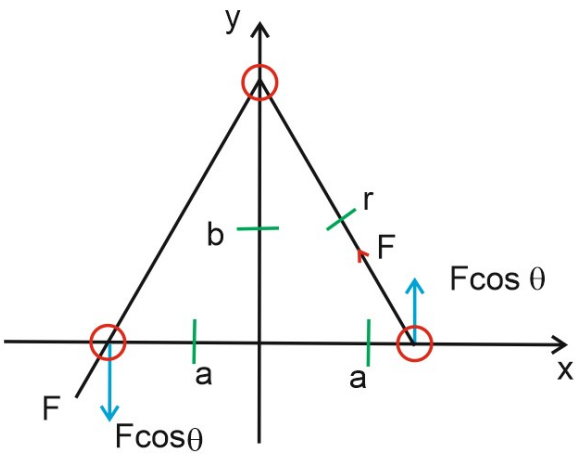

$$F = BI2a = \frac{\mu_0 I}{2\pi r} I \times 2a$$

$$F = \frac{\mu_0 I^2 a}{\pi\sqrt{b^2+a^2}}$$

$$\tau = F\cos\theta \times 2a$$

$$= \frac{\mu_0 I^2 a}{\pi\sqrt{b^2+a^2}} \times \frac{b}{\sqrt{b^2+a^2}} \times 2a$$

$$\tau = \frac{2\mu_0 I^2 a^2 b}{\pi(a^2+b^2)}$$

Q. 16 A fluid is flowing through a horizontal pipe of varying cross-section, with speed $v\ ms^{-1}$ at a point where the pressure is P Pascal. At another point where pressure is $\frac{P}{2}$ Pascal its speed is $V\ ms^{-1}$. If the density of the fluid is $\rho\ kg\ m^{-3}$ and the flow is streamline, then V is equal to :

Option 1:

$$\sqrt{\frac{P}{\rho} + v}$$

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Option 2:

$\sqrt{\frac{2P}{\rho}+v^2}$

Option 3:

$\sqrt{\frac{P}{2\rho}+v^2}$

Option 4:

$\sqrt{\frac{P}{\rho}+v^2}$

Correct Answer:

$\sqrt{\frac{P}{\rho}+v^2}$

Solution:

Applying Bernoulli's Equation

$$P_1+\frac{1}{2}\rho v_1^2+\rho g y_1=P_2+\frac{1}{2}\rho v_2^2+\rho g y_2$$

$$P+\frac{1}{2}\rho v^2=\frac{P}{2}+\frac{1}{2}\rho V^2$$

$$\frac{2P}{2\rho}+\frac{1}{2}\frac{\rho v^2}{\rho}\times 2=V^2$$

$$\sqrt{\frac{\mathrm{P}}{\rho}+\mathrm{v}^2}=\mathrm{V}$$

Q. 17 When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by $y(t)=y_0\sin^2\omega t$, where $'y'$ is measured from the lower end of unstretched spring. Then ω is :

Option 1:

$\frac{1}{2}\sqrt{\frac{g}{y_0}}$

Option 2:

$\sqrt{\frac{g}{y_0}}$

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Option 3:

$\sqrt{\frac{g}{2y_0}}$

Option 4:

$\sqrt{\frac{2g}{y_0}}$

Correct Answer:

$\sqrt{\frac{g}{2y_0}}$

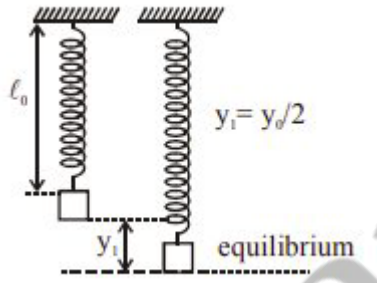
Solution:

$y = y_0 \sin^2 \omega t$

$y = \frac{y_0}{2}(1 - \cos 2\omega t)$

$y - \frac{y_0}{2} = -\frac{y_0}{2} \cos 2\omega t$

$\text{Amplitude} : \frac{y_0}{2}$



$\frac{y_0}{2} = \frac{mg}{K}$

$2\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{2g}{y_0}}$

$\omega = \sqrt{\frac{g}{2y_0}}$

Q. 18 In a dilute gas at pressure P and temperature T, the mean time between successive collisions of a molecule varies with T as :

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Option 1:
T

Option 2:
$\frac{1}{\sqrt{T}}$

Option 3:
$\frac{1}{T}$

Option 4:
$\sqrt{T}$

Correct Answer:
$\frac{1}{\sqrt{T}}$

Solution:
$\tau \propto \frac{1}{\sqrt{T}}$

$\because \quad \ell_{\text{mean}} = \frac{1}{\sqrt{2}\pi n D^2}$ and $V \propto \sqrt{T}$

$\therefore \quad \tau_{\text{mean}} = \frac{\ell_{\text{mean}}}{v}$

$\Rightarrow \quad \tau \propto \frac{1}{\sqrt{T}}$

Q. 19 Assuming the nitrogen molecule is moving with r.m.s. velocity at 400 K, the de-Broglie wavelength (in $\mathring{A}$) of nitrogen molecule is close to :

(Given : nitrogen molecule weight : $4.64 \times 10^{-26} kg$, Boltzman constant : $1.38 \times 10^{-23} J/K$, Planck constant : $6.63 \times 10^{-34} J.s$)

Correct Answer:
0.24

Solution:

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$$v_{rms} = \sqrt{\frac{3KT}{m}}$$

$$m \rightarrow \text{mass of one molecule}(in\ kg) = \frac{\text{molar mass}}{NA}$$

$$\text{de-Broglie wavelenth}, \lambda = \frac{h}{mv}$$

$$given, v = v_{rms}$$

$$\lambda = \frac{h}{m\sqrt{\frac{3KT}{m}}}$$

$$\lambda = \frac{h}{\sqrt{3KTm}} = \frac{6.63 \times 10^{-34}}{\sqrt{3 \times 1.38 \times 10^{-23} \times 400 \times \left(\frac{28 \times 10^{-3}}{6.023 \times 10^{-23}}\right)}}$$

$$\lambda = \frac{6.63 \times 10^{-11}}{2.77} = 2.39 \times 10^{-11}\ m$$

$$\lambda = 0.24\mathring{A}$$

Q. 20 The linear mass density of a thin rod AB of length L varies from A to B as $\lambda(x) = \lambda_0\left(1 + \frac{x}{L}\right)$, where x is the distance from A. If M is the mass of the rod then its moment of inertia about an axis passing through A and perpendicular to the rod is :

Option 1:
$\frac{5}{12}ML^2$

Option 2:
$\frac{7}{18}ML^2$

Option 3:
$\frac{2}{5}ML^2$

Option 4:
$\frac{3}{7}ML^2$

Correct Answer:
$\frac{7}{18}ML^2$

Solution:

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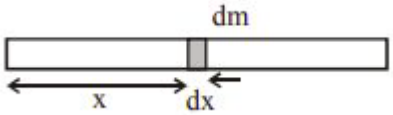

$$I = \int r^2 dm = \int x^2 \lambda dx$$

$$I = \int_0^L x^2 \lambda_0 \left(1 + \frac{x}{L}\right) dx$$

$$I = \lambda_0 \int_0^1 \left(x^2 + \frac{x^3}{L}\right) dx$$

$$I = \lambda_0 \left[\frac{L^3}{3} + \frac{L^3}{4}\right]$$

$$I = \frac{7L^3\lambda_0}{12} \ldots (i)$$

$$M = \int_0^L \lambda dx = \int_0^L \lambda_0 \left(1 + \frac{x}{L}\right) dx$$

$$M = \lambda_0 \left(L + \frac{L}{2}\right) = \lambda_0 \frac{3L}{2}$$

$$\frac{2}{3}M = (\lambda_0 L) \ldots (ii)$$

From (i) & (ii)

$$I = \frac{7}{12}\left(\frac{2}{3}M\right)L^2 = \frac{7ML^2}{18}$$

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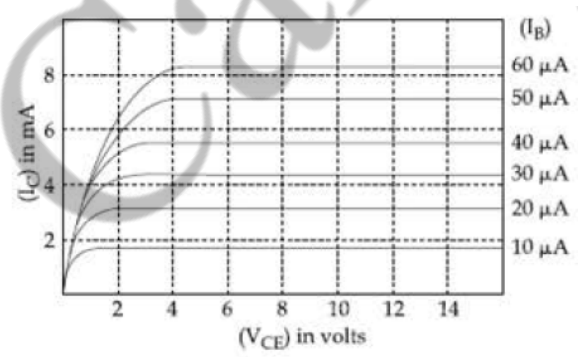
Q. 21 The output characteristics of a transistor is shown in the figure. When V_{CE} is 10 V and $I_C = 4.0\ mA$, then value of β_{ac} is _____



Correct Answer:
150

Solution:

$\Delta I_B = (30 - 20) = 10\mu\text{A}$

$\Delta I_C = (4.5 - 3)\text{mA} = 1.5\ \text{mA}$

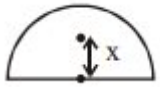
$\beta_{\text{ac}} = \frac{\Delta \text{I}_\text{C}}{\Delta \text{I}_\text{B}} = \frac{1.5\ \text{mA}}{10\mu\text{A}} = 150$

$\beta_{\text{ac}} = 150$

Q. 22 The centre of mass of a solid hemisphere of radius 8 cm is x cm from the centre of the flat surface. Then value of x is _____

Correct Answer:
3

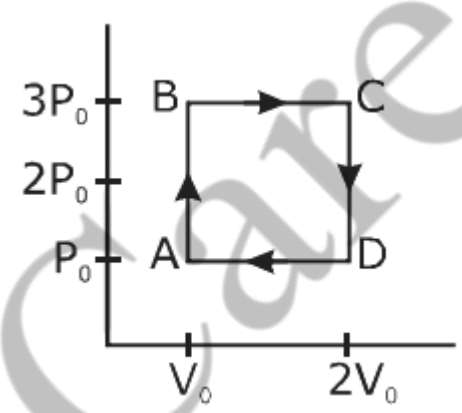
Solution:



$x = \frac{3R}{8} = 3cm$

$x = 3$

Q. 23 An engine operates by taking a monatomic ideal gas through the cycle shown in the figure. The percentage efficiency of the engine is close to _____.



Correct Answer:
19

Solution:

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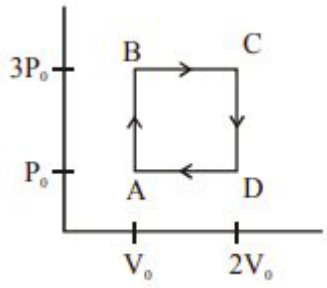

$W_{ABCDA} = 2P_0\ V_0$
$Q_{in} = Q_{AB} + Q_{BC}$

$Q_{AB} = nC\,(T_B - T_A)$

$= \frac{n3R}{2}\,(\,T_B - T_A)$

$= \frac{3}{2}\,(P_BV_B - P_AV_A)$

$= \frac{3}{2}\,(3P_BV_0 - P_0\ V_0) = 3P_0\ V_0$

$Q_{BC} = nC_P\,(T_C - T_B)$

$= \frac{n5R}{2}\,(\,T_C - T_B)$

$= \frac{5}{2}\,(P_CV_C - P_BV_B)$

$= \frac{5}{2}\,(6P_0\ V_0 - 3P_0\ V_0) = \frac{15}{2}P_0\ V_0$

$\eta = \frac{W}{Q_{in}} \times 100 = \frac{2P_0\ V_0}{3P_0\ V_0 + \frac{15}{2}P_0\ V_0} \times 100$

$\eta = \frac{400}{21} = 19.04 \approx 19$

$\eta = 19$

Q. 24 A Young's double-slit experiment is performed using monochromatic light of wavelength λ. The intensity of light at a point on the screen, where the path difference is λ, is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{6}$ is given by $\frac{nK}{12}$, where n is an integer. The value of n is _____

Correct Answer:
9

Solution:

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$I_{max} = k$

$I_1 = I_2 = K/4$

$\Delta x = \lambda/6 \Rightarrow \Delta\phi = \pi/3$

$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$I = \frac{K}{4} + \frac{K}{4} + 2 \times \frac{K}{4}\frac{1}{2}$

$= \frac{K}{2} + \frac{K}{4} = \frac{3\ K}{4} = \frac{9\ K}{12}$

$n = 9$

Q. 25 In a series LR circuit, power of 400 W is dissipated from a source of 250 V, 50 Hz. The power factor of the circuit is 0.8. In order to bring the power factor to unity, a capacitor of value C is added in series to the L and R. Taking the value of C as $\left(\frac{n}{3\pi}\right)\mu F$, then value of n is _____

Correct Answer:
400

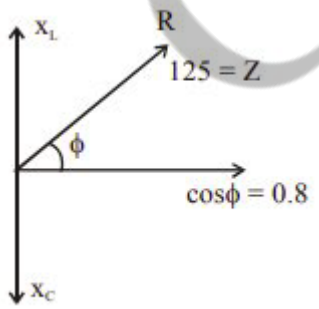
Solution:
$P = \frac{(250)^2}{Z}\cos\phi$

$500 = \frac{(250)^2}{Z}$

$Z = 125\Omega$

$R = 100\Omega$

$X_L = 75\Omega$



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$$75 = \frac{1}{2\pi fC}$$

$$C = \frac{1}{2\pi \times 75 \times 50} = \frac{1}{7500\pi}$$

$$C = \left(\frac{10^6}{2500} \times \frac{1}{3\pi}\right) \mu F$$

$$C = \frac{400}{3\pi} \mu F$$

Chemistry

Q. 1 For a reaction,

$$4M(s) + nO_2(g) \rightarrow 2M_2O_n(s),$$

the free energy change is plotted as a function of temperature. The temperature below which the oxide is stable could be inferred from the plot as the point at which:

Option 1:
the slope changes from negative to positive

Option 2:
the free energy change shows a change from negative to positive value

Option 3:
the slope changes from positive to negative

Option 4:
the slope changes from positive to zero

Correct Answer:
the free energy change shows a change from negative to positive value

Solution:
The free energy change is plotted as a function of temperature, The equation-

$$\Delta \mathrm{G} = \Delta \mathrm{H} - \mathrm{T}\Delta \mathrm{S}$$
$$\Delta \mathrm{G} = -\text{ ve } (\text{ stable oxide })$$
$$\Delta \mathrm{G} = +\text{ ve } (\text{unstable oxide})$$

Slope is entropy change, not free energy change.

The temperature below which the oxide is stable could be inferred from the plot as the point at which the free energy change shows a change from negative to positive value.

Therefore, the correct option is (2).

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Q. 2 The average molar mass of chlorine is $35.5\ g\ mol^{-1}$. The ratio of $^{35}Cl\ to\ ^{37}Cl$ in naturally occurring chlorine is close to :

Option 1:
4:1

Option 2:
3:1

Option 3:
2:1

Option 4:
1:1

Correct Answer:
3:1

Solution:
Given,

The average molar mass of chlorine is $35.5\ g\ mol^{-1}$.

Let , the ratio of $^{35}Cl\ to\ ^{37}Cl$ in naturally occurring chlorine is close to **x : y**

Now, we know

$$\text{Av. molar mass } = \frac{n_1M_1 + n_2M_2}{(n_1 + n_2)}$$

So,

$$35.5 = \frac{x \times 35 + y \times 37}{x + y}$$

$$1.5y = -0.5x$$

$$\frac{x}{y} = \frac{3}{1}$$

Therefore, the correct option is (2).

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Q. 3 Which one of the following statements is not true ?

Option 1:
Lactose contains $\alpha-\text{glycosidic}$ linkage between C_1 of galactose and C_1 of galactose and C_4 of glucose.

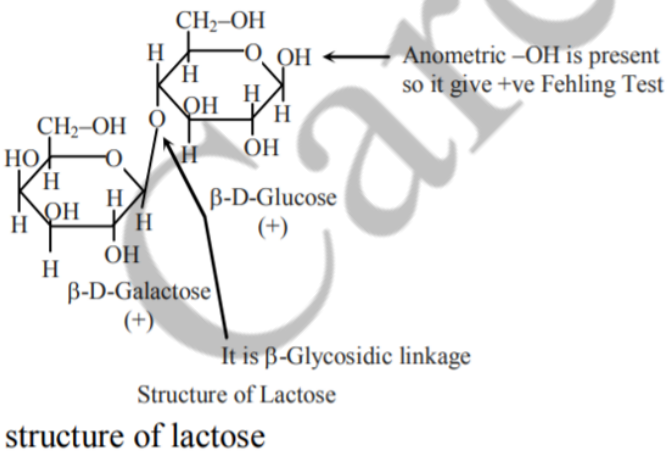
Option 2:
Lactose is a reducing sugar and it gives Fehling's test.

Option 3:
Lactose $\left(\mathrm{C}_{11}\mathrm{H}_{22}\mathrm{O}_{11}\right)$ is a disaccharide and it contains 8 hydroxyl groups.

Option 4:
On acid hydrolysis, lactose gives one molecule of $D(+)-\text{glucose}$ and one moecule of $D(+)-\text{galctose}.$

Correct Answer:
Lactose contains $\alpha-\text{glycosidic}$ linkage between C_1 of galactose and C_1 of galactose and C_4 of glucose.

Solution:



structure of lactose

Lactose contains $\beta-\text{glycosidic}$ linkage not $\alpha-\text{glycosidic}$ between C_1 of galactose and C_1 of galactose and C_4 of glucose.

So, Statement 1 is incorrect but the rest statements are correct.

Therefore, the correct option is (1).

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Q. 4 The value of K_C is 64 at 800 K for the reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The value of K_C for the following reaction is :

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

Option 1:
$\frac{1}{64}$

Option 2:
8

Option 3:
$\frac{1}{4}$

Option 4:
$\frac{1}{8}$

Correct Answer:
$\frac{1}{8}$

Solution:
Given,

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 \rightarrow K_C = 64$$

$$2NH_3 \rightleftharpoons N_2 + 3H_2 \rightarrow K_C = \frac{1}{64}$$

$$NH_3 \rightleftharpoons \frac{1}{2}N_2 + \frac{3}{2}H_2 \rightarrow K_C = \left(\frac{1}{64}\right)^{1/2} = \frac{1}{8}$$

Therefore, the correct option is (4).

Q. 5 Dihydrogen of high purity (> 99.95%) is obtained through :

Option 1:
the reaction of Zn with dilute HCl

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Option 2:
the electrolysis of acidified water using Pt electrodes.

Option 3:
the electrolysis of brine solution.

Option 4:
the electrolysis of warm $Ba(OH)_2$ solution using Ni electrodes.

Correct Answer:
the electrolysis of warm $Ba(OH)_2$ solution using Ni electrodes.

Solution:
High purity (>99.95%) dihydrogen is obtained by electrolysing warm aqueous barium hydroxide solution between nickel electrodes.

Therefore, the correct option is (4).

Q. 6 The reaction of NO with N_2O_4 at 250 K gives:

Option 1:
N_2O

Option 2:
NO_2

Option 3:
N_2O_3

Option 4:
N_2O_5

Correct Answer:
N_2O_3

Solution:
The reaction of NO with N_2O_4 at 250 K

$$2NO + N_2O_4 \xrightarrow{250\ K} \underset{\text{blue solid}}{2\ N_2O_3}$$

Therefore, the correct option is (3).

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Q. 7 The correct match between Item-I (starting material) and Item-II (reagent) for the preparation of benzaldehyde is :

	Item-I		Item-II
(I)	Benzene	(P)	HCl and $SnCl_2, H_3O^+$
(II)	Benzonitrile	(Q)	H_2, Pd-$BaSO_4$, S and quinoline
(II)	Benzoyl Chloride	(R)	CO, HCl and $AlCl_3$

Option 1:
(I) - (Q), (II) - (R) and (III) - (P)

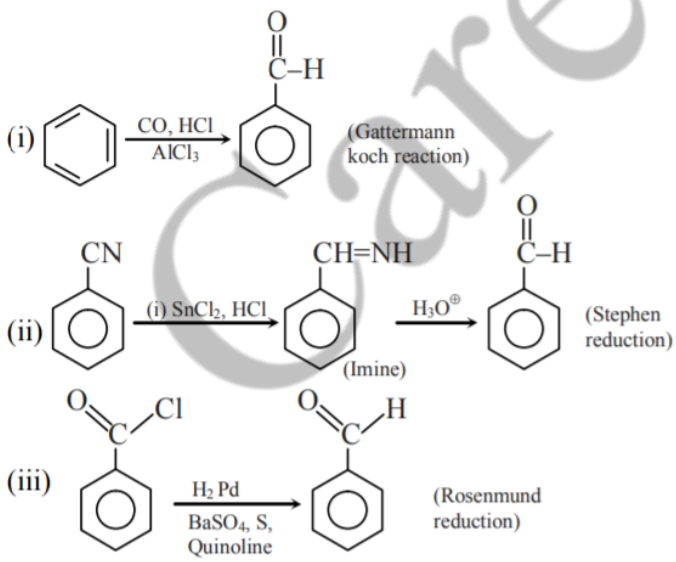
Option 2:
(I) - (P), (II) - (Q) and (III) - (R)

Option 3:
(I) - (R), (II) - (P) and (III) - (Q)

Option 4:
(I) - (R), (II) - (Q) and (III) - (P)

Correct Answer:
(I) - (R), (II) - (P) and (III) - (Q)

Solution:
The preparation of benzaldehyde is -



Therefore, the correct option is (3).

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Q. 8 A crystal is made up of metal ions $'M_1'$ and $'M_2'$ and oxide ions. Oxide ions form a ccp lattice structure. The cation $'M_1'$ occupies 50% of octahedral voids and the cation $'M_2'$ occupies 12.5% of tetrahedral voids of oxide lattice. The Oxidation numbers of $'M_1'$ and $'M_2'$ are, respectively:

Option 1:
$+2, +4$

Option 2:
$+1, +3$

Option 3:
$+3, +1$

Option 4:
$+4, +2$

Correct Answer:
$+2, +4$

Solution:
A crystal is made up of metal ions 'M_1' ana 'M_2' and oxide ions. Oxide ions form a ccp lattice structure.

O2- ions form CCP,

So, there will be a total of 4 Oxygen atoms. (1 corner and 3 face)

Then total charge (Negative) = 4 X (-2) = (-8)

Now,

In the ccp lattice structure, the total number of octahedral voids and tetrahedral voids will be 4 and 8 respectively.

The cation 'M1' occupies 50% of octahedral voids

$$\mathrm{M_1 = 50\% \text{ of O.V.}} \Rightarrow \frac{50}{100} \times 4 = 2$$

The cation 'M2' occupies 12.5% of tetrahedral voids

$$\mathrm{M_2 = 12.5\% \text{ of T.V.}} \Rightarrow \frac{12.5}{100} \times 8 = 1$$

So, Metal ions 'M_1' and 'M_2' are 2 and 1 respectively in oxide.

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So the formula is : $(M_1)_2(M_2)_1O_4$

This must be neutral. Both metals must have a +8 charge in total, to neutralize the Negative (-8) charge.

So, Let's Check Question's Options one by one

1. $(M_1)_2(M_2)_1$ = (+2 X 2) + (+4 X 1) = +8

2. $(M_1)_2(M_2)_1$ = (+1 X 2) + (+3 X 1) = +5

3. $(M_1)_2(M_2)_1$ = (+3 X 2) + (+1 X 1) = +7

4. $(M_1)_2(M_2)_1$ = (+4 X 2) + (+2 X 1) = +1

So, +2 and +4 respectively are correct.

Therefore, the correct option is (1).

Q. 9 The element that can be refind by distillation is :

Option 1:
nickel

Option 2:
zinc

Option 3:
tin

Option 4:
gallium

Correct Answer:
zinc

Solution:
Impure zinc is refined by the distillation method.

Therefore, the correct option is (2).

Q. 10 For a d^4 metal ion in an octahedral field, the correct electronic configuration is :

Option 1:
$t_{2g}^3 e_g^1$ when $\Delta_o < \mathrm{P}$

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Option 2:
$t_{2g}^3 e_g^1$ when $\Delta_o > \mathrm{P}$

Option 3:
$t_{2g}^4 e_g^0$ when $\Delta_o < \mathrm{P}$

Option 4:
$e_g^2 t_{2g}^2$ when $\Delta_o < \mathrm{P}$

Correct Answer:
$t_{2g}^3 e_g^1$ when $\Delta_o < \mathrm{P}$

Solution:
We know

P = Pairing energy

Δ_o = energy separation

(i) If Δ_o < P, the fourth electron enters one of the e_g orbitals giving the configuration $t_{2\ g}^3 e_g^1$. Ligands for which Δo < P are known as weak field ligands and form high spin complexes.

(ii) If Δ_o > P, it becomes more energetically favourable for the fourth electron to occupy a t_{2g} orbital with configuration $t_{2g}^4 e_g^0$. Ligands which produce this effect are known as strong field ligands and form low spin complexes.

But here, the questions does not describe Ligand's spin.

So, Let's check opitons

Option 1) $t_{2g}^3 e_g^1$ when $\Delta_o < \mathrm{P}$

(From Above statements, it is Correct.)

Option 2) $t_{2g}^3 e_g^1$ when $\Delta_o > \mathrm{P}$

(From Above statements, it is incorrect.)

Option 3) $t_{2g}^4 e_g^0$ when $\Delta_o < \mathrm{P}$

(From Above statements, it is incorrect.)

Option 4) $e_g^2 t_{2g}^2$ when $\Delta_o < \mathrm{P}$

(From Above statements, it is incorrect.)

Therefore, the correct option is (1).

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Q. 11 Match the following :

	Test/Method		**Reagent**
(i)	Lucas Test	(a)	$C_6H_5SO_2Cl/aq.KOH$
(ii)	Dumas method	(b)	$HNO_3/AgNO_3$
(iii)	Kjeldahl's method	(c)	CuO/CO_2
(iv)	Hinsberg Test	(d)	$Conc.HCl \text{ and } ZnCl_2$
		(e)	H_2SO_4

Option 1:
(i) - (d), (ii) - (c), (iii) - (b), (iv)-(e)

Option 2:
(i) - (b), (ii) - (d), (iii) - (e), (iv)-(a)

Option 3:
(i) - (d), (ii) - (c), (iii) - (e), (iv)-(a)

Option 4:
(i) - (b), (ii) - (a), (iii) - (c), (iv)-(d)

Correct Answer:
(i) - (d), (ii) - (c), (iii) - (e), (iv)-(a)

Solution:

Test **Correct reagent**

(i) Lucas test $\longrightarrow$ conc. $HCl + ZnCl_2$

(ii) Dumas method $\longrightarrow$ CuO / CO_2

(iii) Kjeldahl's method $\longrightarrow$ H_2SO_4

(iv) Hinsberg Test $\longrightarrow$ $C_6H_5SO_2Cl$ + aq. KOH

Therefore, the correct option is (3).

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Q. 12 Match the following compounds (Column-I) with their uses (Column-II):

S.No	Column-I	S.No	Column-II
(I)	$Ca(OH)_2$	(A)	casts of statues
(II)	$NaCl$	(B)	White wash
(III)	$CaSO_4.\frac{1}{2}H_2O$	(C)	antacid
(IV)	$CaCO_3$	(D)	washing soda preparation

Option 1:
(I) - (D), (II) - (A), (III) - (C), (IV) -(B)

Option 2:
(I) - (B), (II) - (D), (III) - (A), (IV) -(C)

Option 3:
(I) - (B), (II) - (C), (III) - (D), (IV) -(A)

Option 4:
(I) - (C), (II) - (D), (III) - (B), (IV) -(A)

Correct Answer:
(I) - (B), (II) - (D), (III) - (A), (IV) -(C)

Solution:
Compounds and their uses-

(I) $Ca(OH)_2$ is used in white wash
(II) NaCl is used in preparation of washing soda

$$2NH_3 + H_2O + CO_2 \longrightarrow (NH_4)_2CO_3$$
$$(NH_4)_2CO_3 + H_2O + CO_2 \longrightarrow 2NH_4HCO_3$$
$$NH_4HCO_3 + NaCl \longrightarrow NH_4Cl + NaHCO_3(s)$$
$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$$

(III) $CaSO_4.\frac{1}{2}H_2O$ (Plaster of Paris) is used for making casts of statues

(IV) $CaCO_3$ is used as an antacid

Therefore, the correct option is (2).

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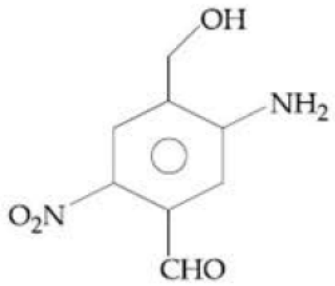
Q. 13 The IUPAC name of the following compound is :



Option 1:
2- nitro-4-hydroxymethyl-5 -aminobenzaldehyde

Option 2:
3-amino-4-hydroxymethyl-2-nitrobenzaldehyde

Option 3:
5-amino-4-hydroxymethyl-2-nitrobenzaldehyde

Option 4:
4-amino-2-formyl-5-hydroxymethyl nitrobenzene

Correct Answer:
5-amino-4-hydroxymethyl-2-nitrobenzaldehyde

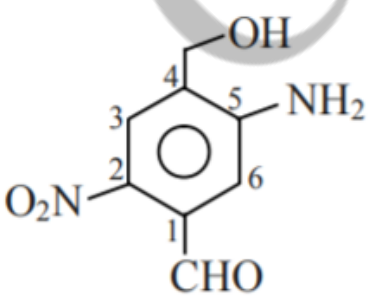
Solution:
Stepwise -

1. CHO comes first in the seniority table.

2. Numbering

Clockwise - 2+4+5 = 11 Minimum (Correct)

AnitClockwise - 3+4+6 = 13



3. Write all substitute Prefix Name with Number

5-**a**mino

4-**h**ydroxymethyl

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2-**n**itro

4. Main group SUffix Name

benzaldehyde

5. write the Prefix name with Number in alphabetical order.

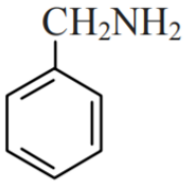
The IUPAC name:

5-amino-4-hydroxymethyl-2-nitrobenzaldehyde

Therefore, the correct option is (3).

Q. 14 Which of the following compounds can be prepared in good yield by Gabriel phthalimide synthesis?

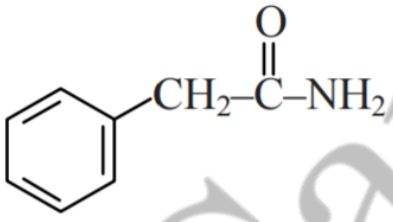
Option 1:



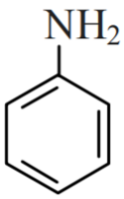
Option 2:

$CH_3 - CH_2 - NHCH_3$

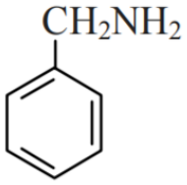
Option 3:



Option 4:



Correct Answer:



Solution:

Gabriel Phthalimide Synthesis is used for the preparation of 1° Aliphatic amine from an Alkyl halide.

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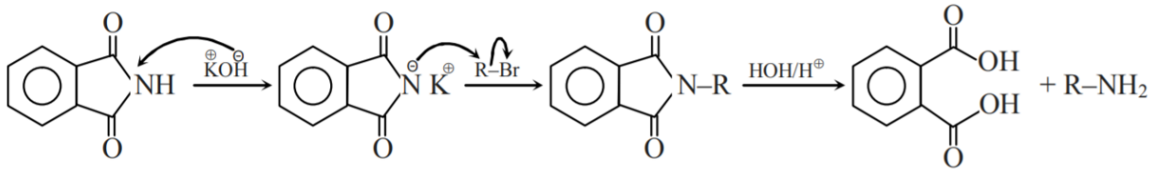
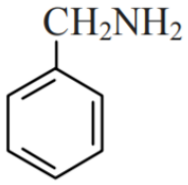
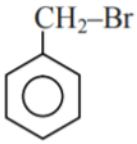
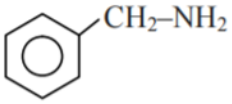


Thus, only Benzylamine can be prepared by Gabriel Phthalimide Synthesis



Here R–Br = 

R–NH2 = 

Therefore, the correct option is (1).

Q. 15 A set of solutions is prepared using 180 g of water as a solvent and 10 g of different non-volatile solutes A,B and C. The relative lowring of vapour pressure in the presence of these solutes are in the order [Given, molar mass of A=100 g mol $^{-1}$; B=200 g mol $^{-1}$; C=10,000 g mol $^{-1}$]

Option 1:
$B > C > A$

Option 2:
$C > B > A$

Option 3:
$A > B > C$

Option 4:
$A > C > B$

Correct Answer:
$A > B > C$

Solution:
Here vapour pressure of water is lowering.

We know this formula,

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Partial Vapour pressure of water $P_{water} = P^oX_{water}$

Lowering Vapour pressure of water = $P^o - P_{water}$

The relative lowering of vapour pressure $= \frac{P^o - P}{P^o} = \frac{\Delta P}{P^o} = 1 - \chi_{water}$

$$\frac{\Delta P}{P^o} = \chi_{solute} = \frac{\frac{m_{solute}}{M_{solute}}}{\frac{m_{solute}}{M_{solute}} + \frac{m_{water}}{M_{water}}}$$

Now,

$$\left(\frac{\Delta P}{P^0}\right)_A = \frac{\frac{10}{100}}{\frac{10}{100} + \frac{180}{18}}$$

$$\left(\frac{\Delta P}{P^0}\right)_B = \frac{\frac{10}{200}}{\frac{10}{200} + \frac{180}{18}}$$

$$\left(\frac{\Delta P}{P^0}\right)_C = \frac{\frac{10}{10,000}}{\frac{10}{10,000} + \frac{180}{18}}$$

From above,

$$\left(\frac{\Delta P}{P^0}\right)_A > \left(\frac{\Delta P}{P^0}\right)_B > \left(\frac{\Delta P}{P^0}\right)_C$$

Q. 16 For the given cell; $Cu(s)Cu^{2+}(C_1M) \parallel Cu^{2+}(C_2M) \mid Cu(s)$ change in Gibbs energy (ΔG) is negative, if :

Option 1:
$C_1 = C_2$

Option 2:
$C_2 = \frac{C_1}{\sqrt{2}}$

Option 3:
$C_1 = 2C_2$

Option 4:
$C_2 = \sqrt{2}C_1$

Correct Answer:
$C_2 = \sqrt{2}C_1$

Solution:

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We know this formula

$\Delta G = -n\,F\,E_{cell}$

ΔG is negative, if E_{cell} is positive

$$\begin{array}{l} \text{Anode : } Cu(s) \longrightarrow Cu^{+2}(C_1) + 2e^- : E^\circ \\ \text{Cathode : } Cu^{+2}(C_2) + 2e^- \longrightarrow Cu\,(S) : -E^\circ \\ \hline \text{Cell reaction : } Cu^{2+}(C_2) \longrightarrow Cu^{+2}(C_1)\ \ E^\circ_{cell} = 0 \end{array}$$

$$E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF}\log Q$$

$$E_{cell} = 0 - \frac{2.303RT}{nF}\log\left(\frac{C_1}{C_2}\right)$$

$$E_{cell} > 0;\ \text{if } \frac{C_1}{C_2} < 1 \Rightarrow C_1 < C_2$$

Only $C_2 = \sqrt{2}C_1$ following the above condition.

Therefore, the correct option is (4).

Q. 17 Reaction of an inorganic sulphite X with dilute H_2SO_4 generates compound Y.Reaction of Y with NaOH gives X. Further, the reaction of X with Y and water affords compound Z. Y and Z, respectively, are :

Option 1:

$SO_2 \ \ and \ \ Na_2SO_3$

Option 2:

$SO_3 \ \ and \ \ NaHSO_3$

Option 3:

$SO_2 \ \ and \ \ NaHSO_3$

Option 4:

$S \ \ and \ \ Na_2SO_3$

Correct Answer:

$SO_2 \ \ and \ \ NaHSO_3$

Solution:

The reaction will be-

$$\underset{(X)}{Na_2SO_3} \xrightarrow{\text{dil. } H_2SO_4} \underset{(Y)}{SO_2} \xrightarrow{NaOH} \underset{(X)}{Na_2SO_3} \xrightarrow{SO_2 + H_2O} \underset{(Z)}{NaHSO_3}$$

So, Y and Z, respectively, are $SO_2 \ \ and \ \ NaHSO_3$

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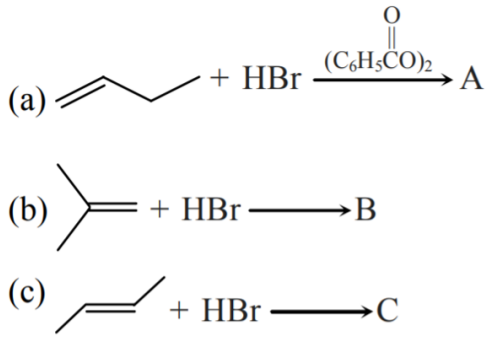
Therefore, the correct option is (3).

Q. 18 The increasing order of the boiling points of the major products A,B, and C of the following reactions will be :



Option 1:
B < C < A

Option 2:
C < A < B

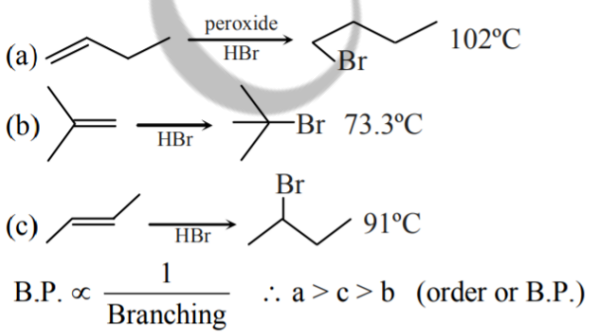
Option 3:
A < B < C

Option 4:
A < C < B

Correct Answer:
B < C < A

Solution:
As we have learnt,

Boiling Point decreases with the increase in branching of the molecule.



Therefore, the correct option is (1).

Q. 19 Mischmetal is an alloy consisting mainly of :

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Option 1:
lanthanoid metals

Option 2:
actionoid and transition metals

Option 3:
Lanthanoid and actionoid metals

Option 4:
actionoid metals

Correct Answer:
lanthanoid metals

Solution:
Mischmetal is an alloy of lanthanoid metals with Fe.

It consists of a lanthanoid metal (~95%) and iron (~5%) and traces of S, C, Ca, and Al.

Therefore, the correct option is (1).

Q. 20 The correct match between Item - I and Item - II is :

	Item - I		**Item - II**
(a)	Natural Rubber	(I)	1, 3-butadiene + styrene
(b)	Neoprene	(II)	1,3-butadiene +acrylonitrile
(c)	Buna - N	(III)	Chloroprene
(d)	Buna-S	(IV)	Isoprene

Option 1:
(a) - (III), (b) - (IV), (c) - (I), (d) - (II)

Option 2:
(a) - (III), (b) - (IV), (c) - (II), (d) - (I)

Option 3:
(a) - (IV), (b) - (III), (c) - (II), (d) - (I)

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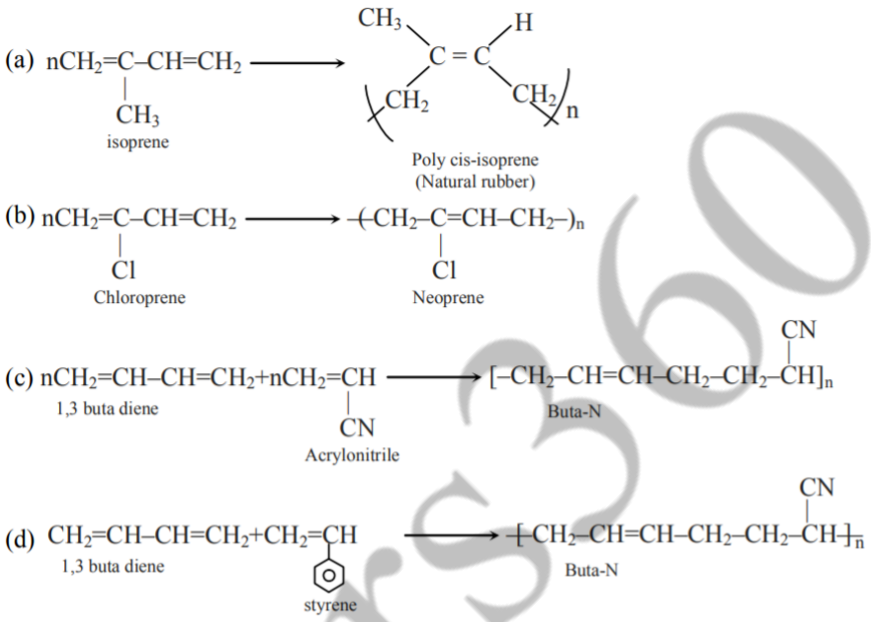
Option 4:
(a) - (IV), (b) - (III), (c) - (I), (d) - (II)

Correct Answer:
(a) - (IV), (b) - (III), (c) - (II), (d) - (I)

Solution:

We know these polymers -



Correct match : (a) - (IV), (b) - (III), (c) - (II), (d) - (I)

Therefore, the correct option is (3).

Q. 21 If the solubility product of AB_2 is $3.20 \times 10^{-11} M^3$, then the solubility of AB_2 in pure water is _____$\times 10^{-4} molL^{-1}$.

[Assuming that neither kind of ion reacts with water]

Correct Answer:
2

Solution:
If Solubility is s then

$$AB_2(s) \rightleftharpoons \underset{S}{A^{+2}_{(aq.)}} + \underset{2S}{2B^{-}_{(aq.)}}$$

$K_{SP} = s^1 \times (2s)^2 = 4s^3$

$3.2 \times 10^{-11} = 4 \times s^3$

$s = 2 \times 10^{-4}$ M/L

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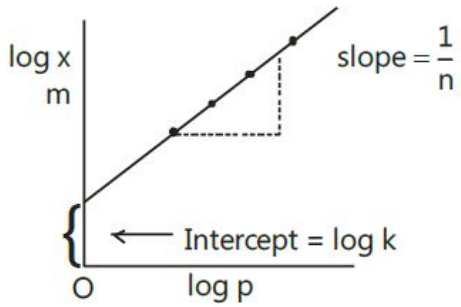
Ans = 2

Q. 22 For Freundlich adsorption isotherm, a plot of log $(\frac{x}{m})$ (y-axis) and log p(x-axis) gives a stright line. The intercept and slope for the line is 0.4771 and 2, respectively. The mass of gas, adsorbed per gram of adsorbent if the initial pressure is 0.04 atm, is _____$\times 10^{-4} g.$ (log 3 = 0.4771)

Correct Answer:
48

Solution:
For Freundlich adsorption isotherm, a plot of log (x/m) (y-axis) and log p (x-axis) gives a straight line.



So, The mass of gas, adsorbed per gram of adsorbent means Need to find x/m which is -

$$\frac{x}{m} = KP^{1/n}$$

$$\log\left(\frac{x}{m}\right) = \frac{1}{n}\log P + \log K$$

$$\text{Slope} = \frac{1}{n} = 2$$

$$\text{intercept} = \log K = 0.4771$$

$$K = 3$$

$$\text{mass of gas adsorbed per gm of adsorbent} = \frac{x}{m}$$

$$\frac{x}{m} = 3 \times (0.04)^2 = 48 \times 10^{-4}$$

Ans = 48

Q. 23 A solution of phenol in chloroform when treated with aqueous NaOH gives compound P as a major product. The mass percentage of carbon in P is _____(to the nearest integer)

((Atomic mass : C=12;H=1;O=16)

Correct Answer:
69

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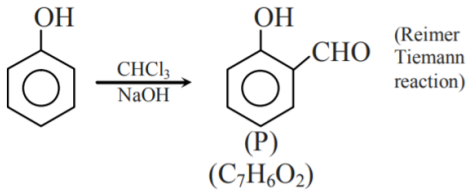
Solution:
A solution of phenol in chloroform when treated with aqueous NaOH gives compound P as a major product.



Now,

The mass percentage of carbon in (P)

= (total mass of C in P) /(total mass of P) X 100

Total mass of P = The molecular weight of $C_7H_6O_2$ = 122

Total mass of C in P = 12 X 7

,So,

$$\%C = \frac{12 \times 7 \times 100}{122} = 68.85 \approx 69$$

Ans = 69

Q. 24 The atomic number of Unnilunium is _____.

Correct Answer:
101

Solution:
Let's break the IUPAC Name-

Unnilunium = Un + nil + Un + ium

Un = 1

nil = 0

un = 1

So, The atomic number will be 101.

Ans = 101

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Q. 25 The rate of a reaction decreased by 3.555 times when the temperature was changed from 40^oC to 30^oC. The activation energy (in kJ mol $^{-1}$) of the reaction is _____.

Take ; R=8314 J mol $^{-1}$ K $^{-1}$ ln 3.555= 1.268

Correct Answer:
100

Solution:
We know this formula for activation energy,

$$\ln\left(\frac{\mathrm{K_{T_2}}}{\mathrm{K_{T_1}}}\right)=\frac{\mathrm{E_a}}{\mathrm{R}}\left[\frac{1}{\mathrm{T_1}}-\frac{1}{\mathrm{T_2}}\right]$$

Given,

$$\mathrm{T_1}=273+30=303\ \mathrm{K};\mathrm{T_2}=272+40=313\ \mathrm{K}$$

$$\frac{\mathrm{K_{T_2}}}{\mathrm{K_{T_1}}}=3.555$$

$$\ln(3.555)=\frac{\mathrm{E_a}}{8.314}\left[\frac{1}{303}-\frac{1}{313}\right]$$

$$1.268=\frac{\mathrm{E_a}}{8.314}\left[\frac{10}{303\times313}\right]$$

$$\mathrm{E_a}=\frac{1.268\times8.314\times313\times303}{10}$$

$$\mathrm{E_a}=99980.7\ \mathrm{J/mol}$$

$$\mathrm{E_a}=99.98\ \mathrm{kJ/mol}$$

$$\mathrm{E_a}=100\ \mathrm{kJ/mol}$$

Ans = 100

Maths

Q. 1 The integral $\int_1^2 e^x.x^x(2+\log_e x)dx$ equals:

Option 1:
$e(4e+1)$

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Option 2:
$4e^2 - 1$

Option 3:
$e(4e - 1)$

Option 4:
$e(2e - 1)$

Correct Answer:
$e(4e - 1)$

Solution:

$$\int_1^2 e^x \cdot x^x \left(2 + \log_e x\right) dx$$

$$\int_1^2 e^x \left(2x^x + x^x \log_e x\right) dx$$

$$\int_1^2 e^x (\underbrace{x^x}_{f(x)} + \underbrace{x^x \left(1 + \log_e x\right)}_{f'(x)}) dx$$

$$(e^x \cdot x^x)_1^2 = 4e^2 - e$$

Q. 2 The area (in sq.units) of the region enclosed by the curves $y = x^2 - 1$ and $y = 1 - x^2$ is equal to :

Option 1:
$\frac{4}{3}$

Option 2:
$\frac{8}{3}$

Option 3:
$\frac{7}{2}$

Option 4:
$\frac{16}{3}$

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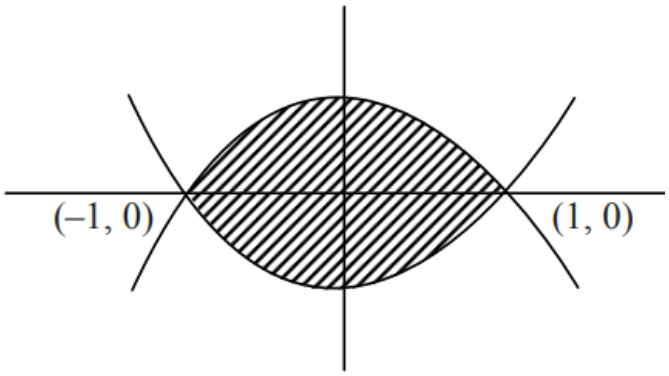
Correct Answer:

$\frac{8}{3}$

Solution:



$$A = \int_{-1}^{1} \left(\left(1 - x^2\right) - \left(x^2 - 1\right) \right) dx$$

$$A = \int_{-1}^{1} \left(\left(2 - 2x^2\right) dx - 4 \int_{0}^{1} \left(1 - x^2\right) dx \right.$$

$$A = 4\left(x - \frac{x^3}{3}\right)_0^1 = 4\left(\frac{2}{3}\right) = \frac{8}{3}$$

Q. 3 The angle of elevation of the summit of a mountain from a point on the ground is 45^o. After climbing up one km towards the summit at an inclination of 30^o from the ground, the angle of elevation of the summit is found to be 60^o. Then the height (in km) of the summit from the ground is :

Option 1:

$\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Option 2:

$\frac{\sqrt{3}+1}{\sqrt{3}-1}$

Option 3:

$\frac{1}{\sqrt{3}-1}$

Option 4:

$\frac{1}{\sqrt{3}+1}$

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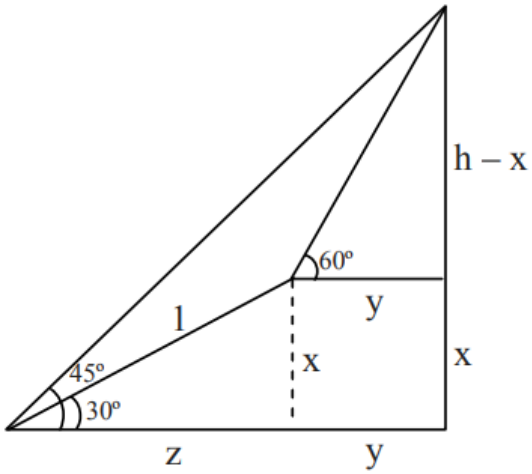
Correct Answer:

$$\frac{1}{\sqrt{3}-1}$$

Solution:



$$\sin 30^\circ = x \Rightarrow x = \frac{1}{2}$$

$$\cos 30^\circ = z \Rightarrow z = \frac{\sqrt{3}}{2}$$

$$\tan 45^\circ = \frac{\mathrm{h}}{\mathrm{y}+\mathrm{z}} \Rightarrow \mathrm{h} = \mathrm{y}+\mathrm{z}$$

$$\tan 60^\circ = \frac{\mathrm{h}-\mathrm{x}}{\mathrm{y}} \Rightarrow \tan 60^\circ = \frac{\mathrm{h}-\mathrm{x}}{\mathrm{h}-\mathrm{z}}$$

$$\sqrt{3}(h-z) = h - x$$

$$(\sqrt{3}-1)h = \sqrt{3}z - x$$

$$\Rightarrow (\sqrt{3}-1)h = \frac{3}{2} - \frac{1}{2}$$

$$\Rightarrow (\sqrt{3}-1)h = 1$$

$$h = \frac{1}{\sqrt{3}-1}$$

Q. 4 The set of all real values of λ for which the function $f(x) = (1-\cos^2 x).(\lambda + \sin\ x)$, $x\epsilon\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, has exactly one maxima and exactly one minima, is :

Option 1:

$$\left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$$

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Option 2:

$\left(-\frac{3}{2}, \frac{3}{2}\right)$

Option 3:

$\left(-\frac{1}{2}, \frac{1}{2}\right)$

Option 4:

$\left(-\frac{3}{2}, \frac{3}{2}\right) - \{0\}$

Correct Answer:

$\left(-\frac{3}{2}, \frac{3}{2}\right) - \{0\}$

Solution:

$f(x) = \left(1 - \cos^2 x\right)(\lambda + \sin x)$

$x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$f(x) = \lambda \sin^2 x + \sin^3 x$

$f'(x) = 2\lambda \sin x \cos x + 3 \sin^2 x \cos x$

$f'(x) = \sin x \cos x(2\lambda + 3 \sin x)$

$\sin x = 0, \frac{-2\lambda}{3}, (\lambda \neq 0)$

for exactly one maxima & minima

$\frac{-2\lambda}{3} \in (-1, 1) \Rightarrow \lambda \in \left(\frac{-3}{2}, \frac{3}{2}\right)$

$\lambda \in \left(-\frac{3}{2}, \frac{3}{2}\right) - \{0\}$

Q. 5 If α and β are the roots of the equation $2x(2x + 1) = 1$ then β is equal to :

Option 1:

$2\alpha(\alpha + 1)$

Option 2:

$-2\alpha(\alpha + 1)$

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Option 3:
$2\alpha(\alpha-1)$

Option 4:
$2\alpha^2$

Correct Answer:
$-2\alpha(\alpha+1)$

Solution:
α and β are the roots of the equation $4x^2+2x-1=0$

$4\alpha^2+2\alpha=1 \Rightarrow \frac{1}{2}=2\alpha^2+\alpha$

$\alpha+\beta=\frac{-b}{a}$

$\alpha+\beta=\frac{-1}{2}$

$\beta=\frac{-1}{2}-\alpha$

Using equation (1)

$\beta=-\left(2\alpha^2+\alpha\right)-\alpha$

$\beta=-2\alpha^2-2\alpha$

$\beta=-2\alpha(\alpha+1)$

Q. 6 For all twice differentiale functions $f: R \to R$, with $f(0)=f(1)=f'(0)=0$

Option 1:
$f''(x) \neq 0$ at every point $x\epsilon(0,1)$

Option 2:
$f''(x)=0$, for some $x\epsilon(0,1)$

Option 3:
$f''(0)=0$

Option 4:
$f''(x)=0$, at every point $x\epsilon(0,1)$

Correct Answer:
$f''(x)=0$, for some $x\epsilon(0,1)$

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Solution:

$f(0) = f(1) = f'(0) = 0$

Apply Rolles theorem on y = f(x) in x $\in [0, 1]$

$f(0) = f(1) = 0$

$\Rightarrow$ f′(a) = 0 where $\alpha \in (0, 1)$

Now apply Rolles theorem on y = f′(x) in x $\in [0, \alpha]$

f′(0) = f′(α) = 0 and f′(x) is continuous and differentiable

Now apply Rolles theorem on y = f′(x) in x $\in [0, \alpha]$

f′(0) = f′(α) = 0 and f′(x) is continuous and differentiable

$\Rightarrow$ f″(b) = 0 for some $, \beta \in (0, \alpha) \in (0, 1)$

$\Rightarrow$ f″(x) = 0 for some x $\in (0, 1)$

Q. 7 If $y = \left(\frac{2}{\pi}x - 1\right)\csc x$ is the solution of the differential equation, $\frac{dy}{dx} + p(x)y = \frac{2}{\pi}\csc x,\ 0 < x < \frac{\pi}{2}$, then the function p(x) is equal to :

Option 1:

$\cot x$

Option 2:

$\csc x$

Option 3:

$\sec x$

Option 4:

$\tan x$

Correct Answer:

$\cot x$

Solution:

$y = \left(\frac{2x}{\pi} - 1\right)\operatorname{cosec} x$

$\frac{dy}{dx} = \frac{2}{\pi}\operatorname{cosec} x - \left(\frac{2x}{\pi} - 1\right)\operatorname{cosec} x \cot x$

$\frac{dy}{dx} = \frac{2\operatorname{cossec} x}{\pi} - y\cot x$

using equation (1)

$\frac{dy}{dx} + y\cot x = \frac{2\operatorname{cosec} x}{\pi}$

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$$\frac{dy}{dx} + p(x) \cdot y = \frac{2\,\text{cosec}\,x}{\pi} x \in \left(0, \frac{\pi}{2}\right)$$

$$\text{Compare : } p(x) = \cot x$$

Q. 8 Let L denote the line in the XY-plane with x and y intercepts as 3 and 1 respectively. Then the image of the point $(-1, -4)$ in this line is :

Option 1:
$\left(\frac{11}{5}, \frac{28}{5}\right)$

Option 2:
$\left(\frac{29}{5}, \frac{8}{5}\right)$

Option 3:
$\left(\frac{8}{5}, \frac{29}{5}\right)$

Option 4:
$\left(\frac{29}{5}, \frac{11}{5}\right)$

Correct Answer:
$\left(\frac{11}{5}, \frac{28}{5}\right)$

Solution:

$$L : \frac{x}{3} + \frac{y}{1} = 1 \Rightarrow x + 3y - 3 = 0$$

$$\text{Image of point } (-1, -4)$$

$$\frac{x+1}{1} = \frac{y+4}{3} = -2\left(\frac{-1-12-3}{10}\right)$$

$$\frac{x+1}{1} = \frac{y+4}{3} = \frac{16}{5}$$

$$(x, y) \equiv \left(\frac{11}{5}, \frac{28}{5}\right)$$

Q. 9 If the tangent to the curve, $y = f(x) = x\ \log_e x, (x > 0)$ at a point $(c, f(c))$ is parallel to the line - segment joining the points $(1, 0)$ and (e, e), then c is equal to :

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Option 1:
$\frac{e-1}{e}$

Option 2:
$e^{\left(\frac{1}{e-1}\right)}$

Option 3:
$e^{\left(\frac{1}{1-e}\right)}$

Option 4:
$\frac{1}{e-1}$

Correct Answer:
$e^{\left(\frac{1}{e-1}\right)}$

Solution:
$f(x) = x\log_e x$

$$f'(x)|_{(c,f(c))} = \frac{e-0}{e-1}$$

$$f'(x) = 1 + \log_e x$$

$$f'(x)|_{(c,f(c))} = 1 + \log_e c = \frac{e}{e-1}$$

$$\log_c c = \frac{e-(e-1)}{e-1} = \frac{1}{e-1} \Rightarrow c = e^{\frac{1}{e-1}}$$

Q. 10 Let $f : R \to R$ be a function defined by $f(x) = max\ \{x, x^2\}$. Let S denote the set of all points in R, where f is not differentiable. Then set S is:

Option 1:
$\{0, 1\}$

Option 2:
$\{0\}$

Option 3:
ϕ (an empty set)

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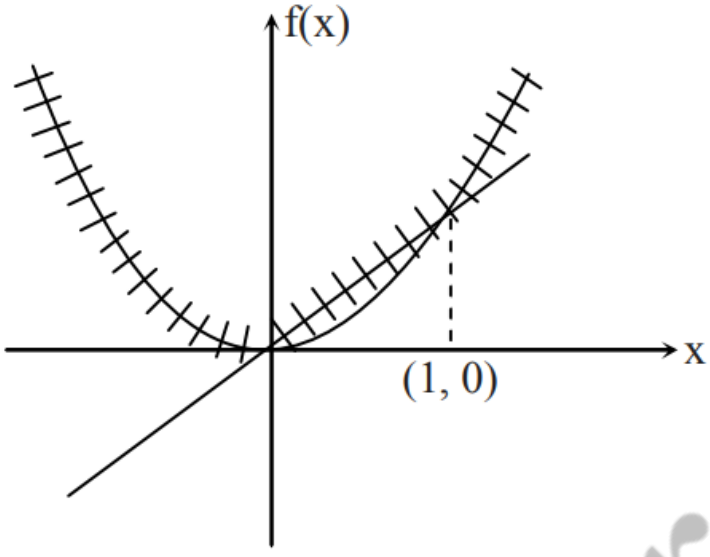
Option 4:
$\{1\}$

Correct Answer:
$\{0, 1\}$

Solution:

$f(x) = \max(x, x^2)$



Clearly f(x) having sharp edge at x = 0 and x = 1

Hence, f(x) is Non-differentiable at x=0,1

S = {0,1}

Q. 11 Let $\theta = \frac{\pi}{5}$ and $A = \begin{bmatrix} \cos\ \theta & \sin\ \theta \\ -\sin\ \theta & \cos\ \theta \end{bmatrix}$. If $B = A + A^4$, then det (B) :

Option 1:
is one

Option 2:
Lies in (2,3)

Option 3:
is zero.

Option 4:
Lies in (1,2).

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Correct Answer:
Lies in (1,2).

Solution:

$$A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$\mathrm{B} = \mathrm{A} + \mathrm{A}^4 = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} + \begin{bmatrix} \cos 4\theta & \sin 4\theta \\ -\sin 4\theta & \cos 4\theta \end{bmatrix}$$

$$\mathrm{B} = \begin{bmatrix} (\cos\theta + \cos 4\theta) & (\sin\theta + \sin 4\theta) \\ -(\sin\theta + \sin 4\theta) & (\cos\theta + \cos 4\theta) \end{bmatrix}$$

$$|\mathrm{B}| = \left(\cos\theta + \cos^4\theta\right)^2 + (\sin\theta + \sin 4\theta)^2$$

$$|\mathrm{B}| = 2 + 2\cos 3\theta, \text{ when } \theta = \frac{\pi}{5}$$

$$|\mathrm{B}| = 2 + 2\cos\frac{3\pi}{5} = 2(1 - \sin 18)$$

$$|\mathrm{B}| = 2\left(1 - \frac{\sqrt{5}-1}{4}\right) = 2\left(\frac{5-\sqrt{5}}{4}\right) = \frac{5-\sqrt{5}}{2}$$

Q. 12 A plane P meets the coordinate axes at A, B and C respectively. The centroid of ΔABC is given to be (1,1,2). Then the equation of the line through this centroid and perpendicular to the plane P is :

Option 1:
$$\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-2}{1}$$

Option 2:
$$\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-2}{2}$$

Option 3:
$$\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{1}$$

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Option 4:

$\frac{x-1}{1}=\frac{y-1}{2}=\frac{z-2}{2}$

Correct Answer:

$\frac{x-1}{2}=\frac{y-1}{2}=\frac{z-2}{1}$

Solution:

$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

$A\equiv(a,0,0), B\equiv(0,b,0), C\equiv(0,0,c)$

$\text{Centroid } \equiv\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)=(1,1,2)$

a=3, b=3, c=6

$\text{Plane: } \quad \frac{x}{3}+\frac{y}{3}+\frac{z}{6}=1$

$2x+2y+z=6$

$\text{Line } \perp \text{ to the plane (DR of line } =2\hat{i}+2\hat{j}+\hat{k})$

$\frac{x-1}{2}=\frac{y-1}{2}=\frac{z-2}{1}$

Q. 13 The common difference of the $A.P.\, b_1, b_2, \ldots b_m$ is 2 more than the common difference of $A.P.\, a_1, a_2, \ldots a_n$. If $a_{40}=-159, a_{100}=-399,$ and $b_{100}=a_{70},$ then b_1 is equal to :

Option 1:

81

Option 2:

-127

Option 3:

-81

Option 4:

127

Correct Answer:

-81

Solution:

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$a_1, a_2, \ldots, an \rightarrow (CD = d)$

$\mathrm{b_1},\ \mathrm{b_2}, \ldots, \mathrm{b_m} \rightarrow (\mathrm{CD} = \mathrm{d} + 2)$

$a_{40} = a + 39d = -159$

$a_{100} = a + 99d = -399$

$\text{Subtract} \ : 60\ \mathrm{d} = -240 \Rightarrow \mathrm{d} = -4$

Using equation (1)

$a + 39(-4) = -159$

$a = 156 - 159 = -3$

$a_{70} = a + 69d = -3 + 69(-4) = -279 = b_{100}$

$b_{100} = -279$

$b_1 + 99(d + 2) = -279$

$b_1 - 198 = -279 \Rightarrow b_1 = -81$

Q. 14 If the normal at an end of a latus rectum of an ellipse passes through an extremity of the minor axis, then the eccentricity e of the ellipse satisfies:

Option 1:

$e^4 + 2e^2 - 1 = 0$

Option 2:

$e^2 + e - 1 = 0$

Option 3:

$e^4 + e^2 - 1 = 0$

Option 4:

$e^2 + 2e - 1 = 0$

Correct Answer:

$e^4 + e^2 - 1 = 0$

Solution:

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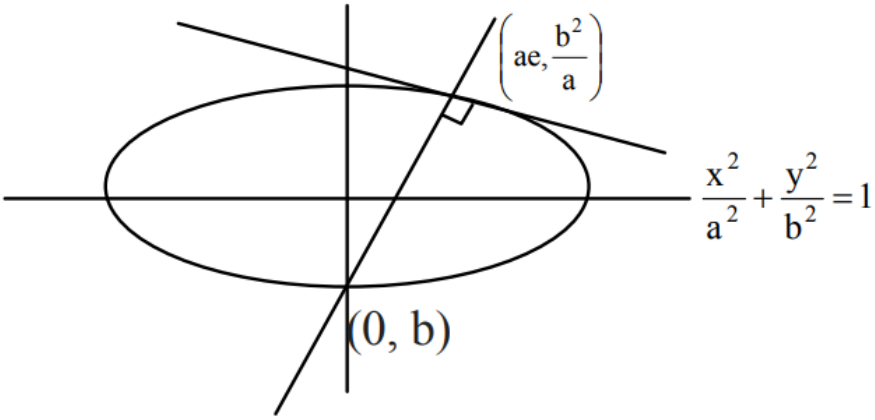

$$\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2e^2$$

$$\frac{a^2x}{ae}-\frac{b^2y}{b^2}\cdot a=a^2e^2$$

$$\frac{ax}{e}-ay=a^2e^2 \Rightarrow \frac{x}{e}-y=ae^2$$

passes through $(0,b)$

$$-b=ae^2 \Rightarrow b^2=a^2e^4$$

$$a^2\left(1-e^2\right)=a^2e^4 \Rightarrow e^4+e^2=1$$

Q. 15 For a suitably chosen real constant a, let a function, $f: R-\{-a\} \to R$ be defined by $f(x)=\frac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and $f(x) \neq -a$, $(fof)(x)=x$. Then $f\left(-\frac{1}{2}\right)$ is equal to :

Option 1:

$\frac{1}{3}$

Option 2:

$-\frac{1}{3}$

Option 3:

-3

Option 4:

3

Correct Answer:

3

Solution:

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$$f(x)=\frac{a-x}{a+x} \quad x \in R-\{-a\} \rightarrow R$$

$$f(f(x))=\frac{a-f(x)}{a+f(x)}=\frac{a-\left(\frac{a-x}{a+x}\right)}{a+\left(\frac{a-x}{a+x}\right)}$$

$$f(f(x))=\frac{\left(a^2-a\right)+x(a+1)}{\left(a^2+a\right)+x(a-1)}=x$$

$$\Rightarrow \left(a^2-a\right)+x(a+1)=\left(a^2+a\right)x+x^2(a-1)$$

$$\Rightarrow a(a-1)+x\left(1-a^2\right)-x^2(a-1)=0$$

$$\Rightarrow a=1$$

$$f(x)=\frac{1-x}{1+x}, \Rightarrow f\left(\frac{-1}{2}\right)=\frac{1+\frac{1}{2}}{1-\frac{1}{2}}=3$$

Q. 16 If the constant term in the binomial expansion of $\left(\sqrt{x}-\frac{k}{x^2}\right)^{10}$ is 405, then $|k|$ equals:

Option 1:
9

Option 2:
1

Option 3:
3

Option 4:
2

Correct Answer:
3

Solution:

$$\left(\sqrt{x}-\frac{k}{x^2}\right)^{10}$$

$$T_{r+1}={}^{10}C_r(\sqrt{x})^{10-r}\left(\frac{-k}{x^2}\right)^r$$

$$T_{r+1}={}^{10}C_r x^{\frac{10-r}{2}}\cdot(-k)^r\cdot x^{-2r}$$

$$T_{r+1}={}^{10}C_r x^{\frac{10-5r}{2}}\cdot(-k)^r$$

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$$\text{Constant term } : \frac{10-5r}{2}=0 \Rightarrow r=2$$

$$T_3 = {}^{10}C_2 \cdot (-k)^2 = 405$$

$$k^2 = \frac{405}{45} = 9$$

$$k = \pm 3 \Rightarrow |k| = 3$$

Q. 17 The centre of the circle passing through the point (0,1) and touching the parabola $y = x^2$ at the point (2,4) is :

Option 1:
$\left(\frac{-53}{10}, \frac{16}{5}\right)$

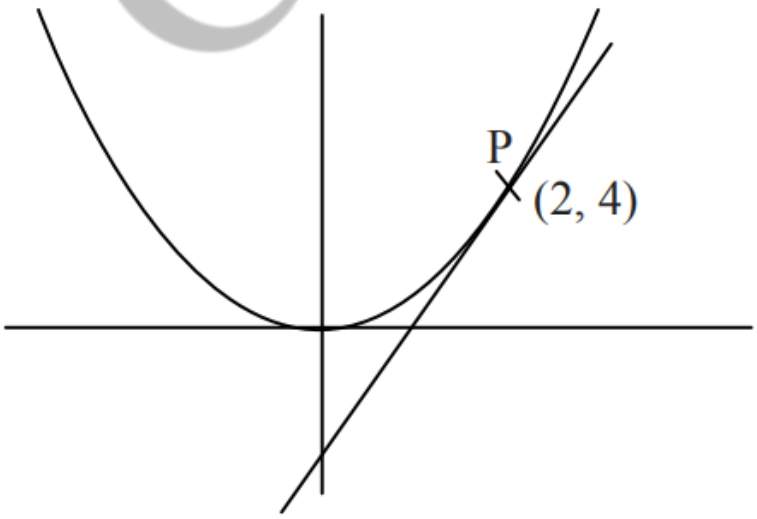
Option 2:
$\left(\frac{6}{5}, \frac{53}{10}\right)$

Option 3:
$\left(\frac{3}{10}, \frac{16}{5}\right)$

Option 4:
$\left(\frac{-16}{5}, \frac{53}{10}\right)$

Correct Answer:
$\left(\frac{-16}{5}, \frac{53}{10}\right)$

Solution:



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$y = x^2$

$\left.\frac{dy}{dx}\right|_P = 4$

$(y-4) = 4(x-2)$

$4x - y - 4 = 0$

$\text{Circle } : (x-2)^2 + (y-4)^2 + \lambda(4x - y - 4) = 0$

$\text{passes through } (0,1)$

$4 + 9 + \lambda(-5) = 0 \Rightarrow \lambda = \frac{13}{5}$

$\text{Circle } : x^2 + y^2 + x(4\lambda - 4) + y(-\lambda - 8) + (20 - 4\lambda) = 0$

$\text{Centre : } \left(2 - 2\lambda, \frac{\lambda + 8}{2}\right) \equiv \left(\frac{-16}{5}, \frac{53}{10}\right)$

Q. 18 Let $z = x + iy$ be a non-zero complex number such that $z^2 = i|z|^2$, where $i = \sqrt{-1}$, then z lies on the:

Option 1:

$\text{line}, y = -x$

Option 2:

imaginary axis

Option 3:

$\text{line}, y = x$

Option 4:

real axis

Correct Answer:

$\text{line}, y = x$

Solution:

$z = x + iy$

$z^2 = i|z|^2$

$(x + iy)^2 = i\left(x^2 + y^2\right)$

$\left(x^2 - y^2\right) - i\left(x^2 + y^2 - 2xy\right) = 0$

$(x-y)(x+y) - i(x-y)^2 = 0$

$(x-y)((x+y) - i(x-y)) = 0$

$\Rightarrow x = y$

$z \text{ lies on } y = x$

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Q. 19 Consider the statement : " For an integer n, if n^3-1 is even, then n is odd." The contrapositive statement of this statement is :

Option 1:
For an integer n, if n is even, then n^3-1 is odd.

Option 2:
For an integer n, if n^3-1 is not even, then n is not odd.

Option 3:
For an integer n, if n is even, then n^3-1 is even.

Option 4:
For an integer n, if n is odd, then n^3-1 is even.

Correct Answer:
For an integer n, if n is even, then n^3-1 is odd.

Solution:
$\text{Contrapositive of } (\text{p} \rightarrow \text{q}) \text{ is } \sim \text{q} \rightarrow \sim \text{p}$

$\text{For an integer } n \text{ , if } n \text{ is even then } \left(n^3-1\right) \text{ is odd}$

Q. 20 The probabilities of three events A, B and C are given by $P(A)=0.6, P(B)=0.4$ and $P(C)=0.5$. If $P(A \cup B)=0.8, P(A \cap C)=0.3, P(A \cap B \cap C)=0.2$, $P(B \cap C)=\beta$ and $P(A \cup B \cup C)=\alpha$ where $0.85 \leq \alpha \leq 0.95$, then β lies in the interval:

Option 1:
$[0.35, 0.36]$

Option 2:
$[0.25, 0.35]$

Option 3:
$[0.20, 0.25]$

Option 4:
$[0.36, 0.40]$

Correct Answer:
$[0.25, 0.35]$

Solution:

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$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$0.8 = 0.6 + 0.4 - P(A \cap B)$
$P(A \cap B) = 0.2$
$P(A \cup B \cup C) = \Sigma P(A) - \Sigma P(A \cap B) + P(A \cap B \cap C)$
$\alpha = 1.5 - (0.2 + 0.3 + \beta) + 0.2$
$\alpha = 1.2 - \beta \in [0.85, 0.95]$
(where $\alpha \in [0.85, 0.95]$)
$\beta \in [0.25, 0.35]$

Q. 21 Suppose that a function $f : R \to R$ satisfies $f(x+y) = f(x)f(y)$ for all $x, y \epsilon R$ and $f(1) = 3$. If $\sum_{i=1}^{n} f(i) = 363$, then n is equal to _____

Option 1:
15

Option 2:
5

Option 3:
25

Option 4:
3

Correct Answer:
5

Solution:
$(x+y) = f(x)f(y)$
put $x = y = 1, \quad f(2) = (f(1))^2 = 3^2$
put $x = 2, y = 1, \quad f(3) = (f(1))^3 = 3^3$
$\vdots$
Similarly $f(x) = 3^x$

$$\sum_{i=1}^{n} f(i) = 363 \Rightarrow \sum_{i=1}^{n} 3^i = 363$$

$$\left(3 + 3^2 + \ldots + 3^n\right) = 363$$

$$\frac{3\left(3^n - 1\right)}{2} = 363$$

$$3^n - 1 = 242 \Rightarrow 3^n = 243$$

$$\Rightarrow n = 5$$

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Q. 22 The sum of distinct values of λ for which the system of equations

$$(\lambda-1)x+(3\lambda+1)y+2\lambda z=0$$

$$(\lambda-1)x+(4\lambda-2)y+(\lambda+3)z=0$$

$$2x+(3\lambda+1)y+3(\lambda-1)z=0,$$

has non-zero solution, is _____

Option 1:
3

Option 2:
0

Option 3:
-3

Option 4:
None of these

Correct Answer:
3

Solution:

$$(\lambda-1)x+(3\lambda+1)y+2\lambda z=0$$
$$(\lambda-1)x+(4\lambda-2)y+(\lambda+3)z=0$$
$$2x+(3\lambda+1)y+(3\lambda-3)z=0$$

$$\begin{vmatrix}\lambda-1 & 3\lambda+1 & 2\lambda\\ \lambda-1 & 4\lambda-2 & \lambda+3\\ 2 & 3\lambda+1 & 3\lambda-3\end{vmatrix}=0$$

$R_1\to R_1-R_2 \quad \& \quad R_2\to R_2-R_3$

$$\begin{vmatrix}0 & 3-\lambda & \lambda-3\\ \lambda-3 & \lambda-3 & -2(\lambda-3)\\ 2 & 3\lambda+1 & 3\lambda-3\end{vmatrix}=0$$

$$(\lambda-3)^2\begin{vmatrix}0 & -1 & 1\\ 1 & 1 & -2\\ 2 & 3\lambda+1 & 3\lambda-3\end{vmatrix}=0$$

$$(\lambda-3)^2[(3\lambda+1)+(3\lambda-1)]=0$$
$$6\lambda(\lambda-3)^2=0\Rightarrow\lambda=0,3$$
$$\text{Sum}\ =3$$

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Q. 23 If $\vec{x}$ and $\vec{y}$ be two non-zero vectors such that $|\vec{x}+\vec{y}|=|\vec{x}|$ and $2\vec{x}+\lambda\vec{y}$ is perpendicular to $\vec{y}$, then the value of λ is _____

Option 1:
0

Option 2:
1

Option 3:
-1

Option 4:
-2

Correct Answer:
1

Solution:

$|\vec{x}+\vec{y}|=|\vec{x}|$

$\sqrt{|\vec{x}|^2+|\vec{y}|^2+2\vec{x}\cdot\vec{y}}=|\vec{x}|$

$|\vec{y}|^2+2\vec{x}\cdot\vec{y}=0 \qquad \ldots(1)$

Now $(2\vec{x}+\lambda\vec{y})\cdot\vec{y}=0$

$2\vec{x}\cdot\vec{y}+\lambda|\vec{y}|^2=0$

From (1)

$-|\vec{y}|^2+\lambda|\vec{y}|^2=0$

$(\lambda-1)|\vec{y}|^2=0$

given $|\vec{y}|\neq 0 \Rightarrow \lambda=1$

Q. 24 Consider the data on x taking the values $0, 2, 4, 8, \ldots\ldots, 2^n$ with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, \ldots\ldots {}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$, then n is equal to _____.

Option 1:
6

Option 2:
7

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Option 3:
8

Option 4:
9

Correct Answer:
6

Solution:

x	0	2	4	8	...	2^n
f	${}^nC_0$	${}^nC_1$	${}^nC_2$	${}^nC_3$	...	${}^nC_n$

$$\text{Mean } = \frac{\sum x_i f_i}{\sum f_i} = \frac{\sum_{r=1}^{n} 2^r \, {}^nC_r}{\sum_{r=0}^{n} {}^nC_r}$$

$$\text{Mean } = \frac{(1+2)^n - {}^nC_0}{2^n} = \frac{728}{2^n}$$

$$\Rightarrow \frac{3^n - 1}{2^n} = \frac{728}{2^n}$$

$$\Rightarrow 3^n = 729 \Rightarrow n = 6$$

Q. 25 The number of words (with or without meaning) that can be formed from all the letters of the word "LETTER" in which vowels never come together is _____.

Option 1:
60

Option 2:
90

Option 3:
120

Option 4:
None of these

Correct Answer:
120

Solution:

$$\text{LETTER}$$

$$\text{vowels } = \text{EE}, \text{ consonant } = \text{LTTR}$$

First place LTTR

$$_\text{L}_\text{T}_\text{T}_\text{R}_$$

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This can be done in 4! / 2! ways

Now, E,E can be places in any 2 of the 5 gaps

This can be done in $^5C_2$ ways

2 Es can be arranged in these 2 places in 2! / 2! ways

So, total= $\frac{4!}{2!} \times {}^5C_2 \times \frac{2!}{2!} = 12 \times 10 = 120$

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